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Editors

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Editors
Mohammad Shorif Uddin
Department of Computer Science
and Engineering
Jahangirnagar University
Dhaka, Bangladesh

Jagdish Chand Bansal
Department of Applied Mathematics
South Asian University
New Delhi, India

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Preface

This book contains high-quality research papers as the proceedings of the International Joint Conference on Advances in Computational Intelligence (IJCACI 2020). IJCACI 2020 has been jointly organized by Daffodil International University (DIU), Bangladesh; Jahangirnagar University (JU), Bangladesh; and South Asian University (SAU), India. It was held on November 20–21, 2020, at DIU, Dhaka, Bangladesh, in virtual mode due to the COVID-19 pandemic. The conference was conceived as a platform for disseminating and exchanging ideas, concepts, and results of the researchers from academia and industry to develop a comprehensive understanding of the challenges of the advancements of intelligence in computational viewpoints. This book will help in strengthening a congenial and nice networking between academia and industry. The conference focused on collective intelligence, soft computing, optimization, cloud computing, machine learning, intelligent software, robotics, data science, data security, big data analytics, and signal and natural language processing. This conference is an update of the first three conferences: (1) International Workshop on Computational Intelligence (IWCI 2016) that was held on December 12–13, 2016, at JU, Dhaka, Bangladesh, in collaboration with SAU, India, under the technical co-sponsorship of IEEE Bangladesh Section; (2) International Joint Conference on Computational Intelligence (IJCCI 2018) that was held on December 14–15, 2018, at Daffodil International University (DIU) in collaboration with JU, Bangladesh, and SAU, India; and (3) International Joint Conference on Computational Intelligence (IJCCI 2019) that was held on October 25–26, 2019, at University of Liberal Arts Bangladesh (ULAB) in collaboration with Jahangirnagar University (JU), Bangladesh, and South Asian University (SAU), India. All accepted and presented papers of IWCI 2016 are in IEEE Xplore Digital Library, and IJCCI 2018 and IJCCI 2019 are in Springer Nature Book Series Algorithms for Intelligent Systems (AIS).

We have tried our best to enrich the quality of the IJCACI 2020 through a stringent and careful peer-review process. IJCACI 2020 received 119 papers from 362 authors from eight countries of the globe. Only 43 papers were finally accepted (acceptance rate 36.13%) for presentation, and the final proceedings contain these 43 papers.

In fact, this book presents the novel contributions in areas of computational intelligence and it serves as a reference material for advance research.

Dhaka, Bangladesh Mohammad Shorif Uddin
New Delhi, India Jagdish Chand Bansal

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About This Book

This book gathers outstanding research papers presented at the International Joint Conference on Advances in Computational Intelligence (IJCACI 2020), which was held on November 20–21, 2020, at Daffodil International University. IJCACI 2020 was jointly organized by the Daffodil International University (DIU), Bangladesh; Jahangirnagar University (JU), Bangladesh; and South Asian University (SAU), India. The conference is conceived as a platform for disseminating and exchanging ideas, concepts, and results of the researchers from academia and industry to develop a comprehensive understanding of the challenges of the advancements of intelligence in computational viewpoints. This book will help in strengthening a congenial and nice networking between academia and industry. It is an update of the previous conferences IJCCI 2019, IJCCI 2018, and IWCI 2016. We are trying our best to enrich the quality of the IJCACI 2020 through a stringent and careful peer-review process. In fact, this book presents the novel contributions in areas of computational intelligence and it serves as a reference material for advance research.

The topics covered include: collective intelligence, soft computing, optimization, cloud computing, machine learning, intelligent software, robotics, data science, data security, big data analytics, and signal and natural language processing.

Dhaka, Bangladesh — Mohammad Shorif Uddin
New Delhi, India — Jagdish Chand Bansal

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About the Editors

Dr. Mohammad Shorif Uddin is Professor at Jahangirnagar University, Bangladesh. He completed his Doctor of Engineering in Information Science at Kyoto Institute of Technology, Japan, in 2002; his Master of Technology Education at Shiga University, Japan, in 1999; and his MBA at Jahangirnagar University in 2013. He is Editor-in-Chief of ULAB Journal of Science and Engineering and Associate Editor of IEEE Access and has served as General Chair of various conferences, including the ICAEM 2019, IJCCI 2018, IJCCI 2019, and IWCI 2016. He holds two patents for his scientific inventions, is a senior member of several academic associations, and has published extensively in international journals and conference proceedings.

Dr. Jagdish Chand Bansal is Assistant Professor (Senior Grade) at South Asian University, New Delhi, and Visiting Research Fellow of Mathematics and Computer Science, Liverpool Hope University, UK. Dr. Bansal received his Ph.D. in Mathematics from the IIT Roorkee. Before joining SAU New Delhi, he worked as Assistant Professor at the ABV-Indian Institute of Information Technology and Management Gwalior and at BITS Pilani. He is Series Editor of the book series "Algorithms for Intelligent Systems (AIS)," published by Springer; Editor-in-Chief of International Journal of Swarm Intelligence (IJSI), published by Inderscience; and Associate Editor of IEEE Access, published by the IEEE. He is the general secretary of the Soft Computing Research Society (SCRS). His chief areas of interest are swarm intelligence and nature-inspired optimization techniques. Recently, he proposed a fission-fusion social structure-based optimization algorithm, Spider Monkey Optimization (SMO), which is now being applied to various problems from the engineering domain. He has published more than 60 research papers in international journals/conference proceedings.

Chapter 1
Fully Blind Data Hiding by Embedding Within DNA Sequences Using Various Ciphering and Generic Complimentary Base Substitutions

Sajib Biswas and Md. Monowar Hossain

1 Introduction

Nowadays, communication with data through Internet is increasing rapidly because of great advancement in technology. Also data security has obtained more concern recently due to the rise in cyber-attack and the huge increment in data transfer rate over the Internet. Security of data means protecting data from unauthorized access, modification, sharing, or even viewing to allow only authorized users for such access. People send sensitive information through public transmission medium where security of data is a great issue and an enormous problem. To solve this problem, cryptography [1] and steganography are the ideal technique.

Cryptography is a technique where a plaintext is changed to a cipher text, so that only authorized user can read it and process it. Though it gives protection during the transmission step, this security is not guaranteed after posterior decryption. Also it is a method which draws suspicion by the attacker. On the other hand, steganography is a method where data is hidden in a digital media and that digital media is called cover media. It is a method which is looking innocent enough to trick the attacker, so it is almost impossible to recognize the message existence.

Day by day, steganographic methods are evolving rapidly because of various transmission digital media. The cover digital media includes picture [2], audio file [3], video streams [4], text file [5], and so on. In recent years, huge number of genetic data are available by public, as a result DNA sequence is now used as a cover media. There are many approach with DNA steganography like [6], where author propose a secure communication by using both cryptography and DNA encryption where key is embedded into DNA sequence, and after the key extracted, a secure communication is established. From that work, we got the idea to embedding the security key into DNA to ensure full blind extraction of message.

S. Biswas (✉) · Md. Monowar Hossain
Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Gopalganj, Bangladesh

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Our proposed method is an extenstion of [7]. In that work, author proposed a method where a keyword must trasfer to the reciver prior to the message, which can be problematic. Also author used default playfair cipher in that work, and as a result, the letters “I” and “J” are treated as same.

In our method, we represent a way which is mainly divided into two parts. One is for ciphering and another one is for embedding. The ciphering steps contain 10×10 Ppayfair cipher [8] and modified Caesar cipher [9]. By using 10×10 playfair cipher, the problem between “I” and “J” is solved, and modified Caeser cipher is used for security. In embedding parts, we perform generic complimentary base substitution and randomly inserts the cover DNA sequences into massage bearer DNA sequences and embed the key randomly. In receiver side, extraction process is done blindly because we also embed the key in sequence so when extraction is performed the receiver does not have to know anything prior like cover sequence or secret key. So we can call it fully blind extraction. Also in ciphering step, we use various ciphering which gives us more security along with the hiding algorithm.

2 Related Work

DNA steganography is evolved over times. Many work has been done in this field. In earliest days, some technique is developed to hide information into live molecular DNA. This opens many doors for experiment with biological sample, but there was biological errors such as mutation. Also few methods are proposed with digital form of DNA from database.

A method is proposed to use amino acid codons in [10]. In the proposed method, author used amino acid codons to represent 26 alphabets. In this work, author also used secret key which will have to send prior to the receiver.

In [11], the author proposed three methods where the sender and receiver have to know the cover sequence prior to the message. After the embedding process by the sender, the receiver will extract message with the help of prior sending cover sequence. A modification of this work is proposed in [12].

A modified table lookup method is developed in [13], where a modified scheme based on the table lookup substitution method is proposed. The proposed method uses an 8-bit binary coding rule to change DNA sequence into a binary format to improve the performance of the original TLSM.

To extract blindly, a method is proposed in [7], where sender and receiver do not need to know the cover sequences but receiver have to know the secret keyword to decipher playfair cipher text, so that keyword has to be sent prior to the encoded sequences. The problem with the scenario is that communicating that type of sensitive information could be seem fishy. Also another work developed in [8] expands the security of playfair cipher. In this work, author proposed a 10×10 model which gives more security than the traditional playfair ciphering which is developed with 5×5 and also cannot distinguished between letters I and J.

3 Biological Preface

DNA contains the biological information which makes every species unique. The information of genes reside in DNA molecules. DNA is made with nucleotides. The four types of nitrogen bases of nucleotide are adenine (A), thymine (T), guanine (G), and cytosine (C).

As shown in Fig. 1, A pairs with T and C pairs with G. So a sequence can be like AGCTAGCATTACGAT. Every three proximate base made a unit called codon. This codon are used to represent amino acids.

For conversion, Fig. 2 shows the adaption rule to change bases to binary and vice versa. According to the rule, A will change in binary as 00, C as 01, and so on. Again in generic complimentary base substitution, it will be differed from general base pairing rule which is also known as Watson-Crick base pairing.

In Fig. 3, it shows the rule for complimentary substitution. In Watson-Crick pairing, A gives T on the other hand for generic base pairing A gives C. In [11], author describes a method by which each base can be assigned a complimentary base as long as it follows a rule.

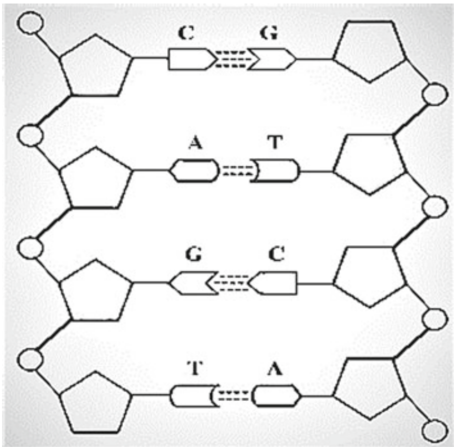

Fig. 1 DNA complimentary base pairing structure

Example:

AGCTAGCATTACGAT

001001110010010011110001100011

Base	Bits
A	00
C	01
G	10
T	11

Fig. 2 Digital coding of DNA bases

General Base Pairing	
Base	**Complement**
A	T
C	G
G	C
T	A

Generic Base Pairing	
Base	**Complement**
A	C
C	T
G	A
T	G

Fig. 3 A general and generic base pairing

$$C(b) \neq C(C(b)) \neq C(C(C(b))),\ b \in \{A, C, G, T\} \tag{1}$$

where b is the base and $C(b)$ is the complement of base. From this property, a different generic rule can be achieved. So it is possible for one to make his own complimentary base pairing rule to utilize his work.

Table 1 showed the mapping of 64 DNA codons into 26 alphabets though in the universal table of codons it does not contain the letters B, J, O, U, Xn and Z. In [10], author suggests a technique where B is assigned to 3 STOP codons, (O, U, X) share two codons from (L, R, S), respectively, and Z share one codon from Y. And START

Table 1 Mapping of DNA codons to alphabet

Alphabet	Codons	Alphabet	Codons
A	GCT, GCC, GCA, GCG	N	AAT, AAC
B	TAA, TGA, TAG	O	TTA, TTG
C	TGT, TGC	P	CCT, CCC, CCA, CCG
D	GAT, GAC	Q	CAA, CAG
E	GAA,GAG	R	CGT, CGC,CGA,CGG
F	TTT,TTC	S	TCT, TCC, TCA,TCG
G	GCT, GGC, GGA,GCG	T	ACT, ACC, ACA, ACG
H	CAT, CAC	U	AGA, AGG
I	ATT, ATC	V	GTT, GTC, GTA, GTG
J	ATA	W	TGG
K	AAG	X	AGT, AGC
L	CTT, CTC, CTA,CTG	Y	TAT
M	ATG	Z	TAC, AAA

codon is neglected by M. Also codon 'AAA' is not introduced in that method as a result many sequence which is formed by this codon will face some great problem when encoding and decoding will happen, so we add it to Z for avoiding any kind of problem that occurs during encoding and decoding process. The author does not suggest anything to replace J because in traditional cipher I and J are supposed to be same. But in our proposed method, we use 10 × 10 playfair cipher where I and J are different. So we make an adjustment by sharing one codon from I to J. After all these replacement, Table 1 is structured.

4 Methodology

4.1 The Encryption Stage

It is the sender stage where sender have to go through some steps to encrypt the message and embedded it into a cover DNA sequence. The description of those steps in Fig. 4 are given in this section.

4.1.1 The Ciphering Module

10 × 10 Playfair Cipher

Playfair cipher is the first step of our proposed method which can also be considered as preprocessing step because it works with encrypts pairs of letters called duplet. Many methods are proposed for this ciphering technique. In [10], a method is suggested

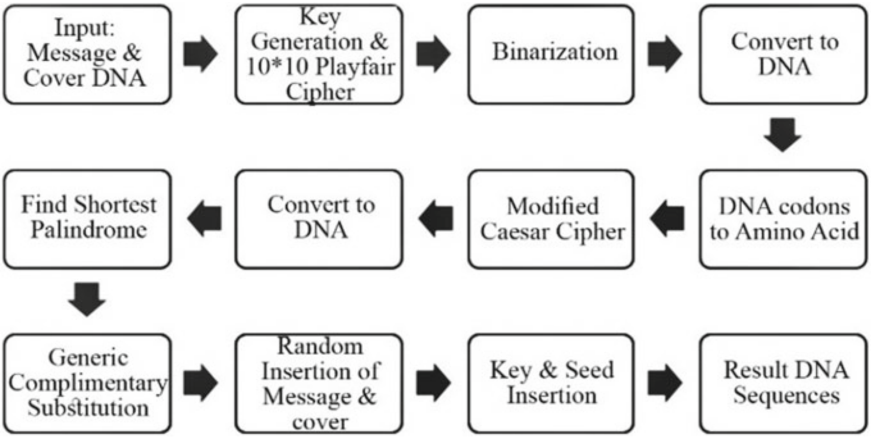

Fig. 4 Encryption stage methodology model

where any kind of binary data such as text, images, and so on can be ciphered. In our suggest method, we are working with text data, so we adapt a method proposed in [8], where the encryption key and message can be uppercase, lowercase, number, and special characters. So the problem of traditional playfair [14] are removed in this method, and it also gives more security from the traditional playfair ciphering.

In our work, we use this method directly in message to make a ciphertext, after that, the DNA coding rule is implemented, from those codons, we obtained uppercase alphabets, and to find those codons, we use ambiguity which is represented as DNA base.

Modified Caesar Cipher

Caesar ciphering [15] is a very popular method where every alphabet in a text is converted by a shifting value. To modify this cipher, many methods are suggested like [9] where author gives a model by which the ciphertext will be readable, so it will not look suspicious. For this work, he used frequency alphabet in Indonesian text. In our work, we take liberty to adapt a method where random numbers are generated with the help of keyword which is used as seed. With every alphabet, we add those random number and get a ciphered alphabet.

From previous steps, we only get uppercase alphabet, so Caesar cipher with only uppercase is more suitable method to save time and space.

4.1.2 The Embedding Module

In this step, we hide the message in cover DNA, and after that stage, the resultant DNA and the cover DNA will be embedded randomly to ensure the blind extraction. Also we will use a DNA palindromic sequence to show end of the message.

Generic Complimentary Base Substitution

This substitution phase converts the cover DNA sequence with respect to message DNA. If the cover sequence ($\mathrm{DNA}_{\mathrm{cover}} = c_1; c_2;\ldots; c_\mathrm{n}$) and the message sequence ($\mathrm{DNA}_{\mathrm{msg}} = m_1; m_2;\ldots.; m_n$) and $|\mathrm{DNA}_{\mathrm{cover}}| > ||\mathrm{DNA}_{\mathrm{msg}}|$ then the resultant DNA sequence ($\mathrm{DNA}_{\mathrm{new}} = c_1'; c_2';\ldots; c_n'$). For this conversion, we follow a rule.

$$\text{Message Base}\begin{cases} A \rightarrow \mathrm{DNA}_{\mathrm{cover}} \\ C \rightarrow \mathrm{C}(\mathrm{DNA}_{\mathrm{cover}}) \\ G \rightarrow \mathrm{C}(\mathrm{C}(\mathrm{DNA}_{\mathrm{cover}})) \\ T \rightarrow \mathrm{C}(\mathrm{C}(\mathrm{C}(\mathrm{DNA}_{\mathrm{cover}}))) \end{cases}$$

To follow this rule, in Fig. 3, generic base pairing will be used. If we have C in message base and G in cover, then the new stego base DNA will be A. For other base, it will change by following the rule. When $|DNA_{cover}| = |DNA_{msg}|$ after that all DNA_{cover} will be append in DNA_{new} without checking with the message because after that length no message will be considered.

End-of-Message Sequence

Finding end of message is very important because after that sequence will not consider the DNA as a message. That is how the time and space will be saved. To solve this problem, we use a palindromic sequence at least of length 8 because almost all DNA sequence have at least a palindrome. But the idea of palindrome here is somehow different than traditional idea. Here a sequence of DNA is called palindrome if the complimentary sequences of the DNA is equals to the DNA backwards.

For example, a sequence TAATTA is palindrome because the compliment of DNA is ATTAAT which is equal to the DNA backward. To make this sequence unique, we use padding with G in both side of DNA. So the example DNA will look like GTAATTAG. This padded DNA will be added in last of the message bearer DNA.

In our work, we use G in both side for padding, but it can be any base not only G. Also we use shortest palindrome which length is at least 8, but anybody can use longest palindrome in sequence but then the length will be larger than this method.

Random Insertion Phase

Before this phase, we have two DNA sequences, one is the real cover, and other is the processed cover. So in this step, we will randomly merge two DNA into one. We generate two seeds with the key, and from the seed, we generate two pseudo-random function to merge. The length of the processed message should be same in this stage. We use the method suggested by [11] for insertion technique. If the length of cover $|DNA_{cover}|$, then the new length of message will be $2 \times ||DNA_{cover}|$. In this process, we use pseudo-random numbe,r so that inverse of this process will be possible. After this step completed, we have to embed message into the DNA sequences.

Key Insertion Phase

Like previous stage, a random function will be generate with a seed. And the key will be converted to DNA and placed inside the previous DNA in random position. To find the seed of this random number, we insert a small DNA sequence in some fixed position, so that we can extract it blindly. After this stage, the length of final DNA will be $2 \times |DNA_{cover}| + |DNA_{key}| + |DNA_{seed}|$. The example of all these phases are shown in Fig. 5.

Input:
Msg: Encrypt
Cover:
CTTAATTATGCTGGTGAAGTCCCGAGAGAGGAGGAAGATGGATTGGTTGCCTGGTTTGCTG

Key = uvEgoq#.i_XnLA]C}1lI
After Playfair = vLbs~hh1
After amino Acid Codon = MZJJHYIZGIZ
After Caesar = DXTQZMDFYUA
Msg DNA = GATAAGTAACTACAAATACCATGAGATATTTATATGAGAGCTA
Palindrome = TAATTA
After GCBS = GTCAAACATAATAGTGGAAGCAGGTGGGGTTATGGTGTTCAGT
After Adding Palindrome =
GTCAAACATAATAGTGGAAGCAGGTGGGGTTATGGTGTTCAGTTTAATTA
TTGGTTTGCTG
After Random Insertion
=GTCAAACATAATAGTGGAACTTAATTATGCTGGTGCGAAGTCCCGAGAGAGGAGGAAGGTGG
GGTTATGGAGATGGATTGGTTGTTCAGTTGGTTGCCTGCCTGGTTTGGTTTGCTGTGCTG
Enter key and seed:
GTCAAACATAATAGTGGAACTTAATTATGCTGGTGCGAAGTCCCGAGAGAGGAGGAAGGTGGG
GCTGGTGCGAAGTCCCGAGAGAGGAGGAAGGTGTTATGGAGATGGATTGGTTGTTCAGTTGGT
TGCCTGCCTGGTTTGGTTTGCTGTGCTGGGGTTATGGAGATGGATTGGTTGTTCAGATG

Fig. 5 A detail example of internal steps of encryption stage

4.2 *The Extraction Stage*

Extraction process is opposite of Fig. 4 process. It starts from finding seed for key and next does the opposite of insertion phase. Then, after doing GCBS and deciphering process, we got the message (Fig. 5).

4.2.1 The Key and Cover Recovery Stage

First in this step from some fixed point, we recover sequence for seed to find the key. After finding the key now, we have two seed from the key which will use to separate DNA_{cover} and DNA_{msg}. Those two sequence shared same length, so the new message length will be $(|DNA_{result}| - |DNA_{key}| - |DNA_{seed}|)/2$.

4.2.2 The Message Recovery Stage

From previous stage DNA_{cover}, we have to find the shortest palindromic sequence. This step is necessary because all DNA_{msg} does not carry message, so if we find end of the message signal, then after all those sequence can be dropped without checking. So that palindromic sequence is checked with the DNA_{msg} when we get the match we will drop all sequence after that, along with the palindromic sequence.

Input:
GTCAAACATAATAGTGGAACTTAATTATGCTGGTGCGAAGTCCCGAGAGAGGAGGAAGGTG
GGGCTGGTGCGAAGTCCCGAGAGAGGAGGAAGGTGTTATGGAGATGGATTGGTTGTTCAGT
TGGTTGCCTGCCTGGTTTGGTTTGCTGTGCTGGGGTTATGGAGATGGATTGGTTGTTCAGATG

After removing seed & key =
GTCAAACATAATAGTGGAACTTAATTATGCTGGTGCGAAGTCCCGAGAGAGGAGGAAGGTG
GGGTTATGGAGATGGATTGGTTGTTCAGTTGGTTGCCTGCCTGGTTTGGTTTGCTGTGCTG
Key = uvEgoq#.i_XnLA]C}1lI
Cover:
CTTAATTATGCTGGTGAAGTCCCGAGAGAGGAGGAAGATGGATTGGTTGCCTGGTTTGCTG
Palindrome = TAATTA
After Removing Palindrome =
GTCAAACATAATAGTGGAAGCAGGTGGGGTTATGGTGTTCAGTTTAATTATTGGTTTGCTG
After Inverse GCBS = GTCAAACATAATAGTGGAAGCAGGTGGGGTTATGGTGTTCAGT
Msg DNA = GATAAGTAACTACAAATACCATGAGATATTTATATGAGAGCTA
After Caesar decipher= MZJJHYIZGIZ
After Playfair decipher = Encrypt

Fig. 6 A detail example of internal steps of extraction stage

$$\text{Stago Base}\begin{cases} \text{DNA}_{\text{cover}} \rightarrow A \\ \text{C}(\text{DNA}_{\text{cover}}) \rightarrow C \\ \text{C}(\text{C}(\text{DNA}_{\text{cover}})) \rightarrow G \\ \text{C}(\text{C}(\text{C}(\text{DNA}_{\text{cover}}))) \rightarrow T \end{cases}$$

The inverse GCBS will be performed according to the rule. It is done by comparing each stego base with corresponded to cover DNA. If we have T in $\text{DNA}_{\text{cover}}$ and A in DNA_{msg}, then after reverse GCBS, we get G.

4.2.3 The Deciphering Stage

After we find new message DNA, we will apply modified Caesar cipher deciphering technique by reducing the random value generated by seed from the secret key. Then from amino acid codon table, we will find the DNA, and after binarization, we are ready for playfair decipher. From the key and encrypted message, we will get plaintext after this stage and that plaintext will be the message send by the sender. In Fig. 6, the extraction process is described briefly.

5 Performance Analysis

5.1 Hiding Efficiency

The performance of steganogarphic method is computed by hiding capacity. In DNA, the capacity is measured by bit per nucleotide (bpn). From our proposed method, we can hide any sequence as long as it is equal or less than the cover DNA length. Also in our method for every three codons, we use an ambiguity to detect the codons in amino acid table. So in every four base, only three contains message, and other is ambiguity. So if the length of cover DNA is $|\mathrm{DNA}_{\mathrm{cover}}|$, then the efficiency of the algorithm will be:

$$\text{Efficiency} = \frac{\frac{3}{4} \times |\mathrm{DNA}_{\mathrm{cover}}| \times 2}{|\mathrm{DNA}_{\mathrm{cover}}|} = 1.5\,\text{bpn}$$

5.2 Security

The overall security of the proposed algorithm is divided in some stage. If any attacker want to discover the message, then he have to know these following information:

1. The randomly generated secret key.
2. The playfair ciphering matrix sequence.
3. The binary coding scheme.
4. The randomly generated number for Caesar ciphering.
5. The complementary rule.
6. The randomly generated number and two seeds for insertion phase.
7. The positions for hiding key in sequence.

This proposed method provide more security than [7]. Because the secret key may be known or missing some value when sending prior to the message, but in our method, it is embedded into DNA sequence itself. Also by using 10×10 playfair cipher t,he security of the data and risk of losing some data during extraction is solved, and for extra security, we use modified Caesar cipher. The detailed description of the security for every point is described below.

Regarding the first point, for every position of the key, maximum number of try will be 94 because it is the total number of alphabets, numbers, punctuations, and special characters used for key. So the total number of guess will be 94^{lk}, where '*lk*' denotes the length of key.

In second point, for first position, maximum number of try will be 100, and for second position, it will be 99 and so on, that is why total number of guess to find correct matrix sequence will be 100!

For point three, author of [7] shows that number of possible try to find correct binary coding scheme is $4! = 24$.

Regarding point four, the maximum number of guess for every position of the message will be 26 because in our work we only used uppercase alphabet in Caesar ciphering. So the total number of guess will be $26^{\,lm}$, where '*lm*' length of message.

For point five, there are only six legal complimentary rules in this method.

For point six, author of [7] showed a method that the number of guesses will be $\left(2^{|\mathrm{cover}|} - 1\right)^2$, where |cover| means length of cover.

Regarding point seven, the key can be hidden in “2 × |cover|” number of random positions. And to represent one length of key , we need three DNA base. So number of guess will be $^{2*|\text{cover}|}P_{3*\text{lk}}$.

Now, the probability of successfully cracking this method will be:

$$P = \frac{1}{94^{lk}} \times \frac{1}{100!} \times \frac{1}{24} \times \frac{1}{26^{lm}} \times \frac{1}{6} \times \frac{1}{\left(2^{|\text{cover}|} - 1\right)^2} \times \frac{1}{\binom{2 \times |\text{cover}|}{3 \times lk}}$$

where ‘*lk*’ can be anything and |cover| may very big like hundred thousand, so it is almost impossible to find the hidden message if anyone do not have information.

6 Result and Comparisons

In this section, we evaluate the performance of our proposed model. In Table 2, we show the payload of seven DNA sequence as a cover sample. Every sequence is identified by an accession number from NCBI database and every sequence is in FASTA format. And a sample text of 25 KB is used as secret message. Also the length, shortest palindrome, maximum capacity of cover, and the payload are shown on the table.

In this experiment, we compare our proposed method with other DNA based hiding method. We follow three attribute to differentiate between methods. They are capacity, blindness, and level of blindness.

From Table 3, our approach shows a greater hiding capacity then [11]. Though capacity of our method is same as [7] but in level of blindness, our method outranks [7].

In Fig. 7, the pictorial description of Table 2 in shown. Here, we can see if the length of cover is much bigger than message, then the actual payload is small, and if both of them are almost same, then payload is maximum.

Table 2 Payload of hiding 25 KB in different sequence

Sequence	Length	Palindrome	Max capacity (KB)	Actual payload (%)
NC_005118.4	164864	TGAATTCA	30.185	82.82
NC_041767.1	170001	GAAATTTC	31.126	80.32
NC_041764.1	172001	AAGCGCTT	31.492	79.38
NC_000016.11	210001	ATTTAAAT	38.449	65.02
NC_000017.11	170557	AGAATTCT	31.228	80.05
NC_000086.7	150750	GGCTAGCC	27.601	90.58
NC_041772.1	171001	TTCATGAA	31.309	79.85

Table 3 A comparison between different hiding methods

Author	Approach	Capacity	Blind?	Fully Blind?
Shiu [11]	Insertion method Complementary method Substitution method	0.58 0.07 0.82	No No No	No No No
Khalifa [7]	Generic complimentary-based substitution	1.5	Yes	No
Proposed method	Generic complimentary-based substitution with key embedding	1.5	Yes	Yes

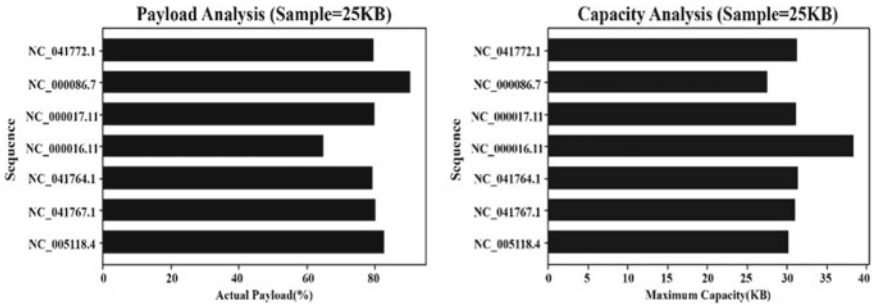

Fig. 7 Payload and capacity analysis

Table 4 Security comparison between different hiding method and proposed method

Approach	Probability of successful attack
Insertion method [11]	$\frac{1}{1.63\times 10^8} \times \frac{1}{n-1} \times \frac{1}{24} \times \frac{1}{2^m-1} \times \frac{1}{2^{\text{cover}}-1}$
Complimentary method [11]	$\frac{1}{1.63\times 10^8} \times \frac{1}{24^2}$
Substitution method [11]	$\frac{1}{6} \times \frac{1}{(2^{\lvert\text{cover}\rvert}-1)^2}$
GCBS [7]	$\frac{1}{(2^{\lvert\text{cover}\rvert}-1)^2} \times \frac{1}{6} \times \frac{1}{24}$
GCBS with key embedding (Proposed method)	$\frac{1}{94^{lk}} \times \frac{1}{100!} \times \frac{1}{24} \times \frac{1}{26^{lm}} \times \frac{1}{6} \times \frac{1}{(2^{\lvert\text{cover}\rvert}-1)^2} \times \frac{1}{\binom{2\times\lvert\text{cover}\rvert}{3\times lk}}$

So from Table 4, we get an overview of security given by different methods where we show probability of an attacker succeeded to find the message.

7 Conclusion

In our work, we manipulate some basic properties of a DNA sequence to hide the data. We mainly do this work in two level, in first, we hide the data into DNA, and in second, we merge the message DNA and cover DNA along with the secret key DNA. Also for giving protection and remove any chance of data loss in decryption, we use 10×10 playfair cipher. We use Caesar cipher to give extra security to data. For blind extraction ,we embedded the secret key in DNA sequence for ensuring fully blind extraction.From experiment result, our proposed method give higher capacity and ensure fully blind extraction than other methods. Also our method gives more security than other blind extraction methods.

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Chapter 2
A Real-Time Health Monitoring System with Wearables and Ensemble Learning

Nishat Ara Tania, Md. Mosabberul Islam, Jannatul Ferdoush, and A. S. M. Touhidul Hasan

1 Introduction

Human life expectancy is increased day by day, and it is increased near about five years with the help of modern health science [1]. Advancement of health science increases the treatment costs significantly, and it is due to the cost addition of disease diagnosis. However, in Bangladesh, many people die without knowing their disease, and especially in rural areas, it is critical because of minimal access to the healthcare service. In Bangladesh, one doctor gives treatment to every 1,847 people [2]. Thus, the people who live in rural areas do not get the quality of medical facilities.

The traditional health services take time to make appointments with a doctor and laboratory test. People cannot go to the hospital for a daily checkup in rural areas, and it is costly. Moreover, it is difficult to predict health disease in real time at home. Therefore, in the paper, we propose a health monitoring system (HMS) to monitor patients with wearable medical sensors (WMSs) and machine learning ensembles. WMSs collect physiological features, i.e., ECG signals, heartbeat, and body temperature from the patient's body and send it to the ensemble engine that predicts the disease. This application will help to identify the disease in real time.

In this proposed HMS, we use machine learning algorithms (MLAs) because it learns to classify if a user is sick or not by measuring the statistical difference between

N. A. Tania · Md. M. Islam · J. Ferdoush · A. S. M. T. Hasan (✉)
Department of Computer Science and Engineering, University of Asia Pacific, Dhaka, Bangladesh
e-mail: touhid@uap-bd.edu

N. A. Tania
e-mail: 16101095@uap-bd.edu

Md. M. Islam
e-mail: 16101055@uap-bd.edu

J. Ferdoush
e-mail: 16101079@uap-bd.edu

M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems,
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a class's values in a given dataset. The system has seven MLAs and four ensemble methods to justify the model's best-fit algorithms and ensemble method.

This research aims to reduce health service costs, overcome traditional health service time, and ensure the quality of healthcare services for rural areas. The proposed system will support real-time health monitoring services for older adults and rural people.

In this paper, we propose an HMS along with WMSs. The main contributions of this paper are summarized as follows:

(1) We exhibit HMS, which is based on WMSs, and the prediction is made from ensemble methods.
(2) We propose a pre-trained model that compares with the received WMSs data and provides the prediction for real-time health monitoring.

The rest of this paper is structured as follows. The studies on the health monitoring system are reviewed in Sect. 2. In Sect. 3, we present the details of the proposed architecture of HMS. The experimental results are discussed in Sect. 4, and the paper is concluded in Sect. 5.

2 Background and Related Work

Android mobile app and wireless body sensors exert to keep track of patients' bodies' physiological signals such as temperature, heart rate, and oxygen saturation level in blood [3]. When the patient is admitted to the hospital intensive care unit or discharges, a doctor can monitor their physical condition. This system sends alerts to the doctor or paramedics. WMSs based health monitoring systems can send messages, emails, emergency calls, and reminders to the care supplier, rely on the intensity of the emergency by using alert systems [4, 5].

In [6, 7], the authors introduced a health monitoring system that collects various sensor data, i.e., ECG signal, situation of oxygen in blood (SPO2), blood pressure, and heartbeat signal. They also developed a Web-based application that communicates with doctors and paramedics. The expert doctor prescribes to the patient. Recently, health monitoring systems integrate the Internet of things (IoT) and machine learning algorithms to predict patients' diseases [8]. However, it does not provide real-time notification to the user.

2.1 Background

In this paper, the proposed health monitoring system uses the ensemble machine learning approach on WMSs data to match the pre-trained model, which gives a real-time prediction about the disease.

2.2 Wearables

WMSs are suitable for fast progress in computing, low-power sensing, and communication. Sensors are worked for receiving data, record, dispatch physiological data orderly, efficiently, and effectively. Business Insider has published a report that wearable sensors were sold more than 56 million in 2015; after three years, the number of sales rose to 123 million [9]. WMSs accumulated physiological signals, i.e., body temperature (BT), blood pressure (BP), heart rate (HR), respiration rate (RESP), electroencephalogram (ECG), galvanic skin response (GSR), oxygen saturation (SpO2), electrocardiogram (EEG), body mass index (BMI), and blood glucose (BG) [10–12].

WMS-based body-area networks (BANs) are drawn much attention in research. Communication protocols, transmission bandwidths, and obstruct unauthorized access have been considered and analyzed for BANs [13, 14]. The mobile-based health projects monitor the patients' health with quality of service, communication, and security [15]. The health monitoring system has been substantiated in many healthcare environments, such as fitness tracking hypertension monitoring by the possibility of using WMSs [16].

Simple multi-threshold mechanisms are used to track extraordinary unwanted signals by alert systems [16–18]. Advanced care and alert portable telemedical monitor called AMON express alerts when the gathered bio-signals are out of the pre-defined range [17]. A threshold-tuning method uses the same technique for real-time physiological exploration systems used for fitness tracking [19].

2.3 Machine Learning Algorithms

Logical models are building by MLAs that algorithm qualifies computers to learn to build analytical models. Successful elimination of instructive features is very challenging but highly crucial for MLAs. For example, in the ECG signal, the multifractal dynamics create very hard to precisely model heartbeats applying the current R–R interval feature [20].

In this paper, we focus on medical decision problems and the usefulness of supervised MLAs for solving these problems. We apply seven MLAs and four ensemble techniques for solving in the healthcare domain decision problems. K-nearest neighbor, radial basis function (RBF) kernel, Naive Bayes, decision tree regression, logistic regression, decision tree classification, random forest regression, and support vector regression (SVR) [21] are applied to make the real-time decision system. We divided these seven MLAs into three categories; there are probabilistic, similarity-based, and error-based. Similarity-based MLAs determined that the incoming data similar to which data elements such as the k-nearest neighbor algorithm predict that the incoming data belongs to the lowest distance from the given data elements. Euclidian distance/Manhattan distance formula can find the nearest distance [22]. Probabilistic

MLAs predict that incoming data instance labels between feature values and labels on their probabilistic relationships [23].

Ensemble methods combine different MLAs for more robust performance than the one MLA can. Naive Bayes, AdaBoost, bagger, and voting classifier ensemble methods are used for the proposed HMS.

3 Hierarchical HMS Structure

In this section, we describe the proposed HMS structure in detail. There are three individual parts, as described in Fig. 1. The first part is collecting data from the patient's body using the WMSs.

The second part of the method is data processing, whereas it has two individual steps; the first step separates the data for finding a specific disease. In the second step, the divided data come into the given numerical range by the system.

The last part is given a prediction of the patient's physiological signals using machine learning ensembles. The prediction table has primary level prediction outputs then finalizes a prediction using ensemble methods. Finally, classify the disease with appropriate medical predictions.

Due to the time limit and some other circumstances, we are using the UCI biomedical dataset [24–26] and Kaggle dataset [27] instead of sensor data in the first part of the proposed HMS, but the framework will be the same.

3.1 Health Monitoring System

The proposed HMS will use WMSs to take multiple disease data. The decision flow of the HMS using five steps, as shown in Fig. 2. The steps are patient physiological signals, storing the signals, data pre-processing, predicting results, and predicting results converts into a binary value. These five steps can identify multiple diseases in parallel.

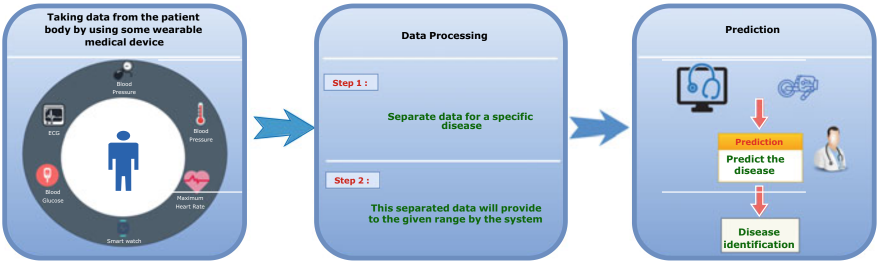

Fig. 1 Schematic diagram of HMS

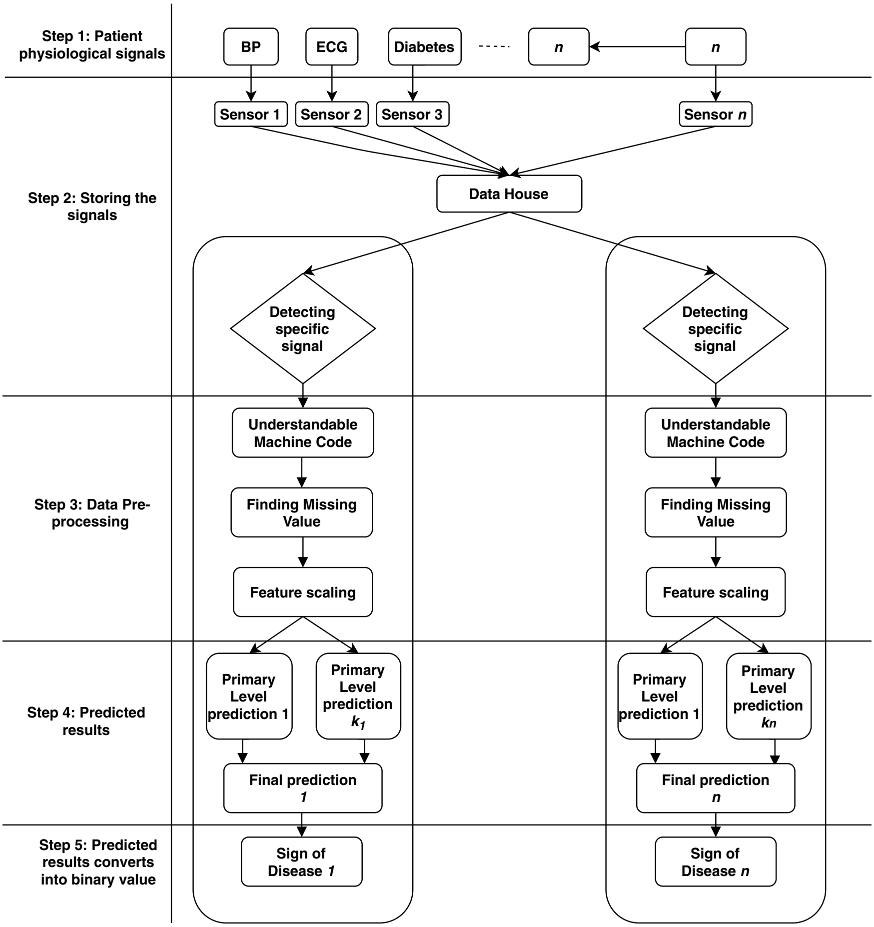

Fig. 2 Framework of HMS

The first two steps collect the physiological signals using WMSs, such as blood glucose, blood pressure, heart rate, body temperature, and ECG signal.

The third step stores the physiological signals in the data house. There is a detecting phase to point out the accurate physiological signals which are needed, and the physiological signal data are pre-processed into a format. String data convert into an integer value that is understandable by machine learning algorithms, and then we find the missing value. If any errors occur, then it corrects it using the mean statistical method. Separating data distributes various physiological data into different groups.

In the fourth step, the pre-processed data trained by machine learning algorithms consist of primary level prediction. Initial prediction is made from different base learners, i.e., K-nearest neighbor and logistic regression. The final prediction consists of a meta learner with an ensemble method, i.e., AdaBoost, and a voting classifier.

The final result is ensemble model prediction based on the primary level prediction of 1 to k_n.

In the fifth step, predicted results convert into binary value for classification of the signals. Sign one (1) indicates the user has a disease, and sign zero (0) shows the user is healthy.

3.2 *Disease Classification*

Figure 3. describes the disease classification of the proposed system. Four individual models are employed to identify disease, and each approach integrates MLAs to train the model to identify the specific disease and ensemble model that predicts the final result. Each model trains and tests three MLAs individually and gives three results; these results send to the ensemble algorithm. The proposed HMS has applied four ensemble methods, where four ensemble methods have produced four outputs for each disease. The best result of these four results selects for the final disease prediction. The final ensemble outputs are combing into a binary result that predicts that a user is sick or not sick.

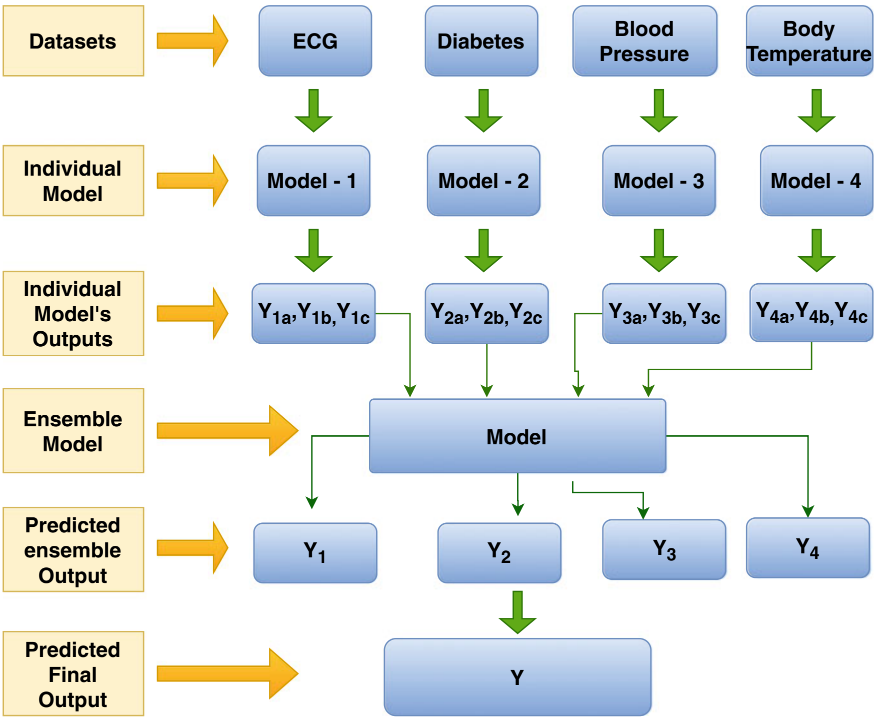

Fig. 3 Ensemble model of HMS

Algorithm of our Ensemble Model:

Input:
D = $((x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4))$
C = number of ensemble method and dataset(C=4)
B = List of base method(sizeof(B) = 7)
E = List of Ensemble method
Declare S = list(), S' = list();
Initialize acc = 0;
Output: A composite model, M
Procedure:
1. for i = 1 to C
2. **for** j = 1 to 127
3. **for** k = 0 to 6
4. **if** j & (1<< k)
5. S = B_k
6. **endif**
7. **endfor**
8. **if** sizeof(S) == 3
9. **if** acc < Acc f(S)
10. S' = S
11. acc = Acc $f(S)$
12. **endif**
13. **endif**
14. **endfor**
15. M = S'
16. ensemble $f(E_i$, M)
17. endfor
18. end procedure

(**Model-1** Arrhythmia Data Classification). This model has three machine learning algorithms for arrhythmia. ECG data are using for this classification.

1. **K-Nearest Neighbors (KNN):** This MLA finds the nearest point of dataset's distance to take decisions [28].
2. **Support Vector regression (SVR):** It is used for continuous values in regression. SVR tries to minimize errors and fit into a threshold line [29].
3. **Radial basis function kernel (RBF):** This method's result depends on the distance from origin or some point. [30].

Model-2 (Diabetes Data Classification). This model has also three machine learning algorithms.

1. **Logistic Regression:** All the hypothesis values convert into 0 and 1 value that makes the result perfect and take it in a range [31].

2. **Decision Tree Regression:** When values are numeric or continuous, then this type of regression is needed [4].
3. **Random forest regression:** It is an ensemble of various nonlinear regression and various regression trees.

Model-3 (Hypertension Data Classification). We use three MLAs; there are decision tree regression, KNN, and decision tree classification. Blood pressure data are using for this classification.

Decision Tree Classification: This algorithm works on smaller subsets and finds the highest gain in those sets [4].

Model-4 (Fever Data Classification). In this model, we used SVR liner kernel, decision tree regression, and logistic regression that we discussed in the previous section. Body temperature data are using for this classification.

Ensemble Model (Final Prediction). In this final model, we used four ensemble methods. Describing them in below:

1. **Voting Classifier:** It is a model in machine learning used in the ensemble of frequent algorithms and predicts output based on the maximum probability of class [32].
2. **Bagger:** This method is constructed to raise the durability and performance of MLA's.
3. **AdaBoost:** It is an ensemble learning technique that makes a robust classifier from multiple weak classifiers [33].
4. **Naive Bayes Classifier:** This ensemble method creates many algorithms where every algorithm shares a general ethics [34]. Every couple of features are categorizing as individual of each other.

4 Experimental Results and Analysis

In this section, the present experiments on real-world datasets to measure the effectiveness of the proposed method.

4.1 Dataset

Table 1 presents the dataset of different diseases with their number of instances, features, and classes. We have used the following datasets, i.e., UCI biomedical dataset [24] for Arrhythmia classification to recognize arrhythmia, Kaggle Pima Indians [27] for diabetes classification, Kaggle cardiovascular disease dataset [25] used in hypertension classification, and UCI wearable stress and affect detection (WESAD) [26] for fever classification. WESAD has 6,30,00,000 instances and 12 features, but we have used 1,00,000 instances and three feature filtering features.

Table 1 Datasets for ensemble model generation

Disease name	Dataset	Instances	Features	Classes
Arrhythmia	Arrhythmia	452	279	16
Diabetes	Diabetes	70	9	2
Hypertension	Cardiovascular disease	70000	13	3
Fever	WESAD	6300000033	3	3

4.2 Experiments

Figure 4 presents the classification accuracy of the ensemble method for ECG classification, diabetes classification, hypertension classification, and fever classification. The proposed HMS uses multiple MLAs, and all these MLAs' results send to the ensemble methods. The ensemble method provides better prediction than individual MLA. That is why the proposed HMS provides a better result. In the four ensemble method, voting classifier gives the best result. The best accuracy of arrhythmia is 97.85%, diabetes is 93.46%, hypertension is 99.85%, and fever is 99.85%. The final prediction of the complete ensemble model accuracy is 96.63%. This final prediction is the status of the patient sick or not.

We take account of the evaluation matrix to check the adequacy of the ensemble model. Table 2 is the evaluation matrix of four disease datasets classification, and the F1 score (F1) dimensions the total performance of precision and recall. Recall indicates an affirmative answer, and how does it predict a positive result.

The recall is capable of measuring the primary learner to identify disease. Precision indicates that it predicts an affirmative answer, and how does it be right.

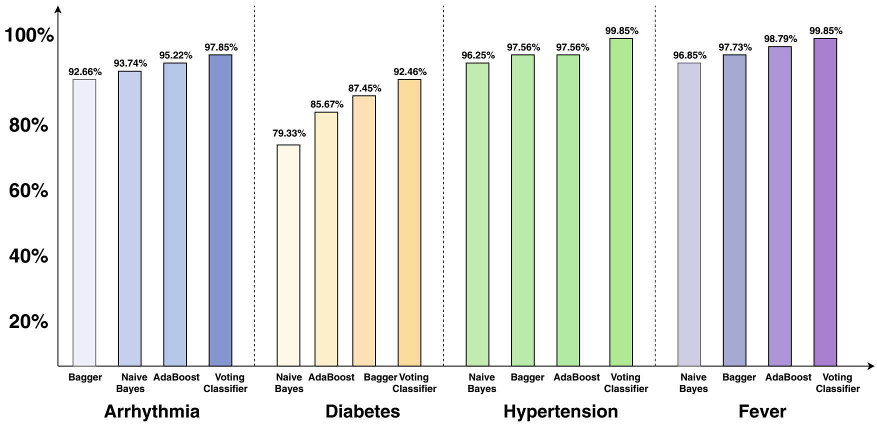

Fig. 4 Accuracy of individual ensemble method for arrhythmia, diabetes, hypertension, and fever

Table 2 Evaluation matrix

Name	Arrhythmia (%)	Diabetes (%)	Hypertension (%)	Fever (%)
F1 score (F1)	97.85	92.46	99.85	99.85
Recall	97.85	92.46	99.85	99.85
Precision	97.85	92.46	99.85	99.85
Accuracy (ACC)	97.85	92.46	99.85	99.85

4.3 Discussion

In the proposed research, several machine learning algorithms and the ensemble method are applied to find the best outcome, whereas the voting ensemble model with feature scaling performs well. We train the model with labeled datasets. Firstly, we embedded our datasets; after that, we find out the part of the empty columns. We marked it with missing data; this empty part fills up with a number that is the average value of that column which is called mean error analysis. Finally, we normalize the data; it is defined as feature scaling. This distinct range is each class of the dataset. Figure 4 presented the result using a heterogeneous ensemble technique with feature scaling based on the voting classifier.

5 Conclusion

In this paper, HMS combines WMS and MLAs with a hierarchical structure supported by robust machine learning ensembles. The proposed HMS can identify various diseases in parallel; people could monitor their physical status at home; it will reduce medical healthcare services' costs, especially in rural areas. The system demonstrated effective disease classification accuracy through physiological data acquired from WMSs: arrhythmia (97.85%), diabetes (92.46%), hypertension (99.45%), and fever (99.45%). The accuracy of the entire model performance is 96.63%, which predicts that a patient is sick or not by converting the result in yes(1) or no (0).

In this research, we accomplished HMS only for four diseases. However, many diseases are essential to recognize in real time. We predict disease status for health monitoring, but we do not have a clinical decision support system (CDSS) to provide more robust patients suggestions after predicting disease. In future research, we will predict various significant diseases and generate automated advice by building CDSS.

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Chapter 3
Statistical Texture Features Based Automatic Detection and Classification of Diabetic Retinopathy

A. S. M. Shafi, Md. Rahat Khan, and Mohammad Motiur Rahman

1 Introduction

Diabetic retinopathy is one of the main causes of global blindness in the economically active population because it affects people between the ages of 20 and 74 [1, 2]. It causes other problems like stroke, cardiovascular disease, diabetic nephropathy, and diabetic neuropathy. The number of people with diabetic retinopathy is growing higher day by day. This number will grow from 126.6 million to 191.0 million by 2030 and the number with vision-threatening diabetic retinopathy (VTDR) will grow from 37.3 million to 56.3 million if no appropriate action is taken [3]. The rapid increase of diabetics especially in low-income countries, shortage of expert ophthalmologists and resources, high cost of treatment, and time spent on labor-intensive screening are some of the main limitations of manual DR detection. This helps to increase the number of patients with DR [4]. Regular eye examination is the only way for the detection and treatment of DR which can reduce the possibility of severe blindness at risk by 50% [5]. There are two types of clinical DR: non-proliferative diabetic retinopathy (NPDR) or background retinopathy and proliferative diabetic retinopathy (PDR). Unfortunately, the DR is usually identified in advanced stages (PDR), with unfavorable forecast even with the right treatment [6]. Therefore, an accurate, automatic, fast, and cost-effective technique is needed to detect DR. Although a high sensitivity is good for screening, a relatively low specificity will result in more false-positive cases in the real world [7]. In present, the clinical experts spend a longer

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A. S. M. Shafi (✉) · Md. Rahat Khan
Khwaja Yunus Ali University, Enayetpur Sirajgonj-6751, Bangladesh

A. S. M. Shafi · M. M. Rahman
Mawlana Bhashani Science and Technology University, Santosh Tangail-1902, Bangladesh

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period of time to decide whether the image is actually normal to minimize false positives, while judging an image with a true abnormality. It is much faster and easier technique. Our main motivation is to develop a system that can distinguish retinal images with and without any DR signs. Therefore, it can filter out the normal images, so that medical specialists may concentrate the abnormal cases and thus saving the screening time, manpower, and costs. In our work, we have developed an automated system that can separate the retinal images as normal and abnormal based on statistical texture features. We have used multiple classifiers to categorize the images into interim classes as single classifier is not able to maintain the accuracy over a large number of diverse datasets. To identify DR in the retinal images, this study utilizes GLCM and GLRLM as a feature extraction technique. The GLCM and GLRLM describe the composition of an image by computing how regularly combines of the pixel with particular qualities and in a predefined spatial relationship happen in an image. The features are learned in different scales to provide scale-invariant features, and then each image in the frame is classified as abnormal (PDR and NPDR) or normal (normal retinal image). This research represents a possible improvement in the detection and classification of diabetic retinopathy. In the rest of the paper, Sect. 2 presents the previous related works on automatic methods for diabetic retinopathy detection, Sect. 3 provides the system architecture of our proposed method in details, Sects. 4 and 5 describe the experimental results and discussion, and Sect. 6 concludes the paper.

2 Literature Review

A brief overview of several DR detection and classification systems have been provided in this section. Liu et al. [8] proposed a novel approach based on weighted path convolutional neural network (WP-CNN) motivated by the ensemble learning for referable diabetic retinopathy identification from eye fundus images. The method achieved an accuracy of 94.23% with sensitivity of 90.94% and specificity of 95.74%. Authors [9] presented an unsupervised coarse-to-fine algorithm for blood vessel segmentation with an average accuracy of 87%. They incorporated multiple concepts, i.e., mathematical morphology, curvature, and spatial dependency to accurately define thin and elongated vessels from vessel pixels. However, the algorithm was unsuccessful in the determination of vessel diameter and also found to be less satisfactory at segmenting vessel structures on low contrast images. Mobeen-ur-Rehman et al. [10] presented a customized five-layered CNN model which consists of two convolution layers and three fully connected neural layers which gave the classification accuracy of 98.15%. The identification of diabetic retinopathy stage based on CNN was developed in [11]. They produced classification accuracy of 83.68% and specificity of 93.65%. Wang et al. [12] developed a method for DR grading and lesion detection via a region-based fully convolutional network (R-FCN). They reported a sensitivity of 92.59% and a specificity of 96.20% when testing on the Messidor dataset. Authors [13] reviewed algorithm for extracting features based on blood vessel area, exudes, hemorrhages, microaneurysms, and texture from digital fundus images for the automated detection of DR. Varun et al. [14] relied on the development and validation of a deep learning algorithm for automated detection of diabetic retinopathy and diabetic macular edema in retinal fundus photographs. Pires et al. [15] built their own CNN architecture to provide a reliable referable DR detector which has 16 weight layers, with about 10 million parameters. The work obtained an area under the ROC curve of 98.2% when testing the Messidor-2. A fully automated DR screening system based on morphological component analysis (MCA) to differentiate between normal and pathological retinal structures was considered in Imani et al. [16]. An SVM classifier was employed to differentiate between normal and abnormal retinal images and achieved 92.01% sensitivity and 95.45% specificity on the Messidor dataset. Goh et al. [17] filtered the retinal healthy images from DR based on the result of exudates detection using the features of local sub-images as the input to multiple classifiers. The experimental results demonstrated that by applying ensemble classifiers they obtained an accuracy of 92% in detecting normal images and 91% in detecting abnormal images. A review of recent development on the detection of DR was presented in [18] that was developed by various computational intelligence and image processing techniques. A combination of fuzzy image processing techniques, circular Hough transform, and several feature extraction methods were proposed by Rahim et al. [19] for automatic screening and classification of diabetic retinopathy and maculopathy. They reached the value of sensitivity, specificity, and accuracy of 92.45%, 93.62%, and 93%, respectively, by using radial basis function SVM (RBF-SVM) classifier. Manjaramkar et al. [20] presented a simple, efficient, and real-time method for segmentation and detection of microaneurysm (MA) which are the clinical signs of DR. For this, they used a novel set of features based on statistics of geometrical properties of connected regions and attained the best performance on DIARETDB1 dataset with a sensitivity of 88.09%.

3 System Architecture

The architecture of the proposed system involves six stages: input module, preprocessing module, segmentation module, feature extraction module, classification module, and post-processing module (Fig. 1). The proposed methodology is summarized as follows.

3.1 Proposed Algorithm

1. Capture the digital fundus image as an input image.
2. Preprocess all input images into a uniform size.
3. Apply Kirsch's template technique to extract blood vessels from the preprocessed image.

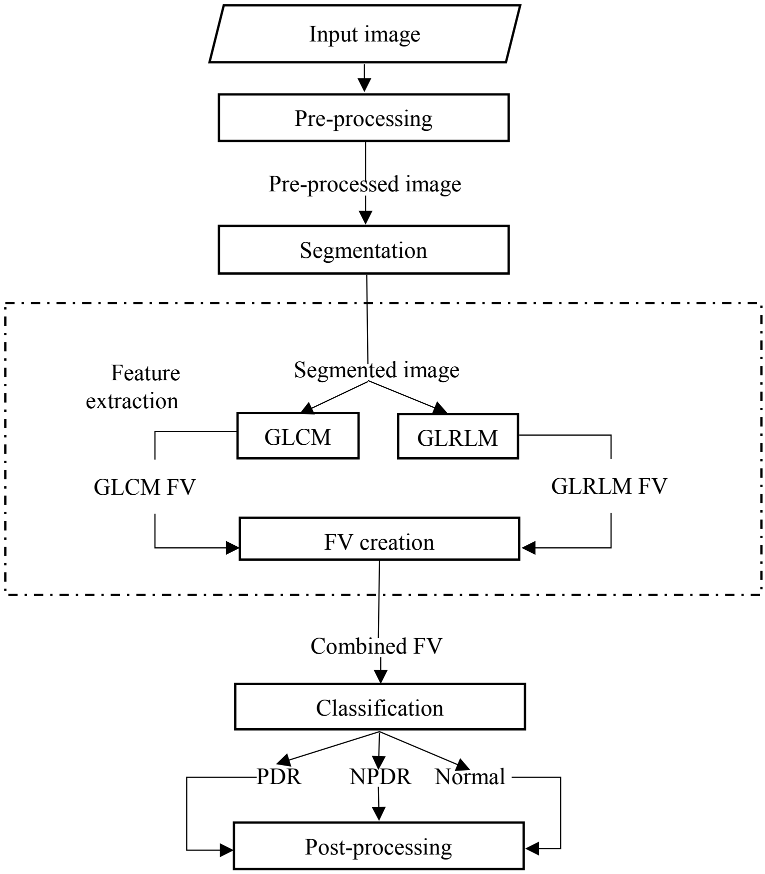
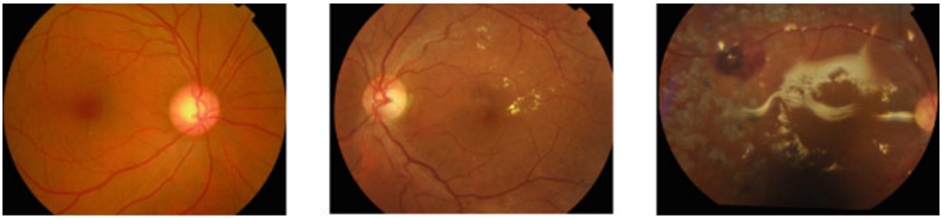


Fig. 1 Detailed diagram of the proposed methodology

4. Execute second-order and higher-order statistical texture features algorithm on the segmented image found from step 3.
5. Generate a feature vector (FV) for each training image.
6. Implement three classifiers (SVM, KNN, and RF) to test the train image for classification and evaluate the performance of the results.

3.2 Acquisition of Retinal Image

The performance of any computer-aided design (CAD) system depends on the training dataset [21]. Our proposed system has utilized more than 600 standard color fundus images from different sources. Most fundus photographs used in this study were DR (> 78%) images, and only 21% were normal retinal images. Sample dataset of two categories of DR along with normal retinal image is shown in Fig. 2.



(a) (b) (c)

Fig. 2 Sample dataset: **a** normal retina, **b** non-proliferative DR, and **c** proliferative DR

Table 1 Our dataset: the distribution of retinal images used in our proposed system

Retinal image type	Short name	Number of images
Non-proliferative diabetic retinopathy	NPDR	167
Proliferative diabetic retinopathy	PDR	341
Normal image	–	136

Important sources of the database include Indian Diabetic Retinopathy Image Dataset (https://idrid.grand-challenge.org/Data/) and Kaggle dataset (https://www.kaggle.com/c/diabetic retinopathy detection). Table 1 shows the distribution of dataset used in the proposed method.

3.3 Pre-Processing Module

An input image is loaded in the computer to find the possible categories of retinal images using the proposed intelligent system. For better and faster calculation, all the input image is resized into uniform size (565 * 375) image.

3.4 Segmentation Module

In our proposed method, we have used Kirsch's template for the extraction of blood vessels from the preprocessed retinal image. Kirsch's template technique uses a single mask of size 3 $\times$ 3 and rotates at 45° increments through each of the eight directions (south, east, north, west, northeast, southeast, southwest ,and northwest) to detect the edge as shown in Fig. 3. It can set and reset the threshold values to find the most appropriate edges in the images [22]. The details procedure can be found in [23]. Figure 4 demonstrates the vessel extraction procedure with Kirsch's template technique.

5	-3	-3
5	0	-3
5	-3	-3

0°

-3	-3	5
-3	0	5
-3	-3	5

45°

-3	-3	-3
5	0	-3
5	5	-3

90°

-3	5	5
-3	0	5
-3	-3	-3

135°

-3	-3	-3
-3	0	-3
5	5	5

180°

5	5	5
-3	0	-3
-3	-3	-3

225°

-3	-3	-3
-3	0	5
-3	5	5

270°

5	5	-3
5	0	-3
-3	-3	-3

315°

Fig. 3 Kirsch's convolution kernels

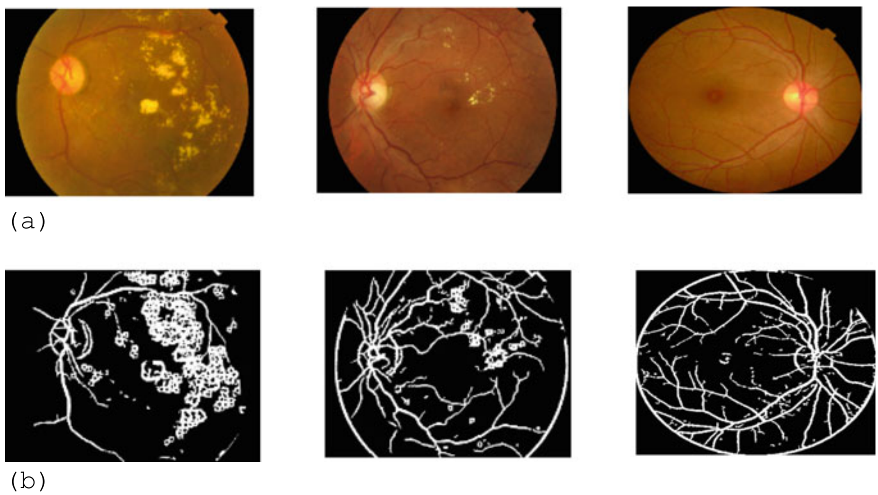


Fig. 4 Resultant image after segmentation: **a** the first row shows the image of PDR, NPDR, and normal retinal image respectively and **b** the last row indicates the blood vessel extraction results

3.5 *Feature Extraction Module*

In feature extraction module, statistical texture features are applied to the segmented image. A feature vector is created by using a large number of training images.

3.5.1 Second-Order Statistical Texture Features

The gray-level co-occurrence matrix (GLCM) also known as spatial gray-level dependence matrix (SGLDM) is a way to gain second-order statistical texture features. The GLCM method analyzes the distribution of gray-level pixel pairs. It provides better results where the textures are visually easily separable and are sensitive to the size of the texture samples being processed. Nine important GLCM features have been selected as a second-order statistical texture feature.

$$\text{Contrast} = \sum_{n=0}^{G-1} n^2 \sum_{i=1}^{G} \sum_{j=1}^{G} P(i, j), |i - j| = n \tag{1}$$

$$\text{Correlation} = \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} \frac{\{i * j\} * p(i, j) - \{\mu_x * \mu_y\}}{\sigma_x * \sigma_y} \tag{2}$$

$$\text{Energy} = \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} \{P(i, j)\}^2 \tag{3}$$

$$\text{Entropy} = -\sum_{i=0}^{G-1} \sum_{j=0}^{G-1} P(i, j) * \log(\mathrm{p}(i, j)) \tag{4}$$

$$\text{Inverse Difference Moment} = \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} \frac{1}{1 + (i - j)^2} P(i, j) \tag{5}$$

$$\text{Variance} = \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} (i - \mu)^2 P(i, j) \tag{6}$$

$$\text{Sum Average} = \sum_{i=0}^{2G-2} i P_{x+y}(i) \tag{7}$$

$$\text{Sum Entropy} = -\sum_{i=0}^{2G-2} P_{x+y}(i) \log(P_{x+y}(i)) \tag{8}$$

$$\text{Difference Entropy} = -\sum_{i=0}^{G-1} P_{x+y}(i) \log(P_{x+y}(i)) \tag{9}$$

where G is the number of gray levels used; μ is the mean value of P; μ_x, μ_y, ∂_x and ∂_y are the means and standard deviations of P_x and P_y; $P_x(i)$ is the ith entry in the marginal matrix obtained by summing rows of co-occurrence matrix, $P(i,j)$.

3.5.2 Higher-Order Statistical Texture Features

The gray-level run length matrix (GLRLM) has been considered for the description of higher-order statistical texture features implemented within the segmented image. The GLRLM method quantifies gray-level runs of consecutive pixels with the same gray-level value, which are defined as the length in number of pixels. The run length features are the least efficient texture features. We have considered seven run length matrix features.

$$\text{Short Run Emphasis} = \frac{1}{n_r}\sum_{i=1}^{M}\sum_{j=1}^{N}\frac{Q(i,j)}{j^2} \tag{10}$$

$$\text{Long Run Emphasis} = \frac{1}{n_r}\sum_{i=1}^{M}\sum_{j=1}^{N}Q(i,j)*j^2 \tag{11}$$

$$\text{Long Gray Level Run Emphasis} = \frac{1}{n_r}\sum_{i=1}^{M}\sum_{j=1}^{N}\frac{Q(i,j)}{i^2} \tag{12}$$

$$\text{High Gray Level Run Emphasis} = \frac{1}{n_r}\sum_{i=1}^{M}\sum_{j=1}^{N}Q(i,j)*i^2 \tag{13}$$

$$\text{Gray Level Non-uniformity} = \frac{1}{n_r}\sum_{i=1}^{M}\left(\sum_{j=1}^{N}Q(i,j)\right)^2 \tag{14}$$

$$\text{Run Length Non-uniformity} = \frac{1}{n_r}\sum_{j=1}^{N}\left(\sum_{i=1}^{M}Q(i,j)\right)^2 \tag{15}$$

$$\text{Run Percentage} = \frac{n_p}{n_r} \tag{16}$$

where $P(i, j)$ is the run length matrix, n_r is the total number of runs, and n_p is the number of pixels.

3.6 Classification Module

This module handles the classification of feature vector into three classes (Normal, NPDR, and PDR). The whole feature vector is divided into two modes: training and testing modes by using K-fold cross-validation [24]. In training mode, GLCM and GLRLM features are extracted from segmented images. The input feature vector consists of 16 different features that are combined together for training with three different classifiers, namely KNN [25], SVM [26], and RF [27]. Our proposed system first detects whether the unknown sample contains DR or not. If it contains DR, then the unknown sample is classified as NPDR or PDR, and it goes to the post-processing module. If it does not contain DR, then the unknown sample is classified as normal.

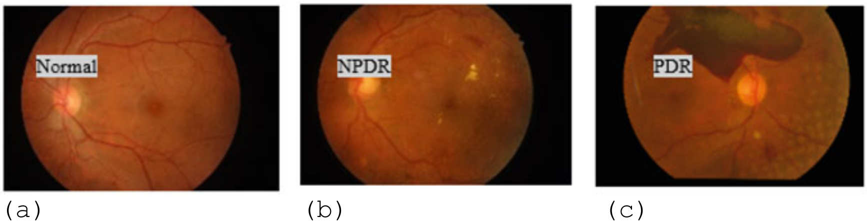


Fig. 5 System's output after post-processing: **a** normal retinal image, **b** NPDR image, and **c** PDR image

3.7 Post-processing Module

The outcome of classification module is used to a produce new image on which possible types of diabetic retinopathy are appropriately labeled. An illustration of this technique is shown in Fig. 5.

4 Results

To assess the efficiency of our proposed system, we evaluated four performance metrics for each class: sensitivity, precision, F1-score, and accuracy. Sensitivity is also known as recall is the ratio of correctly predicted positives cases to all observations in the actual class. The precision metric indicates the correct positive outcomes out of all the positive outcomes. The accuracy of a classifier is simply the ratio of correctly predicted class to total class. F1-score is estimated by applying the weighted average over precision and recall. In case, we have an uneven class proportion, F1-score is generally more valuable than precision because it takes both false positives and false negatives into account. Our main purpose is to construct a model that classifies diabetic retinopathy as accurately as possible. We used DR database with 644 images for the evaluation test, where the entire dataset is divided into training and testing. The performance of detecting and classifying DR using three different classifiers is given in Tables 2, 3, 4, and 5, respectively. Finally, the sensitivity, precision, F1-score, and accuracy across three different classes of each classifier are given in Table 6. In Tables 6 and 7, "Sn" is for sensitivity, "Pre" for precision, and "Acc" for accuracy.

Table 2 Confusion matrix of our proposed system using SVM classifier

	PDR	NPDR	Normal
PDR	326	11	4
NPDR	19	136	12
Normal	4	3	129

Table 3 Confusion matrix of our proposed system using KNN ($K = 3$) classifier

	PDR	NPDR	Normal
PDR	302	23	16
NPDR	21	127	19
Normal	18	14	104

Table 4 Confusion matrix of our proposed system using KNN ($K = 5$) classifier

	PDR	NPDR	Normal
PDR	308	17	16
NPDR	21	136	10
Normal	18	19	99

Table 5 Confusion matrix of our proposed system using RF classifier

	PDR	NPDR	Normal
PDR	333	5	3
NPDR	6	149	12
Normal	2	3	131

Table 6 Sensitivity, precision, F1-score and accuracy of each classifier

Image Type	SVM				KNN ($K = 3$)				KNN ($K = 5$)				RF			
	Sn	Pre	F1	Acc	Sn	Pre	F1	Acc	Sn	Pre	F1	Acc	Sn	Pre	F1	Acc
PDR	93	96	94	94.1	89	89	89	87.89	89	90	90	88.82	98	98	98	97.52
NPDR	91	81	86	93.01	77	76	77	88.04	79	81	80	89.6	95	89	92	95.92
Normal	89	95	92	96.43	75	76	76	89.6	79	73	76	90.22	90	96	93	96.89

From the confusion matrix of Tables 2, 3, 4, and 5, we can observe that the random forest classifier made a correct prediction of 333 PDR, 149 NPDR, and 131 normal images out of 341 PDR, 167 PDR, and 136 normal images which are higher than the other two classifiers.

Table 7 Weighted measure of each classifier

Name of the classifier	Weighted measure			
	Sn	Pre	F1-score	Acc
SVM	91.63	91.89	91.35	94.30
KNN ($K = 3$)	82.93	82.88	83.14	88.29
KNN ($K = 5$)	83.66	84.07	84.45	89.31
RF	95.53	96.45	95.38	95.19

Table 8 Performance comparison among different methods

Authors	Methodology	Dataset	Accuracy (%)
Liu YP et al. [8]	WP-CNN	60,000	94.23
Neto et al. [9]	Unsupervised coarse-to-fine algorithm	60	87
Mobeen-ur-Rehman et al. [10]	Customized CNN	1200	98.15
Garcia et al. [11]	CNN	35,126	83.68
Rahim et al. [19]	Fuzzy image processing techniques	990	93
Proposed method	Statistical texture features	644	95.19

For performance evaluation parameters of each transform individually, we can see that using a random forest classifier, the highest sensitivity is in PDR images which is 98%, and the worst sensitivity is 90% in normal images. Similarly, for both precision and F1-score, we get the highest rate of 98% in PDR images. Our method attains the highest accuracy of 97.52% for the detection of PDR images (Table 6). Table 7 contains the result of weighted measurement of each classifier.

In view of Table 7, the proposed system demonstrates a weighted sensitivity, specificity, F1-score, and accuracy of 95.53%, 96.45%, 95.38%, and 95.19%, respectively, utilizing RF classifier. Table 8 shows the comparison of our proposed method with existing approaches.

5 Discussion

Retinal color photography is an efficient screening method for diagnosing diabetic retinopathy because of its faster with easier acquisition, storage, and transmission of retinal images. Most of the middle- and lower-income countries require a cost-effective retinal imaging framework for regular diabetic retinopathy screening [28]. Therefore, an automatic system for the detection and classification of diabetic retinopathy using statistical texture features has been created. The system proposes a combination of retinal blood vessels segmentation, followed by the feature extraction and, finally, the classification is done with some machine learning algorithms. We have incorporated both second-order and higher-order statistical texture features as a feature extraction tool. Texture features can be extracted using several methods such as structural, statistical, model-based and transform information. We have considered statistical methods. According to non-deterministic properties, statistical methods describe the texture indirectly. We have used tenfold cross-validation for splitting the whole dataset to protect against overfitting. To extract statistical texture descriptors from vessel images, the GLCM and GLRLM features have been used that can demonstrate sound performance in image classification of different categories. An ideal screening test can be measured by its sensitivity, i.e., high probability of detecting disease. The experimental results demonstrate that the three classifiers are able to identify well both categories and particularly random forest classifier performed better in most of the cases than the other two classifiers. The sensitivity of retinal images using RF classifier was high (98%) for PDR and for NPDR was also high (95%). Based on the offered results, it can be seen that the sensitivity for normal cases was lightly low (90%). The explanation may be the functionality of Kirsch's template technique which was used for the detection of blood vessels. Some of the blood vessels were not detected by this technique.

6 Conclusion and Future Work

A large number of diabetic patients and the incidence of DR among them have promoted a great development in automatic DR diagnosing systems. In this study, we developed a computer-aided DR detection and classification system form color fundus images to reduce the workload of ophthalmologists to detect DR at the early stages which may go undetected and evolve into blindness. Weighted value of sensitivity, precision, F1-score, and accuracy of the suggested method, respectively, 95.53% 96.45%, 95.38%, and 95.19% which shows that the proposed method can detect and classify DR well. In the future, these methods will be tested on integrating a larger dataset with high-resolution images. Finally, we will add more classes of non-proliferative diabetic retinopathy viz. low, medium, and severe condition. It will enable the patients to know their condition with better accuracy.

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Chapter 4
Towards an Improved Eigensystem Realization Algorithm for Low-Error Guarantees

Mohammad N. Murshed, Moajjem Hossain Chowdhury, Md. Nazmul Islam Shuzan, and M. Monir Uddin

1 Introduction

Dynamical system is an elegant way of describing a phenomenon. The system appears to be both small and simple for linear problems and gets bigger for nonlinear problems. Although most of the processes around us are nonlinear, we can analyze such systems to identify low-dimensional models for them. We consider the discrete linear system,

$$\mathbf{x}_{i+1} = \mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{u}_i \tag{1a}$$

$$\mathbf{y}_i = \mathbf{C}\mathbf{x}_i + \mathbf{D}\mathbf{u}_i \tag{1b}$$

where $\mathbf{x} \in \mathbb{R}^n$ contains the internal states, $\mathbf{u} \in \mathbb{R}^p$ is the input, $\mathbf{y} \in \mathbb{R}^q$ refers to the output, and i is the time index. A set of inputs can cause a system to react in a particular manner via the internal states to result in the output. $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times p}$, $\mathbf{C} \in \mathbb{R}^{q\times n}$, and $\mathbf{D} \in \mathbb{R}^{q\times p}$ are called the realization of the system.

For many real-life problems, the number of equations in the system is large. An example is the Navier–Stokes equation where the system is large owing to the high

M. N. Murshed (✉) · M. M. Uddin
Department of Mathematics and Physics, North South University, Dhaka 1229, Bangladesh
e-mail: mohammad.murshed@northsouth.edu

M. M. Uddin
e-mail: monir.uddin@northsouth.edu

M. H. Chowdhury · Md. N. I. Shuzan
Department of Electrical and Computer Engineering, North South University, Dhaka 1229, Bangladesh
e-mail: moajjem.hossain@northsouth.edu

Md. N. I. Shuzan
e-mail: nazmul.shuzan@northsouth.edu

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number of spatial nodes in the domain. It is very difficult to extract knowledge from such large systems. Our focus in this work is on eigensystem realization algorithm that makes a model from just impulse response data. We begin by reviewing some of the existing powerful model reduction techniques.

Model order reduction (MOR) is a way of reducing the complexity of models by means of projection. MOR aids in creating a low-dimensional version of the large-scale system and enables a good enough understanding of the phenomenon in terms of fewer dominant states. This notion has been used to analyze randomly generated linear systems, flow past a flat plate, combustion, and many other problems of interest.

Proper orthogonal decomposition (POD) is a statistical method to derive a low rank version of a set of data [1]. The idea can be tailored to obtain a reduced order model, but research has been in progress to find out better projections than the orthogonal ones. An interesting study of POD in the field of turbulence is available in [2]. Balanced truncation (Moore, 1981) and balanced proper orthogonal decomposition (BPOD) are two powerful model reduction methods successfully implemented on CFD problems and randomly generated systems [3]. Rowley et al. used the idea of model reduction and extended it to solve nonlinear complex compressible flows [4, 5]. The available techniques rely on the direct system and a transformed system also known as the adjoint system.

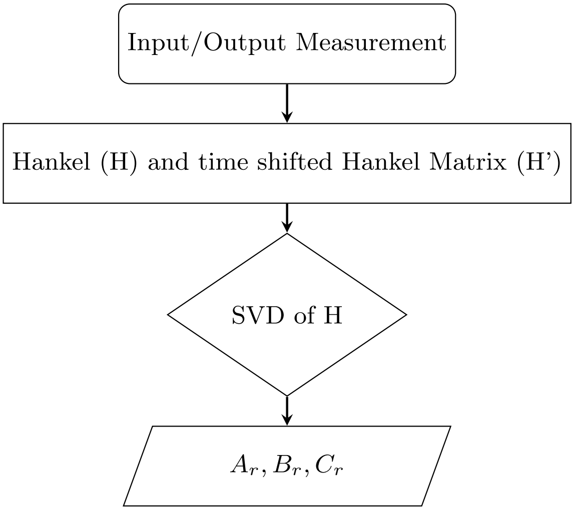
Eigensystem realization algorithm [6, 7] is the only model reduction tool that is based just on the direct system, hence making it applicable on experimental data. It leverages just the input and output measurements to create a reduced order model for a given problem, as shown in Fig. 1. The connection between ERA and BPOD is shown in [8]. ERA has been tested on many occasions. It works well to make low-order models for unstable flows [9], which are then used to design controllers. Tangential interpolation-based eigensystem realization algorithm (TERA) [10], built on ideas of ERA can handle huge amount of input–output data for multi-inputs multi-outputs (MIMO) system. TERA is applied on mass spring damper system and a cooling model for steel. A modified ERA [8] is also developed and compared with the performance of balanced POD on the flow past a flat plate at a low Reynolds number.

Since measured data may contain noise, the way the data gets separated into the signal and the noise has also been a subject under study. A noteworthy work can be found in [11] that discusses the effect of noise, if any, on the modal parameters for a system.

This paper is about the development of an improved version of eigensystem realization algorithm that monitors and empirically determines the rank and the time resolution of the output measurement to produce a reduced order model that is much more reasonable than the one from conventional eigensystem realization algorithm.

The rest of the paper is organized as follows: Sect. 2 discusses the idea of ERA in detail and Sect. 3 contains our main contribution. The numerical tests are shown in Sect. 4, followed by a summary in Sect. 5.

Fig. 1 Eigensystem realization algorithm



2 Background

In this section, we review the eigensystem realization algorithm, its derivation and how it relates to Dynamic Mode Decomposition. These will be used to elaborate on the modified eigensystem realization algorithm in the next section.

Eigensystem Realization Algorithm
Eigensystem realization algorithm is a system identification method that was first used to create models for vibration in aerospace structures. This tool borrows from the idea of Ho's algorithm [12], to find the realizations while keeping the number of internal states to a minimum that is to say that keeping the dimension of matrix A as low as possible. Completely data-driven, ERA uses only impulse response of the system, i.e., just the inputs and the outputs [13].

The discrete, linear time-invariant system in Eqs. (1a) and (1b) can be excited by a pulse defined as

$$u = \begin{cases} 1 & k = 0 \\ 0 & k > 0. \end{cases}$$

For $x_0 = 0$, the system reduces to $x_1 = Ax_0 + B(1)$ to give $x_1 = B$. We can iterate through the system to get,

$$x_2 = Ax_1 + Bu_1 = AB$$

$$x_3 = Ax_2 + Bu_2 = A^2B$$

$$x_4 = Ax_3 + Bu_3 = A^3B$$

and so on, while the outputs appear to be

$$y_0 = Cx_0 = 0$$

$$y_1 = Cx_1 = CB$$

$$y_2 = Cx_2 = CAB$$

$$y_3 = Cx_3 = CA^2B$$

and so on. We observe that $y_k = CA^{k-1}B$ which are also known as Markov parameters. Note that the dimension of Markov parameter is $q \times p$. These are then used to construct the Hankel matrix and the time shifted Hankel matrix,

$$H = \begin{bmatrix} y_1 & y_2 & y_3 & y_4 & \cdots & y_{m-s-1} \\ y_2 & y_3 & y_4 & y_5 & \cdots & y_{m-s} \\ \vdots & \cdots & \cdots & \cdots & \ddots & \vdots \\ y_{s-1} & \cdots & \cdots & \cdots & \cdots & y_{m-2} \end{bmatrix}$$

$$H' = \begin{bmatrix} y_2 & y_3 & y_4 & y_5 & y_{m-s-1} & y_{m-s} \\ y_3 & y_4 & y_5 & y_6 & y_{m-s} & y_{m-s+1} \\ \vdots & \cdots & \cdots & \cdots & \ddots & \vdots \\ y_s & \cdots & \cdots & \cdots & y_{m-2} & y_{m-1} \end{bmatrix}.$$

After performing the singular value decomposition of $H = U\Sigma V^*$, the truncated version of U,V, Σ is computed as:

$$U_r = U(1 : r, :)$$

$$V_r = V(1 : r, :)$$

$$\Sigma_r = \Sigma(1 : r, 1 : r),$$

where r is the rank of H. The reduced order model is then given by,

$$A_r = \Sigma^{-1/2} U_r^* H' V_r \Sigma^{-1/2}$$

$$B_r = \text{the first } p \text{ columns of } \Sigma_r^{1/2} V_r^*$$

$$C_r = \text{the first } q \text{ rows of } U_r \Sigma_r^{1/2}.$$

There are a few important points about the notation used. The output measurement is a function of time and variable s controls the way we stack the time shifted output measurement. The Hankel matrices above refer to a single input and single output system (SISO). The dimensions would change as we deal with a different system, e.g., MISO or MIMO.

Note on Hankel Singular Values. In general, eigenvalues give hint on the system stability. But, Hankel singular values identify the highly energetic states that contribute the most to characterize the system. That means the states with low energy can be truncated to obtain an approximate model. The Hankel singular values are computed from the SVD of the product of the controllability and the observability Gramian.

Connection to Dynamic Mode Decomposition
Dynamic mode decomposition (DMD) is a strategy of creating a model from time series data [14]. It can be viewed as a special case of Koopman operator which is a way of representing a nonlinear dynamical system as a infinite-dimensional linear system. DMD has numerous applications in various disciplines like fluid dynamics, neuroscience, epidemiology, and many others. There are a lot of variants of DMD that are to be utilized based on the type of data. For instance, time delay coordinate-based DMD is an option when the data is highly oscillatory [15].

DMD aims to map the current states to the future states as,

$$X_{i+1} = AX_i.$$

This equation from DMD is essentially Eq. (1a) with no control. This implies that DMD is related to ERA from a dynamical systems point of view.

3 Modified Eigensystem Realization Algorithm

We propose an improved version of eigensystem realization algorithm that searches for the optimal number of temporal nodes, N_t, used to express the output from the actual system and also the optimal rank, r, used in ERA to attain a reduced order model that can predict the output with high accuracy.

The idea behind this modified ERA is to run the conventional ERA multiple times so to identify the best possible number of temporal nodes and the rank that keeps the error,

$$\epsilon = ||\mathbf{y}_{actual} - \mathbf{y}_{ERA}||_2 \tag{2}$$

as low as possible. N_t controls the time resolution: certain N_t values are optimal while others can yield large error. The steps in the middle are the same as the ones in the traditional ERA. At the very end of this modified version, the A_r, B_r, and C_r are computed based on the optimal rank. The routine is provided in Algorithm 1. The specialty of this updated version of ERA is that it uses the appropriate time resolution and identifies the 'best' possible rank to keep the error to a minimum.

Algorithm 1: Modified Eigensystem Realization Algorithm

1 **Pre-run ERA to identify N_t and r that keep the error, (2), as low as possible**
2 Utilize the output measurements, $\mathbf{y}_{actual}$, based on the optimal time resolution, T/N_t
3 Construct the Hankel matrix (H) and the time-shifted Hankel matrix (H')
4 Compute the SVD of the Hankel matrix, $H = U\Sigma V^*$
5 Find the truncated U, Σ, and V using the rank (r) from the pre-run
6 Calculate the reduced system matrices just as in traditional ERA
7 Generate the output from ERA, $\mathbf{y}_{ERA}$, via the reduced order model

We also recommend that a frequency analysis be performed after this modified ERA scheme is enacted. Frequency analysis is often helpful for engineering purposes. A way to do it is by using *tfestimate* function on MATLAB. This function takes in the input and output to generate an approximate transfer function for a certain range of frequencies. Note that *bodeplot* function on MATLAB results in the magnitude and phase of the system, but visual comparison is well done via *tfestimate*.

4 Numerical Results

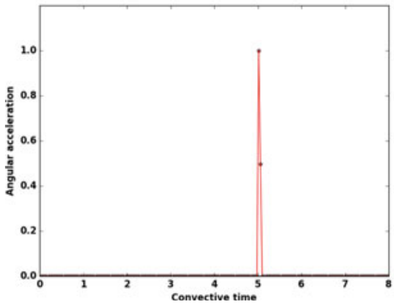
We have tested ERA on four different problems. The first two aims to stress on the model identification function of ERA and the last two prove the ability of ERA to work as a model reduction tool. The results are generated on a personal computer (HP Pavilion 14) with CORE i5 8th Gen processor 1.6–3.4 GHz and RAM of 8 GB via MATLAB version 2019b.

Example 1 Pitch Model (SISO) The 3D motion of an aircraft is governed by the pitch, plunge, and surge models [16]. Many state variables come into play. Velocity, density, temperature, and pressure are a few of them. Computing all these state variables in a grid is not easy since there may be spatial nodes as many as 10^6. In aeronautics, engineers care a lot about what is called the lift coefficient per unit span,

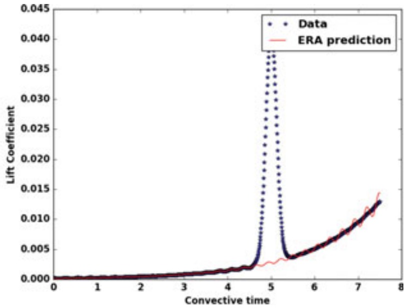
$$C_L = \frac{2L}{\rho U_\infty^2 c}$$

where L is the lift force on the wing, ρ the air density and U_∞ free-stream velocity, and c the chord. The angle of attack, α, is the angle between the airfoil chord and the flow direction. It can be thought of as some angular displacement which automatically makes $\ddot{\alpha}$ the angular acceleration. The pitch motion of an aircraft is the one that is observed with the nose moving up or down. We use the linearized pitch model from [16] that reads:

$$\frac{d}{dt}\begin{bmatrix} \mathbf{x} & 0 & 0 \\ \alpha & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{B}_{\dot{\alpha}} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 0 \end{bmatrix}\begin{bmatrix} \mathbf{x} \\ \alpha \\ \dot{\alpha} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ 0 \\ 1 \end{bmatrix}\ddot{\alpha}$$



(a) Angular Acceleration Variation



(b) Lift coefficient data and ERA prediction for pitch model

Fig. 2 Pitch model input and response

$$C_L = \begin{bmatrix} \mathbf{C} & C_\alpha & C_{\dot{\alpha}} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \alpha \\ \dot{\alpha} \end{bmatrix} C_{\ddot{\alpha}} \ddot{\alpha}$$

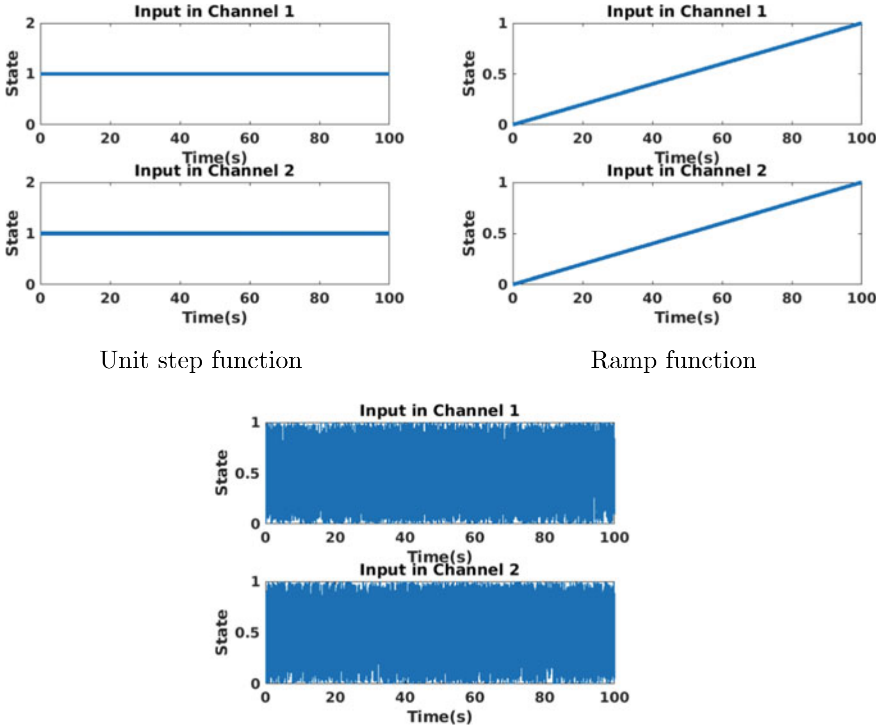
where $\mathbf{x}$ is a vector containing the states. A non-dimensionalized version of time is used, $\tau = t\frac{U_\infty}{c}$. The pitfall in applying ERA on a state space model is that we are bound to use an impulse response. Thus, it is worth correctly identifying the right input for the pitch motion of an aircraft. We examine the behavior of various state variables over time, available in [16]. The angle of attack and angular velocity vary in the form of a ramp and step function, respectively, whereas the angular acceleration Fig. 2a follows a dirac delta function which is the requirement of ERA. Hence, angular acceleration would be a suitable input. Note that we consider the input from $\tau = 5$ upto $\tau = 7.5$. In this example, we use exponential functions to approximate the lift coefficient behavior,

$$C_l = \frac{1}{105}\left(\frac{1}{(\sigma\sqrt{(2\pi)})} e^{-0.5((\tau-5)/\sigma)^2} + 2.2^{0.68\tau-5.8}\right),$$

with σ set at $\sqrt{0.015}$. This signal is fed into ERA to find the reduced order model for the pitch dynamics of an aircraft. It is evident in Fig. 2b that the data and ERA model agree for most of the time except where the lift coefficient has a sudden significant jump followed by a drop at around $\tau = 5$.

Example 2 Second-Order State-Space Model (MISO) We also test ERA on a multi-inputs single-output system and compare the reduced order model to the original model by feeding arbitrary inputs (step, ramp, unit and random). The MISO, in consideration, is defined as per the following matrices,

$$A = \begin{bmatrix} -0.5572 & -0.7814 \\ 0.7814 & 0 \end{bmatrix}; B = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$



Unit step function   Ramp function

Random number

Fig. 3 Arbitrary inputs used to check performance

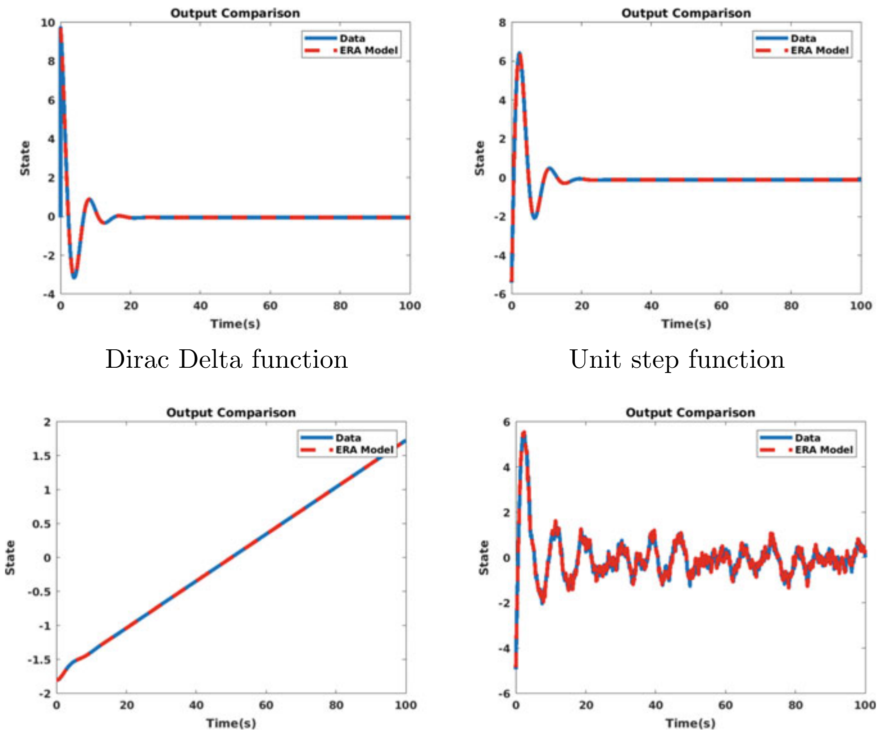
$$C = \begin{bmatrix}1.9691 \ 6.4493\end{bmatrix}; D = \begin{bmatrix}0 \ 0\end{bmatrix}.$$

The reduced order model extracted by ERA came out to be,

$$\hat{A} = \begin{bmatrix} 0.9985 & 0.0137 \\ -0.004 & 0.9959 \end{bmatrix}; \hat{B} = \begin{bmatrix} -1.7916 & -1.9913 \\ -0.5389 & 2.1738 \end{bmatrix}$$

$$\hat{C} = \begin{bmatrix}-2.0568 \ 3.1113\end{bmatrix}; \hat{D} = \begin{bmatrix}1.9691 \ 10.9295\end{bmatrix}.$$

After extracting the Markov parameters and creating a model, we tested the model with different inputs. The outputs Fig. 4, from these different inputs Fig. 3, are compared with outputs from the original model. The error between these outputs was minimal. Even when the input was a randomly generated signal, the model was able to capture the characteristics of the signal. The ERA output resembles the actual output despite the difference in the Markov parameters of the two systems. We can see that ERA found the parameters from the data.



Dirac Delta function

Unit step function

Ramp function

Random number

Fig. 4 Response comparison between ERA and actual system for various inputs

Example 3 Heat Diffusion Equation (Sparse Model) In this example, the heat diffusion equation for a rod of unit length (1D) is defined as,

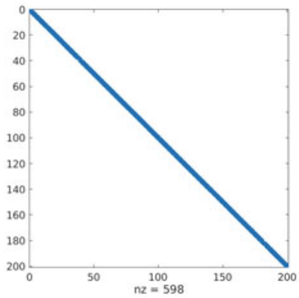
$$\frac{\partial T(x,t)}{\partial t} = \alpha \frac{\partial^2 T(x,t)}{\partial x^2} + u(x,t)$$

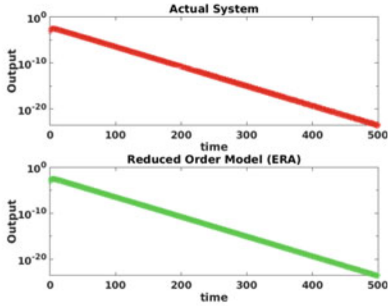
$$T(0,t) = T(1,t) = 0$$

$$T(x,0) = 0,$$

where x refers to space and t is the time. The equations above consist of the partial differential equation showing the evolution of the temperature, $T(x,t)$, the boundary conditions, and the initial condition. The input is controlled by $u(x,t)$.

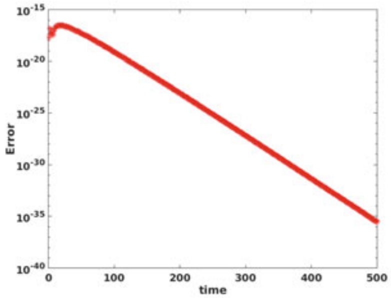
We utilized the system matrices available in [17] (the state matrix shown in Fig. 5a) to compute the output for this problem. The output is then fed into ERA to identify a reduced order model. The crux is to observe the behavior of the norm of the error with the rank set in ERA as shown in Fig. 5d. This allows us to find the optimal number of

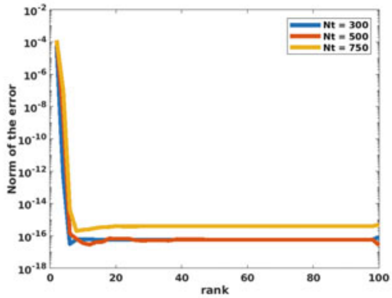




(a) The sparsity pattern in state matrix, A

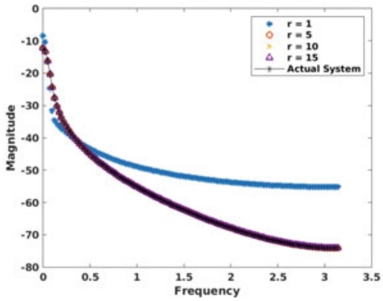
(b) Comparison of the output from the actual system and ERA





(c) Uncertainty with time

(d) Error variation with rank (used in ERA) for three different time resolution



(e) Frequency analysis for different rank used in the ERA

Fig. 5 Heat diffusion model reduction temporal nodes to be used that maintains a low rank for the system. We have identified that $N_t = 300$ works fine for $r = 6$. Thus, the state matrix in the original system is reduced from 200×200 to 6×6 via ERA. The output from ERA, produced by *lsim* function on MATLAB, also agrees well with the output from the actual large system Fig. 5b. The error over time is also displayed in Fig. 5c. The frequency analysis done by *tfestimate* function on MATLAB shows that r greater 1 yields a model as good as the actual system Fig. 5e.

Example 4 Atmospheric Storm Track (Dense Model) The atmospheric storm track is a model from oceanography used to analyze the velocity of the airflow in the zonal (latitude wise) and meridional (longitude wise) setting. We can imagine this of a flow in a channel, the physical domain of which is defined as,

$$0 < x < 12\pi$$

$$-0.5\pi < y < 0.5\pi$$

$$0 < z < 1.$$

In the z-axis, $z = 0$ is the ground level and $z = 1$ is the tropopause.
The mean velocity is set to vary with the altitude,

$$U(z) = 0.2 + z,$$

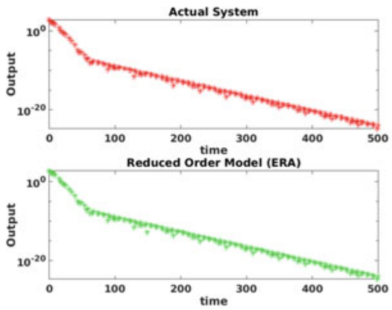
and time is non-dimensionalized as $T = \frac{L}{U_0}$ where $L = 1000\ km$ and $U_0 = 30\ m/s$. The system is thought to have a uniform flow, but can be disturbed by a linear damping at the entrance and the exit of the track. The details of the dynamics can be found in [17]. The governing equation of the states is,

$$\frac{d\psi}{dt} = A\psi, \tag{3}$$

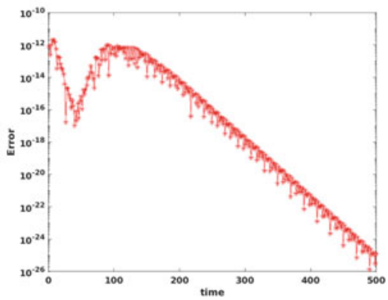
where ψ is the velocity variable.

The output from the actual system is put into ERA. Figure Fig. 6c shows that $r \approx 55$ for $N_t = 500$ and $N_t = 750$, whereas use of 300 temporal nodes allows for $r = 30$. Setting the rank to 30 and number of temporal nodes to 300 results in ERA output that resembles the original output Fig. 6a. The norm of the difference between the approximate output and the actual output is also plotted in Fig. 6b. We observe that the intractable original system of dimension 598×598 gets reduced to a 30×30 system by ERA.

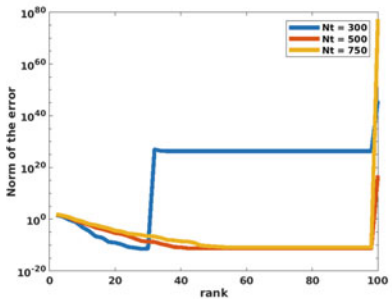
The transfer function estimates for several different rank values which are plotted in Fig. 6d. The ERA model with $r = 1$ is far away from the actual system, $r = 5$ and $r = 10$ show improvement and $r = 15$ enables reduced order modeling that is



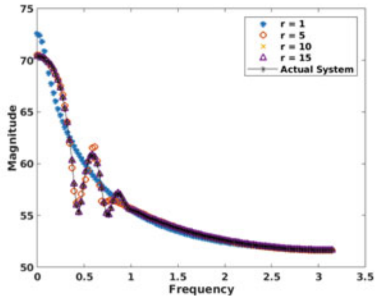
(a) Comparison of the output from the actual system and ERA



(b) Absolute value of the difference in the output from the actual system and ERA



(c) Error variation with rank (used in ERA) for three different time resolution



(d) Frequency analysis for different rank used in the ERA

Fig. 6 Atmospheric storm track model reduction

as efficient as the actual system. Thus, proper selection of the time resolution and rank along with a frequency analysis in the eigensystem realization algorithm shows promise in building reduced order models with low error.

5 Conclusions

In this work, we delineated the steps in our proposed modified eigensystem realization algorithm and implemented this method on four test problems. Modified ERA identifies the model in the first two examples and performs as a tool to reduce the order of the model in the third and fourth examples. By model identification, we mean finding the state, input and output matrix and model order reduction refers to the minimization of the size of the state matrix, also known as the system matrix. The output predicted by ERA agrees well with that from the original system. The first example is concerned about the pitching motion of an aircraft, and the second one a second-order state-space model. The third example is the heat diffusion equation and the last one a model for the airflow velocity when a storm or a cyclone surges. Indeed, the third and the fourth numerical tests demonstrate that the rank should be carefully set at or above 5 to minimize error in the output predicted by ERA and also to get a transfer function estimate that is much close to that of the original system.

We plan to work on a survey of all the model order reduction techniques and apply them on an array of synthetic and practical data and finally weigh the pros and cons of each technique. At the same time, our work would also be to establish any connection between model order reduction method and DMD [18].

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Chapter 5
Skin Lesion Classification Using Convolutional Neural Network for Melanoma Recognition

Aishwariya Dutta, Md. Kamrul Hasan, and Mohiuddin Ahmad

1 Introduction

Skin cancer is a common type of cancer that originates in the skin's epidermis layer by the irregular cells due to ultraviolet radiation exposure [1]. Every fifth person in the USA has a risk of skin cancer in a region under strong sunshine [2]. Among all skin cancer types, melanoma is the nineteenth most frequent cancer, where approximately 3.0 million new cases were identified in 2018. On average, 2490 females and 4740 males lost their lives due to melanoma in 2019 [3] in the USA alone. It is estimated that approximately, in 2020, 1.0 million of newly affected melanoma patients will be diagnosed. An approximated 6850 new cases of deaths due to melanoma are anticipated in 2020 in the USA alone, which will comprise 4, 610 males and 2240 females [4]. Skin cancer deaths in Bangladesh reached 0.04 % of total deaths in 2017, which is ranked 182 in the world [5]. However, a precise and robust early recognition is significant as the survival rate was as high as apparently 90% in advance recognition [6]. Several imaging techniques like dermoscopic image, optical coherence tomography, magnetic resonance imaging, and confocal scanning laser microscopy are currently being used to diagnose skin cancer. Those images are visually inspected by the dermatologists [7], which is often a tedious and time-consuming process. To relieve the dermatologist's tediousness by improving the preciseness in recognition, computer-aided diagnosis (CAD) systems are being used [8]. Currently, CAD is an

A. Dutta
Department of Biomedical Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

Md. Kamrul Hasan (✉) · M. Ahmad
Department of Electrical and Electronic Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh
e-mail: m.k.hasan@eee.kuet.ac.bd

M. Ahmad
e-mail: ahmad@eee.kuet.ac.bd

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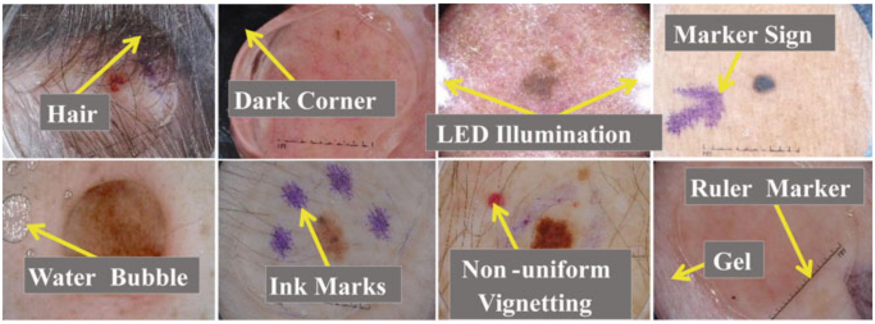


Fig. 1 An example of the challenging images, in the ISIC dataset [11], for the accurate SLC

integral part of the routine health checkup in the clinic, which consists of raw image acquisition, preprocessing, region of interest (ROI) & feature extraction, and finally, recognition [8, 9]. The final step, also known as the classification, is a crucial component, which is a challenging task due to the diverse size of the lesion area, texture, skin color, and the presence of various artifacts like reflections, hair, rolling lines, air bubbles, non-uniform vignetting, shadows, and markers [8, 10], which are shown in Fig. 1.

Nowadays, several methods are being used for the SLC [12, 13]. The sensitivity and specificity of the practitioners are 62%, and 63%, while for the dermatologists are 80% and 60% , respectively, for melanoma recognition [14]. In [15], the authors presented a CAD system for the SLC, where they used boarder and wavelet-based texture features with the support vector machine (SVM), hidden Naïve Bayes (HNB), logistic model tree (LMT), and random forest (RF) classifiers. In [16], the authors proposed a model for the SLC, which comprises a self-generating neural network (SGNN), a feature (texture, color, and border) extraction, and an ensembling classifier. A deep residual network (DRN) was presented in [17] for the SLC, where they demonstrated that DRNs have the capability to learn distinctive features than low-level hand-crafted features or shallower CNN architecture. CNN architecture, along with the transfer learning paradigm, was employed for the SLC in [18]. A 3-D skin lesion reconstruction technique was presented in [19], where depth and 3-D shape features were extracted. Besides, they also extracted regular color, texture, and 2-D shape features. Different machine learning classifiers (SVM and AdaBoost) were employed to classify those features. In [20], the authors proposed an effective iterative learning framework for the SLC, where they designed a sample re-weighting strategy to conserve the effectiveness of accurately annotated hard samples. The stacking ensemble pipeline based on the meta-learning algorithm was proposed in [21], where two-hybrid methods were introduced to combine the mixture classifiers. The effect of dermoscopic image size based on pre-trained CNNs, along with transfer learning, was analyzed in [22], where they resized from 224×224 to 450×450. They proposed a multi-scale multi-CNN fusion approach using EfficientNetB0, EfficientNetB1, and SeReNeXt-50, where three networks architectures trained on cropped images of different sizes. An architecture search framework was presented in [23] to recognize the melanoma, where the hill-climbing search strategy, along with network morphism operations, was employed to explore the search space.

This study proposes a framework for the SLC (multi-class task), where preprocessing, geometric augmentation, CNN-based classifiers, and transfer learning are the integrated steps. We have performed various types of geometric image augmentations for increasing the training samples as in most of the medical imaging domains, a massive number of manually annotated training images are not yet available [24]. A CNN-based classifier has been used to avoid the tedious feature engineering, which can learn features automatically during the forward-backward pass of training images. Transfer learning of CNN is used to initialize all the kernels in convolutional layers by leveraging the previously trained knowledge rather than random initialization. Extensive experiments are conducted to select different hyper-parameters like types of image augmentation, optimizer & loss function, and metric to be maximized for training, the number of layers of CNN to be frozen. We validate our proposed framework by comparing it with several state-of-the-art methods on the ISBI-2017, where our proposed pipeline achieved better results while being an end-to-end system for the SLC.

The remainder of this paper is set out as follows. Section 2 describes the materials and the proposed framework. Section 3 presents detailed results with a proper illustration. Finally, the paper is concluded in Sect. 4 with future works.

2 Materials and Methods

The detailed description of the materials and the methods used in this literature is represented as follows in several subsections.

2.1 Dataset and Hardware

The utilized dataset, for training, validation, and testing, is presented in Table 1, where we use the ISIC-2017 dataset from the ISIC archive [11]. Table 1 presents the class-wise distributions and a short description of the ISIC-2017 dataset.

The proposed framework was implemented with the Python programming language with various Python and keras APIs. The experiments were conducted on a *Windows-10* machine with the following hardware configuration: Intel® Core™ i7-7700 HQ CPU @ 2.80 GHz processor with Install memory (RAM): 16.0 GB and GeForce GTX 1060 GPU with 6 GB GDDR5 memory.

Table 1 Data distribution of the ISIC-2017 dataset

SL #	Class types	Description	Train	Validation	Test
01	Melanoma (Mel)	Malignant skin tumor obtained from melanocytic	374	30	117
02	Seborrheic Keratosis (SK)	Benign skin tumor obtained from keratinocytes	254	42	90
03	Nevus (Nev)	Benign skin tumor obtained from melanocytic	1372	78	393
Total Images			2000	150	600

2.2 Proposed Framework

In the proposed framework, as shown in Fig. 2, the feature extraction and classification of the skin lesion for melanoma recognition have been automated using an end-to-end CNN architecture, where the image augmentation and normalization are the crucial and integral parts of the proposed framework. The deep CNNs are widely used in both medical and natural image classifications, which has achieved tremendous success since 2012 [25]. It often rivals human expertise [26]. In CheXNet [27], a CNN was trained on more than 1.0 million frontal view of the chest X-rays, where it was able to achieve better recognition results than the average recognition by the four experts.

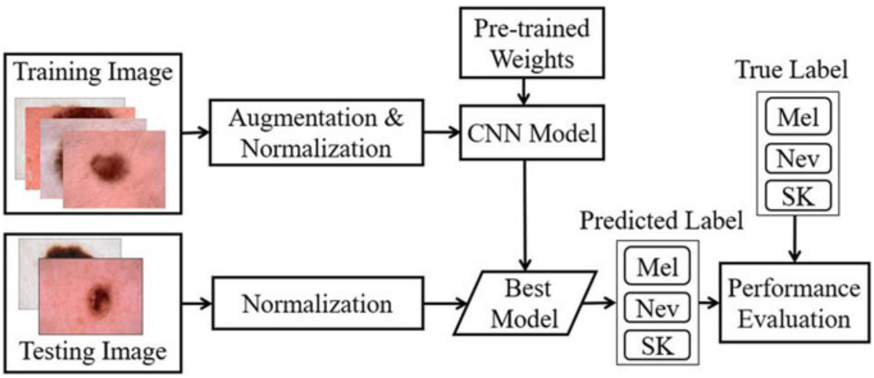


Fig. 2 Block diagram of the proposed framework for an automatic SLC toward melanoma recognition using CNN-based classifiers, preprocessing, and transfer learning

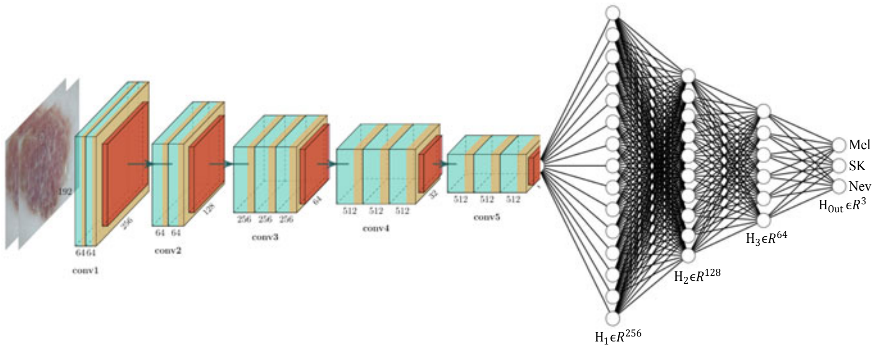


Fig. 3 CNN network for the SLC, where $H_m \in \mathcal{R}^n$ is the mth hidden layer in n-dimensional space. The output layer, H_{Out} lies in 3-dimensional (Mel, SK, and Nev) space

However, in this article, the CNN model is shown in Fig. 3, which has 13 convolutional layers in the 5 convolutional blocks. Each block ends up with a max-pooling layer to lower the computational complexity by reducing the number of connections between convolutional layers, accelerating the CNN models to be generalized by reducing the overfitting. The convolutional layers' output is fed to the fully connected (FC) layers, where we use 3-FC layers and 1-output layer (3 neurons for Mel, SK, and Nev classes). Global average pooling (GAP) was used between convolutional layers and FC layers in place of the traditional flatten layer due to state-of-the-art performance for the image classification [28]. In GAP, only one feature map is generated for each corresponding category, which has an extreme dimensionality reduction to avoid overfitting. In GAP, $height \times weight \times depth$ dimensional tensor reduced to $1 \times 1 \times depth$, where each $height \times width$ feature map transfers to a single number by averaging the $height.width$ values. Additionally, GAP also provides a lightweight SLC network, which makes it suitable for the real-time SLC-CAD systems for the dermatologists. However, for the fine-tuning the convolutional layers, different layers of feature extractor were frozen to select the optimum number layers to be frozen by maximizing the AUC for the SLC.

The scarcity of the training images, especially where the annotation is arduous and costly, can be overcome using a transfer learning [29]. The kernels of the feature extractor (2-D convolutional layers), as shown in Fig. 3, were initialized using the previously trained weights from the ImageNet [30]. The kernels in FC layers were initialized using a glorot uniform distribution[31]. Glorot distribution, also called Xavier distribution, is centered on the mean of 0 with a standard deviation of $\sqrt{(}2/(F_{\text{in}} + F_{\text{out}}))$, where F_{in} and F_{out} are the number of input and output units, respectively, in the weight tensor. Different image augmentations were employed to improve the proposed framework's generalization ability to the unseen test data, which also makes sure that validation error continues to decrease with the training error. Random rotation (0°–90°), width shift (10%), height shift (10%), random shearing (20%), zooming (20%), and horizontal & vertical flipping were used as an image augmentation, in the proposed pipeline, where the outer pixels were filled using a reflection method. The images were also transferred to [0 1] before feeding them into the CNN network. As we saw from the aspect ratio (AR) distribution of the ISIC-2017 images, AR lies in 3:4. So, we resized the input images to 192×256 pixels using the nearest-neighbor interpolation technique [8]. Categorical cross-entropy was the loss function, which was optimized to maximize the average accuracy of lesion classification in our framework. The loss function was optimized using the Adam [32] optimizer with initial learning rate (LR), exponential decay rates ($\beta_1,\ \beta_2$) as $LR = 0.0001$, $\beta_1 = 0.9$, and $\beta_2 = 0.999$, respectively, without AMSGrad variant. The initial learning rate was reduced after 5 epochs by 20.0 % if validation loss stops improving. The proposed pipeline was trained in a machine, as mentioned in Sect. 2.1, with a batch size of 8.0.

2.3 Evaluation Criterion

The proposed pipeline was evaluated using the confusion matrix of true positive (TP), false positive (FP), false negative (FN), and true negative (TN). The recall, precision, and $F1$-score were also used, where the recall quantifies the type-II error (the sample with target syndromes, but wrongly fails to be refused) and the precision measure the percentage of correctly classified positive patients from all positive recognition. The i1-score indicates the harmonic mean of recall and precision, which shows the tradeoff between them. Additionally, how well lesion prediction is ranked rather than their absolute value is measured using the AUC.

3 Results and Discussion

In this section, the results on the ISIC-2017 test dataset (see in Table 1) for the SLC are presented. At the end of this section, state-of-the-art methods are compared against the proposed framework on the same dataset.

A classification report, to visualize the precision, recall, $F1$-score, and support scores per-class basis, is shown in Table 2. A classification report is a deeper intuition for the quantitative evaluation of the classifier, which can also show the weaknesses in a particular class of a multi-class problem. The results, as shown in Table 2, show that the correctly classifying samples of Mel, Nev, and SK are 62.0%, 76.0%, and 73.0% respectively, while the respective type-II error (false-negative rate) is 38.0%, 24.0%, and 27.0%. query Please check the clarity of the sentence 'The support-weighted recall...rejected by the proposed framework'. The support-weighted recall, 73.0% indicates that 27.0% samples having target symptoms, but erroneously fails to be rejected by the proposed framework. The support-weighted precision, 76.0% indicates that only 24.0% samples are wrongly classified among all the recognized

Table 2 Classification report for the SLC, where the weights of the classes were calculated from the supported samples for averaging the metrics

Class	Precision	Recall	$F1$-score	Support
Mel	0.53	0.62	0.57	117
Nev	0.87	0.76	0.81	393
SK	0.55	0.73	0.63	90
Weighted average	0.76	0.73	0.74	600

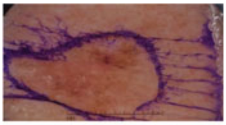
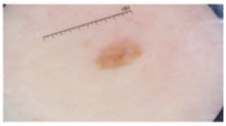
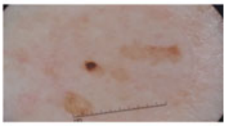
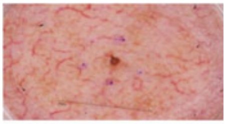
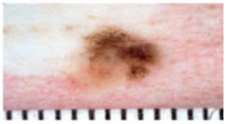
Table 3 Confusion matrix of the test results, where each column and row represent the instances in a predicted and actual class respectively

		Predicted		
		Mel	Nev	SK
Actual	Mel	72	28	17
	Nev	57	300	36
	SK	8	16	66

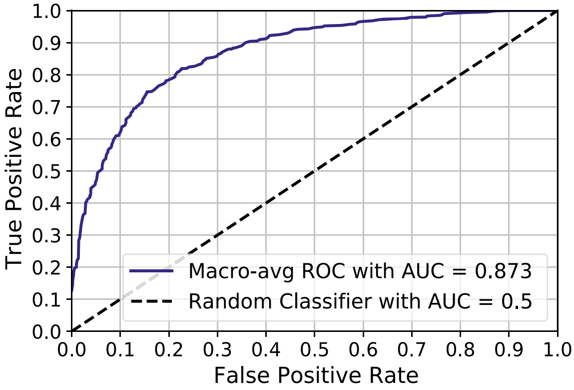
true classes. The $F1$-score reveals that both the precision and recall are better from the proposed framework for the SLC of the ISIC-2017, although the uneven class distribution was being used for training. The more details the class-wise investigation, of the proposed SLC, have been present in the confusion matrix in Table 3. The confusion matrix, as shown in Table 3, presents the number of correct and incorrect predictions by the proposed framework with a count value. Table 3 shows that among 117, 393, and 92 samples, Mel, Nev, and SK classes are classified as (72, 28, and 17), (57, 300, and 36), and (8, 16, and 66), respectively, to Mel, Nev, and SK. Among 117 samples of the Mel classified as the Nev is undesirable FN. Table 1 shows that a more number of samples are in the Nev class in the training dataset, which makes the biased classifier toward Nev. The results, as shown in Table 3, also depict that a significant number of the Mel and SK samples are predicted as Nev. Some of the miss-classified images by the proposed pipeline are shown in Table 4, where we show the difficulties for the correct classifications. From Table 4, it is seen that images (first two rows) having a true class of Mel are classified as SK and Nev with respective confidence probability of 0.934 and 0.88. Similarly, all other images in Table 4 are wrongly classified with a degree of confidence probability. If we visually inspect those images, it seems those images, especially in the fifth row, seem like a Mel as the texture of the lesion area is more complex. However, the dermatologist tells that it is Nev. The probable reason behind such a wrong classification by the proposed framework is the lack of diversity in the training samples, inter-class diversity, and intra-class similarity.

The ROC and precision-recall curves are shown in Fig. 4a, b, respectively. It is observed from Fig. 4a that for a 10.0% false-positive rate, the true-positive rates for the SLC are approximately 62.0%. The corresponding macro-average AUC from

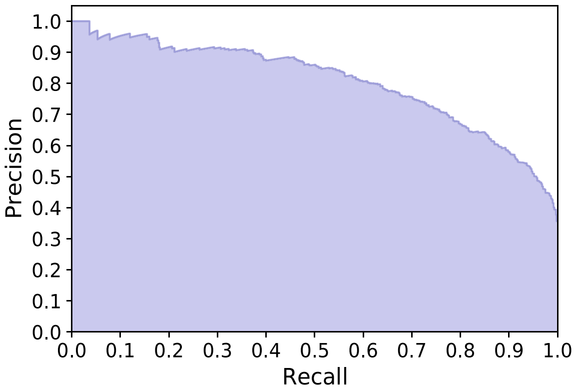
Table 4 Several examples of wrongly classified images, from the proposed framework, with confidence probability

Example Images	Predicted Class with Confidence
	**Actual Class:** Mel **Predicted Class:** SK **Confidence Probability:** 0.934
	**Actual Class:** Mel **Predicted Class:** Nev **Confidence Probability:** 0.880
	**Actual Class:** SK **Predicted Class:** Nev **Confidence Probability:** 0.994
	**Actual Class:** SK **Predicted Class:** Mel **Confidence Probability:** 0.876
	**Actual Class:** Nev **Predicted Class:** Mel **Confidence Probability:** 0.972
	**Actual Class:** Nev **Predicted Class:** SK **Confidence Probability:** 0.941

the ROC curve is 0.873, which indicates that for any given random sample, the probability of accurate recognition as Mel, Nev, and SK is as high as 87.3%. The precision-recall curve, as shown in Fig. 4b, shows the tradeoff between precision and recall for different thresholds, where the high area under the curve represents both high recall and precision. High scores for both show that the proposed pipeline is a blessing with the accurate results as well as returning a majority of all positive results (high recall). The macro-average precision from the proposed pipeline is 80.6%, which indicates that the proposed pipeline is well-suited for the SLC for melanoma recognition.



(a)



(b)

Fig. 4 **a** ROC curve to summarize the tradeoff between the true-positive rate and false-positive rate, and **b** precision-recall curve to summarize the tradeoff between the true-positive rate and the positive predictive value

Table 5 shows the state-of-the-art comparison of our proposed pipeline with recent works, where AlexNet [33] and ResNet-101[34] were implemented in [35] for the SLC. The proposed framework produces the best classification of the skin lesion, as shown in Table 5. Our pipeline produces the best results concerning the recall and precision beating the state-of-the-art works in [36] by a 12.0% margin, and in [34] by a 5.0%, respectively. Our method beats method in [34] by the margin of 39.0%, and 5.0% in terms of the recall and precision, respectively, although the AUC is same in both the methods.

Table 5 State-of-the-art comparison with the proposed framework, which were trained, validated, and tested on the ISIC-2017 dataset

Authors	Year	Recall	Precision	$F1$-Score	AUC
AlexNet [33]	2018	0.34	0.65	–	0.86
ResNet-101 [34]	2018	0.34	0.71	–	0.87
Method-1[a] [36]	2018	0.61	–	–	0.84
Method-2[b] [37]	2018	0.57	0.68	–	0.85
GR[c] [38]	2019	–	–	–	0.78
Our proposed	2020	0.73	0.76	0.73	0.87

[a]ResNet50 + RAPooling + RankOpt [36]
[b]ResNet50 + Eigen Decomposition + SVM [37]
[c]Gabor Wavelet-based CNN [38]

4 Conclusion

In this article, an automatic and robust framework for melanoma recognition has been proposed and implemented. The potentiality of the proposed framework has been validated via several comprehensive experiments. The scarcity of using fewer manually annotated dermoscopic images to build a generic framework was overcome by employing a transfer learning, previously trained weights, and geometric augmentation. Additional tuning of the hyper-parameters and the more related augmentations may yield better recognition results for the SLC. The proposed framework will be tested on other different datasets of dermoscopic images for the SLC in the future. The proposed pipeline will be employed in other domains for recognition to verify the adaptability and generality.

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Chapter 6
A CNN Based Deep Learning Approach for Leukocytes Classification in Peripheral Blood from Microscopic Smear Blood Images

Mohammad Badhruddouza Khan, Tobibul Islam, Mohiuddin Ahmad, Rahat Shahrior, and Zannatun Naiem Riya

1 Introduction

White blood cells, i.e., leukocytes, the fundamental section of the immune system, refer to the mobile factors of blood containing the nucleus barring for the hemoglobin. Moving from blood to tissues, it ensures the body's defense against the invasion of the foreign microorganisms to protect the body from multiple infections and diseases. With a potential production of antibodies, white blood cells hold up a reflection of the immune status of the body [1]. As an indicator device of the presence of hidden infections and ailments in our body, its differential counting performs a great function in assembly up the analysis of hematological elements and alerting medical doctors to undiagnosed scientific prerequisites such as autoimmune diseases, immune deficiencies, and blood problems as properly [2]. And leukemia is one of the major concerns in this aspect as it has one of the highest mortality rates worldwide [3]. To address this sort of disease to the proper treatment plan, the white blood cell count is efficiently required and so the classification of white blood cells as they include a series of variations [4]. WBCs can be categorized into two types, defined by the appearance of the cytoplasm. They are granulocytes including (basophil, eosinophil, and neutrophil) and agranulocytes including (lymphocytes and monocytes). And this variation picks up the complications in the classification process as well as further diagnosis. To build up a prompt guideline for diagnosis and treatment, it takes an enormous wealth of clinical experiences and cognitive knowledge for the pathologists

M. B. Khan · T. Islam (✉) · R. Shahrior · Z. N. Riya
Department of Biomedical Engineering,
Khulna University of Engineering & Technology, Khulna 9200, Bangladesh

M. Ahmad
Department of Electrical & Electronic Engineering, Khulna University of Engineering & Technology, Khulna 9200, Bangladesh
e-mail: ahmad@eee.kuet.ac.bd

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and doctors on which we can supposedly rely on. But this manual process of classification using peripheral blood images under a microscope through observation of matching features appears to be quite tedious and time-consuming [5]. Such kind of manual classification process is not increasingly preferred due to these disadvantages. But this manual classification process is not the only solution to this case, several methods or systems have been proposed till now, such as feature engineering, SVM (support vector machine), and transfer learning. These approaches have achieved a praiseworthy remark, but due to the requirement of multiple steps, i.e., preprocessing, segmentation, feature extraction, and classification, these method methods seem to be quite painstaking, subjective, and complicated. The preprocessing step comes up to generate low noise improve color correction as well as removing the artifacts, so that proper contrast images can be achieved. But no preprocessing algorithms are designed that are conveniently specialized in enhancement. Segmentation refers to the separation of each WBCs components (nuclei, cytoplasm). Since each WBC has its shape and color nature, it seems quite difficult to establish generalized algorithms for different ones to proper segmentation. Even in some papers, color methods are applied to segment different types of WBCs but did not result in any fruitful performance that can be adept enough as the cytoplasm is colorless and boundary of it cannot be detected in most cases [6]. Considering the drawbacks, a demand for a more convenient approach has been arisen to meet the crisis. In recent years, DL has been a vastly promoted method for image recognition, speech detection, and visual analysis which can bring in a promising significance in medical fields to merge the expansion of clinical diagnosis and disease recognition. Due to the insensitivity to the image quality, there is no need for any stage to enhance the image quality regarding deep learning methods. Either any necessity of segmentation stages or handcrafted features are not required to proceed since those terms are dealt with the convolutional concept more precisely [3]. Figure 1 represents our overall WBC subtype classification process. When we put a blood image, it will automatically subtype the WBC by using a customized CNN model.

The convolutional neural network (CNN), an advanced deep learning technology inspired by the natural visual perception mechanism, has been flourished on exponential growth worldwide by performing multi-convolution of the same input through the multi-layers network [2]. Pursuing all this, a deep learning model has been propounded based on a convolutional neural network model with a prospect of confirming a better and wealthier solution in classifying four types of WBCs including eosinophil, neutrophil, monocyte, and lymphocyte in this paper. The proposed model is efficiently susceptible to remove the complicated process in the classification procedure. To improve image recognition, the model adequately enables denoising and removing artifacts to avoid errors in terms. And a stop to quit education, the technique is additionally worried that can be used to perceive the displaced, noisy, and twisted two-dimensional picture besides inflicting on kind of deformity.

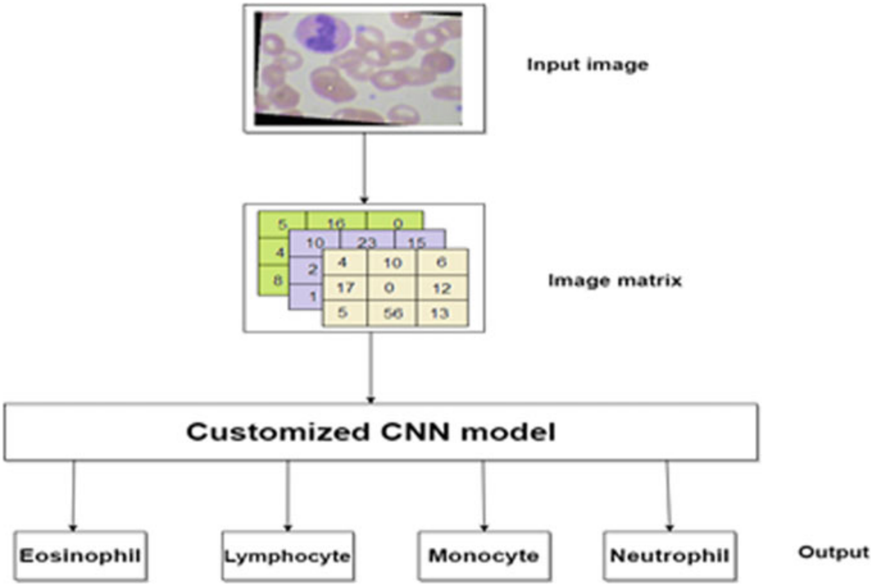

Fig. 1 Proposed WBC subtype classifying CNN model

2 Related Works

Tiwari et al. [7] aim at developing a deep learning framework to solve the concerns of blood cell categorization for blood diagnosis purposes. A CNN based system is designed to categorize the representations of the blood cells into cell subtypes. A CNN architecture is built to do this stuff that includes two convolution layers, two pooling layers, a fully-connected layer: a hidden layer and output layer, and loss function: cross-entropy. SVM and naive Bayes were then eventually used as a basis for comparison with the conceptual CNN based system (DCLNN).

The authors in [8] present a method for classifying RBC's anomalies based on malformed RBC image shapes using the SVM and deep learning approaches in RBC cell classification comparisons. This study reveals the SVM classification algorithm may identify cells in all situations, whether in a small or large dataset, whereas deep learning primarily takes place on a big and very big dataset which RBCs dataset must produce in large quantities to operate with success in case of deformity of RBCs. Vogado et al. [9] aim to identify leukemia utilizing images of blood smear. CNNs had been used to define the input image and the selection and reduction of the functions. The SVM is being used in the classification step to classify images whether it is pathogenic or not? They used two image databases to test the results collected, one containing just one leukocyte per image and another with lots of leukocytes.

Here in [10], deep neural network (DNN) learning was built in this paper to help healthcare professionals in treating patients and to enrich the reliability of detection of heart disease. Deep learning classification and prediction methods were constructed with deep neural networks with linear and nonlinear transfer functions, normalization and dropouts, and a numeric sigmoid categorization using deep learning systems to establish a strong and enriched model of analysis and regression of heart diseases. The authors in [11] have described an automated detecting program based on the deep training algorithm to use human physiological indicators to detect hyperlipidemia. It experimented with data extensions and corrections. By the use of the aforementioned increasing getting to know the algorithm, the proposed framework confirmed greater overall performance with a much less quantity of uncooked facts and labor, hence displaying enormous development in the encoding of clinical text.

This design and implementation are directly related to the obstacle avoidance, to accomplish the objective obstacle detection, hole detection, and the guiding information feedback which are also solely related to each other. Therefore, the design aspects regarding the aforementioned significant criteria are presented with a detail description by the following points.

3 Data Collection and Preprocessing

In this research, a given set of white blood cell images named BCCD data is used to that is quite attainable for the researchers by the authors [12, 13]. The dataset offers approximately 12,500 augmented images of WBCs (JPEG) associated with cell type labels (CSV). Except for basophil, the other four types of WBC images are provided as the percentage of basophil in the blood is very low (0–1%) and each of the four types of white blood cells has around 3000 images grouped into four folders, so far. Among these, about 9957 images of white blood cells are used for training, and the rest (around 2487 images) on testing of the classification of those four types of blood cells. Again, an additional data set containing 410 images of white blood cells (pre-augmentation) is accompanied by the other one. The image served here is of width $\times$ height of 640 $\times$ 480, and the corresponding data acquisition has been attained with the proper consent of authorization. The categorization of data based on their types or classes is given in Table 1 (Figs. 2, 3 and 4).

Table 1 Categorization of WBCs based on their types

Types	Training images	Test images	Image format
Eosinophil	2497	623	JPEG
Neutrophil	2483	624	JPEG
Monocyte	2478	620	JPEG
Lymphocyte	2499	620	JPEG
Total	9957	2487	JPEG

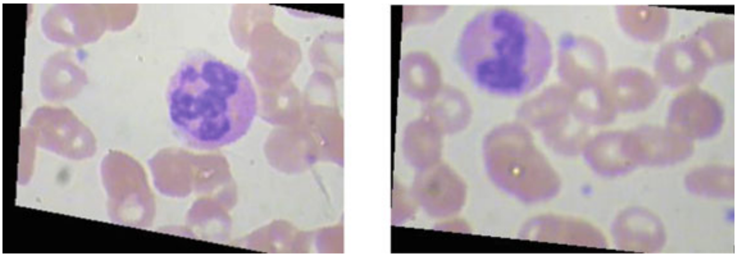

Fig. 2 Eosinophil training images

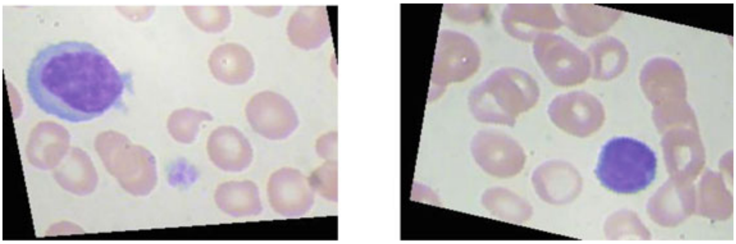

Fig. 3 Lymphocyte training images

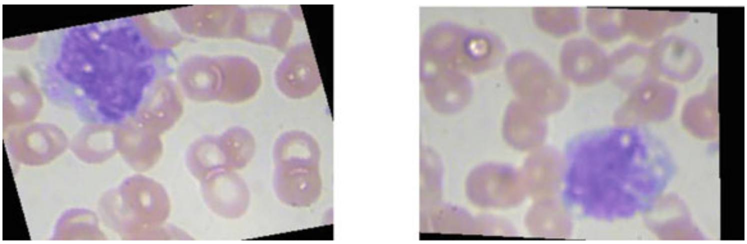

Fig. 4 Monocyte training images

4 Methodology

In this part of this research paper, we represent our proposed customized CNN architecture for blood WBC subtypes classification. Figure 5 represents the proposed CNN model. This computer-aided diagnosis mode is not a pretrained CNN model. It is a customized model, especially for blood WBC subtyping. Convolution neural network (CNN) is one of the environment-friendly frameworks for an object, face, and as properly as picture focus due to the fact of its more than one convolution layer and pooling layers [14]. In our customized model, the first step is the input convolution layer. Our input image is $320 \times 240 \times 3$ where 3 is color channels that go into 2D convolution layers and found output $320 \times 240 \times 12$ matrix, and then it sends to the batch normalization stage. After applying batch normalization, it formed

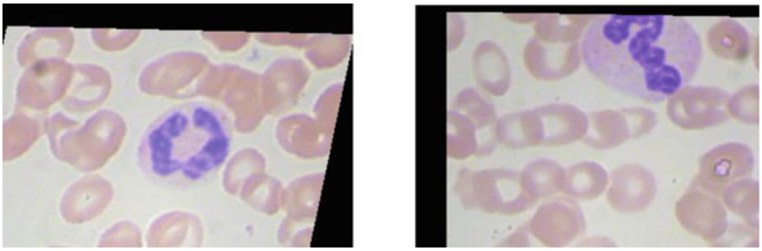

Fig. 5 Neutrophil training images

the $320 \times 240 \times 12$ matrix output on normalized data which significantly decreases the training speed. The same things are done on the second 2D convolution and batch normalization layers.

The 2D convolution is a very simple operation with some kernels which is a weighting matrix. The kernels are applied to the 2D input image matrix which performs element-wise multiplication and sums up the value to produce a single output pixel value. From the second batch normalization, our module is divided into two branches, so that it can work parallel with the layers, and each branch contain again 2D convolution and batch normalization unit. After that, it enters into the concatenate layers. The concatenate layers take the input and concatenate them alongside a definite dimension. After that, max-pooling and 2D convolution are once more carried out in our model. It speeds up the neural community overall performance and additionally stables the network overall performance even though the center and output vector dimension is identified as a $320 \times 240 \times 12$ image matrix. From batch normalization to 2D convolution through different branching, we used six of these same stages in our customized model. After this, global average-pooling is carried out in which output is feed into the dense layer. It finds out the suggestion of all aspects and is fed to the softmax layers. In the end, the dense layer accumulates all the output of preceding layers. In the network, every neuron takes the center of all different neurons and accumulates it and offers a closing prediction (Fig. 6).

5 Results and Discussions

In our custom-made model for leukocytes (WBC) subtype classification using a particular validations value, we found that our model classification accuracy is very high compared with other states of artworks. In Fig. 7, we found the training accuracy of about 98.29% where the validation accuracy was 94.82%. So the proposed model is very much efficient for white blood cell subtypes classification. It also indicates a clear conception that the proposed model is highly applicable in real-life healthcare applications. It can also indicate a CADx based high potential opportunity for future healthcare researches. We evaluated our custom-made model using different

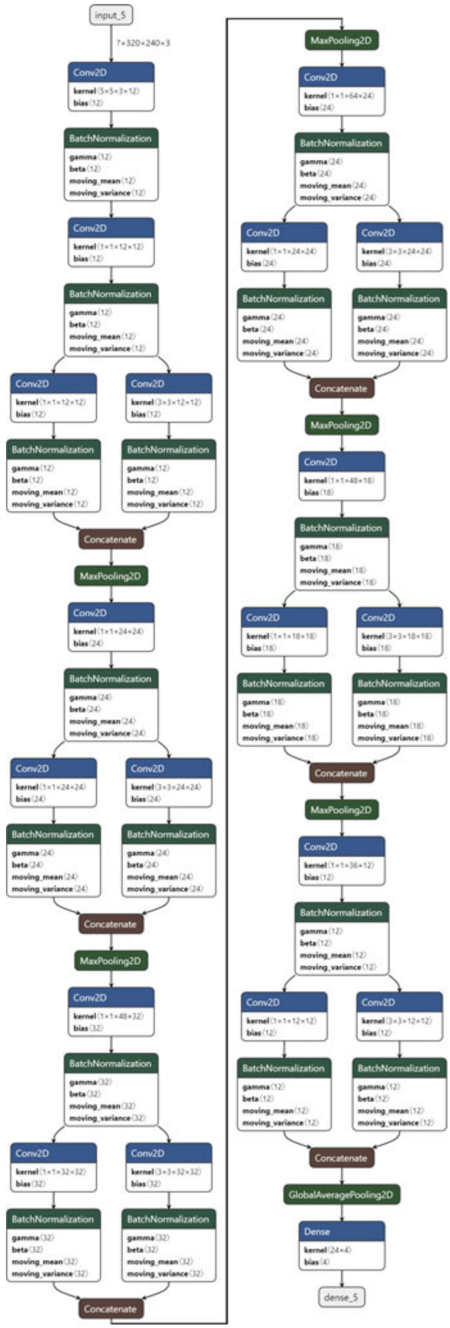

Fig. 6 Proposed CNN architecture for blood subtype training

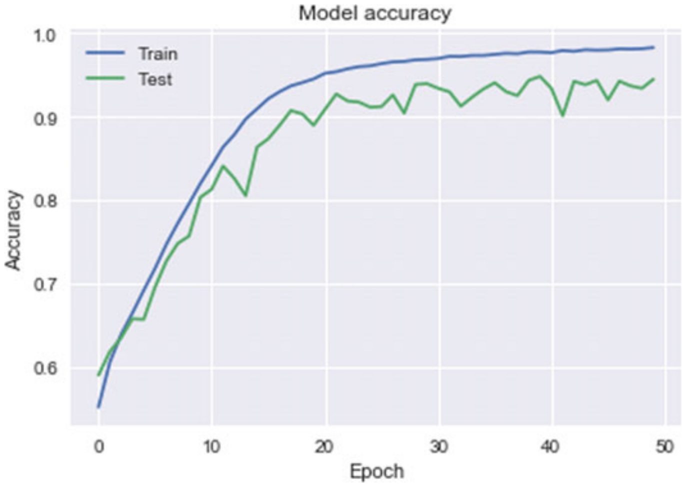

Fig. 7 Train and test accuracy curve of our proposed CNN model

parameters like training accuracy, test accuracy, training loss, validation loss, precession, epoch recall, and so on. Figure 8 represents the training and testing loss of our proposed model. From the figure, we see that training loss is less than 0.12 and test loss is about 0.19 which is very efficient for this kind of CNN model-based research. We have in contrast our proposed customized model for WBC subtype classification with some researchers and that are given in Table 2.

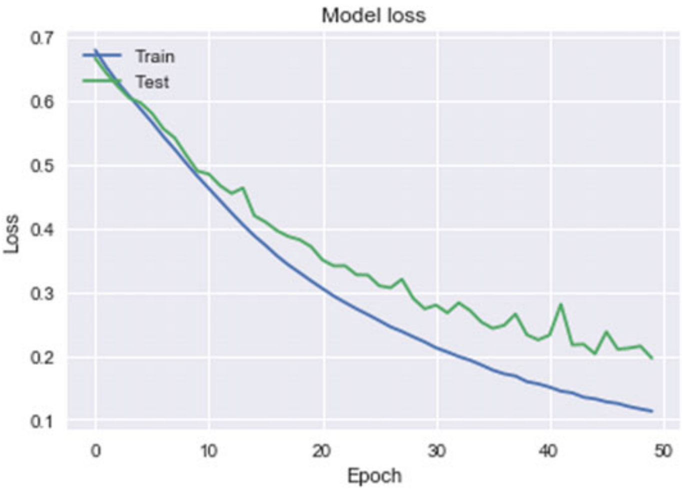

Fig. 8 Train and test loss curve of our proposed CNN mode

Table 2 Comparison with some state-of-the-art researches

References	Research work is done
[8]	Red blood cell classification using SVM and machine learning approach where SVM accuracy was 84% and the machine learning approach was about only 24%
[1]	Identification of blood subtype by using SVM machine learning approach where accuracy was 89%
[5]	Automatic recognition of five types of blood cell using ANN and SVM approach and accuracy was not more than 90%
[6]	Label-free identification of white blood cells using a machine learning approach
[7]	Classification of the cell for diseases diagnosis using different machine learning approach where accuracy was 77.47–82.12%
Proposed model	Leukocytes classification in peripheral blood by using customized machine learning CNN model where accuracy is about 98%

6 Conclusion

From the results of our proposed experiment, it can suggest that our model is highly efficient for classifying leukocytes comparing with other state-of-the-art researches, as we proposed a CNN based model to classify the white blood cell subtype which reduces the segmentation and processing time with a significant validation accuracy of above 94% and a quite high f-score, recall, and precision. As future work, improvement in CNN tuning and enrichment of data amount is proposed, so that our model can be used in real-life medical purposes too. By using this method, we can save medical diagnostic costs as well as the time of diagnosis.

Acknowledgements In our proposed method, we used the BCCD blood dataset for training the customized model. Figs. 2, 3, 4, and 5 and are taken from the dataset. These are one of the author's dataset taken for checking out in our proposed method.

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Chapter 7
PIRATE: Design and Implementation of Pipe Inspection Robot

Md. Hafizul Imran, Md. Ziaul Haque Zim, and Minhaz Ahmmed

1 Introduction

In recent decades, many pipeline inspection robot systems have been developed. The development of pipeline robots began for inspecting large pipelines starting from 100 to 300 mm. These pipelines are commonly utilized in manufacturing sites as sewer pipes and gas and oil pipelines. They are also utilized in atomic power plants. Pipeline inspection robot systems improve safety and reduce the time period. Buried pipes are commonly used for urban gas, sewage, chemical plants, nuclear power plants, etc., which is indispensable in our life. Oil and gas transfer between nearby countries is mostly done by pipelines. However, inspections and maintenance are very costly, whereas using efficient and proper systems are must.

Like a pipeline system is indispensable in our life which is often facing a need for inspection which is a headache for the concerned body. The inspection process requires a lot of groundwork beforehand. Excavation of ground and backfilled after the inspection and maintenance is a lengthy process. Previously this work required a lot of time and money, as efficiency was very less and excavation took most of the time for this work. Now an updated process is in practice to save time and money which is internal inspection systems. This time inspection requires only one ground hole big enough to pass through the inspection system into the sewage. This process

Md. Hafizul Imran (✉)
Department of Software Engineering, Daffodil International University, Dhaka, Bangladesh
e-mail: hafiz33-658@diu.edu.bd

Md. Ziaul Haque Zim (✉)
Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh
e-mail: ziaul15-1133@diu.edu.bd

M. Ahmmed
Department of Information Technologies, Dhaka University, Dhaka, Bangladesh

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does not even create much of a problem for pedestrians as most of the work is done underground and less torn-up roads.

2 Literature Review

Basic technologies for internal pipe inspection are rapidly being developed. Several of these internal pipeline inspection robots for gas, oil, and sewage have made its way to the professional level. But most of them do have limitations with range and steering mechanism.

In-pipe robots can be classified into several elementary forms according to movement patterns. Wheel-type pipeline inspection robots are popular and have been investigated in the laboratory.

Kawaguchi et al. [1] developed a dual magnetic wheel-based pipeline robot that lacks out as the pipes loses magnetic capabilities. Ryew et al. [2] worked on this active mechanism system where two individual vehicles connected through a double active universal mechanism that provides omnidirectional steering where wheels are on 120° apart. Nassiraei et al. [3] developed a fully autonomous robot system dedicated to sewer pipe systems which included an underbody four-wheel mechanism. Suzumori et al. [4] developed a 1-in. inspection robot system which is pneumatic flexible. Its reliability and low speed are not effective enough. The inchworm robot is analogous to the snake-like robot. Zhang and Yan [5] worked on this pipeline diameter adaptive and traction force adjusting system which allows the system to run through different sized pipelines which are not very new but their mechanism was. Li Chen, Shugen Ma et al. [6] developed a snake robot in traveling wave locomotion. Transetha and Patterson [7] suggested a nonsmoothed mathematical model for a wheel-less snake-like robot, which allows the snake robot to push against external obstacles aside from flat ground. Crespi and Ijespeert [8] have shown a lot of snake-like robot simulation on swimming and crawling. Park et al. [9] worked on an actively adaptable pantograph structure-based in-pipe robot where pipes ranging from 400 to 700 mm.

Previously developed pipeline robot systems are tested on straight, slightly inclined, or just curved pipelines because the task space of the developed robot is primarily in the sewer, gas, and oil pipelines. Navigation through a pipeline with multiple curves or T-branches remains a difficult problem. Some robots developed with active joints [2, 10–12] or active universal joints to overcome difficult T, L shapes. But these active joint-based robot systems are very big, complex, and costly to implement.

3 Heuristics

The first problem mostly faced due to conventional steering mechanisms is that it cannot cope-up with pipe bends properly. T shape is one that requires most of the attention, as most of the developed systems fail to turn these bends properly.

Secondly the problem with traveled distance, most of the system is required to have communication and power cable along with it because of which traveled distance is reduced in a bunch. Frictions on these wires face due to L-shaped and T-shaped turns do create unnecessary problems and sometimes which end in damaging the cable.

The first problem is proposed to be solved by the prototyped robot system; as it has the advantage of its separated structure and changeable size and the octopus-like steering mechanism, which can easily grip and move any turns with ease.

4 Proposed Methodology

We proposed our “PIRATE” the pipe inspection robot design and implementation model have many steps as described below.

4.1 Mechanical Structure and Design

The most impressive development of the system is the structure and driving mechanism, which we have named as the Octo-naked structure. A structure is consisting of three parts, where the forward part consists of the primary forward driving mechanism along with all the sensors and camera for the surrounding data, which is also common for the last part. They share the same parts for driving mechanisms. These two parts do have three individual sprung motor mechanisms that allow changeable positioning of the wheels.

Four wheels in the middle section give an extra bit of traction to the driving systems. The middle part consists of the micro-processing system along with communication systems.

As the system consists of three individual parts, we have a chance to fix its structure as such it helps us fit in. As parts are connected to each other with the help of spring, it does allow us to move more flexibly than ever and moving through different special pipe shapes inside sewage, i.e., C shape, T shape, L shape, etc. Getting surrounding the data system can flexibly move its head and tail over these shapes. Other developed systems had problems with one or another shape but with all these kept in mind, we figured this particular structure is the solution to previous systems problems.

Problems with other changeable shaped systems. They have a non-changing volume structure which creates problems in some cases. We managed to put the system in such a structure that it does change its whole volume along with its shape

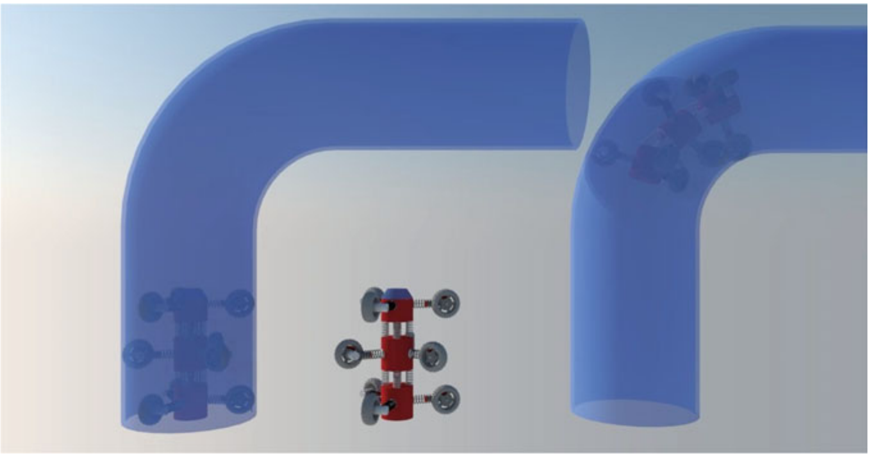


Fig. 1 PIRATE in L-shaped and C-shaped models

changes. Our system can easily move its shape from three structured shapes to one single structure. The total volume of the whole system can easily shift from 1389 to 926 °C, which helps us with fitting the whole system into a very congested structure (Fig. 1).

As the driving wheel along with the wheelbase is changeable as well. It surely gives an advantage over other existing systems in the market.

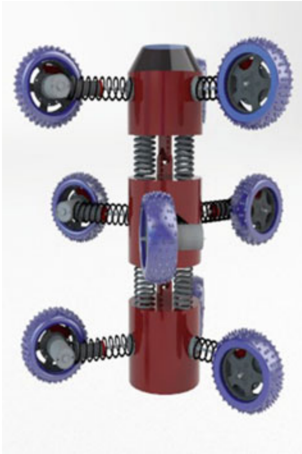
4.2 Helical Spring-Based Structure

When a traditional spring, while not stiffness variability options, is compressed or stretched from its resting position, it exerts associate opposing force close to proportional to its amendment long. The important reason for spring over variably controllable hydraulics is the weight and complex compression and extension firing mechanism which sometimes takes time to recharge. Three springs altogether oppose the rotational forces of each of the springs which prevent the robot from tipping over. Apart from uses of spring in the three-part connectivity, it is also used for the operation of the motor at individual position, which will as well be explained in the next chapter. As helical springs follow Hooke's law, we have made sure that the system does not have extreme force over the springs.

This individual spring mechanism helps with all the usual pull mechanisms while giving the system three degrees of freedom on the movements as well. As mentioned earlier, the problem statements of the previously developed system (Fig. 2).

It is clear to see that we have managed to solve all the previous ones with proposed spring-based structure. This structure solves the C- and T-shaped problems along with

Fig. 2 3D Model



all the other even I-shaped movements which are found as problems of movements that easily venture.

4.3 Communication System: Wi-Fi-Based

This robotic system for sewage pipeline inspection consists of two parts of which inside pipe inspection is one and another is ground workstation monitoring the state of the inspection robot and so is for controlling. Automation was avoided during the building of this prototype as automated may fail anytime and does consume a lot of energy for an intelligent system to run the whole calculation process. But here non-automated systems only will have to get all the sensor's value as well as the CCD camera feed to send to the controller through Wi-Fi system which was carefully set up. Data transfer error rate reduced to have the system run smoothly. The whole controlling and data transfer are done through duplex Wi-Fi communication. Real-time sensor value and camera feed help to track system position and condition.

As the system itself is not intelligent by processing data, so we have provided the system some very important sensing capacity which will help controllers with all the data needed to control at ease. Magnetic flux sensor helps the system to measure pipe thickness, distance sensor for measuring distance, time of flight sensor to detect an incoming collision, measurement of speed, light sensor, tail positioning sensor to have all the data about the environment. The camera feed gives real-time visibility about the surrounding along with all sensor data. All these data are accessible in real time and stored as well in the system for the record.

Having Wi-Fi over other communication protocols was important because of the lack of other systems. Radio frequencies half-duplex system and WiMAX was costly and too much for this project. Wired-based communication has the same problem as any sharp shape or derby can damage the wire. And this lot of wire to run throughout the sewage is costly enough as our primary priority was to cut costs as much as possible.

Wi-Fi consumes almost the same power as of the Ethernet data transfer but much more secure with proper encoding. As the system will not have to be string around the wire it will be weightless. Troubleshoot is easier with its bust availability and updates over the year.

4.4 Robotic Vision and Sensor's

As the robotic system requires a lot of data on the surroundings it is required to have. So, for this purpose, we have used the proper and most needed sensors for the system. So, the controller has the best idea of the surroundings while driving the system. Our sensing mechanism consists of the ToF sensor, electromagnetic Hall sensor, adaptive light mechanism, CCD camera sensor, radiation sensor, ultrasonic sensor.

ToF sensor helps find approaching objects toward the robot. Any object which cannot be detected with the camera can easily be detected with the ToF sensor data. Controlling the robot will be much safer and easier as sewage pipes tend to have other pipes run through and from it.

Magnetic Hall sensors can easily detect damages inside the sewage pipe. An ultrasonic sensor helps to map the route, and this sewage pipe will take and thus helps with mapping the damages inside the pipe and has a better idea of the location of the system if the device is in need of rescue.

For added data collection of the pipe, we have used a radiation sensor which helps with detection of any unnecessary radiation inside the pipe which again can detect electromagnetic radiation as more and more electrical and communication lines tend to pass through/along with sewage pipe to save time and space of underground works. As most of the biochemical waste will pass through sewage pipes along with other wastes. The design was done about the threshold keeping all this in mind.

Finally, the main visual part will give the complete view of the path, which will have the proper visual to be controlled inside the pipe. All the sensor's data and visuals will help with easier control. High-resolution camera technology helps with better visuals which some of the previous systems lack. As data from the high-resolution camera will be hard to pass, an updated compression and encoding system is used to ease up data transfer over the wireless communication system.

5 Development of the Prototype

We have developed a prototype that featuring this newly developed structure which is 500 mm long overall, 100 mm width as the shape is circular in design, and it does have the same height as width. Overtaking all the obstacles like valve location, it can easily run through all the shapes properly with a width of 120 mm in diameter. The front and backside of the robot are cut down and shaped all the parts in a way that it can pass through all the widely used shapes of the pipes. Robot weighs about 10 kg and even with this weight special structure and driving mechanism helps with driving without disturbances. A 10 V 2 A battery system added to the system which can run the system for around 400 m restlessly, which is important to cover a large area in a very short time and reduce the problem to the citizens. And this large battery powers the motor as well as all the electronics. And an added turbo power mechanism is also added for special purposes when the full torque is necessary. Turbo mood supports 15 V 4 A supply. Who efficiency turbo mood is only used when the robot needs an extra bit of push over the normal power supply to the system.

System on chip-based micro-processing is added for onboard computation. As the system supports 64-bit parallel operation, it is needless to say that with this processing power over the main control board system can easily do all the data collection and encoding with very less power input as a system we have used is SoC-based and runs at 5–12 W of power. All these processed data are sent over to the controller over Wi-Fi as we figured this is less costly and easier to use the system, within enclosed space Wi-Fi signals can travel a very long distance. As this Wi-Fi signal, we are using has a very high bandwidth rate. Power consumption, sensing data, data processing, and communication are all solved within the limit to represent this updated model over others.

5.1 3D Structural Diagram

Earlier section mentions that developed project's structure. Now from Fig. 3, we can see the whole system lies as a cylindrical shape. Shape equipped with three individual parts that contain each and every necessary part.

The first section which is uppermost section contains visual and sensory devices. As shown in Fig. 3, the light source illuminates the path for the robot and thus for the visual at the control station. This frontal section does contain few sensors to sense the environment. Time of flight (ToF) sensor gives ideas of the incoming object which might be missed by the operator due to camera quality drop. As we have used CCD camera system which has reputation for quality issue but at the same time, it is reliable and water-resistant. Chances of failing under certain environmental conditions are very rare. The ultrasonic sensor helps with pipe position sensing. As adaptive wheels might not protect the frontal section but sensor values can help the controller decide for the next turn.

The midsection contains mostly microcontroller and communication devices. Similar to the frontal section of the robot, this part again contains ultrasonic sensor and wheel systems.

The third and last section consists of the main driving mechanisms. Whereas other parts wheel does steer as well as driving but this parts wheels only used for forward or reverse movements around 55% forwarding force is sent from this part wheels.

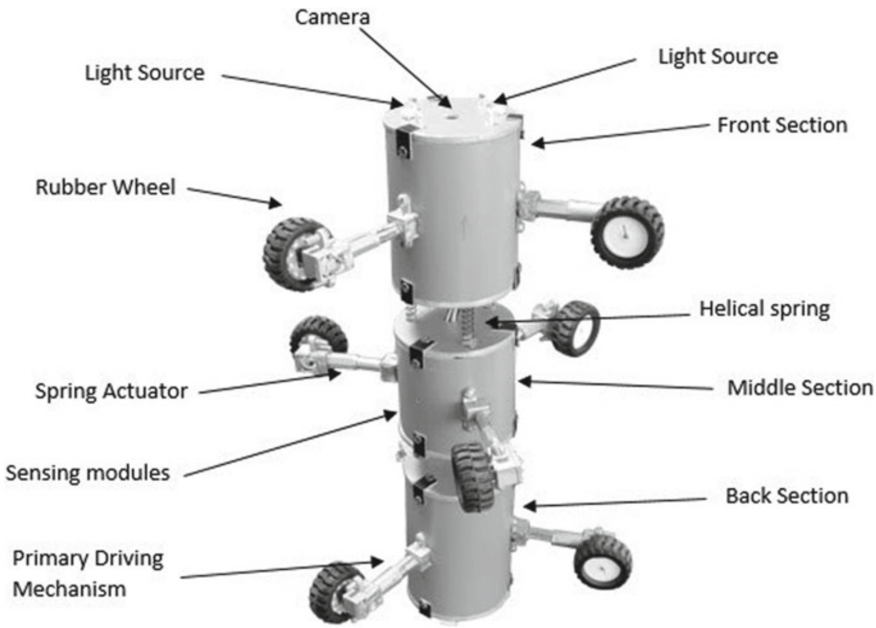


Fig. 3 A 3D structural diagram model of our line inspection robot

5.2 Specifications of the Robot

See Table 1.

Table 1 Specifications of the PIRATE—pipe inspection robot

No.	Spare and parts	Value
1	Weight of each robot module	250 g
2	Motor diameter (L * W * H)	34 * 12 * 10 mm
3	Length of the robot module	5 in.
4	Total length of the robot	19.5 in.
5	Exterior diameter (L * W * H)	19.5 * 5 * 7 in.
6	Max traction force of the robot module	1.088 kN
7	Linear speed	87 rpm
8	Max speed	120 rpm
9	Camera and sensor module length	5 mm
10	Serial communication distance	50 m

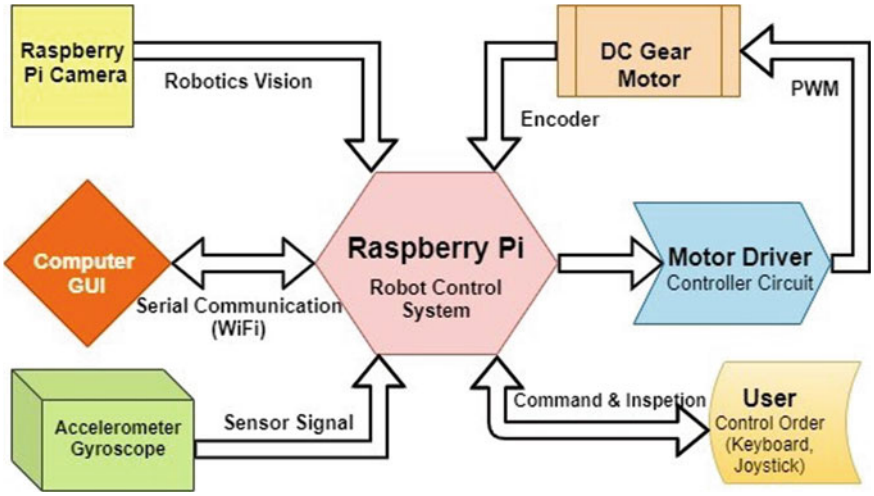


Fig. 4 Working flowchart diagram of our line inspection robot

5.3 *Flowchart Diagram*

The proposed system consists of some major modules, sensors, and computers, namely Gyroscope, Motor Driver Controller, DC Gear motors, Camera, and Raspberry Pi as single board computer system. Below diagram gives a crystal clear idea of full working systems (Fig. 4).

The Raspberry Pi single board computer is used as a robotics control system in "PIRATE" the pipeline inspection robot. This computer system has a camera module to capture the real-time video and stream that to client computer using serial communication. This robot computer system can understand his position using Gyroscope and accelerometer. It also gives command to the motor driver controller circuit for move. Motor driver controller circuit generates PWM signals depending on commands that come from the robot control systems computer.

The user can communicate and gives command to the "PIRATE" through Keyboard, Joystick using the client computer GUI.

5.4 *Motors and Gears*

In this section, we highlight the motor and gear data (Table 2).

Table 2 Specification of the motor and the gearhead

No.	Specification	Value
1	Diameter	34 * 12 * 10 mm
2	Voltage	6–12 V
3	Linear speed	100 rpm @ 6 V
4	Torque	2 kg
5	Current	0.07 A
6	Gearhead reduction	12 V
7	Gearhead maximum continuous torque	16 kg

5.5 3D Model of Robot Chassis

This 3D model in Fig. 5 represents the main developed system. Developed system was a PVC-based body which was low cost and at the same time, lite weight as our one of the main goals was to make this system affordable.

Apart from main body build up it was at the same time important to have a rigid driving mechanism. With that kept in mind stainless steel spring, actuators were used. Same materials were used for springs to hold three individual structures together.

To have maximum possible grip sent to wheels and effective torque uses, rubber wheels were used and avoided magnet-based wheels which are complex in some cases. Plastic wheels lack gripping ability as much as rubber wheels.

This 3D model represents the similar design and materials used in main developed system.

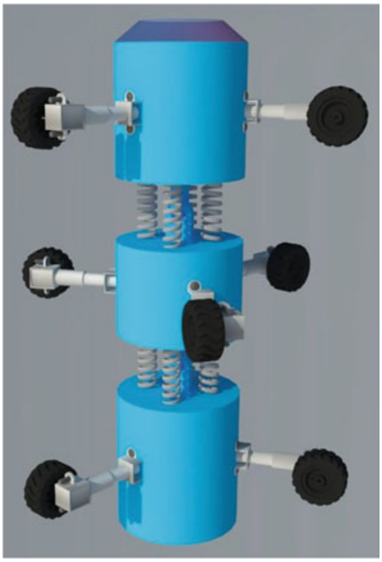


Fig. 5 3D model of PIRATE robot chassis

6 Equations

6.1 Hooke's Law of Helical Spring

In Sect. 4.2, we already described about helical spring-based Structure and Eq. (1) is the mathematical demonstration of that structure actually based on a law of physics and this law stated that force (F) needed to extend or compress a spring by some distance (x) scales linearly with respect to that distance, and that is, $F = kx$, where k is a constant factor characteristic of the spring (i.e., its stiffness), and x is small compared to the total possible deformation of the spring.

$$F = -\frac{d\left(\frac{1}{2}kx^2\right)}{dx} = -kx \tag{1}$$

Equation (2) is for understanding critical damping, and we can compare the characteristic time for the various values of β. For small values of β, we know we have the weakly damped solution.

$$x(t) = (A + \beta t)e^{-\left(\frac{e}{2m}\right)t} \tag{2}$$

6.2 Kinematic Analysis and Optimal Design

The ratio between the input velocity vector and the output velocity vector is expressed as [13]

$$\sigma_{\min}\|\dot{u}\| \leq \left\|\underline{\dot{\theta}}_a\right\| \leq \sigma_{\max}\left\|\underline{\dot{u}}\right\| \tag{3}$$

The static force relation, which is dual to (13), is denoted as

$$\underline{\tau} = \left[G_a^u\right]^T \underline{f} \tag{4}$$

where $\underline{\tau}$ and $\underline{f}$ denote the input torque vector and the operational force vector, respectively. Then, the ratio between τ and f is expressed as

$$\frac{1}{\sigma_{\max}}\left\|\underline{\tau}\right\| \leq \|f\| \leq \frac{1}{\sigma_{\min}}\left\|\underline{\tau}\right\| \tag{5}$$

7 Conclusion

The pipeline inspection robot has been developed that is able to inspect sewage and pipelines. This robot has three major parts, and each consists of parts operated by a micro-DC motor. Independent control of the speed of each consisting part allows steering capability through elbows or T-branches.

Each major part of this robot is foldable by using an embedded four-bar mechanism and a compression spring. This allows the robot to maintain contact and adjust to the inner wall of pipelines with an irregular cross-sectional area. The best strength of each part of this robot is the large contact area for traction over irregular surfaces. They also overcome the sharp corners of branches and elbows.

This study can be applied to design and control of pipeline inspection robots.

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Chapter 8
Wavelet and LSB-Based Encrypted Watermarking Approach to Hide Patient's Information in Medical Image

Faiza Huma, Maryeama Jahan, Ismat Binte Rashid, and Mohammad Abu Yousuf

1 Introduction

Once medical data was stored in paper form, but now, the use of electronic health records has replaced it. According to one recent report, 68% of patients stated they were not confident that their medical records were safe from loss or theft [1]. The security against patient's data leaked or hacked has not been developed yet strongly. Most patient data are available in image formats such as CT scans, MRIs, cardiology videos, and ultrasounds, which holding many confidential safe health information files and can be used for fraud, identity theft, and other fraudulent activities in Medicare. Therefore, to ensure security against identity theft and other fraudulent activities in Medicare and to secure patient's confidential information, we need to figure out techniques of protecting data and medical images. Watermarking and cryptography techniques together are used to ensure the strong security of medical documents and data. Cryptography is a dependable numerical information security method where standard plain content is randomized into an unintelligible configuration (scrambled content) with the goal that it cannot be gotten to by unapproved clients [2]. As one of

F. Huma (✉) · M. Jahan · I. B. Rashid
Information & Communication Engineering, Bangladesh University of Professionals, Dhaka, Bangladesh
e-mail: fhuma46@gmail.com

M. Jahan
e-mail: maryeama02@gmail.com

I. B. Rashid
e-mail: ismatbinterashid99@gmail.com

M. A. Yousuf
Institute of Information Technology, Jahangirnagar University, Dhaka, Bangladesh
e-mail: yousuf@juniv.edu

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the most popular cryptography techniques, nowadays, advanced encryption standard (AES) performs all of its bytes computations instead of bits. AES, therefore, serves as 16 bytes the 128 bits of a plaintext row. Such 16 bytes are structured as a matrix in four columns and four rows. AES uses 10 rounds, 12 rounds for 192-bit, and 14 rounds for 256-bit keys for 128-bit keys. It is an excellent standard for secured electronic communication and also can be applied in many situations where protection of sensitive information is needed. Digital watermarking is a technique where bits of information are embedded in such a way that is completely invisible [3]. Watermarking is done mostly based on the spatial domain or transform domain. The most popular algorithm for watermarking in the spatial domain is the least significant bit modification which is the octet of a multi-byte number which has the least potential value in that place. The least significant bit (LSB) gives the value of the unit and indicates the position of the bit in a binary integer that determines whether or not the number is odd [4]. In the transform domain, The DWT splits the image signal into high and low-frequency parts. The high-frequency part containing information on the edge components, whereas the low-frequency part is again divided into high and low-frequency components. In this research, LSB operation and AES encryption have been applied to the patient's secret data for hiding the information, whereas watermarking a three-layered DWT technique has been applied. Before AES operation, an initialization vector (IV) has been generated using a repetitive multiplication and middle 4-bit selection technique which is to be used as input of AES cryptography. The major contribution of the proposed algorithm has been figured out in the medical field for transmission of different multimedia documents and patient's information over a public network as well as for building a trustworthy system, two-way security providing system, a secured information recovery system. The rest of the paper is organized in the following sections. Section two states related researches on the joint watermarking and cryptography field. Section three gives a brief discussion of the overall proposed watermark embedding and extraction architecture. Section four and five illustrate the performance analysis and conclusion with some discussion of future scopes in this field.

2 Related Works

Fu et al. [5] suggested a cryptographic algorithm based on chaotic maps (Arnold cat map and logistic map–LM that was implemented in medical imaging using the Arnold cat map to change the flat image's 8-bit plane. The LM is used to substitute the pixels. Bigdeli et al. [6] stated that in Arnold's cat diagram there are two flaws. The first is that the iteration times are very brief (usually fewer than 1000 iterations). Secondly, the second is that the height of the flat image is necessarily equal to its distance. In Priya et al. [7], a watermarked medical image encryption technique was proposed using EPR information through integer wavelet transform (IWT) and

LSB technique application. The EPR information considered as a watermark was embedded in a cover medical image. Then, the watermarked image was encrypted as a visually significant encrypted image applying two levels IWT algorithm on the image. Doctor's fingerprint inserted within the LSB bits of wavelet coefficients was used for the authentication purpose of the source. From the experiment, the PSNR value of the encrypted image was achieved above 30. Though the original image and encrypted image were identical because of encryption of the watermarked image, a little amount of distortion was visible in the images. Moniruzzaman et al. [8] focused on DWT and the chaotic medical image watermarking method is to authenticate the medical image by hiding information from the patients within the system. Patient information has been inserted as a binary watermark image into the corresponding medical image. The suggested LSB integration technique, however, achieves a high payload for the various image processing operations at the cost of increased vulnerability. Das and Kundu [9] suggested digital watermarking technology for transforming contourlets. The lower pass sub-bands are selected for the integration of data after three levels of contourlet transformation. Although the payload is less, for less than 1400 bits of data, the perceptual quality recorded by authors as regards the PSNR value does not exceed 35 dB. Nevertheless, the payload is very small the combined crypto-watermarking scheme was introduced by Amutha et al. [10] to protect the medical image by integrating the digital watermark and block cipher algorithms. It combines a replacement watermarking algorithm with an advanced counter-mode encoding standard (AES) encryption algorithm. In this paper, combined encryption and the watermarking system should be ideal for transmitting images in real time. It provides both encrypted and space-specific access to a file. In CTR mode, AES is applied, rendering solutions transparent or DICOM-compliant. This paper found that if a system cannot be used for watermarking, it can decrypt and access the picture if it knows the encryption key AES. A shield-algorithm was proposed by Bansal et al. (2017) [11] to replace LSB in the DCT pixel values and to achieve a PSNR of 29.77 dB. He et al. [12] suggested a fragile watermarking system based on wavelets for safe image authentication. The embedded watermark is created using the discrete wavelet transformation (DWT) in their proposed scheme. Finally, the scrambling of encryption into the least significant bit (LSB) of the host image provides a protection watermark. Boato et al. [13] proposed a private embedding key and an appropriate authentication scheme for the asymmetric watermarking algorithm that achieves a double degree of security for digital data protection. Mokhnache et al. [14] proposed a robust watermarking scheme combing DWT and DCT transform and using a gradient of the image as a measuring tool. In this paper, two-level DWT is applied on the cover image, and then DCT coefficients are calculated from the second level coefficients. Calculation of the gradient of the image is performed because it presents a spatial derivative that gives a topological map of the image which is important for locating the regions where the disturbances are intense and also makes it possible to evaluate the average softness of the image. The robustness of the image is evaluated against various attacks. Mamuti et al. [15] proposed a watermarking scheme using RGB cover image decomposition into R, G, and B channels. In this paper, DWT is applied on R channels, DCT is applied on G channels, and LSB is applied on B channel. Arnold's scrambling process is applied to the watermark image before being embedded. From the proposed algorithm in this paper, the calculated normalized correlation (NC) is 0.9799; PSNR is 55.9463118. All these articles suggest that information transmission be securely transmitted based on PSNR values.

3 Proposed Methodology

In this section, the proposed watermark encryption and embedding algorithms as well as extraction and decryption algorithms have been explained. For the experiment, medical image has been used as the cover image, and the patient's encrypted information is used as the watermark. The proposed algorithm has been divided into five parts.

3.1 Initialization Vector (IV) Generation

In the research, the 64-bit initialization vector (IV) has been used for AES encryption. A seed that consists of 4 decimal numbers is first chosen. The seed number is stored in an undefined size of the array. Then, the seed is reversed and multiplied with the original seed. After multiplication, the result is checked to ensure whether then it has 8 numbers or not and then select the middle 4 bit of this result. This is then used as a new seed and matched with all the previous seeds in the array. If any of the seed in the array matched with the current seed, then the process is stopped. Otherwise, the current seed is stored in the array and the process is repeated in the above way. When the process is stopped, the array of seeds is filtered by selecting only the seed which is less than the starting seed and stored in another array. This is done only when the array contains more than four seeds of four decimal numbers otherwise the seeds in the array construct initialization vector (IV). If further the new array contains more than four seed, then the first 4 seed of the new array is selected and used to construct the IV. The selected four seed must contain four decimal numbers to construct 64 bit IV. These four seeds of four decimal numbers construct a new number having a length of 16, and the equivalent hexadecimal value of this number of length 16 is achieved. The hexadecimal value of length 16 is converted to a 4 * 4 matrix. This matrix is used as an initialization vector for AES encryption. The initialization vector generation algorithm is described in Algorithm-1. Figure 1 illustrates the block diagram IV generation.

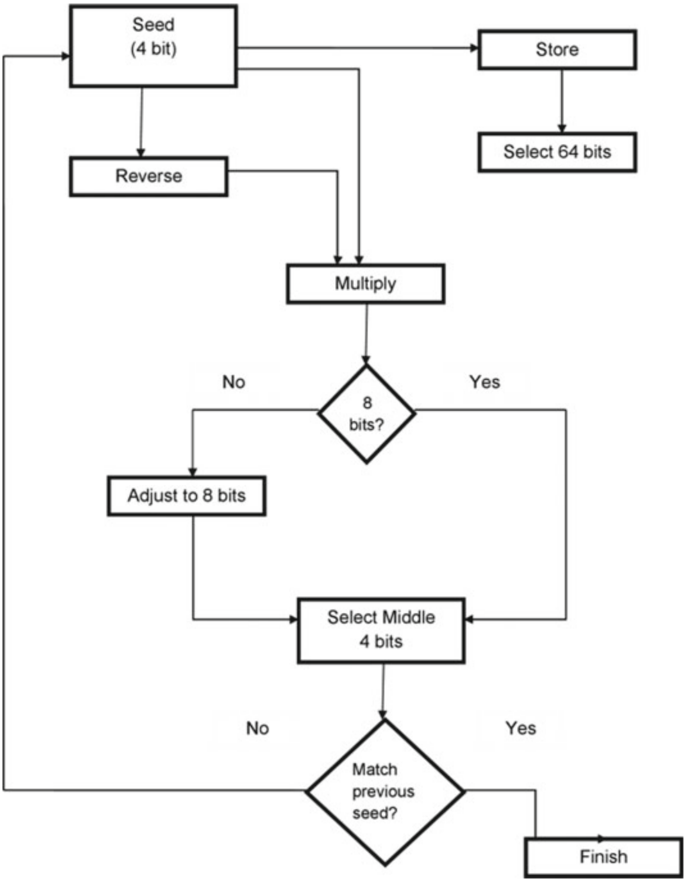

Fig. 1 Initialization vector (IV) generation block diagram

3.2 AES Encryption of Patient's Secret Information in OFB Mode at the Sender Side

In the proposed method, AES encryption has been implemented in output feedback mode. AES being a symmetric cryptography algorithm utilizes the same key both in the source and destination side. At first, the input plain text (secret message) is converted into blocks of 128 bits. It requires feeding the consecutive output blocks from the underlying block cipher back to it. These feedback blocks give a string of bits to feed the encryption algorithm which behaves as the key-stream generator. The created mainstream is XOR-ed with 128-bit plaintext blocks. The OFB mode includes IV as the initial random n-bit input block, a 64-bit input block in our case.

The first input block is the IV and the previous output block is each following input block. This produces ciphertext utilizing IV, round keys, and blocks of patient

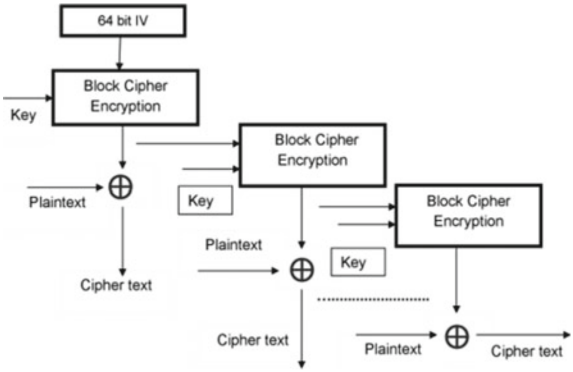

Fig. 2 AES encryption in OFB mode for encrypting patient's information at the sender side

information. Figure 2 illustrates the block diagram of AES encryption in OFB mode for encrypting patient's information at the sender side.

3.3 Embedding of Patient's Encrypted Information as the Watermark Using LSB Replacement

Grayscale MRIBarin.bmp image (Fig. 3a) has been selected as the cover image and the pixel values are obtained from the cover image. Three-level DWT is implemented on the cover image to split it into high and low-frequency parts as shown in Fig. 3b illustrates the block diagram of DWT of the cover image. In the LSB replacement step, an LSB is replaced in values of the LL2 sub-bandwidth a bit of watermark. The LSB technique is implemented based on a formula. If LSB of sub-band value $S(i,$

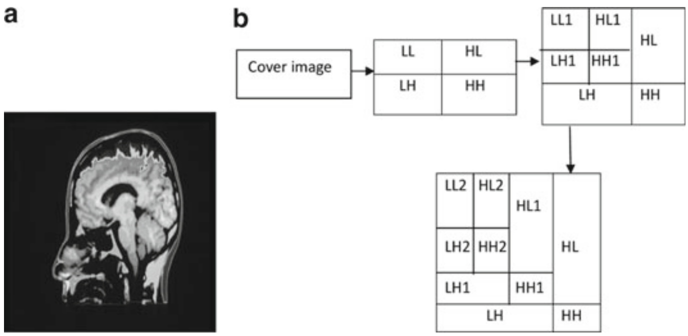

Fig. 3 **a** Cover image in (MRIBrain), **b** three-level DWT transformation of cover image

j) is equal to watermark bit SM of the watermark to be embedded. $S(i, j)$ remains unchanged if not set of $S(i, j)$ to SM. Here, SM refers to the next watermark bit to be embedded. According to the proposed algorithm, after applying LSB substitution between the LL2 sub-band of the cover image and watermark bits, some pixel values of the sub-band have become changed depending on the least significant bit of sub-band pixel value and watermark bits. The algorithm for embedding the watermark including the patient's secret information is described in Algorithm-2.

Algorithm 1: Initialization vector (IV) Generation

```
Input: Initialize a decimal 4-bit seed and t=1. Maximum Iteration occurs till t=0
Output: 64-bit IV_value and 4×4 IV_matrix.
begin
  while ( t==1) do
    If (Size (seed) is not equal to 4)
      Step-1. Adjust seed length so that Size (seed) =4.
    end
      Step-2. Initialize a decimal array m=seed, where m∈ℝand
              m∈Z +.
      Step-3.rev_seed=Reverse (seed).
      Step-4. r= (seed) (rev_seed).
    if (Size(r) not equals to 8)
      Step-5. adjust Size(r) to 8 by adding leading zeroes.
    end
      Step-6. Initialize Current_seed=middle 4 decimal
              number of r.
    for (i=0 to Size (m))
        if (Current_seed is equals to m (i))
            Step-7. Set t=0 and break the process.
        end
    end
    t=1.
            Step-8. Seed=Current_seed.
  end
        if (Size(m)>4)
          Step-9. Initialize starting_seed=the value of seed
                  considered at starting.
          Step-10. m=m (m<starting_seed).
        end
        if (Size(m)>4)
          Step-11. m=first 4 decimal seed value of this array.
        end
          Step-12 .IV_dec= number sequence containing 4
                   decimal  value of array   m
          Step-13 .IV_value=hexadecimal (IV_dec).
          Step-14. IV_matrix=4×4 matrix contains
                   the values in IV_value.
end
```

Algorithm 2: Embedding of the watermark

```
Input: Read cover=a Grayscale image and w=watermark message including ci-
pher text c where length (w) =152 or length (binary (w)) =1216 bit.
Output: s= LSB replaced sub band of cover and wmImg=watermarked image.
begin
        Step-1. make length(cover)=300×300 pixel.
        Step-2. [LL,LH,HL,HH]=dwt (cover).
        Step-3. [LL1,LH1,HL1,HH1]=dwt(LL).
        Step-4: [LL2, LH2, HL2, HH2] =dwt (LL1).
        Step-5. binary_bit=binary (ascii (w)).
    for (i=1 to height(LL2))
        for (j=1 to width(LL2))
            if (LSB(LL2(i,j)= binary_bit)
                    Step-6. s(i,j)=LL2(i,j)
            end
            else
                if (LSB(LL2(i,j)=1 and binary_bit=0)
                        Step-7. s (i,j)=LL2(i,j)-1.
                end
                else
                        Step-8. s (i, j) =LL2(i,j)+1.
                end
        end
    end
                Step-9. ILL1=IDWT(s, LH2, HL2, HH2).
                Step-10. ILL=IDWT (ILL1, LH1, HL1, HH2).
                Step-11. wmImg=IDWT (ILL, LH, HL, HH).
end
```

3.4 Extraction of the Watermark Using LSB Extraction Algorithm at the Receiver Side

Using the seed sent with the watermarked image, IV generation is done at the destination receiver side similarly at the sender side. To get the accurate patient's information embedded by the sender, the IV produced at the destination site must be similar to that was at the sender side. At the receiving end, the watermarked image is read. In order first, to remove the watermark, the lower frequency sub-band of the watermarked image must be used to convert a distinct wavelet applying three-level DWT on the watermarked image. At the starting of extraction at first, the LL2 sub-band's pixel values are extracted. Now, starting from the first-pixel value extraction of the watermarked key character from the first component of the pixels is done. This is followed up to terminating symbol otherwise if the extracted key matches with the key entered the receiver at the destination. If the extracted key corresponds to the key entered by the destination user, this is then followed to the termination symbol elsewhere. This process is done if the keys correct otherwise previous step is followed again. Then, the selection of the next pixel value and extraction of watermark data from the first component of the next pixel value is done. This step is followed up to terminating symbol otherwise extraction of the message is finished here. Suppose, Pixel of LL2 sub-band of watermarked image: 11111001 11001000 0000001111111000. After LSB extraction, the watermark bits: 1010. The algorithm for extracting the watermark from the watermarked image is described in Algorithm-3.

Algorithm 3: Watermark extraction

```
Input: wmImg= the watermarked image and Initialize
char=length(watermark)=152 or 152*8=1216 bit.
Output: w= extracted watermark and c=encrypted portion of watermark.
begin
    Step-1. [LL,LH,HL,HH]=dwt(wmImg).
    Step-2. [LL1,LH1,HL1,HH1]=dwt(LL).
    Step-3. [LL2,LH2,HL2,HH2]=dwt(LL1).
    for (i=1 to height(LL2))
        for (j=1 to width(LL2))
            Step-4.w_bits=LSB(LL2(i,j).
        end
    end
    Step-5. w=char(w_bits).
    Step-6. Initialize c=encrypted portion of w.
end
```

3.5 Decryption of Patient's Secret Information at the Receiver Side

After extraction of the watermark data, the encrypted portion of the watermark data has to be identified and the portion has to be decrypted to get the patient's secret information. At first, the encrypted portion of watermark data has to be converted into blocks of 128 bits. Each block having 4 rows and block length 32 = 4 column as AES-128 is being used. So, the 31 block size is 4 × 4, i.e., each block carries 16, 8 bits, or 1 byte of data. It has also 10 rounds and for every 10 rounds, 10 round keys are required. Generally, in the case of AES encryption in OFB mode, the decryption of a data is done in the same way as encryption. The generated IV is required to decrypt the data. After the rounds of decryption process on the encrypted portion of the watermark using the generated IV and the same cipher key is done, then we get the original patient's information that is sent by the sender and required to verify the patient. Figure 4 illustrates the block diagram of AES encryption in OFB mode for decrypting patient's information.

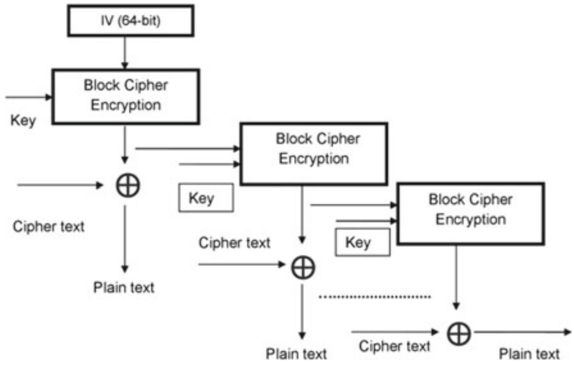

Fig. 4 AES encryption in OFB mode for decrypting patient's information

4 Performance Analysis

As a cover image, gray images, and color images from different formats were used for our experiment with the proposed algorithm. Until image operations are continued to do, colored images are transformed into grayscale, of which the resolution is more than 300 × 300. In Table 1 the images and their details have been specified.

4.1 Visual Comparison

Imperceptibility usually refers to the fact that the perceived content cover image should not be adversely affected by watermark presence. To human observation, the watermark ought to be barely detected while the cover is contaminated with the sensitive information of patients. Figure 5a, b illustrates the cover image and the LL2 sub-band of the cover image, respectively. Figure 5c, d illustrates LSB replaced LL2 sub-band of watermarked image and watermarked image after inverse DWT. The difference between the cover image and watermarked image can also be shown in another convenient way using a difference image which has been observed by subtraction operation between the cover image and watermarked image. The difference image achieved from the observation is completely black which is shown in Fig. 6c. This indicates the watermarked image is rarely distinguishable from the

Table 1 Definitions of images used

Image	Grayscale/color	Resolution (pixel)	Format
MRIBrain	Color	600 × 401	Bmp
Skeleton	Grayscale	329 × 291	Tiff
MRIScan	Grayscale	211 × 234	Bmp
ScannedBrain	Color	539 × 645	Png
Sonography	Grayscale	539 × 645	Jpg

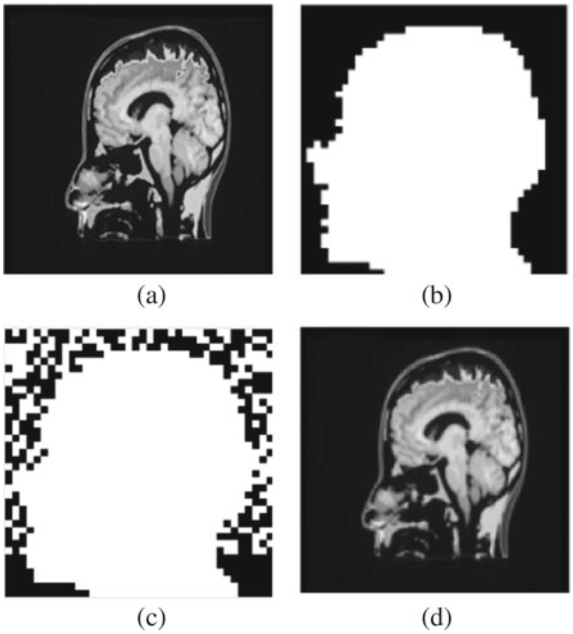

Fig. 5 MRIBrain **a** cover image, **b** LL2 sub-band of cover image, **c** LSB replaced LL2 sub-band, **d** watermarked image

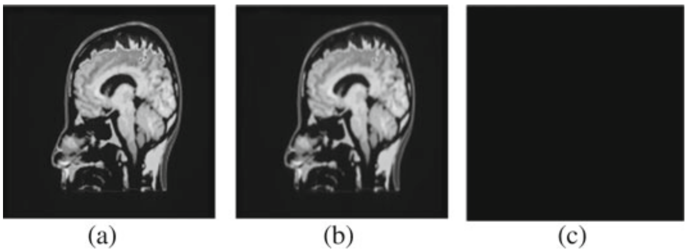

Fig. 6 MRIBrain **a** cover image, **b** watermarked image, **c** difference image

original cover image. This experiment was also done with other images which are shown in Fig. 7a, b.

4.2 *Histogram Analysis*

A further important criterion for evaluating a watermarking algorithm's efficiency and quality is to carry out a histogram analysis of both the cover image and the watermarked image. The cover image histogram and watermarked image histogram

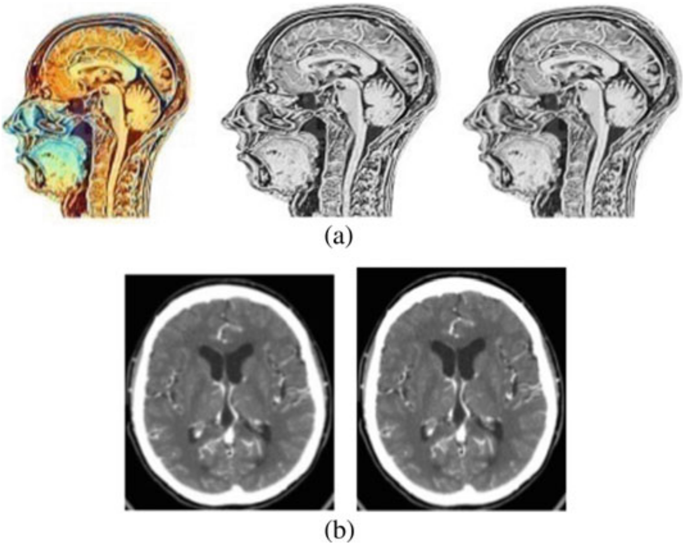

Fig. 7 **a** ScannedBrain colored cover image versus grayscale cover image versus grayscale watermarked image, **b** MRIScan cover image versus watermarked image

have been observed to expose that if there is a dramatic difference in the histograms, an image has embedded data. In the case of the planned histogram algorithm, it was discovered that the cover image and the watermarked image are much related and that the transition cannot be perceived. Figure 8 illustrates histogram representation of the cover image and watermarked image.

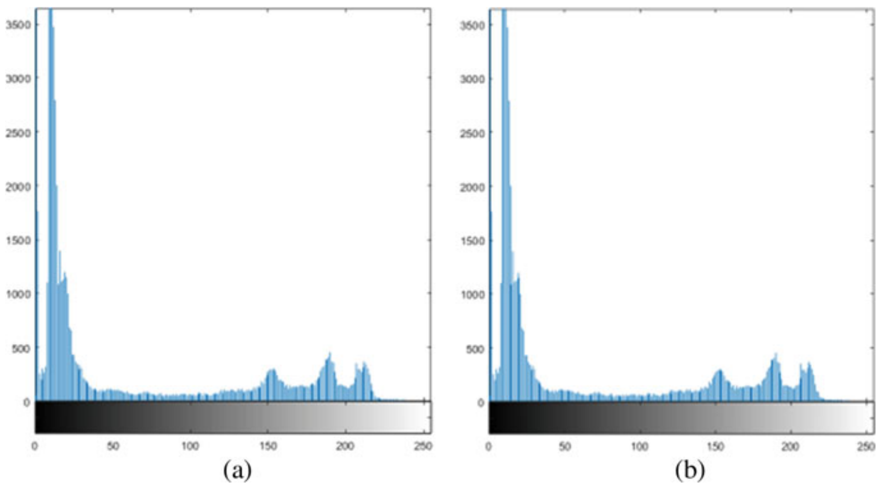

Fig. 8 MRIBrain **a** histogram of the cover image, **b** histogram of watermarked image

4.3 Mean Square Error (MSE)

MSE has been used to differentiate between the cover images and the watermarked images. The low value of MSE indicates that the watermarked image is almost similar to the original image.

$$\text{MSE} = \frac{1}{m \times n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)] \tag{1}$$

where m = height of the cover image, n = width of the cover image, $I(i, j)$ = pixel value before embedding data, and $K(i, j)$ = pixel value after embedding data.

4.4 Peak Noise to Signal Ratio (PSNR)

In the proposed algorithm, the PSNR has been observed to analyze the visual quality of the watermarked image in comparison with the cover image.

$$\text{PSNR} = 10 \times \log_{10} \frac{C_{\max^2}}{\text{MSE}} \tag{2}$$

where $C_{\max}$ = 255 for an 8-bit image. The observation of PSNR and MSE value for different images is illustrated in the following Table 2 for comparative analysis. The observation of PSNR value for the different number of bits to be emended is illustrated in the following Table 3.

Table 2 Observed PSNR values for different watermarked images

Image	MSE	PSNR (dB)
MRIBrain	0.1101	57.7114
MRIScan	0.2491	54.1664
ScannedBrain	0.06481	60.014

Table 3 Calculated PSNR values for different no. of watermark bits embedded

Image name	No. of bits to be embedded	MSE	PSNR (dB)
MRIBrain	1216	0.1101	57.7114
MRIBrain	1400	0.11005	57.614
MRIBrain	1432	0.11102	57.705
ScannedBrain	1216	0.0642	60.0536
ScannedBrain	1400	0.0634	60.053

Table 4 Performance of normalized correlation (NC) against attacks

Attack type	MRIBrain	MRIScan	ScannedBrain
	NC	NC	NC
No attack	1	1	1
Salt and pepper noise (0.05)	0.9544	0.9563	0.8703
Motion filter attack	0.9654	0.9789	0.9805
JPEG compression	0.999	0.9995	0.9991
Gaussian noise (0.02)	0.9165	0.914	0.8729

4.5 *Normalized Correlation and Time Complexity*

For statistical measurement of similarities between cover images and watermarking images, correlation analysis has been done which is an important parameter to understand the quality of a watermarking algorithm. Correlation (r) between the two images is calculated by implementing the following formula.

$$r = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n}(I(i,j) - \text{avg}(I)) \times (K(i,j) - \text{avg}(K))}{\sqrt{(\sum_{i=1}^{m}\sum_{j=1}^{n}(I(i,j) - \text{avg}(I)^2) \times (\sum_{i=1}^{m}\sum_{j=1}^{n}K(i,j) - \text{avg}(K)^2)}}. \quad (3)$$

The observations from correlation analysis for the proposed algorithm with various cover and watermarked images are listed in Table 4 and a graphical representation has been shown in Fig. 9a. The observation of time complexity has been represented in Fig. 9b.

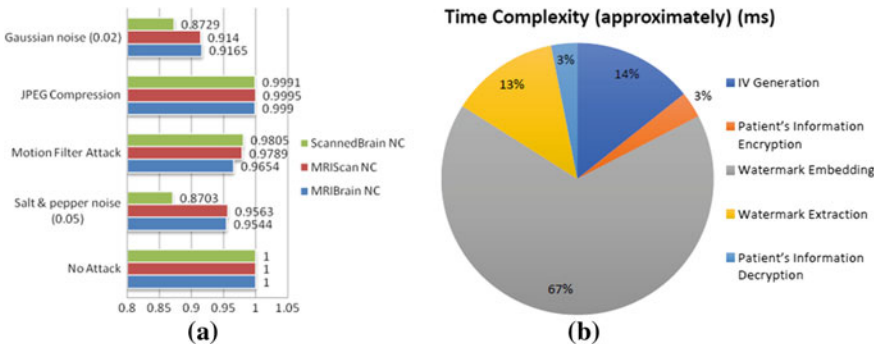

Fig. 9 **a** Normalized correlation (NC) observation against different attacks, **b** time complexity (approximately) (ms) for each of section of proposed algorithms of the experiment

5 Conclusion

In this paper, there are two levels of security maintained by using watermark and encryption where both methods ensure data security independently in their own ways. An IV generation that is used in AES encryption OFB mode ensures extra security for patient's information as this algorithm will only be known by the sender and receiver. AES encryption has been used to hide patient's information where DWT and LSB are used for watermarking images. In the case of DWT operation because of its compressing pixel quality, the distorted pixel cannot be distinguished by necked eyes. So the watermarked image cannot be different from the original image. For the proposed technique, a good range of PSNR, MSE, correlation values for different images have been also observed to ensure the imperceptibility of watermarked images. So, from the experimental result showed that the proposed joint watermarking and encryption method keeps not only a good quality watermarked image but also ensures strong security for patient's information.

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Chapter 9
Comparative Study of Different Implicit Finite Difference Methods to Solve the Heat Convection–Diffusion Equation for a Thin Copper Plate

Nihal Ahmed, Ashfaq Ahmed, and Muntasir Mamun

1 Introduction

Finite difference method (FEM) has been frequently used for solving the convection–diffusion equation [1]. The convection–diffusion equation generally discusses the heat and energy flow in a system and vastly used for 2D and 3D systems. Both implicit and explicit schemes have been used to solve this equation [2–5]. The explicit method can be defined as the process when a state of the system is calculated for a later time from the state of the system at a current time [6]. In the unsteady state problems, explicit methods have been frequently used, but the problem of instability in the large-scale time-dependent problems has always been there [7]. In this paper, we have analyzed three different implicit schemes for the heat transfer analysis of a thin plate. The schemes are Crank–Nicolson method, alternating direct implicit (ADI) method, and alternating direct semi-implicit method. The Crank–Nicolson method despite being unconditionally stable provides some errors such as method error, representation error, and rounding error. But as time increases, these errors also tend to neutralize due to its unconditional behavior [8]. An ADI method uses a direct, non-iterative method to solve a small set of simultaneous equations. The advantage of this method is that it is stable for any time step [9]. The semi-implicit

N. Ahmed (✉)
Department of Material Science and Nanotechnology, Bilkent University, Ankara, Turkey

A. Ahmed
Department of Mechanical Engineering, University of California Riverside, Riverside, California, USA
e-mail: ashfaq.ahmed@okstate.edu

M. Mamun
Department of Mechanical Engineering, Khulna University of Engineering and Technology, Khulna, Bangladesh
e-mail: mamun1605099@stud.kuet.ac.bd

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scheme discretizes some terms into implicit and the remaining into explicit of a time-dependent problem [10]. But it has a problem that it may give errors in one direction [11].

All of these schemes have been heavily used by the scientific community for solving various practical problems. Lozinski et al. [12] use the Crank–Nicolson method and finite element discretization to derive two posterior upper bounds for the heat equation. Jin et al. [13] analyze the sub-diffusion equation and uses the Crank–Nicolson time-stepping scheme. Geoola et al. [14] studied a solid sphere indulged in an incompressible fluid and its time-dependent convective heat transfer using the ADI method. Karaa [15] solved a three-dimensional convection–diffusion problem using a higher-order compact alternating direction scheme. Banaszek et al. [16] and Ramaswamy et al. [17] investigated the incompressible flow using a semi-implicit finite element method.

Previous works regarding the discussed implicit methods were mainly focused on the application of these schemes for solving various parabolic equations and phenomena. Their main objective was to apply the schemes to solve complex numerical equations or to improve the efficiency of the solution by introducing high-order schemes. Although diffusion convection of thin plate is a very common heat transfer-related problem and explicit methods have been previously used to solve them, but no impactful work using the mentioned implicit scheme has been done. In this paper, the heat convection–diffusion analysis has been done using the Crank–Nicolson method, ADI method, and the alternating direction semi-implicit method. The methods have been used to solve the two-dimensional convection–diffusion equation. Then, the set of equations achieved from the solution has been solved using MATLAB programming language. The results obtained from the set of equations have been used to compare the three different schemes, their stability, accuracy, and efficiency. Our observations indicate that the alternating direction implicit method provides the most stable and accurate solution.

2 Numerical Solution

The governing equation of heat energy of a body is given by

$$\begin{aligned} &K\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)\mathrm{d}x\mathrm{d}y\mathrm{d}z+[\pm\text{Convection}]+[\pm\text{radiation}]\\ &\quad+\left[+\frac{\text{Generation}}{-\text{Absorbtion}}\right]\mathrm{d}x\mathrm{d}y\mathrm{d}z=\rho c_p\frac{\partial T}{\partial t}\mathrm{d}x\mathrm{d}y\mathrm{d}z \end{aligned} \tag{1}$$

Here, K indicates the thermal conductivity of the material, ρ indicates the density of the material, c_p indicates the specific heat at constant pressure, and T indicates the temperature profile.

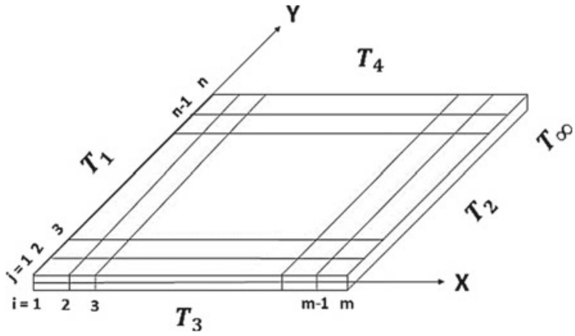


Fig. 1 Heat convection–diffusion diagram of a thin plate

For the problem discussed in this paper, the radiation is negligible so that can be ignored. Convection can be considered as heat carried away. Here, the generation and absorption are considered zero. Hence, Eq. (1) can be simplified as

$$K\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)\mathrm{d}x\mathrm{d}y\mathrm{d}z-2h(T-T_\infty)\mathrm{d}x\mathrm{d}y=\rho c_p\frac{\partial T}{\partial t}\mathrm{d}x\mathrm{d}y\mathrm{d}z \tag{2}$$

Now both sides of Eq. (2) have been divided by dxdydz, and thus we get a simplified form of Eq. (2)

$$K\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)-\frac{2h(T-T_\infty)}{\mathrm{d}z}=\rho c_p\frac{\partial T}{\partial t} \tag{3}$$

Figure 1 indicates the problem domain that was selected for this analysis.

2.1 Solution Using the Crank–Nicolson Method

Crank–Nicolson method is a very popular finite difference approach to solve heat conduction equations and similar partial differential method. Here, this method will be used to solve the heat convection of a thin plate.

According to the FEM, Eqs. (4), (5) and (6) are discretized equations

$$\left.\frac{\partial T}{\partial t}\right|_{i,j,t/2}=\frac{T_{i,j,t+1}-T_{i,j,t}}{\Delta t} \tag{4}$$

$$\left.\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)\right|_{i,j,t}=\frac{T_{i-1,j,t}-2T_{i,j,t}+T_{i+1,j,t}}{\Delta x^2}+\frac{T_{i,j-1,t}-2T_{i,j,t}+T_{i,j+1,t}}{\Delta y^2} \tag{5}$$

$$\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)\Bigg|_{i,j,t+1} = \frac{T_{i-1,j,t+1}-2T_{i,j,t+1}+T_{i+1,j,t+1}}{\Delta x^2}+\frac{T_{i,j-1,t+1}-2T_{i,j,t+1}+T_{i,j+1,t+1}}{\Delta y^2} \tag{6}$$

By modifying Eq. (3), the following equation was obtained.

$$\frac{\partial T}{\partial t}_{i,j,t/2} = \frac{1}{2}\left[\frac{K}{\rho c_p}\left\{\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)-\frac{2h}{\rho c_p dz}(T-T_\infty)\right\}_{i,j,t+1} + \frac{K}{\rho c_p}\left\{\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)-\frac{2h}{\rho c_p dz}(T-T_\infty)\right\}_{i,j,t}\right] \tag{7}$$

Now, to simplify the equation, $\Delta x = \Delta y = \Delta$ has been assumed, and the discretized equations have been implemented in Eq. (7)

$$\frac{T_{i,j,t+1}-T_{i,j,t}}{\Delta t} = \frac{1}{2}\left[\frac{K}{\rho c_p}\left\{\frac{T_{i-1,j,t+1}-2T_{i,j,t+1}+T_{i+1,j,t+1}}{\Delta x^2} + \frac{T_{i,j-1,t+1}-2T_{i,j,t+1}+T_{i,j+1,t+1}}{\Delta y^2} - \frac{2h}{\rho c_p dz}(T-T_\infty)\right\}_{i,j,t+1} + \frac{K}{\rho c_p}\left\{\left(\frac{T_{i-1,j,t}-2T_{i,j,t}+T_{i+1,j,t}}{\Delta x^2}+\frac{T_{i,j-1,t}-2T_{i,j,t}+T_{i,j+1,t}}{\Delta y^2}\right) - \frac{2h}{\rho c_p dz}(T-T_\infty)\right\}_{i,j,t}\right] \tag{8}$$

Now, by bringing $t + 1$ related term on the left-hand side, the final form of the solvable differential equation has been obtained

$$\left(\frac{K\Delta t}{\rho c_p \Delta^2}\right)\left(T_{i-1,j,t+1}+T_{i+1,j,t+1}+T_{i,j-1,t+1}+T_{i,j+1,t+1}-4T_{i,j,t+1}\right) - 2T_{i,j,t+1} - \frac{h\Delta t}{\rho c_p dz}T_{i,j,t+1} = -2T_{i,j,t} - \left(\frac{K\Delta t}{\rho c_p \Delta^2}\right)\left(T_{i-1,j,t}+T_{i+1,j,t}+T_{i,j-1,t}+T_{i,j+1,t}-4T_{i,j,t}\right) - \frac{h\Delta t}{\rho c_p dz}T_{i,j,t} - \frac{2h\Delta t T_\infty}{\rho c_p \mathrm{d}z} \tag{9}$$

To simplify the equation, let us assume $\frac{K\Delta t}{\rho c_p \Delta^2} = r\&\mathrm{d}z = d =$ thickness. Hence, Eq. (9) becomes

$$r\left(T_{i-1,j,t+1}+T_{i+1,j,t+1}+T_{i,j-1,t+1}+T_{i,j+1,t+1}\right) - 2\left(1+2r+\frac{h\Delta t}{2\rho c_p d}\right)T_{i,j,t+1}$$

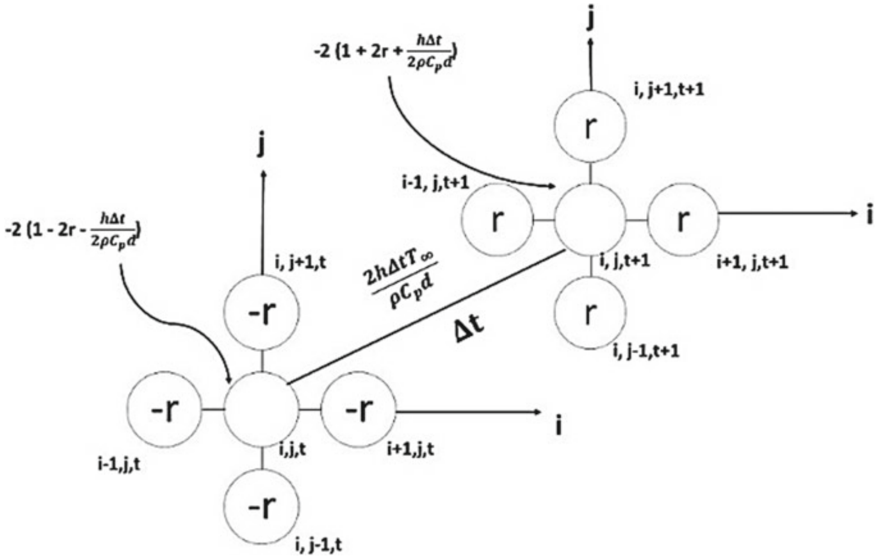


Fig. 2 FEM heat transfer grid for Crank–Nicolson method

$$
\begin{aligned}
&= -r\left(T_{i-1,j,t} + T_{i+1,j,t} + T_{i,j-1,t} + T_{i,j+1,t}\right) \\
&\quad - 2\left(1 - 2r - \frac{h\Delta t}{2\rho c_p d}\right)T_{i,j,t} - \frac{2h\Delta t T_\infty}{\rho c_p T}
\end{aligned}
\tag{10}
$$

Here, the value of i ranges from 2 to $(m - 1)$ and the value of j ranges from 2 to $(n - 1)$. Where m and n indicate the number of nodes in x and y-direction, respectively. By implementing the values of i and j, a system of linear equations will be obtained. By solving those systems of linear equations, the temperature for different nodes of a thin plate can be obtained. The mesh grid system of analysis for heat conduction–convection analysis of thin plate has been shown in Fig. 2.

2.2 Solution by Alternating Direction Implicit Method

The conventional two-dimensional ADI method has been used for solving parabolic and elliptic partial differential equation and heat transfer problems [18]. The alternating direction implicit method contains two-directional solving method in one-time step. Hence, convection in one direction is considered as half of the total convection.

So similar to Eq. (8), we have

$$
\frac{T_{i,j,t+1} - T_{i,j,t}}{\Delta t} = \frac{K}{\rho c_p \Delta^2}\left(T_{i-1,j,t+1} - 2T_{i,j,t+1} + T_{i+1,j,t+1} + T_{i,j-1,t} - 2T_{i,j,t} + T_{i,j+1,t}\right)
$$

$$- \frac{h}{2\rho c_p \mathrm{d}z}\left[\left(T_{i,j,t+1} - T_\infty\right) + \left(T_{i,j,t} - T_\infty\right)\right] \tag{11}$$

By assuming $\frac{K\Delta t}{\rho c_p \Delta^2} = r$ and modifying Eq. (11) the following horizontal traverse equation is obtained

$$r\left(T_{i-1,j,t+1} + T_{i+1,j,t+1}\right) - \left(1 + 2r + \frac{h\Delta t}{2\rho c_p d}\right)T_{i,j,t+1} = -r\left(T_{i,j-1,t} + T_{i,j+1,t}\right) - \left(1 - 2r - \frac{h\Delta t}{2\rho c_p d}\right)u_{i,j,t} - \frac{h\Delta t T_\infty}{\rho c_p T} \tag{12}$$

Here, the value of i ranges from 2 to $m - 1$ and the value of j ranges from 2 to $n - 1$. Where m and n indicate the number of nodes in x and y-direction, respectively.

Similarly, for vertical traverse, Eq. (13) can be obtained

$$r\left(T_{i,j-1,t+1} + T_{i,j+1,t+1}\right) - \left(1 + 2r + \frac{h\Delta t}{2\rho c_p T}\right)T_{i,j,t+1} = -r\left(T_{i-1,j,t} + T_{i+1,j,t}\right) - \left(1 - 2r - \frac{h\Delta t}{2\rho c_p T}\right)T_{i,j,t} - \frac{h\Delta t T_\infty}{\rho c_p d} \tag{13}$$

Here, the value of i ranges from 2 to $m - 1$ and the value of j ranges from 2 to $n - 1$. Where m and n indicate the number of nodes in x and y-direction, respectively. By solving these systems of the linear equation, the nodal temperatures can be obtained (Fig. 3).

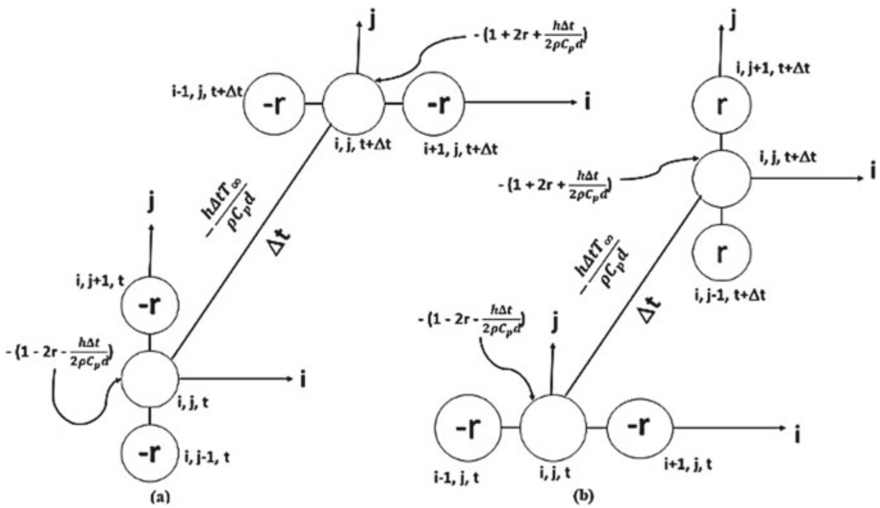


Fig. 3 FDM 2D heat transfer Stencil for ADI method: **a** horizontal traverse, **b** vertical traverse

2.3 Boundary Conditions

In this paper, we have considered Dirichlet boundary condition.

Nodes along y-axis:

$$T(0, y, t) = T_1 \tag{14}$$

$$T(a, y, t) = T_2 \tag{15}$$

Nodes along x-axis:

$$T(x, 0, t) = T_3 \tag{16}$$

$$T(x, b, t) = T_4 \tag{17}$$

Plate temperature at t = 0:

$$T(x, y, 0) = T_\infty \tag{18}$$

3 Comparison of Three Different Solution Schemes

This section is dedicated to comparing the obtained results by Crank–Nicolson, alternating direction implicit, and ADI semi-implicit method and has been analyzed and compared. Mousa et al. [1] designed an algorithm to solve the heat equation of a 2D plate. His algorithm used similar "Dirichlet conditions" and an initial temperature at all nodes. He used the local fractional Euler method and the second-order Runge–Kutta method to solve the heat equation. His analysis focused on determining the nodal temperature at various points of the plate for different time steps. Figure 4 indicates the points whose temperature was considered during analysis, and Table 1 compares the result between the different algorithms for aluminum and boundary condition of $T_1 = T_2 = T_4 = 0$ K, $T_4 = 200$ K, and an initial temperature of 0 K.

Euler method is an explicit method and the methods used in this paper are implicit methods. Table 1 indicates that when applied to two-dimensional heat transfer problems, all four methods exhibit similar results.

For further analysis, copper material has been used. Table 2 indicates the properties that were used in the final analysis.

Fig. 4 2D plate considered by Mousa [1]

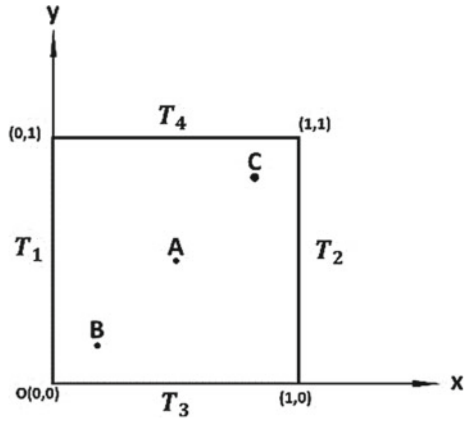


Table 1 Temperature at point *A*, *B*, *C* for steady-state time

	Euler method [1]	Crank–Nicolson method	Alternating direct implicit	ADI semi-implicit
Point A (Kelvin)	53.84	53.84	53.89	53.91
Point B (Kelvin)	88.82	88.82	88.97	89.43
Point C (Kelvin)	16.23	16.23	16.56	16.79

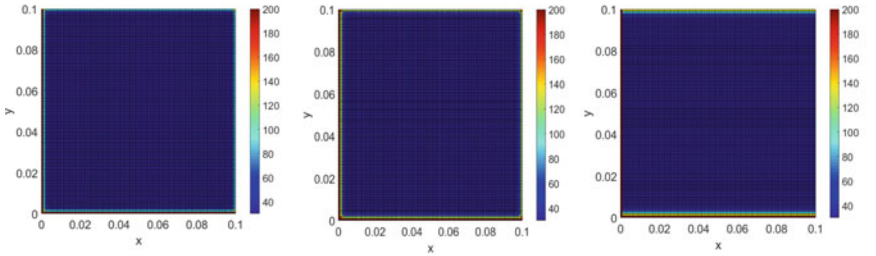
Table 2 Material properties

Property	Numerical value
Thermal conductivity (W/mK)	385
Density (Kg/m^3)	8960
Specific heat capacity (J/KgK)	376.812
Convective heat transfer coefficient of air	10

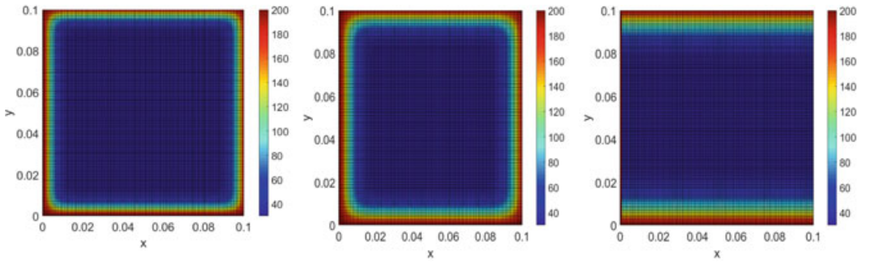
Figure 5 indicates the temperature profile for the boundary condition of $T_1 = T_2 = T_4 = T_4 = 200$ K, an initial temperature of 30 K and length and width of 0.1 m. Here, the temperature profile for four different time steps has been analyzed.

Figure 5 stipulates the temperature profile of a thin plate for different times. The contours denote that the Crank–Nicolson and alternating direction implicit method show similar results for different times. But the alternating direction semi-implicit method provides wrong values row-wise. Semi-implicit methods are considered unconditionally stable.

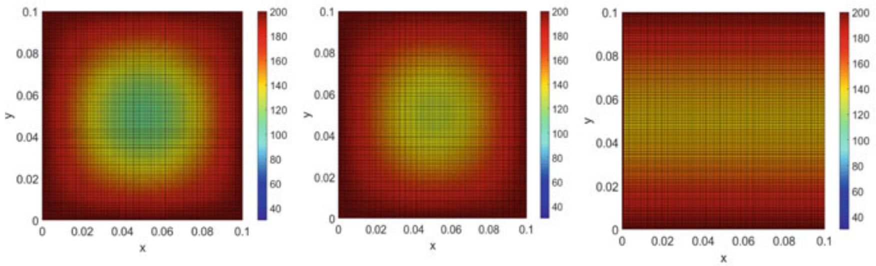
In this paper, the system of implicit equations is complex second-order differential equations, and if we focus on the numerical solution section, then it can be seen that the horizontal traverse equations are solved explicitly. Hence, there are errors in the horizontal direction. Figure 5 also depicts that as time increases the difference between the solutions obtained by Crank–Nicolson and alternating direction scheme are converging.



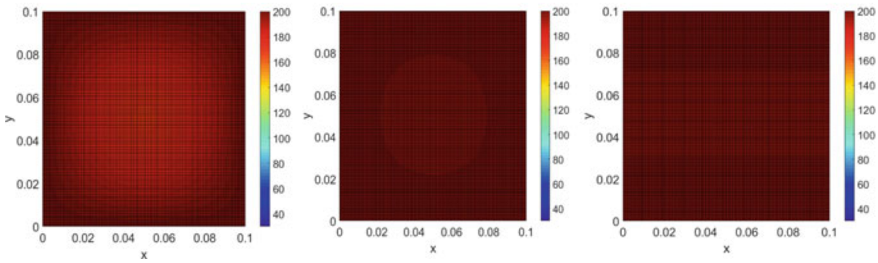
1(a) Crank Nicolson method 1(b) ADI 1(c) ADI semi implicit



2(a) Crank Nicolson method 2(b) ADI 2(c) ADI semi implicit



3(a) Crank Nicolson method 3(b) ADI 3(c) ADI semi implicit



4(a)Crank Nicolson method 4(b) ADI 4(c) ADI semi implicit

Fig. 5 Temperature profile for heat conduction and convection of thin plate (1) $t = 0$ s, (2) $t = 0.1$ s, (3) $t = 5$ s, (4) $t = 10$ s

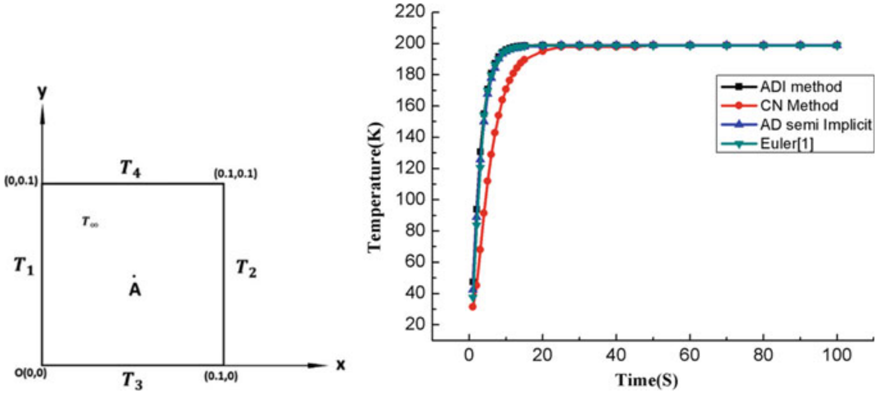
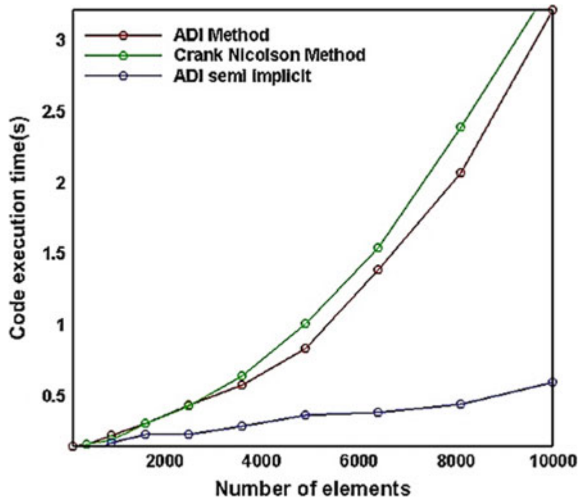


Fig. 6 Change in temperature in point *A* with time for four different solution schemes

Figure 6 gives an elaborated idea about the differences between the two schemes. Although the Crank–Nicolson method is considered stable and a very simple system of linear equation solution is required for every time step, but the system is not impervious to errors. That is the situation that has happened in this paper. Although the problem can be solved by introducing a mesh size that is smaller than a certain critical value, but that cannot be done due to the limitation of computational power. As a result, there is a slight difference between the explicit method and the Crank–Nicolson method. On the contrary, the alternating direct implicit method has unconditional stability which means that the time step size does not have any effect on the result. Hence, it provides a more accurate result for both unsteady and steady state.

Figure 7 illustrates the computational time required for three different schemes. For this paper, a processor of 2.20 GHz, random access memory of 8 gigabytes, and

Fig. 7 Computation time analysis for three different solution schemes



an M2 2242 solid-state drive have been used for computation. The processor was maintained at a constant temperature for stable performance for all three schemes. The Crank–Nicolson method took the highest time to execute the program, although the alternating direction implicit method provided more values.

The alternating direction semi-implicit method took the least amount of time to execute the code. This is because semi-implicit schemes split the terms in such a way that explicit discretization time step is significantly smaller than the largest stable semi-implicit discretization time step.

4 Conclusion

In this work, we have used the Crank–Nicolson, ADI, and alternating direct semi-implicit to solve the heat convection–diffusion phenomenon of the copper thin plate. Obtained results show that the Crank–Nicolson method is unstable when the time step is large and analyzed over a small period. It requires the highest computational time among the three schemes. The ADI method provides the best results for this problem. Although the computation time for the alternating direction semi-implicit scheme is much less than the two other methods, its solution is unstable in the horizontal direction. Considering all the facts, it is suggested that ADI scheme should be used for such problems if the computation power is limited and the problem is time dependent.

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Chapter 10
An Expert System to Determine Systemic Lupus Erythematosus Under Uncertainty

Shakhawat Hossain, Md. Zahid Hasan, Muhammed J. A. Patwary, and Mohammad Shorif Uddin

1 Introduction

SLE is an illness of the insusceptible framework, which is evaluated to influence over 20,000 individuals in Australia and New Zealand. Side effects can be unclear and shift from individual to individual, and therefore analysis can be troublesome. Be that as it may, once analyzed, a mix of endorsed treatment and way of life alterations empowers a great many people with lupus to appreciate a relatively typical life. People usually die of SLE because SLE is not detected in time.

Therefore, an early suspicion of SLE is a must to save thousands of lives. This paper represents the development of an expert system to determine SLE at its initial stage. The proposed system solely considers the signs and symptoms of the SLE to establish its knowledge base. The rule-based inference methodology using evidential reasoning (RIMER) [1] approach is used to establish the basic. A significant challenge for any expert system which accomplishes a medical suspicion is handling uncertainties [2, 3]. Uncertainties can include origin during knowledgebase construction

S. Hossain
Department of Computer Science and Engineering, International Islamic University Chittagong, Chattogram, Bangladesh
e-mail: shakhawat.cse@outlook.com

Md. Zahid Hasan (✉)
Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh
e-mail: zahid.cse@diu.edu.bd

M. J. A. Patwary
Big Data Institute, College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China
e-mail: jamshed@szu.edu.cn

M. S. Uddin
Department of Computer Science and Engineering, Jahangirnagar University, Dhaka, Bangladesh

M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems,
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or input acquisition [4, 5]. For the capturing of all sorts of uncertainties to establish expert systems free from all types of flaws, scientists for more than four decades have been conducting researches on it [6]. The proposed RIMER approach takes care of all kinds of uncertainties while calculating the suspicion outcomes. Some IF–THEN rules are used to capture human knowledge. These rules are assigned with some weight to show relative importance. The sign-symptoms and patient's histories are qualitative and hence measured using three linguistic terms: high, medium and low. This research is needed to build an expert system that determines SLE without any uncertainty.

An expert system simplifies the healthcare process and makes the clinical practice easier. It lessens the medical errors [7] which have been started in 1970 [8]. It is designed to work just like a human expert. To infer just like many social mechanisms are used in the expert systems like forwarding chaining, backward chaining [9], Bayesian theory [10], evidential reasoning [5, 11], fuzzy method [12–14], etc. A large number of expert systems are built-in health care which used machine learning techniques [15–18] to improve the quality of health care. But no expert system has been developed to determine SLE. However, there are some systems that diagnosis SLE based on some laboratory tests. These tests are very costly and time-consuming. So, developing an expert system to determine SLE is very important.

The remaining part of the paper is arranged as follows: The second section briefly describes the systemic lupus erythematosus (SLE). The third section elaborately discusses the methodology of the belief rule-based inference methodology that uses the evidential reasoning (RIMER) approach for SLE determination. The fourth section gives a detailed implementation of the system. The fifth section analyzes the experimental results, and finally, Sect. 6 concludes the paper.

2 Systemic Lupus Erythematosus (SLE)

Systemic lupus erythematosus (SLE), otherwise known as lupus, is an autoimmune disease that attacks the body's multiple organs within a brief period [19]. In the case of SLE disease, the body's immune system attacks the own body as it becomes confused and considers multiple organs as the foreign body. Though SLE can be diagnosed with several symptoms, it is very often determined with oral ulcer, photophobia, alopecia, arthralgia, and malar rash [19]. However, these symptoms are exposed when SLE reaches a severe level, and it turns out to be practically inconceivable for a patient to endure. So, scientists have been conducting researches to find out the symptoms of SLE at its initial stage. Researchers are also focusing on the causes of SLE to identify its remedy. Tragically, the specific reason behind SLE is still vague to researchers. As a result, the diagnosis of SLE at the early development stage is almost a challenge for the physician. Physicians in maximum cases fail to determine SLE as it is challenging for a physician to analyze all the underlying indications of SLE, which puts an enormous amount of people to death every year [20]. However, the expert system proposed in this paper presumes SLE at its underlying stage by dissecting all the signs and symptoms of SLE.

3 Methodology

The proposed expert system is developed using belief rule-base inference methodology using the evidential reasoning (RIMER) approach [1]. The RIMER approach is an extended version of evidential reasoning (ER) approach [21] which is built based on the Dempster–Shafer theory of evidence. In the ER approach, Dempster–Shafer theory is well formulated to aggregate uncertainties that result in the representation of total uncertainties [21] in the final assessment.

3.1 Basic Knowledge Base Structure

Construction of a knowledge base structure depends on some specific If–Then rules collected from some experts. Domain knowledge also has a contribution; in this case, attribute weights and rule weights are also taken into account.

The basic rule base is represented as follows:

$$R_k : \text{if } A_1^k \wedge A_2^k \wedge A_3^k \ldots \ \ldots A_T^k \ \text{ then } \left\{\left(D_1, \overline{\beta}_{1k}\right), \left(D_2, \overline{\beta}_{2k}\right) \ldots\ldots \left(D_N, \overline{\beta}_{3k}\right)\right\} \tag{1}$$

where, A_1^k ($i = 1, 2,\ldots, T_k$) represents the antecedent-attribute of the kth rule, D_i is the consequents and β_{ik} ($i = 1, 2, \ldots, N$) is the degree of belief regarding the consequent.

For example, R_1 : If safety is good and fire protection is excellent and sea worthiness is good then, ship performance is estimated as {(good, 0.6), (average, 0.4), (poor, 0)}.

Here, {(good, 0.6), (average, 0.4), (poor, 0)} is the belief distribution for ship performance consequent. This distribution describes that the system is exactly 60% sure the performance of the ship is good and 40% sure that the ship performance level is average. This belief rules in the RIMER approach are represented in matrix format (Table 1).

Inference

In RIMER approach, the input provided by the user is transformed into some referential values based on the pre-determined belief degrees. For example, three referential values Poor, Average and Good are considered then for a certain input, the input transformation process will be,

Table 1 Belief rule expression matrix

Output	A^1 (w_1)	A^1 (w_1)	A^1 (w_1)	A^1 (w_1)
D_1	β_{11}	β_{12}	β_{1K}	β_{1L}
D_1	β_{21}	β_{22}	β_{2K}	β_{2L}
D_1	β_{31}	β_{32}	β_{3K}	β_{3L}
D_1	β_{N1}	β_{N2}	β_{NK}	β_{NL}

$$\begin{cases} \left.\begin{array}{l} \text{Average} = \dfrac{\text{Good} - \text{Input}}{\text{Good} - \text{Average}} \\ \text{Good} = 1 - \text{Average} \\ \text{Poor} = 0 \end{array}\right. ; \text{when, Good} \geq \text{Input} \geq \text{Average} \\ \left.\begin{array}{l} \text{Poor} = \dfrac{\text{Average} - \text{Input}}{\text{Average} - \text{Poor}} \\ \text{Average} = 1 - \text{Poor} \\ \text{Good} = 0 \end{array}\right. ; \text{when, Average} \geq \text{Input} \geq \text{Poor} \end{cases}$$

In RIMER approach, each antecedent of a rule is activated by assigning a weight to it which justifies the credibility of the rule. Weight activation process can be accomplished by using the following equation.

$$w_k = \frac{\theta_k \alpha_k}{\sum_{i=1}^{L} \theta_i \alpha_i} \tag{2}$$

$$\theta_k = \prod_{i=1}^{T_k} \left(\alpha_i^k\right)^{\overline{\delta}_{ki}} \tag{3}$$

And

$$\overline{\delta}_{ki} = \frac{\delta_{ki}}{\max\{\delta_{ki}\}} i = 1, \ldots, T_k \quad \text{and} \quad 0 \leq \overline{\delta}_{ki} \leq 1 \tag{4}$$

Here, w_k represents activation weight and θ_k presents the relative activation weight for the kth rule. α_i is the belief degree and δ is the relative weight for antecedent attribute.

To capture the uncertainties emerged from an incomplete input, the initial belief degree of an attribute is updated by using the following formula.

$$\beta_{ik} = \overline{\beta}_{ik} \frac{\sum_{t=1}^{T_k} \left(\tau(t, k) \sum_{j=1}^{J_t} \alpha_{tj} \right)}{\sum_{t=1}^{T_k} \tau(t, k)} \tag{5}$$

where

$$\tau(t,k) = \begin{cases} 1 \quad \text{if} \quad U_i \quad \text{is used in defining} \quad R_k (t = 1, \ldots, T) \\ \qquad\qquad 0 \quad \text{otherwise} \end{cases}$$

When the weight activation process is properly accomplished, the belief rules are aggregated by utilizing the evidential reasoning (ER) approach which consequently produces the final outcomes. The final outcomes the system can be presented as,

$$R(U) = (D_j, B_j);\ j = 1, \ldots, N \tag{6}$$

The above equation demonstrates the outcomes D_j along with its degree of belief B_j for any given utility factor U.

4 System Implementation

4.1 System Structure

A computer-based system encompasses at least three essential components to be adequately functional as represented by the following blueprint. The proposed expert system combines the three necessary layers of an expert system—presentation layer, application layer and data management layer. The presentation layer represents the interface of the system, which is responsible for input acquisition and output presentation. The presentation layer interacts with application logic, and the application logic interacts with the database and makes decision through the decision-making system. The application logic consists of a knowledge base which is most likely utilized by the inference engine to deduce the final output. The data management layer contains all sorts of system data as well as expert knowledge (Fig. 1).

Here, the application layer comprises of the RIMER approach and SLE determination system. The RIMER approach has portions: BRB knowledge base and inference engine, which is based on evidential reasoning. Sign-symptoms and experts' knowledge that represents the initial rule base constitute the data management layer.

4.2 Knowledge-Base Construction

The basic knowledge of the proposed system is constructed with signs and symptoms of the SLE. The complete domain knowledge for assessing the SLE has been described in a three-layer hierarchical architecture by the domain experts where the sign—symptoms of the patients construct the primary layer of the architecture. The intermediate level is constructed with more generic symptoms expression areas. The top level of the architecture represents the outcomes of the system (Fig. 2; Table 2).

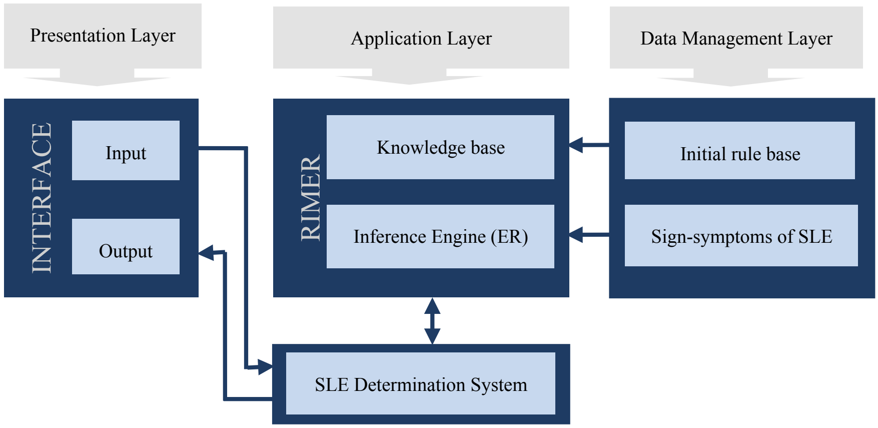

Fig. 1 Architectural view of the proposed system

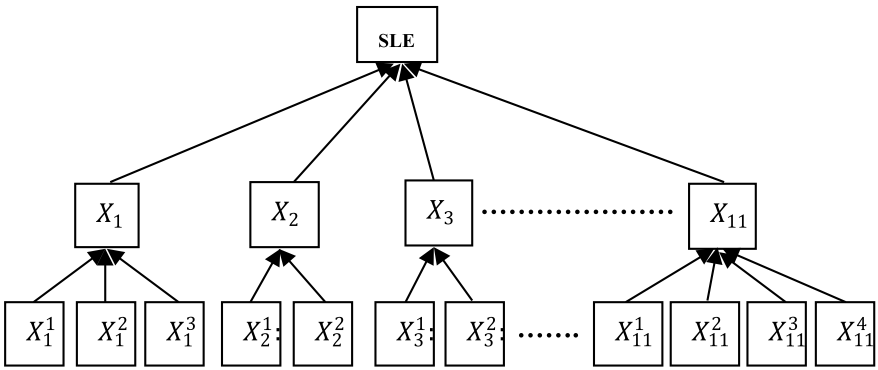

Fig. 2 Architectural theory diagram of the SLE suspicion system

The overall knowledge base was constructed with 21,343 sub-rules. This rule base was established with the help of several domain experts from different hospitals in Bangladesh. The additional knowledge representation parameters like rule weight, antecedent weights are also assigned by these domain experts (Table 3).

4.3 Inference Engine Development

The interface of the proposed system has been developed by using the evidential reasoning (ER) approach, which accomplishes the mathematical calculations to aggregate established initial rules. The proposed method considers three referential

Table 2 SLE assessment parameters

Intermediate level attribute	Basic/root level attribute
X_1: Skin	X_1^1: Malar Rash
	X_1^2: Discoid Rash
	X_1^3: Skin Ulcer
	X_1^4: Vasculitis Lesion in Fingure Tip and Nail
X_2: Lungs	X_2^1: Pneumonitis
	X_2^2: Pleurisy
X_3: Heart	X_3^1: Chest Pain
	X_3^2: Respiratory Distress
X_4: Vascular	X_4^1: Small Red or Purple Dot on the Skin
	X_4^2: Pain and Ulcer
X_5: Eyes	X_5^1: Conjunctivitis
	X_5^2: Scleritis
X_6: Kidneys	X_6^1: Hematuria
	X_6^2: Oliguria
X_7: Gastrointestinal	X_7^1: Oral Ulcer
	X_7^2: Vomiting
	X_7^3: Diarrhoea
	X_7^4: Abdominal Pain
	X_7^5: Nausea
X_8: Hematological	X_8^1: Anemia
X_9: Central Nervus System	X_9^1: Headache
	X_9^2: Poor Concentration
	X_9^3: Psychosis
	X_9^4: Seizure
X_{10}:General	X_{10}^1: Fever
	X_{10}^2: Arthralgia
	X_{10}^3: Arthritis
	X_{10}^4: Myalgia
	X_{10}^5: Alopecia
	X_{10}^6: Lupus Hair
X_{11}:Other Symptoms	X_{11}^1: Lymphadenopathy
	X_{11}^2: Abortion
	X_{11}^3: Menorrhagia
	X_{11}^4: Infertility

Table 3 Sample initial belief rules

Rule No.	Rule weight	Antecedents(If) [S = Severe, M = Medium, L = Low]	Consequences (Then)
1	1.0	$(X_1^1, S)\wedge(X_1^2, S)\wedge(X_1^3, M)\wedge(X_1^4, S)$	$X_1\{(S, 1.0), (M, 0.0), (L, 0.0)\}$
2	0.90	$(X_1^1, M)\wedge(X_1^2, S)\wedge(X_1^3, L)\wedge(X_1^4, L)$	$X_1\{(S, 0.3), (M, 0.3), (L, 0.4)\}$
3	0.95	$(X_1^1, L)\wedge(X_1^2, S)\wedge(X_1^3, M)\wedge(X_1^4, M)$	$X_1\{(S, 0.3), (M, 0.4), (L, 0.3)\}$
4	1.0	$(X_1^1, M)\wedge(X_1^2, S)\wedge(X_1^3, M)\wedge(X_1^4, \mathrm{M})$	$X_1\{(S, 0.0), (M, 1.0), (L, 0.0)\}$
5	1.0	$(X_1^1, S)\wedge(X_1^2, S)\wedge(X_1^3, M)\wedge(X_1^4, M)$	$X_1\{(S, 0.5), (M, 0.5), (L, 0.0)\}$
6	0.85	$(X_1^1, M)\wedge(X_1^2, S)\wedge(X_1^3, S)\wedge(X_1^4, L)$	$X_1\{(S, 0.5), (M, 0.25), (L, 0.25)\}$
7	1	$(X_1^1, L)\wedge(X_1^2, L)\wedge(X_1^3, M)\wedge(X_1^4, L)$	$X_1\{(S, 0.0), (M, 0.0), (L, 1.0)\}$
8	1.0	$(X_1^1, L)\wedge(X_1^2, S)\wedge(X_1^3, M)\wedge(X_1^4, M)$	$X_1\{(S, 0.2), (M, 0.8), (L, 0.2)\}$
9	1.0	$(X_1^1, M)\wedge(X_1^2, S)\wedge(X_1^3, S)\wedge(X_1^4, L)$	$X_1\{(S, 0.8), (M, 0.2), (L, 0.2)\}$
10	0.9	$(X_1^1, M)\wedge(X_1^2, L)\wedge(X_1^3, L)\wedge(X_1^4, L)$	$X_1\{(S, 0.0), (M, 0.1), (L, 0.9)\}$

values—Severe, Medium and Low to describe a patient's condition. The proposed system updates antecedent's belief degree to capture uncertainties (Table 4).

After the belief degree is updated, the activation weight is generated, and finally, the initial belief rules are aggregated. While aggregating rules, the ER approach calculates the uncertainties in each step which have a direct impact on the final uncertainty assessment process. The uncertainty in the ER approach is calculated for each antecedent separately. The final results in the proposed system are displayed using an evaluation grade along the calculated uncertain degrees (Fig. 3).

Table 4 Belief degree update

	Severe	Medium	Low
Initial belief	0.6	0.1	0.3
Updated belief	0.43	0.2	0.37

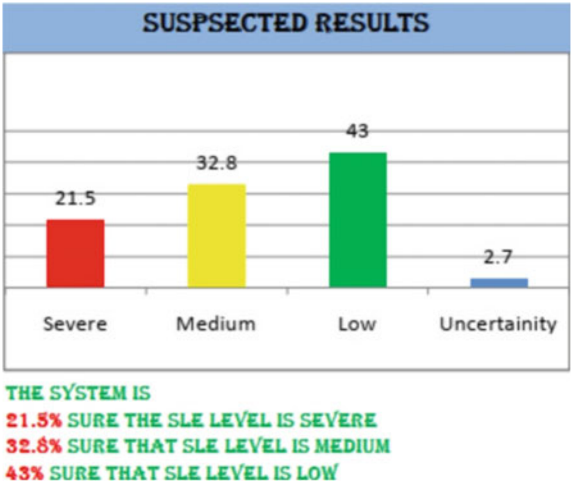

Fig. 3 Results display page

4.4 Software Development for SLE Suspicion

The expert system for determining SLE has been developed using Java (8.0) in Spring Boot framework. The front end of the system has been designed with HTML and CSS, where JSP interacts with the main application for output and output processing. The system model has been developed using Oracle (10 g) for storing and processing all sorts of data. The interface of the system (Fig. 4) accepts inputs as some evaluation grades against each attribute which has been developed using drop-drown buttons. The interface displays the results in appraisal grades, along with the assessed belief. A graphical representation is also generated by the system for an easy understanding of the suspected results (Fig. 3).

5 Result Analysis

The proposed system has been tested for 1210 times with a dataset of around 170 SLE patients under different condition. The system predicted SLE correctly almost every time. The results generated by the expert system have been presented using a confusion matrix (Table 5).

So, from the test results, the sensitivity and specificity of the proposed expert system can be calculated as follows:

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} = \frac{897}{897 + 1} = 0.9989 \quad \text{or, } 99.89\%$$

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}} = \frac{309}{309 + 3} = 0.9904 \quad \text{or, } 99.04\%$$

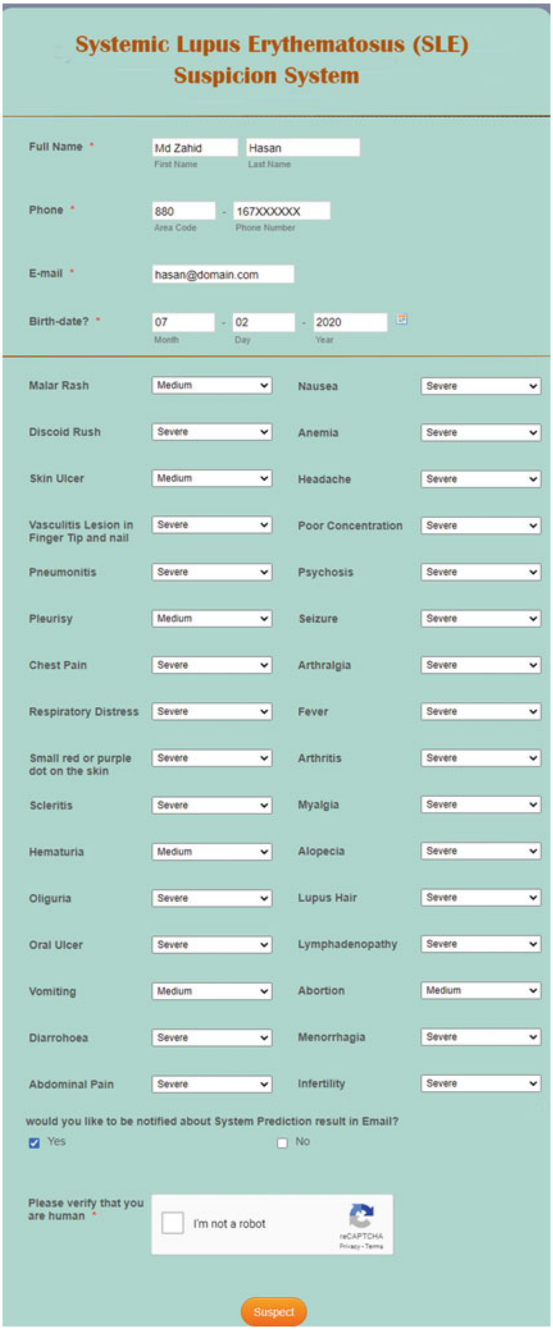

Fig. 4 System interface

Table 5 Confusion matrix of the system-generated results

$N = 1210$	Predicted Yes	Predicted No
Actual Yes	897	1
Actual No	3	309

So, the accuracy of the system is calculated as,

$$\text{Accuracy} = \frac{\text{True positives} + \text{True negatives}}{\text{Total number of test}} = \frac{897 + 309}{1210} = 0.9967 \text{ or, } 99.67\%$$

From the above experiment, sensitivity and specificity of the proposed system are very close to 100%. Similar is the case with accuracy parameters which is 99.67%.

To validate the proposed system performs better than the manual method, a comparison between the manually calculated results and expert system-generated works is conducted carefully. The next table demonstrates a comparative study between the manual experimental results and the expert system. The performance of both manual and expert system is measured based on the benchmark results. The benchmark results are determined by analyzing clinical historical data by the domain experts (Table 6).

Data generated from both manual and the expert system has been tabled and compared in the above table, which demonstrates the expert system-results almost duplicated the benchmark one. In contrast, the manual system results lie far away. This tabular demonstration indicates the preference of the proposed expert system in a case determining SLE at its initial stage.

The above receiver operating characteristic (ROC) curve (Fig. 5) illustrates the performance of the manual and expert system using two distinct colored curve red for manual system and black curve for the expert system. The ROC curve shows that the expert system covers the 0.851 unit area under the curve (AUC), which superior to the manual system measuring AUC as 0.776. So the performance of the expert

Table 6 Comparisons of the results from manual and expert systems along with benchmark results

Test No.	Benchmark results (%)	Manual system results (%)	Expert system results (%)
1	69.42	60.61	63.33
2	72.02	58.45	69.76
3	89.55	83.02	86.23
4	71.45	79.79	79.88
5	63.78	59.95	61.65
6	84.86	80.18	82.78%
7	70.54	63.43	67.32
8	68.87	60.12	63.98
9	64.73	69.66	60.43
10	51.45	42.77	46.65

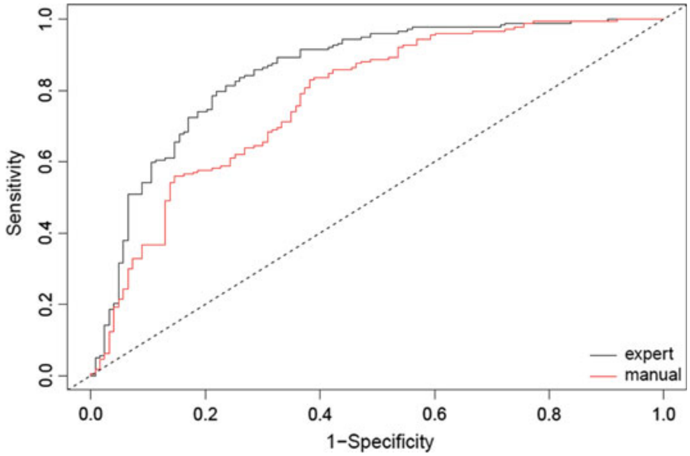

Fig. 5 Performance comparison between the manual and expert system using ROC curve

system is excellent to that one of the manual method. The diagonal line in the ROC demonstrates the reference line. The ROC has been generated using SPSS ZV.20.

6 Conclusion

This study focuses on developing an expert system that determines SLE accurately by capturing all possible uncertainties. SLE is considered as the most dangerous disease as it is still untraceable at its early development stage. Moreover, it is expensive to determine SLE through laboratory tests. The physicians need to wait for the results of laboratory tests to provide a treatment which worsens the conditions of the patients. So, developing an expert system that affirms an early suspicion of SLE has been a demand for time. This study intends to create such an accurate and robust system. In the developing and populated counties, an expert system that determines SLE is more critical because the traditional approach is expensive and time-consuming.

Therefore, by this work, an expert system is established, which determines SLE under uncertainties. The system for its user-friendly interface appears comfortable for the physician to use. It is also convenient for the patients as they can use the system without taking any help from the physicians. For these, the patients need to provide the inputs based on how they feel. Using the system by the patients will give the most relevant results as the patients can give the input according to how they think. The appropriate way of using the system is to use the system by the physicians in the presence of the patients. At that case, signs and symptoms are correctly observed. In future, the limitations that come out from this system will be removed, and the design will be revised. The framework in the following adaptation will be developed with a more enriched knowledge base comprised of an adequate rule-base which will ensure the robustness as well as the potentiality of the system. The system errors of this research will be removed in the future system by a training module. An online BRB base expert system will be developed in future.

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Chapter 11
Bengali Stop Word Detection Using Different Machine Learning Algorithms

Jannatul Ferdousi Sohana, Ranak Jahan Rupa, and Moqsadur Rahman

1 Introduction

Worldwide, approximately 6500 languages are spoken [1], and each language is different from the other language. So, stop words also differ from one language to another language. These language-specific stop words are used only for sentence formation because they do not have any useful value and are used frequently in a document.

Therefore, documents can contain two types of words, one is the stop word that does not contribute anything, another one is the content word that holds major informative value, and has high contribution in a document. Content words are used mostly as index terms in query or indexing, as stop words rarely used, due to its very little contribution to query.

The Bengali language is the 5th most widely used writing system in the world and with around 265 million speakers, the Bengali language is the 7th most spoken language in the world [1]. But Bengali language is not well structured like western and a very few researches have been done in the field of Bangla Language Processing (BLP). For this reason, a new method is proposed to work on the Bengali language to detect Bengali stop word by using a sequentially created dictionary for the string to numeric conversion to make Bangla text processing easier. Proper Bengali stop word detection method is necessary for the betterment of the system's accuracy. In this paper, 61 Bengali available characters are assigned with unique numerical value ranges from 0 to 60 in the dictionary to generate the final unique float type numerical value for each word. Each word is checked individually in a sentence but instead of the word, the position of the corresponding word is checked for the detection of

J. F. Sohana · R. J. Rupa
Sylhet Engineering College, Sylhet, Bangladesh

M. Rahman (✉)
Shahjalal University of Science and Technology, Sylhet, Bangladesh
e-mail: moqsad-cse@sust.edu

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stop word and different supervised machine learning classifiers are used. Here, the position of the corresponding word means the stored word index value. Checking the position of the word for stop word detection purpose reduces the chance of incorrect detection of stop words.

Let us assume a Bengali sentence, কি এবং কী এর পার্থক্য কি (What is the difference between কি (what) and কী (what)). In this sentence, the 1st, 3rd, 4th, 5th word are content word and the 2nd and 6th word are stop word. In this sentence, the same word কি (what) is used as a content word and stop word which creates confusion. But this paper solved this problem as each word is checked according to their position individually. So, the position of the 1st word কি (what) is '-0′ and detected as a content word but the position of the 6th word is '-5′ according to their stored index value in the dataset and detected as a stop word.

Let us assume another Bengali sentence, আমার বাস ছাড়ার আগেবাসস্ট্যান্ডে পৌঁছাতে হবে (I have to reach at the bus stand before the bus departure). Here, 4th আগে (before) used as a content word but by using a pre-defined stop word list to detect stop word, it cannot identify the word আগে (before) as a content word for this sentence, as a word are not analyzed contextwise, so আগে (before) detected as a stop word according to the pre-defined stop word list. But this problem is solved by the proposed method as all words are checked individually according to their indexed value as position and labeled contextwise.

2 Literature Review

Very few works have been done for stop word detection in the Bengali language. In paper [2], an automatic method of identifying stop word by using the vector space model is proposed, based on document–document similarities. This method may fail to achieve expected accuracy without using document–document similarities. There are no standard Bengali stop word lists available for the Bengali language. A computer-based system to automatically identify high scored function words from Technology Development for Indian Languages (TDIL) Bengali corpus is proposed in [3] and they used the AFWI algorithm and Z-score method to score each generated optimized word. A total of 290 standard function words are achieved by the proposed method.

The first attempt of detection and classification of the Bengali stop word [4] and the Bengali Stop phrase [5] by using a corpus-based method are proposed in the Bengali language. But they failed to detect stop word which used as a content word sometimes because they used a corpus-based approach and pre-defined stop word list to detect Bengali stop word, but in this paper, Bengali stop word is detected from sentences and every word is checked according to their position which is the word indexed value, and pre-defined stop word list is not used, so this problem is solved by the proposed method. Various methods can be used for stop word detection but commonly corpus-based methods, frequency-based methods, and finite automata-based methods are used mostly. The authors in [6] proposed an approach for detection and classification of Bengali stop word and stop phrase combinedly by using both the corpus-based method and finite automata-based method, and in both methods, the corpus-based method outperformed with 94% accuracy whereas the accuracy of the finite automata-based method is 88%.

Text classification is an ongoing research field in information retrieval and Natural Language Processing (NLP). Paper [7] created a Chinese stop word list by using a statistical approach based on weighted chi-squared statistics on the 2 * p contingency table in Chinese text classification.

Upon the entropy of each visual word, a method presented in [8] for automatic detection of stop words in visual vocabularies from Maya hieroglyphs and compared with the frequency-based method and PCA-based method for dimensionality reduction and outperformed than other two methods by removing 30% of visual words without decreasing in retrieval precision. In [9] only 1.78 s takes to remove stop words from the document, with comparing existing pattern matching approaches that take 3.3 s to remove stop words by using deterministic finite automata from 300 English documents.

Different approach lexical classes used in [10] to construct and categorize stop word list for the Gujarati language and 1125 unique stop words are achieved.

After reviewing many research papers on Bengali stop word detection, no work has been found as the proposed method. One major contribution of this paper is the creation of the Bengali dictionary for the word to numeric value conversion that contains a unique integer value for each available 61 Bengali character and detection of stop word from sentences.

3 Methodology

In the proposed dictionary-based approach, unique integer numeric value ranges from 0 to 60, sequentially assigned to each available 61 Bengali character for sentence conversion to a unique numeric value. The main goal of this paper is to detect the Bengali stop words from Bengali sentences by applying different supervised machine learning classifier algorithms.

The following steps are followed for the proposed method (Fig. 1).

3.1 Dataset Preparation

The data are manually collected from various online newspapers, blogs, and labeled. From 10,000 sentences, 80% is used for training purposes and the rest of the 20% is used for testing purposes. In the proposed method, only less or equal twenty-five lengths of sentences are considered for the implementation and disregarded those sentences whose lengths are more than twenty five.

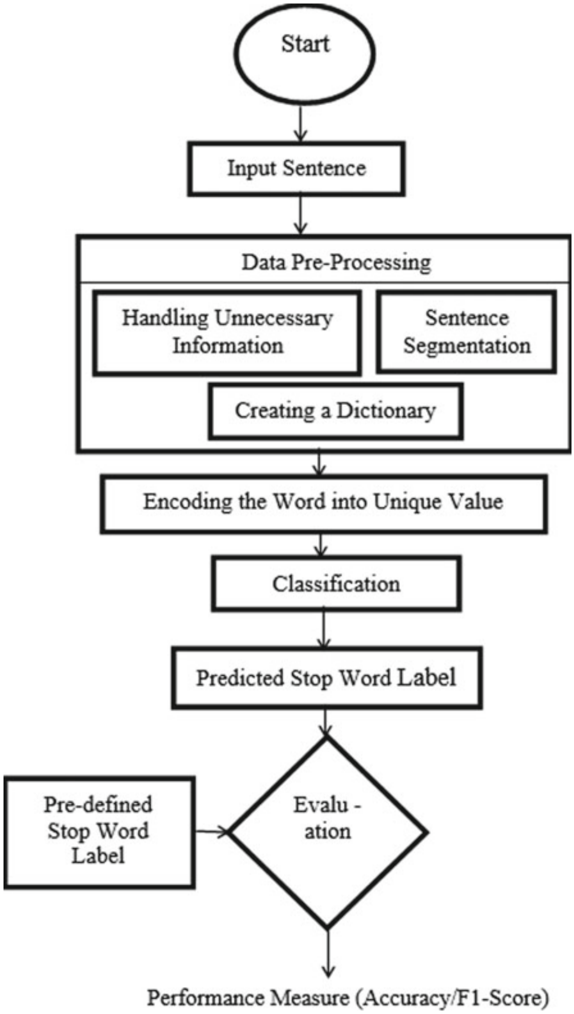

Fig. 1 Architecture of the proposed method

Many words occurred very frequently in the dataset. These words are counted as a stop word in the pre-defined stop word list according to their frequency value, which is not an appropriate approach to detect stop word from sentences, as frequent occurred words may be used in a sentence as a content word. For this reason, in the proposed method, frequency values of a word are not considered for implementation (Tables 1 and 2).

Table 1 Top 10 most frequent words in the dataset

Words	Frequency
না	1685
আমি	1027
করে	859
এই	734
আমার	543
হয়	506
ও	464
কি	426
থেকে	376
জন্য	328

Table 2 Environment and dataset analysis

Library	Scikit-learn
Language	Python
Development environment	Jupyter-Notebook
Total number of sentences	10,000
Total number of sentences with stop word	6547
Total number of sentences without stop word	3453
Maximum sentence length	25

3.2 Data Pre-processing

The dataset should be pre-processed for the implementation. The pre-processing dataset can increase the performance of the proposed method.

Handling Unnecessary Information. At first, all kinds of punctuation, numerical digits, and symbols need to remove as they do not provide any useful information in the proposed method for better performance but can decrease the performance score (Table 3).

Table 3 Unnecessary information in a sentence

Some Example of unnecessary information
‘ : , - -- `
. … ! & ? ~
% # | ; “ (
$ ^ @ € =)
০ ১ ২ ৩ ৪ ৫
৬ ৭ ৮ ৯ < >
+ * \ /

Let us assume a Bengali sentence before pre-processing is তারা দু-জন খুবই ভালো বনধু (They both are very good friends). But after pre-processing, the sentence looks like as তারা দুজন খুবই ভালো বন্ধু (They both are very good friends), because the unnecessary information '–' (hyphen) is removed from the sentence to increase the speed of the performance.

Sentence Segmentation. All sentences are segmented into words then words to a character by using the string split function of Python to make the encoding process easier and each word is considered as a feature. Labeling of the dataset is done by checking every word position where the position is the stored index value.

For example, a Bengali sentence is বাবা বাজারে গেছেন (Dad went to the market) (Table 4).

Here, in check '0' means, not stop word and instead of the word, the index value is used as a position to check the stop word.

For example, another Bengali sentence is আমি বেশ ক্লান্ত (I am very tired) (Table 5).

Here, বেশ (very) is a stop word as the check is '1' for index 1. Also without the word বেশ (very), the sentence is syntactically and semantically correct, so the word বেশ (very) is used for the formation of the sentence which holds no significant value.

After sentence segmentation, the dataset looks like as shown in Fig. 2.

Creating a Dictionary. A dictionary is created sequentially for each available 61 Bengali character with unique numeric value ranges from 0 to 60 for encoding the

Table 4 Array index representation of example in the dataset

Example				
0	1	2	Position	Check
বাবা	বাজারে	গেছেন	0	0
বাবা	বাজারে	গেছেন	1	0
বাবা	বাজারে	গেছেন	2	0

Table 5 Array index representation of example in the dataset

Example				
0	1	2	Position	Check
আমি	বেশ	ক্লান্ত	0	0
আমি	বেশ	ক্লান্ত	1	1
আমি	বেশ	ক্লান্ত	2	0

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	position	check
আনারের	জুস	ত্বকের	যত্নে	দারুণ	উপকারী	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	0	0
আনারের	জুস	ত্বকের	যত্নে	দারুণ	উপকারী	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	1	0
আনারের	জুস	ত্বকের	যত্নে	দারুণ	উপকারী	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	2	0
আনারের	জুস	ত্বকের	যত্নে	দারুণ	উপকারী	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	3	0
আনারের	জুস	ত্বকের	যত্নে	দারুণ	উপকারী	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	4	1
আনারের	জুস	ত্বকের	যত্নে	দারুণ	উপকারী	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	5	0

Fig. 2 Representation of sentence segmentation

Table 6 Created a dictionary

Representation of a dictionary									
অ : 0	আ : 1	ই : 2	ঈ : 3	উ : 4	ঊ : 5	ঋ : 6	এ : 7	ঐ : 8	ও : 9
ঔ : 10	ক : 11	খ : 12	গ : 13	ঘ : 14	ঙ : 15	চ : 16	ছ : 17	জ : 18	ঝ : 19
ঞ : 20	ট : 21	ঠ : 22	ড : 23	ঢ : 24	ণ : 25	ত : 26	থ : 27	দ : 28	ধ : 29
ন : 30	প : 31	ফ : 32	ব : 33	ভ : 34	ম : 35	য : 36	র : 37	ল : 38	শ : 39
ষ : 40	স : 41	হ : 42	ড় : 43	ঢ় : 44	য় : 45	ৎ : 46	◌ং : 47	◌ঃ : 48	◌ঁ : 49
◌া : 50	◌ি : 51	◌ী : 52	◌ু : 53	◌ূ : 54	◌ৃ : 55	◌ে : 56	◌ৈ : 57	◌ো : 58	◌ৌ : 59
◌্ : 60									

word to a unique float type numeric value as computers can only understand and process numerical value, not raw texts (Table 6).

3.3 Encoding the Word into Unique Value

At the very first, every sentence is segmented into words, then every word into character. Each character is assigned with integer-type unique numeric values that are pre-defined in the dictionary. For example, আমি খাব না (I will not eat) Bengali sentence.

Firstly, a sentence is segmented into words (Table 7).

Secondly, words are segmented into character along with the unique value according to the pre-defined dictionary (Table 8).

Thirdly, the pre-defined dictionary value of each character of a word is multiplied with the power of base value 61 where the power of base value is increased in ascending order (i.e., power = 0, 1, 2, 3, 4, 5, …), and then a unique integer value is generated by summation (Tables 9 and 10).

Finally, the words are represented as a float type of unique value as in many cases generated integer encoding value may get bigger and take more space by adding more digits, also can be problematic to work with large integer-type encoding values in pre-processing. For example, the Bangla word রাজপরিবার (Royal family), after

Table 7 Segmentation of a sentence

Example of sentence segmentation		
আমি	খাব	না

Table 8 Segmentation of words into character with pre-defined dictionary value in a sentence

Example of a segmented word to character conversion		
আ ম ◌ি	খ ◌া ব	ন ◌া
1 35 51	12 50 33	30 50

Table 9 Multiplication and summation of segmented words with base value 61

Example of a segmented word for integer conversion		
আ ম ি	খ া ব	ন া
$1*61^2+35*61^1+51*61^0$	$12*61^2+50*61^1+33*61^0$	$30*61^1+50*61^0$

Table 10 Generated unique integer value after summation

Example of the generated unique integer value of words		
আমি	খাব	না
5907	47735	1880

Table 11 Float type of final unique value of words in a sentence

Example of float type word value representation		
আমি	খাব	না
5907.0	47735.0	1880.0

multiplying the word with 61 base value the final value generated from summation is 7,251,261,795,922,304 which is quite large, take more space, also difficult to work with that large encoding value in pre-processing step. But using float-type encoding value instead of integer number, the difficulties are less a little bit as the word রাজপরিবার (Royal family) final encoding value is 7.251261e+16, which is easy to work compared with integer values (Table 11).

4 Classification

The entire dataset is split into the ratio of 80% and 20% for the training and testing phases. The various classifier is used to train the proposed method.

The following classifiers are used to evaluate the proposed method:

Gaussian Naïve Bayes Classifier. Gaussian naïve Bayes is a probabilistic classifier and useful when continuous data are used and it assumes that features follow a normal distribution. As it required less training data, so it is easy and fast to predict the test dataset. By finding the mean and standard deviation of every data point, this model can be fit.

Support Vector Machine Classifier (SVM). SVM is a supervised machine learning algorithm used for classification and regression problems. It is mostly used in classification problems. Each data sample is plotted as a point in n-dimensional space with the value of each feature being the value of a particular coordinate and differentiates two classes by finding the hyper-plane. SVM is effective in high dimensional spaces and also memory efficient as it uses a subset of training points in the decision function. It uses a technique called the kernel trick to transform the data and then based on these transformations it finds an optimal boundary between the possible outputs. Simply SVM kernel converts non-separable problems to separable problems and most useful in nonlinear separation problems.

Decision Tree Classifier. The goal of the decision tree classifier is to predict the output based on the input features. Decision tree classifier used for classification and regression problems and also known as a tree-based classifier. It is simple and easy to understand. It starts from the root node of the decision tree to predict the class of the input and compare the value of the root attribute with the original input, then follows the branch and jumps to the next node based on the comparison, the process keeps continuing until it reaches the leaf node of the tree and it required less data cleaning.

Random Forest Classifier. A random forest classifier is a collection of multiple decision trees used for classification and regression problems. It is a type of ensemble learning method, as it reduces the over-fitting by averaging the result, creates decision trees on inputs, predicts results from every decision tree, and selects the best solution by voting. It maintains high accuracy even a large proportion of the data is missing, also very flexible, and does not require scaling of data.

Logistic Regression. Logistic regression is used for classification and works with binary data, where output is represented as either '0' or '1'. Logistic regression is an estimation of logit function where logit function is a log of odds in favor of the event and also a predictive analysis algorithm that performed on the concept of probability and utilizes sigmoid function to map the predicted values to probabilities. By passing the inputs, it gives a set of outputs based on probability through a prediction function and returns a probability score between 0 and 1.

5 Evaluation and Result

Four metrics, precision, recall, F1-Score, and accuracy are considered to evaluate the proposed method performance.

Precision. The precision determines the actual true positive predictions from the total predicted positive.

$$\text{Precision} = \frac{\text{True Positive}}{\text{Total Predicted Positive}} \tag{1}$$

Recall. Recall determines the actual true positive predictions from the total actual positive.

$$\text{Recall} = \frac{\text{True Positive}}{\text{Total Actual Positive}} \tag{2}$$

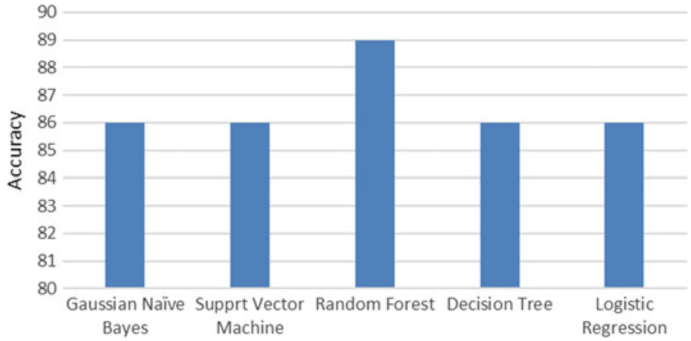

Fig. 3 Accuracy score of Gaussian Naïve Bayes, support vector machine, random forest, decision tree, logistic regression classifiers

Table 12 Comparison based on various classifier result

Classifier	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
Gaussian Naïve Bayes	74	86	80	86
SVM	74	86	80	86
Random forest	88	89	88	89
Decision tree	86	86	86	86
Logistic regression	74	86	80	86

F1-Score. F1-Score is a combination of precision and recall.

$$\mathrm{F1} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Accuracy. Accuracy determines the correctly predicted observation from the total observation (Fig. 3; Table 12).

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total Number of Predictions}} \tag{4}$$

A pre-defined Bengali stop word list found in [11] is used in the proposed method to detect Bengali stop word in a sentence and 72% accuracy is achieved. So, the proposed method performed better than the pre-defined Bengali stop word list method (Fig. 4).

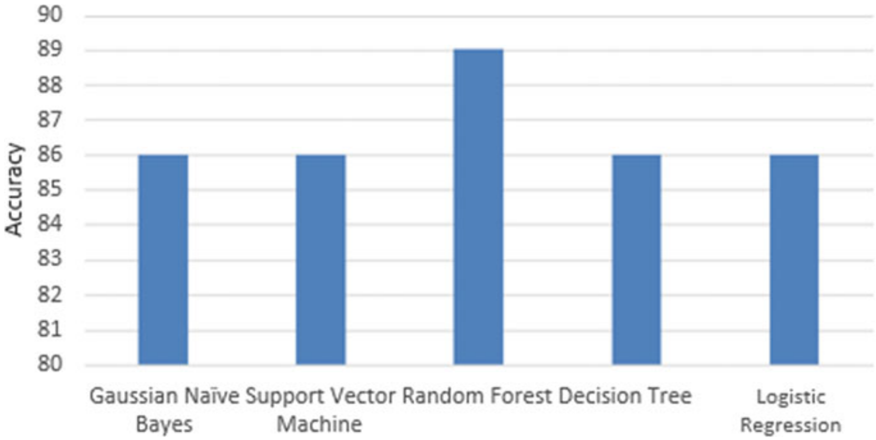

Fig. 4 Accuracy results of using a pre-defined Bengali stop word list and random forest classifier model

6 Conclusion

In this paper, the Bengali stop word is detected by using a dictionary-based approach from sentences which is the first attempt in the field of Bengali stop word detection. A total of 10,000 datasets are used for the implementation and data are pre-processed for better accuracy score as unnecessary information can decrease the accuracy score. By comparing various classifiers, the overall accuracy is 89% achieved by random forest classifier as it splits a node by considering the best feature among a random subset of the features and does not consider the most important feature. The performance can be enhanced by increasing the amount of dataset and used a more balanced dataset for the implementation.

But in the proposed method, one sentence is checked several times and for this reason, a total of 68,153 rows and 27 columns are created after data pre-processing against 10,000 real input data. So, in the future work, this issue needs to be solved that one sentence is checked several times but counted as one sentence in the performance measure. Also, the future plan is to detect stop words more accurately and also remove stop words properly by using the deep learning method.

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Chapter 12
Data Mining and Visualization to Understand Accident-Prone Areas

Md. Mashfiq Rizvee, Md Amiruzzaman, and Md. Rajibul Islam

1 Introduction

Injuries and loss of life caused by road traffic crashes are a global problem and remarkably affect socioeconomic growth and social prosperity [6]. Road traffic injuries are estimated to be the eighth driving reason for death worldwide for all age groups and the main motive of death for kids and youngsters 5–29 years old. Road traffic crashes are causing a projected 1.35 million passing, and daily, right around 3700 individuals are put to death globally in road traffic accidents associated with busses, cars, bicycles, motorcycles, vans, or people on foot [10]. In the USA, road traffic crashes are a leading purpose of dying for humans aged 1–54 and the pre-eminent source of non-normal demise for sound US residents dwelling or touring abroad [12]. Various vehicles and transportation such as trucks, bikes, mopeds, people on foot, creatures, taxis, and different voyagers are the users of roads all through the world. Travel made conceivable by motor vehicles helps monetary and social advancement in numerous nations. Yet every year, vehicles associated with crashes are accountable for many passing and injuries.

Road and traffic crashes are characterized by a lot of factors that are for most of the discrete nature [1]. The serious issue in the investigation of crashes information is its heterogeneous nature [13]. Although investigators utilized segmentation of the crashes data to diminish this heterogeneity applying several estimates, for example,

Md. Mashfiq Rizvee
North South University, Dhaka 1229, Bangladesh
e-mail: mashfiq.rizvee@northsouth.edu

Md Amiruzzaman (✉)
Kent State University, Kent, OH 44242, USA
e-mail: mamiruzz@kent.edu

Md. R. Islam (✉)
University of Asia Pacific, Dhaka 1205, Bangladesh
e-mail: md.rajibul.islam@uap-bd.edu

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expert knowledge, there is no assurance that this will produce an optimum segmentation which comprises homogeneous sets of road crashes [4]. For that reason, cluster investigation which is a significant data mining technique can help the segmentation of road and traffic crashes and such can be utilized as the earliest assignment to accomplish numerous objectives.

Road and traffic crashes are sometimes found to be increasingly repeated in certain specific areas. The investigation of these areas can assist in understanding certain features of accident occurrences that make a road crash to take place repeatedly in these areas. Interestingly, time analysis is found to be another significant road accident feature to understand accident-prone areas. Various data mining techniques have been used in literature to conduct these analyses. Data mining utilizes various algorithms and procedures to find the correlation in a huge volume of data which is viewed as one of the most significant tools. DBSCAN algorithm is one of the common density-based clustering algorithms of data mining that makes a group of abstract objects into classes of similar objects. This clustering algorithm is used in our study on pedestrian crashes data that has been taken from the US government's national database for public use (i.e., https://www.data.gov [16]). The quality of the clusters obtained by DBSCAN is then verified using a heatmap analysis. A past study indicated the importance of time and day which can be helpful to understand which days and times have a higher chance of accident [1]. So, in this study, we ran statistical analysis, and from the descriptive statistics, we found which month, day, and times are highly accident-prone.

The purpose of this study was three folds. First, to understand and identify the accident-prone places based on pedestrian crashes data. Second, identify most accident-prone time, day, and month which could help pedestrians to be more cautious during those times. Third, find appropriate visual techniques to present the findings so that both non-experts (e.g., pedestrian) and experts could comprehend the information better.

Below we list contributions of this study:

- We identify the accident-prone areas using the density-based spatial clustering of application with noise (DBSCAN) clustering algorithm.
- We generate a "heatmap" showing the number of accidents that happened between January to December. The heatmap shows which places had the most number of accidents.
- We further analyze the data by breaking it down to the most accident-prone month, day, hour, and location using different visualization techniques.
- We conducted a user study survey to understand which visualization figures help to understand the findings better.

2 Related Work

Literature based on road and traffic accidents has been of interest to many. For example, Kumar and Toshniwal [7] analyzed the main factors associated with a road and traffic accident. In their study, the authors used association mining rules to identify the circumstances that are correlated with the occurrence of each accident. As the number of road accidents is increasing day by day, road and traffic safety is becoming a priority. Therefore, a study could focus on finding ways to improve road traffic safety and manage it efficiently.

Tian et al. [15] analyzed the causes of road accidents based on data mining and provided a method of traffic data analysis which can improve road traffic safety management effectively. Continuing the journey to find the factors that make more damages in road accidents. Amiruzzaman [1] found patterns referring to specific days or times (e.g., the time of most accidents, the area with the most consistent number of accidents). However, no attempt was made to understand whether the results are conveyable to non-expert users.

While road and traffic safety is an important and interesting challenge in the research domain, hence several studies have been focused on pedestrian safety [6, 8, 9]. For example, Mohamed et al. [9] used a clustering regression approach to analyze the severity of pedestrian-vehicle crashes in large cities. Their use of the clustering approach helped to examine how the segmentation of the accident datasets help to understand complex relationships between injury severity outcomes and the contribution of the geometric, built environment, and socio-demographic factors.

Previous research studies show a gap in finding pedestrian-oriented accident places. Although Mohamed et al. [9] found that most pedestrian accidents occur in large cities, however, where exactly those accidents occurred in a large city is yet to be a priority in a research study. Moreover, there is a clear gap between research findings and the presentation of those to non-expert users. Perhaps, a study should focus on these aspects and help non-expert users to identify accident-prone areas within a city. Visualization techniques and user study could be a solution to this problem.

This paper will provide a straightforward visualization-based presentation of the findings to the non-expert users and show how the data mining features can significantly help to discover accident-prone areas. In this study, we are going to identify some of the fatal areas of pedestrian road accidents using unsupervised learning algorithms. Then, we will verify the clusters using a heatmap.

The visualization of statistical analysis will help us more to identify the most dangerous time, day, or month of the year the accidents took place. Also, the road features where the number of accidents is high will give us intuition about the mindset of pedestrians.

3 Method

3.1 Data

For this study, we downloaded the data from a national database (i.e., https://www.data.gov) [16]. The original database contains 71 columns with different attributes. The initial database consisted of 33,707 rows. However, some values were missing. Also, some duplicate values might have occurred due to human errors. Moreover, some of the columns were unnamed. Those columns also had a lot of Not a Number (NaN) values. We used "pandas" (i.e., Python library) data frame to import the data. The database included population-related information such as driver injury, crash locations, the gender of the pedestrian/driver, city/county names, etc.

3.2 Data Preprocessing

Initially, we identified important columns that needed to be processed. Then, we considered the most meaningful columns, e.g., latitude and longitude. In terms of latitude and longitude, each pair of the columns of each row indicated one crash record. Furthermore, we removed the missing values of "latitude" and "longitude," before plotting the coordinates into a map view. Because the main objective was to identify accident-prone places, and missing latitude and longitude values were not helpful to achieve that research goal. Following steps were taken during the data preprocessing:

- The "duplicated().values.any()" and "isna().values.any()" methods of Python programming language helped us to determine if we had any duplicates and missing values in our selected feature columns.
- We again used a method "dropna()" that helped us drop the missing values of our "latitude" and "longitude" columns.
- The duplicate values were not removed because multiple accidents may have occurred in the same location.

3.3 Data Analysis

Aggregate Accidents After the preprocessing, we ran a descriptive statistical analysis, and aggregated accidents by days, months, and time to find the number of crashes on each day of each month (Table 1).

DBSCAN Algorithm. We used DBSCAN to cluster the co-ordinate points on a leaflet map generated from Python's "Folium" library. With a given set of data points,

Table 1 Describes the number of accidents each day

Months	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Total cases
January	246	352	469	379	406	438	317	2607
February	259	327	351	371	374	402	362	2446
March	268	398	380	374	419	413	402	2654
April	257	367	386	401	391	425	405	2632
May	309	333	382	381	445	437	381	2668
June	287	352	345	328	368	416	373	2469
July	303	340	363	355	341	383	356	2441
August	342	378	346	382	382	495	402	2727
September	306	408	446	425	414	519	487	3005
October	340	507	534	578	506	599	511	3575
November	333	528	489	532	480	508	448	3318
December	276	452	445	478	503	566	444	3164

the DBSCAN clustering algorithm groups together the neighboring points within a given distance [5].

The *DBSCAN* is a density-based clustering algorithm [5]. It allows us to find k number of incident locations that are with the proximity of a given distance of ϵ. It also allows us to find scattered locations or outliers. Suppose there are n number of points (i.e., also referred as a location in this paper), such as $p_1, p_2, \cdot, p_n$. If there are at least k (i.e., the minimum number of points) number of points within a given distance of ϵ, then those points make a cluster together. Also, note that a point p is known to be a core point if and only if there are k points are within ϵ distance. Points that are not within ϵ distance and do not form a cluster with k number of points, then those points are called outliers.

To find if a point q is in the neighborhood of point p, a distance function $d(p, q)$ can be used, if the distance function returns a value less than or equal to the given distance ϵ for the point q, then p and q form a neighborhood which can be expressed as,

$$N_\epsilon(p): \{q|d(p, q) \leq \epsilon\} \tag{1}$$

In this study, we used the DBSCAN clustering algorithm to identify high accident-prone areas. In this study, ϵ was 0.05 (50 m) and k value was 300.

Heatmap Analysis. We wanted to make sure that our findings from the DBSCAN clustering algorithm were accurate, so, we tried to find the optimal number of clusters using the "Silhouette Score." The *Silhouette Score* helps to explore how many clusters are optimal for a dataset [11].

There are more methods of measuring the quality of a cluster, for example, Davies–Bouldin Index (DB), the Calinski–Harabasz Index (CH), and Dunn Index [2, 17]. However, the notion of a "Good Cluster" is relative. So, we have further tried to verify the quality of the clusters found in our analysis using a heatmap. The generated heatmap on the map provided us the number of accidents.

To generate the heatmap, we used the Gaussian kernel. The Gaussian kernel provides a better result in heatmap analysis [18], which can be expressed as,

$$e^{-\frac{(x-x_0)^2+(y-y_0)^2}{\alpha}} \tag{2}$$

where α represents influences of each data points on its surroundings, x_0 is the longitude and y_0 is the latitude of the current location, and x is the longitude and y is the latitude of a neighboring data point.

The generated heatmap complements our findings from the clusters (see Figs. 1 and 4). It also verifies that the quality measurement (Silhouette Score) of the clusters showing that the quality was good. The map gives us an idea about some of the densest areas of the accidents, as well as the numbers, represent the number of crashes happening around that area. Also, we compared our results and found that

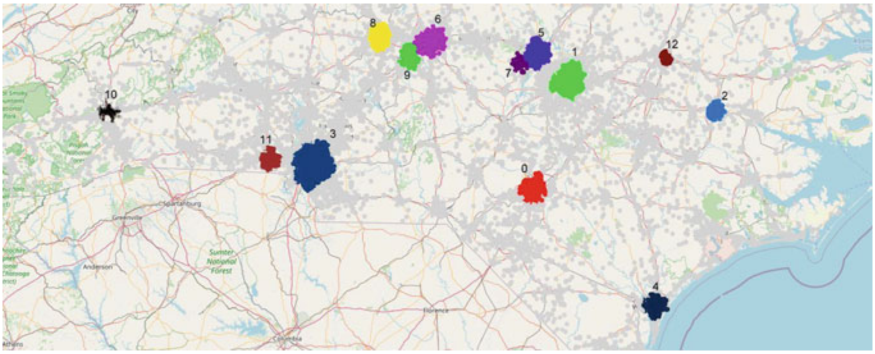

Fig. 1 DBSCAN results on the map to show accident-prone areas. There are 13 clusters where at least 300 accidents occurred

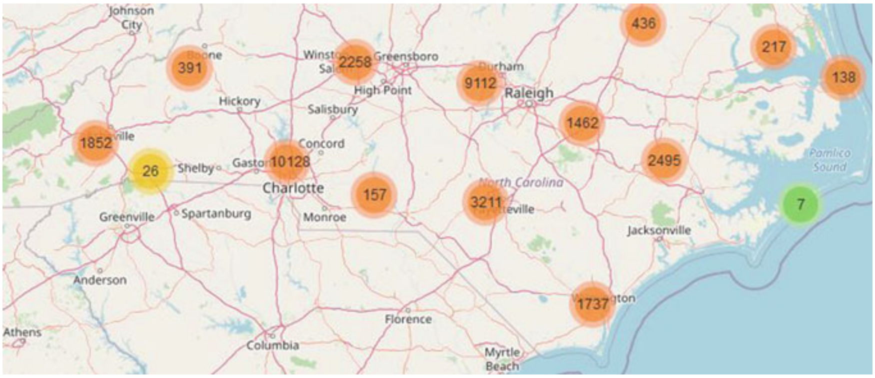

Fig. 2 "Marker Cluster" is showing us the number of accidents in our region of interest. We could see clusters nearly in similar areas of Fig. 1

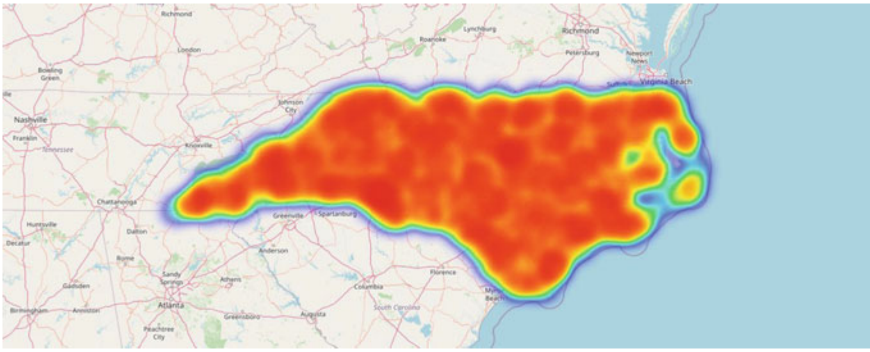

Fig. 3 Heatmap to show concentration of incidents in an area

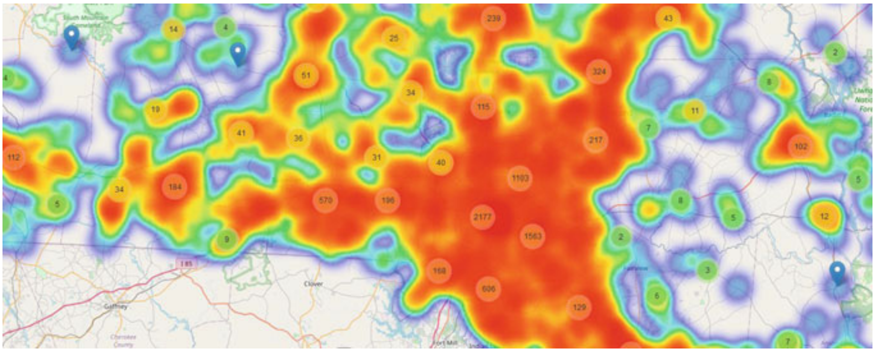

Fig. 4 Heatmap to show concentration of incidents in an area. This figure is a zoomed version of Fig. 3

the numbers of the accidents correspond to each of the clusters using marker cluster as well (see Figs. 2 and 4).

The marker cluster was generated using following equation,

$$(\sqrt{(x_0 \times C - x \times C)^2 + (y_0 \times C - y \times C)^2}) >> (Z_m - Z_c) \tag{3}$$

where C is a constant, x_0 is the longitude and y_0 is the latitude of the current location, and x is the longitude and y is the latitude of a neighboring data point. Z_m is the maximum zoom value (predefined to 22), and Z_c is the current zoom value.

Analysis of Visualization Techniques. Finally, we tried to extract some key points from the research data that can provide us a broad picture of the accidents (e.g., the months/days/time of the highest number of crashes or the road features of the crashes). We tried to represent the results by visualizing it with advanced, yet very simple to interpret visualization techniques. We took a survey of 50 participants who are not involved in the visualization development process and tried to find out the best visualization techniques among several graph/pie charts representations. We also tried to see if non-expert users could interpret the data from the figures (see Figs. 6, 7, and 8).

4 Results and Discussions

On the marker cluster map, the cluster with 10,128 accidents indicated a significantly higher number of accidents than other clusters (see Fig. 2). Similarly, our results indicated that another cluster with 9112 number of accidents. The marker cluster map presents all the accidents grouped with their locations and neighboring location as well. Figure 2 indicates that most of the accidents took place in the middle of the city. Perhaps this was the case because the population density in the middle of the city is comparatively higher than the rest of the places.

The Heatmap we generated was a two-layer visual map. The first layer was the "heat," and the second one was the marker. If we enable both layers, then we get a better picture of the accidents-prone areas (see Fig. 4). When we zoom in the map or select a specific cluster, then cluster areas with detailed information will appear. The numbers are broken down into smaller ones. Also, the heatmap view could help to understand which areas are more accident-prone than others. These visualization results could help non-expert users to decide in which areas they have to be more cautious or pay more attention.

4.1 Crash Analysis by Month

The most number of crashes have happened during the month, October (Fig. 5 (left)). Our results indicated that more than 3500 accidents occurred during this period. One of the plausible explanations for this case is that October lies in the middle of autumn in North Carolina [3]. The second most number of accidents has taken place in the month, November with 3318 cases. Then comes the month December with 3164 filed cases of accidents. The month where the least number of accidents have happened is July with 2441 cases. July is usually the warmest weather of the year in North Carolina and is the rainiest month in Chapel Hill. July month is in the middle of the summer season in North Carolina [3]. So, perhaps the presence of natural light helps to prevent accidents in July.

4.2 Crash Analysis by Time

The number of accidents happening on each hour could be explained by Fig. 6. The most number of accidents have taken place between hours 18 and 19. More than

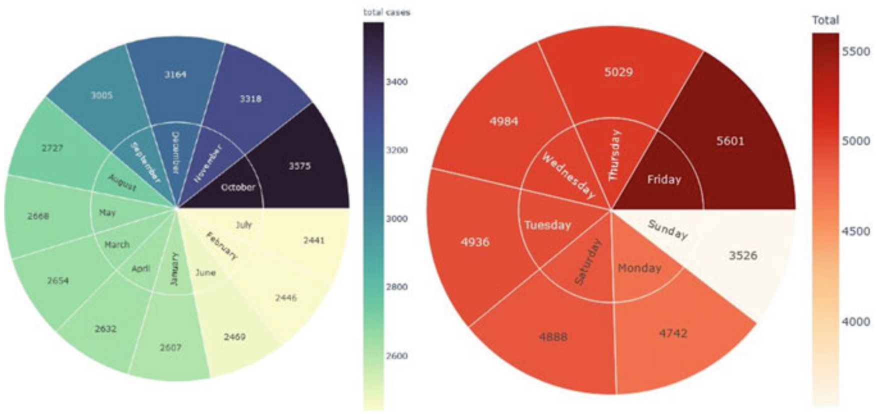

Fig. 5 (Left)Sunburst chart gives us a depiction of the month with the most number of accidents, October. (Right) Showing us the day with the most number of accidents. Friday has the most number of total cases with over 5600 crashes

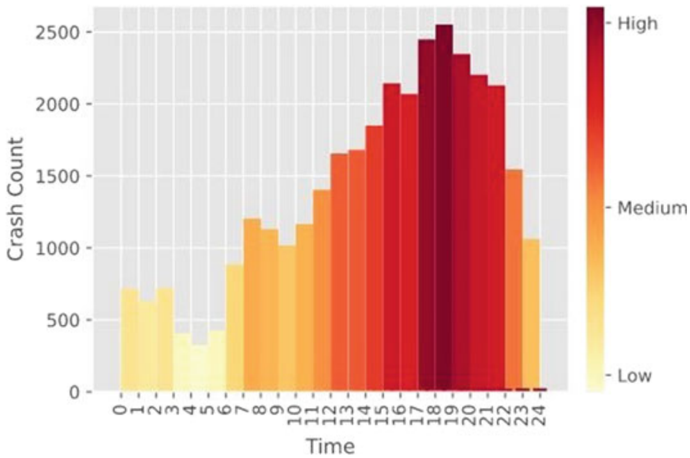

Fig. 6 Showing us accidents each hour. The most number of accidents have happened between hours 18 (i.e., 6 pm) and 19 (i.e., 7 pm)

2500 accidents happened during that time. The least number of accidents have taken place during the midnight and post-midnight hours, from the 0th hour to 6th. The higher number of accidents occurred starting from the 15th hour till the evening at 21st. The least number of accidents happened within the hour 4 and 5 (see Fig. 6). This could be because this is the time most people leave their workplace and rush to go home. As a result, most accidents occur during the hours of 18 and 19. Similar findings were reported by Amiruzzaman [1], where the author found most traffic accidents occurred in DC after office hours.

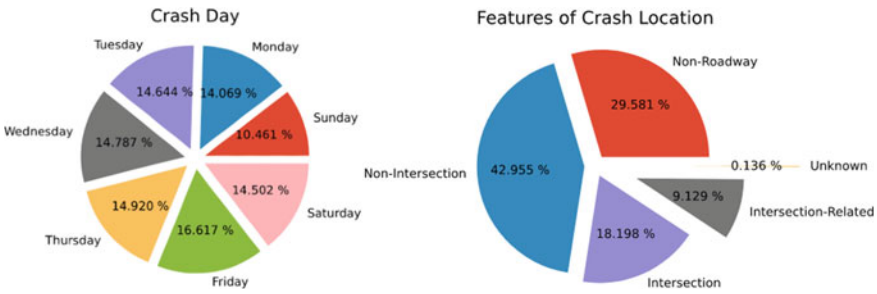

Fig. 7 (Left) Exploded pie chart is showing us the most number of accidents on each day of the week by percentage. (Right) The exploded pie chart is showing us the most number of accidents happening on the "non-intersection" roads with 42.955% of the total accidents

4.3 Crash Analysis by Day

The percentage of accidents happening on different days of the week are shown in Fig. 5 (right) and Fig. 7 (left). The evidence from the results which is presented in the exploded pie chart (Fig. 7 (left)) and the sunburst (Fig. 5 (right)) shows that most of the accidents have taken place on Fridays. This may be because Friday is the end of weekdays in North Carolina, and most people are eager to relax and pays less attention. Evidence suggested that more than 5600 crashes or 16.617% of the total crashes have happened during the Fridays. It is surprising that the most number of accidents happening on the last day of the working week. The day with the least number of accidents is Sunday with 10.461% of the total accidents. Similar findings were reported in a previous study [14], our findings confirm those findings.

4.4 Crash Location Features

It is an astounding fact that the highest number of accidents has emerged from the "non-intersections" (Figs. 7 (right), and 8). The number is seemingly higher than the other ones. More than 14,000 accidents took place at the non-intersection locations. The second place where the accident numbers are high is where the "non-roadways" are. Around 10 thousand accidents took place. The "intersection" location is where about 6 thousand accidents took place. It is a bit surprising fact too because usually, the intersections points are where the number of transports could be found even more.

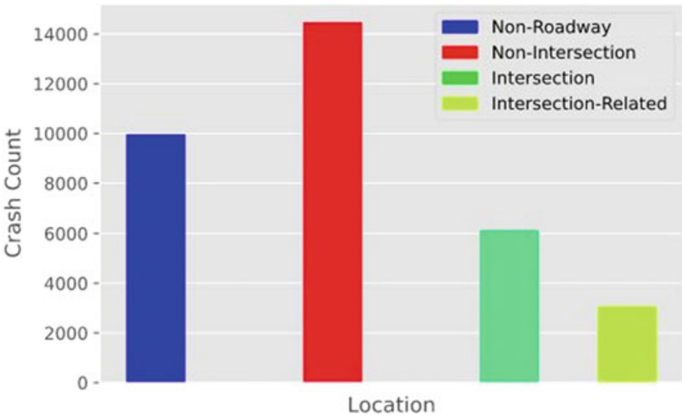

Fig. 8 Showing us the number of accidents happening on different roads. Non-intersections have more than 14,000 accidents. The intersection, in comparison, has the least number of accidents with a number of around 3000

5 Conclusion

Obtained results from data mining and data analysis suggested that there is a higher rate of accidents in the middle of the city. The most number of accidents have occurred in Charlotte. Also, the most dangerous time to go out is within the hour 18 and 19. The accident numbers are significantly higher than usual during the time between the afternoon and evening, from 3:00 pm to 8:00 pm. We also found that the most number of accidents happened in October, in the middle of autumn. And a higher percentage of accidents happened on Fridays.

The most number of accidents happened on the non-intersection roads and the least number of accidents happened on Intersections. It can be said that people tend to be less cautious about the environment if it is not an intersection and increase the chances of getting involved in crashes. However, this tendency needs to be changed. Overall, people should be more cautious if they are going out during these specific times.

Perhaps, analyzing more data from the updated database from the law enforcement agencies could help us to find more interesting information. Also, more data mining techniques could be used to predict the accident-prone regions. As for the future study, we can suggest that different supervised learning algorithms can be used to predict the accident-prone areas. Moreover, we can use deep learning techniques to classify areas based on other features as well.

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Chapter 13
A Novel Deep Convolutional Neural Network Model for Detection of Parkinson Disease by Analysing the Spiral Drawing

Md. Rakibul Islam, Abdul Matin, Md. Nahiduzzaman, Md. Saifullah Siddiquee, Fahim Md. Sifnatul Hasnain, S. M. Shovan, and Tonmoy Hasan

1 Introduction

PD is one of the top familiar neurological dysfunctions. The death of dopaminergic neurons in the substantia nigra of the brain in the human body causes PD. The numerous cell types all over the central and peripheral involuntary nervous systems are also responsible for PD. Most importantly, it is associated with movement disorder indications. It is a cumulative adult-onset disease and affects more than 2% of the population over the age of 65 [1]. Fifty to less than three hundred fifty new patients per million individuals are affected yearly around the globe by PD [2].

PD is associated with clinical features such as tremor, bradykinesia, hypokinesia, akinesia, and rigidity [3]. Bradykinesia and rigidity have a distinguishable change on the sketching and handwriting capabilities of PD patients. For this reason, micrographia has been utilized for diagnosis of PD (primary stage) [4]. Moreover, the utilization of sketching or handwriting for PD detection is the need for a suitable person to clarify the sketches or handwritings, particularly in the primary stages of the disorder. The kinematics of wave or spiral drawing specifies physiological limiting factors such as the magnitude of tremor and expanse of dyskinesia and bradykinesia [3]. It can be successfully differentiated between the drawings of PD patient and unaffected person. And finally, we have come to the point that PD can

Md. R. Islam (✉) · A. Matin · Md. Nahiduzzaman · Md. S. Siddiquee · F. Md. S. Hasnain · T. Hasan
Department of Electrical and Computer Engineering, Rajshahi University of Engineering and Technology, Rajshahi, Bangladesh

S. M. Shovan
Department of Computer Science and Engineering, Rajshahi University of Engineering and Technology, Rajshahi, Bangladesh

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be detected by analysing the spiral and wave sketching image of PD subjects and healthy subjects.

Our motivation behind this paper is to establish a reliable deep ConvNet-based classification model by extracting the features of the spiral drawing of healthy subjects and PD patients for detecting PD at the early stages.

2 Literature Review

The prime challenges of designing a ConvNet are associated with the performance metrics of the model. Types of data also crucial in the field of PD detection at an early stage. In this segment, we have focused on the leading researches which have been carried out by many eminent researchers in this regard.

- Canturk et al. [5] focused on voice signals for PD detection, and they proposed a reliable and inclusive machine learning method for their research purpose.
- Kotsavasiloglou et al. [6] introduced an inquiry of machine learning-based automated system for PD detection from the simple drawing of healthy persons and Parkinson's affected persons.
- Drotár et al. [7] used SVM classifier for classification purpose on PD sketching dataset by analysing the kinematic and pressure attributes.
- Hariharan et al. [8] also focused on Parkinson-related voice dataset and imposed a composite automated system with the collaboration of depletion and selection feature techniques.
- Moetesum et al. [9] proposed a model with visual attributes instead of kinematic features of spiral drawings for detection Parkinson's at an early stage.

Here, our intention was going through this literature review, the spiral drawing, and wave drawing image processing-based deep ConvNet classification model design for PD prediction at an early stage.

3 Convolutional Neural Networks

Nowadays, ConvNet has become most famous for solving image-related data challenges. Computer vision is a renowned field of artificial intelligence (AI) where a deep learning method became popular when ConvNet appeared as a classification model for better results in the ImageNet competition in 2012 [10]. ConvNet is biologically persuaded as an alternative form of multilayer perceptron. ConvNet learns attributes directly from the specified input data by circumventing the hand-crafted feature [10], and it is very advantageous in the field of image processing. The most beneficial thing is no requirement of any kernel which is designed manually for feature extraction of images. ConvNets have mainly convolutional, pooling, and fully connected (FC) layer; after getting the input image, feature maps are created by the first convolutional layer (CL), then feature's maps size will become smaller in pooling layer, afterwards for image classification. Finally, the FC layer is responsible, and it has seemed that the image size is reducing in every layer for the final classification.

3.1 Convolutional Layer

Convolutions are used in ConvNet because it is dominating for parameters sharing as well as for sparsity of connections in image classifications [11]. CL is a group of feature extraction filters, and those filters are used to slip across the whole spiral drawing images in our proposed model. The convolution operation is performed all over the input image by the filters and CL maps a specific matrix. The calculation in CL is represented in Eq. (1)

$$y = W^T x + b \tag{1}$$

where y, x, w, b correspond to the output of CL, input image, filter weights, bias, respectively.

3.2 Pooling Layer

Insertion of a pooling layer (PL) in the middle of two consecutive convolution layers is a recognized procedure for designing deep ConvNet. It is used to avoid overfitting problems by diminishing the depth of the feature maps while training is executed for the model. In PL, a window function is used in the input patch, and it determines the maximal value in the neighbourhood. For minimizing the intricacy of the image processing, the size of the feature map is decreased, and that is the main purpose of using PL. We have used max pooling or average pooling in ConvNet model. But when we have concern about real-life applications, then max pooling is our best option [12].

3.3 Fully Connected Layer

The FC layer has been evinced very strongly in recognizing and classifying the image-related problems for computer vision technology. The input to the FC layer is the output from the endmost pooling layer. It flattened the matrix into a vector and nourished it into the FC layer. The final FC layer of any model generates the final output with an activation function which drives them to take the final decision for classifications.

3.4 Activation Function

Activation function (AF) is a mathematical approach for determining the output of a deep neural network. By calculating the weighted sum of a network, AF decides the neuron should be activated or not. It also assists in normalizing the output of neurons and also computationally efficient. Mainly, ReLU is applied in most real-life binary classification problems as it is efficient and faster than most other AF [12] and Eq. (2) expresses the ReLU function. We have used sigmoid AF for final fully connected layer, and the calculation is represented in Eq. (3).

$$y = \max(0, x) \tag{2}$$

where x corresponds to the input of ReLU.

$$S(x) = \frac{1}{1 + e^{-x}} \tag{3}$$

where x corresponds to the output of the FC layer.

3.5 Dropout

Dropout is a familiar and powerful regularization procedure for deep neural networks. It is used for removing the overfitting complexity of a deep learning model. Traditionally, dropout is used on the fully connected layers, but it is also acceptable to initiate after the max-pooling layers by generating noise augmentation of image. Dropout arbitrarily zeroes some of the portions of neurons. As a result, in that iteration, the network uses a subset of the whole model.

4 Dataset Collection

The data is obtained from Kaggle's dataset, which was introduced by Zham et al. [13] and some self-labelled public spiral drawing data of Parkinson's subjects and healthy subjects in Bangladesh. The dataset have consisted of two classes of images and they were Parkinson's affected and not affected spiral drawings. Our collected dataset have consisted of 660 images in total, and we have selected 459 images randomly for training, 82 images for validation, and 119 images for testing. For training purpose, the healthy subjects and Parkinson's subjects have been kept almost balanced. In this study, whole training data, validation data, and testing data have been chosen randomly using machine learning library (Scikit-learn). And the whole data distribution is presented briefly in Fig. 1.

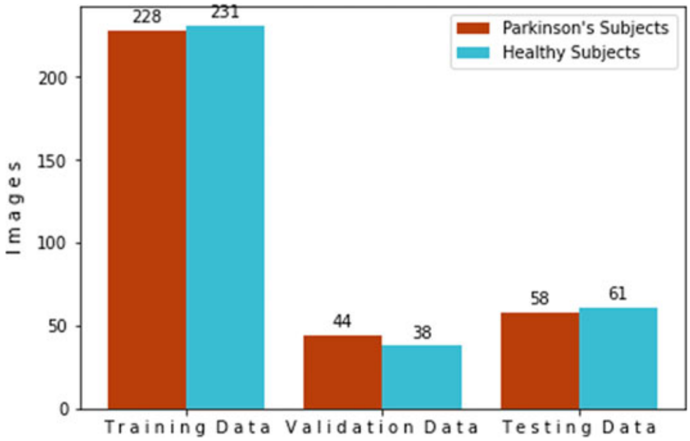

Fig. 1 Data distribution

Table 1 Image data augmentation

Serial numbers	Parameters	State of parameters
1	Range of rotation	10
2	Range of zooming	0.2
3	Range of width Shifting	0.1
4	Range of height shifting	0.1
5	Range of shearing	0.1

5 Data Processing

5.1 Data Augmentation

Image number in our collected dataset for training purposes is not sufficient for a deep ConvNet, and that is why we have to find a way to avoid overfitting problems in our proposed classification model. For this reason, the size of our collected dataset have been expanded artificially employing image augmentation. And finally, the training set have become larger without getting any new images. The image augmentation parameters for our model are given in Table 1.

5.2 Data Preparation

Our studies have lead us to classify the spiral drawing between the drawing of healthy subjects and PD subjects. Here, our main concern was the sketch of an image, and we have designed the model for a grey-scale image, and we have re-scaled the image

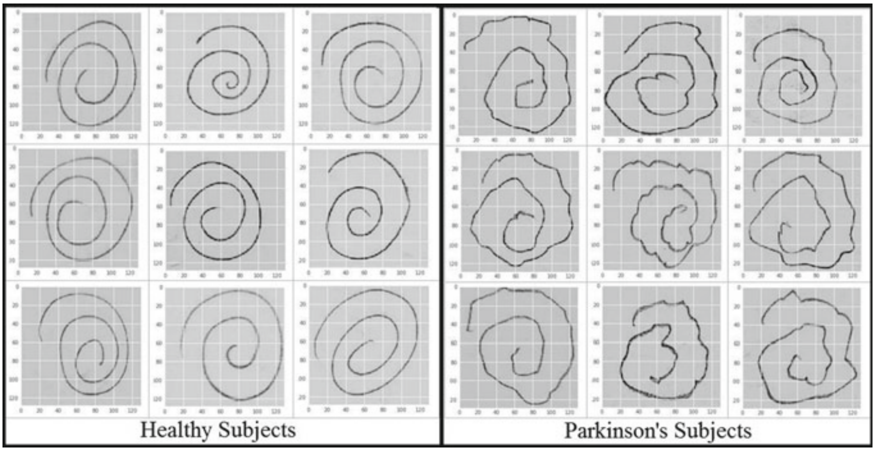

Fig. 2 Spiral drawings

pixel value from 0 to 1. Another question arises, as the size of every image is not the same, then a large image takes more time for the training process of a deep learning model. And that is why after image data augmentation and re-scaling images, we have set the input image shape to 130 × 130 pixels (Fig. 2).

6 Model Architecture

The ConvNet architecture is accountable for features generation, which is represented by the image, and furthermost that features are passed through the FC dense layer for solving the classification problems of the precise type of images. In our study, the model has comprised four CLs, and a max-pooling layer have also introduced after two successive CL. The number of kernels at the first and second CL was 64, and the size of each kernel was (5,5), respectively, the third and fifth convolution layer has contained 32 kernels, and the size of each kernel is (3,3). AF ReLU nonlinearity has been used in each and every CL. The pool size of the max-pooling layer after the second and fourth convolution layer was (2,2). A dense hidden layer has also been inserted before the final FC layer, and ReLU nonlinearity has also been used here as an activation function. Afterwards, the latest and the last FC layer incorporate with one neuron, and the sigmoid AF has also been used for generating the expected class score. Not only in the CL but also in the FC layer, a dropout with a 0.5 probability has also been used. For compilation purposes, Adam optimizer with standard parameters (learning rate = 0.001), which is the most preferable for binary classification, has been used. We have trained the model, and the batch size was 64 for training purposes. The training set have also inflicted by the image data augmentation with 10-degree random rotation, shearing, shifting, and zooming. And that is why the

Table 2 Model summary

Layer	Output shape	Parameter No.
Input	(130,130,1)	0
Convolution1	(126,126,64)	1664
Convolution2	(122,122,64)	102,464
Max Pooling1	(61,61,64)	0
Convolution3	(59,59,32)	18,464
Convolution4	(57,57,32)	9248
Max Pooling2	(28,28,32)	0
Dropout1(0.5)	(28,28,32)	0
Fully Connected	(512,1)	12,845,568
Dropout1(0.5)	(512,1)	0
Sigmoid	(2)	1026

model have overcome the overfitting problem and become a genuinely robust and generalized PD prediction model.

In our proposed model, the input image shape was 130 × 130 × 1 (grayscale), and the output image shape after the first and second CL was, respectively, 126 × 126 × 64 and 122 × 122 × 64. Afterwards, the first max-pooling layer the image shape became 61 × 61 × 64, and the rest of the output image shapes of our proposed deep ConvNet model are given in Table 2. The entire trainable parameters in our proposed model were 12,978,434.

7 Experimental Results

The experimental results of this PD prediction model are based on the validation accuracy, test accuracy, validation loss, and f1 score [12]. The accuracy and the f1 score can be described by Eqs. (4) and (5), respectively.

$$\text{Accuracy} = \frac{\text{tp} + \text{tn}}{\text{tp} + \text{tn} + \text{fp} + \text{fn}} \tag{4}$$

$$\text{f1 score} = 2 * \frac{\text{P} * \text{R}}{\text{P} + \text{R}} \tag{5}$$

where tp = true positive, tn = true negative, fp = false positive, fn = false negative, P = precision, R = recall.

The training accuracy and validation accuracy are presented in Fig. 3, and the loss (training and validation) of our PD prediction model are also shown in Fig. 4. Area under the ROC curve is also presented in Fig. 5. Most precisely, the confusion matrix is even presented in Fig. 6. By using Eqs. (4) and (5), we can calculate our model accuracy and f1 score, and our model has reached 96.64% accuracy and 96.55% f1

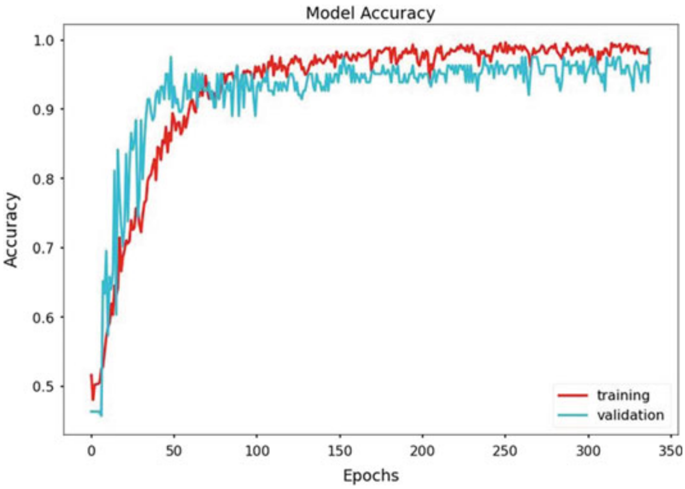

Fig. 3 Accuracy

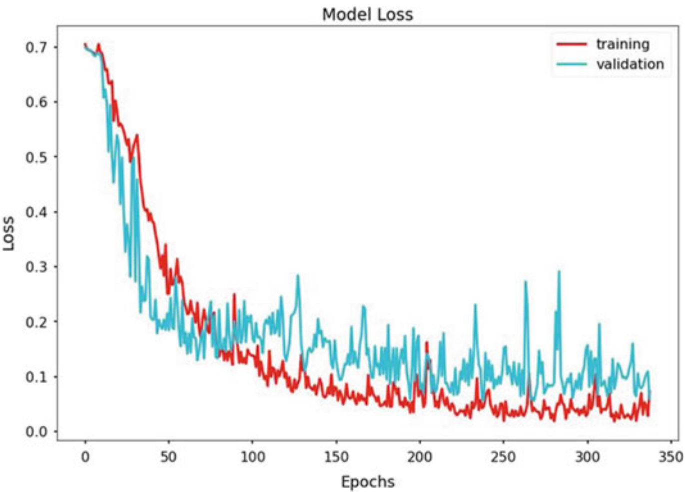

Fig. 4 Loss

score. In contrast with [14–16], a remarkable improvement is found in test accuracy as well as in f1 score and other performance matrix calculation in our imposed model. Researchers assessed an AlexNet-based transfer learning model [14] with 80% of training data (MR image) and 20% of test data (MR image) and achieved an accuracy of 88.9%, AUC of ROC curve was 0.962. Then, another research group imposed a simple ConvNet (two convolution layers, two fully connected layers) [15] with 88% accuracy on a small dataset (Parkinson's drawings). So, it is clear that the performance matrix of our designed model is efficient enough. A more recent work

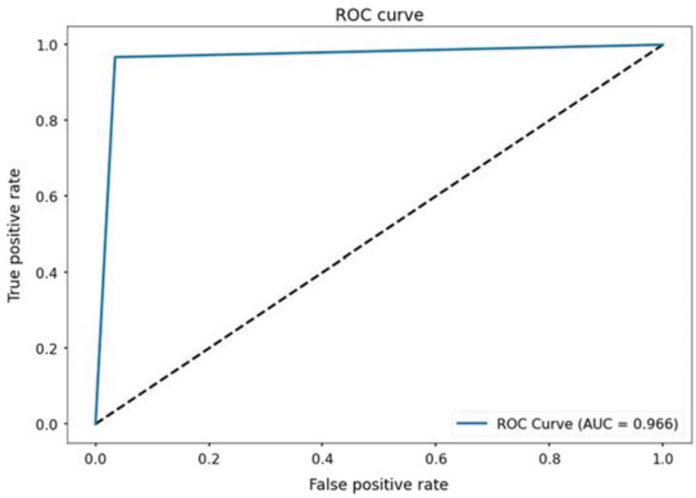

Fig. 5 AUC of ROC curve

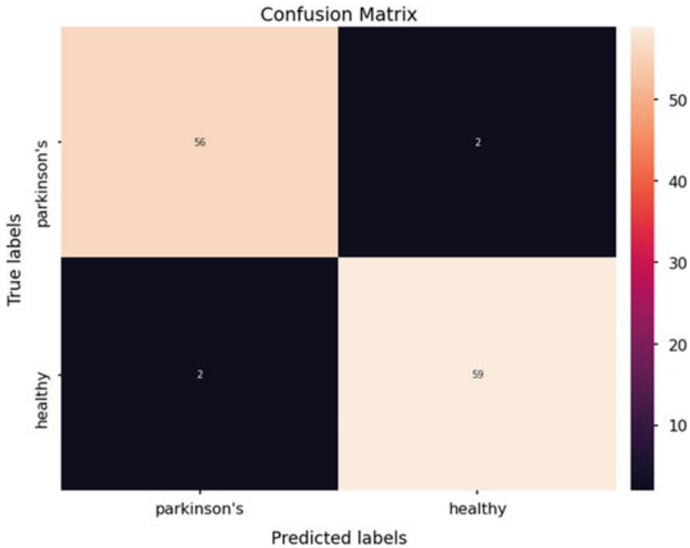

Fig. 6 Confusion matrix

[16] had a model accuracy of 93.3%, and an adequate limit of 3.34% even overcame that by our proposed model.

8 Conclusion

The PD detection deep ConvNet model, which has been proposed here is quite decent in classification between the spiral drawing of Parkinson's affected subjects and healthy subjects. A great question and difficulties will arise when the dataset is extended reasonably. Afterwards, we have to concern about improving the performance of the proposed model with big data. Further, different types of drawings or sketches can be introduced to design a robust and efficient model for PD prediction as to future work. Besides, we have planned to introduce different deep ConvNet architecture for the large dataset as well as transfer learning and so on.

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Chapter 14
Fake Hilsa Fish Detection Using Machine Vision

Mirajul Islam, Jannatul Ferdous Ani, Abdur Rahman, and Zakia Zaman

1 Introduction

Hilsa fish is known as one of the tastiest and favorite fish in the world. It is named as Ilish in Bangladesh. Every year Bangladesh produces 75% of the total Hilsa fish production in the world [1]. And the other 25% produced in India, Myanmar, Pakistan, and some countries along the Arabian sea. Three kinds of Hilsa fishes are available in Bangladesh. These are Tenualosa Ilisha which is called Padma Ilish and it is the most popular Hilsa fish in the world. Another is Nenuacosa Toli, which is called Chandana Ilish. And the last one is Hilsha Kelle that is known as Gurta Ilish. Hilsa is known as saltwater fish, but it lays eggs in freshwater rivers (Padma, Meghna, Jamuna) delta at the Bay of Bengal. The demand for Hilsa fish in Asian countries exists throughout the year. However, it increases a lot especially in festivals like Pohela Boishakh (Bengali new year), Saraswati Puja, etc.

In many countries, the production of Hilsa fish is now declining. But its production in Bangladesh is increasing every year. By exporting these fishes in many countries of the world, a large amount of foreign currency is earned every year. It adds 1% of the total GDP in Bangladesh [1]. For several years, in many countries, including Bangladesh, India, some unscrupulous traders are selling fake Hilsa fish. With a large portion of original Hilsa fish, they added part of fake Hilsa fish inside it then exported

M. Islam (✉) · J. F. Ani · A. Rahman · Z. Zaman
Department of Computer Science and Engineering, Daffodil International University, Dhaka 1027, Bangladesh
e-mail: merajul15-9627@diu.edu.bd

J. F. Ani
e-mail: jannatul15-10483@diu.edu.bd

A. Rahman
e-mail: abdur15-10022@diu.edu.bd

Z. Zaman
e-mail: zakia.cse@diu.edu.bd

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to many countries. This tarnishes the image of our country and in return lowering the revenues.

These fake Hilsa look a lot like the original one, but there are some differences in shape, mostly in the head and tail, and also in the taste and smell. The Sardine, Sardinella, Cokkash, Chapila, Indian oil sardine, and all of these look like original Hilsa fish. The body of the Hilsa fish is equally long toward the abdomen and back, but the abdomen is higher than the back in Sardine. The head of the Hilsa fish is slightly longer, on the other hand, the head of the sardine fish is slightly smaller and the front side is slightly blunt. Fake Hilsa's eyes are bigger than the real Hilsa fish. The back of the original Hilsa is bluish green. Sardine has black dots at the base of the dorsal fin, but Hilsa fish has black dots on its gill. The front of the sardine's back fins and the edges of the tail fins are blurred but the hind fins of Hilsa are whitey.

But the sad truth is, fake Hilsa cannot be easily detected by looking at it many times. To solve this problem, we use machine vision-based deep learning models to identify the original fish. Our proposed method can easily classify Hilsa fish and fake Hilsa fish with high accuracy. We used Xception, VGG16, InceptionV3, NASNetMobile, and DenseNet201 to find out which model gives better performance.

The remainder of this article is organized as follows: Sect. 2 describes some related work. Section 3 describes the research methodology and a very brief study on the used models. Section 4 shows the result analysis. And Sect. 5 concludes the study with future direction.

2 Background Study

We studied some notable research work on fish classification and also on image processing. Pavla et al. [2] used a modified Rosenblatt algorithm to classify six species of fish. They have used 2132 silhouettes images. Of these, 1406 images were used for model training, and 726 used for model testing. Out of 726 images, their modified model was able to classify 386 images correctly. Israt et al. [3] used SVM, KNN, and ensemble-based algorithm to recognize six types of local fish. They used a histogram-based method to segment the gray-scale fish images. In their research, SVM achieved the highest accuracy of 94.2%. In the paper [4], the authors used pre-trained VGG16 for feature extraction of eight species of fish images and logistic regression has been used for the classification. Then, they achieved 93.8% accuracy. Rowell et al. [5] used the InceptionV3 model to classify Nile tilapia is harvested alive or Hibay. They used Adam optimizer in the last two layers of the model for increasing the accuracy rate. After 1000 iterations, their model achieved the best accuracy. Hafiz et al. [6] used 915 fish images from six different classes. Six different types of CNN model have been used in this research, namely VGG-16 for transfer learning, one-block VGG, two-block VGG, three-block VGG, LeNet 5, AlexNet, GoogleNet, ResNet 50, and a proposed 32 Layer CNN with 404.4 million parameters, which gives the best accuracy than the rest. Xiang et al. [7] used transferred DenseNet201 for the diagnosis of breast abnormality. Here, 114 abnormal tissue images have been used for classification. Then, their model gave 92.73% accuracy. Jing et al. [8] used a multiclass support vector machine (MSVM) algorithm to classify six species of freshwater fish. They have captured all the images in 1024 × 768 size. Then, multiple features have been extracted from these images. Ogunlan et al. [9] used a support vector machine (SVM)-based technique for classifying 150 fish. Of these, 76 have been used for model training and 74 for testing the model. They have used three more models in their research. Those are artificial neural networks (ANNs), K-means clustering, and K-nearest neighbors (KNN). And principal component analysis (PCA) is used to reduce the dimensionality of a dataset. SVM achieved the highest accuracy of 74.32%. Mohamad et al. [10] used the SVM, KNN, and ANN models to classify Nile tilapia fish. They used the scale invariant feature transform (SIFT) and speeded up robust features (SURF) algorithms for feature extraction. SURF achieved the highest accuracy of 94.44%. However, no prior work has been done on identifying fake Hilsa fish, and the image dataset used in this study is also not available before our research.

3 Research Methodology

Figure 1 shows the working strides of our study. In brief, after assembling the datasets, it is labeled, pre-processed, and divided into the train and validation set. Using the datasets, five different CNN models are trained and validated to evaluate model performance. Their performance has been visualized through the confusion matrix and several evaluation metrics such as accuracy, precision, recall, and F1 score are used to compare among different models.

3.1 Image Data Collection and Pre-process

There have been 16,622 different sizes of images that have been categorized into two groups named real Hilsa and fake Hilsa. Those different sizes of images are taken

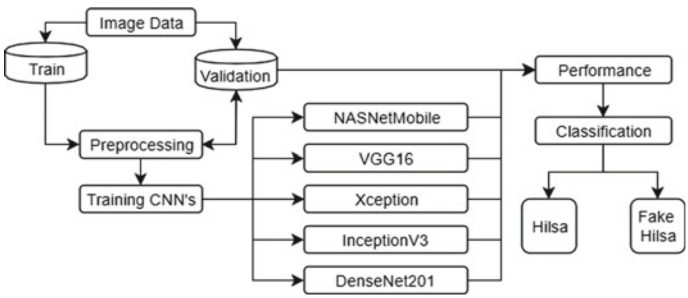


Fig. 1 Proposed classification procedure

from the endemic fish market and many alternative sources. There are three different types of Hilsa fish in the 8722 images: Tenualosa Ilisha (Padma Ilish), Nenuacosa Toli (Chandana Ilish), and Hilsha Kelle (Gurta Ilish). The other 7900 images are of different types of fake Hilsa fish. Mostly Sardine, Sardinella, Indian oil sardine, Cokkash, Chapila, and few other fishes that look similar to Hilsa. All 16,622 images were resized into $224 \times 224 \times 3$ to train the model. All the collected images are divided into training and validation dataset. 80% (13,301) images are in the training set and another 20% (3321) images are in the validation (Fig. 3). We have named the test set as a validation set. And labeled our dataset into two classes, namely Class 0 (fake Hilsa fish) and Class 1 (Hilsa fish) are shown in Fig. 2.

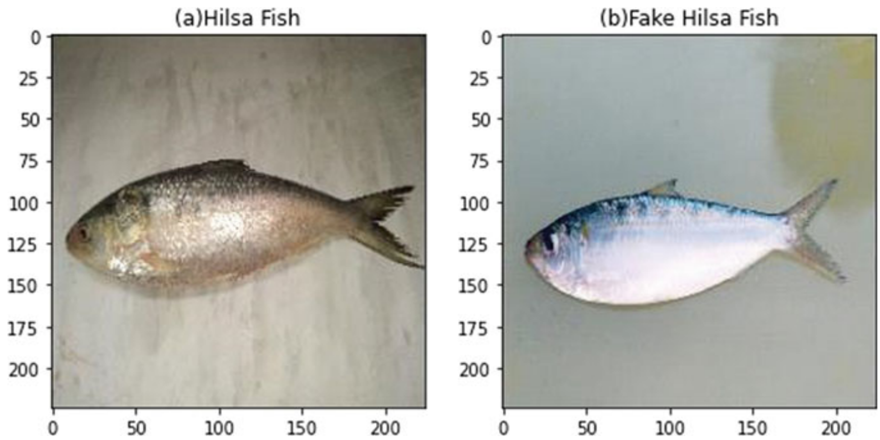


Fig. 2 Images of two classes **a** Hilsa fish, **b** fake Hilsa fish

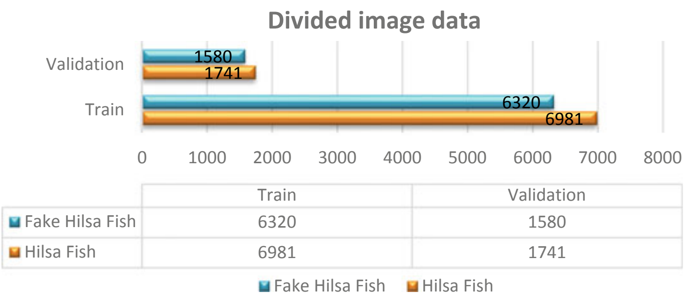


Fig. 3 Image dataset is divided into training and validation sets

3.2 Model Generation

We use five different types of CNN models to identify the original Hilsa fish and the fake one. The characteristics of these CNN models [11] are shown in Table 1. A brief discussion of these models is given below.

Convolution neural network (CNN) is a type of deep neural network (DNN) which applies for image processing, natural language processing, and many other classifications, recognition, and detection tasks. It includes an input layer, some hidden layers known as convolution layers, pooling layers, flattening layers, and fully connected layers. The convolution layer is the most significant layer for extracting features, edges, colors, shape, pattern, etc., from the images. The output of this layer is the feature map. Pooling layers reduce the size of an image. Mainly, three types of pooling layers are used: max, average, and sum. It removes 3/4% of the activation, seen in the previous layers. Fully connected layers (FC) are called dense layers [12]. Fully connected layers mean all nodes in one layer connected to the output of the next layers.

Xception [13] was invented by Google researchers. It is an architecture based on depthwise separable convolution layers. It has 36 convolution layers into 14 modules. Parameters are quite similar to InceptionV3. It has been trained with millions of images from the ImageNet database [14].

VGG16 [15] is sometimes called OxfordNet because it was invented by a visual geometry group from Oxford in 2014. It was first used to win the ILSVRC (ImageNet [14]) competition. The input shape is fixed for the conv1 layer ($224 \times 224 \times 3$). Although its size is quite larger than the rest, it is very useful for learning purposes and easy to implement.

InceptionV3 [16, 17] was invented in 2015 by Google Inc. It has a total of 48 deep layers. It has been trained with millions of images from the ImageNet database [14]. InceptionV3 is the most used convolution neural network model for image recognition.

NASNetMobile [18] is another type of convolution neural network (CNN) that is divided into two cells: a normal cell that returns a feature map in the same aspect, and a reduction cell which reduces the height and width of a feature map. It has been trained with millions of images from the ImageNet database [14]. It has three input channels, height, width, and RGB color channel. It requires $224 \times 224 \times 3$ input shape for images.

Table 1 Model characteristics in details

Model	Size (MB)	Parameters	Depth
Xception	88	22,910,480	126
VGG16	528	138,357,544	23
InceptionV3	92	23,851,784	159
DenseNet201	80	20,242,984	201
NASNetMobile	23	5,326,716	–

DenseNet201 [19] has a total of 201 deep layers. We achieved the highest accuracy by using this model. The input shape is 224 × 224 × 3 for "channel-last" data format and 3 × 224 × 224 is for "channel-first" format. The summary of DenseNet201 CNN model is too large which cannot be fully disclosed in this article. That is why we have visualized here some beginning layers (Fig. 4a) and some ending layers (Fig. 4b).

To train those models, we require to resize the images as 224 × 224 × 3 and also rescale them into 1/255 pixel values. Because customarily the pristine pixel values of the images are integers with RGB coefficients between 0 and 255. This slows down the learning process because the range of integer values is too large. For this, we need to normalize our pixel value between 0 and 1. This method is known as min–max normalization [20]. All the models are trained and the results are visualized using Scikit-learn, Keras, OpenCV, Matplotlib, and TensorFlow as backends. Google Colab [21] is used to execute all the processes and experiments.

```
Model: "model_1"
__________________________________________________________________________________________________
Layer (type)                    Output Shape         Param #     Connected to
==================================================================================================
input_1 (InputLayer)            (None, 224, 224, 3)  0
__________________________________________________________________________________________________
zero_padding2d_1 (ZeroPadding2D (None, 230, 230, 3)  0           input_1[0][0]
__________________________________________________________________________________________________
conv1/conv (Conv2D)             (None, 112, 112, 64) 9408        zero_padding2d_1[0][0]
__________________________________________________________________________________________________
conv1/bn (BatchNormalization)   (None, 112, 112, 64) 256         conv1/conv[0][0]
__________________________________________________________________________________________________
conv1/relu (Activation)         (None, 112, 112, 64) 0           conv1/bn[0][0]
__________________________________________________________________________________________________
zero_padding2d_2 (ZeroPadding2D (None, 114, 114, 64) 0           conv1/relu[0][0]
__________________________________________________________________________________________________
pool1 (MaxPooling2D)            (None, 56, 56, 64)   0           zero_padding2d_2[0][0]
__________________________________________________________________________________________________
conv2_block1_0_bn (BatchNormali (None, 56, 56, 64)   256         pool1[0][0]
__________________________________________________________________________________________________
conv2_block1_0_relu (Activation (None, 56, 56, 64)   0           conv2_block1_0_bn[0][0]
__________________________________________________________________________________________________
```

(a) Beginning Layer

```
__________________________________________________________________________________________________
batch_normalization_1 (BatchNor (None, 1024)         4096        dense_1[0][0]
__________________________________________________________________________________________________
activation_1 (Activation)       (None, 1024)         0           batch_normalization_1[0][0]
__________________________________________________________________________________________________
dropout_1 (Dropout)             (None, 1024)         0           activation_1[0][0]
__________________________________________________________________________________________________
dense_2 (Dense)                 (None, 1024)         1049600     dropout_1[0][0]
__________________________________________________________________________________________________
batch_normalization_2 (BatchNor (None, 1024)         4096        dense_2[0][0]
__________________________________________________________________________________________________
activation_2 (Activation)       (None, 1024)         0           batch_normalization_2[0][0]
__________________________________________________________________________________________________
dropout_2 (Dropout)             (None, 1024)         0           activation_2[0][0]
__________________________________________________________________________________________________
dense_3 (Dense)                 (None, 2)            2050        dropout_2[0][0]
==================================================================================================
Total params: 21,348,930
Trainable params: 21,115,778
Non-trainable params: 233,152
__________________________________________________________________________________________________
```

(b) End Layer

Fig. 4 Model summary of DenseNet201

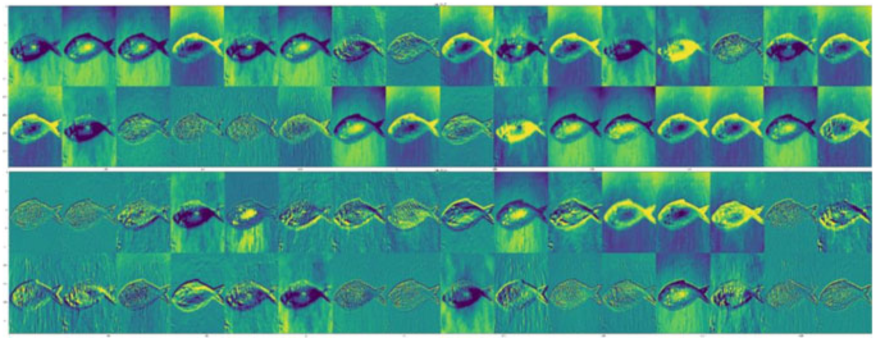


Fig. 5 Visualize the features extraction from the first two conv2d (DenseNet201)

3.3 Features Extraction

There are various convolution layers in each CNN model. Convolution layers that process the two-dimensional images are known as conv2d layers. Each conv2d gets input images, having three color channels. Then, this input image is processed through a convolution filter. This filter is known as a convolution kernel (extract features of the input image) or a feature detector. Each input image is arranged in a matrix which consists of combining the value of three color channels. The matrix tends to get smaller for each filter to read the features of the input image, therefore, there are many differences to see the same image in each filter. High-level filters read the special pattern of input images and low-level filters read the normal features of the input image [22]. Each filter normally carries pre-defined weights. The term feature extraction means the reduction of features and RGB from the raw input image in every conv2d layer. Each conv2d has some arguments [23] or parameters for image processing, those are filter, kernel-size, strides, padding, data format, dilation-rate, groups, activation, use bias, and a few more. These arguments are different in each CNN model which we have used in our research to identify Hilsa fishes.

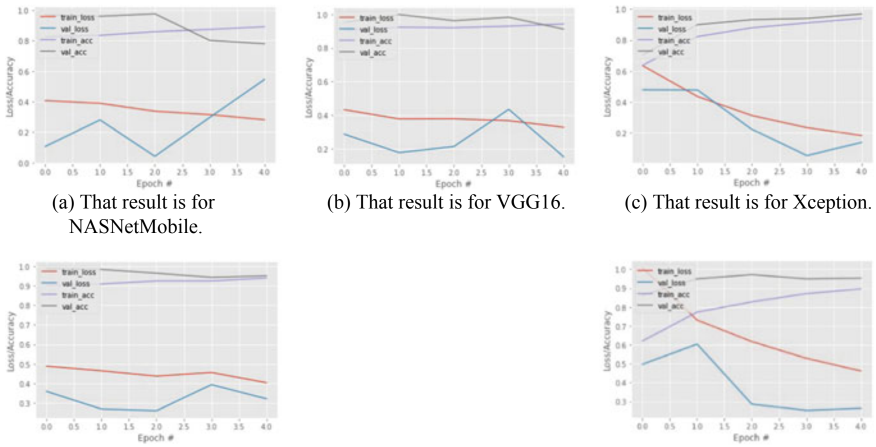
In our research, 32 feature maps are extracted by each conv2d layer. We can see from Fig. 5, many changed images of the Hilsa fish after applying filters in each conv2d. Among these filters, some have highlighted the edges, some have highlighted the shape, and some have highlighted the background and heat map. CNN reads the images in each conv2d layer and learns the features of every input image for doing better classification.

4 Result Analysis

This section discusses the experimental results of the five CNN models. We train those models with 6981 original Hilsa and 6320 fake Hilsa fish images. Models are tested with 1741 Hilsa fish and 1580 fake Hilsa images to evaluate performance. After that NASNetMobile gave the lowest accuracy of 86.75% and DenseNet201 gave the highest accuracy of 97.02%.

As shown in Fig. 6, it shows the training accuracy and loss as well as the validation accuracy and loss in each epoch of these five models. In Fig. 6a, training accuracy in the last epoch 89.1% on the other side last epoch validation accuracy 78.02% which is less than the first epoch accuracy 96.98%. The training loss at the last epoch is the lowest at 28.12% and the validation loss is 54.56%. The obtained result of this graph is for the NASNetMobile model. In Fig. 6b, train accuracy after all epoch is 94.25%, and validation accuracy is 91.19%. At the same time, training and validation loss in the last epoch, respectively, 32.79% and 15.20%. That result is for VGG16. In Fig. 6c, 93.95% training accuracy, 96.77% validation accuracy got from the last epoch and 18.24% training loss, 13.82% validation loss got from the last epoch for the Xception model. In Fig. 6d, got 93.88% training and 95.01% validation accuracy in the last epoch. Also, got 40.42% training and 32.21% validation loss for the InceptionV3 model. From Fig. 6e, 89.62% of training accuracy and 95.36% validation accuracy got from the last epoch. With this, 46.16% training and 26.36% validation loss are obtained from the last epoch. That results are for the DenseNet201 model. Considering the average results of these five graphs, Fig. 6e is better than the others.

Table 2 lists the results of the confusion matrix for the five CNN models. A confusion matrix is the measure and visualized the total right and wrong predictions made by the classifiers. When the model predicts it is a Hilsa fish and the actual output is also Hilsa fish, then it is a true positive (TP). When the model predicts it is a fake Hilsa fish and the actual output is also fake Hilsa fish, then that term is true negative (TN). When the classifier predicts it is a Hilsa fish but the actual output is



(a) That result is for NASNetMobile.

(b) That result is for VGG16.

(c) That result is for Xception.

(d) That result is for InceptionV3.

(e) That result is for DenseNet201.

Fig. 6 Classification performance

Table 2 Confusion matrix result for five CNN models

Model	True positive (TP)	True negative (TN)	False positive(FP)	False negative (FN)	Accuracy (%)
NASNetMobile	1331	1550	30	410	86.75
VGG16	1568	1578	6	173	94.61
Xception	1666	1542	38	75	96.60
InceptionV3	1652	1565	15	89	96.87
DenseNet201	1659	1563	17	82	97.02

fake Hilsa fish, then it is a false positive (FP). And when the classifier predicts it as a fake Hilsa fish and the actual output is also a Hilsa fish, then it is a false negative (FN).

According to Table 3, we found in NasNetMobile that the difference between the false positive rate and the false negative rate is high. DenseNet201 shows a significant improvement because the difference between FP and TP rate is low rather than the NasNetMobile. The accuracy measure of Xception, Inceptionv3, and DenseNet201 are very close to each other. The accuracy of Xception is 96.60% and InceptionV3 is 96.87%. Here, the accuracy measured difference between them is very low which is 0.27%. The accuracy of DenseNet201 is 97.02%, and the difference of its accuracy with InceptionV3 is only 0.15%. Based on the data in Table 2, it can be stated that the DenseNet201 is giving better classification performance than the other model.

From Table 3, we observe that the F1 score is increasing along with the accuracy from top to bottom. Here, one noticeable thing is that NASNetMobile has a difference between sensitivity and specificity, which is also visible in Fig. 7a. Apparently for Xception (Fig. 7c), InceptionV3 (Fig. 7d), and DenseNet201 (Fig. 7e), the graphs seem to look the same, but we can see from Table 3 that there are some differences between their sensitivity and specificity.

In Fig. 8, the receiver operating characteristics (ROC) curve and area under the curve (AUC) are visualized. ROC is a probability curve and AUC represents the degree or measure of separability between classes. When the ROC is higher that means the model is performing well. The ROC curve is plotted with TPR against the

Table 3 Classification report for five CNN models

Model	Accuracy (%)	Precision (%)	F1 (%)	Sensitivity (%)	Specificity (%)	FPR (%)	FNR (%)
NASNetMobile	86.75	97.80	85,82	79.08	97.79	1.89	23.54
VGG16	94.61	99.62	94.60	90.09	99.61	0.37	9.93
Xception	96.60	97.77	96.72	95.36	97.76	2.40	4.30
InceptionV3	96.87	99.10	96.95	94.61	99.10	0.94	5.11
DenseNet201	97.02	98.99	97.10	95.01	98.98	1.07	4.79

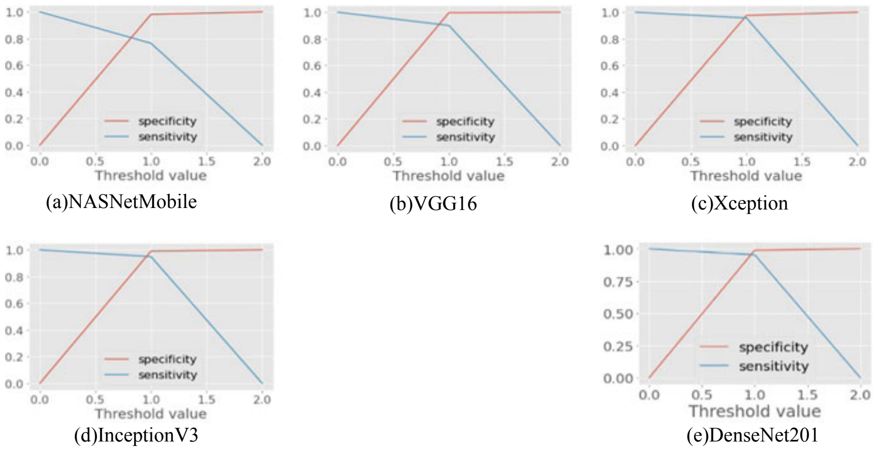


Fig. 7 Visualized sensitivity and specificity for five CNN models. **a** NASNetMobile, **b** VGG16, **c** Xception, **d** InceptionV3, **e** DenseNet201

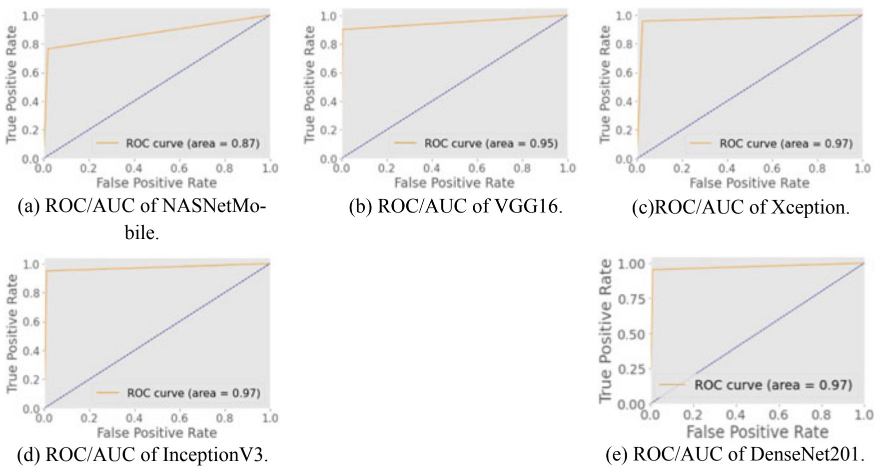


Fig. 8 Receiver operating characteristics (ROC) curve and area under the curve (AUC) for five CNN models

FPR where TPR is on the y-axis and FPR is on the *x*-axis [24]. The ROC/AUC for Xception (Fig. 8c), InceptionV3 (Fig. 8d), and DenseNet201 (Fig. 8e) is 0.97.

From the intensive result analysis, we can conclude that DenseNet201 gives a better classification performance in our experiment.

5 Conclusion

The main objective of this research describes the proposed methodology to identify original Hilsa fishes. This is the first study of original Hilsa and fake Hilsa identification using convolutional neural network (CNN). We used five CNN models in our experiments and observed their performance. Then, we have also performed an observational comparative analysis of our obtained results. Here, NASNetMobile shows the least performance with an accuracy of 86.75%, and DenseNet201 produces a very good performance with an accuracy of 97.02%. We hope that this study will be beneficial for researchers who will research-related topics. In future, we want to enrich our dataset with more images and develop a mobile application that can be used to identify the original Hilsa fish in real time.

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Chapter 15
Time Restricted Balanced Truncation for Index-I Descriptor Systems with Non-homogeneous Initial Condition

Kife I. Bin Iqbal, Xin Du, M. Monir Uddin, and M. Forhad Uddin

Abbreviations

X	Time restricted controllability Gramian
Y	Time restricted observability Gramian
X_∞	Infinite controllability Gramian
Y_∞	Infinite observability Gramian
x_0	Non-homogeneous initial condition
φ	Eigenvector set
λ	Eigenvalue set
Z_{c_∞}	Low-rank controllability Gramian on infinite time domain
Z_{o_∞}	Low-rank observability Gramian on infinite time domain
Z_c	Low-rank controllability Gramian on finite time domain
Z_o	Low-rank observability Gramian on finite time domain
Σ	Singular Value set
B_0	Vector of null space

K. I. Bin Iqbal (✉) · M. F. Uddin
Department of Mathematics, Bangladesh University of Engineering & Technology, Dhaka 1000, Bangladesh
e-mail: farhad@math.buet.ac.bd

X. Du
School of Mechatronic Engineering and Automation, Shanghai University, Shanghai 200072, China
e-mail: duxin@shu.edu.cn

Key Laboratory of Knowledge Automation for Industrial Processes, Ministry of Education, Beijing 100083, China

M. M. Uddin
Department of Mathematics and Physics, North south University, Dhaka 1229, Bangladesh
e-mail: monir.uddin@northsouth.edu

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B_{aug} Augmented B matrix
$\text{Im}(B_0)$ Image of extension matrix B_0
δ Dirac delta distribution function

1 Introduction

It is mandatory to solve time restricted Lyapunov equation pair for balancing-based model reduction on certain time interval $[t_0, t_f]$ which are of the form [1]

$$AXE^T + EXA^T = e^{E^{-1}At_f}BB^Te^{A^TE^{-T}t_f} - e^{E^{-1}At_0}BB^Te^{A^TE^{-T}t_0} \quad (1a)$$

$$A^TYE + E^TYA = e^{A^TE^{-T}t_f}C^TCe^{E^{-1}At_f} - e^{A^TE^{-T}t_0}C^TCe^{E^{-1}At_0} \quad (1b)$$

where $E,\ A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{m\times n}$, and $t = [t_0, t_f]$ is the nominated time interval starting with the nonzero initial time t_0, and ending at the final time t_f. The matrices X and Y are known as, respectively, the time-limited controllability and observability Gramians of the generalized state-space system with non-homogeneous initial condition [2, 3]

$$\begin{aligned} E\dot{x}(t) &= Ax(t) + Bu(t), \quad x(t_0) = x_0 \\ y(t) &= Cx(t) + Du(t), \end{aligned} \quad (2)$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^p$, and $y(t) \in \mathbb{R}^m$ are state, input, and output matrices respectively. Since balanced truncation model reduction only considers the structure of the matrices A, B, and C, no effect of initial condition x_0 is imposed during model reduction [4]. Therefore, there is a necessity arisen to modify the traditional BT technique what led Beattie et al. [3], Heinkenschloss et al. [4] to reform the BT technique to reduce the model on infinite time domain. However, with a view to simplifying the operation of the actuators, and sensors of large-scale systems, it is essential to evaluate X, and Y on restricted time interval $[t_0, t_f]$. In [1, 5, 6], the state-space system (2) has been reduced on limited time interval with homogeneous initial condition $x(t_0) = 0$ by solving the generalized time restricted Lyapunov equation [1, 7].

On the other hand, in this paper we modify the BT technique for reducing the order of large-scale index-I descriptor systems on considered time interval $[t_0, t_f]$ with non-homogeneous initial condition of the form [8]

$$\begin{aligned} \underbrace{\begin{bmatrix} E_{11} & E_{12} \\ 0 & 0 \end{bmatrix}}\underbrace{\begin{bmatrix} \dot{x_1}(t) \\ \dot{x_2}(t) \end{bmatrix}} &= \underbrace{\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}}\underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}} + \underbrace{\begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix}}u(t), \quad x(t_0) = x_0 \\ y(t) &= \underbrace{\begin{bmatrix} C_{11} & C_{12} \end{bmatrix}}\underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}} + D_a u(t), \end{aligned} \quad (3)$$

what can be written in the form of differential algebraic equations as

$$\begin{aligned} E_{11}\dot{x_1}(t) + E_{12}\dot{x_2}(t) &= A_{11}x_1(t) + A_{12}x_2(t) + B_{11}u(t) \\ 0 &= A_{21}x_1(t) + A_{22}x_2(t) + B_{21}u(t) \\ y(t) &= C_{11}x_1(t) + C_{12}x_2(t) + D_a u(t). \end{aligned} \tag{4}$$

where $x_1 \in \mathbb{R}^{n_1}$ and $x_2 \in \mathbb{R}^{n_2}$. Considering $x(t) = x_1(t) + E_{11}^{-1}E_{12}x_2(t)$, this system can be convert into the generalized form in (2), in which [9]

$$\begin{aligned} E &:= E_{11}, \\ A &:= A_{11} - (A_{12} - A_{11}E_{11}^{-1}E_{12})\Gamma A_{21}, \\ B &:= B_{11} - (A_{12} - A_{11}E_{11}^{-1}E_{12})\Gamma B_{21}, \\ C &:= C_{11} - (C_{12} - C_{11}E_{11}^{-1}E_{12})\Gamma A_{21}, \\ D &:= D_a - (C_{12} - C_{11}E_{11}^{-1}E_{12})\Gamma B_{21}. \end{aligned} \tag{5}$$

where $\Gamma := (A_{22} - A_{21}E_{11}^{-1}E_{12})^{-1}$. For modification purpose, we create a basis matrix B_0 spanning the null space to extend the B matrix as $B_{\text{aug}} := [B\ B_0]$, and solve the Lyapunov equation of the form

$$AXE^T + EXA^T = e^{E^{-1}At_f}B_{\text{aug}}B_{\text{aug}}^T e^{A^T E^{-T}t_f} - B_{\text{aug}}B_{\text{aug}}^T, \tag{6a}$$

$$A^T YE + E^T YA = e^{A^T E^{-T}t_f}C^T Ce^{E^{-1}At_f} - C^T C, \tag{6b}$$

using RKSM method to find out low-rank controllability Gramian factor $Z_c \in \mathbb{R}^{n\times k}$ and the observability Gramian factor $Z_o \in \mathbb{R}^{n\times k}$ without occupying full-rank Cholesky factorization such that

$$X \approx Z_c Z_c^T \quad \text{and} \quad Y \approx Z_o Z_o^T.$$

what are efficient for calculation and are indispensable to establish balancing and truncating transformations for BT [2, 10] in order to construct a reduced state space of the underlying descriptor system (3). Giving supreme priority on error minimization, we conduct our numerical experiments on several Brazilian interconnected power system (BIPS) data models to demonstrate the efficiency of our reduced balanced systems on limited time intervals with non-homogeneous initial condition comparing with the reduced models using infinite Gramians while the interval outside the particular time boundary is out of our attention.

2 Preliminaries

Balanced truncation is a proficient technique for model order reduction (MOR) what makes infinite controllability, and observability Gramians balanced, and diagonal

,i.e., X_∞= Y_∞ constructing balancing transformation matrices T_{L_∞}, and T_{R_∞}. The matrices X_∞ and Y_∞ are found from solving Lyapunov equation pair written as [11]

$$\begin{aligned} AX_\infty E^T + EX_\infty A^T + BB^T &= 0, \\ A^T Y_\infty E + E^T Y_\infty A + C^T C &= 0. \end{aligned} \tag{7}$$

The RKSM is one of the most popular projection-based methods to solve (7) introduced in [12] in which a projection $\mathbb{V}$ is created at each iteration performing on the Krylov subspace

$$K_m(A, E, B, s) = \text{span}\left\{\prod_{j=1}^{m}(A - s_j E)^{-1}B\right\}$$

where s_j is a set of shift parameter calculated from [13]. Therefore, the large-scale Lyapunov equation (7) can be easily converted into small-scale Lyapunov equation projected by $\mathbb{V}$ as following

$$\tilde{A}\tilde{X}\tilde{E}^T + \tilde{E}\tilde{X}\tilde{A}^T + \tilde{B}\tilde{B}^T = 0, \tag{8}$$

$$\tilde{A}^T\tilde{Y}\tilde{E} + \tilde{E}^T\tilde{Y}\tilde{A} + \tilde{C}^T\tilde{C} = 0. \tag{9}$$

where

$$\tilde{E} := \mathbb{V}^T E\mathbb{V},\ \tilde{A} := \mathbb{V}^T A\mathbb{V},\ \tilde{B} := \mathbb{V}^T B,\ \tilde{C} := C\mathbb{V},$$

Algorithm 1: RKSM for solving (7) of generalised system (2)

Input: E, A, B, m (no.of iteration)
Output: $Z_{c_\infty} \in \mathbb{R}^{n\times k}$ such that $Z_{c_\infty} Z_{c_\infty}^T \approx X_\infty$, where $k \ll n$.

Find initial basis matrix v_1 by solving the linear system $\big(A - s_1 E\big)\, v_1 = B$
Initial orthonormal vector set $\mathbb{V}_1 := qr(\mathit{full}(v_1))$
while $j \leq m$ **do**
  Find the next basis matrix by solving linear system $\big(A - s_j E\big)\, v_j = v_{j-1}$
  Construct orthonormal vector set $\mathbb{V}_j := qr(\mathit{full}(v_j))$
  Solve small-scale Lyapunov Equation (7)
  Operate eigenvalue decomposition

$$\tilde{X}_\infty = [\varphi_1\ \varphi_2]\begin{bmatrix}\lambda_1 & \\ & \lambda_2\end{bmatrix}[\varphi_1\ \varphi_2]^T \tag{10}$$

  Establish low-rank factor

$$Z_{c_\infty} = \mathbb{V}_j \varphi_1 \lambda_1^{\frac{1}{2}}$$

  after truncating less effective eigenvalues λ_2
  Stop Rational Krylov iteration.

what can be solved effortlessly by any direct solver method like [14] to find out low-rank controllability, and observability Gramian factors Z_{c_∞}, Z_{o_∞} required for BT.

Now using the Gramian factors Z_{c_∞} and Z_{o_∞}, reduced order model is constructed by applying the following steps :

- Perform Singular-value decomposition (SVD)

$$Z_{o_\infty}^T E Z_{c_\infty} = \begin{bmatrix} U_r \ U_{n-r} \end{bmatrix} \begin{bmatrix} \Sigma_r & \\ & \Sigma_{n-r} \end{bmatrix} \begin{bmatrix} V_r^T \\ V_{n-r}^T \end{bmatrix}. \tag{11}$$

- Form left and right balancing and truncating transformations

$$T_L : = Z_{o_\infty} U_r \Sigma_r^{-\frac{1}{2}} \quad \text{and} \quad T_R : = \tilde{Z}_{c_\infty} V_r \Sigma_r^{-\frac{1}{2}}. \tag{12}$$

- Finally construct the reduced order model [15]:

$$\begin{aligned} \hat{E}\dot{\hat{x}}(t) &= \hat{A}\hat{x}(t) + \hat{B}u(t), \\ \hat{y}(t) &= \hat{C}\hat{x}(t) + \hat{D}u(t) \end{aligned} \tag{13}$$

  where $\hat{x}(t) \in \mathbb{R}^r$, $\hat{y}(t) \in \mathbb{R}^m$, and $r \ll n$. The matrices $\hat{E}$, $\hat{A}$, $\hat{B}$, $\hat{C}$, $\hat{D}$ are constructed as follows

$$\begin{aligned} &\hat{E} := T_L^T E T_R, \quad \hat{A} := T_L^T A T_R, \\ &\hat{B} := T_L^T B, \qquad \hat{C} := C T_R, \quad \hat{D} := D. \end{aligned}$$

The error of the reduced order system can be calculated as [16]

$$\left\| G(s) - \hat{G}(s) \right\| \tag{14}$$

where

$$\begin{aligned} G(s) &= C(sE - A)^{-1}B + D, \\ \hat{G}(s) &= \hat{C}(s\hat{E} - \hat{A})^{-1}\hat{B} + \hat{D}. \end{aligned} \tag{15}$$

represents the transfer function matrices of the original and reduced order systems, respectively, and $\|.\|$ denotes system norm.

Despite BT works perfectly on time domain $[0, \infty]$, it fails by giving poor approximation of full systems when time domain consists of non-homogeneous initial condition i.e. $[t_0, \infty]$. Therefore, in [4], the authors introduced modified balanced truncation technique, where they created an augmented matrix $[B\, B_0] \in \mathbb{R}^{n \times p+q}$ by extending B. The nonzero initial value $x_0 \in \mathrm{Im}(B_0)$ contains in the subspace of $Q_0 \subset \mathbb{R}^n$ spanned by $B_0 \in \mathbb{R}^{n \times q}$. Finally, they converted the state-space system (2) as

$$
\begin{aligned}
E\dot{x}(t) &= Ax(t) + \underbrace{\left[B\ B_0\right]}\begin{bmatrix} u(t) \\ u_0(t) \end{bmatrix}, \quad x(t_0) = x_0 \\
y(t) &= Cx(t) + Du(t),
\end{aligned} \tag{16}
$$

what was used for MOR using BT. On the contrary, the authors in [3] reformed the non-homogeneous initial condition $x(t_0) = x_0$ as $x_0 = B_0 Q_0$ where Q_0 was spanned by B_0, and evaluated the output $y(t)$ of (16) explicitly using the Duhamel formula as

$$
y(t) = Ce^{E^{-1}At} B_0 Q_0 + \int_0^{\infty} Ce^{E^{-1}A(t-\tau)} Bu(\tau)\mathrm{d}\tau. \tag{17}
$$

It is closely observed that the first term of the right side of (17) is the response of the system to the initial condition x_0 with $u(t) = 0$, and the second term is the response of the system to the $u(t)$ with homogeneous initial condition, i.e., $x(t_0) = 0$. In addition, they spitted the state-space equation (16) based on (17) as

$$
\begin{aligned}
E\dot{w}(t) &= Ax(t) + B_0 u_0(t), \quad w(t_0) = 0 \\
y(t) &= Cw(t) + Du_0(t),
\end{aligned} \tag{18}
$$

and

$$
\begin{aligned}
E\dot{x}(t) &= Ax(t) + Bu(t), \quad x(t_0) = 0 \\
y(t) &= Cx(t) + Du(t),
\end{aligned} \tag{19}
$$

where $u_0(t) = Q_0 \delta(t)$, and $\delta(t)$ is the Dirac delta distribution. Finally, they applied balanced truncation technique on them individually to construct balanced reduced models. But this technique needs much calculation time since it is required to imply BT twice separately, and approximation of full model depends on the perfect value of the distributive function $\delta(t)$.

3 Main Work

To cope with the realistic problem, we develop our algorithm on restricted time frame $[t_0, t_f]$ with nonzero initial condition instead of flourishing on infinite time domain. With a view to efficiently dealing with the non-homogeneous initial condition, i.e., $x(t_0) = x_0$, we build an orthogonal basis matrix B_0 such that $B_0 Q_0 = 0$ what spans null space. Consequently, the first term of the right-hand side of Eq. (17) is annihilated, and only the term responding to input $u(t)$ remains in the form of

$$
y(t) = \int_0^{t_f} Ce^{E^{-1}A(t-\tau)} Bu(\tau)\mathrm{d}\tau \tag{20}
$$

Therefore, BT needs to be implied on (19) only in lieu of occupying twice. We develop our technique to reduce the order of large-scale index-I system (3). Since B_{11} and B_{21} are linearly independent submatrices of B, we assume two basis submatrices B_{x_1} and B_{x_2} as

$$B_{x_1} := e^{B_{11}t_0}, \quad B_{x_2} := e^{B_{21}t_0} \tag{21}$$

and extract orthonormal basis vector conducting QR decomposition of B_{x_1}, and B_{x_2} spanning the null space what extend B_{11}, and B_{21} through forming two augmented submatrices like (16) as follows

$$B_{\text{aug}_1} := \begin{bmatrix} B_{11} & B_{x_1} \end{bmatrix}, B_{\text{aug}_2} := \begin{bmatrix} B_{21} & B_{x_2} \end{bmatrix} \tag{22}$$

Algorithm 2: RKSM for solving (6a) of index-1 descriptor system (25)

Input: $E_{11}, E_{12}, A_{11}, A_{12}, A_{21}, A_{22}, B_{aug_1}, B_{aug_2}, t_0, t_f 0 < tol \ll 1$ (tolerance value).

Output: $Z_c \in \mathbb{R}^{n \times k}$ such that $Z_c Z_c^T \approx X$

1 Find initial basis matrix v_1 by solving the linear system

$$\begin{bmatrix} A_{11} - s_1 E_{11} & A_{12} - s_1 E_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ * \end{bmatrix} = \begin{bmatrix} B_{aug_1} \\ B_{aug_2} \end{bmatrix}$$

2 **while** $j \leq m$ **do**

3 Find the next basis matrix by solving linear system

4

$$\begin{bmatrix} A_{11} - s_j E_{11} & A_{12} - s_j E_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} v_j \\ * \end{bmatrix} = \begin{bmatrix} v_{j-1} \\ 0 \end{bmatrix} \tag{23}$$

5 Construct orthonormal vector set

$$\mathbb{V}_j = qr(full(v_j))$$

6 Compute $B_j := \mathbb{V}_j \tilde{B}_j$

7 **if** $\|B_j - B_{j-1}\| / \|B_j\| < tol$ **then**

8 Solve the small-scale Lyapunov equation (26a) for finding X

9 Establish low-rank factor as

$$Z_c = \mathbb{V}_j \varphi_1 \lambda_1^{\frac{1}{2}} \tag{24}$$

after truncating less effective eigenvalues λ_2 performing eigenvalue decomposition as (10).

10 Stop Rational Krylov iteration.

Therefore, the index-I system (3) can be written after modification as

$$\begin{aligned} \underbrace{\begin{bmatrix} E_{11} & E_{12} \\ 0 & 0 \end{bmatrix}}\underbrace{\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_1(t) \end{bmatrix}} &= \underbrace{\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}}\underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}} + \begin{bmatrix} B_{\text{aug}_1} \\ B_{\text{aug}_2} \end{bmatrix} u(t), \\ y(t) &= \underbrace{\begin{bmatrix} C_{11} & C_{12} \end{bmatrix}}\underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}} + D_a u(t). \end{aligned} \tag{25}$$

what can be easily expressed as (2) through replacing B_{11}, and B_{21} by B_{aug_1}, and B_{aug_2} in (5). Since the matrix extension is only occupied in matrix B, and the output (20) of the system responses on time interval $[0, t_f]$, the observability Lyapunov equation of index-I systems (3), and (25) remains same, and it is enough to form time restricted Lyapunov equations as (6a), and (6b) on restricted time interval $[t_0, t_f]$ plugging the boundary time value t_f only. Now, they can be solved by index-I RKSM Algorithm 2 converting into small-scale equations projecting by $\mathbb{V}$ formed as

$$\tilde{A}\tilde{X}\tilde{E}^T + \tilde{E}\tilde{X}\tilde{A}^T = e^{\tilde{E}^{-1}\tilde{A}t_f}\tilde{B}_{\text{aug}}\tilde{B}_{\text{aug}}^T e^{\tilde{A}^T\tilde{E}^{-T}t_f} - \tilde{B}_{\text{aug}}\tilde{B}_{\text{aug}}^T \tag{26a}$$

$$\tilde{A}^T\tilde{Y}\tilde{E} + \tilde{E}^T\tilde{Y}\tilde{A} = e^{\tilde{A}^T\tilde{E}^{-T}t_f}\tilde{C}^T\tilde{C}e^{\tilde{E}^{-1}\tilde{A}t_f} - \tilde{C}^T\tilde{C} \tag{26b}$$

where

$$\tilde{E} := \mathbb{V}^T E\mathbb{V},\ \tilde{A} := \mathbb{V}^T A\mathbb{V},\ \tilde{B} := \mathbb{V}^T B,\ \tilde{C} := C\mathbb{V}.$$

Algorithm 3: BT for (25) on certain time interval $[t_0,\ t_f]$

Input: sub-matrices $E_{11}, E_{12}, A_{11}, A_{12}, A_{21}, A_{22}, B_{aug_1}, B_{aug_2}, C_{11}, C_{12}, D_a$

Output: Matrices $\hat{E}, \hat{A}, \hat{B}, \hat{C}, \hat{D}$ of the reduced system.

1 Compute low-rank solution factors Z_c, Z_o and partition their product by singular value decomposition as (11)

2 Establish left, and right balancing transformation matrices

$$T_L := Z_c U_r \Sigma_r^{-\frac{1}{2}}, \quad T_R := Z_o V_r \Sigma_r^{-\frac{1}{2}} \tag{27}$$

3 Generation of the reduced order model as following

$$\begin{aligned}
\hat{E}&: = T_L^T E_{11} T_R, \quad \hat{A}: = \hat{A}_{11} - \hat{A}_{12}\Gamma\hat{A}_{21},\\
\hat{B}&: = \hat{B}_{11} - \hat{A}_{12}\Gamma B_{aug_2},\\
\hat{C}&: = \hat{C}_{11} - (C_{12} - C_{11}E_{11}^{-1}E_{12})\Gamma\hat{A}_{21},\\
\hat{D}&: = D_a - (C_{11}E_{11}^{-1}E_{12})\Gamma B_{aug_2}
\end{aligned} \tag{28}$$

where

$$\begin{aligned}
\Gamma&: = (A_{22} - A_{21}E_{11}^{-1}E_{12})^{-1},\ \hat{A}_{11}: = T_L^T A_{11} T_R,\\
\hat{A}_{12}&: = T_L^T A_{12} - T_L^T(A_{11}E_{11}^{-1}E_{12}), \hat{A}_{21}: = A_{21}T_R,\\
\hat{B}_{11}&: = T_L^T B_{aug_1},\ \hat{C}_{11}: = C_{11}T_R.
\end{aligned}$$

We set a tolerance value $0 < tol \ll 1$ in Algorithm 2 for finding the fastest convergence solution of (26a) by reducing the computational time. Since (6a) and (6b) are dual to one another, we can use the same Algorithm 2 for finding low-rank observability Gramian Y factors by solving small-scale equation (26b) taking only the transposes of the system blocks as $E_{11}^T, A_{11}^T, A_{21}^T, A_{12}^T, A_{22}^T, C_{11}^T, C_{12}^T$ what is equivalent to solve the linear system [17]

$$\begin{bmatrix} A_{11}^T - sE_{11}^T & A_{21}^T \\ A_{12}^T - sE_{12}^T & A_{22}^T \end{bmatrix} \begin{bmatrix} v \\ * \end{bmatrix} = \begin{bmatrix} C_{11}^T \\ C_{12}^T \end{bmatrix}$$

After computing the low-rank controllability, and observability Gramian factors Z_c, and Z_o by Algorithm 2, we construct the balancing and truncating transformation matrices T_L, and T_R as (27) using time restricted BT Algorithm 3, and then, the reduced order index-I descriptor model on finite time interval $[t_0, t_l]$ are formed as (13) by constructing balanced reduced state, input, and output matrices as (28). After that, we generate the transfer function of the reduced order system as (15) to find out the error bound calculating Euclidean norm as (14).

4 Numerical Results

Several large-scale power system model is chosen for numerical experiment where E_{12} block of the descriptor system (25) is treated as zero. First of all, we reform the index-I descriptor system as (25) suitable to BT with non-homogeneous initial condition $x(t_0) = x_0$. Then, time restricted Lyapunov equations (6) are solved on restricted time interval $[t_0, t_f]$ limited from non-homogeneous initial condition $x(t_0) = 1$ to boundary condition $x(t_f) = 3$ using RKSM Algorithm 2 to extract low-rank factors what are used in BT Algorithm 3 at final stage for MOR constructed as (28) (Table 1).

Looking at Fig. 1b, c, it is clearly visualized that time restricted reduced model minimizes both absolute and relative errors on restricted time interval [1, 3] comparing with the reduced model on infinite time domain. Furthermore, TRBT also gives better approximation of original model on nominated time interval what is demonstrated in Fig. 1a. Eventually, the data presented in Table 2 reflects the same image as the visual representation Fig. 1 shows. Hence, the goal, we mentioned earlier, to achieve better accuracy at the low-rank in any limited time interval is completely fulfilled.

Table 1 Dimensions of the selected BIPS model

Model	Differential	Algebraic	Full	Inputs/outputs
I	606	6529	7135	4/4
II	1693	11582	13275	4/4

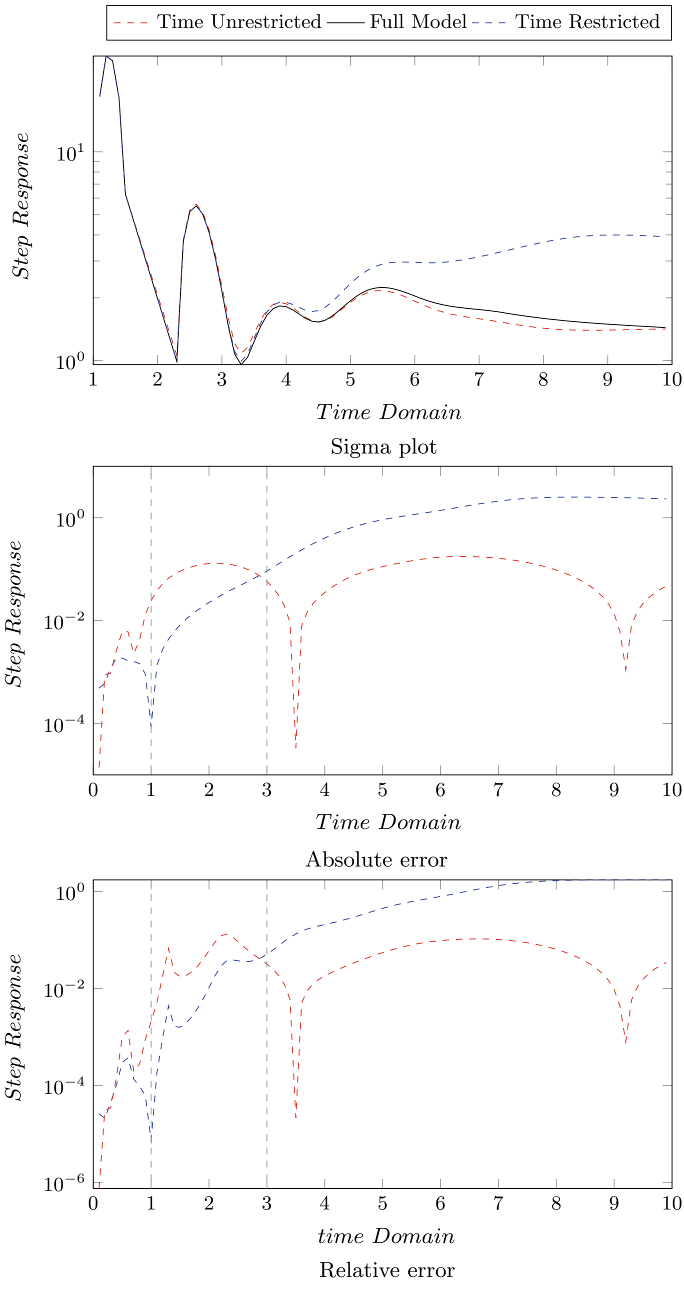

Fig. 1 Error minimization of time-restricted and time-unrestricted reduced systems of Mod-II BIPS model on time range [1, 3]

Table 2 Competitive numerical analysis on absolute & relative errors of the time-limited and infinite Gramians under nominated time interval [1, 3]

Model	ROM	Error			
		Absolute		Relative	
		t	∞	t	∞
I	50	0.0086	0.0274	7.0873×10^{-4}	0.0023
II	40	4.8918×10^{-4}	0.0125	1.3194×10^{-4}	0.0034

5 Conclusions

It has been shown in this paper that system balancing as well as model reduction can be performed efficiently in finite time intervals which impacts crucially in practical cases due to the availability of data usages in limited time intervals. Since the technique proposed here deals with non-homogeneous initial condition, the observer can choice any limited interval from any segment of the time domain using our developed technique. Excepting few cases, most of the situation the numerical efforts of TRBT is better than standard BT and works comparatively well in the small time domain what impacts highly on the contemporary practical issues arisen in our daily life.

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Chapter 16
Deep Transfer Learning-Based Musculoskeletal Abnormality Detection

Abu Zahid Bin Aziz, Md. Al Mehedi Hasan, and Jungpil Shin

1 Introduction

Musculoskeletal conditions affect more than 1.7 billion people around the world [1]. Its symptoms include pain, joint noises, decreased range of motion, etc. Poor work practices and postures, repetitiveness, force of movements have been identified as the major risk factors behind these symptoms [2]. Rotator cuff tendinitis, carpal tunnel syndrome, tension neck syndrome, epicondylitis, etc., are the most common disorders associated with musculoskeletal conditions [3]. Doctors usually use laboratory and radiographic tests to diagnose musculoskeletal conditions. Laboratory tests generally include blood tests to check ESR (erythrocyte sedimentation rate), level of creatine kinase, cyclic citrullinated peptide antibody, HLA-B27, and so on. Different imaging tests are performed to diagnose musculoskeletal conditions like X-ray, computed tomography (CT), magnetic resonance imaging (MRI), bone scanning, dual-energy X-ray absorptiometry (DXA), and ultrasonography. But there are many obstacles in the way of effective diagnosis for these conditions such as the surgical burdens in low- and middle-income countries, diagnostic errors by misreading radiographs [4, 5]. These problems encouraged the increasing use of computational methodologies in recent years.

Detecting abnormalities from radiographic images falls under image classification tasks. Various computational methods including both deep learning and machine learning methodologies have been introduced over the years. Among machine learning techniques, Criminisi et al. applied a decision forest-based model in their work where Ricci et al. used an SVM classifier for the segmentation of the blood vascula-

A. Z. B. Aziz (✉) · Md. A. M. Hasan
Department of Computer Science & Engineering, Rajshahi University of Engineering & Technology, Rajshahi 6204, Bangladesh

J. Shin
School of Computer Science and Engineering, University of Aizu, Aizuwakamatsu, Japan
e-mail: jpshin@u-aizu.ac.jp

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ture of the retina [6, 7]. Ng et al. employed both clustering techniques and the watershed algorithm for image segmentation [8]. Among deep learning methodologies, convolutional neural networks (CNN) have been very fruitful in image classification and segmentation tasks [9]. Shin et al. detected thoraco-abdominal lymph node and classified interstitial lung disease in their work by considering multiple CNN architectures [10]. Bullock et al. provided a useful framework (Xnet) for image segmentation using CNN [11]. Cohen et al. provided a web-based system for predicting chest diseases from X-ray images [12].

Rajpurkar et al. provided a comprehensive collection of musculoskeletal radiographs in their work which is publicly available for researchers at stanford machine learning group's website [13]. This dataset has become a vital resource for abnormality detection tasks. Pradhan et al. used this dataset for human bone classification in their work [14]. They did not take account of abnormalities in their work. Shubhajit et al. employed a deep CNN for detecting abnormalities [15]. Although their work produced decent accuracies in the testing datasets, they did not categorize their findings according to the study types (elbow, finger, humerus, etc.). They also did not provide any comparison with the existing work of Rajpurkar et al.

Transfer learning has become a successful medium in various classification tasks in the past few years. It particularly provided efficient results when there's not enough data available or when there's any limitation in training large datasets. In this work, we provided an effective method for utilizing transfer learning by selecting a portion of layers from pre-trained models. After tuning our model, we checked the performance of our model in the testing dataset and found decent results. Our results project that our method can be a beneficial tool for detecting abnormalities from bone X-ray images.

2 Materials and Methods

2.1 Dataset's Description

We considered five study types (elbow, finger, forearm, humerus, and shoulder) from the MURA dataset for detecting abnormalities in our work. The MURA dataset was collected from Stanford Hospital and provided by Stanford University's machine learning group. Each of the studies is either normal or abnormal. The ratio of training and testing datasets was about 10:1 in terms of the number of images.

There were a total of 8744 studies, among which 7949 were used in training and 795 in testing. We noticed patient overlapping among the studies which means some of the studies were from the same patient. A summary of the number of studies is shown in Table 1. Our training dataset contained 21213 images and testing dataset

Table 1 A summary of our dataset with respect to the number of studies

Study type	Training dataset		Testing dataset	
	Pos. Studies	Neg. Studies	Pos. Studies	Neg. Studies
Elbow	660	1064	66	92
Finger	655	1280	83	92
Forearm	287	590	64	69
Humerus	271	321	67	68
Shoulder	1457	1364	95	99
Total	3330	4619	375	420

Table 2 A summary of our dataset with respect to the number of images

Study type	Training dataset		Testing dataset	
	Pos. Images	Neg. Images	Pos. Images	Neg. Images
Elbow	2006	2625	230	235
Finger	1968	3138	247	214
Forearm	661	1164	151	150
Humerus	599	673	140	148
Shoulder	4168	4211	278	285
Total	9402	11811	1046	1032

contained 2078 images. They were collected from the studies of Table 1. The number of images per study type is shown in Table 2. In the tables, positive studies/images suggest abnormal and negative studies/images suggest normal cases. We used the training and testing datasets as Rajpurkar et al. to compare our model's performances. That is why we did not apply any additional data splitting techniques.

2.2 Data Preprocessing

Data preprocessing is very crucial in CNN's overall performance. In our work, there were three steps involved in data preprocessing: data augmentation, resizing, and cropping and normalization.

- Data augmentation: Data augmentation is an effective approach to increase the diversity of data. Among a variety of considerable augmentation techniques, we employed five in our data. They are horizontal flip, random contrast, random gamma, random brightness, and shift scale rotate. We also converted the pixel values to floating-point numbers to ease computation in the later steps. All of these techniques are shown step by step in Fig. 1.

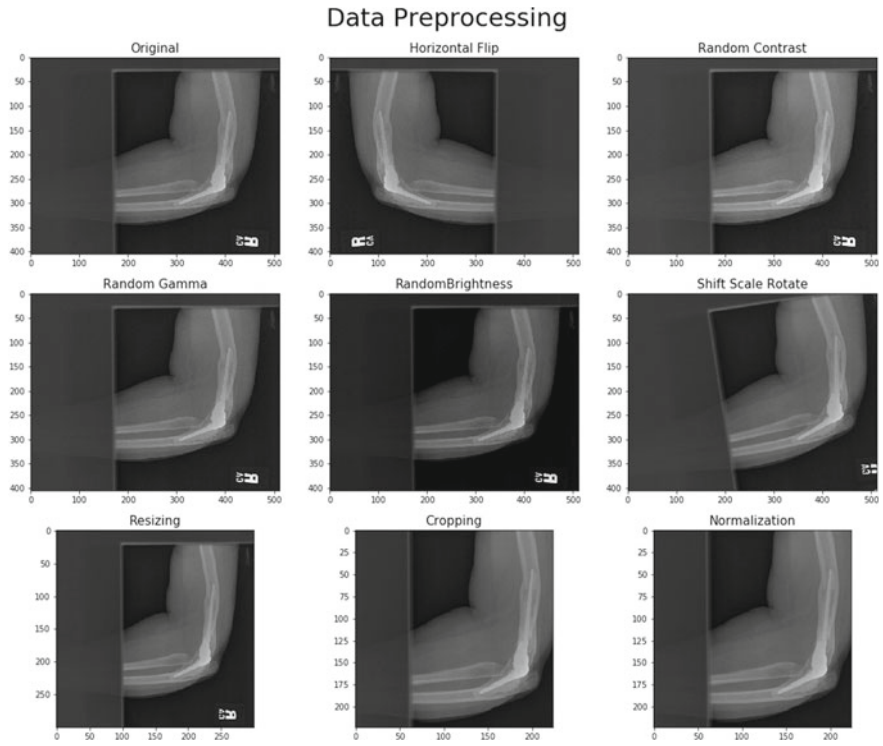

Fig. 1 Data preprocessing steps for a positive study of elbow. First two rows are the selected data augmentation techniques. In the last row, first two images are shown after resizing and cropping. And the last image is shown after normalization

- Resizing and cropping: One of the prerequisites of deep learning is to make sure all data has the same shape. But in our dataset, the image size was different for different patients. That is why, the input shape we chose for our work is (224, 224, 3). We did it in two steps. First, we resized the image to (300, 300, 3). Then, we cropped the image from the center in such a way that the output would be (224, 224, 3). The reason behind this approach was to eliminate unnecessary pixels from the image.
- Normalization: After cropping, we normalized the images to redistribute the pixel values of our images between zero to one. In our case, we applied the min-max normalization. We applied the following equation for this task:

$$\text{NormalizedImage} = \frac{\text{Image} - \text{Min}}{\text{Max} - \text{Min}} \tag{1}$$

  Here, Image= input image, Normalized Image= image after normalization, Max= maximum pixel value in the input image and Min= minimum pixel value in the input image.

2.3 Architecture of Our Model

After preprocessing, we fed the images to our CNN model. The pre-trained model we used to build our model was DenseNet169 because of its increased number of direct connections among convolution layers [16]. It also helped us to make sure that the flow of information was maximum within the layers. We selected a part of the densenet model in our model. This portion of the densenet model is then connected to a global average pooling layer to get the feature maps from the pre-trained model. Then, the pooling layer is fed to a fully connected layer with 256 nodes. After that, the fully connected layer is connected to the output layer which is actually a softmax activation function. The softmax activation function returned a probability distribution of having abnormality in the input image. The basic structure is illustrated in Fig. 2.

Now let us see how we selected layers from the pre-trained model. First, we used the training data on the whole model with all the layers. Then, we removed two layers from the end and trained the model with the remaining layers. We continued this process until we got decent accuracy. As we trained individual models for each study type, the number of eliminated layers was different in all cases. The number of eliminated layers varied from sixteen to twenty across all study types. Therefore, the total number of parameters was not identical for all study types. We employed categorical cross-entropy as the loss function during training which is defined as follows:

$$\text{Loss} = -\sum_{i=1}^{n} Ya_i log Yp_i \tag{2}$$

Here, n = the number of classes, Ya = actual probability of class i and Yp = predicted probability of class i.

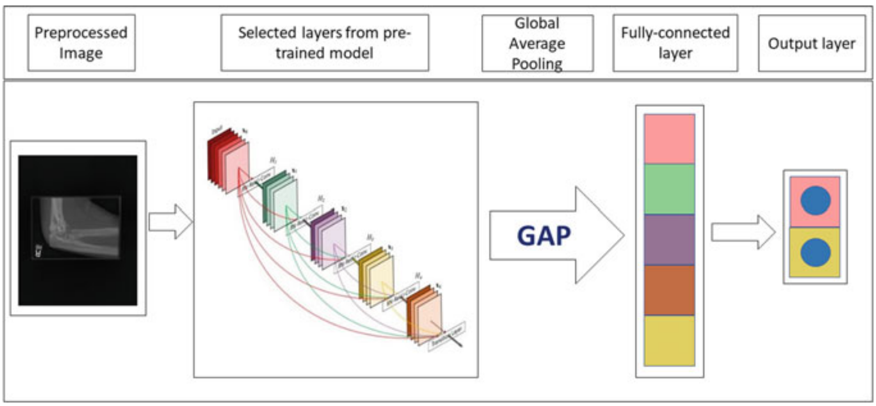

Fig. 2 Architecture of our model

To compute the probability for a study, we predicted the probability of abnormality for each image first. Then we calculated the mean value of all the probabilities. This mean value is considered as the final probability of abnormality in that study.

2.4 Method Evaluation Metrics

In this work, we considered sensitivity (SN), specificity (SP), Matthews correlation coefficient(MCC), and accuracy (ACC) for our method evaluation. To calculate these metrics, we needed four parameters: true positive (TP), true negative (TN), false positive (FP), and false negative (FN). We can calculate these metrics by using the following formulas:

$$\mathrm{SN} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{3}$$

$$\mathrm{SP} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{4}$$

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{5}$$

$$\begin{aligned}\mathrm{MCC} = (\mathrm{TP} * \mathrm{TN} - \mathrm{FP} * \mathrm{FN})/[(\mathrm{TP} + \mathrm{FP}) *\\ (\mathrm{TP} + \mathrm{FN}) * (\mathrm{TN} + \mathrm{FP}) * (\mathrm{TN} + \mathrm{FN})]^{1/2}\end{aligned} \tag{6}$$

We also calculated the Cohen’s kappa(k) statistic to compare with the existing model [17]. For this, we needed two parameters: accuracy (ACC) and the hypothetical probability of chance agreement (Pe). We used the following equation to calculate this metric:

$$\mathrm{k} = \frac{\mathrm{ACC} - \mathrm{Pe}}{1 - \mathrm{Pe}} \tag{7}$$

3 Results and Discussion

3.1 Hyperparameter Tuning

Hyperparameter tuning can increase a model’s performance significantly [18, 19]. We tuned a number of hyperparameters to fine-tune our model: pre-trained models, learning rate, and optimizer.

First, we selected a suitable pre-trained model for our task. We investigated a few pre-trained models for our classifier like DenseNet169, ResNet50, Inception-V3. For all of these models, we applied the layer selection technique explained earlier. Among them, DenseNet169 provided the best accuracy.

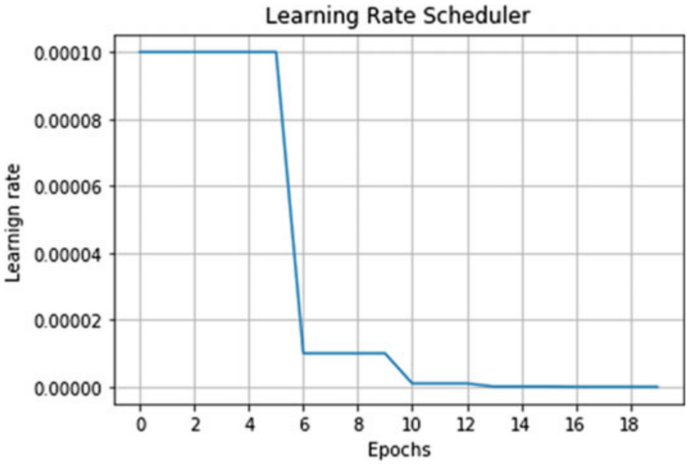

Fig. 3 Changes of values of the learning rate with respect to the number of epochs

We employed a learning rate scheduler for our model rather than using a fixed value to converge faster. Initially, we chose the learning rate as 0.0001. Then, we decreased the learning rate every time, the validation loss plateaued. We set the value of patience parameter three which meant if validation loss did not reduce for three consecutive epochs, the learning rate would be decreased (see Fig. 3).

We considered four different optimizers for our model. They were Adam, RMSprop, stochastic gradient descent (SGD), and Adagrad. Although both SGD and RMSprop provided good training accuracy, RMSprop was better in the testing dataset. So we selected RMSprop as the optimizer for our classifier.

3.2 Result Analysis

After preprocessing the radiographs and tuning the hyperparameters, we trained our models on the training dataset using the selected values. As we built different models for each study type, we trained the models separately. The models were implemented using the Keras framework with TensorFlow as backend [20]. If we look at Table 1, the number of positive images is slightly lower for all of the studies. So we updated the loss functions by emphasizing more weights on the positive images to balance the overall prediction. Furthermore, to avoid overfitting, we employed dropout regularization in the fully connected layer. The built-in functions of the Keras framework helped with these implementations. We implemented the training process with the help of Google Colaboratory (Colab). It took about 2–3 h to complete the whole training process for each study type. After training, we tested the performance in the testing datasets. As we can see from Table 3, our model performed pretty well.

Table 3 Performance of our model for each study type on the testing datasets

Study type	ACC (%)	SN (%)	SP (%)	MCC
Elbow	87.18	82.74	93.77	0.750
Finger	82.5	75.68	90.31	0.621
Forearm	87.06	80.83	93.33	0.688
Humerus	85.76	87.14	84.46	0.715
Shoulder	86.56	86.49	86.64	0.695

Table 4 Comparison between our model and existing model in the testing datasets using Cohen's kappa statistic

Study Type	Rajpurkar et al.'s model(Cohen's kappa) [13]	Our model(Cohen's kappa)
Elbow	0.710	**0.756**
Finger	0.389	**0.650**
Forearm	0.737	0.732
Humerus	0.600	**0.715**
Shoulder	0.729	**0.731**

To perform a proper evaluation of our classifier, we made a comparison with our model's results to Rajpurkar et al.'s results [13]. Since they used Cohen's kappa statistic as evaluation metrics, we also calculated it for our model. The comparison is shown in Table 4. We can see from Table 4, our model performed better in every study type except for the elbow. To be more specific, the performance of our model improved by 6.47%, 67.05%, 19.16%, and 0.27% for the elbow, finger, humerus, and shoulder study, respectively. Only for the forearm study, our model's performance decreased by 0.67%. This increased performance proved our model's effectiveness.

We also wanted to represent our model's performance graphically. To do so, we plotted the receiver operating characteristic (ROC) curve based on the testing performance of our model. ROC curve is a line plot of true positive rates over false positive rates to find corresponding thresholds. It tells us about our model's ability to differentiate between class labels. This representation is shown in Fig. 4. We also calculated the AUC value which is the area under the ROC curve. It returns a value between zero and one. The AUC values were 0.890, 0.809, 0.846, 0.880, and 0.854 for the elbow, finger, forearm, humerus, and shoulder, respectively.

So after considering all these evaluation metrics, we can say that our model's performance was satisfactory. Although there is still room for improvement, our model's capability to detect abnormalities is better compared to the existing methodologies.

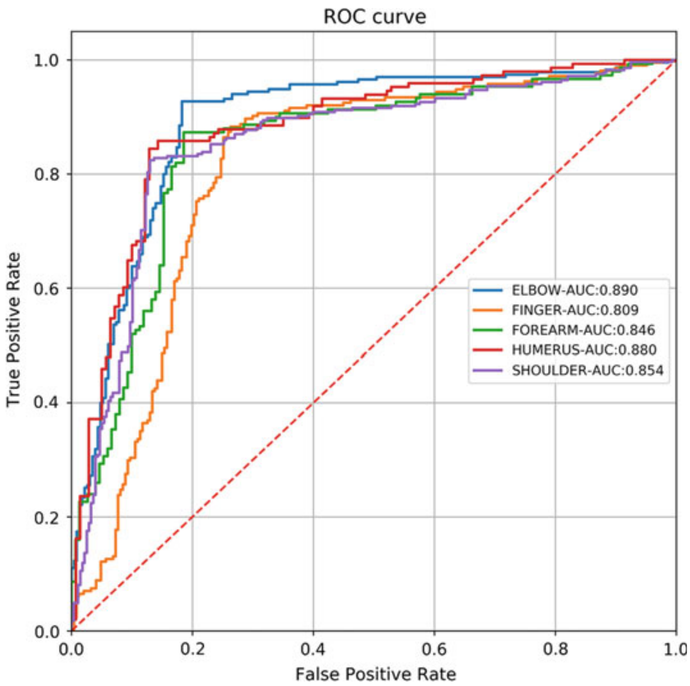

Fig. 4 Performance of our model using the ROC curve

3.3 Discussion

Our abnormality detection model can be taken to the next step by providing a web-based application. We considered five types of study for detecting abnormalities in our work. We can look for other study types and configure models using our technique to investigate its performance. In this study, we applied a different method for choosing layers from pre-trained models. We can use other techniques to utilize the pre-trained models in the future. CNN's performance depends a lot on hyperparameter tuning. There is a lot of other hyperparameters than can be tuned in the future to increase our model's performance. Due to hardware limitations, we built different models for each study which reduces the amount of samples for each model. A single model using images of all study types can be investigated in the future.

Nevertheless, our classifier's overall predictive performance is better than the existing work in terms of Cohen's kappa statistic. Among other metrics, the sensitivity was a bit down. We think it is because of the fewer number of positive images in the training datasets. We tried to make up for the imbalances by adding extra weight to the positive samples. Although the sensitivity was a bit low, other evaluation metrics (specificity, MCC, accuracy) were pretty satisfactory of our model. It is our belief that our model can be useful for not only academic researches but also practical uses.

4 Conclusion

The purpose of our study was to find a suitable computational method for detecting abnormalities from musculoskeletal radiographs. We particularly employed transfer learning for our work. We applied data augmentation, resizing & cropping and normalization for data preprocessing. We also took care of the class imbalance problem by redistributing the weights while computing losses. Our layer selection process from pre-trained models can be an effective technique for utilizing transfer learning. Furthermore, we improved our model's performance by tuning a number of hyperparameters. We also took the necessary precautions to avoid overfitting. After training, we evaluated the performance of our classifier with the existing models. The comparison demonstrated that our model yielded satisfactory results in the evaluation metrics. We believe that our methodology can be a great resource for future researchers. Our work also validates the increased use of computational methods for medical image analysis.

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Chapter 17
User-Centred Design-Based Privacy and Security Framework for Developing Mobile Health Applications

Uzma Hasan, Muhammad Nazrul Islam, Shaila Tajmim Anuva, and Ashiqur Rahman Tahmid

1 Introduction

The fast-growing mobile technology often facilitates healthcare for both patients and medical professionals. In present days, mobile health and wellness applications commonly referred to as mHealth applications have a greater impact on how medical services are delivered and how patient data is handled. These applications assist users in self-management of their overall health, disease detection, and precautionary treatment [1, 2]. The number of mHealth applications available for consumers in major application stores nearly stands to 318,000, since 2015 which has almost doubled due to increased adoption of smartphone and a large amount of investment in the digital health market [3].

Mobile health applications deal with a large amount of patient data. Thus, privacy and confidentiality are two important aspects to consider when dealing with medical data of consumers. Privacy generally refers to patients having substantial control over the extent, timing, circumstances and sharing of information about oneself with others [4]. As per [5], it has been reported that HealthEngine, Australia's most popular medical appointment booking application, routinely shared 100s of users' private medical information to personal injury law firms as part of a referral partnership contract. In [6], authors discussed that the media are increasingly reporting instances of security breaches of large amounts of electronically stored personal data. If mobile health data are stolen, consumers trust in such applications would diminish and they can be exposed to social or economic risks [7]. Also if medical data is tampered, decisions made upon those would be inaccurate and risky. Thus, it is essential to regularly review technology, tools, rules and regulations to protect patients' health data. In most cases, patient health data could be protected if the application developers

U. Hasan (✉) · M. Nazrul Islam · S. Tajmim Anuva · A. Rahman Tahmid
Department of Computer Science and Engineering, Military Institute of Science and Technology, Mirpur Cantonment, Dhaka 1216, Bangladesh
e-mail: nazrul@cse.mist.ac.bd

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adopt the minimum security measures while developing the applications and also while their maintenance in long term.

The user-centred design (UCD) approach is a very effective method of system development where special attention is given to develop a system following users' needs. Consideration of UCD in developing mHealth applications is of great importance [8]. This way developers will be aware of securing user data which is a basic concern of the mHealth application users. Again multiple security issues could be addressed at different stages of mHealth applications development following the UCD approach. In other words, where application developers tend to design their applications randomly without being careful about application security and user experience, UCD could be a useful approach to reduce the privacy/security risks of the mHealth applications and thus enhance user satisfaction and adoption of mHealth applications. Also no study exists that describes a UCD-based approach to enhance privacy and security of mHealth applications. Hence, the primary focus of our research includes exploring the security and privacy issues of mHealth applications in Bangladesh, proposing a UCD-based framework for enhancing privacy and security of mHealth applications and finally asses the applicability of the proposed framework to develop mHealth applications.

The organization of the paper is as follows: Sect. 2 highlights the works related to privacy and security of mobile health applications. The entire research methodology is briefly explained in Sect. 3. The ways of exploring the security and privacy threats is discussed in Sect. 4 and the conceptual framework development in Sect. 5 followed by discussing the evaluation of the proposed framework in Sect. 6. Finally, a brief concluding remark is presented in Sect. 7.

2 Literature Review

The current scenario of mHealth applications' security and the works related to this research is discussed in this section. Though many researches have been conducted on the occurring security threats, only a few study discusses ways to mitigate those issues. In [9], Naveed et al. presented a series of three studies of the mHealth applications and show several mHealth applications make widespread use of unsecured Internet communications and third-party servers. Huckvale et al. [10] revealed that 89% ($n = 70/79$) of applications transmitted information to online services and no application encrypted personal information stored locally. Neisse et al. [11] proposed a privacy enforcing framework which dealt with the unsecured permissions and found that almost all the top applications request network-related permissions and while 80% of the applications request permissions to write to external storage. Altuwaijri et al. [12] conducted a detailed study on the android data storage model and security issues and mentioned that the main reasons for building over-privileged applications are as developers may copy the code and apply it without understanding it, may request permissions that they thought it is mandatory and related to the functionalities they designed but it is not. Briggs et al. [13] conducted a review on mHealth applications and found that as many mobile devices are sold as consumer goods, the "rush to market" may also cause short-cuts in design that eventually results compromise of security.

Some of the studies have proposed some frameworks for enhancing security of mobile health applications. Vithanwattana et al. [14] examined individual information security frameworks for mHealth applications to provide a detailed domain analysis and identified the additional needs to provide mechanisms to support non-repudiation, accountability, and auditability. Hussain et al. [15] proposed a security framework that provides protection mainly through a set of security checks and policies and comprises of two layers: a security module layer (SML) and a system interface layer (SIL). Safavi et al. [16] formulated a framework that consists of ten principles and nine checklists, capable of providing complete privacy protection package to wearable device owners, while Munim et al. [17] proposed a conceptual framework for improving end user privacy by analysis the mobile data flow, resources sharing and mobile OS security.

A number of works highlighted the challenges faced to preserve security and privacy. Kotz et al. [18] expressed that a current challenge is to develop mechanisms that can automatically turn sensors on and off to preserve user privacy and can be personalized to minimize user burden while maximizing utility. Plachkinova et al. [19] grouped the security challenges into the following four categories: authentication and authorization (AA); integrity and accountability (IA); ease of use and availability (EUA); and confidentiality, management, and physical security (CMPS).

Some of the studies focused on the importance of meeting user needs. In [20], Mirkovic et al. mentioned understanding and addressing users' requirements is one of the main prerequisites for developing useful and effective technology-based health interventions. The results of this study outlined different user requirements related to the design of the mobile patient support application for cancer patients. Poole's [21] research highlighted that collaboration between behavioural science and human–computer interaction researchers may lead to more successful mobile health interventions, and for enhancing the acceptance of mHealth technologies, it is important to understand how they use mobile devices in daily life. McCurdie et al. [22] stated that the implementation of a UCD process in the development of mHealth tools is critical in ensuring user engagement, and consequently application effectiveness in terms of sustained behavioural change of the application users. To improve quality of the mHealth applications, Wicks et al. [23] discovered five potential approaches which are boosting application literacy, application safety consortium, enforced transparency , active medical review and government regulation. In [24], Biagianti et al. discussed that mHealth interventions are likely to bring good in long-term only if they provide intrinsic value for the user in managing their condition or improve critical aspects of their functioning and well-being and thus involving end-users in the co-development of digital interventions is very important to achieve this goal.

To summarize, the prior literature showed that security and privacy issues are of great concern in developing and adopting mHealth applications. There are no explicit set of guidance for application developers to be adopted during the application development phase to avoid security breaches as much as possible. Also no such work is found that adopted UCD-based approach to mitigate the privacy and security issues of mHealth applications. Thus, a UCD-based framework that addresses privacy and security needs of the users could be adopted by the developers of mHealth applications.

3 Research Methodology

The following steps were followed to conduct this research:

i. *Selection of mHealth Applications*: As example cases, mobile health applications developed in Bangladesh that are mostly downloaded or reviewed by the users in Google Playstore are considered in this research. This study is limited to Android applications only excluding the IOS or other operating system-based applications.

ii. *Finding Security and Privacy Threats*: Next, the security and privacy threats of the mHealth applications were explored in two ways. Firstly, a security testing tool *DROZER* was used to test the selected applications. Secondly, user reviews related to security and privacy of the selected applications were analysed using the noticing, collecting and thinking (NCT) model [25].

iii. *Developing a Framework*: After analysing the discovered security and privacy threats, the ones among those that are solvable by application developers during mHealth application development were listed out. Next those solvable issues were made to fall under the three elements—confidentiality, integrity and availability (of the CIA model). Next the listed out issues were studied to realize in which stages of UCD they can be addressed by a developer in the process of application development.

iv. *Framework Evaluation*: Finally to evaluate the effectiveness of the proposed framework, an mHealth application was developed following the guidelines and stages of the framework. The developed application and another existing application from the Google Playstore having similar functionalities were tested with 15 users following a *within subject test* approach to find out which application has better management of patient data in terms of privacy and security and to show the effectiveness of the proposed framework.

4 Exploring Security and Privacy Threats

For exploring the privacy and security issues of mHealth applications, 20 applications have been studied from the health domain developed in Bangladesh. The selected applications have been listed in Table 1. After selecting the applications, we inspected the privacy threats with “Drozer” which is a penetration testing tool that allows to check for vulnerabilities in Android applications. After analysis with Drozer, following two parameters were found:

Table 1 Summary of application testing results using Drozer

Application names	Sensitive permission used	Content providers exported	Broadcast receivers exported	Services exported
Doctorola	Camera, SD card, Bluetooth	(0)	(4)	(5)
Patient aid	Internet, SD card	(0)	(4)	(7)
MAYA	Bluetooth, SD card	(1)	(7)	(5)
DIMS	SD card	(0)	(3)	(6)
Amar Doctor	Read/write external storage, internet	(0)	(4)	(3)
MediInc	Camera, read/write external storage, internet	(0)	(2)	(1)
HealthBD	Read/write external storage, internet	Null	(0)	(0)
PlexusD	Read/Write external storage, internet	(0)	(3)	(4)
Shahstho Calculator	Read/write external storage, internet	(0)	(3)	(4)
Medicine App Bangla	Read/write external storage, internet	(0)	(3)	(0)
MSDacter	Internet	Null	Null	Null
DacterAchen	Read/write external storage	Null	Null	(1)
Medicine Directory BD	Read/write external storage	(0)	(3)	(2)
Dacterbhai	Read/write external storage	(0)	(6)	(4)
My Health	Read external storage, internet	(0)	(4)	(4)
IbnSina	Read/write external storage, internet	Null	Null	Null
Drugbook	Read/Write External Storage, Internet	(1)	(6)	(4)
BD Drug Directory	Internet	(0)	(2)	(1)
Pharmacy BD	Internet	(0)	(2)	(3)
PDM	Read/Write External Storage, Internet	(0)	(3)	(0)

i. *Sensitive permissions used by the application*: When an application is tested through Drozer, it gives data about the sensitive permissions used by the application. Some of the sensitive permissions are access to camera, voice recording, message, etc., taken from user during application installation leading towards a breach of privacy. Some irrelevant sensitive permissions used by an mHealth application detected by Drozer is marked in Fig. 1.

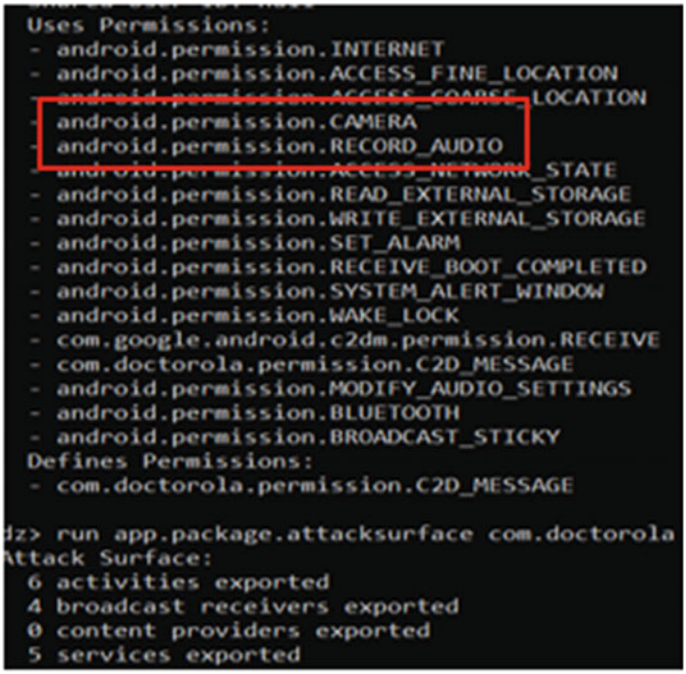

Fig. 1 Result of an application testing with Drozer

ii. *Attack prone areas*: From Drozer testing, we also found four attack-prone areas of the target applications, namely activities exported, broadcast receivers exported, content providers exported and services exported.

The summary of testing each of the 20 applications using *Drozer* is presented in Table 1. The results showed that each of the applications have security issues that falls under a single criteria (HealthBD) to multiple/all criterion (Drugbook). Here Null means those components were not used in the applications.

Again, user reviews of those applications were analysed based on the noticing, collecting and thinking (NCT) model [25]. Generally, in *NCT model*, at first the relevant data are collected carefully and then it is meticulously studied to notice related concepts/theme. In our study, user reviews related to privacy and security issues of the selected applications were collected from Google Playstore at first and then analysed based on NCT model. Reviews related to security and privacy like breach of trust, asking unnecessary permissions, faulty authentication, lack of access control, asking irrelevant personal information, UI loading issues, system login error or code bugs, etc., were primarily considered and analysed through NCT model. The security and privacy issues discovered through analysis of user reviews are presented in Table 2.

Table 2 Results of user review analysis (privacy/security related)

App name	Thematic (security/privacy issue related) code	Examples of user reviews
Patient aid	Sensitive permission	"The app wanted access to my media and photos unnecessarily"
DIMS	Leakage of sensitive information, Open for all, faulty registration	"App contents should have been restricted only to certified professionals"
Medinc	Faulty login	"Every time while I login, it fails making the app stop"
MEDshr	Private info demanded, lack of trust	"Medical certificates were demanded which could be used by the developers for illegal causes"
PDM	Password handling	"I can't reset password and also can't recover old password"
Daktarbhai	Verification issue	"OTP code is unavailable for doctor registration"
Bd drug directory	Misleading information	"It is mentioned in the app that SACMO are not medical person"
My health	Verification problem, misinformation, not reliable	"I can't add otp code. Also the sent code has no option for input, nearby hospitals not shown"

5 Proposed Framework

The proposed framework is developed based on the basic CIA and UCD model. The CIA model is designed to guide policies for information security based upon three fundamental terms including confidentiality, integrity and availability. Thus, the importance of considering its basic guidelines to develop a more specific framework to ensure privacy and security of mobile health applications is of no bounds. Since focus of this research is also to ensure user satisfaction of the application usage besides the security of user data, relying on basic user-centred design (UCD) model is also necessary to address the user needs at every stage of the development process. A broad description of how the proposed framework has been developed which is given below along with a brief idea of the basic CIA and UCD model.

Basic CIA Model [26]: The confidentiality, integrity and availability, also known as the CIA triad, is a model that guides the design/development of security principles. Confidentiality refers to hiding the sensitive data or keeping it private from unauthorized people. A common type of security attack is to intercept any sensitive data in illegal ways and make unexpected changes to it before sending it to a receiver. Integrity is to make sure that the data is never altered and is in its accurate form of the original secured information. Thus, consistency, accuracy and trustworthiness of data are integrity. It is important to ensure that the information needed is readily accessible to the authorized person at all times. Availability is to make sure that user information is available to the user anytime he wants. Often due to system crashes or failures, the required data is not accessible in the time of need.

Basic UCD Model [27]: A basic user-centred design model has the following stages for developing a system. Firstly, to develop a system it is essential to understand the purpose of use of a product or in what situation it is going to be used. That is to clarify the domain of the product to restrict the design for the intended users only. Secondly, it is necessary to identify the specific requirements of the users before prototyping or developing an actual design. Thirdly, the most crucial part of UCD approach is to design and develop the intented system as per the collected user requirements. Finally, it comes the evaluation phase. The developed system is tested by actual users to verify if all the requirements have been met or not. Another important stage of UCD is the post-development maintenance phase to ensure long-term wellness of the system.

However, the proposed framework consists of two parts which all together forms the entire security and privacy framework for mHealth applications. While developing the application, a developer needs to follow the step-by-step process of the second part. And to get an insight about the category (confidentiality, integrity and availability) of each of the security features to be ensured in the application development process (part two), he/she needs to look at the broad categorization in the first part of the model.

In the first part (Fig. 2) of the model, among the security aspects that are solvable by a developer during application development phase, they are classified under the components of the CIA model. Among those it is seen that some security and privacy aspects fall under multiple elements and some can be placed under a definite element. Figure 2 shows that which security aspects fall under which elements of a CIA model. Like encryption, privacy policy, etc., are under confidentiality solely. But authentication and access control fall under both confidentiality and integrity.

Next part of the model (Fig. 3) includes how developers can address these security features at different stages of the UCD process. Depending on the type of security aspect and the nature of the stages of UCD, following is the discussion of why each of these security/privacy aspects are to be handled in the respective stages of UCD:

Stage 1—Context of Use: The security feature *Access control* based on role (such as doctors, patients and visitors) should be considered during this phase. If an application has multiple users like doctors, patients, and nurses, then the access of all types of data is not required by every users of the application. The information concerned only should be exposed to the respective type of user. Hence in this phase, depending on the type of application to be developed, the developer needs to identify exactly what data to be shared with each type of users.

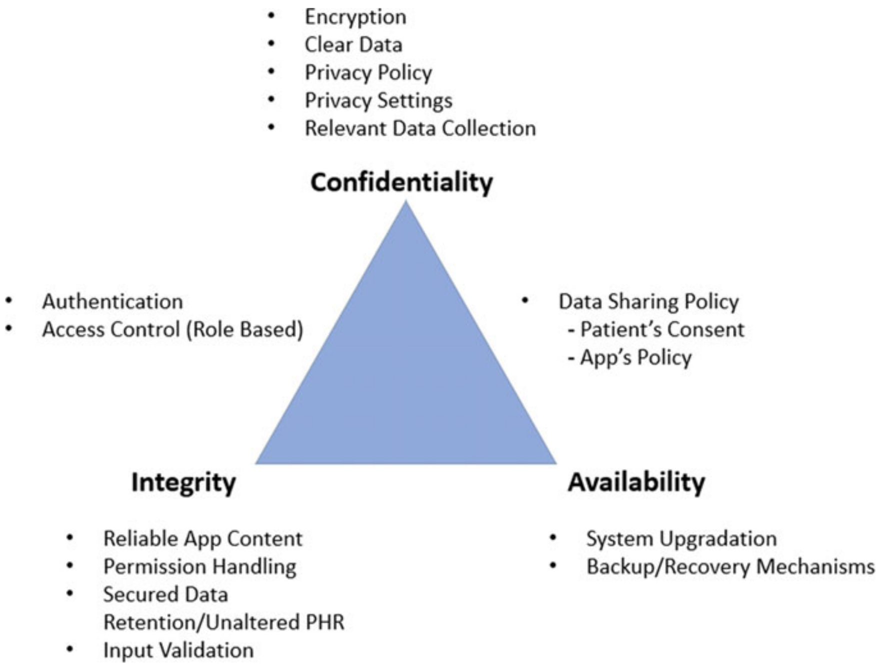

Fig. 2 Categorizing the security issues based on CIA model

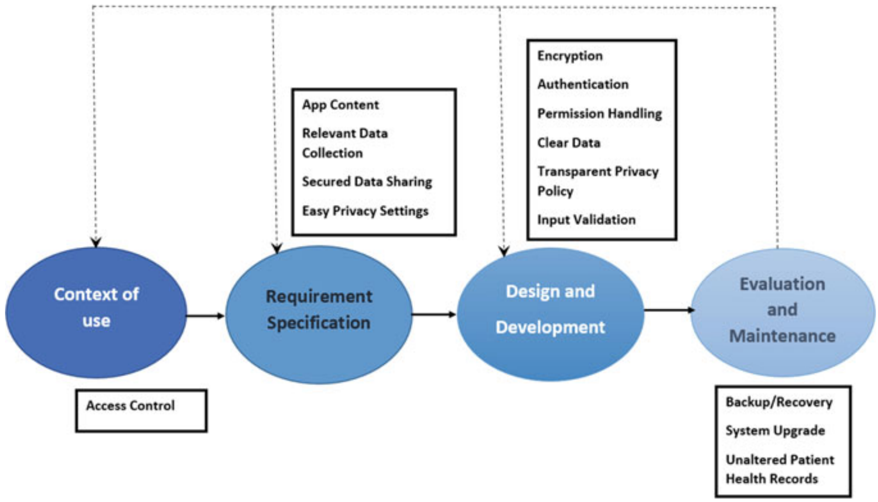

Fig. 3 Security features to be solved in different stages of UCD method

Stage 2—Requirements Specification: The following security features need to be handled in this stage. The feature *App Contents* (authentic application contents) means the contents or information provided by the application needs to be very transparent and authentic based on the exact user requirements. For this, the best way is to survey from the respective users what data they are comfortable with while using an application of a particular health domain. *Relevant demographic data collection* is another security feature which involves collecting only those personal information of a patient through the application which is required for any functionality of the application or any treatment purpose. At this phase of UCD, it is very important for the developers to identify the relevant data (of patient) as per the application's purpose. In case of any *Data Sharing,* which is another crucial feature, a developer must make it clear to the users about the data sharing policy of the application and take user opinion into account regarding how much comfortable they are about the data being shared by the application. *Flexible privacy settings* feature should be incorporated in any mHealth application to facilitate the user so that he/she can modify the data to be shared or hidden anytime he/she wants. These include options to share data to a doctor or reliable person, change recovery email/account anytime, clear data in the application, etc.

Stage 3—Design and Development: The main features to be considered in this stage are as follows: *Encryption* of sensitive information is a very crucial security feature and is an effective way to prevent misusing patient data. Developers need to make sure while coding to incorporate efficient encryption algorithms for hiding private information. *Authentication* which is another most important feature to ensure privacy is necessary while opening an application to make sure that the person is the actual user. It can be ensured using methods like fingerprint, password or pattern lock. In case of password lock, strong password requirements must be set during application development. Also, email authentication can be used. *Permission handling* feature during the development phase should be carefully done. Often many applications ask irrelevant permissions after downloading them which in the background keeps collecting users' personal information on the phone secretly which is an absolute breach of privacy. Also, the *Privacy policy* of an application should be clearly written and exposed to users during installation.

Stage 4—Evaluation and Maintenance: *Backup/Recovery* is a very crucial task in the UCD maintenance phase of mHealth application development to guarantee retrieval of data in emergency cases, thus ensuring availability need of an user. Backup can be either in the form of secured cloud storage or there can be options to recover the data if it timely gets restored in another reliable recovery account. *Unaltered PHR* (patient health record) maintenance means to make sure that the patient health data stored in the application is not changed by any intruder or malicious application. Also, *System upgrade* from time to time is needed to make sure that the login system works properly as well as there is no fault in the system usage and overall application performance to provide good user experience.

6 Framework Evaluation

For evaluation of the proposed framework, a reference application was selected from the applications' that we had analysed for exploring the privacy and security issues. The selected reference application is basically a medicine information and reminder application which uses some unnecessary permissions while installation. It belongs to an important category of mHealth applications and needs much improvement in terms of security and user experience. So we developed a similar category application (MediAid) following the proposed UCD-based framework.

In total, nine out of fourteen security and privacy features of the framework have been considered to develop the application. The other features were not included as the application did not need those features because of the category it was. A *Within-Subject* experiment was conducted to evaluate both the existing and the developed application. For participant selection, we selected users of age group 22–24. All the participants were from same educational background and were android phone users. There were in total 15 participants. From them, 8 of them were male and 7 of them were female. At first, a short briefing was given to the users about the experiment. Then we gave them an application to test randomly (Fig. 4).

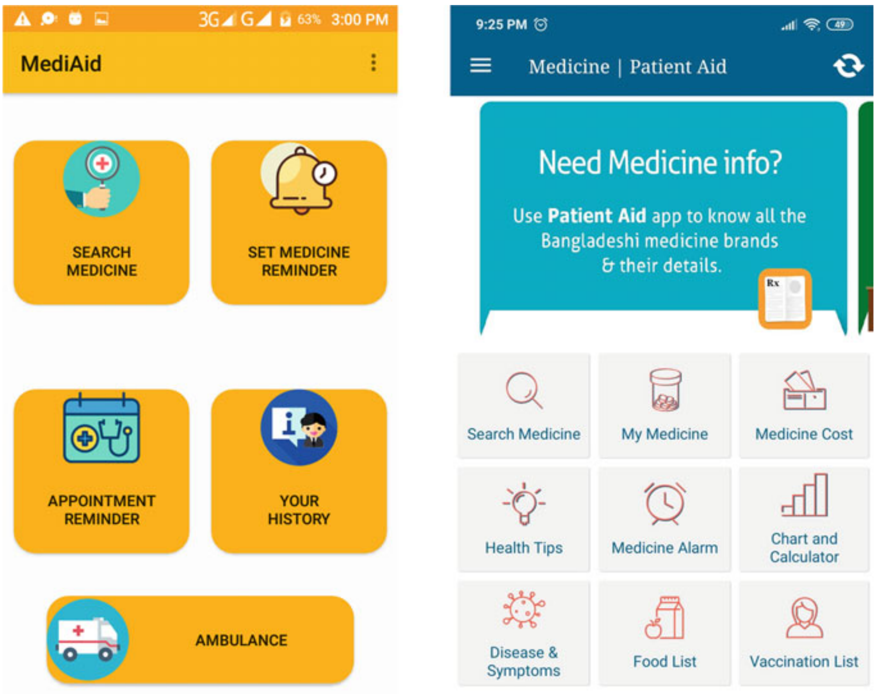

Fig. 4 Application interface of MediAid (left figure) and Reference application (right figure)

Table 3 Security and privacy related questionnaire

Qs. no.	Question description
1.	The app ensures confirming user identity through proper authentication method
2.	The app has a well defined privacy policy that clearly mentions how it handles patient data sharing policy
3.	The app allows it's users to have control over his/her data privacy through adjustable app settings
4.	No irrelevant or unnecessary data is collected from the user by the app
5.	No irrelevant permissions (e.g. Camera, Video/Audio record, etc.) is used by the app
6.	Patient data is backed up by his/her consent and can be recovered easily in case of any system failure
7.	The app has scope for cleaning previous patient data ensuring no data retention
8.	The contents and information available on the app are reliable and relevant

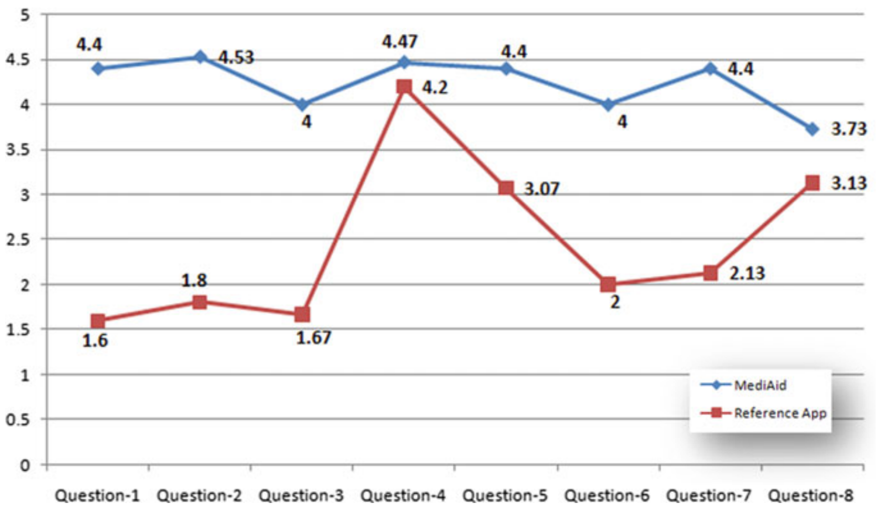

Fig. 5 Mean curve of participants responses

A total of eight post-test questionnaires (Table 3) were designed for both the applications based on privacy and security issues and one more question about the overall privacy aspect of the application. The eight questions were designed as per the *Likert Scale* having five options starting from strongly agree, agree, neutral, disagree to strongly disagree.

The study results (Fig. 5) showed that the participants agreed most of the time with the privacy and security aspects handled by the developed *MediAid* application than the reference application. The results thus indicated that the privacy and security issues have been better maintained by the *MediAid* application developed based on the proposed framework compared to the reference application.

7 Conclusions

A practical and user-dependent framework for mHealth applications is proposed here that ensures involving users at every stage of the application development process to best realize their needs of privacy and security. The proposed framework helps to check the privacy and security issues at different stages of the UCD-based development cycle for enhancing the security, privacy, and user satisfaction which in turn will improve the over all user acceptance and long-term usage of mHealth applications. This research has a few limitations. The application developed following the proposed framework has been tested in a restricted academic environment with a limited number of users (15). Thus in future, it needs to be tested by sufficient number of participants of varied age group to see its efficiency in a broader scale. Again the developed application could handle nine out of fourteen security heuristics in the framework due to the fact that this application did not had to include those rest features because of the category it was.

In future, this framework can be incorporated with some usability checklists to measure its usability and user experience-related satisfactions along with ensuring the security and privacy factors. This will be a challenge to bring a good trade-off between usability and security. Also, this framework should be tested on applications that can include all the mentioned security and privacy checklists to understand its effectiveness better.

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Chapter 18
Improved Bengali Image Captioning via Deep Convolutional Neural Network Based Encoder-Decoder Model

Mohammad Faiyaz Khan, S. M. Sadiq-Ur-Rahman, and Md. Saiful Islam

1 Introduction

Automatic image captioning is the process of generating a human-like description of an image in natural language. It is a significantly challenging task as it requires identifying salient objects in the image, understanding their relationships, and generating relevant descriptions of these image features in natural language. The process of generating captions of the images can be applied to automate self-driving cars, implement facial recognition systems, aid visually impaired people, describe CCTV footage, improve image search quality, etc. Despite numerous research attentions in encoder-decoder-based image captioning in the English language, a few works are done in BIC. Researches in these languages can have far-reaching consequences in solving many region-based socio-economic problems.

The fields of computer vision and natural language processing have seen significant progress due to the recent development in deep learning. The task of image captioning lies at the intersection of these two. Most notable works are based on the encoder-decoder framework, which is very similar to the sequence-to-sequence model for machine translation [1]. The framework contains a CNN-based image feature extractor and a recurrent neural network (typically LSTM [2]) based caption decoder to generate the relevant words iteratively. The decoder's job is to take in the caption generated so far and predict the next word with the highest probability among

M. Faiyaz Khan · S. M. Sadiq-Ur-Rahman · Md. Saiful Islam (✉)
Dept of Computer Science & Engineering, Shahjalal University of Science and Technology, Sylhet, Bangladesh
e-mail: saiful-cse@sust.edu

M. Faiyaz Khan
e-mail: mfaiyazkhan@student.sust.edu

S. M. Sadiq-Ur-Rahman
e-mail: shifathrahman472533@gmail.com

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all the vocabulary words until an ending token is generated. All the existing works in the Bengali language [3, 4] follow the same architecture, as mentioned above. In [5], there is a comparison between the two architectures of image captioning. The first one is the **inject** architecture, where the RNN is used as a caption generator conditioned by the image features. The second one is the **merge** or **mixture** architecture, where the RNN is primarily used for encoding linguistic representation only. The encoded linguistic features and the image features are then merged and passed as input to a multimodal layer that performs the word by word prediction of the caption. The comparative study shows that models with merge architecture serve better than models with inject architecture. In natural language processing tasks like chunking, part-of-speech tagging, and named entity recognition, CNN-based models have provided faster and accurate results [6]. Also, in [7], it has been shown that CNN-CNN models are competitive in performance with CNN-RNN models in terms of image captioning.

Inspired by the aforementioned successes of the fusion model and CNN in NLP, we propose an encoder-decoder-based model following merge architecture for Bengali image captioning. We used ResNet-50 [8] for encoding image features and a one-dimensional CNN for encoding the linguistic features. Unlike [7], the CNN used in our work is followed by a pooling layer for capturing meaningful and significant features. Later, both the image and language features are mingled and passed to the decoder to generate the image's caption (Fig. 1). We evaluate our work on the BanglaLekhaImageCaptions dataset [10]. The experimental results show that our model performs better than all the existing models in the Bengali language. We also conducted a qualitative and quantitative comparison between our CNN-CNN based mixture model and the CNN-LSTM based mixture model proposed in [5]. The experimental results confirm that the CNN-based language encoder is responsible

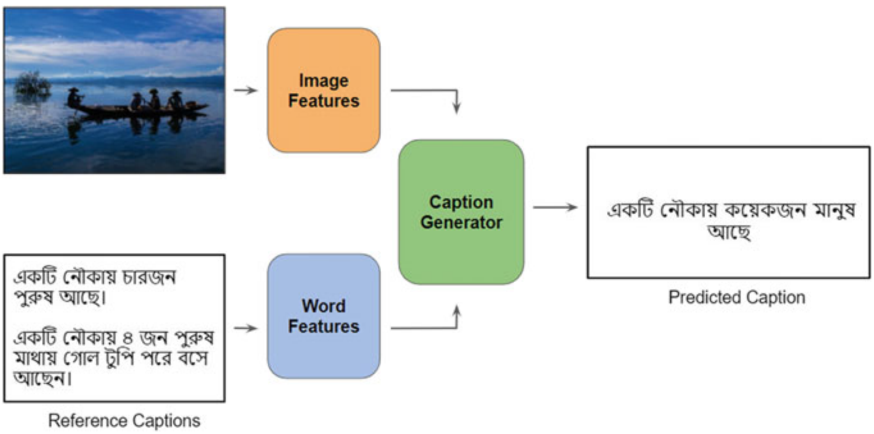


Fig. 1 Illustration of sequence-to-sequence basic architecture where image and linguistic features are merged to generate meaningful captions in the Bengali language

for the proposed model's overall better performance. In summary, the main technical contributions of this paper are the following:

- We present CNN instead of regular LSTM to learn linguistic information and use it for word prediction during the caption decoding phase. Meanwhile, we use ResNet-50 [8] architecture as an image feature extractor.
- We qualitatively and quantitatively evaluate our approach on the BanglaLekhaImageCaptions dataset.
- Our model achieves SOTA performance on the BanglaLekhaImageCaptions dataset and outperforms the existing encoder-decoder models while describing complex scenes. We also present the human evaluated score for qualitative evaluation of the generated captions. The code is available on Github.[1]

2 Related Works

In the early years of research in image captioning, many complex systems consisting of primitive visual object identifiers and language models were used [11]. These systems were rule-based and predominantly hand-designed. Moreover, these systems worked only on a limited domain of images. In [12], image captioning was treated as a machine translation task. But this system failed to capture the fine-grained relationship among the objects in the image. Along with the advancements of deep learning methods, image captioning systems produced considerably improved performances following the same deep learning based architecture as machine translation [1, 13]. These works adopted the same encoder-decoder framework [14–16] and framed the idea of image captioning as translating the image into text. These systems used CNN for encoding images and RNN for decoding the images into sentences. Later, attention mechanisms were introduced to mimic the human behavior of capturing only the important features in an image and translating them into a natural language description [17, 18]. These systems generated the captions conditioned by the attention at a specific place of the image at each time step. Most of the systems built for English language are evaluated on MSCOCO [19], Flickr30k [20] and Flickr8k [21] datasets.

Researches on other languages like Japanese [22], Chinese [23], German [24], Arabic [25], etc. have also been performed. Most of these research works are experimented and evaluated on the translated versions of the MSCOCO [19] and Flickr8k [21] datasets in their respective languages. In [3], a Bengali image captioning dataset [10] was introduced along with a model which is very similar to [14]. While the results generated are not accurate enough, but it surely instigated further research works in Bengali. In [4], a comparative analysis of the existing encoder-decoder LSTM decoder based models was presented. The models were evaluated on a trimmed down, machine-translated version of the Flickr8k [21] dataset in Bengali. The Bengali captions generated, however, do not maintain the typical Bengali sentence structure and

[1] https://github.com/FaiyazKhan11/Improved-Bengali-Image-Captioning-via-deep-convolutional-neural-network-based-encoder-decoder-model.

lacks usability. In this work, we present an encoder-decoder model with a CNN language encoder. We also provide experimental results on the BanglaLekhaImageCaptions dataset comparing our work with the existing LSTM based models.

3 Dataset

We trained and evaluated our model on the BanglaLekhaImageCaptions dataset [10]. It is a modified version from the one introduced in [3]. It contains 9,154 images with two captions for each image. The captions are generated by two native Bengali speakers. While this data set is not big in volume compared to the existing datasets in the English language, it maintains relevance with the Bengali culture to some extent. But the dataset also has a considerable amount of human bias. This bias hinders any model's ability to describe non-human subjects. Also, the captions are not detailed in some cases, which causes the training and evaluation of any model to be not as accurate as expected.

To train our model, we divided the data set into three parts, which are train, test, and validation. For training, we used 7154 images. 1000 images were used during validation, and the rest 1000 images were used during testing.

4 Model

The model is based on encoder-decoder architecture. A two-dimensional convolutional neural network is used to encode the image features, and a one-dimensional convolutional neural network is used to encode the word sequences of the caption data. Later, both the encoded image and text features are merged and passed to a decoder to predict the caption in a word by word manner (Fig. 2)

The model is divided into three parts, which are given as follows:

- **Image Feature Encoder**: We used pre-trained ResNet-50 [8] as image feature extractor. It is trained on ImageNet dataset [9]. Traditionally, neural networks with many layers tend to perform well in recognizing patterns. However, they also suffer from overfitting issues and are not easy to optimize. Residual CNNs are designed to have shortcut connections between layers. These connections perform identity mapping. ResNets are easy to optimize, and their performance increase with increasing network depth. We discard the final output layer of the ResNet-50 as it contains the output of image classification and used only the encoded image features produced by the hidden layers.
- **Word Sequence Encoder**: Two-dimensional convolutional neural networks have been extensively used in pattern recognition, image classification, and time series forecasting. The same property of these networks can be used in sequence processing. In our model, we used one-dimensional CNN for extracting one-dimensional

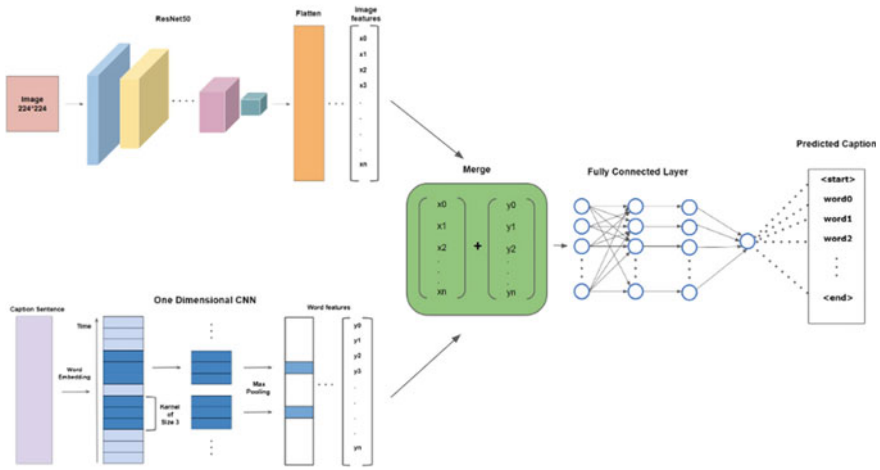


Fig. 2 The overview of the framework of our proposed CNN-ResNet-50 merged model, consisting of a ResNet-50 as image feature extractor and 1D-CNN with word embedding for generating linguistic information. Later, these two features are given as inputs to a multimodal layer that predicts what to generate next using this information at each time step

patches from a sequence of words. The CNN has 512 filters with a kernel size of 3. The activation used is Rectified Linear Units (ReLU). The CNN is followed by a Global Max Pooling Layer, which captures critical features from the convolutional layer's output.

- **Caption Generator**: The caption generator is a simple decoder containing a Dense 512 layer with ReLU activation. The output of the image feature encoder and word sequence encoder are combined by concatenation and used as input to the dense layer. The dense layer generates a softmax prediction for each word in the vocabulary to be the next word in the sequence, and the word with the highest probability is selected. This process continues until an ending token is generated.

The caption generator's output is transformed into a probability score for each word in the vocabulary. The greedy method chooses the word with the highest probability for each time step. This method may not always provide the best possible caption as any word's prediction depends on all the previously predicted words. So, it is more efficient to select the sequence with the highest overall score from a candidate of sequences. So we use the beam search technique with a beam size of 5. It considers the top five candidate words at the first decode step. For each of the first words, it generates five-second words and chooses the top five combinations of first and second words based on the additive score. After the termination of five sequences, the sequence with the best overall score is selected. This method allows the process to be flexible and generate consistent results.

5 Result and Analysis

This section provides a quantitative and qualitative analysis of the performance of our model. While the evaluation metrics give a numeric idea about the captions' correctness, they can sometimes misinterpret any result. Qualitative analysis can evaluate the subtle difference in the generated captions compared with the natural human language description.

5.1 Quantitative Analysis

Predicted captions from the models were evaluated using the existing evaluation metrics **BLEU** [26] (Bilingual Evaluation Understudy), **METEOR** [27] (Metric for Evaluation of Translation with Explicit Ordering), **ROUGE** [28] (Recall-Oriented Understudy for Gisting Evaluation), **CIDEr** [29] (Consensus-based Image Description Evaluation) and **SPICE** [30] (Semantic Propositional Image Caption Evaluation).

A comparison among our model, inject architecture-based model with CNN language encoder, mixture architecture-based model with LSTM language encoder, and the model proposed in [3] (Bi-directional LSTM language encoder with inject architecture) can be found in Table 1. We replicated the model of [3] using the same ResNet-50 as image feature extractor instead of the VGG-16 [34] used in the original work to make sure that the better performance of our model is not only due to the better image model. We also present the scores of both the greedy and beam search method. From Table 1, it can be seen, our model based on CNN word sequence processor has achieved better results in all the metrics than the traditional LSTM-based models with both mixture and inject architectures and the CNN-based models with inject architecture. Our model's superior performance can be attributed to the one-dimensional CNN model we used for sequence processing with the merge architecture. We used a window of size 3 for CNN. This window size with merge architecture enabled our model to learn words or word fragments of size 3. As a result, the fine-grained information present in the captions is learned during training. Following the CNN layer, the pooling layer filters only the significant features, which means the correlation between the words is stored better. Besides, this combination of one-dimensional CNN as a sequence processor with merge architecture can remember more diversified words while generating captions. These are evident in the comparison of the quality of the captions generated by the models in Fig. 3. The scores with the highest accuracy have been shown in table 1 with boldface. Among the evaluation metrics the most important metrics for evaluating image captions are **CIDEr** [29] and **SPICE** [30] since these are specially prepared for evaluating image captions. Better scores in these two metrics indicate the quality of performance of our model. The scores of the evaluation metrics were calculated using pycocoevalcap[2]

[2]https://github.com/salaniz/pycocoevalcap.

Table 1 Quantitative analysis of performances among different models

Search type	Models	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE
Greedy	Our model	**0.651**	**0.426**	**0.278**	**0.175**	**0.297**	**0.417**	**0.501**	**0.357**
	CNN + LSTM [mixture] [5]	0.632	0.414	0.269	0.168	0.291	0.395	0.454	0.350
	CNN + Bi-LSTM [inject] [3]	0.619	0.403	0.261	0.163	0.296	0.380	0.433	0.344
	CNN + CNN [inject]	0.538	0.347	0.228	0.145	0.250	0.378	0.318	0.334
Beam	Our Model	**0.589**	**0.395**	**0.267**	**0.175**	**0.294**	**0.434**	**0.572**	**0.353**
	CNN + LSTM [mixture][5]	0.562	0.381	0.257	0.166	0.286	0.423	0.558	0.345
	CNN + Bi-LSTM [inject][3]	0.575	0.374	0.241	0.149	0.286	0.412	0.532	0.349
	CNN + CNN [inject]	0.433	0.287	0.185	0.113	0.255	0.386	0.328	0.324

library for python 3 available in github[3] which is a support for MS COCO caption evaluation tools [31].

The performance comparison among our model and other models can be seen in Fig. 3. We present the predicted Bengali caption and corresponding English-translated caption for non-native Bengali speakers. Our model performed better not only in the scores but also in the quality of captions. In Fig. 3a, it is observable that our model predicted the most relevant caption compared to other models describing the gender of the human subject and the work he is doing in the image. In Fig. 3b, our model detected the gender correctly and captured the person's age range. It is also noticeable that all the metrics' score, including the SPICE, is better, which indicates the better quality caption.

5.2 Qualitative Analysis

Some samples along with the English translation are depicted in the Fig. 4 to represent the quality of captions generated. For qualitative evaluation, two native Bengali speakers were asked to give each caption a score between 0 and 1 from a set of sample images randomly selected from the test set. The scores are then averaged to generate the overall human evaluated score. In Fig. 4a, the model generated a decent caption correctly predicting the number of humans considering the human subjects' small size. In Fig. 4b, our model described the expression of face and the person's age range, i.e., child correctly. The caption is also well detailed. Also, in Fig. 4c, d, the model described the content of the image in a very similar way to human description and hence, has achieved good evaluation scores.

In Fig. 5, some wrong predictions made by our model are shown. In Fig. 5a, it could not correctly describe the content of the image at all, and in Fig. 5b, it misinterpreted the birds as people.

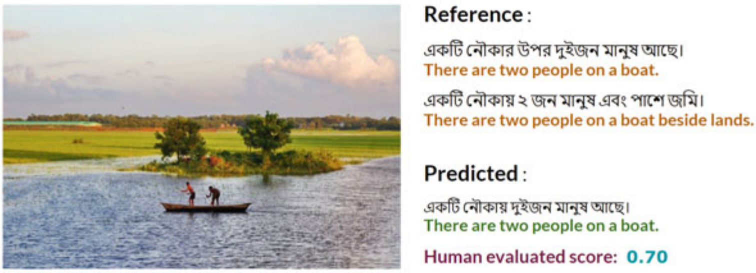
The faulty predictions made by our model can be attributed mainly to the dataset introduced by [3]. In this dataset, most of the images contain human subjects which have almost similar types of captions. As a result, the model is trained with a massive amount of similar types of human subjects and hence fails to detect and describe non-human subjects during testing. Besides, number of captions for each image is only two compared to five in the widely used English datasets. The dataset contains a significant amount of spelling mistakes in Bengali caption, which is also evident in the examples above. All of these are indicative of lacking details and variety in the dataset.

[3]https://github.com.

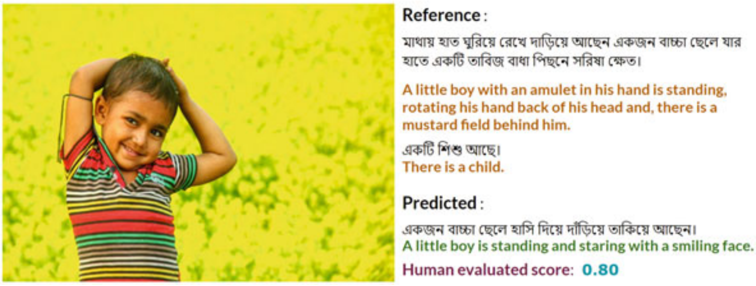
Fig. 3 Comparison among the captions generated by different models

6 Conclusion

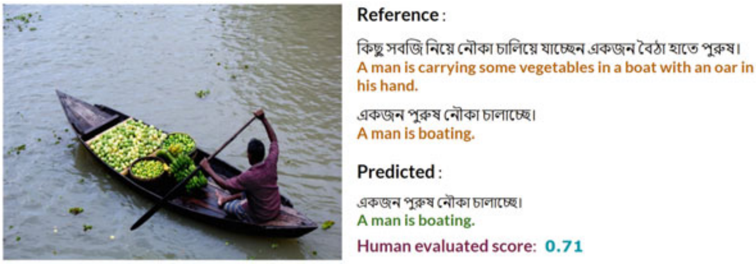
This paper presents a CNN-CNN merged encoder-decoder-based image captioning system instead of a traditional sequence-to-sequence model. A substantial test conducted on the BanglaLekhaImageCaptions dataset with superlative performance validates the efficacy of our proposed model. Additionally, experimental results show that the CNN language model, combined with the merge architecture, captures the fined-grained sentence structure information with better linguistic diversity and produces more accurate and humanoid captions than the traditional LSTM. Nevertheless,



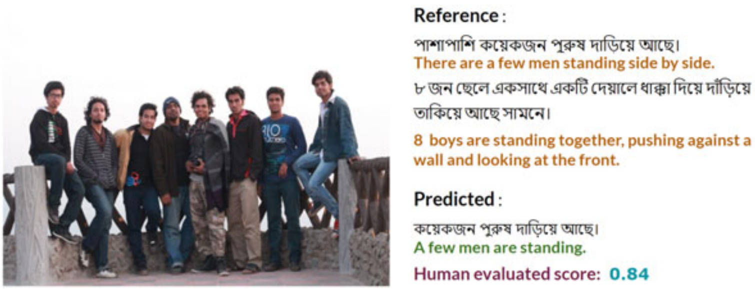
(a)



(b)



(c)



(d)

Fig. 4 A glimpse of captions predicted by our model with qualitative evaluation

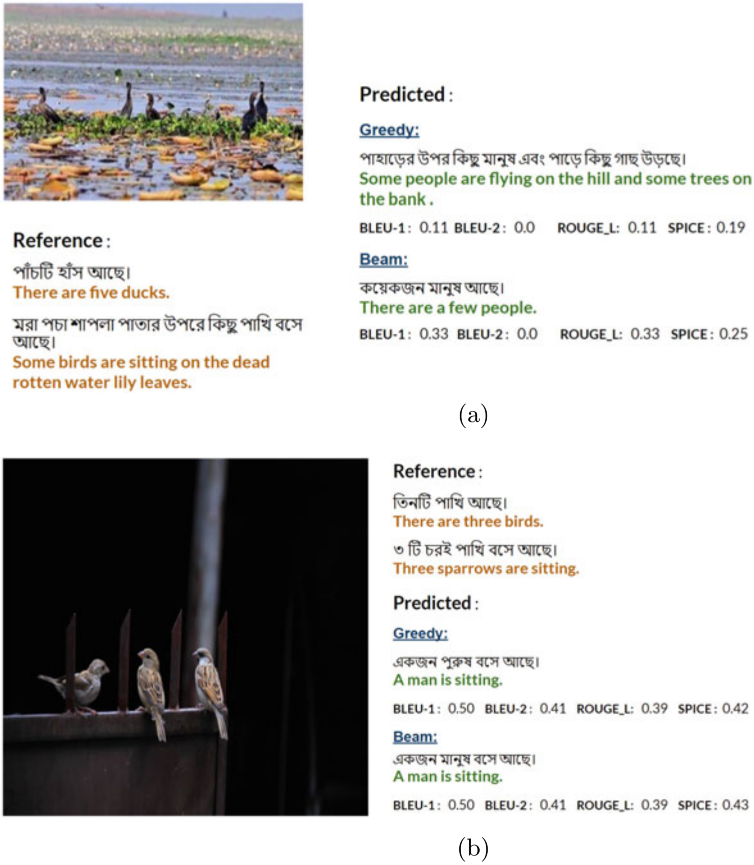


Fig. 5 Few incongruous captions generated by our model

the proposed model suffers from recognizing non-human subjects as the dataset is biased towards human subjects. This leaves us desired for a well-varied and detailed captioned dataset for Bengali Image Captioning. Therefore, we are motivated to develop a gold standard image caption dataset for Bengali for future work. Using other search methods such as constrained beam search in the decoding phase can be an area of future work. Besides, multilingual transformers available in NLP (Natural Language Processing) tasks like BERT [32], XLM [35], XLNet [33], etc. can generate promising results in the Bengali language.

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Chapter 19
Extract Sentiment from Customer Reviews: A Better Approach of TF-IDF and BOW-Based Text Classification Using N-Gram Technique

Tonmoy Hasan and Abdul Matin

1 Introduction

Customer satisfaction reflects the success of any business organization and from customer reviews, the organizations can know how much their customers are satisfied [1]. For restaurants or food businesses, it is difficult to make space in the business market without considering customer's feedback. Organization should pay attention to customer's reviews and improve their services in order to allure more and more customers. But it is a very onerous and lengthy job for the human to scrutinize a huge amount of reviews. However, sentiment analysis can easily handle this task. Opinion mining is another introduction of sentiment analysis that aims to attempt the determination of polarity (i.e., positive, negative, or neutral) of text, document, or paragraph [2].

In this paper, we have used three feature analysis techniques such as BOW, TF-IDF and N-Gram. Different N-Gram techniques, for instance, unigram, bigram, trigram, and incorporation of unigram and bigram (Unigram + Bigram), incorporation of bigram and trigram (Bigram + Trigram), and incorporation of unigram, bigram, and trigram (Unigram + Bigram + Trigram) are applied for both BOW and TF-IDF. To get classification result, RF, SVM, NB, and voting ensemble have been employed for sentiment extraction of food reviews. Precision, recall, f1-score, and accuracy are considered as the performance parameter of the techniques.

T. Hasan (✉) · A. Matin
Department of Electrical & Computer Engineering (ECE), Rajshahi University of Engineering & Technology (RUET), Rajshahi 6204, Bangladesh

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1.1 Research Objectives

In this paper, we have proposed an efficient method to classify sentiment of customer reviews to mitigate the following objectives:

(i) Identify the best N-Gram feature for customer sentiment classification.
(ii) Finding out the better one between TF-IDF and BOW feature extraction technique.
(iii) Impacts of voting ensemble classifier on overall performance.

2 Literature Review

Fouad et al. [3] have analyzed sentiment classification and considered for four datasets. BOW, Lexicon-based features (Lex), and Emotion-based features (Emo) have been applied for these datasets. Different machine learning algorithms such as SVM, Logistic Regression (LG), and NB are considered for different combinations of feature extraction techniques. Positive and negative tweets can be detected through their machine learning model. Furthermore, they have also claimed that their proposed system would be effective for marketing, political polarity detection, and reviewing products.

Qamar et al. [4] have considered tweets of various Saudi telecommunication companies such as STC, Zain, etc. for analyzing sentiment. These tweets have been labeled as positive, negative, and neutral through different supervised learning classification algorithms and feature selection formulas. They have gone through NB, K-Nearest Neighbor (KNN), Artificial Neural Network (ANN) as machine learning algorithms and CfsSubsetEvaluation, N-Gram, Info Gain for feature selection techniques.

Elghazaly et al. [5] have used SVM and NB as machine learning techniques in order to determine positive and negative sentiment for Arabic text classification. Political tweets of presidential elections in Egypt 2012 have been analyzed to compare the results of SVM and NB classifiers using several performance parameters (F-measure, Recall, and Precision). They have gone through TF-IDF model as feature extraction technique and their main concerns are accuracy and time. In their research, however, they have found that NB is a more suitable classifier for their main aspects.

Bespalov et al. [6] have analyzed sentiment depending on deep neural network technique for two datasets of online product reviews concerning higher-order N-gram. BOW, N-Gram, and Inverse Document Frequency (IDF) are used for Amazon and TripAdvisor customer reviews. 20,000 and 3000 reviews from Amazon and TripAdvisor ,respectively, have been used and a proportion of 70%/30% has been maintained for train and test purposes.

Zul et al. [7] have explored social media for sentiment analysis conducting K-Means and NB algorithms. Facebook and Twitter are the primary sources for their data. The opinions collected from social media have been labeled by Sentiwordnet. In their research, BOW is used as the feature selection technique. They have found that better accuracy can be achieved through NB alone rather than a combination of NB and K-Means.

3 Proposed Framework and Methodology

The overall view of our proposed system is portrayed in Fig. 1. Initially, data has been collected and preprocessing is accomplished undergoing various steps such as removal of URLs, converting to lowercase, tokenization, removal of expressions, removal of punctuation, removal of stopwords, and stemming. Then the text data has been converted to numeric data through different feature extraction techniques (i.e., TF-IDF, BOW, and N-Gram). After the conversion, individual classifiers such as RF, SVM, NB, and voting ensemble classifiers are applied in an attempt to get classification results. According to the outcomes, business policy and product quality would be updated by business organizations having identified negative reviews.

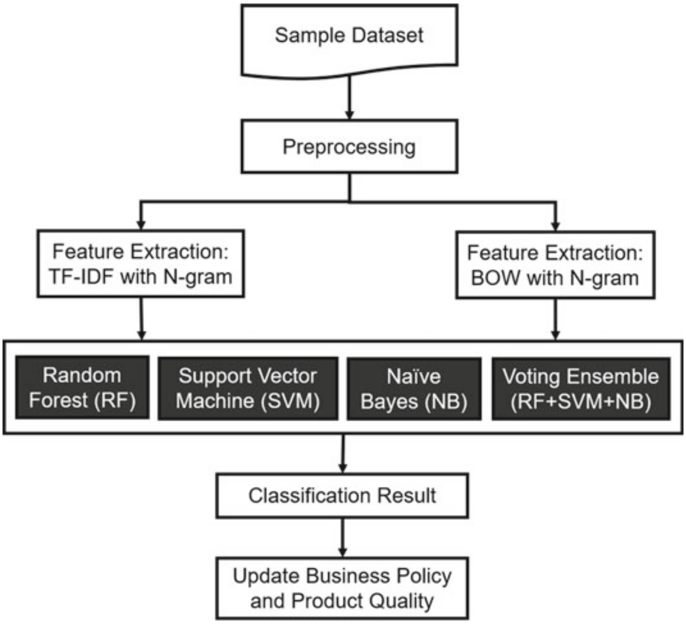

Fig. 1 Proposed system framework

3.1 Dataset

Dataset has been collected from Kaggle named "Sentiment Analysis Classification" [8]. The dataset contains different customer's reviews of foods incorporating total 18,532 reviews for training and testing. We have randomly taken 10,000 reviews for the experiment where 7105 labeled as positive and rest 2895 labeled as negative.

3.2 Preprocessing

Data preprocessing is important to get cleaned data from raw information. Some important steps of preprocessing are:

- Removal of URLs: It is the process of removing all the URLs from the text as they are unnecessary for sentiment classification.
- Converting to lowercase: Converting to lowercase all the text data significantly helps with the consistency of desired outcomes. In natural language processing(NLP), it is one of the most common text processing phases used in research purview.
- Tokenization: It is the step of splitting the reviews into smaller units. These units can be as part like a word is a unit in a sentence and a sentence is a unit in a paragraph. For sentiment analysis, normally word tokenization technique is followed, for example like I enjoyed the food very much to be [I, enjoyed, the, food, very, much].
- Removal of expressions: Expressions don't contain valuable information for sentiment classification. So, removal of expressions is performed as a preprocessing step in order to prepare more clear data.
- Removal of punctuation: Another common preprocessing step in natural language processing is removal of punctuation. It is the process of removing all the punctuation from the text.
- Removal of stopwords: Stopwords are commonly used words in a language like "a", "is", "the", etc. Removal of stopwords from the text doesn't affect the meaning as they don't provide significant information.
- Stemming: Stemming is the preprocessing of shortening the modified words to their word stem, base, or root form. For example, playing can be play, cooked can be cook, disappointed can be disappoint, etc. (Table 1)

3.3 World Cloud Generation

It is a leading process used in NLP for representing text data in a cloud of words [9]. The word clouds have been shown in Figs. 2 and 3 for positive and negative reviews

Table 1 Review before and after Preprocessing

Before preprocessing	After preprocessing
I was very disappointed, I didn't think the coffee lived up to its description. I won't buy it again	Disappoint think coffee live descript buy

Fig. 2 Word cloud for positive reviews

Fig. 3 Word cloud for negative reviews

respectively. In this cloud, the size of the word demonstrates the frequency of the word. More frequently appeared words are larger than less frequent words. It should be noted that only positive words have not appeared in the positive cloud and only negative words have not appeared in the negative cloud rather than the words which are more frequently used in the positive and negative reviews appeared in the positive and the negative word cloud, respectively.

3.4 Feature Extraction

Term Frequency-Inverse Document Frequency (TF-IDF): TF-IDF is one of the most observable feature extraction techniques of NLP which is especially for text modeling [10]. Term frequency indicates the frequency of a specific term or word occurs in a document whereas inverse document frequency indicates the occurrences of the word in all documents. The formula for TF-IDF follows as:

$$tfidf_t = f_{t,d} \times \log \frac{N}{df_t} \tag{1}$$

where,
$tfidf_t$ = weight of term t
$f_{t,d}$ = frequency of term t in document d
N = total number of documents
df_t = number of documents containing the term t.

Bag of Words (BOW): BOW is a very common technique of extracting features from text data [11]. It helps to alter from text to number by counting the occurrence of words within a document and machine learning algorithms work with these numerical data. For example,

- Text 1: They have the best price and best service.
- Text 2: Best service ever.

The vocabulary consists of these eight words: "They", "have", "the", "best", "price", "and", "service", "ever" (Table 2).
Vector of Text 1: [1, 1, 1, 2, 1, 1, 1, 0]
Vector of Text 2: [0, 0, 0, 1, 0, 0, 1, 1]

N-Gram: In natural language processing, N-Gram technique is a frequently used method where n indicates a continuous number of terms or words[10]. For n=1, it indicates unigram. Similarly, bigram and trigram can be indicated for $n = 2$ and $n = 3$ respectively. If we look upon a text such as "The foods are mind blowing" then N-Gram
Unigram: "The", "foods", "are", "mind", "blowing".
Bigram: "The foods", "foods are", "are mind", "mind blowing".
Trigram: "The foods are", "foods are mind", "are mind blowing".

Table 2 Bag of words representation

	1. They	2. have	3. the	4. best	5. price	6. and	7. service	8. ever
Text 1	1	1	1	2	1	1	1	0
Text 2	0	0	0	1	0	0	1	1

Different N-Gram schemes have been used along with TF-IDF and BOW feature extraction techniques separately in order to fit supervised machine learning models.

3.5 Classifier

Random Forest (RF): RF is a supervised classifier in which decision tree is the elementary unit aiming to perform both classification and regression jobs [12]. It is based on ensemble learning and is made up of a wide number of single decision trees. For the final output, the prediction of each decision tree is considered and depending on the maximum number of votes, result with the highest performance is predicted.

Support Vector Machine (SVM): SVM is one of the most prominent classification algorithms among all the supervised machine learning algorithms that explore data and helps to acknowledge patterns. In this method, binary classification has been accomplished by drawing a hyper-plane that cuts apart the document vectors and maximum distance is maintained between hyper-planes [13]. This is the primary objective of SVM.

Naive Bayes (NB): NB is one of the most powerful and straightforward supervised machine learning algorithms in which classification is accomplished by adopting the Bayes Theorem [14]. It is not only very fast but it also boosts up the performance surprisingly. Moreover, it is a probabilistic algorithm which gives very efficient result for both binary as well as multiclass classification.

Voting Ensemble Classifier: Ensemble classifier is a method of combining multiple individual models in order to secure better performance. Voting ensemble is one of the simplest methods of ensemble learning. In this technique, individual classifier makes a prediction which is considered as a vote and according to the number of majority vote, the final prediction is accomplished [14].

3.6 Performance Parameters

For a classifier, four possible assessments are true positive (TP), false positive (FP), true negative (TN), and false negative (FN). TP denotes the number of reviews that are classified as positive and their actual class is also positive. FP expresses the number of reviews which are classified as positive but their actual class is negative. For the negative reviews, TN and FN are stated similarly. Again, performance parameters, e.g., precision, recall, f1-score, and accuracy are calculated in virtue of these outcomes.

Precision: It is known as the proportion of the number of items exactly identified as positive divided by the total number of items that are classified as positive [12].

Recall: It is the proportion of the number of events exactly identified as positive divided by the total number of exact positive events [12].

F1-Score: It can be achieved from the following equation [12]:

$$\text{F1-score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Accuracy: Accuracy can be obtained from the following equation [12]:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

4 Result and Discussions

In this section of our study, a comparative result has been pictured in order to look out the performances of different techniques. Again, a comprehensive graph analysis has been shown to visualize more deeply. Tables 3 and 4 summarizes accuracy and f1-score respectively for different N-Gram approaches using both TF-IDF and BOW separately. However, the dataset we have used has two labels: positive and negative. In our experiment, total 2500 words are considered as features for all the feature

Table 3 Accuracy of different classifier using features combination

N-Gram	TF-IDF				BOW			
	RF	SVM	NB	Ensemble	RF	SVM	NB	Ensemble
Unigram	0.846	0.861	0.8095	0.8525	0.854	0.8545	0.8505	0.8675
Bigram	0.7835	0.796	0.8015	0.806	0.723	0.7735	0.7945	0.7945
Trigram	0.737	0.746	0.7395	0.7475	0.7135	0.729	0.7275	0.731
Unigram + Bigram	0.8505	0.867	0.8345	0.8575	0.8555	0.851	0.861	0.87
Bigram + Trigram	0.778	0.7925	0.7965	0.801	0.7265	0.7695	0.791	0.793
Unigram + Bigram + Trigram	0.8425	0.864	0.821	0.8555	0.8515	0.8455	0.8615	0.8655

Table 4 Average F1-score of different classifier using features combination

N-Gram	TF-IDF				BOW			
	RF	SVM	NB	Ensemble	RF	SVM	NB	Ensemble
Unigram	0.8575	0.8705	0.84	0.8637	0.86	0.864	0.8579	0.873
Bigram	0.801	0.8345	0.8283	0.8266	0.715	0.8217	0.8137	0.806
Trigram	0.78	0.809	0.8172	0.8103	0.7522	0.8229	0.814	0.8092
Unigram + Bigram	0.8626	0.88	0.861	0.8684	0.8732	0.8608	0.8615	0.8848
Bigram + Trigram	0.7912	0.823	0.8268	0.834	0.722	0.8243	0.8055	0.8114
Unigram + Bigram + Trigram	0.8464	0.871	0.839	0.875	0.858	0.862	0.864	0.8702

extraction methods. To evaluate the performance, 80% of reviews are used for training purpose and rest 20% of reviews are used for testing purpose (Tables 3 and 4).

It is obvious from Table 3 that accuracy becomes the lowest 71.35% when BOW has been used for RF classifier along with trigram feature and the highest 87% when ensemble classifier has been applied for the incorporation of unigram and bigram (Unigram + Bigram) features along with BOW technique. For Table 4, f1-score becomes the lowest 71.5% for RF classifier when BOW has been applied along with bigram feature and f1-score reaches a maximum 88.48% when ensemble classifier has been applied for BOW using the incorporation of unigram and bigram (Unigram + Bigram) features.

4.1 Feature

We have considered unigram, bigram, trigram, and incorporation of them, e.g., Unigram + Bigram, Bigram + Trigram, and Unigram + Bigram + Trigram for both TF-IDF and BOW and then applied for RF, SVM, NB, and voting ensemble classifiers in order to investigate classification result.

Unigram (1,1): Firstly, for unigram feature, it has been observed that ensemble classifier achieves the best result rather than all other classifiers by securing accuracy of 86.75% (Fig. 4) and average f1-score of 87.3% for BOW.

Bigram (2,2): For this feature, it has been noticed that ensemble classifier achieves the highest accuracy by securing 80.6% for TI-IDF (Fig. 5) and SVM classifier achieves the highest average f1-score by securing 83.45% for TF-IDF.

Trigram (3,3): This time, trigram feature has been applied and ensemble classifier gains the highest accuracy for TF-IDF by securing 74.75% (Fig. 6) and SVM gains the highest f1-score by securing 82.29% for BOW.

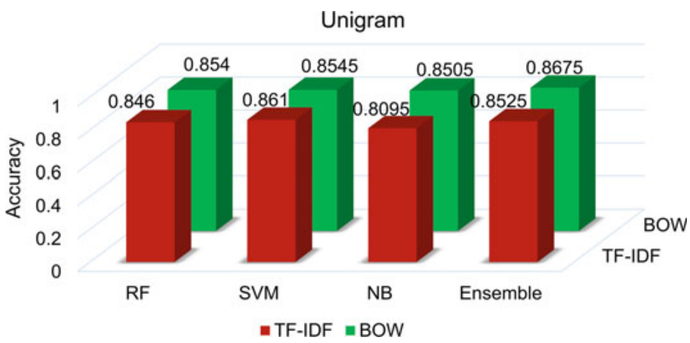

Fig. 4 Accuracy for unigram feature

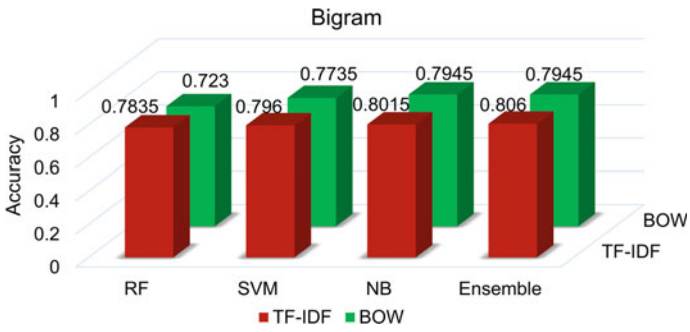

Fig. 5 Accuracy for bigram feature

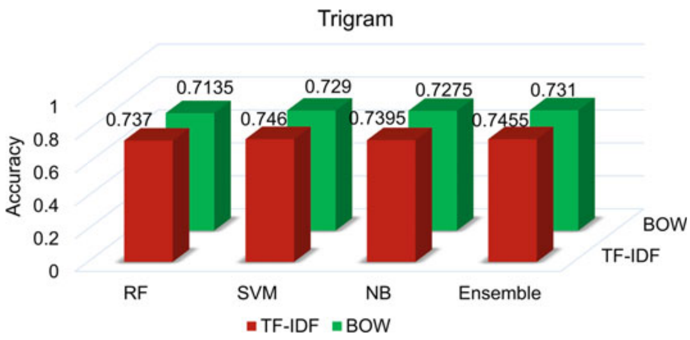

Fig. 6 Accuracy for trigram feature

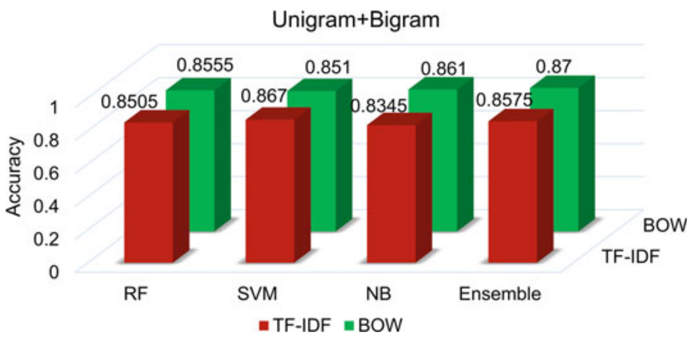

Fig. 7 Accuracy for unigram + bigram feature

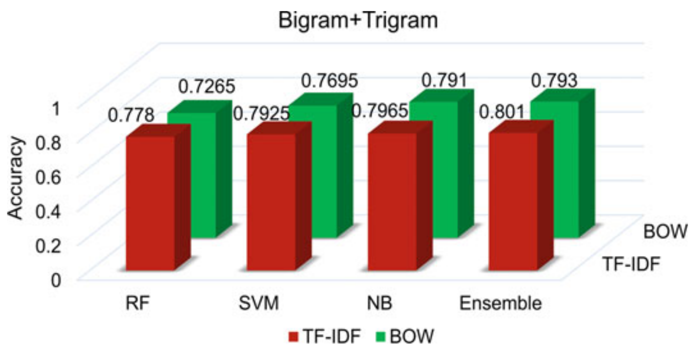

Fig. 8 Accuracy for bigram + trigram feature

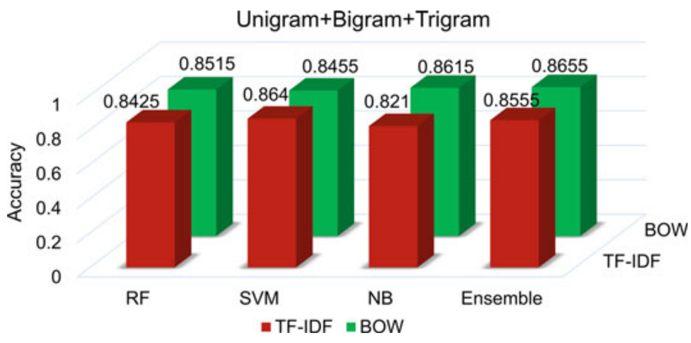

Fig. 9 Accuracy for unigram + bigram + trigram feature

Unigram + Bigram (1,2): When we have used the incorporation of unigram and bigram features, ensemble classifier results in the highest accuracy of 87% for BOW (Fig. 7) and for f1-score, the highest value is 88.48% obtained by ensemble classifier for BOW.

Bigram + Trigram (2,3): For the incorporation of bigram and trigram features, ensemble classifier performs the best accuracy by securing 80.1% for TF-IDF (Fig. 8). For f1-score, ensemble classifier performs the best by securing 83.4% for TF-IDF.

Unigram + Bigram + Trigram (1,2,3): At last, we have applied the incorporation of unigram, bigram, and trigram features and it is notable that ensemble classifier results in the highest accuracy of 86.55% (Fig. 9). For f1-score, the highest score 87.5% which is achieved by ensemble classifier for TF-IDF.

Again, it is very obvious in Fig. 10 that RF classifier performs worse than other classifiers in terms of precision when bigram (2,2), trigram (3,3), and incorporation of bigram and trigram (2,3) features are used separately for TF-IDF technique. From another point of view, in Fig. 11, all the classifiers perform very nearly to each other in terms of precision for BOW technique and any large deviation is not noticeable. From Fig. 12, it is clear that there is no significant deviation in the recall score and

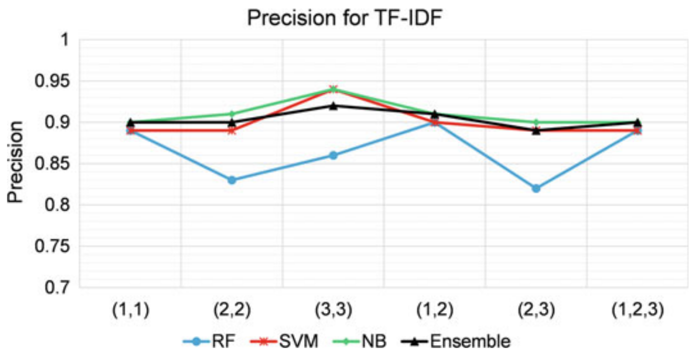

Fig. 10 Average precision for TF-IDF

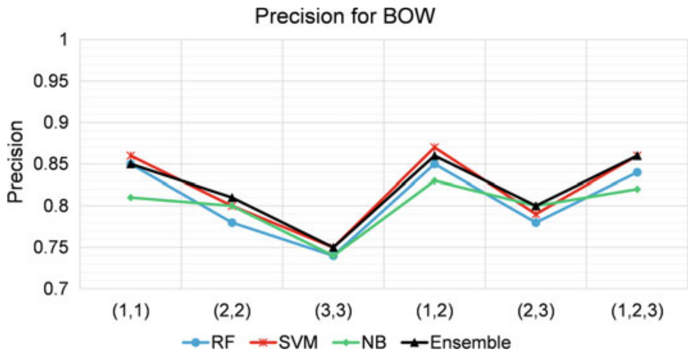

Fig. 11 Average precision for BOW

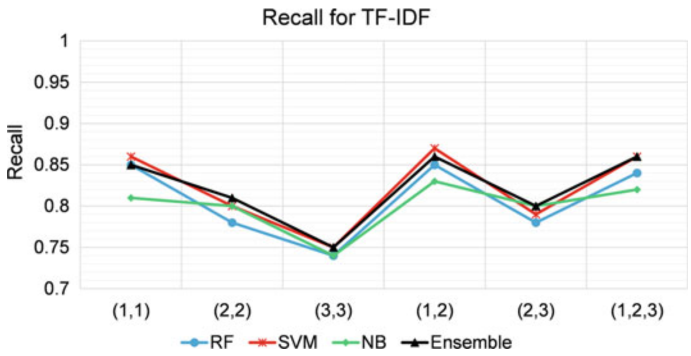

Fig. 12 Average recall for TF-IDF

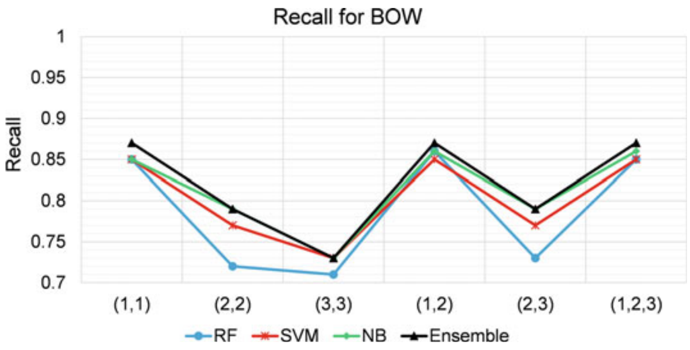

Fig. 13 Average recall for BOW

all classifiers give very close values for TF-IDF. On the other hand, Fig. 13 shows that RF classifier becomes unable to attain better recall score comparing with other classifiers when bigram (2,2), trigram (3,3), and incorporation of bigram and trigram (2,3) features are employed separately for BOW feature extraction technique.

The promising outcomes of our experiment reveal that our proposed approach can perform sentiment classification with good accuracy. There is no doubt that business organizations might be able to identify negative reviews efficiently. After having an inspection over the negative reviews, it is very possible to find out customer demand in a short time and business organizations can reshape their products and policies.

5 Conclusion

In this paper, a powerful system has been introduced for customer reviews sentiment analysis. Our main research objectives have been fulfilled perfectly by the outcomes of our experiment. In our methodology, TF-IDF and BOW feature extraction techniques have been conducted separately for variant N-Gram features and their combination to classify customer reviews. By exploring the outcomes, it is observed that unigram takes advantage of the dataset and boosts up the performance. For higher value of n, the accuracy decreases significantly and the highest accuracy is 87% obtained by voting ensemble classifier for BOW feature extraction method using the amalgamation of unigram and bigram (1,2) features. Again, in the comparison of TF-IDF and BOW techniques, TF-IDF gives better performance for most of the techniques. It is also explored that better accuracy can be achieved through voting ensemble classifier rather than individual classifiers. In future, emoticons that also contain customer sentiment can be considered and hybrid approach may be applied to improve the performance.

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Chapter 20
Analyzing Banking Data Using Business Intelligence: A Data Mining Approach

Anusha Aziz, Suman Saha, and Mohammad Arifuzzaman

1 Introduction

At present, the banking field plays significant roles in a country's economic activities [1]. As a financial institution, a bank gives loans and receives deposits. As an unremittingly expanding financial field, it becomes risky to give loans to the bank customers due to not repaying the amount of loans with interests. Not repaying loans with interests in time or credit risks also known as non-performing assets (NPA) have a vast impact on the bank's financial statements [2]. However, in this modern era, as a subject-situated, coordinated, time-variation, and non-unpredictable assortment of information, a data warehouse can store and integrate historic banking data and support business intelligence on these data for decision making. Several data mining algorithms can be applied on those warehouse data to extract information for making business decision.

Currently, business intelligence (BI) approach has a great impact on banking field. This approach consolidates reporting, business analytics, data visualization, visual analysis, descriptive analytics, statistical analysis, data tools and infrastructure, and best practices to assist associations with making more information-driven choices. It has two phases: BI creation and BI consumption. BI creation is the most tedious stage, and this stage requires a lot of future money-related and workforce assets. This stage comprises of a few phases of BI business definition, BI framework improvement system, recognizable proof and information source, select BI tool, how to make and

A. Aziz (✉) · S. Saha
Department of Computer Science Engineering, Bangladesh University of Business and Technology (BUBT), Dhaka, Bangladesh

M. Arifuzzaman
Department of Electronics and Communication Engineering, East West University, Dhaka, Bangladesh
e-mail: mazaman@ewubd.edu

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execute BI, and search and data about new needs and different applications. In the opposite side, BI consumption phase identified with the end client [3].

In paper [4], a new BI system is proposed in order to support decision making. In case of financial institutions, the decision making becomes more critical due to huge amount of data, several parameters, incomplete knowledge, etc. In order to take a decision based on these different issues, a data warehouse is required which could handle these several parameters at a time. In banking, need is given to the customers who as of now have existing credit and reimbursement history is acceptable. Basically, this research work focuses on the decision-making capability in loan distribution which indicates to whom a bank should give a loan and to whom it doesn't. By analyzing the banking data of customer's loan collection performance can also make decision of giving loan to the customers who has no records of loan. Here, Kimball lifecycle methodology is used in this research. This technique is demonstrated and still utilized by a great deal of research in different fields [5, 6]. It has become the best-stream business best practices [7]. SQL server-2019 is used as database to store the bank data, and visual studio-2019 is used to load data to make OLAP cube.

The major contributions of this research are summarized below:

- We have implemented the Kimball lifecycle eight steps, and by using this lifecycle, the dimension tables have been transferred from OLTP data into OLAP data in visual Studio.
- We have designed a star schema for banking data warehouse and constructed data warehouse. ETL process is implemented to transfer data from OLTP database to OLAP database.
- We have implemented the ETL process to transfer data from OLTP
- Power BI tool has been used to analyze banking data. For example, performance of loan collection per branch helps to make decision on giving loans to customer.

The analysis of bank loan collection has been carried out with many parameters, but in this article, it focuses mainly on the parameters like purposes of loans, ownership of homes, annual incomes, years in job, terms of loan, loan history of customers.

The remainder of this paper is composed as follows: Sect. 2 portrays the literature review. In Sect. 3, proposed methodology has been presented and analysis result has been presented in Sect. 4. Finally, the paper has been concluded in Sect. 5.

2 Literature Review

Already many researches have made some good contributions on analyzing banking data. Some of them are highlighted here. Yadav and Thakur [8] proposed big data approach on the customer usage bank data and analyzed bank loan. The authors [8] used Hadoop analysis on this research work and focused on analyzing credit risk and loan performance of the 'online credit cards'. In [9], Chui and Ding constructed a bank customer data warehouse and applied association rule mining technique to analyze the bank customer data. Here, the authors focused on analyzing and forecasting the real needs of the customers to provide efficient supports to the bank operator. On the other hand, Das et al. [10] proposed an information distribution center dependent on blueprint configuration on credit dispensing information for dynamic. The authors focused on making new schema for loan dispensing data to include potential borrowers who might not be considered for greater sum. Authors in [11] evaluated loan collection performance using BI. They mainly focused on evaluating the loan performance per branch to decide which bank needs attention to make profit. Khaled proposed a high-performance predictive model for helping credit managers in taking sound and safe individual advance choices [12]. Authors in [13] decided the basic factors related with BI selection in the ERP framework.

In view of these previous research works, a data warehouse on banking data is constructed here. The data is analyzed in order to categorize the bank customers in different groups and has made some decisions based on analysis. This paper is different from others previous research in the sense that data warehouse is constructed followed by ETL implementation. Then after extraction, reconstruction, and loading, OLAP is created. Afterward BI approach is applied to analyze these data. Mainly we focused on analyzing the performance of loan collection based on the customer's loan information and making decision whom to a bank should give a loan and to whom it doesn't.

3 Proposed Methodology

In this section, we have described our proposed approach. Firstly, to carry out this research, we have collected a dataset 'Dataset for Bank Loan Prediction' from Kaggle [14]. The data are processed for OLTP database. Figure 1 represents our proposed system. In this approach, datasets are gathered from different sources to database. After ETL process, these data are transferred to data warehouse. Using business intelligence technique, data in the data warehouse are analyzed, making reports to whom to give loan or whom doesn't. BI approach helps the system to make reports thus giving meaningful information to end users like bank managers. For actualizing the new framework, first the data warehouse would be structured by utilizing the bottom-up approach of Kimball lifecycle technique. In this bottom-up approach of Kimball lifecycle, data marts are first made to include revealing and scientific capacities for explicit utilitarian procedures and afterward information bazaars are included to make an extensive endeavor data warehouse. Implementation of Kimball lifecycle's eight steps has been described as below:

i. Determine the Main Subject
   The main subject of this idea is to analyze the customers banking data and make decision on whom to give loan, credit cards or not in order to avoid credit risk.

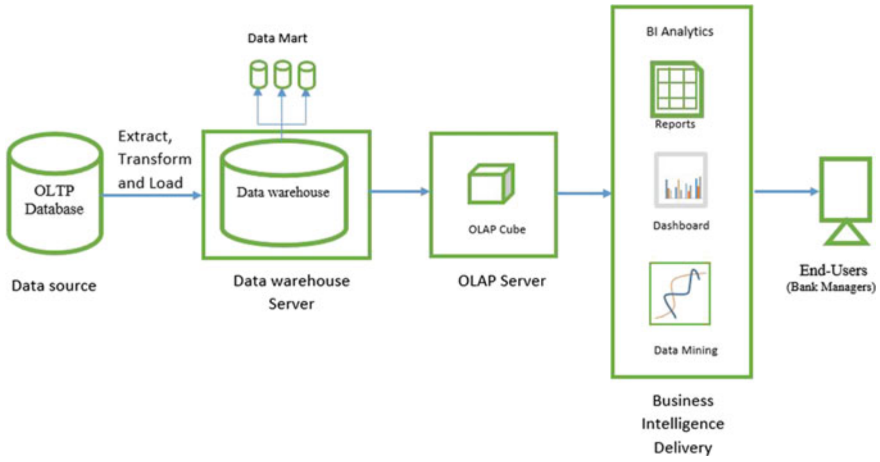


Fig. 1 Proposed approach

ii. Determine the Representation by a Fact Table
In this paper, the fact table would represent the loan status. Distinguish and connect the dimensions: dimension table and plan data warehouse center beginning from the star schema. Here, there are four-dimensional tables and 1 fact table. Figure 2 depicts the star schema for banking data with the dimension tables and a fact table.

iii. Determine the Fact
The fact chosen for this paper is the loan collection fact (Fact_Loan_Collection). Here, Loan_Status is classified and current loan amount is counted.

iv. Store Data into the Database
Data should be stored in the database before calculating result in order to avoid errors.

v. Balance the Dimension Table
From the dimensions that have been perceived, a delineation is made containing sorted out information about the qualities in the dimension table. Here, there are four-dimensional tables. The dimensions tables are Dim_Customer, Dim_Loan, Dim_Credit_History, Dim_EMI (Tables 1, 2, 3, and 4).

vi. Determine the Gradually Changing Dimensions
The dimensions can be changed gradually with time.

vii. Determine the Physical Design
This step shows the physical design of a data warehouse.

viii. This stage distinguishes and relates fact table with the dimension tables

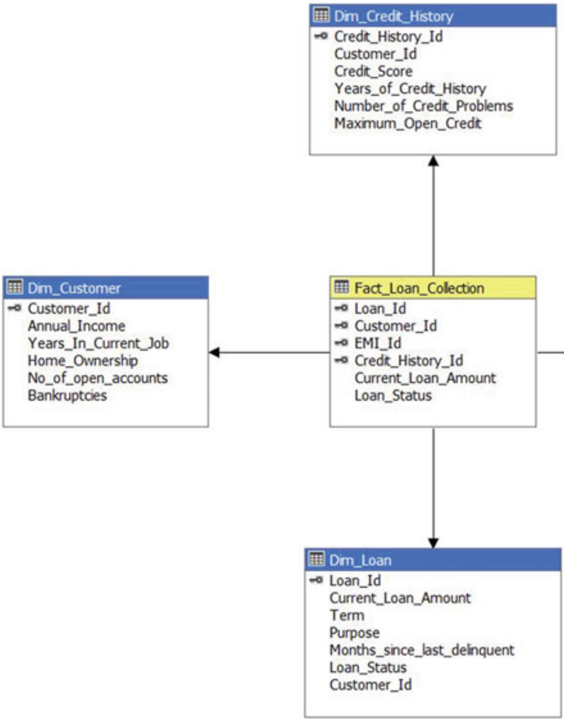


Fig. 2 Star schema for banking data

Table 1 Dim_Customer

Attribute	Description
Customer_Id	Primary key of customer dimension
Annual_Income	Total income of a customer in a year
Years_In_Current_Job	Total years in current job
Home Ownership	4 types: home mortgage, rent, own
No_of_open_accounts	Total no of open accounts in bank
Bankruptcies	Have 2 types value: 0 for who can repay debt; 1 for who cannot repay

4 Experimental Result and Analysis

After making the star schema using bottom-up of Kimball lifecycle, now the physical design has to be done. The data have been transferred from Online Transaction Processing (OLTP) to Online Analytical Processing (OLAP) using Extract, Transform, & Load (ETL). We have used Microsoft SQL Server-2019 as the OLTP database. Microsoft SQL Server-2019 is a relational database management system

Table 2 Dim_Loan

Attribute	Description
Loan_Id	Primary key of loan dimension
Current_Loan_Amount	Current debt amount
Term	2 types: long term, short term
Purpose	Purpose for loan
Months_since_last_delinquent	Last repayment
Loan_Status	2 types: fully paid, charged off
Customer_Id	Id of a customer who has the loan account

Table 3 Dim_Credit_History

Attribute	Description
Credit_History_Id	Primary key of credit history dimension
Customer_Id	Id of a customer who has at least a credit history
Credit_Score	Range from 585 to 7510
Years_of_credit_history	Total times in years in credit history of a customer
Numbers_of_credit_problems	Total credit accounts of a customers
Maximum_Open_Credit	The maximum limit of a credit

Table 4 Dim_EMI

Attribute	Description
EMI_Id	Primary key of EMI dimension
Monthly_debt	Monthly repayment amount with interest
Current_Credit_Balance	Remaining balance of a credit
Tax_Liens	Overdue of a loan tax
Loan_Id	Id of a loan of a customer

developed by Microsoft, and Microsoft visual Studio 2019 is used for ETL process. Microsoft Visual Studio 2019 is an integrated development environment (IDE) also from the Microsoft, and it has been the market leader in OLAP technology for many years. Generally, OLAP technology also known as multidimensional model organizes summary data into multidimensional structures. Aggregations are stored in the multidimensional structure in cells at coordinates specified by the dimensions. For banking data analysis, Microsoft Power BI is used here.

Power BI is a business analytics service by Microsoft. It aims to provide interactive visualizations and business intelligence capabilities with an interface simple enough for end users to create their own reports and dashboards. The ETL process has been shown in Fig. 3. This process is implemented by using the visual studio-2019's Integration Services Project. Here, first the information of all dimension is gathered and transferred the location of the database and then all the data are inserted into the fact table and then transacted the data from the database fact table to the cube. Thus, the data are transformed from OLTP to OLAP. In our work, 100,000 data are transferred from OLTP database to OLAP and in OLAP cube all data are integrated and stored.

Figure 4 displays the customer's information data which is transferred from OLTP database to OLAP according to the star schema.

Figure 5 is representation of the ETL process of loan dimension, loan from OLTP database which are filtered in order to get loan information. ETL process of credit

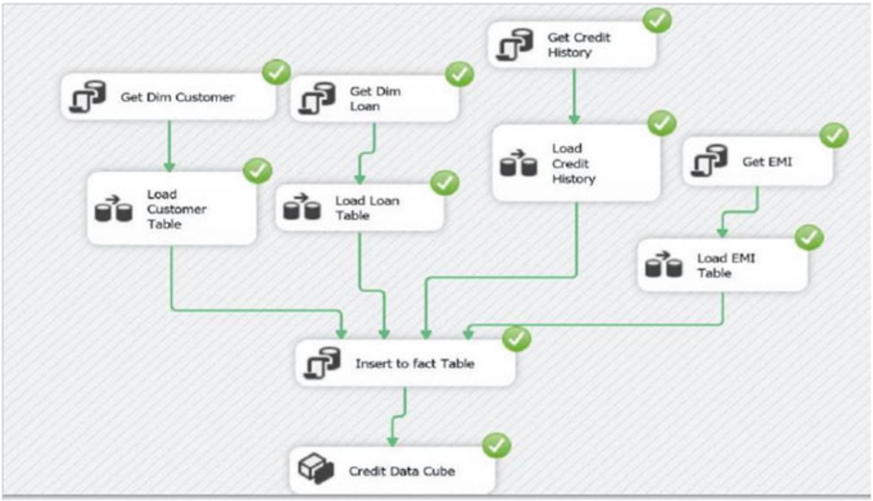


Fig. 3 ETL process for the bank collection fact

Fig. 4 Load data from customer source to destination

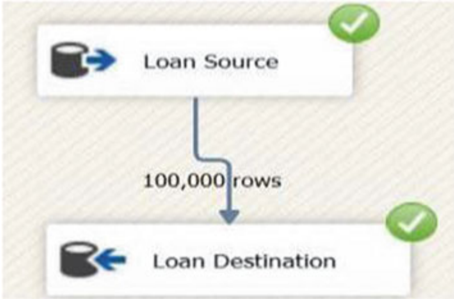


Fig. 5 Load data from loan source to destination

history dimension and the history of the credit accounts of customers are transferred from the OLTP to OLAP which has been presented in Fig. 6.

Figure 7 shows the ETL process of EMI dimension, and EMI information of loan is transferred according to the star schema.

After loading all the above dimension's rows, the fact table can be filled with key of every dimensions. Then, OLAP cube is deployed by using visual studio's multidimensional analysis services project. Figure 8 shows the OLAP cube done in visual studio. This OLAP cube has four-dimensional tables and a fact table. Each dimension table contains 100,000 data. These data have been transferred to fact table. By doing these, we have constructed the data warehouse on banking data. Here the data are transferred from OLTP to OLAP by using ETL process. Here the OLAP cube of the banking data based on the home ownership, terms, and loan status.

Fig. 6 Load data from credit history source to destination

Fig. 7 Load data from EMI source to destination

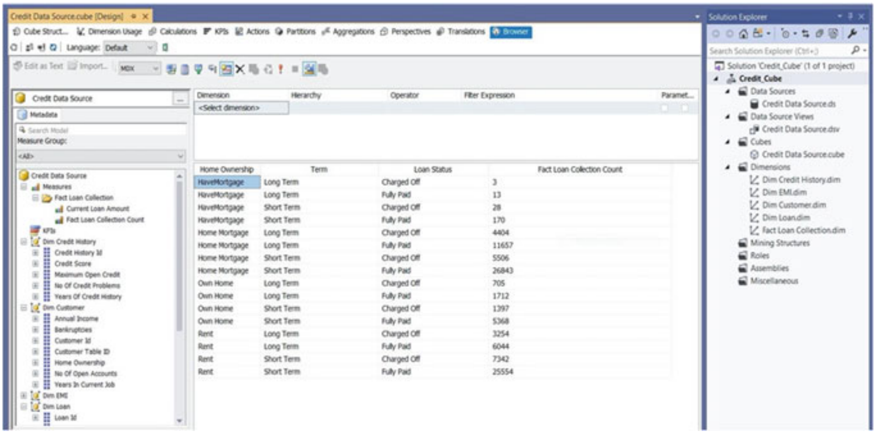


Fig. 8 OLAP cube in visual studio

OLAP analysis in Microsoft Power BI has been presented in Fig. 9. Here the analysis report is generated based on OLAP cube of the banking data, home ownership, terms and loan status. After analyzing the report, we found that more loans can be given for short terms than for long terms. Also, home ownership also has impacts on the analysis. Customers who have home mortgage higher tendency to pay the loans fully than the customers who rent home. The customers who own home or have mortgage have lower tendency for pursuing loans. This report would help the bank managers to know the progress of the branch by this report.

From this report, they are also able to take decisions on whom to give credits or not. It would also help the bank managers to include new customers for loan as well as to include the possible borrowers who are not considered for bigger amount of

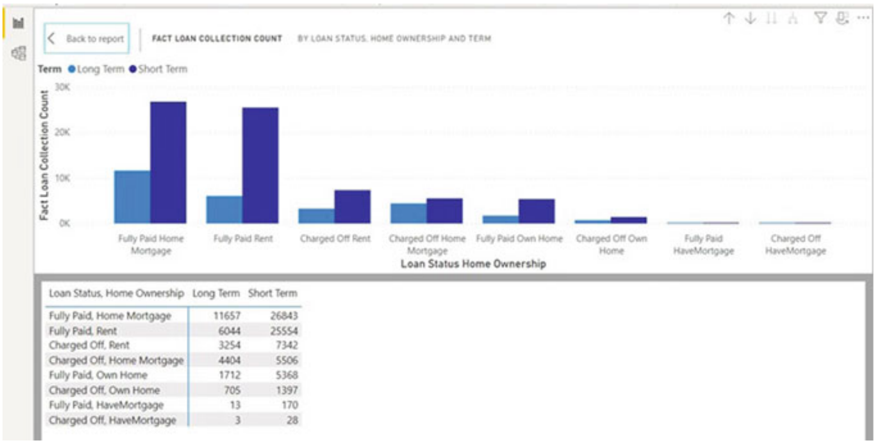
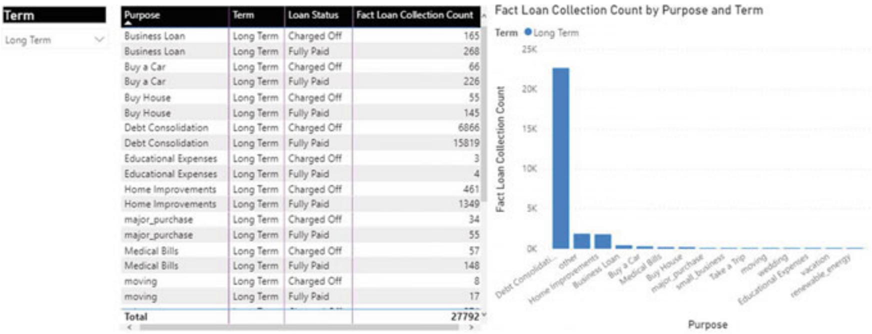


Fig. 9 OLAP analysis with power BI

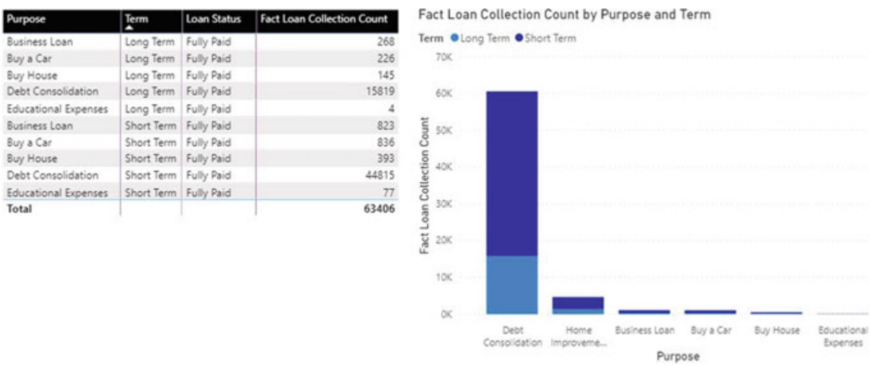
loan and thus increase bank customers. It would also help bank managers to reduce the credit risk. Multidimensional query and processing of banking data regarding loan collection are as follows:

i. Slicing: Slice is an OLAP operation that process information by selecting a specific dimension from an OLAP cube. Figure 10 shows the formation of a slice report where it shows the total loan collection statistics of long term.
ii. Dicing: Dice is an OLAP operation that process information by selecting two or more dimensions from an OLAP cube. As shown in Fig. 11, it selects fully paid loan collection data for the purposes of business loan, buy a car, buy house, debt consolidation, and educational expenses.
iii. Drilling: Drilling is an OLAP operation that process information by analyzing the data from a higher level of aggregation and then drilling down to a lower level to analyze it in multiple aspects. Figure 12 shows the loan analysis report which is drilled based on purpose and terms. From Fig. 12, it can be seen that more loans are given for debt consolidation than other purposes. So, the customers who apply for loans with the purpose of debt consolidation have



Purpose	Term	Loan Status	Fact Loan Collection Count
Business Loan	Long Term	Charged Off	165
Business Loan	Long Term	Fully Paid	268
Buy a Car	Long Term	Charged Off	66
Buy a Car	Long Term	Fully Paid	226
Buy House	Long Term	Charged Off	55
Buy House	Long Term	Fully Paid	145
Debt Consolidation	Long Term	Charged Off	6866
Debt Consolidation	Long Term	Fully Paid	15819
Educational Expenses	Long Term	Charged Off	3
Educational Expenses	Long Term	Fully Paid	4
Home Improvements	Long Term	Charged Off	461
Home Improvements	Long Term	Fully Paid	1349
major_purchase	Long Term	Charged Off	34
major_purchase	Long Term	Fully Paid	55
Medical Bills	Long Term	Charged Off	57
Medical Bills	Long Term	Fully Paid	148
moving	Long Term	Charged Off	8
moving	Long Term	Fully Paid	17
Total			27792

Fig. 10 OLAP analysis sheet of slice



Purpose	Term	Loan Status	Fact Loan Collection Count
Business Loan	Long Term	Fully Paid	268
Buy a Car	Long Term	Fully Paid	226
Buy House	Long Term	Fully Paid	145
Debt Consolidation	Long Term	Fully Paid	15819
Educational Expenses	Long Term	Fully Paid	4
Business Loan	Short Term	Fully Paid	823
Buy a Car	Short Term	Fully Paid	836
Buy House	Short Term	Fully Paid	393
Debt Consolidation	Short Term	Fully Paid	44815
Educational Expenses	Short Term	Fully Paid	77
Total			63406

Fig. 11 OLAP analysis sheet of dice

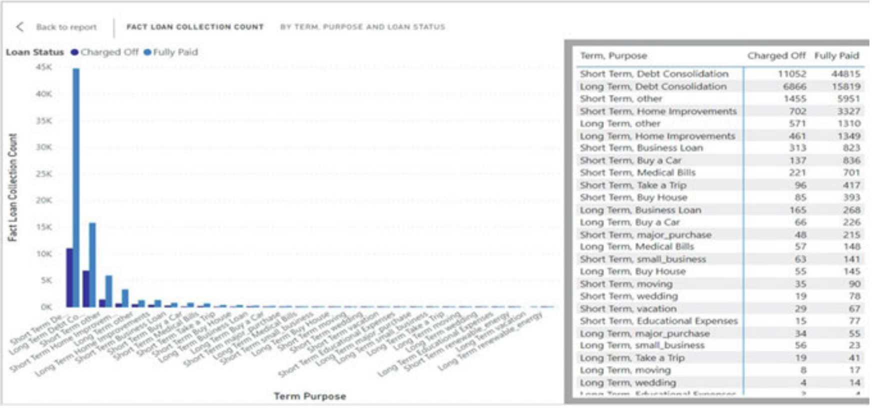


Fig. 12 Loan analysis report based on purposes and terms

higher tendency to have the loans than the customers with other purposes. This report would help bank managers to categorize customers in different groups. It would also help bank managers to evaluate the banking data for loan and thus provide banking decision-making knowledge to bank managers and help to make reports efficiently with in less time.

iv. Pivot: Pivot is an OLAP operation that rotates the direction of the display dimensions of a report. Dimensions include term, purpose, home ownership, and loan status can be rotated according to bank manager's choice.

5 Conclusion

In this paper, at first, we have designed a star schema for banking data by using visual studio followed by constructing data warehouse. We have analyzed the banking data by using Power BI and make out the decision on whom to give loan. This report can help bank managers to analyze customer's loan information data and to make decision on loan collection performance and also make the decision to give credits who are eligible. Customer's data such as term of loan, purposes of loan and what type of ownership does the customer has are mainly used to analyze the loan collection. Mainly, we focused on the customer's data, and from their data, we have extracted knowledge to categorize the customers. It also has contribution on reducing credit risk.

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Chapter 21
A Proposed Home Automation System for Disable People Using BCI System

Tashnova Hasan Srijony, Md. Khalid Hasan Ur Rashid, Utchash Chakraborty, Imran Badsha, and Md. Kishor Morol

1 Introduction

According to World Health Organization (WHO), almost 15% of the world's population lives with disability [1]. Disability is an abnormal health condition that arise for defect or failure of a body part. Disabled people cannot lead a normal, independent life because of their bad health condition. Home care service can help in their daily activities and help them to lead an ordinary life.

Home care is long-term support for disabled people's everyday life. It is hard for an individual to live with a disability. These people also need some nursing as well as additional health care monitoring services. Disable people should have special attention from their families. But to live a standard life in our society no one can be with these disabled people all the time. It is challenging to understand how to take special care of a disabled person.

Homecare is helpful for disabled adults and their families, Home Care Association of America believes that and explaining that "caregivers serve as another set of 'eyes and ears' in the home" [2]. Disabled people need some nursing just as extra social insurance checking administrations.

BCI (Brain-Computer Interface) is a system that establishes a connection between humans and several devices. BCI framework use brain signals for controlling several devices rather than use body parts. This interface framework is particularly useful for severely disabled, or locked-in individuals with no dependable strong authority over their body parts to cooperate with surrounded peripherals [3]. Utilization of BCIs in healthy individuals can comprise the two prospects to persistent mind preparing, and, if an effective, tendency to overuse BCIs even in children, constituting severe

T. H. Srijony · Md. K. H. U. Rashid (✉) · U. Chakraborty · I. Badsha · Md. K. Morol
Department of Computer Science, American International University Bangladesh, Dhaka, Bangladesh

Md. K. Morol
e-mail: kishor@aiub.edu

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medical and ethical problems [4]. In the past, researchers developed BCI system only for biomedical applications [5]. However, nowadays, it has been used for nonmedical applications also BCI framework contains different strategies for the different framework. EEG (Electrochephalography) is one of the fundamental parts of the BCI framework. EEG is used to capture the brain signal using multiple electrodes placed in the human scalp [6]. The human brain emits signals to control its movement, i.e., hand gesture, walking, etc. EEG sensor captures the change of the human's brain signal using the electrodes placed on their scalp and converted it into commands. EEG is the most familiar signal used to stimulate such Brain-computing frameworks in light of its low rate and helpful technique for activity [7].

Ghodake et al. [3] classified the BCI system in two ways, dependent and independent. When the BCI system depends on brain signals and other robust signals then that is called dependent BCI system. To originate electroencephalogram (EEG) signals on the dependent system it needs more operations rather than an independent BCI system. However, independent BCI framework is something that isn't reliant on brain signals.

Non-invasive (e.g., EEG) or invasive (e.g., intracortical), two types of procedure can classify Electrophysiological BCI [8]. In the Invasive method, necessary sensors need to put into the brain by using surgical operation. On the other hand, the non-invasive method reduces the risk for users because electrodes sheltered over the scalp area without any surgical operation [3].

The remainder of the paper is sorted out as follows: In Sect. 2, we talk about the past research works in this area. Next in Sect. 3, we describe our proposed method for home automation system for disabled people and summarize the functionality of the proposed system in this section. Section 4 finishes up the paper and gives an idea about future work.

2 Related Work

In recent years, different types of BCI signal has been used by authors. P300, and SSVEP (Steady-state visual evoked potential) signal was widely used. P300 means, this signal was captured after 300 ms of flashing. P300 (P means positive and 300 means 300 ms) produced a positive signal.

Rebsamen et al. [9] proposed a P300-based BCI system and build a fully autonomous wheelchair for indoors. In that system, a robotic wheelchair has been created to travel from a particular distance in an office or a home. A-C programming language based application has been used. In that application, there was an odd number of buttons placed on the screen. That system used 15 electrodes EEG to receive signals from the subjects. Those were P300 signals. Filtered and clean signal were fed to a Support Vector Machine algorithm that produced outputs of a new score for all the buttons. The highest score acted as a targeted button. In the C-based system, predefined paths were set. But that path was not hardcoded, so users could change the path in the future if they want. Also, some obstacle detection sensor has been used to detect an obstacle. That wheelchair had some artificial intelligence, because of that continuous data from the EEG was not required. Similar types of P300 based BCI system has been used by Kim et al. [5]. In that study, P300 has been used to build a system that controls TV channels. For data collection, 16 electrodes using antiCAP active electrodes and actiCHamp amplifiers at a sampling rate of 500 Hz and 46-inch TV were used. Raw data was initially captured in MATLAB by using BrainVision [10] remote data access protocol. Then these data were filtered using Common Average reference with the bandpass filtered to 0.1–8 Hz. To test the accuracy, tenfold cross-validation has been applied using Support Vector Machine (SVM) with a linear kernel. To reduce the EEG channels from 16 electrodes without losing the accuracy, SVM-based recursive feature elimination has been used. Hence, they tried to remove the lowest rank at each elimination. They used tenfold cross-validation. In similar type of work, seven sets of distinct channel combinations were created. In Lin et al. [6] used an SSVEP (Steady-state visually evoked potentials) based BCI to control home appliances for disabled patients. On that system, there were several flickering occurs. NeuroSky EEG chips captured those flickering. That EEG sensor used to capture the SSVEP signal from three locations OZ, FP2, and A2. Oz caught the stimulated flickering blocks signal with different frequencies, location FP2 used for eyes winking, and A2 used as a reference. In that study, an android based application was the central controller of that system. EEG sensor captured the signal from three locations. Those EEG signals were extracted and handled by NeuroSky EEG chips. At that point, these signs were transmitted through a Bluetooth transmitter to the Bluetooth beneficiary of an android device. That android device occupied an algorithm called FFT algorithm. That algorithm used to convert the spatial domain to the frequency domain for recognizing the suitable frequency. According to these frequencies, different types of command were transmitted using IR for TV and AC, Bluetooth for Light, etc. Equivalent sorts of P300 based BCI framework has been used in by Masud et al. [11]. That paper proposed a smart home system utilizing the P300-BCI Framework. For controlling the smart home environment, symbols based P300 was used here. With the assistance of the P300 BCI system, users could type with the help of brain signals and control the smart home system. The Random Forest classifier was used for classification. That system consisted of a 6×4 matrix with a classification unit that contains numbers and symbols. Symbols or numbers were shown to users randomly and then count the number of times the symbol or number startled and include them. With the help of P300, objective symbols or numbers can be identified by classifying the row columns. In that system, the upper three rows were for control smart home, for example, turn "on" or "off" TV/lights/music, increment or lessening volume, emergency alert, and call and the several rows hold numbers for making a call, etc. That system utilized a 32-channel Brain Amp EEG framework with inspecting 250 Hz. For classifying P300, these six channels Pz, P3, O1, Cz, P4, and O2, were utilized, and besides, a group of data was used to prepare the RF classifier. That smart home framework could control home apparatuses without moving anyone's body parts. Such work has also been done by Shivappa et al. [12]. That paper worked with a home automation system using BCI. That system consisted of many parts like Arduino boards, speakers, ultra—cortex headset, personal computer with the processing IDE, SD card module, proximity sensors, a smart bulb, a fan attached to a smart plug, and so on and for voice command, Alexa was used. Home automation relayed on Auditory Steady-State Response (ASSR). Auditory signals were generated by users, which used for the home automation system with the assistance of the BCI framework. That framework began with Arduino's assistance with a nearness sensor, and the sound was created, relying upon the separation with a scope of the client from the sensor. Then electrodes collected EEG signals and filtered to visualize the command for acting. A 8-channel BCI with a 32-bit processor sampling rate of 250 Hz was used. Fast Fourier transform (FFT) algorithm was used for converting the signal from the time domain to the frequency domain, which helped to get the pick values from a signal. Signal to noise ratio (SNR) and threshold value involved for considering a signal. A signal would be considered when the threshold value gets higher than the SNR. A similar type of work done by Masood et al. [13], that paper developed a BCI-based smart home control system to facilitate the disabled and needy persons. That system used EEG sensors for gathering brain signals produced during eye blinking. Using the eye blinking signal pattern, a person could turn on or off the light and fan. Another study by Gupta et al. [14] claimed that processing brain signals through wearable or wireless EEG devices could result in very helpful for the development of medications as well as healthcare monitoring.

Ullah et al. [15] an EEG-based BCI framework was built, offering a low cost that could be affordable and suitable for use. That system aimed to launch a cheap and straight forward system that could be affordable for everyone. The framework was built with a single channel BCI framework as opposed to utilizing the multi-channel BCI framework. Homemade silver electrodes were used for the obtained signal to make it affordable, which was compiled to a computer for processing the signal by soundcard in exchange for using other outer devices. The hardware comprised of electrodes, EEG speakers, and AID converter, and so on. Different theological tasks like relaxing left and right-hand movement, consciousness, etc. mixed EEG signals were captured, and also, the changes were observed. EEG signals are depending on electrodes. To decrease the expense of wires, a distance across 11 mm, an unadulterated silver plate, was utilized with the high-quality silver-covered link. A basic, standard, compact bio-enhancer was fabricated and used for a low scope of EEG signals. In a PC, raw EEG signals were put away for examination and preparation. MATLAB was being used for raw information preparing where the Fast Fourier Transform was applied for breaking down the signal frequency pattern. It was structured such that correspondence with others should be possible by text. The characters were shown in the 6×6 matrix, and users could choose any characters among the areas in the matrix. With one-second postpone individually, each row and column was featured. For line section choice, the client could utilize right-hand symbolism development. Finishing a sentence, a send alternative was accessible, which should be chosen. A GSM modem or cell phone was associated with sending the message. In that paper, single-channel EEG was utilized for diminishing the expense, yet it tended to be extendable to more than one channel.

Another SSVEP (Steady-state visually evoked potentials) dataset used by Anindya et al. [16] to build a prototype of home appliances control system. In that study, two types of datasets had been used. One was eight frequencies within 6–12 Hz, and another one was 8, 14, and 28 Hz. In that paper, the entire system was divided into two parts. One was a signal processing block and a control block. In the signal processing block, a python based computer program has been used. Window-sinc digital filter with black window used to pre-processing of the EEG signal. To reduce the different characteristics of the datasets, two filters were designed in this study. To find the highest amplitude FFT algorithms used on the filtered datasets. Once the signal has been obtained, then that feature has been classified using SVM to the captured label of the associated device that to be controlled. In that study, they tested two kernels, one was linear, and another was the radial-base function. In the control block, the targeted label was sent to Arduino UNO from the computer using serial communication. Based on that label, Arduino UNO toggled the state of its digital ports to activate or deactivate the associated optocoupler and LED circuit. Ghodake et al. [3] used an independent non-invasive synchronous BCI system to build a home automation system. In that system, the NeuroSky Mindwave sensor has been used to collect the EEG data from the subject's brain. That Mindwave sensor can provide raw data along with two custom values. One was attention level meter values, and another one was meditation meter values. These attention values indicate the mental focus and meditation values show the user's mental calmness. In that system, users could operate two home appliances, bulb, and fan. To turn on these home appliances, the user should put their attention towards objects. A total of 20 number of attentions have been taken. To get the average values, the first five values have been taken to make ordinary. The rest 15 values average has been taken in 3 sets. Each includes five numbers. If the final cost was greater than 40, which was the threshold values, this system created a command to turn on the Bulb or the Fan. Another NeuroSky Mindwave sensor has been used by Bhemjibhaih et al. [17]. In that paper, authors have introduced a system for paralyzed people. That system would help paralyzed people to control home appliances. In that system, the Kinect motion sensor has been used to capture the image and locate it to the user in a 2-D plane. Those data were processed using MATLAB's image processing toolbox to detect user identification and centroid calculation. An HMC5883L triple axis compass module from Honeywell was used as a digital compass. That compass recognized the angle of the user's, where they were looking at. That device sent the measured angle to Raspberry pi that was acted as a device control module in that system. In that Raspberry pi, Python three based application has been used to collect the data and make the required decision. That device transferred the command from Raspberry pi to home appliances using WIFI. Then responding to the command, the devices turned ON or OFF.

Molina et al. [18] claimed that it's possible to demonstrate a framework to accommodate or develop BCI operation by taking advantage of the information of a person's emotional state, which can be identified from EEG signals. Brain activity patterns corresponding to specific generated visual images could be identified using EEG. Bobrov et al. [19] was represented as a phase of the method, where seven persons aged from 23 to 30 with normal vision and right-handed have participated in the experiment, and the survey continued using non-invasive commercially-available EEG devices. The Bayesian approach and MCSP method did EEG pattern classification.

The patterns of brain activity by imagining pictures could be classified based on EEG recordings gained by both used headsets and are Emotive EPOC (Emotiv Systems Inc., San Francisco, USA) and Brain Products ActiCap (Brain Products, Munich, Germany). Abiyev et al. [20] claimed that a person's emotional and muscular status could be evaluated for controlling a wheelchair. The BCI design depended on FNN (Fuzzy Neural Network), which was utilized for the classification of brain signals. BCI design was composed to activate a brain-controlled wheelchair adopting six mental operations of the user. The operations were moving forward and backward, turn right and left, turn on the chair, stop and start the action. EEG signals were classified using the help of FNN, where tenfold cross-validation data set was used by Abiyev et al. [20]. The appropriate design of the FNN system was completed by fuzzy means classification and gradient descent algorithm. The gained outcomes of 100% classification verify that the applied techniques were for a capable candidate for the EEG signals classification.

In Abdulkader et al. [21], the BCI system records the brain waves and sends them to the computer to perform the determined task. Waves that were transmitted from BCI generally applied to elicit a concept or control a purpose. It is possible to identify the application areas that could benefit from brain waves in simplifying or gaining expected objectives. Jarkko et al. [22] claimed that different values from brainwaves could be used as a controller in games where beta waves reading was only useable for controlling games. In that paper, those players who have attention deficit disorder can't play a game with Senzeband. It also demonstrates the experiment with different brain waves' values and builds a game that takes advantages as many as possible. Programming development of the card game begins with the logic that turns cards with the player's concentration.

Achanccaray et al. [23] proposed a P300-based BCI, relies upon Hoffman's approach, the six figures were used as an improvement. Eight subjects had partaken in this assessment, P300 acknowledgment was assisted through a blend of Adaptive neuro-fuzzy inference systems (ANFIS) classifiers by throwing a polling form, using four EEG channels. A 16-channel electroencephalograph (EEG) structure was used to record cerebrum development. Each subject performs four detached gatherings for getting ready. For data analysis, cross-validation was applied to survey the classifier execution. The acknowledgment of the P300 occasion using a blend of four (ANFIS) classifiers by throwing a voting form had gained basic results for strong subjects and post-stroke patients. The display of this P300-set up BCI depends on the getting situation; i.e., hurt degree in central tactile framework additionally, impacted cerebrum territories by stroke. Despite the way that a couple of instructional courses give a sensible arranged classifier, and cross-validation was useful to find it. An essential patient condition could be compensated by a progressively drawn-out planning time. The delayed consequences of typical precision were more important than 75% for all subjects, very same results were gotten for strong subjects and post-stroke patients. Yet, the better classifiers for each subject had achieved correctnesses more conspicuous than 80%. Similar types of P300 based BCI system has been used by Aydin et al. [24]. In that paper, a locale-based interface was proposed to control a smart home condition. The requests at the interface were picked by considering the potential needs of an injured subject in a home area. The interface involves two levels, and there were seven districts at each level. The chief degree of the redesign perspective was that constrained air framework, TV, control contraptions, remote control, and telephone were structure packs in the principle level. Five healthy male subjects were dealt with the attempts ensuing to checking instructed consent. The total of the BCI exercises as overhaul presentation, data variety, and sign taking care of philosophy were executed using MATLAB stage. The data recorded for a character was updated as the going with cross-section: Channel $\times$ Window Size $\times$ Region $\times$ Number of Flashes ($16 \times 240 \times 7 \times 15$). Linear Discriminant Analysis (LDA) classifier had been used in P300 based BCI considers adequately. In that assessment, the precision was surveyed from two different edges. As such, the target decision exactness was dictated by keen AND operator. In this examination, the area based lift perspective was successfully used for controlling a wise home condition with typical 95% exactness at 5 flashes. The author proposed the RBP interface to extend the number of things to be controlled without extending the interface's multifaceted idea. In any case, for a home control application, exactness had a higher need than speed. So zone-based interface was especially sensible for control applications that require strong decisions. The author would not revolve around nonstop nonconcurrent execution of zone-based perspective for home computerization structures. Similar types of P300 based BCI system has been used by Cipresso et al. [25]. BCI uses neurophysiological signals as data requests to control external devices, while Eye Tracking (ET) licenses the estimation of eye position and improvements. Of all advancements, eye improvements are defended the longest in Amyotrophic Lateral Sclerosis (ALS). The inspiration driving the eBrain adventure was to survey the usage of a BCI P300 theory and an eye-following system, both as an Augmentative and Alternative Communication (AAC) device also, an ideal assessment instrument with ALS patients. A smooth emotional obstacle, generally including official limits, had been portrayed in 10–50% of ALS patients, while a slight degree of patients (5%) present clinical evidence of frontotemporal dementia (FTD). The eye-following development could measure stubbornly visual improvement control of ALS patients, like this delivering an Augmentative and Alternative Communication Systems (AAC). The essential hindrance in the usage of ET structures was that they require full visual movability and the nonattendance of critical visual deficiencies; the past may be lost or balanced in the last periods of ALS, and the last maybe accessible in ALS patients of impelled age, along these lines prohibiting the usage of this device. A congruity among speed and exactness should be recognized. A starting late sponsored adventure, "eBrain: BCI-ET for ALS," proposed to survey BCI P300 strategy with the eye-following development, for instance, AAC structures. The BCI device module would be established on the g.USBAmp biosignal intensifier, related with a working cathode head sleeve. The biosignal speaker would be related to a minimal PC, running Windows 7 64 bits. That PC would be related to an outside screen, where the upgrades would be acquainted with the customer. The Eyelink-1000 would be used for the eye-tracker, including a quick infrared camera and the related illuminator, arranged just underneath the Display Screen. The eye-tracker has PC to make sure about eye-head information through the camera and method them logically. The proposed examination was depicted by the proximity of a couple of imaginative perspectives: (1) Comparison between two promising advances, one broadly examined (ET), the other an extraordinarily promising contender (P300 BCI), (2) affirmation of an automated scholarly battery, concentrating on the neuropsychological examination of higher solicitation scholarly works in ALS patients and (3) synergic appraisal of clinical, exploratory and inquire about office data will give a more exhaustive perspective about the infection. Another type of P300 based BCI system has been used by Corralejo et al. [26]. In that paper, target making and looking over an assistive contraption for working electronic devices at home by strategies for a P300-based brain-computer interface (BCI). It considered courses through ten menus and to administer up to 113 control orders from eight electronic devices. Ten out of the fifteen subjects had the alternative to work the mentioned instrument with exactness of more than 77%. Eight out of them showed up at correctness higher than 95%. Also, bit rates up to 20.1 pieces/min were practiced. The peculiarity of that assessment lies in the use of a space control application in real circumstances, authentic devices directed by potential BCI end-users. Yet hindered customers would most likely not have the choice to set up this structure without the assistance of others. This examination makes a momentous move to survey how much such people could work an autonomous system. Their results suggest that neither the sort nor the degree of impediment is relevant to suitably works a P300-based BCI. In extension, the assistive BCI gadget could be easily changed to increase the degree of necessities and essentials that can be fulfilled.

Other types of BCI system has been used by Lin et al. [27]. In that paper, a brain-computer interface based smart living environmental auto-adjustment control system (BSLEACS) was proposed. They put a cost-effective, from a general perspective extendable and easy-to-use BSLEACS to control electric home machines reliant on the capability in the customer's scholarly state (slowness or then again status). In BSLEACS, a far off physiological sign obtaining module and a presented signal dealing with modules were besides proposed. In wireless physiological sign obtaining module and installed sign arranging module contained the focal core interests of a little volume and low force utilization and were more reasonable for objective application. To assess the structure execution of BSLEACS for controlling home machines, an aggregate of 75-groundwork framework responses and review results from 15 subjects were cross-referenced and evaluated. For dissected, they utilized the restrictions of binary classification. The wireless physiological sign gaining module is little enough to be brought into a headband as a wearable EEG contraption. It gives the upsides of adaptability and significant lot EEG viewing (more than 33 h by utilizing 1100 mA Li-particle battery). The embedded sign preparing module, which gives astonishing estimations, was proposed to see the client's academic state and was moreover executed as a UPnP control contraption. BSLEACS not proportionate to other BCI-based control structures, considering the way that BSLEACS needs a single EEG channel to recognize mental state by checking the EEG signal in the domain of Oz of the overall 10–20 EEG framework. For 75-preliminary test outcomes, the PPV and affectability of BSLEACS for controlling home contraptions are 70 and 81.40%, autonomously. BSLEACS has been checked in a reasonable region and showed that the lights/light could be reasonably and along these lines balanced progressively subject to the qualification in the client's academic state.

Table 1 represents the overview of different types of BCI system and their result. From that table, it is clear, P300 based BCI system was used by many authors. Also, P300 based BCI has the highest accuracy rate.

3 Proposed Method

In this paper, an independent non-invasive BCI system will be used to create a home automation system for disabled people. In this process, the EEG sensor from Brain Product GmbH, Germany [28], will be used to collect data from the user. This data will be stored in MATLAB using BrainVision remote data access protocol [10]. A windows application will be used as a system that shows the options to the user. Initially, users can choose from four different options, i.e., Fan, Bulb, TV, and AC. In a computer monitor, we are going to divide the screen into four quadrants. Each quadrant will have one option. Sequentially a green circle will be flashed inside each option. This flashing should be a minimum of 250 s. Because we will be using a P300 signal, this signal generates a positive signal after 300 ms of flashing. Then that initial data will be filtered using CAR (Common average reference). This data will be filtered and averaged in every epoch. Then this average epoch will be the classification of our data. This classification will be evaluated using tenfold cross-validation using SVM (Support Vector Machine).

Using this SVM's recursive feature elimination, we will eliminate the lowest rank of each round for a specified period. Then this approach will give us a set of distinct channels that will be used to get a P300 signal accurately. Then this signal will be converted as a command using the windows application. This command will be given to an Arduino UNO that will work as a hub. This Arduino UNO will connect our windows application to home appliances. An IR sensor will be used to control the TV and AC. Bluetooth technology will be used to communicate with FAN and Bulb. This IR and Bluetooth sensor will be attached to the digital ports of Arduino UNO.

A short description of our proposed system (Fig. 1) is given below:

Step 1: We will set EEG sensor on a patient's head, then collect the signal data.
Step 2: Upon successfully collecting the signal data, we will store in MATLAB using remote data access protocol.
Step 3: Then from that we will filter the signal using CAR(Common Average Reference) and also in this stage classifier will be created.
Step 4: In this stage, we will evaluate the classifier with K fold validation using SVM.
Step 5: After this, we will try to create an expected set of values to find out the P300 signal correctly.

Table 1 Overview of related work

S. No.	Name	Number of subjects	System based on	Number of channels used	Result
1	Toward realistic implementation of brain-computer Interface for TV channel control	8 subjects 7 Male, 1 Female	P300 based BCI	3 and 16	92.3% using 16 EEG channels, 89.6% using 3 EEG channels
2	A wireless BCI-controlled integration system in smart living space for patients	7 subjects 7 Male	SSVEP (steady-state visually evoked potentials)-based	3	For light control 100%, For TV control 96.29%, For Air-condition control 94.87%
3	A prototype of SSVEP-based BCI for home appliances control	–	SSVEP(steady-state visually evoked potentials)-based	–	83.26 and 71.67% using SVM with RBF kernel
4	Low-cost single-channel EEG based communication system for people with lock-in syndrome	5 subjects	P300 based BCI	1	87% using 1 EEG channel
5	A P300 brain-computer interface based intelligent home control system using a random forest classifier	3 subjects	P300 based BCI	6	87.5% using 6 EEG channel
6	Home automation system using brain-computer interface paradigm based on auditory selection attention	4 subjects	OpenBCI	8	For light control 92, 50, 59, 67% for ON, OFF, brighten, dim states For Fan control 100% for ON, 92% for OFF

(continued)

Table 1 (continued)

S. No.	Name	Number of subjects	System based on	Number of channels used	Result
7	Brain-computer interface based on generation of visual images	7 male	Brainproducts ActiCap and Emotiv EPOC	–	Up to 68%
8	A cooperative game using the P300 EEG-based brain-computer interface	Healthy subjects (12) and subjects with disabilities (13)	P300 based BCI	10 dry electrodes	For 7 Questions, percentages are 92.3, 61.5, 69.2, 92.3, 76.9, 92.3, 100
9	Brain-computer interface based smart home control using EEG signal	Subject with disabilities	Arduino Uno	–	Accuracy 80–100%
10	Brain-Computer Interface for Control of Wheelchair Using Fuzzy Neural Networks	Unable to walk or move	Emotiv EPOC	14	FFT and FNN with tenfold cross-validation data

Step 6: When we get the expected set of values we will convert this signal to a command using Windows application.
Step 7: Command will be sent to an Arduino UNO. This Arduino UNO will work as a HUB.
Step 8: Arduino UNO will extract this command in this stage.
Step 9: In last stage, Arduino UNO will send the command to the exact home appliances.

4 Conclusion

The fundamental aim of this paper was to review the BCI framework related work and proposed a system for disabled people. Our proposed methodology includes a BCI sensor that helps disabled people to control Bulb, TV, AC, and Fan. An independent non-invasive BCI system has been used. BCI sensor captured the signal from the scalp of disabled people. This proposed system converted this signal into a command using a windows-based application. This command was initiated by an Arduino UNO. In

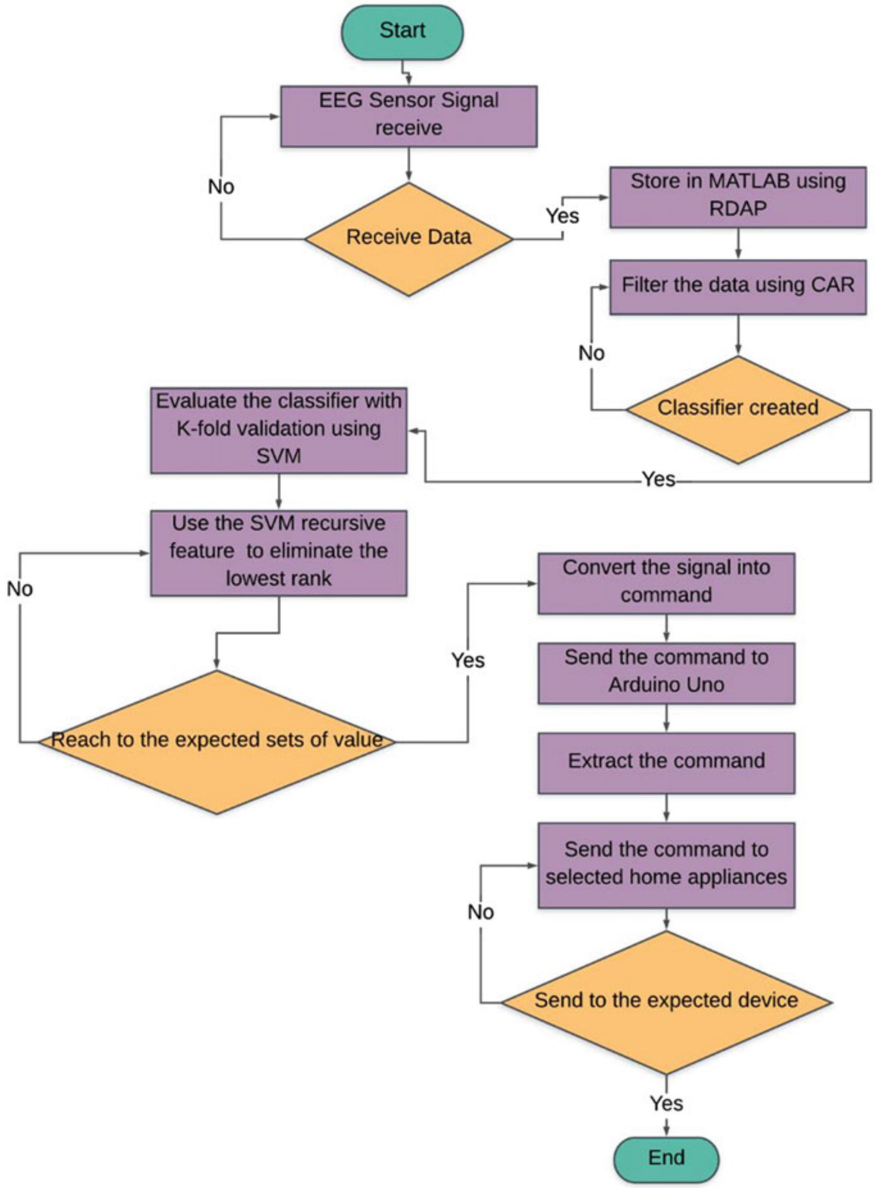

Fig. 1 Proposed system

future this proposed system will be practically implemented and also, we will try to reduce the number of EEG electrodes in this system.

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Chapter 22
Smartphone-Based Heart Attack Prediction Using Artificial Neural Network

M. Raihan, Md. Nazmos Sakib, Sk. Nizam Uddin, Md. Arin Islam Omio, Saikat Mondal, and Arun More

1 Introduction

Because of coronary artery disease (CAD), heart attack takes place in the human body. As it has turned out to be one of the leading causes of death throughout the world, CAD is a hot topic in medical science and public health issues. On average, the mortality rate has gone down sharply between the last couple of decades due to several non-communicable disease (NCD). Researchers of the World Health Organization (WHO) have uncovered that nearly 17.9 million people died from CVDs only in 2016 [1]. This figure represents 31% of all global deaths. Moreover, it has been estimated that this figure will grow sharply, and low and middle income countries like Bangladesh will suffer the most. Bangladesh has higher rates of CVD than any other country in Asia, and nonetheless, it has been the least studied [2]. In the global combat against CVD, Bangladesh is a country missing in action. In 2018, WHO published that because of CVD 118,287 people have died in Bangladesh which is 15% of total deaths [2]. Currently, there is no population-based monitoring system present in Bangladesh to trace chronic diseases. Besides, it is cumbersome to find accurate data on the prevalence of diseases in Bangladesh as there is no government or non-government organization that collects health data. Health professionals and policymakers have failed to perceive the magnitude of the problem because of the absence of information on the prevalence of CVD. But still, it is not too late. It is possible to cure a heart attack (AMI) when it is in its early stage. The best approach is

M. Raihan (✉) · Md. Nazmos Sakib · Sk. Nizam Uddin · Md. Arin Islam Omio
North Western University, Khulna, Bangladesh
e-mail: raihan1146@cseku.ac.bd

S. Mondal
Khulna University, Khulna, Bangladesh

A. More
Department of Cardiology, Ter Institute of Rural Health and Research, Murud, India

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to prevent the development of AMI in the human body by controlling the risk factors. Using the android software that we have developed, a user can anticipate if he is at risk of developing AMI at the early stage. Users have to answer 14 questions, and the application can predict AMI with high accuracy. Using this application, people can get instant help. The main objectives of this project are to analyze the clinical data and the risk factors and get a better understanding of ANN. But above all, the whole point of our study is to spread awareness and to motivate people. By using our mobile app, people can identify themselves if they are at risk of having heart diseases (HD) can avoid sudden death.

The remaining part of the manuscript has arranged as follows: in Sects. 2 and 3, the related works and methodology have been elaborated. In area 4, the experimented aftereffects of this undertaking have been chaired with the dictation to legitimize the oddity of this investigation work. At last, this exploration paper has ended with Sect. 5.

2 Related Works

Several studies have been conducted regarding heart disease by using different techniques and algorithms. Jayshril et al. have proposed a system using learning vector quantization (LVQ) algorithm to predict if heart disease (HD) is present or absent in a patient [3]. They collected 303 medical records with each having 13 features. Using different numbers of neurons and different learning rates, their highest accuracy was 85.55% [3]. Using MATLAB to implement this system, they achieved 89% accuracy to predict heart disease. A similar data mining techniques were also used to study coronary heart disease (CHD) and to predict heart attack [4]. This research emphasizes that a neural network system is better at predicting heart disease than decision tree (C4.5), SVM, and Naive Bayes. In another research, a smartphone-based application was developed that predicts the chance of getting a heart attack using statistical analysis and data mining techniques [5]. They categorized the chances of heart attack into three (high risk, medium risk, and low risk). Using the C4.5 DT, accuracy, and sensitivity were 86% and 91.6%, respectively. Another system was developed to forecast the risk of ischemic heart disease using backpropagation having 84.47% accuracy [6]. Another study explained how the machine learning technique has the proficiency to anticipate heart disease (acute coronary syndrome) [7]. They used random forest, AdaBoost, SVM, bagging, and K-NN algorithms and compared the outcomes. They gained the highest accuracy 76.28% from bagging [7]. In [8], another investigation was conducted to predict heart attack using smartphone and chi-squared test. They found significant correlation of having a cardiac event when low and high category of risk was compared. The p-value was 0.0001.

As the uses of smartphone increases day by day, our application can provide instant help to the population with more efficient and user-friendly way.

3 Methodology

Figure 1 shows the workflow to train our model. Our workflow is segregated into six parts:

- Identification of AMI risk factors
- Data collection
- Data preprocessing
- Data training
- Integrate the artificial neural network (ANN) model
- Code implementation.

Each part of the workflow is discussed as follows.

3.1 Identification of AMI Risk Factors

To predict AMI using neural networks, at first, we needed to consider a dataset. Before that, we needed to identify the potential causes that cause AMI. To identify these factors, we found 202 causes (24 British town follow-up of 4–2 years) of HD [5]. Among them, we have identified 14 major causes of heart attack (AMI). Those factors are age, gender, hypertension (HTN), diabetes mellitus (DM), dyslipidemia (DLP), smoking habit, physical exercise, family history, drug history, stress, chest pain, dyspnea, palpitation, and ECG. These are the most important and common factors that can cause heart attack (AMI).

3.2 Data Collection

To develop this system, we collected 917 cases with ECG results where 506 data from AFC Fortis Escorts Heart Institute, Khulna Bangladesh, and 281 data from Rural Health Process Trust, India [5–8]. Among this 917 instance, only 835 data were selected because of having proper diagnosis results and the ECG report. In our survey, we found patients aged from 15 to 100. Most of the patients were male about 76% (Table 1).

3.3 Data Preprocessing

In the real world, it is tough to accrue raw data without missing values. For better training a model, the dataset should be clean and authentic. Therefore, data preprocessing plays the most vital role in training a model. In our dataset, all the data was

Fig. 1 Workflow of the analysis

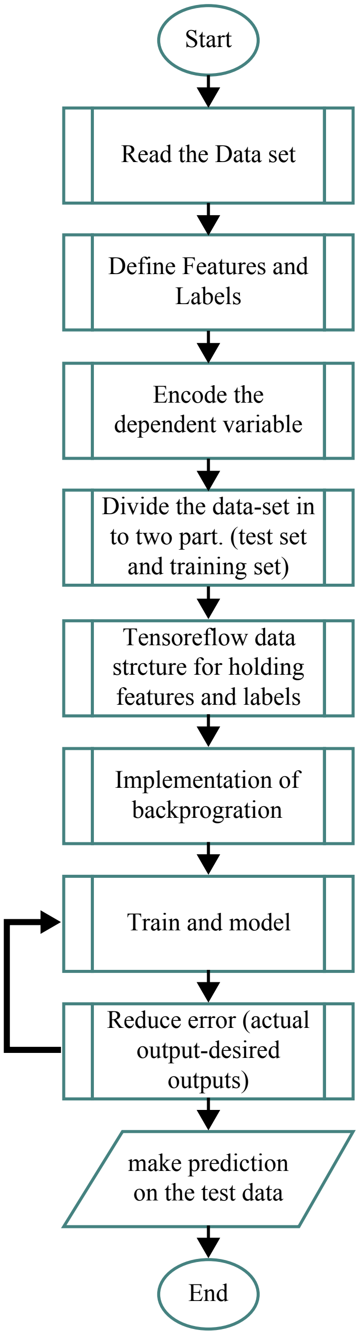

Table 1 Features list

Features name	Sub-category	Data distributions
Sex	Male	76.5%
	Female	23.5%
Age	Maximum value	100
	Minimum value	15
	Mean	53.32
	Median	53
Smoking	Yes	29.7%
	No	60.1%
	Ex	10.2%
HTN	Yes	55.6%
	No	44.4%
DLP	Yes	11.5%
	No	89.5%
DM	Yes	34%
	No	66%
Physical exercise	Yes	8.5%
	No	91.5%
Family history	Yes	34%
	No	66%
Drug history	Yes	64%
	No	36%
Psychological stress	Yes	20.7%
	No	79.3%
Chest pain	Yes	58.8%
	No	41.2%
Dyspnea	Yes	35%
	No	65%
Palpitation	Yes	17.8%
	No	82.2%
ECG	Abnormal	62.2%
	Normal	37.8%
AMI (Heart attack)	Yes	61.2%
	No	38.8%

string type. So we needed to convert them into numeric value before applying them in an algorithm. All our feature value was limited between YES and NO except smoking. Smoking had three different values YES, NO, and EX. We replaced YES numeric value 1 and NO with numeric value 0. We gave EX value 2. After replacing all string values with numeric value, our dataset was ready to feed into the algorithm. However, there were a few missing values in our dataset. Therefore, we needed to replace those messing values with some legit numeric values. We used Python Imputer function from sklearn library and the median strategy to replace missing values.

3.4 Data Training

We used K-fold cross validation for training and testing our model. The number of folds in our project was 10. That means our dataset was partitioned into 10 equal subsets. When one subset was considered as test data, the remaining 9 subsets were treated as training data by the algorithm.

3.5 Integrate the Artificial Neural Network (ANN) Model

Multilayer perceptron (MLP) is a feed-forward ANN that generates a set of outputs from a set of inputs. MLP uses backpropagation for training the network [10]. The backpropagation algorithm performs gradient descent to try to minimize the error between the network's output values and the given target values. There are several different optimizers and error functions that backpropagation used to minimize error. In our project, we used **mean squared error** as an error function and Nadam as an optimizer. Nadam combines adaptive moment estimation (Adam) and Nesterov-accelerated gradient (NAG). The benefit of using this optimizer is that we do not need to change the learning rate [11] again and again to get higher accuracy. We initialized our model's learning rate at 0.002. To reach the minima, Nadam adjusts the learning rate. Nadam optimization can be done by the following equation:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}\left(\beta_1 \hat{m}_t - \frac{(1-\beta_1)g^t}{1-\beta_1^t}\right)$$

Backpropagation also needs an activation function. The most popular activation function is sigmoid. However, we used the rectified linear activation unit or ReLU for its liner characteristic in the hidden layers, and for the output layer, we used softmax function.

3.6 Code Implementation

We implemented our neural network using the Python programming language. In our project, we used several machine learning packages. For computing tensor data, we used the **TensorFlow** library along with **keras** for neural network implementation. For data preprocessing, we used **pandas** library. Besides these, we also used **matplotlib**, **numpy**, and **sklearn** to plot graph. For server-side implementation, we used **Flask** for backend, **MySQL** for database, **Nginx** for load balancer, and **Gunicorn** for **WSGI HTTP** server. Our android application was implemented using the ANDROID STUDIO IDE and JAVA.

4 Experimental Analysis and Discussions

In Table 2, we can see the performance of our model. Tenfold produced ten different results. Fold eight gave the best result. Therefore, we used this model in our android application.

4.1 Confusion Matrix

The most eligible tool to justify a model is the confusion matrix (CM) [12]. Table 3 shows the CM for our cross-validation best model. Using these values, we can easily calculate accuracy, sensitivity, precision, specificity, and F1-score [13]. In Table 4, we have the performance result of our best-trained model. We can see how well

Table 2 Cross-validation outcome

Fold number	Training		Validation		Figure number
	Started from	Ended at	Start from	Ended at	
1	58	84	64	82	Fig. 2a
2	60	85	61	81	Fig. 2b
3	62	85	57	84	Fig. 3a
4	63	86	52	76	Fig. 3b
5	53	85	54	87	Fig. 4a
6	59	84	69	86	Fig. 4b
7	60	85	65	81	Fig. 5a
8	44	84	61	91	Fig. 5b
9	58	83	50	87	Fig. 6a
10	60	85	50	75	Fig. 6b

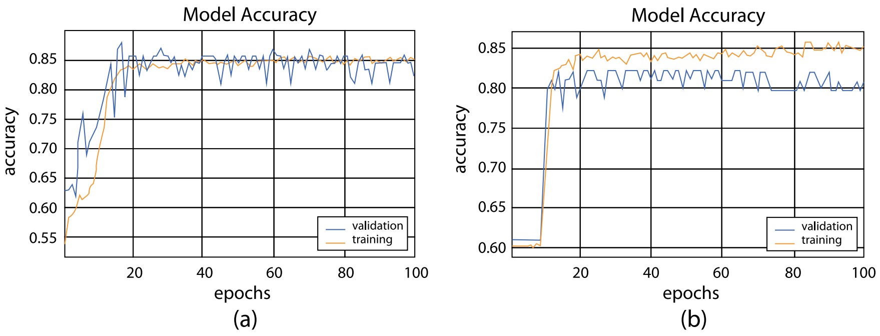

Fig. 2 Accuracy versus epoch graph for fold 1 (**a**) and fold 2 (**b**)

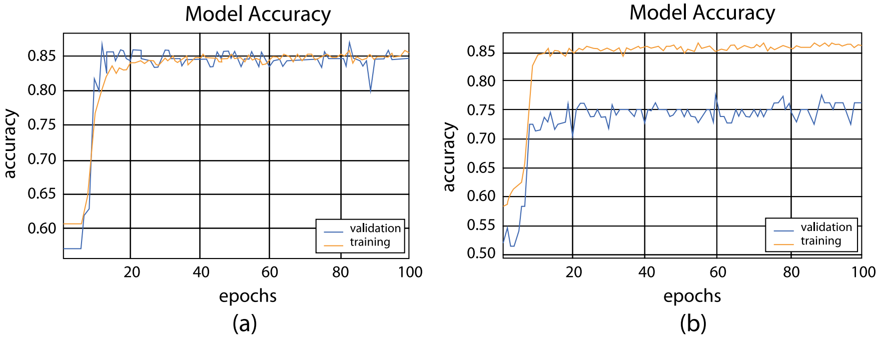

Fig. 3 Accuracy versus epoch graph for fold 3 (**a**) and fold 4 (**b**)

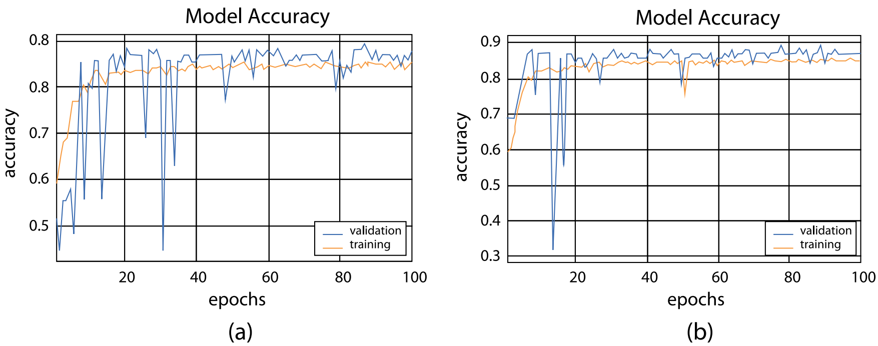

Fig. 4 Accuracy versus epoch graph for fold 5 (**a**) and fold 6 (**b**)

our model was performed by looking at the performance parameter. We get 91.6% accuracy, sensitivity 81.2%, specificity 98%, precision 96.2%, and F1-score is 88.1%.

In Table 4, we have the performance result of our best-trained model. We can see how well our model performed. We get 91.6% accuracy, sensitivity 81.2%, specificity 98%, precision 96.2%, and F1-score is 88.1%.

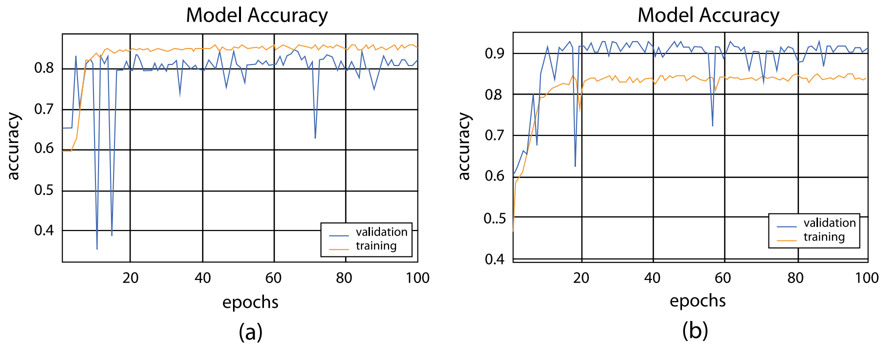

Fig. 5 Accuracy versus epoch graph for fold 7 (**a**) and fold 8 (**b**)

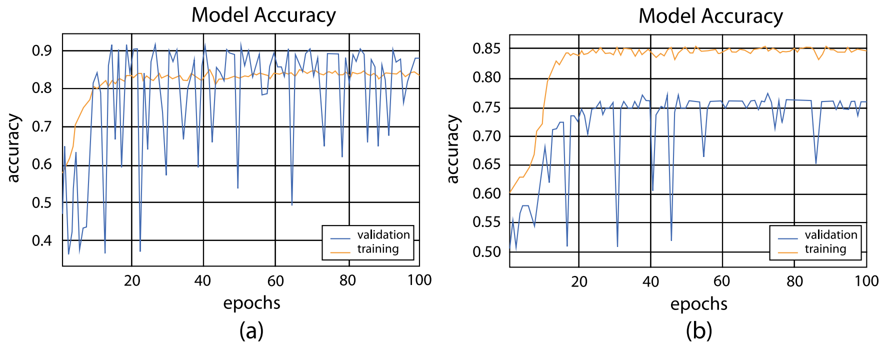

Fig. 6 Accuracy versus epoch graph for fold 9 (**a**) and fold 10 (**b**)

Table 3 Confusion matrix

Predicted class			
Actual class		No	Yes
	No	True negative (TN) = 50	False positive (FP) = 1
	Yes	False negative (FN) = 6	True positive (TP) = 26

4.2 Comparison

We can compare our work with existing similar work. Table 5 has a clear view of the performance of similar work that has been done. Our algorithm has performed better than most of the other algorithms.

Table 4 Performance of the model (ANN)

Evaluation matrix	Outcome (%)
Accuracy	91.6
Misclassification rate	8.4
Sensitivity/recall	81.2
Specificity	98
Precision	96.2
Prevalence	38.5
F1-score	88.1

Table 5 Performance analysis of the algorithms on same AMI dataset

<table>
<tr><th>References</th><th>Number of features</th><th>Algorithm name</th><th>Accuracy</th><th>Simulation tool and programing language</th></tr>
<tr><td>[5]</td><td>14</td><td>C4.5 Decision tree</td><td>86%</td><td>WEKA tool and Java</td></tr>
<tr><td>[6]</td><td>13</td><td>Backpropagation</td><td>84.47%</td><td>Python machine learning libraries</td></tr>
<tr><td rowspan="6">[7]</td><td rowspan="6">14</td><td>AdaBoost</td><td>75.49%</td><td rowspan="6">WEKA tool and Java</td></tr>
<tr><td>ANN</td><td>70.40%</td></tr>
<tr><td>Bagging</td><td>76.28%</td></tr>
<tr><td>K-NN</td><td>72.33%</td></tr>
<tr><td>Random forest</td><td>75.30%</td></tr>
<tr><td>SVM</td><td>72.72%</td></tr>
<tr><td>Our research study</td><td>14</td><td>ANN</td><td>91.6%</td><td>Python machine learning libraries integrated in smartphone</td></tr>
</table>

4.3 Receiver Operator Characteristic (ROC) and Area Under the Curve (AUC)

For classification problems in machine learning, we use receiver operator characteristic (ROC) and area under the curve (AUC) to visualize the performance of a trained model which was shown in Fig. 7. The higher the AUG of a model, the better the model is efficient to discriminate among the classes. As for our model, our ROC curve has a significantly higher value. We can say it by looking at the AUC, We have a high AUG value which is 95.

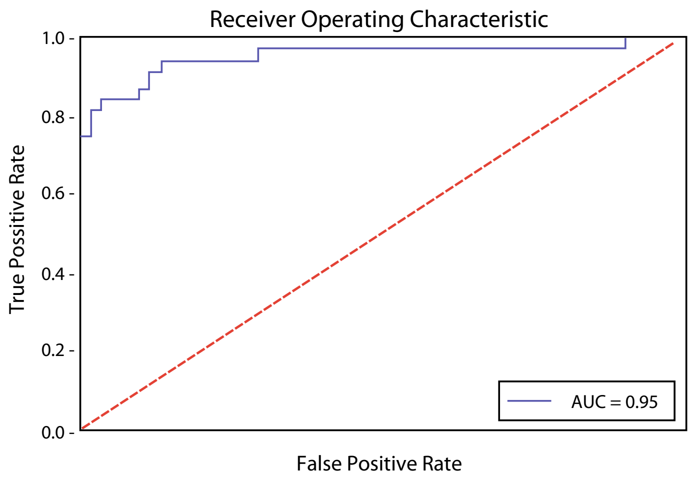

Fig. 7 Receiver operating characteristic (ROC) curve

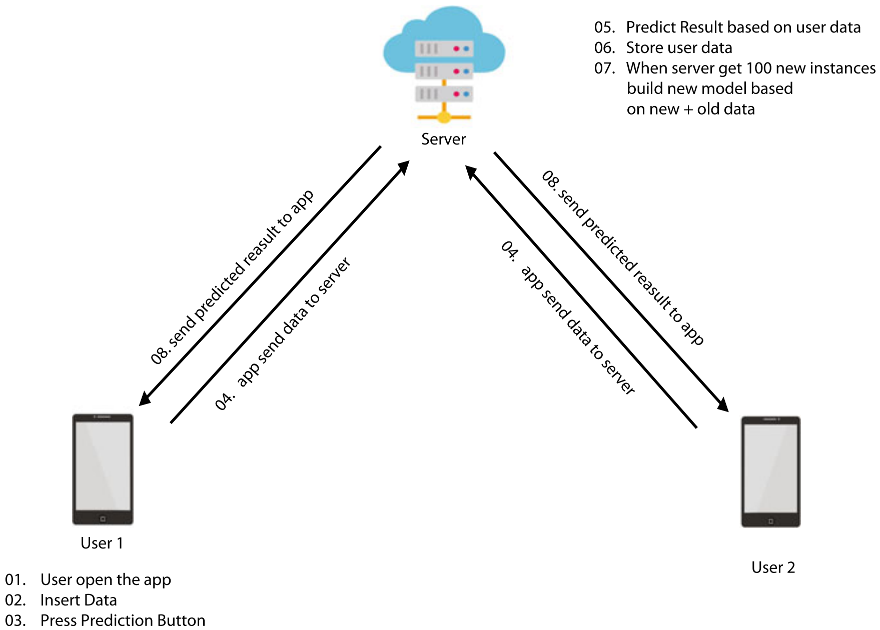

Fig. 8 Workflow of mobile application

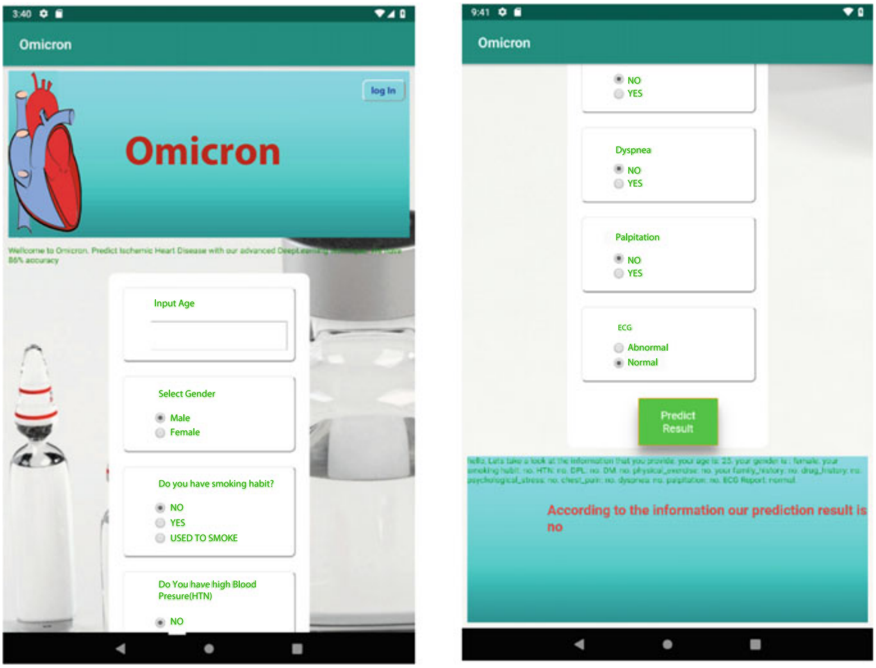

Fig. 9 Android application screenshot

4.4 *Workflow of Mobile Application*

The workflow of the android application was shown in Fig. 8. The screenshots of the android application can be seen in Fig. 9. At first, the user needs to answer all the mandatory questions, and after then, they will click the result button. A few second later, the user will get his result.

5 Conclusion

In a developing country like Bangladesh, most of the people live below the poverty line and the expense of medical diagnosis is very high. Therefore, most people do not feel necessary to diagnose the early development of CVDs. Using our free mobile application, people can have a clear idea of their health status against CVD, especially AMI. If they are predicted as having a chance of getting AMI, then they can immediately contact a doctor and get cured at its early stage. Our ANN model achieved **91.6%** accuracy. This trained model was used to implement a mobile application. Despite having high accuracy, we have some limitations. We were able to collect a limited number of data (835). The more data can be fed to the algorithm, the better accuracy we can get. Apart from that, the application needs an Internet connection to give a prediction as the server does all the heavy neural networking calculation. According to the current situation in Bangladesh, everyone does not a regular Internet connection. Even there is some remote place where there is no mobile network. We have some different ideas for the future. We will work with more real-time data. In future, our mobile application will have more features like a user will take a picture of their ECG report and it will show if the ECG report is abnormal or normal. There will be also a feature where a user can directly chat with doctors 24 hours for a medical emergency.

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Chapter 23
Design and Development of a Gaming Application for Learning Recursive Programming

Md. Fourkanul Islam, Sifat Bin Zaman, Muhammad Nazrul Islam, and Ashraful Islam

1 Introduction

Programming or computer programming is the process of designing and building an executable computer program for accomplishing a specific computing task [1]. It is a way of giving instructions to a computer about what it is supposed to do. Programming is important for learning to innovate and create effective solutions for difficult problems to enhance and increase the power of computers and the Internet through the development of software and applications. Thus, learning computer programming is a dire need for high school, college, undergraduate, and postgraduate students enrolled in computer science programs. Among the different concepts of programming, recursion is considered as a higher-level concept that has great importance. It is an important problem-solving strategy, which helps to break down big problems into smaller ones. The recursion can be considered as the way of thinking about problems, which can lead to simple and elegant solutions to certain problems that would otherwise be practically very difficult with an iterative algorithm. For some problems, a recursive algorithm can be much easier to develop, code, and comprehend. Also, in many cases, recursive algorithms resemble more closely to the logical approach that would be taken to solve a problem [2].

Most of the students fear to take the next step of learning higher-level programming concepts such as recursion, though it is one of the most powerful problem-solving approaches. McCauley et al. [3] found that educators have identified recursion as a challenging topic for the new students to learn to program. Recent studies showed that enrollment of new students in computer science (CS) related programs is decreasing [4] due to difficulties in learning the concepts of programming and maintaining interest in them. Improved teaching techniques should be applied to address these

Md. Fourkanul Islam (✉) · S. Bin Zaman · M. Nazrul Islam · A. Islam
Department of Computer Science and Engineering, Military Institute of Science and Technology, Mirpur Cantonment, Dhaka 1216, Bangladesh
e-mail: nazrul@cse.mist.ac.bd

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learning difficulties. When students start learning computer programming, they find other fundamental concepts of programming to be comparatively understandable. But, when they move forward and start learning advanced topics like recursion, they find it pretty difficult and thus start losing interest in programming. Nowadays, teaching techniques are being upgraded, and reinforcement-based techniques are being introduced to teach students. Among the existing (programming) teaching methods, computer and mobile gaming applications are prominent [5].

A serious game or applied game is a game designed for a primary purpose other than pure entertainment [6]. In other words, games that do not have entertainment, enjoyment, or fun as their primary purpose are known as serious games [7]. The primary 'serious' purposes can be to teach or train in areas such as education, health care, advertising, politics, etc. In essence, 'serious games' can be applied as an umbrella term for any game-based initiative that has an additional, 'serious' agenda. Serious games provide a way to create and share educational content while also making students feel more enthusiastic about their computing education by making it more relevant to real life [8]. It also provides a means for visualizing what is happening while learning the concept. Even the research conducted by Hasan [9] showed that games can help developing cognitive abilities of autistic children.

Therefore, the objectives of this research are to design and develop a serious game-based application to teach the concept of recursion in programming language and to find the impact of the developed application in terms of improving the learning ability of programming concepts.

The remaining sections of this article are organized as follows. Section 2, theoretical background on recursive programming and the use of games in learning purposes is discussed. In Sect. 3, designing the proposed game design along with its development is discussed. Section 4 presents the study procedure, participants profiles, data analysis, and outcomes of the evaluation study. Finally, the limitations, future work, and concluding remarks are presented in Sect. 5.

2 Literature Review

The necessity of computer programming is growing day by day. In recent years, a number of studies used serious games to find the effectiveness of those games in terms of learning basic and advanced programming concepts. For example, to teach java programming robocode [10] was developed as an open-source educational game where players can code their own robot to fight against opponent robot.

Kazimoglu et al. [4] developed a game named *Program your own robot* in adobe flash CS5 using action script 3. They found out that most of the serious games were built mainly focusing on learning programming not on developing computational thinking. As students are becoming less interested in science and engineering M. Muratet et al. [11] developed a game to make complex education interesting and counter the threat of decreasing students in computer science. They had constructed a game named Prog&Play using an open-source strategy game named kernel panic.

They also built an API that hides the game synchronization complexity and gives access to a subset of the game data. From their theoretical concept and survey, they showed that it helps students to improve their programming skills and their satisfactory levels were increasing.

A study conducted by Coelho et al. [12] showed that serious game is effective in learning introductory programming concepts. They found that problem solving through game is interesting, and the participant feedback was positive. Barnes et al. [13] also did a similar to find the feasibility of serious games to strengthen programming skills and to attract students and keep them in computer science. They found that serious game in programming enables the students to develop their own programs which helps the students to get more interest in learning programming. Wang and Chen [14] conducted a study to determine the effectiveness of helping learners to construct knowledge by game-based learning. Their study was intended to examine the effects of type of game strategy and preference-matching on novice learners. From their study, they have suggested that during concept clarification and consolidation phases, mismatching challenging game preference can be employed as an effective strategy to enhance learning.

There is potential to implement programming learning more smoothly by increasing people's knowledge and interest in programming by using various programming learning games as suggested by Mitamura et al. [15]. For this, they have developed four different games focusing on teaching basic syntax and concepts of programming and took feedback from users. Though user feedback indicates that it is difficult to learn programming by this game is difficult but the game is interesting who have prior knowledge in programming.

Most of the studies were on teaching the basics of programming but Chaffin et al. [16] had focused on teaching more advanced topics like recursion. They emphasized on specific algorithms like DFS and built a game like application to visualize the concept with the help of interactive visualization and simulation. Their game shows tree traversal of the recursive call in the DFS algorithm in a better graphical way also with the visualization of 'Depth First Search' stack showing in each node. Zhang et al. [17] also focused on specific recursion problems like finding the *n*th fibonacci number and finding the factorial of a number. They developed a game named *Recursive Runner* and showed a flow of the execution of the recursive function using the game. The evaluation study showed that visualizing fibonacci and factorial functions flow of execution was easy for the students after playing this game.

3 Design and Development of a Serious Game

A gaming concept has been proposed that requires the user must use the concept of recursion for solving different challenges using the proposed game. According to this concept, a character has to traverse through a 2D grid in a way that it can color the marked cells with the required color. There are some basic blocks present which help in controlling the movement of the character. There are panels for the main function and the user-defined function. The user has to design a code flow using various programming-related blocks to make the character color the required cells of the grid. By using a mouse click, the user will be able to add the blocks to the respective panels for designing the code flow. There are blocks that require an understanding of basic control statements of programming.

The proposed game consists of three levels. Level-1 is aimed at making the user familiar with the interface of the gaming application. Level-2 stands for getting the user familiarized with the recursive programming concept, while level-3 is designed to understand whether the user got a better grasp on the recursive programming concept. Successful completion of level 1 leads the user to play level 2, while level 2 completion leads the user to play level 3.

Running the application would at first open the home page of the game, which consists of a menu where the user has to enter his/her name to start playing the game with level 1. The main components of the levels of the game can be seen in Fig. 1.

3.1 Level 1

In this level, the character has to traverse through a fixed path using the blocks by placing them on the main function panel. There is no procedure panel available at this level. The user can just use the forward and rotate blocks to traverse the required path.

After building the main function in the required manner, if the user clicks the 'Play' button, then the character will traverse the path according to the directions provided by the user. While this path is being traversed, the cells will get colored. If all the required cells have been colored by the character according to the fixed path, then the user will successfully complete this level, and a button will pop up which will allow him to progress to the next level. If the player makes any mistake or sees that his designed function could not provide a correct answer, the player can clear all the blocks by pressing the 'Clear' button and then try again.

Figure 2 shows the starting and ending scenes of level 1. This level involves basic instructions without the use of any kind of recursion. The concept of recursion is introduced in the second level.

3.2 Level 2

In this level, the task of the user again colors some marked cells of the 2D grid.

But in this level, there is a cell containing a coloring brush and a cell containing a color bucket. Besides, this level contains an extra panel for a user-defined function, named as Proc (in short for Procedure, which basically means a function). The user has to direct the character in such a manner that the character picks up the brush at first, then moves toward the color bucket to make the brush colored.

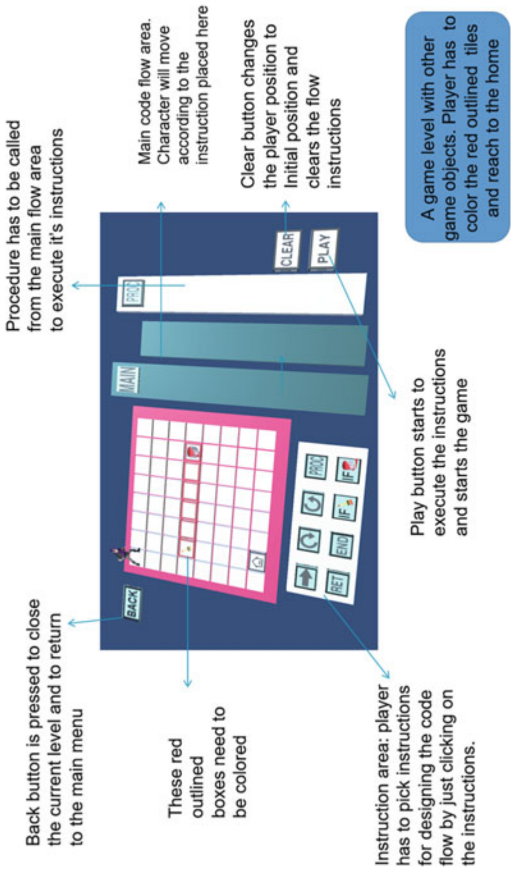
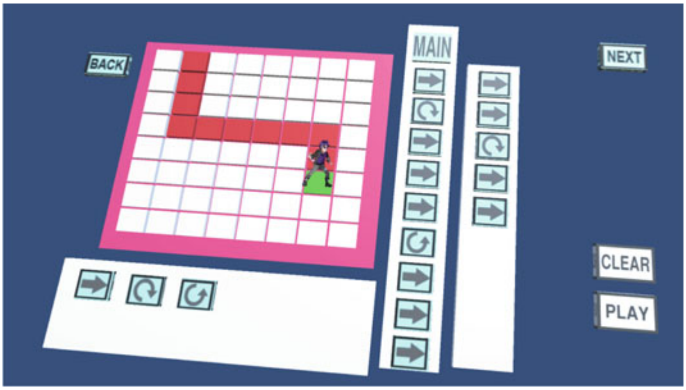


Fig. 1 Basic game components

(a)



(b)

Fig. 2 Screenshot of level 1: **a** Starting scene, **b** success scene

Finally, goes back in the direction of the coloring brush so that all cells in that path become colored. Once the character reaches the cell containing the coloring brush for the second time, the color is removed from the brush, and the brush is dropped. After that, the character will have to reach the ending point to complete the level. If all the required cells are colored and the character managed to reach the ending point, then a button will pop up which will allow the user to move to the next level in a similar way to that in level 1. Figure 3 shows the starting and ending scenes of level 2.

In this level also, the player may just try to use the forward and rotate blocks and traverse the required cells. But the player will not be able to complete it since there is only a limited space for adding blocks in the main function and the user-defined function panels. For this reason, the player is bound to use the IfBrush and the IfBucket blocks in the proper manner and call the function (named as Proc) from

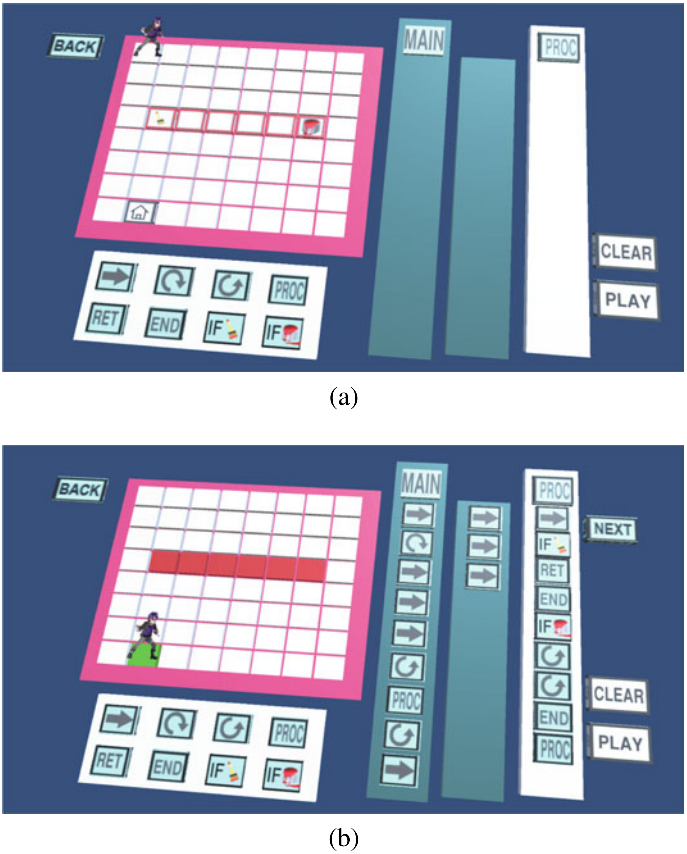


Fig. 3 Screenshot of level 2: **a** starting scene, **b** success scene

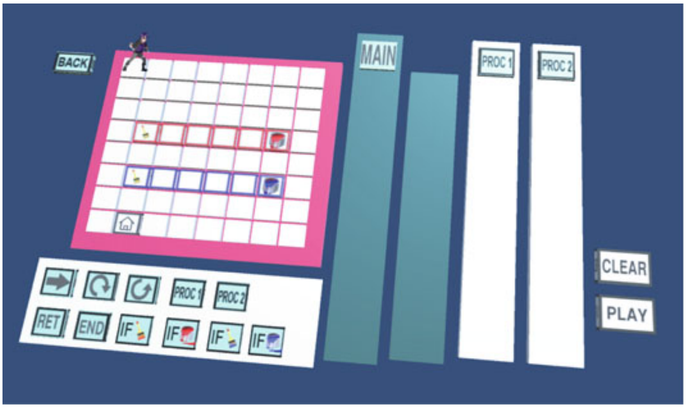
within its own scope so that all necessary blocks can be placed within the limited space provided. This concept of calling a function from within the same function is actually recursion. So, at this level, the player will get familiarized with the recursive programming concept.

3.3 Level 3

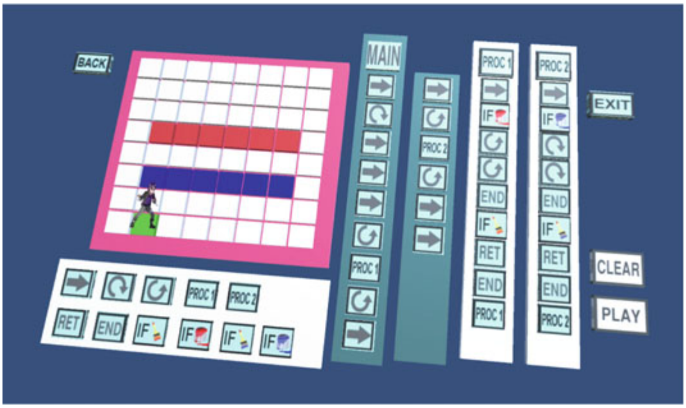
This level will start once the user completes level 2. It is a similar level to the second level. The only difference here is that there are two separate groups of cells that are to be colored by two separate colors (red and blue). Another difference is the presence of a new panel for a second user-defined functions (Proc2) along with the panel for the first user-defined function (Proc1) and the main panel.

Similar to the second level, the user must direct the character in such a manner that the character picks up the brush at first, then moves toward the red color bucket near it to make the brush red-colored. After that, it has gone back in the direction of the coloring brush so that all cells in that path become colored in red. Once the character reaches the cell containing the coloring brush for the second time, the color is removed from the brush and the brush is dropped. Then, the character will have to go for coloring the second group of cells with blue color. For that, it has to pick up the second brush and in a similar fashion, color the cells with the blue color.

Finally, the character will have to reach the ending point to complete the level. If all the required cells are colored and the character managed to reach the ending point, then this level will be completed and thus the game ends. Figure 4 shows the starting and ending scenes of level 3.



(a)



(b)

Fig. 4 Screenshot of level 2: **a** starting scene, **b** success scene

The third level is an extended version of the second level. It has been designed to understand whether the user/student got a grasp on the recursive programming concept. By performing two similar tasks by two recursive function designs, the challenge will be solved with much fewer steps, and it will be a fun experience for the user. Along with the fun aspect, the student will also learn the recursive concept of programming.

4 Evaluating the Serious Game

4.1 Study Objectives

The objective of this evaluation study was to measure the effectiveness of the serious game in terms of improving the learning ability of programming concepts for different skill level people in learning programming.

4.2 Study Procedure

A light-weighted evaluation was carried out at the software engineering lab of the authors' institute. The evaluation study was replicated with sixteen students of different age groups and knowledge levels. According to the participant's profile, we divided the skill level of the participants into three categories as *Expert*, *Moderate* and *Novice*. Participants with prior knowledge in both basic programming and recursion are categorized as *Experts*. *Moderate* groups have prior knowledge only in basic programming, while *Novice* participants have no knowledge in programming and recursion at all. There were four students in expert, eight in moderate, and four in novice category.

A pretest and a posttest questionnaire were designed. The pretest questionnaire was given to the students before playing the game. The students were asked to solve the pretest questionnaire first, and then, a short presentation about the game was given to the participants. After that, the students were asked to play the actual game. The time taken to solve the tasks of every level, as well as the number of attempts required for solving each level, were noted down. After that, they were asked to take part in answering the post-test questionnaire, which had questions very similar to the pre-test questionnaire. Question 1 was worth 2 points and the other two were worth 3 points each for both pretest and posttest questionnaires. The accuracy of the answers and the time taken from the pretest and posttest questionnaires were analyzed to determine the effectiveness of the gaming application.

4.3 Data Analysis and Findings

Based on the collected data, the points for the students of a specific category were calculated. Formula used to measure the obtained points was

$$p = 1/n * \sum_{i=1}^{n} x * t/t_i$$

where p is the average points obtained on a question by the students of a category, x is the points for that question, t is the minimum time taken by a student to solve it, t_i is the time taken by the ith student to solve it, and n is the total number of students in that category. So, the recorded points are out of 2 for question 01 and out of 3 for questions 2 and 3. A brief summary of the recorded data is given in Table 1.

After that, a comparison was made between the pre- and posttest results by summing up the values for all the three questions for each category. Thus, all the values obtained were points earned out of 8. The formula used for this calculation is as follows:

$$P = \sum_{i=1}^{n} x_i$$

where P is the points scored out of 8, x_i is the points scored for the ith question for either pretest or posttest.

In this way, Table 2 was formulated. The results showed the progress of the students of mid-level was significant compared to the other levels.

A bar chart showing the progress of all skill-based categories in the post test in comparison to the pretest is given in Fig. 5.

Table 1 Summary results of the evaluation study

Test phase	Ques no.	Expert	Moderate	Novice
Pre-test	1	1.0	0.68	0
	2	2.5	1.0	0
	3	2.1	1.42	0
Post-test	1	1.89	1.62	0
	2	2.75	2.09	1.5
	3	2.84	2.41	1.5

Table 2 Pre- and posttest points based on students' skill level

Student type	Pre test	Post test
Expert	5.6	7.48
Moderate	3.1	6.12
Novice	0	3

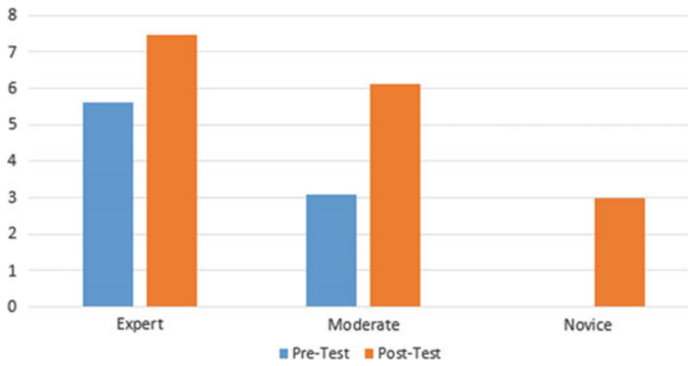


Fig. 5 Points scored based on skill level

Table 3 Overall comparison of pre and post test results

Test phase	Total points	Average	Std. Dev.	Variance
Pre test	41.92	2.62	2.12	4.49
Post test	81.12	5.07	2.82	7.95

Then, an overall analysis of the pretest and posttest results were conducted using the total points earned by all the 16 students and computing their average, standard deviation, and variance. Table no 3 showed an overall increase in performance in the posttest results compared to the pretest results.

In case of understanding the recursion, it was found out that the students who had moderate level of prior knowledge in basic programming (moderate group) gained a lot through the use of this game, rather than those who had good level of prior knowledge in (advanced) programming concepts and those who had no knowledge in programming. So, the results showed that the serious game developed for learning recursive programming is much more effective for students with prior moderate knowledge group in basic programming.

5 Conclusions

In this research, a serious gaming application was developed and the effectiveness of using serious games in learning the recursion concept of programming language was explored. The gaming application was evaluated with participants having different level of familiarity with programming, and the results showed that the participants who had prior knowledge in basic programming gained a lot through the proposed game, rather than those who had prior knowledge in advanced programming concepts and those who had no knowledge in programming. Some limitations of this research were that it did not cover all types of recursion, and limited number of levels were made. Besides, the number of participants for each of the participants group was not adequate. In future, the proposed gaming application will be extended to cover all types of recursion including other programming concepts and will be evaluated with a larger number of participants. Besides, another study will be conducted that will show the comparison of the learning effectiveness between the traditional teaching methods and the uses of the proposed gaming application.

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Chapter 24
Alcoholism Detection from 2D Transformed EEG Signal

Jannatul Ferdous Srabonee, Zahrul Jannat Peya, M. A. H. Akhand, and N. Siddique

1 Introduction

Alcoholism is a term for any alcohol consumption leading to mental or physical health problems [1]. Long-term alcohol consumption can cause a wide range of psychological issues. Excessive consumption of alcohol causes damage to the activity of the brain, and over time, mental health can become severely affected [1, 2]. Nowadays, alcohol misuse is related to increased risk of crime, including child neglect, domestic violence, burglary, fraud, road accidents, and assault [3]. Alcoholism is the most common psychological condition in the general population and the lifetime prevalence of alcohol dependence is 8–14% [4, 5]. Seeing alcoholism as a major burden for any modern society, physiological and neurological information relating to alcoholism is needed for the care of alcoholic patients.

Alcoholism detection is an important issue nowadays. There exist a number of approaches to detect alcoholism. Among those methods, screening is a well-established method and recommended for over the age of 18 for the diagnosis of alcoholism [6]. Blood alcohol content (BAC) is another common test for actual alcohol use [7]. Screening and BAC tests do not provide any discriminating factors

J. F. Srabonee · Z. J. Peya · M. A. H. Akhand (✉)
Department of Computer Science and Engineering, Khulna University of Engineering and Technology, Khulna 9203, Bangladesh
e-mail: akhand@cse.kuet.ac.bd

J. F. Srabonee
e-mail: srabonee1507090@stud.kuet.ac.bd

Z. J. Peya
e-mail: jannat.kuet@gmail.com

N. Siddique
School of Computing, Engineering and Intelligent Systems, Ulster University, Londonderry, Northern Ireland, UK
e-mail: nh.siddique@ulster.ac.uk

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between alcoholics and non-alcoholics [8]. On the other hand, clinical alcohol tests are carried out using different brain signals such as magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI), and electroencephalography (EEG). Several studies are available for alcoholism detection using such brain signals [9–11].

Among different brain signals, EEG is considered realistic and effective for alcohol detection because it has many advantages over other methods when studying brain function [9]. EEG alludes to the examination and investigation of electrical fields of the brain recorded by enhancing voltage contrasts between anodes set on the skull. Clinically, EEG refers to monitoring the spontaneous electrical movement of the brain over a while as estimated by a few terminals on the scalp. EEG is portable and non-invasive and provides low-cost, high-temporal resolution data [12, 13]. As EEG signals monitor brain activity and an alcoholic person's brain activity is different from that of a normal person, there is a space for the detection of alcoholism from EEG signals. Therefore, EEG is used as an informative indicator of the alcoholic brain condition [10, 11].

Different machine learning (ML) algorithms are utilized for the automatic classification of EEG signals for alcoholism detection. In ML, intelligent systems are trained with real-world data. Bae et al. [9] proposed a system to evaluate the distinctions in the status of the mind-dependent on the EEG signal of typical subjects and heavy drinkers; the authors used support vector machine (SVM) classifier in their experiment. Gökşen et al. [11] used relative entropy and shared knowledge to pick the optimum channel configuration to maximize alcoholic and control classification. In their work, K-nearest neighbors (KNN) classifier accompanied by the Euclidean distance metric plus mutual knowledge for the channel selection process were used. Shri et al. [14] proposed the utilization of gamma-band entropy for the recognizable proof of drunkards utilizing multilayer perceptron (MLP), probabilistic neural system (PNN), and SVM classifiers to recognize the control subjects from their alcoholic partners.

Sriraam et al. [15] studied the impact of liquor on visual occasion-related possibilities of EEG utilizing spectral entropy (SE) parameters in the beta band. The SE highlights were added to a KNN classifier to separate visual event-related potentials produced in heavy drinkers and controls inside the beta band. Shooshtari et al. [16] proposed a method for selecting an optimal subset of EEG channels to identify alcohol perpetrators from normal subjects focusing on the application of model-based spectral analysis and matrix correlation. Ong et al. [17] implemented the principal component analysis (PCA) as a feature selection technique when selecting a subset of visual-evoked potentials (VEP) channels. The selected channels preserve as much information as possible compared with the full set of channels.

Recently, deep learning (DL) algorithms, advanced ML methods, are found to be the most effective for classification and other perceptual tasks [18]. Among different DL methods, convolutional neural network (CNN) is one of the most dominant techniques for image analysis and classification and has been investigated in this study. When employing a CNN, the input is a tensor with shape (number of images) $\times$ (image height) $\times$ (image width) $\times$ (image depth). An innovative approach is used in this study to transform 1D EEG data into a 2D form. Pearson's correlation coefficient is used to determine the correlation between the EEG channels which was represented in the correlation matrix. Further, the matrix was plotted using heatmap and transformed into 2D. The transformed 2D EEG data were prepared to feed into CNN, and thus, the classification of alcoholic and control subjects was performed. The proposed method has been investigated on the benchmark dataset and found to be an effective alcoholism detection method compared with other existing methods.

The rest of this paper is organized as follows. Section 2 explains the proposed method. Section 3 presents experimental studies and discussion. The paper ends with Sect. 4.

2 Alcoholism Detection from Transformed 2D EEG Signal

This study aims to use a CNN-based method for alcoholism detection from the EEG signal. For this purpose, the EEG signals, i.e., data from different channels', were converted into 2D representations which were fed into the CNN classifier. The methodology was divided into three phases—EEG data preparation, transforming into 2D, and finally classifying with CNN. Figure 1 illustrates the proposed method and the following subsections briefly explain the steps of the procedure.

2.1 EEG Data Preparation

EEG data collected through placing electrodes on the skull and dimension of the dataset depends on the number of electrodes uses which refers to channels. A universal 10–20 framework is well known to portray and apply the area of scalp cathodes with regard to an EEG. The EEG dataset used in the study contained 64 channels among which 61 were active EEG channels, and the rest of the three channels were used for reference. The position of active 61 channels with channel identification is shown in Fig. 2. All scalp terminals alluded to Cz.

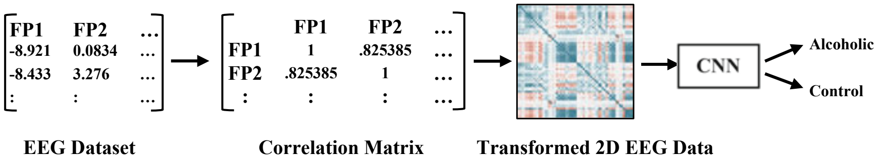


Fig. 1 Schematic view of the proposed method

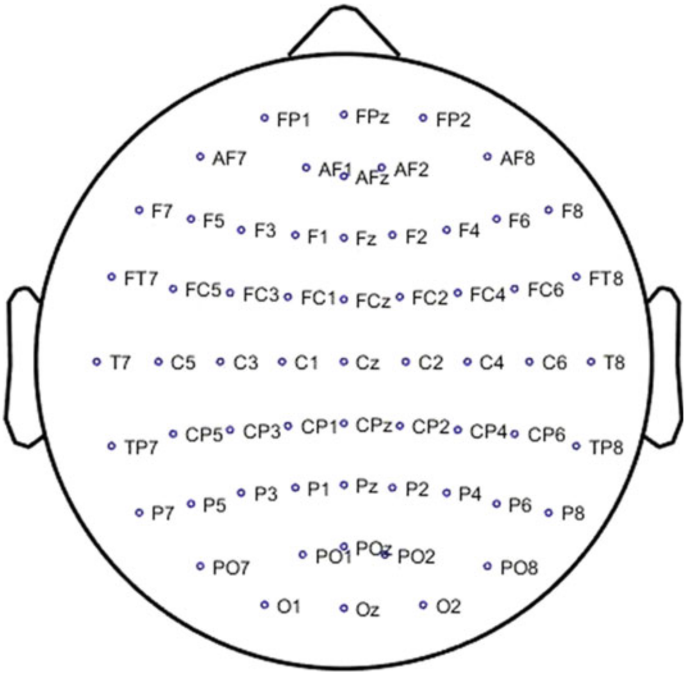


Fig. 2 Location of EEG electrodes of 61 active channels

2.2 Correlation-Based 2D Transformation of the EEG Data

In this study, Pearson's correlation coefficient (PCC) is employed to transform 1D EEG data into a 2D form. PCC or bivariate relationship is a proportion of direct relationship between two factors of one measurement. Numerically, it is the covariance proportion of two exhibits to the whole of their standard deviation. The worth lies somewhere in the range of -1 and $+\ 1$, where $+\ 1$ infers a 100% positive relationship, 0 does not show a connection, -1 means a 100% negative connection [19]. The PCC is usually denoted by ρ. For a set of random variables (X,Y), the ρ is defined by,

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \tag{1}$$

where cov is the covariance, $\sigma_{X,}$ and σ_Y are the standard deviations of X and Y respectively. The formula for ρ can be expressed in terms of mean and expectation. Since,

$$\text{cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] \tag{2}$$

ρ can also be written as:

$$\rho_{X,Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y} \tag{3}$$

where μ_X and μ_Y are the means of X and Y respectively and E is the expectation [20].

PCC applied on all the active EEG channels' (i.e., 61 channels discarding three reference channels) data to proceed for the classification task. Firstly, the PCC was determined between the combination of each two channels to formulate the representations. The coefficients were later rearranged into a matrix form, and hence, a 2D data array was obtained. The size of the PCC matrix was 61 × 61 for 61 active EEG channels. The matrix was then plotted as a color image using the seaborn heatmap library of Python, and the outcome is shown in Fig. 3. Each tiny square in the image represents the correlation between two individual channels. The intensity of the red or blue color indicates the transition of the correlation level. Red color indicates a negative correlation (−1), blue indicates a positive correlation (+1) and white indicates no correlation (0). As each channel always perfectly correlates with itself, the line of blue squares (indicating positive correlation) going from the top left to the bottom right is the main diagonal. The image is symmetrical above the main diagonal being a mirror image of those below the main diagonal.

Figure 4 shows two sample 2D images where Fig. 4(a) shows the imge of alcoholic patient and Fig. 4(b) shows the image of control subject. As the electrodes are placed on different locations of the scalp, the values in the correlation matrix show the effect of alcoholism over different regions of the brain. The colored plot helps

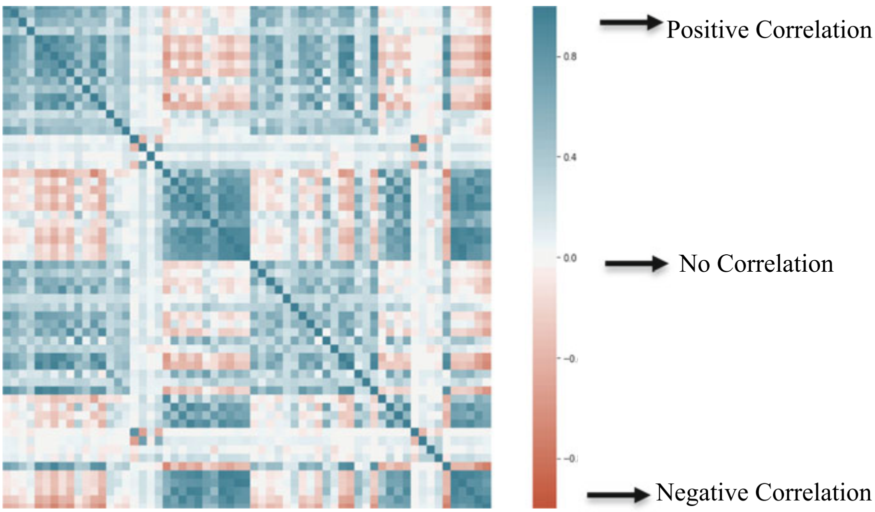
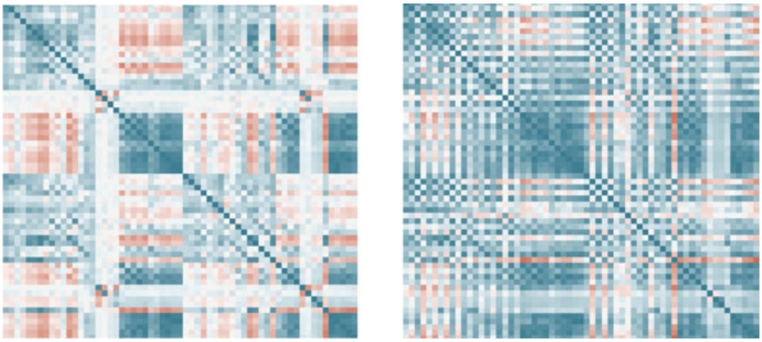


Fig. 3 Image representation of correlation-based 2D transformed EEG channels' data



(a) Alcoholic (b) Control

Fig. 4 Significance of 2D representation of EEG data on alcoholism detection: **a** Alcoholic and **b** Control

to visualize and understand that variation. The 2D representation in Fig. 4 shows a distinct variation for alcoholic and control subjects and is considered suitable for classification using CNN.

2.3 Classification with CNN

A CNN is a deep learning method equipped with catching an input picture, relegating essentialness (learnable weights and biases) to various parts of the picture, and having the option to recognize them [21]. Compared with other classification algorithms, the preprocessing needed for CNN is much less. CNN comprises various layers that incorporate image layer, convolution layer, pooling layer, flatten layer, fully connected layer, and output layer. In CNN architecture, pooling activity follows every single convolution activity and a few sets of convolution-pooling layers may be used to get authentic results for a specific issue.

Figure 5 is the architecture of a standard CNN used in this study; there are four convolution layers followed by four max-pooling strategies. The input to CNN is the PCC -based 2D EEG data (as like Figs. 3 and 4), i.e., RGB images. The images were resized to 64×64 and used as input to the first convolution layer (conv2d_1). The Rectified Linear Unit (ReLu) activation function was used in each convolution layer. In the conv2d_1 layer, 32 feature maps of 62×62 dimensions were produced using 32 kernels of size 3×3. The feature maps were further fed into the next layer which was the max-pooling layer (max_pooling2d_1) for feature extraction with a pool size of 2×2. This layer down-sampled the input representation by taking the maximum value by pool size for each dimension along with the features axis, and the resulting shape was (31, 31, 32) where 32 was the number of feature maps. The kernel size was the same in the rest of the convolution layers, but the number of feature maps or kernels had been increased to 64 in the second convolution layer (conv2d_2), 128 in

Layer (Type)	Output Shape (Rows, Columns, Filters)
Input shape	(64, 64, 3)
conv2d_1 (Conv2D)	(62, 62, 32)
max_pooling2d_1 (MaxPooling2)	(31, 31, 32)
conv2d_2 (Conv2D)	(29, 29, 64)
max_pooling2d_2 (MaxPooling2)	(14, 14, 64)
conv2d_3 (Conv2D)	(12, 12, 128)
max_pooling2d_3 (MaxPooling2)	(6, 6, 128)
conv2d_4 (Conv2D)	(4, 4, 128)
max_pooling2d_4 (MaxPooling2)	(2, 2, 128)
flatten_1 (Flatten)	(512)
dense_1 (Dense)	(64)
dense_2 (Dense)	(1)

Fig. 5 Architecture of the convolutional neural network

the third (conv2d_3), and the fourth (conv2d_4) layer of convolution. Max pooling method with a pool size of 2×2 was used after each convolution layer. The resulting matrices after the four convolution layers, and four max-pooling layers were divided into a single-layer or a single-column matrix, containing all the pixel values of these matrices in a single column that is also known as the flattening layer. This flattening layer having 512 nodes was being used to serve the next two dense layers (fully connected artificial neural network), and finally, the output of CNN comes from the dense_2 layer having a single node.

3 Experimental Studies

This section describes the experimental outcomes of the proposed alcoholism detection system on the benchmark dataset. The performance of the proposed method is also compared with existing methods.

3.1 Experimental Data

The EEG dataset used for the proposed study of alcoholic and control subjects is collected from the UCI Machine Learning Repository [22]. The data incorporates 64 channels mounted on the subject's scalp that were tested for 1 s at 256 Hz (3.9-ms age). A universal 10–20 framework is used to portray and apply the area of scalp cathodes with regard to an EEG. There are a total of 64 channels, and the position of active 61 channels (three for reference) with channel identification is shown in Fig. 2.

The dataset consists of EEG signals of 10 drunkards (i.e., alcoholic persons) and 10 control subjects (i.e., non-alcoholic persons); and each subject contains data of 10 individual runs. A separate test set is provided with the same 10 drunkards and 10 control subjects as the preparation. The training set contains 468 data samples, and the test set contains 480 data samples. The dataset is used in several recent studies considering it as a standard benchmark dataset for alcoholism detection [9, 11].

3.2 Experimental Setup

The proposed model was implemented in the Python environment using Keras, TensorFlow, pandas, NumPy, sklearn, matplotlib, and pyplot libraries. The device in which the model was implemented had the following configuration: processor of 4th Generation Intel® Core™ i7-4720HQ CPU @ 2.60 GHz, GPU of NVIDIA Ge-Force GTX 850 M, 4 GB.

3.3 Experimental Result and Analysis

We have analyzed the performance of the proposed method rigorously and finally compared with existing methods. To evaluate the performance of the CNN model, various metrics were employed which are accuracy, sensitivity, specificity, and $F1$ score. Accuracy is specified by the classifier as the percentage of the correct estimates, either for the input of the training set or the test set input. Loss is characterized as a feasible error, which represents the price paid in classification problems for the inaccuracy of predictions. For the measurement of losses, the binary cross-entropy method is used. This loss is measured using the following:

$$H_p(q) = -\frac{1}{N}\sum_{i=1}^{N} y_i \,.\, \log(p(y_i)) + (1 + y_i).\log(1 - p(y_i)) \tag{4}$$

where y is the label and $p(y)$ is the predicted probability of the point being that label for all N points [23]. Figure 6 shows loss curves for both training and test sets for 100 training epochs. It is observed that the loss of both train and test sets decreased rapidly up to 20 epochs and reached an almost steady-state position. After 40 epochs, the loss of the test set seems to increase slightly indicating overfitting.

Figure 7 shows the accuracy curves of both training and test sets for the same 100 epochs of Fig. 6. Notably, the test set accuracy was increasing gradually as well as rapidly with training set accuracy up to 20 epochs. Training set accuracy reached a stable state after 23 epochs and test set accuracy reached the stable state after 41 epochs. Finally, the system reached 98.125% test set accuracy.

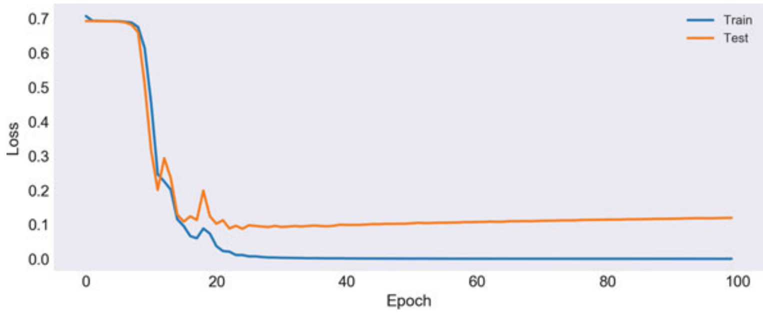


Fig. 6 Loss curve for epoch varying up to 100

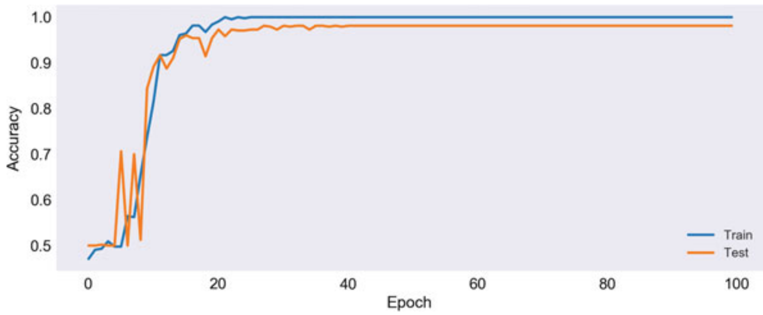


Fig. 7 Accuracy curve for epoch varying up to 100

Sensitivity is a measurement for the amount of actual positive cases that have been predicted to be positive (or true positive). Sensitivity can be measured mathematically using Eq. (5). The higher sensitivity value would indicate a higher value of the true positive and a lower value of the false-negative and vice versa [24]. The sensitivity of the model reached a score of 98%.

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \tag{5}$$

Specificity is characterized as the percentage of actual negatives that have been predicted as negative (or true negative). The higher value of specificity would mean a higher value of true negative and lower false-positive rate [24]. Specificity is measured using Eq. (6), and the sensitivity of the model reached a score of 97%.

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}} \tag{6}$$

The $F1$ score (also F-score or F-measure) in the statistical analysis of the binary classification is a measure of the accuracy of a test. $F1$ Score is needed when a balance between sensitivity and precision [25]. The $F1$-score of the model reached a value of 98%.

$$\text{Precision} = \frac{\text{True Positive}}{\text{Total Predicted Positive}} \tag{7}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{8}$$

Table 1 compares the test set alcoholism detection accuracy achieved by the proposed method with other existing methods on the same UCI benchmark dataset. A brief description of the individual methods is also presented in the table for a better understanding of the methods. Existing methods used Granger causality-based feature extraction with support vector machine (SVM) classifier, K-nearest neighbor (KNN), mean gamma-band power-based correlation coefficient measurement between channels with SVM, gamma sub-band power with principal component analysis (PCA), and neural network classifier. Among the existing methods, the pioneering method by Ong et al. [17] is shown to achieve the highest accuracy of 95.83%. The method seems relatively complex combining γ sub-band power, PCA, neural network. The recently proposed method considering Granger causality feature extraction with SVM by Bae et al. [9] is shown to achieve an accuracy of 90.00%. On the other hand, the proposed method outperformed the existing methods with an accuracy of 98.125%. The performance comparison indicates that PCC-based 2D

Table 1 Performance comparison of proposed alcoholism detection method with existing ones

Sl. No.	Work Ref., Year	Methods		Accuracy (%)
1	Bae et al. [9], 2017	Granger causality feature extraction + SVM (polynomial)		90.00
		Granger causality feature extraction + SVM (linear)		85.00
2	Gökşen and Arıca [11], 2017	KNN + Euclidean distance	Relative entropy	73.67
			Mutual information	82.33
		KNN + Mahalanobis distance	Relative entropy	80.33
			Mutual information	80.00
		Linear Discriminant Analysis	Relative entropy	74.67
			Mutual information	76.67
		SVM	Relative entropy	76.00
			Mutual information	75.33
3	Shooshtari and Setarehdan [16], 2010	Mean γ band power + orrelation coefficient measure between channels + SVM		80.00
4	Ong et al. [17], 2005	γ sub-band power + PCA + Neural Network		95.83
5	Proposed method	PCC-based 2D transformation + CNN		98.13

representation of EEG data and classification with CNN is an effective method of alcoholism detection.

4 Conclusions

The neural activity of an alcoholic person is different from a normal person. EEG signals can capture the variations of neural activities in the human brain through electrodes that are placed on different regions of the scalp. In this study, the correlation among different EEG channel signals is measured and transformed into 2D image form as input to CNN to classify the signals for alcoholism detection. The proposed approach is shown to outperform existing methods tested on a benchmark dataset.

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Chapter 25
Numerical Study on Shell and Tube Heat Exchanger with Segmental Baffle

Ravi Gugulothu, Narsimhulu Sanke, Farid Ahmed, and Ratna Kumari Jilugu

1 Introduction

In industries shell and tube heat exchangers are the most wanted, popular equipment like petrochemical, food industry, oil and gas refinery etc [1, 2]. So Nowadays, the development of heat exchangers with high thermal efficiency is great significant task for energy saving purpose. For many years in oil refining, power plants, chemical engineering processes and milk processing units shell and tube heat exchangers are cavort key role listed by Ravi et al. [3]. Shell and tube heat exchangers are added extensively operates engineering equipment in perseverance. Baffles are one which is most important components in shell and tube heat exchanger devices [4]. Shell and tube heat ex-changers with segmental baffles which shown in Fig. 1 may have many drawbacks as listed by pioneers [4–6]. For several years, various types of baffles (segmental baffle, helical baffle, ladder type baffle etc.) have been used in order to enhance heat transfer with allowing a reasonable loss of pressure throughout the heat exchanger. The flow through out the shell side is having complex orientation with single sengemtal baffle arrangement in STHE (Shell and Tube Heat Exchanger), where the segmental baffles lead to a streamline inside the shell which is partially parallel and partially normal to the bank of tube [7]. According to Dogan Eryener [4]

R. Gugulothu · N. Sanke
Department of Mechanical Engineering, University College of Engineering, Osmania University, Hyderabad, India

F. Ahmed (✉)
Department of Nuclear Science and Engineering, Military Institute of Science and Technology, Dhaka, Bangladesh

R. Kumari Jilugu
Quality Control, 02-Electrical Machines, BHEL Ramachandrapuram, Hyderabad, Telangana State, India

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baffle arrangement is one of the method for enhancing the heat transfer coefficient as well as reduction of pressure drop. Owing to its extensive applying, abundance of effort has been reveal in shell and tube heat exchangers.

2 Literature

Gaddis et al. [7] presented an evaluation procedure for evaluation of pressure drop in shell side in a STHE amid segmental baffles. Huadong Li et al. [1] experimentally studied the pressure drop and local heat transfer of a STHE amid segmental baffle by changing the detriment between baffles and shell. Kamel Milani Shirvan et al. [8] designed and experimentally studied on tube structure with smooth tube and cosine wave tube for water to water loops, found that larger thermal performance factor. Mellal et al. [2] investigated the heat transfer coefficient, turbulence in baffle spacing using CFD simulation in STHE baffle spacing of 106.6, 80 and 64 mm, while varying six baffle angles in geometrical configuration of 450, 600, 900, 1200, 1500 and 1800, respectively and found that the segmental baffle orientation angle 1800 at 64 mm baffle spacing is the efficient orientation in geometry of STHE for flow mixing and highest thermal performance factor. Roetzel and Lee [9] experimentally studied the detriment flow of shell and tube heat exchangers amid segmental baffles; author found the higher coefficient of heat transfer is observed the value of Reynold number increases in tube side. Soltan et al. [10] reviewed 20–100% baffle space of shell diameter and studied an impact of baffle spacing on heat transfer area and pressure drop in single phase shell and tube heat exchangers amid segmental baffle. Reppich and Zagermann [11] designed and developed a computer based code to calculate pressure drop for shell side as well as tube side for single phase for applying in the stream of liquid/gaseous state. Saffar Avval and Damangir [12] optimized the spacing between the baffles as well as sealing strip numbers for each type of STHEs. Resat Selbas et al. [13] designed a STHE oriented with segmental baffle of 35 and 45% cut baffle using genetic algorithm.

3 Methodology

Figure 1 shows the investigated shell and tube heat exchanger with segmental baffle. In an experiment/study the cold fluid is passing through shell side and hot water is flowing through tube side were considered by Ravi et al. [3, 5, 6]. In this research work tube side fluid flow is considered as 1.69 kg/s and shell side fluid flow is varying from 1.96 to 4.34 kg/s at a temperature of 300 K in cold fluid and 350 K in hot fluid. The fluid wall was considered to be no slip and the fluid is assumed to be incompressible Newtonian fluid with with constant properties.

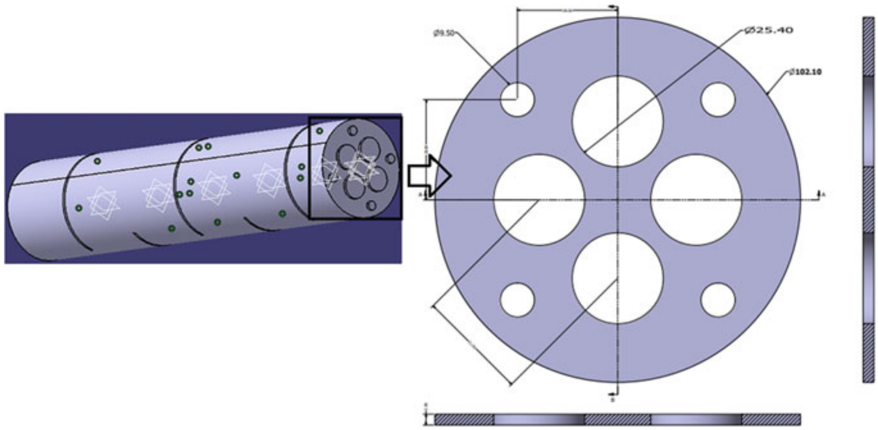

Fig. 1 Illustration of computational domain

4 Governing Equations

Numerical simulations were investigated using ANSYS Fluent.SIMPLE algorithm with 2nd order upwind discrimination was selected to resolve pressure and velocity coupling. In order to capture the streamline curvature of fluid precisely, RNG k-ϵ model with standard wall function is used as turbulence model. The convergence criteria was set to e-4 for continuity, momentum and turbulence equations whereas for the energy equations it was set to e-8. The governing equations for computational domain are expressed as follows:

Continuity equation is

$$\rho\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}\right)=0 \tag{1}$$

Momentum equation:

$$\begin{aligned}\rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)&=\frac{\partial}{\partial x}\left(\mu\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial u}{\partial z}\right)-\frac{\partial p}{\partial x}\\ \rho\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)&=\frac{\partial}{\partial x}\left(\mu\frac{\partial v}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial v}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial v}{\partial z}\right)-\frac{\partial p}{\partial y}\\ \rho\left(u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)&=\frac{\partial}{\partial x}\left(\mu\frac{\partial w}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial w}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial w}{\partial z}\right)-\frac{\partial p}{\partial z}-\rho g\end{aligned} \tag{2}$$

Energy equation:

$$\rho\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}\right)=\frac{\partial}{\partial x}\left(\frac{k_f}{c_p}\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{k_f}{c_p}\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{k_f}{c_p}\frac{\partial T}{\partial z}\right) \tag{3}$$

$$\frac{\partial}{\partial t}(\rho k) + \frac{dy}{dx_i}(\rho \varepsilon u_i) = \frac{d}{dx_j}\left(\alpha_k \mu_{eff} \frac{dk}{dx_j}\right) + G_k + G_b - \rho\varepsilon - Y_M + S_k \quad (4)$$

$$\frac{\partial}{\partial t}(\rho \varepsilon) + \frac{dy}{dx_i}(\rho \varepsilon u_i) = \frac{d}{dx_j}\left(\alpha_k \mu_{eff} \frac{dk}{dx_j}\right) + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon}\rho \frac{\varepsilon^2}{k} - R_\epsilon + S_\varepsilon \quad (5)$$

where, $C_{1\varepsilon}$ and $C_{2\varepsilon}$ are the model constants have values deduced methodically by the RNG theory. The chosen values are $C_{1\varepsilon} = 1.42$, $C_{2\varepsilon} = 1.68$, which is depicted by ANSYS theory guide.

5 Grid Generation and Validation Test

Because of having complex configuration on geometry, tetrahedral cells were generated over the cold flow domain. Hexahedral meshes were selected for hot fluid zone. For achieving better accuracy in computation, tetrahedral cells are converted to polyhedral meshes. Denser and smooth grids were generated near the contact regions in order to capture the velocity gradient properly. Besides, considering the computational time and requirement of CPU, the focus was maintained to create optimized mesh over the flow domain. For ensuring accuracy of CFD simulation, evaluation of Mesh Independency to implore the optimized grid is essential. Therefore, 4 numbers of grid systems (G1, G2, G3 and G4) created with four unusual sizes (4, 3, 2 and 1 mm) with mass flow rate 3.38 kg/s to obtain the mainly unswerving outcome as 0.99% of pressure drop accuracy between G3 and G4. The adopted grid size (G4) and results of mesh independence test for simulation one is given in Fig. 2 (Table 1).

6 Result and Discussion

6.1 Analysis of Flow Structure

Due to the obstruction of the segmental baffles, the variation of velocity occurs over the fluid domain. Figure 3 is showing the variation of the velocity over the plane for different mass flow rate. It is obvious that due to the hindrance created by baffles, the velocity increases while crossing baffles because of having smaller area. The flow gets stopped near the baffle's wall which is evident from Fig. 3.

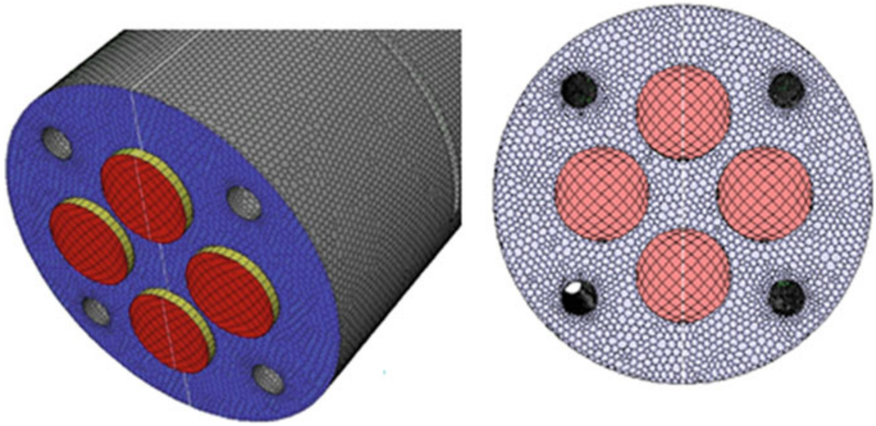

Fig. 2 Visualization of grid generation

Table 1 Comparisons of different grid systems to ensure independency test

Grid	Element numbers	Pressure drop (P)	δP (%)
G4	3220435	958.84723	0.994
G3	989376	968.47929	4.07
G2	428860	929.84602	1.64
G4	220594	945.42583	1

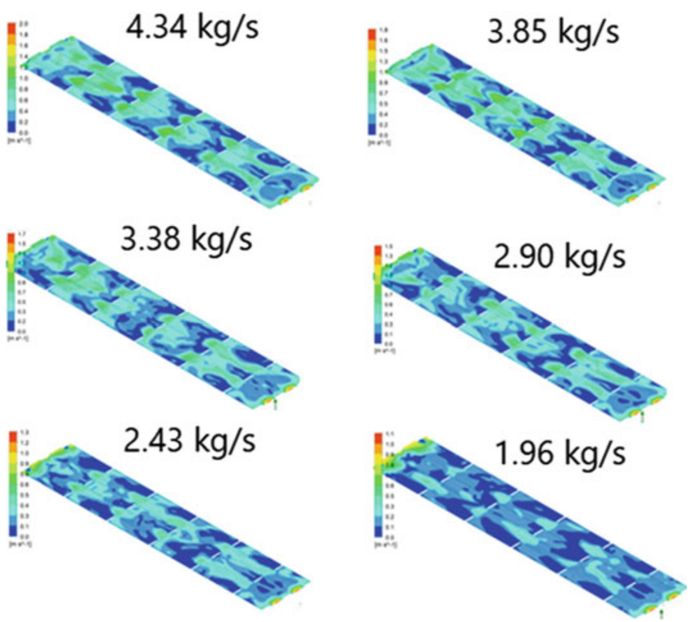

Fig. 3 Velocity at the planar region of domain for different mass flow rate

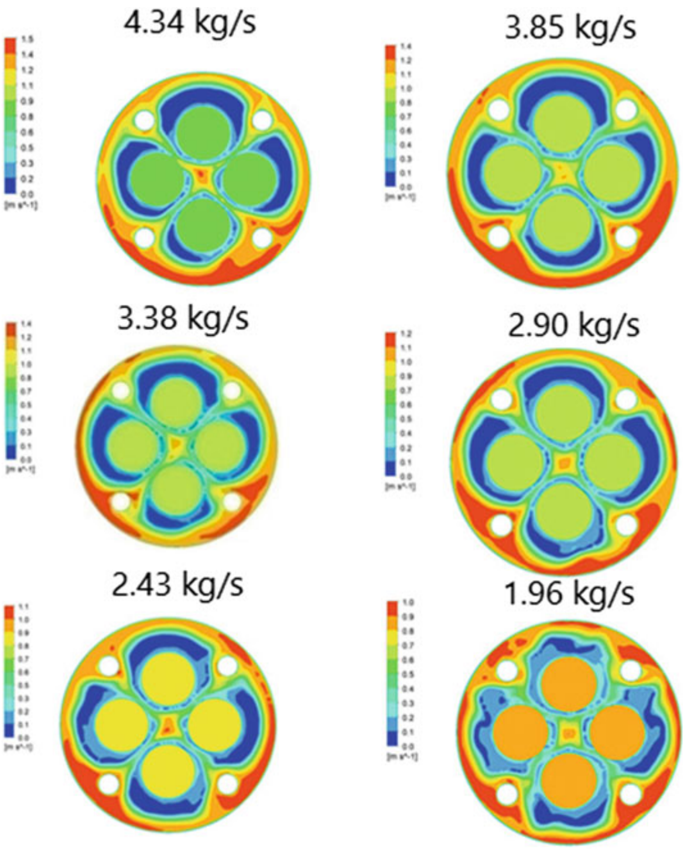

Fig. 4 Variation of velocity at the cold fluid outlet for different flow rates

Besides, it is evident from Fig. 4 that the formation of vortex at outlet of cold fluid exhibits the heat transfer performance. Since the formation of vortex leads to secondary flow which helps to increase heat transfer, it is evident that segemental baffle with 40% baffle cut forms swirling and vortex due to turbulence created by baffles. Figure 5 represents the velocity along the centerline of the computational domain.

6.2 Analysis of Pressure Drop

Figure 6 shows the variation of pressure drop over the direction of flow for different mass flow rates. It is evident from the figures that along the direction of flow, the

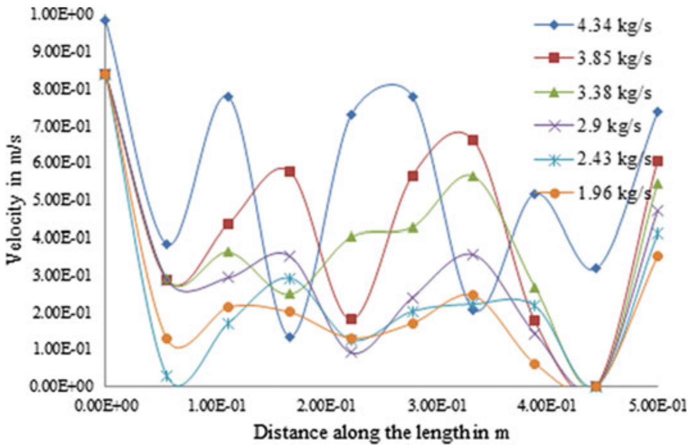

Fig. 5 Velocity along the centerline of flow domain for different mass flow rate

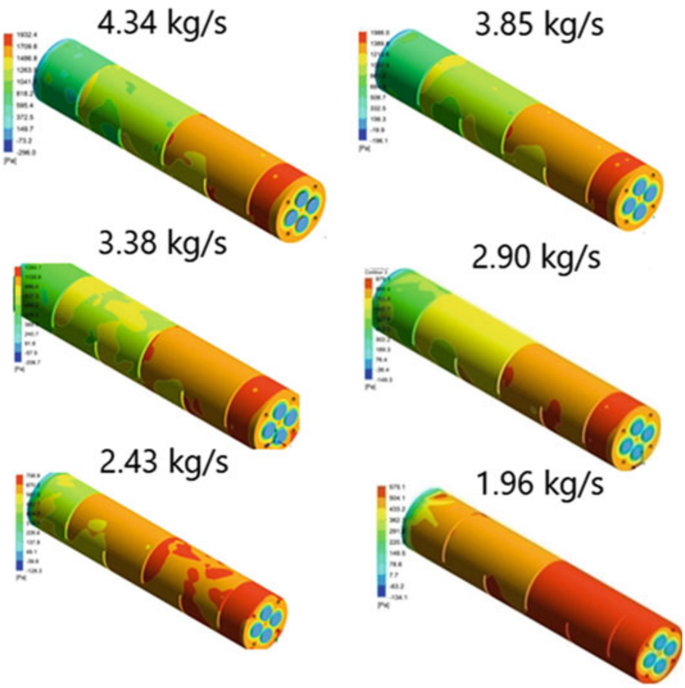

Fig. 6 Variation of pressure over the wall of computational domain

pressure drop decreases and reaches to minimum at the outlet of the flow domain. Besides, the figure evaluated that for the lower mass flow rate the variation of pressure drop is found to be less compared to higher mass flow rate of the cold fluid.

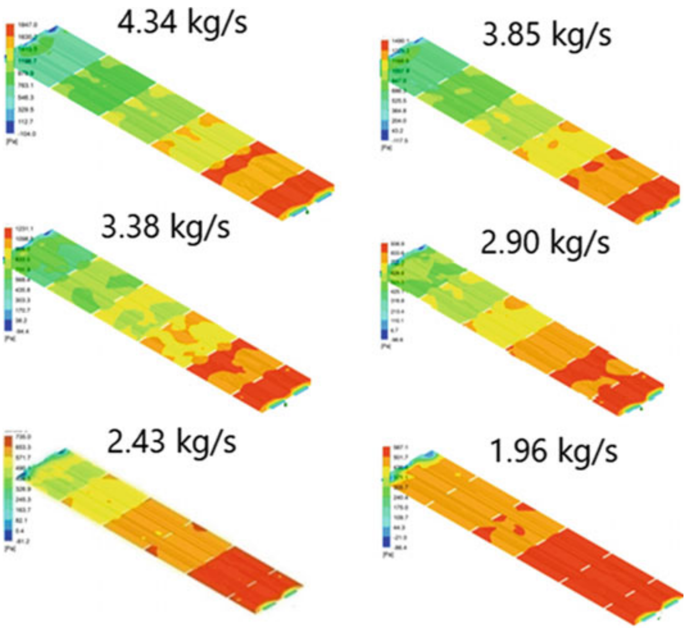

Fig. 7 Variation of pressure at the planar region of computational domain

The pressure allocation of heat exchangers over the plane can reveal the resistance characteristic of baffles and heat transfer performance. And Fig. 7 shows the pressure drop distribution along the plane surface. Figure 7 evaluates that near the baffle region the flow gets more obstructed as observed in Fig. 3 which results in the higher pressure drop near the segmental baffle region. Pumping costs are majorly depends on the pressure drop, while designing a heat exchangers. Therefore, higher pressure drop leads the higher operating costs and vice versa. Net pressure drop decreases slowly 1072.41–377.54 Pa for shell side mass flow rate 4.34–1.96 kg/s.

Figure 8 shows variation in pressure drop along with mass flow rate. For higher mass flow rate higher the pressure drop and vice versa. It is observed from the Fig. 9 that increase in mass flow rate increases the pressure drop. Highest pressure drop of 27.91% is found for the increase of 1.24 times in flow rate and the trends gradually decrease.

7 Conclusion

The present research study evaluates the 3D simulations of shell and tube heat exchangers amid segmental baffle at different mass flow rates at cold fluid zone inlet and constant mass flow rate for hot fluid zone inlet. Different hydrodynamic charac-

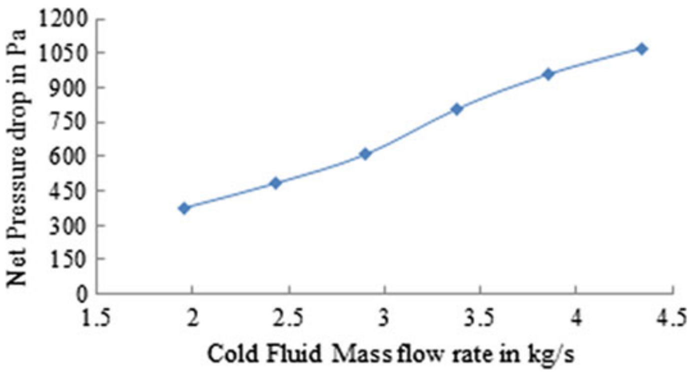

Fig. 8 Analysis of pressure at different mass flow rates of cold fluid

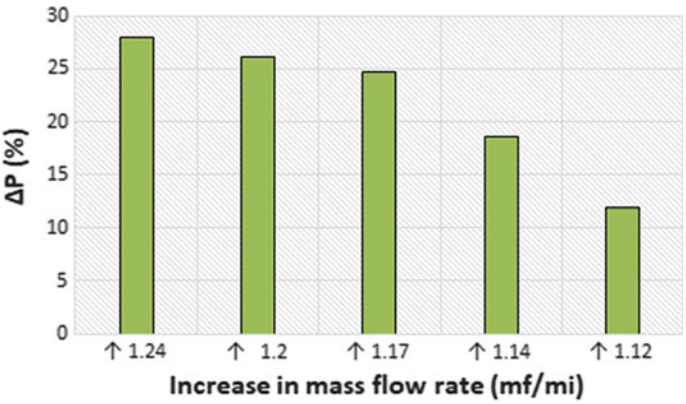

Fig. 9 Increase in pressure due to increase in flow rate

teristics were evaluated locally in order to illustrate the velocity and pressure drop over the whole geometry. Author noticed for increase in flow rate of about 1.24, 1.2, 1.17, 1.14, and 1.12 times, the pressure drop consecutively increased 27.91, 26.10, 24.72, 18.52, and 11.85%. Hence the present study evaluates the impact of local hydrodynamic parameters to improve the design of shell and tube heat exchanger.

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Chapter 26
A Rule-Based Parsing for Bangla Grammar Pattern Detection

Aroni Saha Prapty, Md. Rifat Anwar, and K. M. Azharul Hasan

1 Introduction

Parsing is an activity of breaking down sentences into parts to check the grammatical correctness of a given sentence using specific algorithm for a given context-free grammar (CFG). Predictive parsing is one of the most used parsing algorithms [1, 2] that has some issues on ambiguity, and left factoring is needed when the given grammar is left recursive. Shift-reduce parsing overcomes those issues but still conflicts when a state requests for both shift and reduce actions or more than one distinct reduce actions [3]. Therefore, the backtracking is necessary. Ambiguity is an inherent issue for a natural language grammar because of the large number variants of a single sentence. As a result, more than one parse tree is generated that leads to ambiguity. For avoiding the above important issues, namely left recursion, backtracking and ambiguity, and for checking the correctness grammar pattern for Bangla sentence, we introduce the idea of the CYK parsing algorithm [4, 5]. CYK is a dynamic programming-based algorithm, and it has better running time in the worst-case scenario than shift-reduce and predictive parsers. For CYK parsing, firstly, CFG rules need to be converted to Chomsky normal form (CNF). Then, it considers each every possible substring from a given string (st). It sets S[len, st, vt] to be true if there is a non-terminal R and a substring of length [len] starts from st, can be generated from R. Then, it considers all the substrings that have the length one. Next, it takes all the substrings of length two and so on. For the substring of length greater than two, it divides the string into two parts and checks for some production rules S → QR so that Q matches one and R to the next. If it matches, it considers S as the match as the full substring.This process is challenging for the Bangla language because one sentence which has the same semantic meaning can be represented in a different pattern as

A. Saha Prapty (✉) · Md. Rifat Anwar · K. M. Azharul Hasan
Department of Computer Science and Engineering, Khulna University of Engineering and Technology, Khulna 9203, Bangladesh
e-mail: az@cse.kuet.ac.bd

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there is no fixed structure and even the same string can be represented in different ways [6], and for this, the meaning of the sentence will remain same. We select air traffic information system (ATIS) as our domain to check the grammar pattern. We have developed the CFGs for the ATIS domain in the way that users of the ATIS are asking questions and the validity of the sentences is to determine. The scheme is applicable to recognize grammar [1], text conversion [7], machine translation [8, 9], automatic question answering [10, 11] and so on.

2 Related Works

For parsing Bangla sentence, predictive parser is applied in [1, 2] that cannot work for ambiguous grammar. For mapping and translating Bangla language with other natural languages, predicate preserving parser is discussed in [7], but in order to generate a parse tree, grammar should be unambiguous and left recursion must be avoided. Using the basics of linguistic knowledge builder, the head-driven phrase structure grammar is described in [12] that can handle the semantics of Bangla sentence. To check the rule probability and word probability of a Bangla sentence, CYK parsing algorithm is used in [4, 6] to fit it the most probable structure of the sentence. But to be successful, the probabilistic parsing, the corpus that generates the probability, should be we justified and a large dataset is needed for the better accuracy of the statistical parsing. A morphological parsing is proposed in [13], an unsupervised way for parsing a sentence in Bangla language. Syntactical analysis of a context-sensitive Bangla text is done in [8] for machine translation using the NLP conversion toolkit. LR parsing is used in [14] for parsing Bangla grammar which is time-consuming and cannot handle all left recursive grammar. Shift-reduced parser is used in [3] but conflicts for a state that requests more than one specific reduce actions or both shift and reduce actions. To avert this problem, we should develop an unambiguous grammar which is implemented in [5].

3 Domain-Specific Constituents

Context-free grammer (CFG) is extensively used for modeling the formation of any natural language. It comprises a set of production rules, each of which explicit form that the tokens of the language can be systematized. Any sentence in the formal language that can be obtained from production rules defined by CFG is referred to as a grammatical sentence. It is difficult to explain the whole structure of a natural language, but a simplified model can be generated based on the context. By utilizing formal language to build up natural languages is defined as generative grammar as the sentences of the language is generated by CFG. In this section, we develop the constituents for the ATIS domain. The goal of the users of the

ATIS is to ask domain-related information as a form of a descriptive sentence. For example, আমাকে একটি টিকিট দিন যেটা সকালে ছাড়বে, আমার চারটি টিকিট চাই, আপনার ভ্রমন শুভ হোক etc. Tables 1, 2 and 3 show some specimens of a noun phrase and verb phrase and the POS tag description with an example for the ATIS domain. This POS tag symbols that are shown in Table 3 are used to combine for generating CFGs for parsing the Bangla sentences of the domain.

Table 1 Noun phrase identification

Rank	Noun Phrase	Identifying Noun
1	ঢাকার টিকেট	টিকেট
2	ঢাকা থেকে খুলনা	ঢাকা, খুলনা
3	যাতায়াতের মালামাল	মালামাল

Table 2 Verb phrase identification

Rank	Verb Phrase	Identifying Verb
1	টিকিট চাই	চাই
2	পাসপোর্ট দেখান	দেখান
3	বিমান দেরিতে ছাড়বে	ছাড়বে

Table 3 POS tag description

Rank	Parts-of-speech	Symbol	Example
1	Noun	Noun	বিমান, মালামাল
2	Pronoun	Pronoun	আমি, আপনি
3	Adjective	Adjective	ভালো, অতি
4	Transitive, intransitive verb	TV, IV	যাবো, যেতে
5	Adverb	Adv	দ্রুত, ধীরে
6	Conjunction	Conj	এবং, ও, কিন্তু
7	Preposition	Pre	থেকে, হতে
8	Modifiers	Mod	একটি
9	Negative	Neg	না, নেই
10	WH Question	Wh	কখন,কবে
11	Yes–No question	Aux	কি
12	Relational operator	RelOp	সেটা, যা, যেটা

4 Sentence-Level Context-Free Grammar Generation

For the specific domain ATIS, the sentences in the Bangla language can be classified into three categories, namely simple, complex and compound. These sentences can be represented as different forms such as declarative, imperative Yes–No questions, WH questions.

Simple Sentence A simple sentence is a sentence that contains single independent clause having a subject and verb that represents a complete sense. We present the CFGs using the production rules in the following using specific example.

Declarative Sentence: The type of sentence that describes a fact is known as declarative sentence. The statement is made of describing through speech or actions. Example: আপনার ভ্রমন শুভ হোক। The corresponding CFG is
S → NP VP;
NP → Noun | Pronoun Noun | Noun Adverb | Pronoun Noun Adverb;
VP → Adverb TV;
Pronoun → আপনার
Noun → ভ্রমন;
Adjective → শুভ;
TV → হোক;

Imperative Sentence: This type of sentence is used to convey a request or order, and most of the time subject is hidden. Example: টিকিট দেখান। The corresponding CFG is
S → NP VP;
NP → Noun;
VP → TV;
Noun → টিকিট;
TV → দেখান;

Yes–No Question: The sentence deserves an answer of type "Yes/No". Example: আপনি নাস্তা নেবেন কি? The corresponding CFG is
S → NP VP Aux;
NP → Pronoun| Noun| Pronoun Noun;
VP → TV| Noun TV| Pronoun Noun TV| TV Aux;
Pronoun → আপনি;
Noun → নাস্তা;
TV → নেবেন;
Aux → কি;

WH Questions: In this type of sentence, the given answer should be descriptive rather than answering with yes or no. Example: আপনি কোথায় যাবেন? The corresponding CFG is
S → NP Wh VP
NP → Pronoun;
VP → TV;
Pronoun → আপনি;
Wh → কোথায়;
TV → যাবেন;

Complex Sentence: Complex sentence is a sentence that contains one independent clause having at least one subordinate clause. The independent clause can make sense by standing alone, whereas the subordinate clause is dependent on the independent clause. Example: আমাকে একটি টিকিট দিন যেটা সকালে ছাড়বে। Here, আমাকে একটি টিকিট দিন is the main clause and যেটা সকালে ছাড়বে is the subordinate or dependent clause. Therefore, RelClause → RelOp VP; RelOp → যেটা;

Compound Sentence: A sentence that contains an independent clause having one or more dependent clause is known as a compound sentence. Even the dependent clause has an individual subject and verb, but it cannot express a proper thought. Example: আমি ঢাকা যাব এবং আমার চারটি টিকিট চাই। Here, আমি ঢাকা যাব। is the main clause where এবং আমার চারটি টিকিট চাই। is the dependent clause.

5 The Parsing Methodology

Parse tree is a tree that graphically represents the derivation of the grammar of the input string and the process is called parsing. The representation of the proposed framework is shown in Fig. 1.

Tokenization is a function that takes a sentence as input and breaks down input as a distinct unit that is addressed as a token. These tokens are assembled as a list for finding the pattern of sentences during parsing. For the given input, the system is compliant with the tokens that are described as: Token = আপনার, ভ্রমন, শুভ, হোক After analyzing the simple, complex and compound sentences, complete production rules are generated by rule generator and described by CFG. It comprises set of production, each of which explicit the form that tokens of the language can be systematized. The rules are designed based on the corpus and the parse tree is generated by the rules. For building up the model, CFG is developed for the specific domain, and by using this, there are two parse trees generated for the given input. Figure 2 shows the parse tree as an example.

The CYK parsing algorithm is implemented using the rules that are converted into CNF. In CNF, the non-terminal is produced by two non-terminals or single terminal symbol. For example, $P \rightarrow MN$ or $P \rightarrow x$; where P, M, *and* N are non-terminal symbols and x is a terminal symbol. For each sub-sentence, it is divided into two parts except the bottom row to match the CNF rule. During parsing, terminal symbol

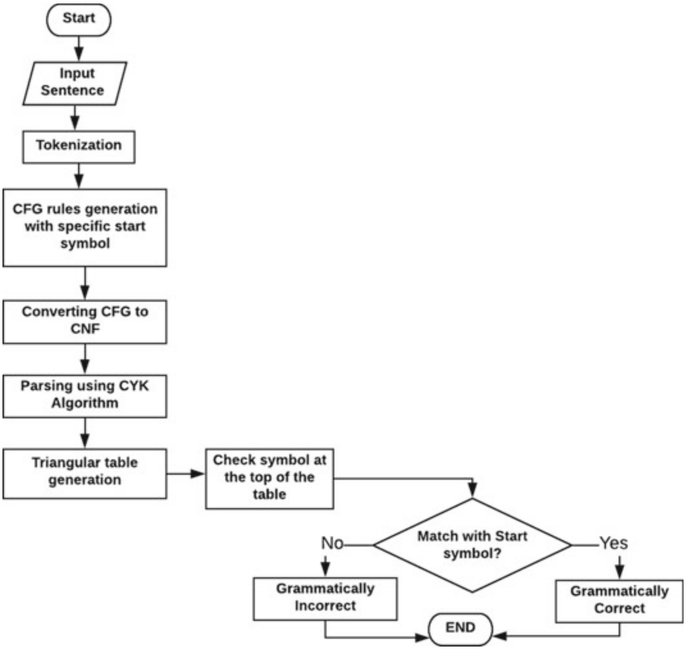
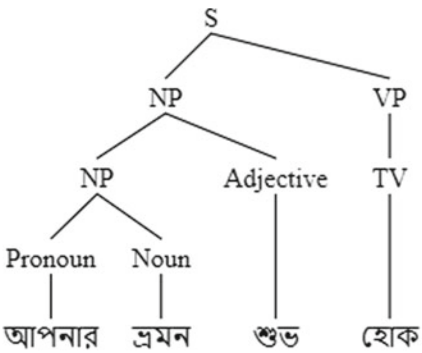
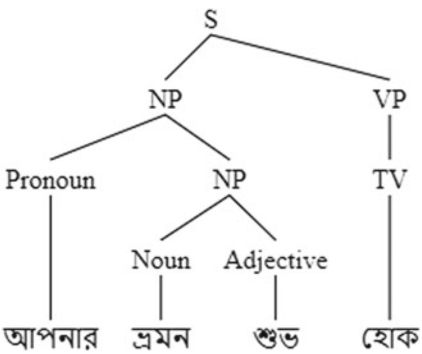


Fig. 1 Parsing Technique



(a) Parsing example by parse tree



(b) Parsing example by parse tree

Fig. 2 Parsing example

is replaced by the left side of the produced grammar and the number of replacement of the terminal symbol depends on the number is a row. After passing through the entire input for each row, one column is reduced from the last, and at the very last, if the terminal symbols can be replaced by the start symbol, then it is considered that the given input is valid. The major factors of start symbol are noun phrase and verb phrase for declarative sentence. The production rule of the input sentence আপনার ভ্রমন শুভ হোক is given as:
S → NP VP;
NP → Pronoun Noun| Noun Adjective| Pronoun Noun Adjective;
VP → TV;
Pronoun → আপনার*;*
Noun → ভ্রমন*;*
Adjective → শুভ*;*
TV → হোক;

Here, NP →Pronoun Noun Adjective is not in the CNF form. After converting this part of the CFG into CNF the form the formal grammar is

NP → Pronoun Noun| Noun Adjective| Pronoun NP. VP → TV;
Pronoun → আপনার*;*
Noun → ভ্রমন*;*
Adjective → শুভ*;*
TV → হোক;
Now, the triangular table is constructed by using the rule of CYK parsing algorithm.
$Z_{m,n} = (Z_{m,n}, Z_{m+1,n}), (Z_{m,m+1} Z_{m+2,n}) \ldots (Z_{m,n-1}, Zn, n)$
$Z_{m,n}$ is the group of variables B such that $B \rightarrow P_i$ is a production rule of grammar G. We are considering the n pairs that are previously calculated.

Illustrative Example: We are going to parse the Bangla sentence আপনার ভ্রমন শুভ হোক using the above-generated grammar. Figure 3 shows the complete triangular parse table. The first row (or bottom row) is generated by the terminal symbol of the generated CFG. To generate the other rows, it successively follows the function

S (1,4)			
NP (1,3)	S (2,4)		
NP (1,2)	NP (2,3)	Φ (3,4)	
Pronoun (1,1)	Noun (2,2)	Adjective (3,3)	TV,VP (4,4)
আপনার	ভ্রমন	শুভ	হোক

Fig. 3 Illustrative example for a simple sentence

$Z_{m,n}$. The second row is calculated as

$$\begin{aligned} Z_{1,2} &= Z_{1,1}, Z_{2,2} \\ &= \{Pronoun\}\{Noun\} \\ &= \{PronounNoun\} \\ &= \{NP\} \end{aligned}$$

that matches the production rule: NP → Pronoun

$$\begin{aligned} Z_{2,3} &= Z_{2,2}, Z_{3,3} \\ &= \{Noun\}\{Adjective\} \\ &= \{NounAdjective\} \\ &= \{NP\} \end{aligned}$$

that matches the production rule: NP → Noun Adjective.

$$\begin{aligned} Z_{3,4} &= Z_{3,3}, Z_{4,4} \\ &= \{Adjective\}\{TV, VP\} \\ &= \{AdjectiveTV, AdjectiveVP\} \\ &= \{\Phi\} \end{aligned}$$

As it has no production to match, it produces nothing. For third row of Fig. 3, parsing continues as follows. $Z_{m,n} = (Z_{m,m}, Z_{m+1,n})(Z_{m,m+1}, Z_{m+2,n})$

$$\begin{aligned} Z_{1,3} &= (Z_{1,1}, Z_{2,3})(Z_{1,2}, Z_{3,3}) \\ &= \{Prnoun\}\{NP\} \cup \{NP\}\{Adjective\} \\ &= \{PrnounNP, NPAdjective\} \\ &= \{NP\} \end{aligned}$$

that matches the production rule: NP → Pronoun NP (Fig. 4)

$$\begin{aligned} Z_{2,4} &= (Z_{2,2}, Z_{3,4})(Z_{2,3}, Z_{4,4}) \\ &= \{Noun\}\{VP\} \cup \{NP\}\{TV, VP\} \\ &= \{NounVP, NPTV, NPVP\} \\ &= \{S\} \end{aligned}$$

that matches the production rule: S → NP VP for 4th row generation CYK parsing rule follows as $Z_{m,n} = (Z_{m,m}, Z_{m+1,n})(Z_{m,m+1}, Z_{m+2,n})(Z_{m,m+2}, Z_{m+3,n})$

S → NP VP| NP VP Aux| NP Wh VP| WH NP VP| Adverb VP| NP WH| S RelClause

NP → Pronoun Noun| Det Noun| Pronoun NP| Pronoun Aux|Adjective Noun| Adjective NP| NP NP| Noun Noun| Noun| NP Noun| Noun Aux| NP Aux| Noun Aux Noun| NP Aux Noun| Pronoun NP| NP Adjective| Pronoun| Pronoun Aux Pronoun| Pronoun Aux Adjective Noun| Pronoun Aux NP| Pronoun Wh| Wh Noun| Pronoun Wh Noun| Wh Noun Noun| Pronoun Wh NP| Wh Noun Pronoun| Pronoun Adjective Noun| Modifier Noun| Pronoun Modifier Noun| Noun Adjective Noun| Noun Modifier Noun| Noun Modifier Noun| Noun NP| Noun Aux NP| Noun Aux Adjective Noun| Adjective Det Noun| Pronoun Aux Pronoun| Pronoun Aux Pronoun Noun| Pronoun Noun Noun| Pronoun Noun Wh| NP Wh| Pronoun Noun Wh| Noun Conj Noun|Pronoun Conj Noun| Pronoun Conj Pronoun| Pronoun Noun Conj Noun| Adjective Noun Wh|Adjective Noun Conj| Adjective Noun Adjective Noun| Noun Adjective Noun Conj| Noun Pronoun Conj Pronoun| Noun Pronoun Adjective| Noun Noun Conj Pronoun| Noun Noun Conj Pronoun| Pronoun Adjective| Pronoun Pronoun conj Pronoun| Modifier Conj Adjective| Noun Pre Noun| Noun Pre| AP Noun| Pronoun Aux Adverb| Adverb Noun| Noun Adverb| Pronoun Aux Pronoun Noun| Pronoun Pronoun| Pronoun Pronoun Noun| Pronoun Noun Adjective Noun| Pronoun Aux NP| Pronoun Aux Noun|Pronoun| Noun| Noun Adjective

VP → IV TV| Noun IV| TV| Adverb TV| TV Aux| Adjective Noun TV| Noun TV| Noun Noun TV| Adverb IV| IV IV TV| Noun Aux Adverb IV| Adverb IV IV TV| Adverb IV IV| Adjective Noun Aux Adverb IV| Noun Aux Adverb IV IV| Noun Aux Adverb IV| Pronoun IV| Wh Noun Pronoun IV| Noun TV| Adjective Noun TV| Modifier Noun TV|Adjective Noun IV|Noun VP| Noun IV TV| Noun Adjective Noun IV| TV Aux| VP Aux| IV TV Aux| Adverb TV Aux| Noun Adjective Noun IV| Wh TV| Adverb TV| Noun Conj Noun IV| Pronoun Aux VP| Noun IV TV Neg| Noun TV Neg| TV Neg| Noun TV Adverb| IV TV Adverb| Pronoun Adjective TV| Pronoun IV TV Neg| Noun TV TV| Pronoun TV TV| Pronoun TV TV Neg| TV Adjective Noun|IV TV Adjective Noun|TV Adjective Pronoun| IV TV Adjective Pronoun| Adjective Noun TV Neg|Adjective TV| Adverb IV TV|Modifier Noun IV| Det Noun IV| IV TV Neg

AP → Adjective Noun| Adjective Pronoun| Adjective Neg| AP Adjective

RelClause → RelOp VP| Conj S

Fig. 4 Production rules for Bangla grammar pattern detection

$$
\begin{aligned}
Z_{1,4} &= (Z_{1,1}, Z_{2,4})(Z_{1,2}, Z_{3,4})(Z_{1,3}, Z_{4,4}) \\
&= \{Pronoun\}\{S\} \cup \{NP\}\{VP\} \cup \{NP\}\{TVVP\} \\
&= \{PronounS, NPVP, NPTV\} \\
&= \{S\}
\end{aligned}
$$

that matches the production rule: S → NP VP. As it reaches to the start symbol, the given sentence is grammatically correct. The process of parsing the complex sentence is illustrated in Fig. 5 of the sentence এবং আমার চারটি টিকিট চাই। For the complex sentence in Fig. 5, at the index (1,4), it generates start symbol (S) as it stands for the main clause that expresses a specific meaning, and at index (5,7), it represents relative clause that does not make any sense, but at the top row, it produces start symbol that indicates the given sentence is grammatically correct. Figure 6 shows a parsing example for a compound sentence আমি ঢাকা যাব এবং আমার চারটি টিকিট চাই. At index (4,8), the relative

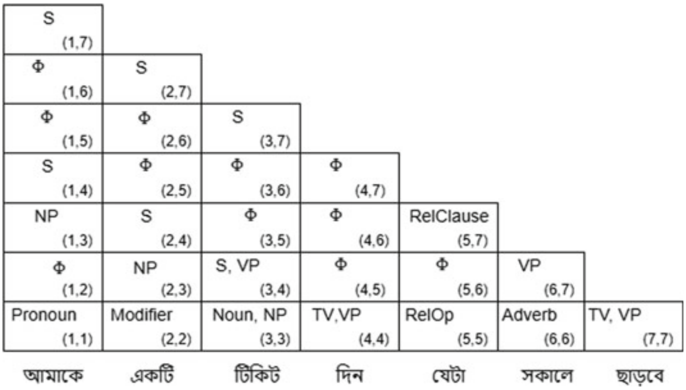


Fig. 5 Illustrative example for complex sentence

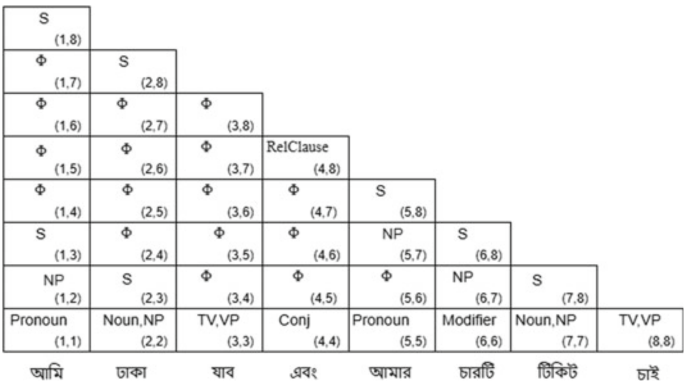


Fig. 6 Illustrative example for compound sentence

clause has subject and verb but still cannot stand for individual meaning, whereas at index (5,8), it individually stands for meaningful thought. Analyzing the simple, complex and compound sentences, complete production rules are generated and applied to detect the appropriateness of given input sentence in the proposed system. The following is the summary of the production rules:

6 Experiment Results

The main focus of this work is parsing Bangla sentences for analyzing whether the sentence is grammatically appropriate or not. We create individual 80 structures for different types of sentences for our domain-based system. These patterns cover sentences like declarative, imperative, Yes–No questions, and WH questions, complex and compound sentences. For testing purposes, we collect around 500 unique sentences from street people (mostly university students), and the result is presented in Table 4.

To implement statistical parsing, ambiguity is not a vital issue but we need a large dataset to generate a probability for each rule. But in CYK parsing algorithm, for better performance, more rules should be generated rather than the size of the dataset. In shift-reduce parsing, ambiguity and shift-reduce conflict create issues for parsing input sentences. Shift-reduce conflict stands for where a state can apply both reduce and shift options. Even if there is more than one rule for the same state to reduce the following state, it creates reduce–reduce conflict. To solve these problems, the grammar should be unambiguous and left factoring is needed to solve the left recursion problem which is very time-consuming. The advantages of the CYK parsing algorithm over shift-reduce parsing are CYK parsing can parse the sentence whether the grammar is ambiguous or not and it is faster than the shift-reduce parser. In the predictive parsing, left recursion and ambiguity should be eliminated for the parsing input sentences which is not required in CYK parsing, and for this reason, it is faster than other parsing approaches. To correctly recognize the Bangla sentences, CFG is developed for parsing which is more generalized and appropriate than the CFG . In this work, the CFG for verb phrase is more precise by using the transitive

Table 4 Performance analysis

S. No.	Sentence type	Correctly detected (%)
1	Declarative sentence	87
2	Imperative sentence	83
3	Yes–No answer type question	78
4	WH Question	75
5	Complex sentence	65
6	Compound sentence	82

verb as TV and intransitive verb as IV. Analyzing the criteria, performance is better than other existing works. The result is good for simple and compound sentences, but for complex sentence, the result is not up to the mark. By adding more production rules in G, the result can be improved.

7 Conclusion

Parsing is an effective and most used technique for generating and verifying language patterns. Parsing Bangla grammar is difficult as there are no fixed rules, ambiguity issues, and a huge collection of words, and undefined structures of sentences. Having these complex issues, we build our CFG based on a specific ATIS domain. The CFG can cover various aspects of Bangla sentences related to the ATIS domain even non-traditional format. The scheme can be extended to other domains by more appropriate production rules for other domains to detect Bangla grammar patterns. Although we consider a specific domain, the same structure can be applied to any domain as the grammar structure is unique for Bangla language. We have a plan to be executed to develop a process to preprocess sentences for questions answering, language modeling, prediction of the next word analyzing the previous words, and also used in text mining using our developed methodology for Bangla language. In the future, by producing more rules, the detection of Bangla grammar patterns will be more generalized for traditional and non-traditional form.

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Chapter 27
Customer Review Analysis by Hybrid Unsupervised Learning Applying Weight on Priority Data

Md. Shah Jalal Jamil, Forhad An Naim, Bulbul Ahamed, and Mohammad Nurul Huda

1 Introduction

In recent years, online shopping is popularizing day by day. An estimation of 1.79 billion people buys digital goods worldwide in 2018. The forecast says the number of online buyers will increase by over 2.14 billion in 2021 [1]. But online merchant sometimes fails to maintain the quality of digital goods. As a result, E-commerce customer becomes deceived. They don't have an option to check products before unpacking since they have been purchasing over the internet. So, they have to return the products if it is faulty. This process of returning products is complex and consumes time. According to the Investopedia info-graphic Ecommerce, customer returns at least 30% of their buying products [2].

Sentimental Analysis, also known as, opinion mining is the process of evaluating text to determine the fine-grained emotions that are expressed [3]. Sentiment Classification helps people to visualize the expression of the narrator whether the narrator expresses a negative or positive opinion to a particular bound on a given text [4].

Bangla is treating as the seventh most spoken language over the world. Almost 200 million people use Bangla to express their expression and emotion [5]. Generally, people give their reviews after purchasing online products in their mother tongue. So a large number of reviews (comments, hashtags) generate every day in Bangla. When people decide to buy any digital products, first of all, they review other feedback to conclude whether products are good or bad. Several authors have done various sentiment analysis research on different languages such as English, Chinese, Urdu,

Md. S. J. Jamil (✉) · F. A. Naim · M. N. Huda
United International University, Dhaka, Bangladesh
e-mail: mjalal162017@mscse.uiu.ac.bd

M. N. Huda
e-mail: mnh@cse.uiu.ac.bd

B. Ahamed
Sonargaon University, Dhaka, Bangladesh

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etc. [6]. But in contrast, sentiment analysis research works in Bangla language are improvised.

However, the goal of this paper is to extract the hidden emotions conveyed by the customer from reviews in Bangla of the online shop. During the experimental period, we used several clustering methods such as K-Means, Mean Shift, DBSCAN, Agglomerative clustering, etc. We pick up the clustering method with the best accuracy for linear polarity. A new technique named PWWA (Priority Word Weight Assignment) applies to the dataset for preparing the datasets to the clustering method. After that, the K-means clustering method ensemble with the Agglomerative hierarchical clustering method applied to the dataset. The study aims to increase accuracy from other clustering methods by at least 5–10% after applying PWWA to the dataset.

2 Related Works

In 2016, few researchers of Sylhet Engineering College researches on Bangla Sentiment Analysis. The author gathers the most frequent words of the corpus using the frequency distribution of words to pull out the hidden sentiment in Bangla text. The author takes the last four or more backward from a sentence. The author calculates the TF-IDF weight value of positive and negative lexicon. Using the TF-IDF classifier, the author effectively identifies the sentiment of a sentence [4].

In 2016, few researchers of Shah Jalal University of Science and Technology (SUST) study the process of identification of stems or root of Bangla words. They state if two unigram texts present a certain percentage of similarity in spelling and have a certain percentage of contextual similarity in many sentences, then these words have a higher probability of originating from the same root. The words from the same stem are identical in terms of meanings [7].

In 2017, researchers of Waikato University propose two different methods. They induce Twitter-specific opinion lexicon and develop distant supervision methods for generating synthetic training data. For the first method, they state lexicon produce significant result over manually annotated lexicon. The second method performs better over the classifier that trained from tweets annotated with emoticons [8].

In 2016, few researchers of Cairo University propose a new technique named SAOOP that automatically determines human sentiment from text. They use a hybrid model consisting of Bag-of-Words and Parts-of-Speech model. They state that SAOOP has higher accuracy than others in three different perspectives [9].

Another research author builds the Twitter-specific Bangla polarity lexicon. They use a hybrid approach to construct a vocabulary that automatically labeled small training data. The author uses a semi-supervised learning method to train data due to the lack of large Bangla tweet corpus. The author uses the SVM and Maximum Entropy to show how effective categorization studies. The author states emotions increase the accuracy level by 36% when emoticon's mixes with unigram text tweets [10, 11].

On the other hand, in the proposed research work we have applied various unsupervised machine learning techniques such as K-Means, Density-Based Spatial, Mean Shift, Agglomerative, and ensemble approach combined with different clustering methods. We pick up the clustering method with the best accuracy for linear polarity. An innovative method named PWWA applies to the dataset of 2000 Bangla reviews for preparing the datasets to the clustering method. After that, the K-means clustering method ensemble with the Agglomerative hierarchical clustering method applied to the dataset.

From the experiments, we have observed that the proposed method provides 98% accuracy for the experimental corpus by employing PWWA.

3 Procedures and Prepare Dataset

3.1 Corpus Collection

The dataset that has been used in the experiments is collected manually. The E-commerce site is one of the trusted sources to gather the feedback on Bangla Language. For our study, we gather customer feedback of Bangla Language from some popular E-commerce sites such as daraz.com, picakaboo.com, ajkerdeal.com. For smooth analysis, we collect simple Bangla reviews. We collect more than 2000 simple Bangla reviews for our corpus.

3.2 Pre-Processing

Bangla reviews gathered from the online site are full of garbage. We preprocess the dataset to make it useful for experiments. First of all, we remove stop words from dataset such as Bangla Dari ('।'), comma (','), colon (';'), etc. For Tokenization, we use KERAS Tokenizer [12]. After tokenizing the dataset, we check each token to determine whether it is unigram text, emoji or emoticon. If it is emoji or emoticon, we decode the emoji or emoticon in the alphabetic form using Emoji library [13] for smooth training on the dataset or corpus collection. 👌→: ok_hand: 👍→: thumbs up: 👊→: oncoming fist.

3.3 Priority Word Weight Assignment

Evolution of PWWA. After gathering the corpus we analyze the structure of Bangla sentences. The basic sentence structure of Bangla is subject + object + verb. In most cases, the polarity of sentences relies on the adjective in the sentences. An adjective is a crucial term in terms of determining the polarity of any sentence. In contrast, a negation word with an adjective can change the polarity from positive to negative.

আমি দারাজ থেকে পণ্য কিনে সন্তুষ্ট → Positive
Polarity → Positive, Adjective → সন্তুষ্ট
আমি দারাজ থেকে পণ্য কিনে সন্তুষ্ট নই → Negative
Polarity → Negative, Adjective → সন্তুষ্ট, Negation → নই

We examine our corpus collection manually. As a result, we find a word structure responsible for the polarity of simple Bangla sentences. We find that unigram words in some specific positions in a sentence influenced the polarity of the sentence than other unigram words. Also, emoticon influenced the polarity of the sentence by twice than unigram words. In our study, we named those unigram words and emoticons as Priority words.

Discovering Priority Word. A structured sentence can express an opinion easily. Similarly, on the other hand, a single word or emoticon can express the opinion. In the beginning, we separate emoticon from the sentences because every emoticon itself is a priority word. After analyzing our corpus, we discover the adjective stays on the first and last two words of a sentence in maximum simple Bangla sentences. The structure narrates the polarity of the simple Bangla sentence can determine from those four unigram words. The polarity of the sentence also depends on the emoticons present in the sentence. We propose the maximum number of priority words in a sentence is four unigram words and the number of emoticons, and the minimum is one.

আমি দারাজ থেকে এই পণ্য কিনে সন্তুষ্ট
আমার জীবনে দেখা সবচাইতে খারাপ পণ্য
অসাধারণ একটি পাওয়ার ব্যাংক পেয়েছি
খুব বাজে কোয়ালিটির হেডফোন পাঠিয়েছে
মাত্রই হাতে পেলাম নতুন মডেলের হেডফোন: thumbs up

In the above first four examples, we have shown that each sentence polarity can be determined correctly from the Priority word discussed previously. But the polarity of the last sentence cannot be determined from unigram word except emoticons present in the sentence. So, both four unigram words and emoticons in a sentence named Priority words play a vital role in determining the polarity of a sentence. Words in bold letters and emoticons are the Priority word in the above examples.

Weight Assignment. The most crucial step to develop the PWWA method is weight assignment. We assign a weight for each unigram word or emoticon depending on some criteria. At first, we make a dictionary for each word in the datasets. The key to the dictionary is each word in sentences and the value of the dictionary is a decimal value represented as word position in the sentence.

Sentence, S: আমি এই পণ্য কিনে সন্তুষ্ট: thumbs up:

Dictionary, D: {"আমি":1, "এই":2," পণ্য":3," কিনে":4," সন্তুষ্ট":5,": thumbs up:":6}.

Then we apply the PWWA assignment on each word such as $W_{\text{সন্তুষ্ট}}$, $W_{\text{:thumbsup:}}$, in every sentence in the datasets. We convert words of the sentences to its stem because words from the same stem are identical in terms of meanings. Previously we discuss that the unigram word stays on first or last two words in the sentences are priority words. The initial weight of each word is zero. Initially, $Wi_{\text{সন্তুষ্ট}} = 0$.. If any word places in one of those positions we increase weight by one. For the above sentence, the word 'সন্তুষ্ট'is in the priority position. So, $Wi_{\text{সন্তুষ্ট}} = Wi_{\text{সন্তুষ্ট}} + 1$.

Then we find the position in the respective sentence of all stems of that word. We determine the highest common position among the stems. If the position of the word and the highest common position is similar, then we increase the weight by one. The position of $Wi_{\text{সন্তুষ্ট}}$is four and the highest common position is also four. So, $Wi_{\text{সন্তুষ্ট}} = Wi_{\text{সন্তুষ্ট}} + 1$.

If the word itself is an emoticon, then we increase the weight by two. The reason behind this, nowadays people express their opinion by emoticon than plain text. In the customer's feedback, emoticon puts a higher weight than the unigram word [9]. Here, $W_{\text{:thumbsup:}}$ is an emoticon itself. So, $Wi_{\text{:thumbsup:}} = Wi_{\text{:thumbsup:}} + 2$.

After completing the previous processes we check the current weight of the word. If the weight is still zero, then we mark it as less important. We assign a negative weight. The weight is equal to the negative value of the position of the word in the dictionary. Here, the weight of $Wi_{\text{পণ্য}}$is zero and the position of $Wi_{\text{পণ্য}}$in the dictionary is 3. So, $Wi_{\text{পণ্য}} = -1 * 3$[Here, 3 is the position of $Wi_{\text{পণ্য}}$in the dictionary].

PWWA Algorithm. Input: a tokenize dataset.

Output: a tokenize dataset with weight.

1. Initial dataset, DS
2. Dictionary, D = make _ dictionary (DS)
3. For each word W, in each sentence S, in dataset DS
4. Initial weight for each W, $W_i = 0$
5. Prepare Stem dictionary, D_{stem} = make_ stem _ dictionary (W, D)
6. Prepare array of each stem respective position in S, A[] = value of each D_{stem} item
7. Highest common position of stems, Max_{stem} = highestCommonPosition(A)
8. If Position(W) in S $\in P_{\text{Priority}}$ then, $W_i = W_i + 1$ [P_{Priority} = Priority Position]
9. If Position (W) in S == Max_{stem} then, $W_i = W_i + 1$
10. If W $\in D_{emoticon}$ then, $W_i = W_i + 2$ [D_{emoticon} = Emoticon dictionary]
11. If $W_i == 0$ then, $W_i = -1 * \text{Position(W) in D}$
12. $DS(W) = W_i$
13. Go to step 2 until end of DS
14. DS is the tokenize dataset with PWWA weight.

4 Proposed Method

For this research, we have gathered 2000 Bangla customer reviews. To reduce data redundancy and inconsistency and to improve accuracy we have applied preprocessing method on the dataset. The proposed model or procedures of this research has been exposed (see Fig. 1). After the emoji transformation of data according to the model, we use several clustering methods such as K-means, Mean Shift, DBSCAN, Agglomerative Hierarchical, etc. to establish boundaries that separate similar collections of objects as a cluster. For our study, the number of clusters is two. K-Means ensemble with Agglomerative delivers the best analysis result among those.

K-Means split corpus collection into a given number (k) of clusters. Initially, the chosen number of centroids is k. Assume that, the initial centroid is a data point at the center of a cluster. Then calculate the Euclidean distance of each corpus from the centroids. Each corpus data point assigned to cluster depending on the minimum Euclidean distance from cluster centroid. After assigning corpora to a cluster centroid, each centroid is re-calculated. The technique of clustering and centroid adjustment is repeated until the values of the centroids stabilize.

K-Means Algorithm:
Input: k (the number of clusters), D (a dataset).
Output: a set of clusters.
Method: Randomly choose k data points from D as initial cluster centroids.
Repeat:
Step 1: Calculate the Euclidean distance of the data point from centroids.
Step 2: Assign data point to the cluster with minimum Euclidean distance from cluster centroid.
Step 3: Re-calculate cluster centroid and update the cluster centroid.

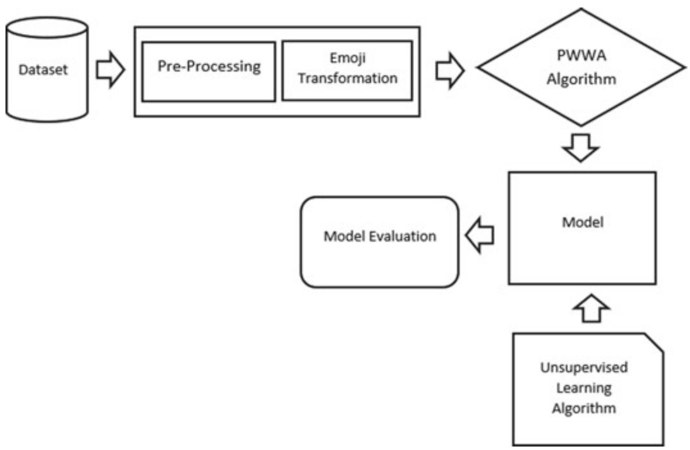

Fig. 1 Proposed model of this research

Step 4: Until no change in cluster centroid.

The agglomerative clustering works in a bottom-up manner. Initially, each corpus data point is considered as a single cluster. In the process of clustering, the most similar two clusters merge to form a new big cluster depending on the minimum Euclidean distance. The process repeats until all single cluster converts to the given number of clusters.

Agglomerative Clustering Algorithm:
Input: k (the number of clusters), D (a dataset).
Output: a set of clusters.
Step 1: Make each data point of the dataset as a single cluster.
Step 2: Compute Euclidian distance of each data point from clusters.
Step 3: Merge the most similar cluster depending on minimum Euclidean distance.
Step 4: Continue to step 2, until termination criteria is satisfied.

5 Experimental Result and Analysis

5.1 Training and Test Dataset

Earlier, we said that we collect almost 2000 Bangla reviews for the corpus. For better experimentation, we randomly shuffle the corpus collection. We use the corpus collection to fit and to predict the clustering method.

5.2 Experimental Clustering Methods

We used several clustering methods for the experiment. Those are K-Means, Mean Shift, DBSCAN, Agglomerative, and The PWWA method ensemble with the K-Means and Agglomerative clustering method.

5.3 Method Evaluation

For the experiment, we use 888 negative sentences and 1209 positive sentences to evaluate the accuracy of the proposed model. To obtain a proper result the main task is to identify priority words correctly discussed in Sect. 3. In the process of evaluation, we use four terms such as true positive, true negative, false positive, and false negative .

True Positive (TP) defines that output is positive but the clustering model predicts it as positive. True Negative (TN) defines that output is positive but the clustering model predicts it as negative. False Positive (FP) defines that output is negative and the clustering model predicts it as positive. False Negative (FN) defines that output is negative and the clustering model predicts it as negative. Accuracy is the percentage of sentences that the proposed model correctly classified. That is,

$$\text{Accuracy } \% = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100 \quad (1)$$

To evaluate our proposed model, we demonstrate the result of different clustering methods without the PWWA method in Table 1 and with the PWWA method in Table 2.

Table 1 shows the experimental results of different clustering methods before applying the PWWA method on corpus collection. The results explain that the proposed method is discussed in Sect. 4 gives better accuracy than others. The proposed method correctly identifies 1860 sentences out of 2097. The accuracy is approximately 88%. Table 2 shows the experimental results of different clustering methods after applying the PWWA method on corpus collection. We compare the results of the two tables that have been depicted in Fig. 2. The results explain that the proposed method correctly identifies 2058 sentences out of 2097 sentences. The accuracy of the proposed method is discussed in Sect. 4 is approximately 98%. In both tables, the proposed method gives the highest accuracy. Finally, the result concludes that the accuracy of the proposed method [with the PWWA method] in Table 2 is approximately 10% greater than from Table 1 [without the PWWA method].

Table 1 Result using clustering techniques without the PWWA method where total sentences are 2097

Clustering method	TP	TN	FP	FN	Accuracy (%)
K-Means	1126	838	50	83	56.08
Mean shift	0	805	83	1174	38.38
DBSCAN	0	888	0	1209	42.43
Agglomerative	252	814	74	957	50.83
K-Means ensemble agglomerative	**1126**	**734**	**154**	**83**	**88.69**

Bold value indicates Numerical values of total and average

Table 2 Result using clustering techniques with the PWWA method where total sentences are 2097

Clustering method	TP	TN	FP	FN	Accuracy (%)
K-Means	1198	886	2	11	57.22
Mean shift	42	483	405	1167	25.03
DBSCAN	15	536	352	1194	37.48
Agglomerative	1	858	30	1208	40.96
K-Means ensemble agglomerative	**1198**	**860**	**28**	**11**	**98.14**

Bold value indicates Numerical values of total and average

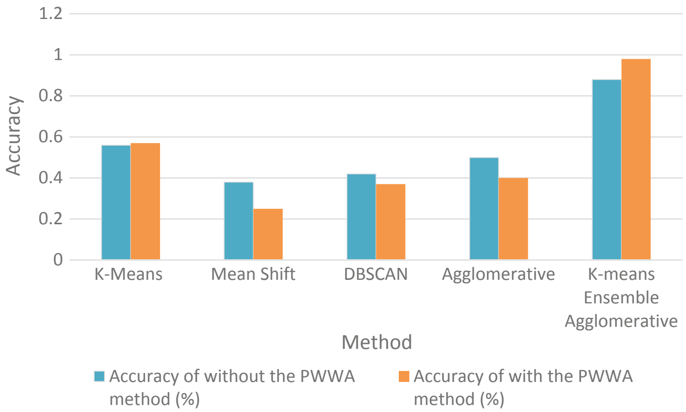

Fig. 2 Accuracy comparison with/without using the PWWA model

6 Conclusion

The proposed method improves E-commerce product quality by analyzing the customer reviews using unsupervised machine learning algorithms. It concludes the following.

- Uses Priority Word Weight Assignment in data processing stage
- Uses an ensemble clustering algorithm in post-processing stage by embedding K-Means and Agglomerative Hierarchical approaches
- Improves the accuracy 5–10% because of PWWA
- Deals with Bangla Sentiment Analysis
- Achieves total 98% accuracy.

In near future, we would like to do experiments that consider more than two polarities.

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Chapter 28
Machine Learning and Deep Learning-Based Computing Pipelines for Bangla Sentiment Analysis

Md. Kowsher, Fahmida Afrin, and Md. Zahidul Islam Sanjid

1 Introduction

Sentiment classification is based on text polarity, which classifies text/sentences into positive, negative, or neutral sentiments [1]. Sentiment extraction from sentences has become one of the most notable researches in the field of natural language processing. Work on sentiment or opinion mining has grown with the growth of the exchange of textual data over the globe online. Now people share their opinion electronically on different aspects such as online product feedback, book or movie review and political comments. Thus, analysis of these opinions gains importance to understand people's aspirations. Sentiment extraction drives the work on automatic sentiment or opinion detection. Generally, sentiment depicts two types of opinion, positive or negative. In various platforms where the opinion of mass people matters, the correspondents tend to utilize the opinions to improve their respective domains. Like online product sellers, food suppliers update their service on a regular basis based on public opinion. In Bangladesh, recently, the motor vehicle sharing services Uber or Pathao can be a burning example in this context. Their services gradually improved based on passenger feedback. But the challenging thing is to traverse through the feedbacks manually as it is a time-consuming and complex task. Automatic sentiment detection (ASD) can thus help in this circumstance which will detect the sentiment polarity from one's opinion. This can help in better decision making according to the context of one's feedback. Moreover, it can be used in various NLP applications like chatbot [2, 3], information retrieval [4].

Md. Kowsher (✉)
Noakhali Science and Technology University, Chattogram, Bangladesh

F. Afrin
Daffodil International University, Dhaka, Bangladesh

Md. Zahidul Islam Sanjid
BRAC University, Dhaka, Bangladesh

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In this research work, the authors have worked with a sufficiently large sentiment dataset that contains reviews from various domains like restaurants, cricket, products and other individuals. Ten diverse machine learning classifiers logistic regression, multinomial Naive bayes, k-nearest neighbors, support vector machines, AdaBoost, gradient boosting, decision trees, random forest, impact learning and three deep neural network techniques ANN, LSTM, CNN were used for foreseeing the sentiment level from a text. A pretrained model named BnSentiment is also proposed that can be easily used by non machine learning practitioners to classify a sentence. Our work in this research would help the marketers to easily implement a sentiment classifier in their respective systems to monitor their customer feedback.

2 Related Works

Sentiment analysis (SA) is a classical problem that is affiliated with determining the polarity of one's opinion. The workflow of SA involves feature extraction from text corpus followed by a classifier model training [5]. This kind of typical workflow has been used in numerous sentiment classification problems such as movie review classification [6] or online product review classification [7] and Twitter tweets classification [8]. A positive and negative sentiment detection model for restaurant reviews is done by Akshay et al. [9] obtaining the highest accuracy of 94.5%. Another work shows 81.77% accuracy on cell phone reviews where the authors used SVM as a classifier [10]. Sentiment analysis for the Portuguese language on Twitter data has been delineated in [11]. Ombabi et al. laid down a novel Arabic sentiment classifier with 90.75% accuracy that outperforms several well-established state-of-the-art approaches [12]. In the domain of natural language processing mixing of two languages to form a new one is frequently seen. Such a work on colloquial Singaporean English which is a coalescence of Chinese and Malay language has been done in [13].

However, most of the works on sentiment classification are concentrated on English and foreign languages. The scarcity of proper resources and data become the primary obstacle in the Bengali research field. Different machine learning and deep learning approaches have been taken on different domains like micro-blogs, product reviews, movie reviews, etc. SVM with maximum entropy [14], multinomial Naïve Bayes (MNB) [15] has been used to classify sentiment polarity on those domains. In [16], Hossain et al. developed a Bangla book review dataset and performed several machine learning techniques and observed MNB gives 88% accuracy. Another work on the Bangladesh cricket dataset, based on SVM, achieved 64.59% accuracy [17]. Sarker et al. proposed a sentiment classifier for Bengali tweets where the SVM classifier shows 45% accuracy over n-gram and SentiWordnet features. A classification on the Bengali movie review sentiment shows different evaluation metrics on different machine learning algorithms. Among them SVM and LSTM model gives 88.90% and 82.42%, respectively [18].

Reviewing the related works, it is observed that there are numerous opportunities to develop the works on Bangla natural language processing.

3 Methodology

In order to figure out the right sentiment from collected sentences utilizing machine learning and deep neural network techniques, we adopted six significant phases such as corpus creation, label verification, data prepossessing, training data incorporating ML and DNN algorithms and predictions. The high number of tests and appropriate assorted variety is guaranteeing the quality of data. Data prepossessing makes the information valid and special cases free. We have executed the ten reasonable machine learning classifiers and profound three deep neural systems on the pre-handled data. We also proposed an architecture of convolutional neural network (CNN) that is a variant of DNN. Figure 1 depicts the whole workflow of this work.

3.1 Data Sources

The corpus we created has data from disparate sources. As we observed there have been many online web portals and social sites where people share their valuable opinions, we collected data from Facebook, Twitter, YouTube comments. Nowadays, online stores have become a large sector of digital marketing. So, we collected from online store product reviews. Other than that, movie reviews, book reviews were also incorporated in this corpus. A group of three people was formed to alleviate the data collection process. The corpus contains 40k sentences. Table 1 gives a glimpse of the collected data.

3.2 Labelling and Verification

Collected data were primarily labelled by the collectors and thereafter labelled by 10 native speakers through an online annotation tool doccano [19]. Each speaker individually annotated 30% of the whole dataset by tagging them positive or negative. Figure 2 shows a snapshot of the annotation tool.

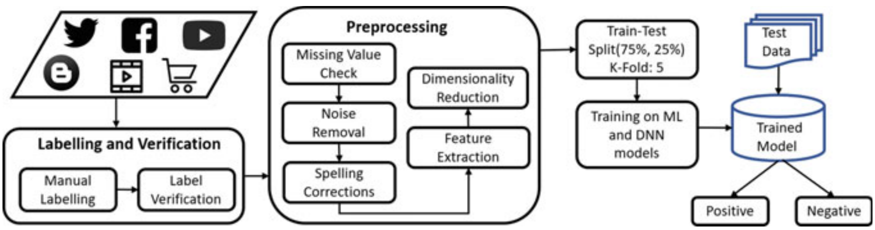

Fig. 1 Workflow of sentiment analysis

Table 1 Sample of collected Bengali sentimental texts

Text	Sentiment
জয় বাংলা কাপ! তাও আবার স্বাধীনতার মাস মার্চে। যার মাথা থেকে এমন চমৎকার আইডিয়াএসেছে তালে স্যালুট (Joy Bangla Cup! And March is the month of independence. Salute to the one who came up with such a wonderful idea)	Positive
ক্লিনিকের মালিককে গণদোলাই দেওয়া দরকার (The owner of the clinic needs to be whitewashed)	Negative
কি বলব সমস্যা আছে উভয় দিকে (What can I say, there are problems on both sides)	Negative
সুন্দর একটি সামাজিক পেজ। লাইক দিয়ে পাশে থাকুন। (A Beautiful social page. Stay by liking the page)	Positive
দালালের বাচ্চারা নিজ দেশের গণতন্ত্র নিয়ে লেখ। (Children of brokers, write about the democracy of your own country.)	Negative

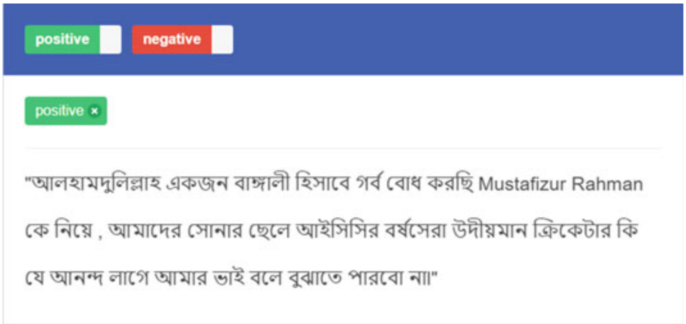

Fig. 2 Snapshot of the text annotation tool

Kappa statistics was then applied on the labelling of data collectors and the majority voting of the labelling by the native speaker group.

3.3 *Data Preprocessing*

Data preprocessing is that progression where the information gets changed or encoded to carry it to such an express that it becomes compatible with the machine. Information pre-handling is a basic development in the data mining process. It incorporates changing over the rough data from various sources into a prominent plan. Pre-processed data helps the algorithms to access and interpret the data easily. To make the system progressively reliable, we had multi-sorted out a preprocessing scheme of the dataset.

3.4 Missing Value Check

There could be two sorts of missing values like missing total information or inadequate information. In the case of missing information, we alleviated the row. For inadequate information, if the values were manually correctable, we corrected them manually. Otherwise, the whole row was alleviated in this case too.

3.5 Noise Removal

By noise removal, this paper indicates the removal of punctuations, stop words, non-Bangla characters, special symbols and emoticons. Though emoticons can express a lot of emotions, it was observed that a relatively small fraction of data contains emoticons. So the cleaning procedure contains emoticon removal as well. The processing steps along with an example is shown in Table 2.

3.6 Spelling Corrections

As the collected data come from various classes of people, there is a great chance of some words being typed wrong or misspelled. We took Bangla Academy supported accessible dictionary (AD) database [1] for finding the appropriate form of a word. From the sentiment corpus $SC = \{d_1, d_2, d_3, \ldots d_n\}$ where d_i is a text data. Each text $d_i = \{w_1, w_2, w_3, \ldots w_n\}$ where w_i is a word, specifically space-separated text element. For each word w_i in d_i if w_i is not in AD, then w_i is considered to be a misspelled word. Then, the appropriate word was searched in AD and replaced with

Table 2 Noise removal from data

Processing	Text
Original	আমরা কাজ করবো কিভাবে!! 😫😫 । Document তৈরী করতে আমাদের সবাইকে কি করতে হবে? ৫-৬ জন আমরা, কঠিন হবে :(
Removing Punctuation	আমরা কাজ করবো কিভাবে 😫😫 Document তৈরী করতে আমাদের সবাইকে কি করতে হবে ৫ ৬ জন আমরা কঠিন হবে
Removing Digits	আমরা কাজ করবো কিভাবে 😫😫 Document তৈরী করতে আমাদের সবাইকে কি করতে হবে জন আমরা কঠিন হবে
Removing Non-Bangla Characters	আমরা কাজ করবো কিভাবে 😫😫 তৈরী করতে আমাদের সবাইকে কি করতে হবে জন আমরা কঠিন হবে
Removing Emoticons	আমরা কাজ করবো কিভাবে তৈরী করতে আমাদের সবাইকে কি করতে হবে জন আমরা কঠিন হবে
Removing StopWords	করবো কিভাবে তৈরী সবাইকে কঠিন

Table 3 Spelling corrections example

Processing	Text
Original	আমরা কাজ করবো কিভাবে!! 😫😫 । Document তৈরী করতে আমাদের সবাইকে কি করতে হবে? ৫-৬ জন আমরা, কঠিন হবে :(
Removing Punctuation	আমরা কাজ করবো কিভাবে 😫😫 Document তৈরী করতে আমাদের সবাইকে কি করতে হবে ৫ ৬ জন আমরা কঠিন হবে
Removing Digits	আমরা কাজ করবো কিভাবে 😫😫 Document তৈরী করতে আমাদের সবাইকে কি করতে হবে জন আমরা কঠিন হবে
Removing Non-Bangla Characters	আমরা কাজ করবো কিভাবে 😫😫 তৈরী করতে আমাদের সবাইকে কি করতে হবে জন আমরা কঠিন হবে
Removing Emoticons	আমরা কাজ করবো কিভাবে তৈরী করতে আমাদের সবাইকে কি করতে হবে জন আমরা কঠিন হবে
Removing StopWords	করবো কিভাবে তৈরী সবাইকে কঠিন

the wrong one. The workflow applied to sample data is given in Table 3. The bold word in the original text was not found in AD thus corrected manually.

3.7 Feature Extraction

Feature extraction, more precisely word embedding is a very powerful method to represent words in a way that comparable words are translated correspondingly. Among the popular word embedding methods like one-hot encoding, bag of words, TF-IDF, count vector, word2vec, etc., we used Word2Vec as a strategy for representing the words for each sentence. Word2Vec is a novel and widely used word embedding model. It uses neural networks to find the semantic similarity of the context of the words. There are two inversely related architectures such as continuous Bag of Words (CBOW) and skip-gram; in our work, we used CBOW.

Continuous Bag of Words (CBOW) is another architecture highly used in language processing tasks. Unlike Skip-Gram, CBOW predicts a target word based on the context of some given words as input. CBOW uses continuous distributed representations of the context. In CBOW, a fixed window is constructed with some sequence of words and the model tries to predict the middle word of the window based on the future and history words using log-linear classifier. The model tries to maximize Eq. 1 [20] to achieve the predicted word wt.

$$\frac{1}{V}\sum_{v\in V}\log\left(p\left(w_v \mid\mid w_{v-c},\ \ldots w_{v-2}, w_{v-1},\ w_{v+1},\ w_{v+2}\ldots w_{v+c}\right)\right) \qquad (1)$$

where V and c denotes the same as Skip-Gram model. Figure 3 [21] represents both the model in a nutshell.

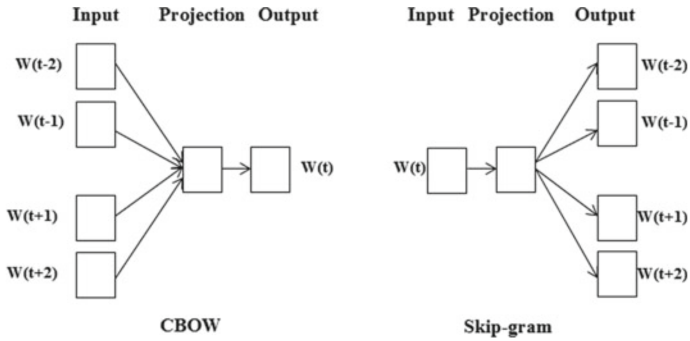

Fig. 3 Skip-gram and CBOW architecture

3.8 Dimensionality Reduction

Dimensionality reduction is a methodology for reducing dimensions of the dataset without hampering the ML model performance. Basically, the NLP problems deal with a huge dimension. The features were reduced to 5000 by using dimension reduction methods so that unnecessary information can be alleviated and the execution time of selective algorithms can be reduced. There exist heaps of techniques for decreasing measurement estimation along with LDA, SVD, NMF, PCA, and afterward forward. Here, we applied the Principal Component Analysis (PCA) to reduce dimension.

4 Algorithms

The training data that has been prepared for the machine learning environment at the initial phase was incorporated into each algorithm to figure out the target variable, and the model execution was overviewed by getting exactness. To execute the most suitable techniques for gender recognition, we have executed ten machine learning classifier algorithms, for instance, logistic regression, SVM, naive Bayes classifier, k-nearest neighbors, support vector machines, AdaBoost, gradient boosting, decision trees, random forest, impact learning [22] and three deep neural networks such as ANN, LSTM and CNN. Among them, impact learning is an as of late gathered machine learning that works for both classification and regression calculations. The principal nature of this algorithm is it works better for the competitive assessment and fits bend by the impacts of features from the characteristic pace of natural increase (RNI) and back forces or impact from different self-ruling features. Mathematically, it can be expressed as

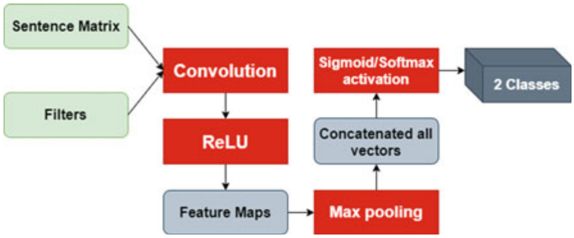

Fig. 4 CNN configuration for information extraction

$$x\,(t) = \frac{k\sum_{i=1}^{n} c_i y_i}{r - ak} + b \tag{2}$$

In this work, we also report on a progression of investigations with CNNs prepared on the counter vectorizer for the English and Bangla character-based sexual orientation name arrangements. Let $x\ i \in R\ ,\ k$ be the k-dimensional word vector corresponding to the ith word in the sentence. A sentence of length n is addressed as $x1 : n = x1 \oplus x2 \oplus \cdots \oplus xn$, where $\oplus$ isthe concatenation operator. In general, let $xi : i + j$ refer to the concatenation of words and a feature ci is generated from a window of words $xi : i + h - 1$ by

$$ci = f(w \bullet xi : i + h - 1 + b) \tag{3}$$

For a whole feature detector, one can write

$$c = [c1, c2, \ldots .cn - h + 1] \tag{4}$$

Then max-overtime pooling operation $c = \max c$ is applied. In one of the model variants, the authors experimented with having two ‘channels’ of word vectors. For all datasets, they used: rectified linear units, filter windows (h) of 3, 4, 5 with 100 feature maps each, dropout rate (p) of 0.5, l2 constraint (s) of 3 and mini-batch size of 50. Figure 4 illustrates the used CNN configuration.

Long short-term memory (LSTM) networks are a type of recurrent neural network capable of learning order dependence in sequence prediction problems [17]. Since every word in a sentence is related, every slice of a sentence is a sequence of previous slices. So, LSTM is a good choice in this system and is described in Fig. 5.

5 Experiment, Installation and Usage

To explore different aspects regarding our proposed methodology, we first fabricated the model and trained it. Thirteen classifiers including machine learning and deep neural networks were utilized to foresee the most appropriate method of finding the

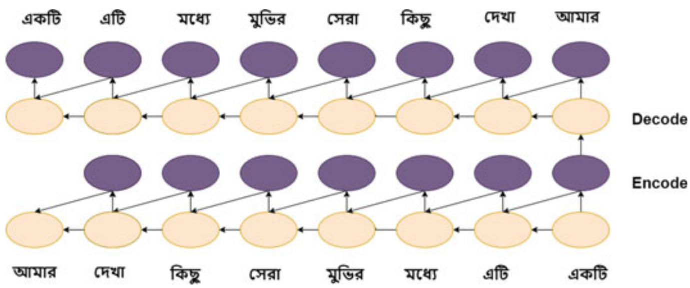

Fig. 5 LSTM structure for the sequence of word segments

sentimental state of the text ;the preprocessed data was feed to these models, and evaluation metrics were observed. Model with the high evaluation metric scores has been released for public usage.

5.1 Experimental Setup

The whole project was done in the Python programming language (3.7). As the data set was pretty much larger than average and we needed to implement some of the deep learning architectures, Google Collaboratory was used for GPU support. For ML and DNN frameworks, scikit-learn and Keras (with TensorFlow backend) were used, respectively. Besides that, another ML framework impact learning was also incorporated.

5.2 Prediction and Performance Analysis

The training set was ready at this point and significant hyper-parameters were tuned to find a fine-tuned model with relatively comparable evaluation metrics. The test set was used for evaluating the models. For each calculation, the evaluation metrics considered were accuracy, f1 score, Cohen's kappa, recall, precision, and ROC AUC. All results are summarized in Table 4.

5.3 Installation and Usage

A Python library called BNSentiment was introduced with the trained model for public usage. To install the BNSentiment, you need to follow the command "pip install BNSentiment". The usage is very simple as follows (Fig. 6).

Table 4 Performance metrics for Bangla sentiment analysis

Algorithms	Accuracy	F1 score	Cohens Kappa	Recall	Precision	ROC AUC
Random forest tree	0.818451	0.800837	0.661362	0.854811	0.878045	0.898489
SVM	0.838308	0.828754	0.711194	0.869203	0.879148	0.878984
Naive Bayes	0.781637	0.755234	0.487592	0.641848	0.621684	0.777619
Logistic regression	0.802508	0.792736	0.691527	0.862527	0.882975	0.883157
KNN	0.794679	0.734254	0.713177	0.842274	0.868984	0.844483
Decision trees	0.806126	0.755986	0.725985	0.874074	0.852637	0.856149
LDA	0.813893	0.813758	0.683734	0.874274	0.853834	0.843815
ANN	0.828134	0.798389	0.758389	0.844814	0.861488	0.828483
Impact learning	0.823539	0.819752	0.769752	0.891294	0.911493	0.893959
AdaBoost	0.818834	0.807437	0.737437	0.831425	0.843895	0.899935
Gradient boosting	0.768458	0.782975	0.712975	0.862522	0.842658	0.821308
CNN	0.791955	0.753177	0.661334	0.824883	0.839277	0.836293
LSTM	0.845308	0.834939	0.632378	0.704893	0.813829	0.813543

```
import BnSentiment as bs
s = "যতবারই এখানে আসি খুব ভালো লাগে"
s = bs.predict(s)
print(s)
s = "শারীরিক অসুস্থতার কারনে যাওয়া হয়নি"
s = bs.predict(s)
print(s)
Output:

Positive
Negative
```

Fig. 6 Usage example of BnSentiment

6 Conclusion and Future Works

This paper shows some analysis on different machine learning and deep learning approaches over sentiment detection from text. A CNN architecture was also proposed for classification purposes. A pretrained model with the highest accuracy was prepared for easy usage which is made available as a python open-source package. As deep learning requires relatively more data, we shall continue our work on enlarging the data-set. Also, we target to make a python API that can be easily incorporated in any web framework. We have also plan to use a different word extraction algorithm to analyze the ruth word extraction system on Bangla sentiment analysis.

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Chapter 29
Estimating ANNs in Forecasting Dhaka Air Quality

Mariam Hussain, Nusrat Sharmin, and Seon Ki Park

1 Introduction

Air pollution in Bangladesh is increasing in these days due to recent development activities. The capital city experiences heavy concentrations of air pollution by two significant sources: emissions from vehicles and industries. Additionally, overcrowded population in the city also accelerate pollutant concentrations. Air pollutants are described as fine and dust particles by their sizes which are mixtures of solids and liquids. The particulate matters (PM) are described by mass concentrations with a coarse size 10 μ (PM10), whereas fine particles are diameters of 2.5 μ (PM2.5). For instance, brick industries operate their kilns using coals and wood as primary energy that produce air pollutants [3, 4, 10].

These air pollutants are also growing concerns for several diseases. Respiratory diseases are broadly termed as chronic obstructive pulmonary disease (COPD) such as bronchitis, emphysema causing by air pollutants [7]. Statistics reports that 7 million people in Bangladesh are affected by asthma. The World Health Organization (WHO) reported 9660 deaths due to lung cancer occurred in Bangladesh in 2014 [14]. In 2016, WHO estimated air pollution cause deaths about 195,000 people each year [10]. An epidemiological study showed that PM2.5 exposure during pregnancy is linked with

M. Hussain · S. K. Park (✉)
Department of Climate & Energy System Engineering, Ewha Womans University, Seoul, South Korea
e-mail: spark@ewha.ac.kr

M. Hussain
e-mail: mariam.hussain9@ewhain.net

N. Sharmin
Department of Computer Science & Engineering, Ewha Womans University, Seoul, South Korea
e-mail: nusrat.sharmin@ewhain.net

M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems,
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preterm birth, lower weight, and post-neonatal infant mortality [7]. Premature infants are born in Bangladesh about 4, 38,800 every year, whereas 23,600 deaths include respiratory problems [14].

Therefore, air quality for Dhaka and nearby cities has grown focuses on various researches. For example, [2] evaluated trends and spatial analyses for air pollution and proved that 30–50% PM2.5 indicating the highest concentration in winter. Their measured data (2002–2007) in various instruments and quality control by gravimetric and descriptive statistics analyzed spatial–temporal trends relating to meteorological parameters such as winds. Furthermore, trans-boundary and meteorological variables were also scrutinized and found higher PM transports in dry seasons for Dhaka [12]. Their paper performed statistical computations on backward trajectories to find air pollution relationships to (climatology) seasonal variations. Begum et al. [3] also studied air pollutants from 2007 to 2009 and concluded brick-kiln industries as one of the largest sources. A recent study [4] compared ambient air quality for two decades (1996–2015) and demonstrated annual concentrations of air pollutants in Dhaka. However, meteorological variables and various air pollutants information might be challenging to study as they are not freely accessible.

Eventually, there are several studies have utilized machine learning (ML) techniques for forecasting various meteorological variables: rainfall and temperature including air pollutants. By using these art of artificial neural network (ANN), specifically multi-layer perception (MLP) is tested for 4 years (hourly) from 75 gauge observed rainfalls [9]. Based on their developed methods, the results for rainfall predictability were found satisfactory forecasts by incorporating multiple meteorological parameters. Back-propagation neural network (BPNN) has also greater forecasting performances around 99.79% using large datasets of 100 years [17]. Their methods considered hidden layers (HL) 7 with 10 years of datasets and predicted rainfall between 95 and 99%. Furthermore, [18] built an artificial neural network (ANN) for enumerating monthly precipitation in Ampel, Indonesia, and found 98% model accuracy. Their methods are tested on input layers, data sensitivity, optimal model parameters such as learning rate (LR), HL, and maximum epoch. BPNN was found optimized with 4000 epoch at LR = 0.4 and HL = 3 for rainfall prediction. Eventually, [11] found 98% accuracy for monthly rainfall rate estimation while preprocessed for 12 years data using only an in Matlab R2009a. The above researches and their results encourage in applying ML algorithms for predicting air quality in Dhaka.

A research question is raised how to maximize ML techniques for predicting air quality based on above findings. As computational resources are also limitations for advanced modelling, applications of ML algorithms using limited parameters might provide flexibility in air quality forecasting. For instance, [15] applied several ML algorithms and suggested ensemble learning for predicting Dhaka air quality. Even though there are studies incorporated meteorological data, ML techniques also allow predicting with minimum data (but large size) for future projections as well as history matching. Thus, in context of Dhaka, Bangladesh, the current paper aims to examine several artificial neural network (ANN) algorithms to test their predictability and accuracy that might be tuned as an efficient tool for future operational forecasting technology. The paper is outlined as experimental methodology in Sect. 2, brief summary of chosen ANNs with an optimization technique in Sect. 3, results and discussion in Sect. 4, with final remarks and future directions in Sect. 5.

2 Methodology

The methodology section includes study site, data, and ML experiments. Figure 1 shows the study site (star) and zoomed for Dhaka. The capital was designated because this only city is observed PM2.5 in concentration ($\mu g/m^3$) as well as air quality index (AQI) by US Dept. consulates. These datasets are observed, given an open access, and found in the link (https://aqicn.org/sources/). As there is missing data greater than 50% found in 2016, the hourly observed raw concentration is considered for 2017 and 2018 with total points around 33,690. The experiments are designed using the art of artificial neural networks (ANN) by incorporating time series similar to [9, 11, 15, 17, 18]. The experimental methodology in the present study is shown in a flowchart Fig. 3. Data processing is carried to clean missing values before ANN

Fig. 1 Study sites: Dhaka (capital denoted by star) metropolitan, Bangladesh. Modified based on Mapdata (2020)

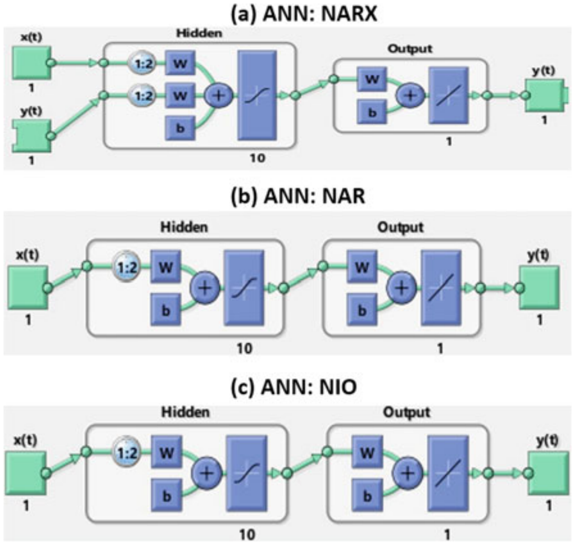

Fig. 2 ML architectures: **a** NARX, **b** NAR, and **c** NIO available in Matlab R2018b

experiments. By averaging, the hourly data is smoothed to produce daily, monthly, and seasonal, and descriptive statistics (mean, minimum, and maximum) summaries.

Several previous papers are also found on data pre-processing [17] and [9]. The present study utilized only hourly data in Matlab R2018b for finding the best ANN performance. In ML applications, there are several time series (ANNs) that are employed for understanding their predictability for PM2.5. These ANNs are nonlinear autoregressive with external input (NARX), nonlinear autoregressive (NAR), and nonlinear input–output (NIO) that are tested with Levenberg–Marquardt (LM) optimization techniques (shown in Fig. 2) based on Matlab 2018b Neural Network Time Series Toolbox. The data is divided for training, testing, and validating by 60, 20, and 20%, respectively, for each algorithm and initialized with a random generation. The experimental design is summarized in Table 1. The assumptions and architectures for each ANN are also described in Sect. 3.

3 Artificial Neural Network with Levenberg–Marquardt (ANN-LM)

To observe the accuracy and predict air quality, the techniques in ML are formulated by a system of artificial neural networks. Artificial neural network is popularly known as a biologically inspired network, a form of machine learning where the system learns to identify an output based on a series of input variables [16]. These advanced analytical methods can model highly complex nonlinear functions instead of the linear correlation [6]. It is composed of several simple neuron-like processing units

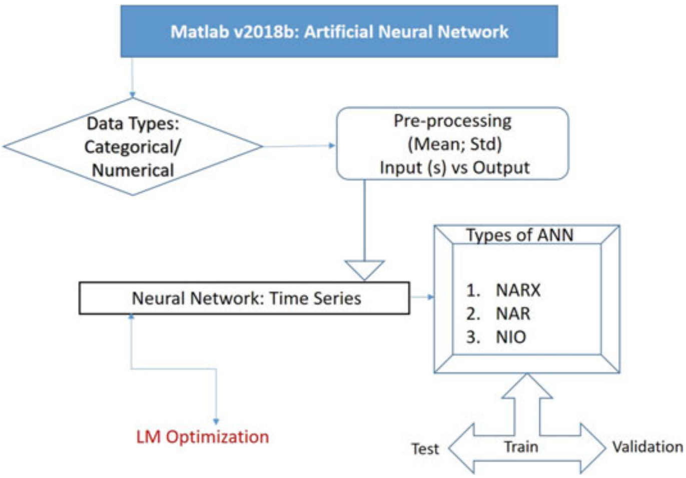

Fig. 3 A brief flowchart for experimental methods in the present study

Table 1 A detailed summery for experimental methodology

Tool	Matlab R2018b
Types of ANN	NARX, NAR, and NIO
Optimization technique	Levenberg Marquardt
Study site	Dhaka
Data source	US Embassy, Bangladesh
Air pollutant	PM2.5 ($\mu g/m^3$)
Data	2016, 2017 and 2018 (hourly)
Missing data	1461, 271 and 2903
Data distribution	Train: 60%; test & valid: 20%
Initial condition	Random distribution
Hidden layers	10
Number of delays	2

and organized in primarily three layers, namely the input layer, few hidden layers, and output layers Fig. 4 s [16].

These interconnected layers are responsible for data processing that assimilates as several interconnected “neurons.” “Synaptic connections” are created from this interconnected architecture from input nodes between the layers before approaching to the output neurons. Training procedures try to find out a set of weights that produces a mapping that fits well with the training set from this generated network architecture [16, 21]. These statistical weights are associated with each input and hidden layer (HL) that are capable of adaptations by an algorithm. During network constructions, these weights are activated to form synaptic connections among neurons. These could

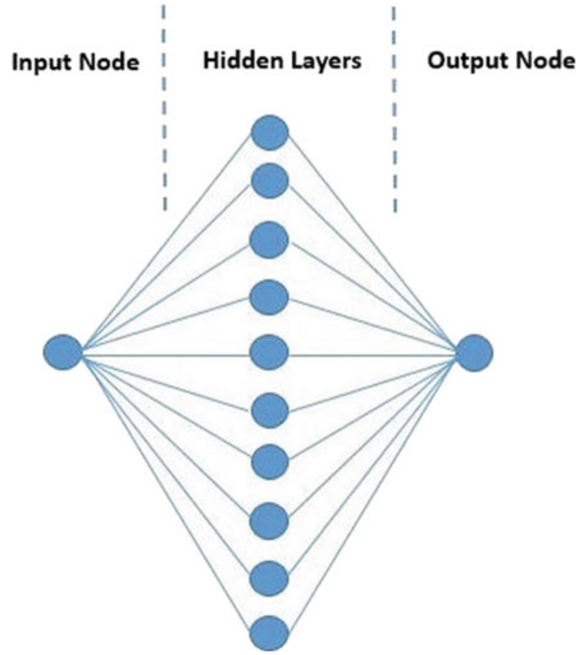

Fig. 4 Typical model of neural network

define the network as an arbitrary nonlinear function [20]:

$$f(w, x) = y \tag{1}$$

Here x is input vector in the network, w are the weights of the networks and y is the output vector approximated or predicted by the network. This arrangement of connections between the layers in the network carries information to the output layers [21]. There are also types of networks where information directs backward is known as recurrent neural networks (RNNs).

The nonlinear autoregressive network with exogenous inputs (NARX) Fig. 2a is a special type of recurrent dynamic neural network in which there are feedback connections enclosing several layers of the network. The NARX model is a combination of neural networks and the linear ARX model. This combined dynamic network tool popularly used in time series model analysis or nonlinear filtering tasks [19]. Therefore, the model can learn to predict one time series based on past values of the same time series, the feedback input, and another time series. This is known as the external or exogenous time series. The equation for NARX model based on the Matlab 2018b Neural Network Time Series Toolbox is given below:

$$y(t) = f(y(t-1), y(t-2), \ldots, y(t-n_y), v(t-1), v(t-2), v(t-n_v)) \tag{2}$$

In the above equation, the next value of the dependent output signal $y(t)$ is regressed on former values of the output signal and former values of an independent (exogenous) input signal. In this way, NARX provides better results than conventional RNN. There is a problem of vanishing gradients, i.e., difficulties in mapping long-term dependencies. NARX model could be implemented by using a feedforward neural network to approximate the function f.

Nonlinear autoregressive network Fig. 2b is another popularly known RNN commonly used in multistep ahead time series forecasting [1]. This dynamic network is based on the combination of a linear autoregressive model with feedback connections including several layers of the networks. It uses the past history of the time series to predict the next values by using Eq. (3) [5]:

$$\hat{y}(t) = f(y(t-1), y(t-2), \ldots, y(t-d)) \tag{3}$$

Here, f is a nonlinear function and the future value depends only on the regressed d former value of the output value. Hence, the historical combination of input and output value of the system formulates an input vector in the network that represent any standard feedforward neural networks [5].

To solve the nonlinear time series for predicting the future values, other dynamic neural network is nonlinear input–output Fig. 2c. This nonlinear network is a combination of two series, an input series $x(t)$ and an output series $y(t)$. The network predicts the value of series $y(t)$ based on the past value of series $x(t)$ but without utilizing the past history of $y(t)$. Based on the Matlab 2018b, Neural Network Time Series Toolbox could represent as:

$$y(t) = f(x(t-1), x(t-2), \ldots, x(t-d)) \tag{4}$$

Neural networks could be represented as highly nonlinear functions. Therefore, networks can be envisaged as an ordinary function of an optimization problem. The objective of the optimization problem is to minimize the global error, E between the network output and the desired output by adjusting the parameters of weight and biases in the network defined as [20, 21]:

$$E = \frac{1}{N}\sum_{n=1}^{N}(y-t)^2 \tag{5}$$

Here, N is the total number of training patterns, $(y-t)$ yields the error for each pattern n and t is the target. Levenberg–Marquardt (LM) algorithm is applied to train our above chosen ANNs. The purpose of the Levenberg–Marquardt (LM) is to solve the nonlinear least-squares problem [8]. This is a well-known curve fitting method which is also a fast and reliable second-order local method [19]. The method is a combination of two methods: gradient descent (first-order) and Gauss–Newton (second-order). Both of these two methods are iterative algorithms where a series of calculations are utilized assuming values of x to find a solution. Therefore, the two iterative methods are complementary according to the advantages they provide [21]. The proposed algorithm by LM is based on the observation. The update rule of the algorithm is a conjoining of the two above-mentioned algorithm and is represented as

$$x_{i+1} = x_i - (H + \lambda I)^{-1} \nabla f(x_i) \tag{6}$$

where H is the Hessian matrix evaluated at x_i. The update rule works as if the error goes down by following an update it connotes that the quadratic assumption on $f(x)$ is working and λ should reduce to alleviate the influence of gradient descent. On the contrary, if the error goes up it implicates to follow gradient more and to increase λ by same factor. Hence, the Levenberg algorithm works as [13]:

1. Apply an update by following the above mentioned rule.
2. Evaluate the error at the new parameter vector.
3. The weight is reset to their previous value if the error has increased for the update and increase λ by a factor of 10 or some such significant factor. Then repeat (1) and try an update again.
4. Set the weights at the new value if the error has decreased for the update and decrease λ by a factor of 10 or so.

4 Result Analysis

ML applications are simplified by data pre-processing which is a key requirement for data-driven models. According to expected outcomes, performance qualities, and computational resources, ML algorithms are employed for better performances and interpretations from model results. In numerical modeling, trading of calculations and outputs are limited by allocated computational resources [15]. Eventually, missing values in datasets might also result in ambiguities. As the present study incorporates US Embassy data, i.e., available since 2016, the contained missing values could also impact in ML algorithm and its prediction with poor accuracy. For instance, Fig. 6 indicates PM2.5 hourly time series of observed PM2.5 concentration (μg/m^3). The vacuum dots demonstrate all missing values for study periods (Fig. 5).

Later, these hourly data are utilized to filter for daily summaries by ignoring unobserved data. Figure 6 is classified for months by several seasons. For instance, Fig. 6 is shown pre-monsoon (March, April, and May) for chosen periods in (a), (b), and (c). Similarly, monsoon (June, July, and August) and post-monsoon (September, October, and November) seasons are shown in Fig. 6d–i, respectively. From hourly to daily, these summaries clearly imply data gaps which infer to data limitations for producing winter season out of the given datasets. Thus in seasonal summary, winter season is not summarized. For obtaining a general information, averaging from hourly, daily, and monthly is significant [4]. However, in these ways, smoothing extreme values also limit data points for short datasets (shown in Fig. 8). These are limitations for short period of data analysis as well as ANN predictions (Fig. 7).

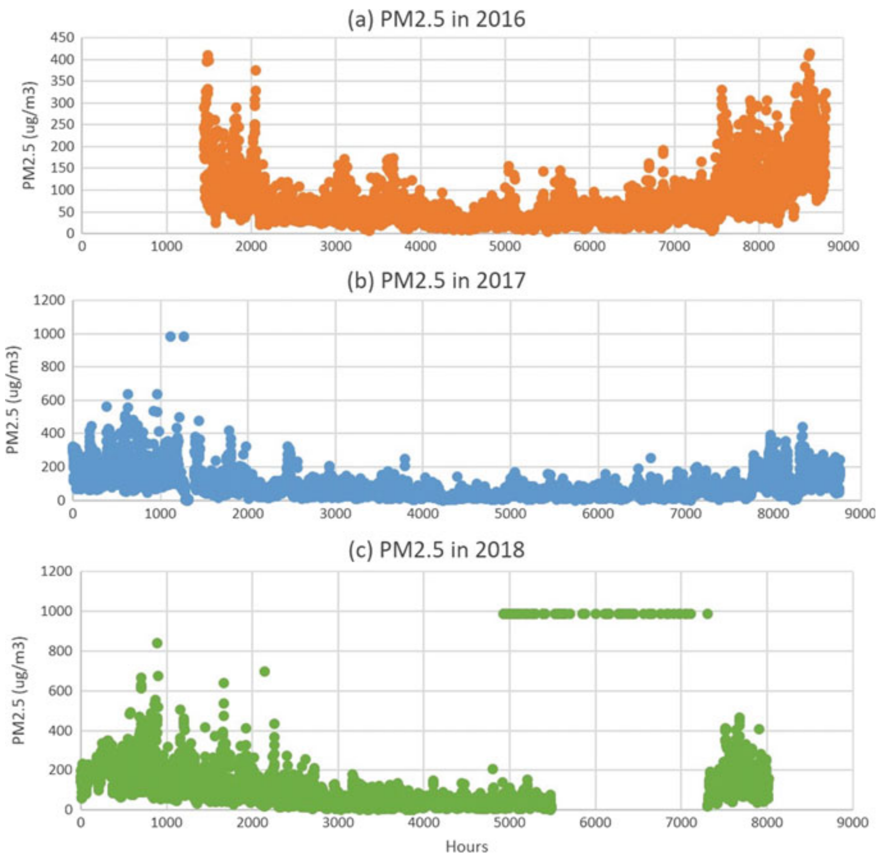

Fig. 5 Observed PM2.5 (μg/m^3) data for hourly time series [Data: US-Embassy, Bangladesh]

The data processing provides the base for applying ML techniques to test ANNs. After analyzing hourly, daily, and monthly time series, ANN tests are carried out using hourly data because of large data size. The visualization of data provides a ground to scrutinize predictability and accuracy of ANN modeling. These experiments incorporate only 2017 and 2018 data to estimate best ANN for air quality prediction and to evaluate ANN performance. As the chosen ANNs are time series data, the inputs are PM2.5 (raw concentration in) and the output are provided as AQI index based on PM2.5. By the architectures of these time series, ANN algorithms are tested to predict AQI. The experiment evaluated according to the ANN models described in Sect. 3 based on these variables. Moreover, this time series algorithms might overcome challenges due to missing values when datasets are large. Therefore, the predictability and its accuracy are significant in training, testing, and validating periods for obtaining insights for future predictions.

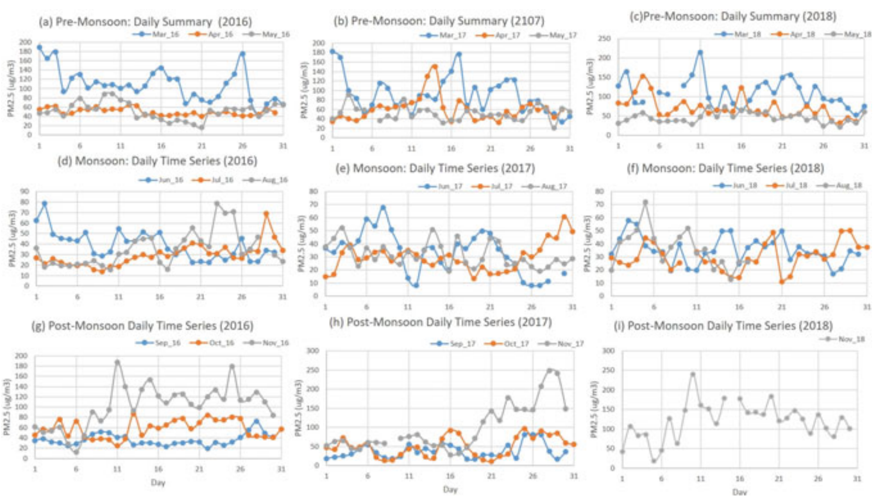

Fig. 6 Similar to Fig. but valid for daily time series

NARX, NAR, and NIO are possible to provide weekly, monthly, sub-seasonal or forecasts because of time series functions. Previous studies show how time series algorithm such as BPNN and MLP performed forecasting 98% accuracy based on large size data [9, 11]. The current ANN model can also be used for multiple purposes because of their time series architecture (shown in Sect. 3). Therefore, the current paper also tests prediction performances and their accuracy for hourly datasets. The selected ANN models are evaluated by two methods: mean square error (MSE) and linear regression between outputs and targets to find accuracy similar to [9, 17]. The MSE is calculated by Eq. 7 where i is no. of observations. Table 2 shows that MSE and the obtained gradient from optimization technique evaluating for target values. NARX algorithm indicates an overall lower MSE for each evaluation process (training, testing, and validating) than NAR and NIO.

$$\mathrm{MSE} = (\mathrm{Output}(i) - \mathrm{Target}(i))^2 \tag{7}$$

Moreover, the optimization algorithm, i.e., LM and ANN model, demonstrates a lower gradient for NARX than the other two algorithms (Table 2). Both MSE and gradient result show that NARX has outperformed the other two algorithms in terms of MSE because of the NN architecture (See Sect. 3 for ANN details). Since NARX incorporates inputs with an extra weights as well as number of delays (see Fig. 2) during the computation, MSE and gradient consist of smaller number of points unlike NAR. The regression analysis is given for each ANN in Fig. 9. These forecasting accuracies are found similar to [9, 11, 15, 17, 18].

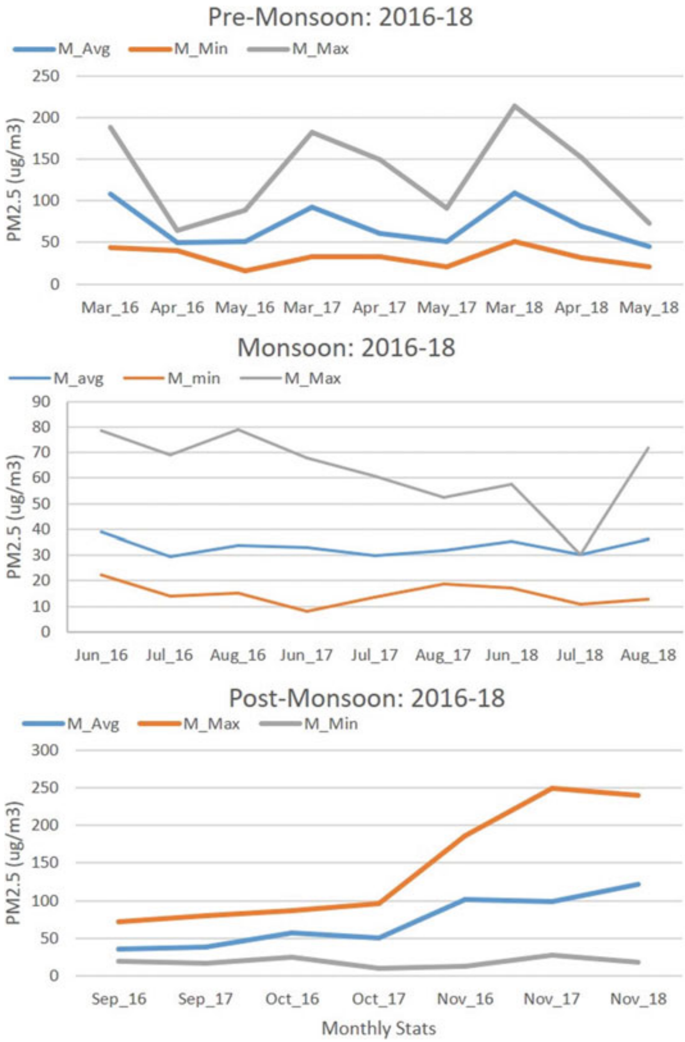

Fig. 7 Similar to Fig. but valid for monthly time series and descriptive statistics

More explicitly, optimization roles in ANN models are to optimize the data during the training period. LM is illustrated in Fig. 10 that NAR is the fast optimized value within in only 4 iterations with 2338 points than NIO (iteration = 38 and points = 4001), and NARX (iteration = 40 and points = 2262). These results suggest that LM optimization is more computationally efficient in Matlab (2018b) for these time series than BPNN found in [11]. Here, NARX validation might take iterations which refers to require less data points. This result might infer how NARX is flexible due to intrinsic weight and biases in ANN architectures assist in better predictions with

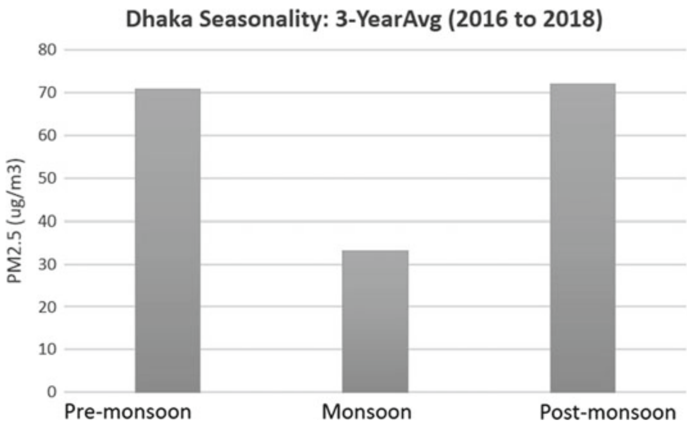

Fig. 8 Similar to Fig. but valid for seasonality for chosen periods

Table 2 ANNs evaluation by MSE via optimization

MSE	Targets	NARX	NAR	NIO
Train	20214	1323.44	1621.08	3398.84
Test	6738	1934.22	2338.19	4001.12
Valid	6738	2262.93	2407.73	3348.66
LM	Gradient	144.31	1430.16	129.43

limited data. Though each ANN model for train, test, validation and their overall assessment show forecasting accuracy about 99%, the numbers of iterations for each ANN, regression weights, and biases have varied significantly. This result suggests that the random initialization for ANN might be influenced for placing weights and biases according to algorithm architectures. However, the overall predictability for ANN did not show any influences by the random initialization.

5 Conclusion

The hourly air quality data is examined for predicting air quality in Dhaka, Bangladesh. Pre-processing data is significantly utilized in ANN time series experiments for saving computational resources. The ANN experiments prove data-driven models have higher predictability in air quality forecasting. The results also suggest ANN results can be sensitive for initial conditions, data sizes and data segregation for train, test, and validate. The ANN architectures were influenced strongly by predicting air quality. Still, these results are found similar accuracy to previous studies. As the model performances were evaluated by regression analysis and were optimized by the LM algorithm where NARX has outperformed the other two, i.e., NAR and

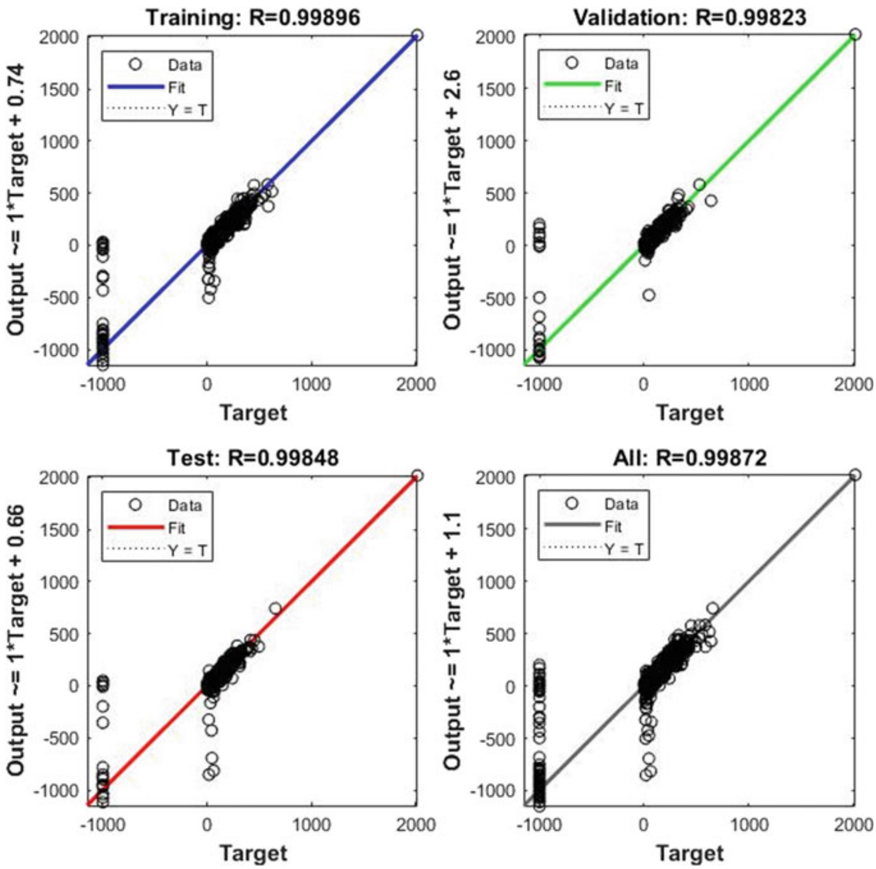

Fig. 9 Regression analysis: NARX and LM

NIO. In terms of computational resources, the NARX model shows higher efficiencies than NAR and NIO. As the computational resources are one of the challenges in numerical modeling, NARX is mostly preferred as an efficient algorithm for developing as a future forecasting tool. The pre-processing of large (air quality) datasets is equally significant to establish ANN as an operational tool for predicting air quality in Bangladesh. Eventually, this results provide key future directions to explore sensitivity tests sub-seasonal and seasonal forecasts. The researchers also aim to study multi-ANN to test air quality forecasting in relation to meteorological variables like wind, temperature, and rainfall.

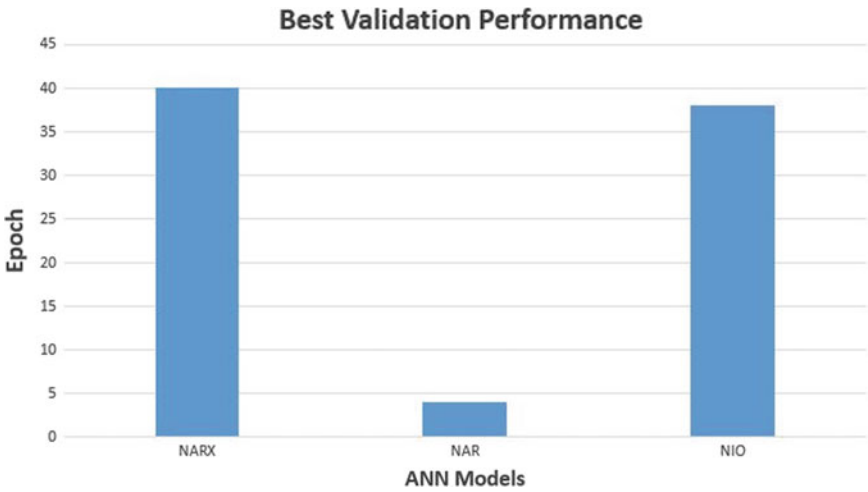

Fig. 10 Three ANNs: best validation performances

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Chapter 30
Bit Plane Slicing and Quantization-Based Color Image Watermarking in Spatial Domain

Md. Mustaqim Abrar, Arnab Pal, and T. M. Shahriar Sazzad

1 Introduction

Digital watermarking is considered as one of the most widely used approaches to protect the ownership rights of digital images, audios and videos. The success of internet and advancement of technology has ensured digital content to be delivered instantaneously with high efficiency and accuracy. Protection of multimedia content has recently become an important issue as illegal activities has increased a lot all over the digital platforms. Hence, to protect the copyright of the multimedia contents, digital watermarking technology has been introduced. Embedding watermark signal into a host signal is called digital watermarking. The watermark signal can be extracted using decryption keys to compare with the original watermark signal. The watermark signal is inseparably concealed into the host image. Additionally, it also protects the host data from different attacks.

In this paper, color image watermarking technology has been discussed. For performing watermarking schemes, a host image, a watermark image and an encryption key are needed. An image where the watermark is embedded using various algorithms is called the host image. A watermark image is the image which will be stored or transmitted through a host image. An encryption key may be necessary to embed a watermark image into a host image, and it may be needed to extract the watermark data later.

Generally, image watermarking technique is performed in two domains: spatial domain and frequency domain. In spatial domain, the watermark is embedded by modifying the pixel values of the original image [1]. In this technique, the robustness is weaker but the invisibility of the watermark is quite good and the run time is much

Md. Mustaqim Abrar (✉) · A. Pal · T. M. Shahriar Sazzad
Department of Computer Science and Engineering, Military Institute of Science and Technology (MIST), Dhaka 1216, Bangladesh

T. M. Shahriar Sazzad
e-mail: shahriar@cse.mist.ac.bd

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shorter [1]. Image watermarking in frequency domain results in the host image being transformed into its frequency domain. The watermark embedding into the frequency coefficient is highly robust, though it has higher run-time complexity [2].

The rest of the paper is organized as follows. Section 2 presents related works on different color image watermarking methods. The preliminaries are described in Sect. 3. Section 4 illustrates the process of the proposed watermarking scheme. The performance analysis of the embedded watermark in host image is evaluated in Sect. 5. Section 6 concludes the proposed watermarking approach.

2 Related Works

In this section, different existing methods related to image watermarking technology are discussed. Sharma et al. [1] used the least significant bit (LSB) to embed watermark into a host image. Multiple-parameter discrete fractional Fourier transform (MPDFRFT) with random DFRFT is incorporated to enhance the robustness and the security for watermark embedding. Additionally, inverse MPDFRFT along with random DFRFT and LSB extraction mechanism is applied to reconstruct the original image but it has high time complexity.

Chou and Wu [3] used quantization technique to embed color watermark signal in the spatial domain. CIE-Lab color space and RGB color space forward and inverse transformation were required in [3] which resulted in higher time complexity in spite of being a spatial domain technique. The performance evaluation of this method showed that it has weaker robustness than other frequency domain watermarking techniques.

A lot of research works has already been carried out, but it is still a big challenge to incorporate the benefits of frequency domain over spatial domain. An amalgamation domain watermarking method was presented in the spatial domain based on the direct current (DC) coefficient in discrete cosine transform (DCT) domain [4]. The method [4] had strong robustness and lower time complexity which was tested in the experimental results. The main drawback indicates that this approach is only suitable for binary images.

Licks and Jordan [5] discussed the well-known geometric attacks and tested the existing approaches using various types of geometric attacks. A blind image watermarking re-synchronization scheme against local transform attacks was proposed by Tian et al. [2]. For embedding watermark information into the host image, the binary space partitioning tree and local daisy feature transform were applied in [2] and [5]. Though the frequency domain watermarking scheme is highly robust, it has higher time complexity.

Jia [6] illustrated a novel color image watermarking technique based on singular value decomposition (SVD) for copyright protection of color image. In this process [6], 4×4 non-overlapping pixels block in host image are processed through SVD followed by embedding and modifying the values of orthogonal matrix in the color watermark. In extraction process, the modified values of orthogonal matrix are used to extract watermark. Experimental results indicate that the proposed scheme has strong robustness against most common attacks although the time complexity was found to be very high. This is due to the fact that the SVD and the inverse SVD operations required a higher computation complexity of O(n^3) [6].

Su and Chen [7] introduced a blind watermarking scheme in the spatial domain where to resolve the problem of protecting copyright by embedding the binary watermark image into the blue component of a RGB image was incorporated. The experimental results indicate better invisibility of watermark and stronger robustness. The main drawback of this approach includes the use of binary image as watermark which restricts to embed 32 × 32 color images into 512 × 512 host color image.

An algorithm for copyright protection of digital images was proposed by Munoz et al. [8], where an invisible color watermark was embedded in the middle discrete cosine transform (DCT) coefficients while incorporating the quantization index modulation (QIM) [9] with its variant dither modulation (DM). While embedding the watermark into the host image, the host image is converted to YCbCr from RGB color space and separation of the luminance channel from YCbCr was performed for further processing. Later on, the DCT is applied in each non-overlapped 8 × 8 pixels block. Finally, the inverse discrete cosine transform (IDCT) is incorporated on each watermarked block, and the Cb and Cr parts are reassembled with the modified luminance channel Y. High robustness against JPEG compression, impulsive and Gaussian noise is observed through experimental results, but its time complexity is high to perform DCT and inverse DCT along with QIM-DM insertion or extraction [8].

A new fast and robust color image watermarking method in spatial domain has recently been introduced by Su et al. [10] where the DC coefficient of two-dimensional discrete Fourier transform (2D-DFT) was obtained in the spatial domain without using the true 2D-DFT. In watermark embedding process, the host and watermark image was pre-processed with Arnold transform followed by obtaining the watermarked pixel block by calculating the boundary values of DC coefficient of 2D-DFT. It was observed that this technique can minimize the time complexity as the inverse DFT was not utilized to obtain the watermarked image. The running time can be minimized, and the robustness can be increased by making some changes in the above-mentioned approach which will be discussed in this paper. A comparison table is shown in Table 1.

Table 1 indicates that a number of methods have good robustness [except 2] while some have high time complexity compared to other methods. This paper proposes a technique which integrates some concepts of the above-mentioned methods to gain more invisibility, robustness and shorter running time compared to others.

Table 1 Performance comparison of existing schemes

Existing schemes	Working domain	Execution domain	Latency	Robustness
Scheme[1]	Spatial and frequency	Long	–	–
Scheme[2]	Spatial	Long	–	×
Scheme[3]	Spatial	Short	–	–
Scheme[5]	Frequency	Long	–	–
Scheme[6]	Frequency	Long	×	–
Scheme[7]	Frequency	Short	–	–
Scheme[8]	Spatial	Long	–	–
Scheme[10]	Spatial	Short	–	–

3 Preliminaries

3.1 *Arnold Transformation*

Arnold transformation is one of the most used scrambling techniques to permute watermark image for watermark security [10]. The equation of Arnold transformation can be written as:

$$\begin{bmatrix} i' \\ j' \end{bmatrix} = \mathrm{mod}\left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} i \\ j \end{bmatrix}, M\right) \tag{1}$$

where M is the dimension of the square matrix, mod(.) stands for the modular function, i and j are the original location of the pixel value in the matrix, and i' and j' are the new location in the scrambled matrix. Additionally, the iteration number can be used as a secret key in Arnold transform. The original image can be restored using the following inverse Arnold transformation equation:

$$\begin{bmatrix} i \\ j \end{bmatrix} = \mathrm{mod}\left(\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} i' \\ j' \end{bmatrix}, M\right) \tag{2}$$

3.2 *Bit Plane*

A bit plane of an image is a set of bits corresponding to a given bit position in each of the binary numbers representing the image. In 24-bit color image each red, green and blue channel contains pixel value from 0 to 255 which can be represent using 8 bits and thus for a 24-bit image there can be 8 bit planes. Each bit plane has different significance depending on its position in the binary representation. Bit plane 0 contains the set of least significant digits where bit plane 7 contains the set of most significant digits. The upper bit planes hold the major information and the lower bit planes hold the subtle information about the image. Using only upper bit planes, it is possible to approximate an image of the original image. Increment of number of upper bit planes increases the possibility to obtain an image similar to the original image. This approach is considered as a lossy process which provides the approximate image of the real image compared to the use of less number of bit planes.

If upper 4 bit planes is used to represent the pixel value then the maximum difference with the original pixel value will become:

$$1111_2 = \left(1 \times 2^3\right) + \left(1 \times 2^2\right) + \left(1 \times 2^1\right) + \left(1 \times 2^0\right) = 8 + 4 + 2 + 1 = 15$$

The minimum difference will become zero when the last 4 bits of original pixel value are 0000_2. This maximum difference can be brought down to 8 from 15 if an additional 8-pixel value is used during the reconstruction phase. This is considered as a lossy data compression technique.

3.3 Quantization Process

To reduce the number of set of amplitudes of analog signal, quantization process is used in signal processing. It is always possible to use any amplitude of a pulse for the maximum amplitude $V_{\max}$ and minimum amplitude $V_{\min}$ of the signal.

Existing research works indicate that quantization process in watermarking by using two different sets of quantized values for watermark bit 0 and 1 has already been used. There are two different formulas that can be used to calculate the quantized values. There is a vulnerability as it uses fixed quantization step T. Study by [10] used the value of $T = 125$ as it is the correlation between the latency and robustness of the watermark.

3.4 DC Co-Efficient (DCC)

In [10], the DC coefficient of two-dimensional discrete Fourier transform (2D-DFT) is calculated in the spatial domain using following formula

$$\text{DCC} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \tag{3}$$

where x and y spatial domain variables, $x = 0, 1, \ldots, M - 1$, $y = 0, 1, \ldots, N - 1$, $f(x, y)$ is the pixel value in (x,y) position, M and N are the width and height of the image and DCC is called the 2D-DFT DC coefficient of function $f(x, y)$.

The modification of each pixel in the image block after watermark embedding is found in [10] by using following formula

$$\Delta f(x, y) = \frac{1}{\mathrm{MN}} \cdot \left(\mathrm{DCC}^* - \mathrm{DCC}\right) \tag{4}$$

where DCC and DCC* are the 2D-DFT DCC coefficient of the original image and the watermarked image, respectively, and $\Delta f(x, y)$ is the change of the pixel value in (x, y) position.

4 Proposed Watermarking Technique

This study proposed approach has two phases which include pre-processing and post-processing. Pre-processing phase includes Arnold transformation followed by compressed bit plane sequencing of the watermark image. Post-processing phase includes de-compression of the watermark image followed by inverse Arnold transformation. The upper 4 bit planes are used to embed into the luminance channel of host image shown in Fig. 1. One bit from the upper 4 bit planes is embedded into a 4×4 block of the luminance channel.

4.1 Pre-processing of the Host Image and Watermark Image

In watermarking, it is required to use two different images that include an original host image and a watermark image. A 24-bit color RGB host image H with a size of $M \times M$ was converted to YCbCr color model. The luminance channel Y from YCbCr color model was used to embed the watermark image. A 24-bit color watermark image W with size of $N \times N$ was split into three different channels (red, green and blue). For each color channel W_i Arnold transformation was incorporated to permute the position of the pixel values by Eq. (1). Additionally, to perform Arnold Transform a unique secret key iter_i is used as iteration number for each color channel. Finally, each pixel value of each color channel (red, green and blue) consisting of unsigned integer was converted into its binary representation of 8 bits. The four least significant bits were eliminated from the above-mentioned 8 bits of each channel and a $\mathrm{sequence_i}$ of length of $N \times N \times 4$ was created containing only 0 and 1, where $i = 1$, 2, 3 for each color channel (red, green and blue). The bit_seq was formed combining the $sequence_i$ of three-color channels and the length of bit_seq $(N \times N \times 4 \times 3)$.

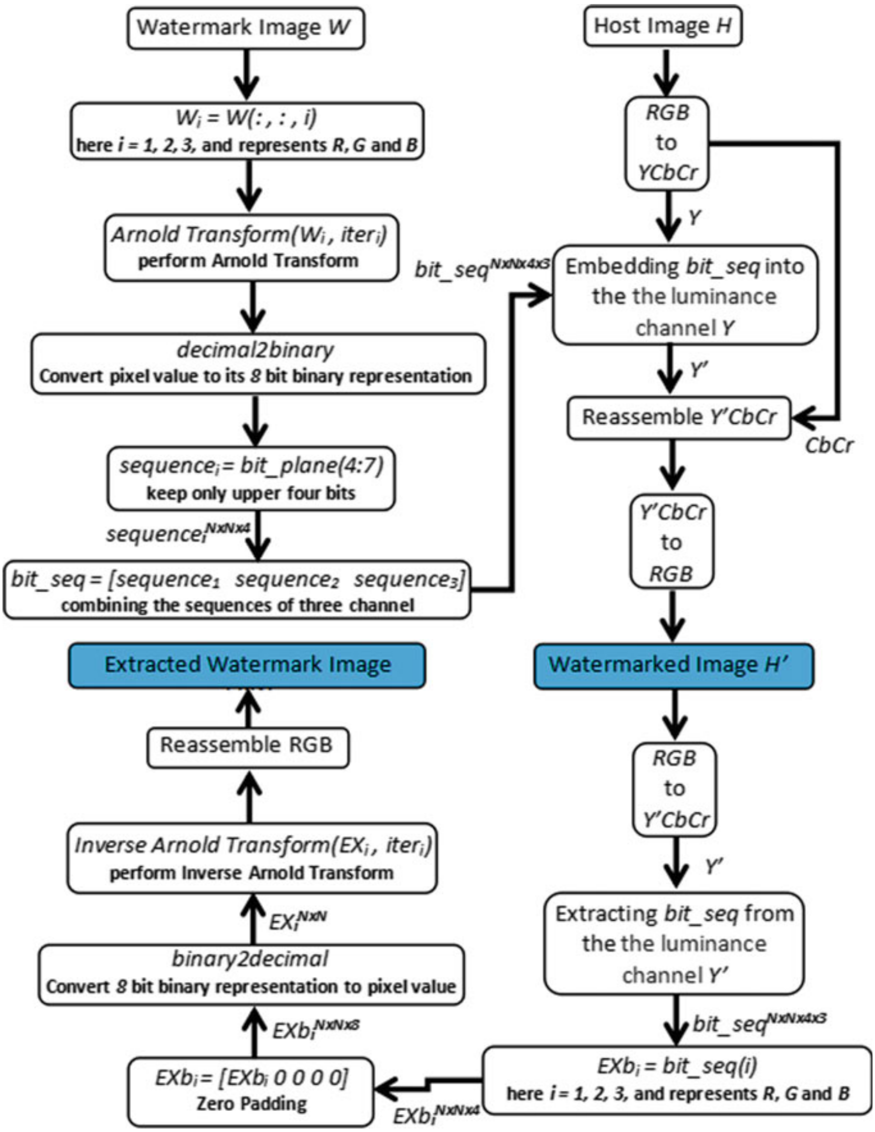

Fig. 1 Block diagram

4.2 Watermark Embedding Process

The bit_seq was embedded into the luminance channel Y using the algorithm shown in Fig. 2 which is based on the quantization technique [10]. In quantization technique, each calculated value (DCC) is located to its closest quantized value using quantization formula.

Fig. 2 Algorithm for watermark embedding

```
function y = embedding (y, bit_seq, seed1, seed2)
    [M M] := size(y)
    [N] := size(bit_seq)
    m := 4
    //choose N integers pseudo-randomly from 1 to (M*M)/(m*m)
    //without repeating using seed1 as secret key
    Bk := randperm((M*M)/(m*m), N, seed1)
    //choose N integers pseudo-randomly
    //using seed2 as secret key
    T := randperm(N, seed2)
    for w := 1 to N
        //start index of the block Bk(w)
        [i j] := start_position(Bk(w))
        //sum of the entire block
        DCC := sum(y(i to i+m-1 , j to j+m-1))
        //Calculating the lower and upper quantized value
        if (bit_seq(w) == 1)
            Cl := floor(DCC/T(w))*T(w)+0.25*T(w)
            Ch := (floor(DCC/T(w))+1)*T(w)+0.25*T(w)
        else
            Cl := (floor(DCC/T(w))-1)*T(w)+0.75*T(w)
            Ch := floor(DCC/T(w))*T(w)+0.75*T(w)
        end
        //choose the nearest quantized value of DCC
        Q_Value := nearest_one_to_DCC(Cl,Ch)
        //modifying the luminance channel, y
        y(i to i+m-1 , j to j+m-1) += ((Q_Value-DCC)/(m*m))
    end
```

The 2D watermark image was compressed using bit plane slicing. The generated bit planes were reconstructed from 2 to 1D bit_seq. Each bit from 1D was embedded into one of the 4×4 blocks of Y. For each bit, different quantization step was used to calculate quantized value. For all bits of bit_seq to be embedded in Y, the length of the bit_seq must be less than equal to the total number of blocks $(M \times M)/(m \times m)$. The blocks and the quantization steps are pseudo-randomly chosen based on the secret keys seed1 and seed2. The DCC of the block was calculated using Eq. (3) followed by calculation of the lower quantized value Cl and higher quantized value Ch for each DCC of the block as shown in Fig. 2. Two different formulas were used to calculate these quantized values where 0 and 1 were incorporated.

The DCC of the block was shifted to its nearest quantized value by adding $\left(\frac{(\text{nearest\_quantized\_value}-\text{DCC})}{(4\times4)}\right)$ to each pixel of the block where nearest_quantized_value was considered Cl or Ch. The same process was repeated until all the bits were embedded into Y. The watermarked image Y′CbCr was reassembled using the modified Y' followed by conversion to RGB color model.

Fig. 3 Algorithm for watermark extraction

```
function y = extracting (y, seed1, seed2)
    [M M] := size(y)
    N := 32*32*8
    m := 4
    //choose N integers pseudo-randomly from 1 to (M*M)/(m*m)
    //without repeating using seed1 as secret key
    Bk := randperm((M*M)/(m*m), N, seed1)
    //choose N integers pseudo-randomly
    //using seed2 as secret key
    T := randperm(N, seed2)
    for w := 1 to N
        //start index of the block Bk(w)
        [i j] := start_position(Bk(w))
        //sum of the entire block
        DCC := sum(y(i to i+m-1 , j to j+m-1))
        if (mod(round(DCC),T(w))<0.5*T(w))
            w(i) := 1
        else
            w(i) := 0
        end
    end
```

4.3 Watermark Extraction Process

A blind extraction technique was incorporated where the watermarked image and the secret keys are essential. The bit_seq of watermark image from the luminance channel Y' of the watermarked image is extracted by using the algorithm as shown in Fig. 3 [10]. The blocks and the quantization steps are chosen pseudo-randomly based on the secret keys seed1 and seed2 (as used in the watermark embedding process).

4.4 Post-Processing the Bit Sequence

The extracted bit_seq was separated into three sequences EXb_i for each red, green and blue color channels. In the pre-processing phase, lower four bits (watermark image) were ignored for each pixel values but in the post-processing phase to generate the watermark image, it is essential to consider all bits (lower four bits as well from the host image). These generated 8 bits (binary pixel values) were further converted to decimal values (EX_i) of length $N \times N$ using EXb_i. In each color channel EX_i, inverse Arnold transformation was applied to permute the position of the pixel values corresponding to their original position (position of the original watermark image) using Eq. (2). The same unique keys iter_i were used as iteration number for each color channel. These three-color channels were combined together to generate a 24-bit extracted watermark image EXW.

Fig. 4 Host images: **a** Avion, **b** Baboon, **c** Lena, **d** Peppers and **e** Sailboat

Fig. 5 Color watermark image

5 Experimental Results

5.1 Dataset

The digital images used to evaluate the performance of the proposed method were adopted from CVG-UGR image dataset [11] as shown in Fig. 4. All the host images were reshaped to 512 × 512 pixels, and the watermark image as shown in Fig. 5 was resized to 32 × 32.

5.2 Impact of Embedded Watermark in Host Image (Invisibility Test)

Table 2 indicates a comparative result of this study proposed approach and existing method [10].

Table 2 Invisibility comparison

PSNR/SSIM		
Host Image	Method [10]	Proposed method
Avion	38.1039/0.9353	42.2413/0.9965
Baboon	37.9085/0.9794	42.9327/0.9970
Lena	38.0535/0.9414	42.6550/0.9931
Peppers	37.6262/0.9231	42.3636/0.9929
Sailboat	36.4809/0.9250	42.3983/0.9944
Average	37.5788/0.9332	42.5182/0.9940

Table 2 indicates that this study proposed approach has better result compared to existing work [10] (Peak signal-to-noise ratio (PSNR) and the Structural Similarity Index (SSIM)).

5.3 Robustness Test

A number of common attacks (i.e., JPEG, Gaussian noising, salt & pepper noising, Butterworth low-pass filtering, median filtering) were applied to the watermarked images to evaluate the robustness of the proposed approach in terms of normalized correlation (NC).

Figure 7 shows the average NC of the extracted watermark images after different attacks.

Figures 6 and 7 indicate clear that this study proposed an approach which has better robustness against JPEG, Gaussian noise and median filtering attacks compared to method [10].

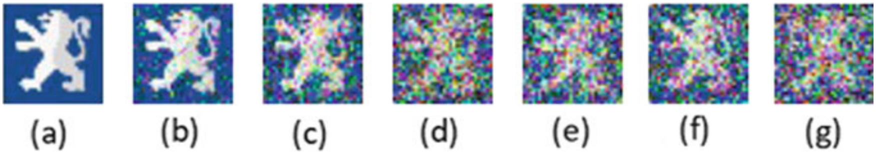

Fig. 6 Extracted watermark images from watermarked images after different attacks: **a** No attack NC = 0.9996, **b** JPEG (QF = 75) NC = 0.9712, **c** JPEG (QF = 50) NC = 0.9681, **d** Gaussian noise $\left(0, \sqrt{0.001}\right)$ NC = 0.8529, **e** Salt & Peppers noise (2%) NC = 0.8505, **f** Butterworth low-pass filtering ($D_0 = 100$, $n = 10$) NC = 0.8870 and **g** Median filtering (3 × 3) NC = 0.8818

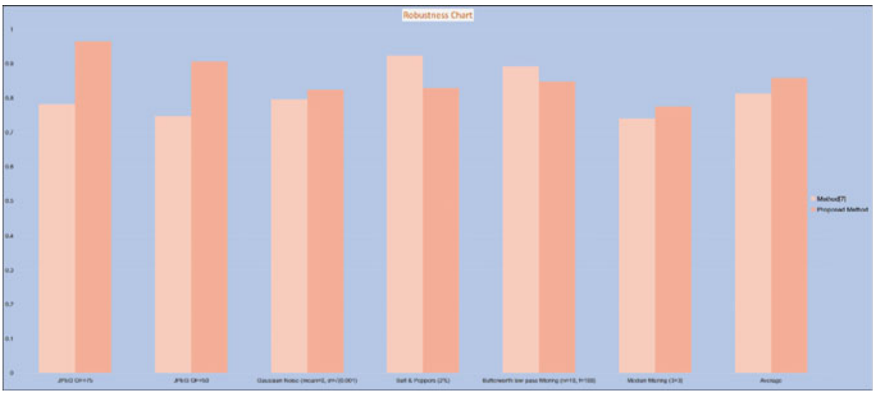

Fig. 7 Robustness comparison between method [10] and proposed method

Table 3 Time complexity comparison

Watermark embedding/Extraction time		
Host image	Method [10]	Proposed method
Avion	0.3853/0.3407	0.3465/0.2693
Baboon	0.3698/0.3395	0.3457/0.2713
Lena	0.3903/0.3381	0.3474/0.2691
Peppers	0.3878/0.3376	0.3381/0.2702
Sailboat	0.3773/0.3423	0.3452/0.2696
Average	0.3832/0.3406	0.3445/0.2700

5.4 Time Complexity

Table 3 compares the time complexity of watermark embedding and extraction of the proposed method and the existing method [10].

6 Conclusion

In this study, a technique for copyright protection of digital color images is presented to provide better invisibility and robustness against different attacks. The proposed technique is a combined approach based on the compression and quantization process which provides high robustness and invisibility against the common image processing attacks. It is still a challenge to overcome different type of attacks. This study has tried to overcome different type of attacks while increasing the visibility and robustness. However, there are some areas to improve which will be investigated in future work.

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Chapter 31
BAN-ABSA: An Aspect-Based Sentiment Analysis Dataset for Bengali and Its Baseline Evaluation

Mahfuz Ahmed Masum, Sheikh Junayed Ahmed, Ayesha Tasnim, and Md. Saiful Islam

1 Introduction

Newspapers provide a fantastic source of knowledge of different domains. With the extensive growth of the web platform, people express their opinions on various topics, complain and convey sentiment on different social websites or news portals. In this paper, we are going deep down to classify people's subjective opinions based on Bengali newspaper comments. Sentiment analysis [1] tries to extract the biased information from a user-written textual content by classifying it into one of the predefined sets of classes. Aspect-based sentiment analysis (ABSA) is a text classification system that aims to detect the aspect terms in sentences and predict the associated sentiment polarities. For example, the following sentence is taken from a Bengali news portal's comment section:

"Ronaldo and Messi are very good footballers."

This sentence has the aspect of sports and the polarity is positive. Aspects and sentiments may or may not be explicitly mentioned in a sentence. It is the engagement of a machine learning algorithm, to identify the implicitly stated aspect term and sentiment polarity.

The ABSA task can be divided into two sub-tasks: aspect term extraction and sentiment classification. The principal responsibility of opinion mining is aspect term

M. Ahmed Masum (✉) · S. Junayed Ahmed · A. Tasnim · Md. Saiful Islam
Shahjalal University of Science and Technology, Kumargaon, Sylhet 3114, Bangladesh
e-mail: mahfuz27@student.sust.edu

S. Junayed Ahmed
e-mail: sheikh44@student.sust.edu

A. Tasnim
e-mail: tasnim-cse@sust.edu

Md. Saiful Islam
e-mail: saiful-cse@sust.edu

M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems,
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finding [2]. Identification of the terms of aspect is often referred to as aspect term or opinion target extraction. We have used supervised learning-based approaches in this paper for both aspect term extraction and classification of sentiments. The supervised approach to learning is one that establishes a relationship between inputs and outputs, based on examples.

The Bengali language has suffered from a lack of research on the study of ABSA. We devoted our work to the Bengali language by contributing as follows:

- We created a benchmark dataset, BAN-ABSA, that contains 9009 unique comments collected from some popular Bengali news portals. We classified a sentence into one of four aspects: Politics, Sports, Religion, and Others. Again for every aspect, we use respective ternary sentiment classification: Positive, Negative, and Neutral.
- We made the dataset publicly available.[1]
- We also conducted a baseline evaluation on our dataset by implementing some state-of-the-art models to find the one that gives the best performance in terms of Bengali ABSA.

2 Related Work

We list the most remarkable researches pertaining to both SA and ABSA throughout this section. Due to the deficiency of researches regarding this topic in Bengali, we also mention some works conducted in English.

Sentiment Analysis can be considered as a problem of classification in NLP. Methodologies based on lexicon were used primarily in earlier researches [3, 4]. They involved creating customized features that were expensive. The feature quality significantly influenced the outcome.

ABSA is a fine-grained version of SA that seeks to detect not only the polarity but also the aspect. Key tasks of ABSA include aspect extraction and polarity detection [5]. Deep neural networks are used recently in ABSA with great success. Long Short Term Memory (LSTM) [6], which is a variation of Recurrent Neural Network (RNN) with feedback connections, is being used in many models with promising results [7–9]. Convolutional Neural Network (CNN) is a neural network that can assign weights and biases to various aspects of the input data using convolutional layers. CNN based models are found to be a good performer in this task [10, 11]. Clause level classification of a sentence can also show significant performance [12]. A language model can distinguish between phrases in a sentence by using probability distribution over the sequence of words. The use of language models have introduced a new dimension to the sentiment analysis [13–16].

Several datasets have been proposed for ABSA in English and other languages. A dataset was created in [17], namely SentiHood, which contains 5215 sentences col-

[1] https://www.kaggle.com/mahfuzahmed/banabsa.

lected from Yahoo! question answering system of urban neighborhoods. The dataset has 11 aspects and 3 sentiments. They conducted a baseline evaluation on the dataset and got best accuracy using Logistic Regression (LR) based model LR-Mask and LSTM in the sub-tasks of ABSA respectively. Reference [18] proposed a dataset containing 5412 Hindi product review sentences of 12 domains and 4 polarities. Conditional Random Field (CRF) and Support Vector Machine (SVM) performed the best in their dataset. 39 datasets containing 7 domains of 8 languages were introduced in SemEval-2016 Task 5 [19]. The available languages were English, Arabic, Chinese, Dutch, French, Russian, Spanish, and Turkish. Another ABSA dataset containing 2200 reviews of the Czech language was crafted in [20]. They achieved best performance by utilizing linear-chain CRF.

ABSA is relatively new in Bengali language. Two datasets were introduced in [21], a cricket and a restaurant dataset. The cricket dataset contains 2900 comments collected from online news portals and the restaurant dataset contains 2800 comments translated from an English corpus. The research was taken further in [22] by utilizing CNN to extract the aspect from a sentence. Our research is the first one in Bengali which completes both sub-tasks, aspect extraction, and sentiment classification using Bi-LSTM neural network.

3 BAN-ABSA Description

There is always a scarcity of the Bengali benchmark dataset for Aspect-Based Sentiment Analysis. Though two datasets were presented in [21], there were very few comments in both of the datasets. So we created another benchmark dataset for this task. Our BAN-ABSA dataset contains a total of 9009 comments collected from some prominent Bengali online news portals. It contains four aspects: politics, sports, religion, and others and every aspect have their associated polarity: positive, negative, and neutral. The comments are collected by the authors of this paper and annotated in collaboration with nine annotators.

3.1 Data Collection

Our target was to collect comments from various news portals where enormous people express their opinions on different topics. Among several online portals, we opted for the most popular ones. The data collection process is described below:

Daily Prothom Alo [23]: Prothom Alo is a very popular Bangladeshi newspaper. Authentic news about daily affairs are published here. On their website, there is a comment section where people post their opinions about ongoing events of all aspects. But very few comments can be found on their website. We head over to their Facebook official page [24] and collected 16,735 comments of different aspects.

Daily Jugantor [25]: Daily Jugantor is another popular Bangladeshi newspaper but due to the unavailability of comments on their website, we reached out to their Facebook page [26] and collected 14,389 comments.

Kaler Kantho [27]: Lots of people in Bangladesh read Kaler Kantho online. Very few of them throw their opinions on the website. A total of 8062 comments were collected from the website and Facebook official page [28].

After the data collection process, there were a total of 39,186 comments to pre-process.

3.2 Annotating Data

After the collection of comments, we removed multi-lined comments from the dataset. In some cases, we trimmed multi-lined comment into single-line comment. We removed emojis from the comments. These were done manually by the authors. After that, 9009 comments remained for annotating.

For the purpose of annotating, we split the data into three parts and divided between 9 annotators. All of them are undergraduate students. Each comment was annotated by 3 annotators for both aspect and sentiment polarity. Any comment could have one of 3 aspects: politics, religion, and sports. The comment that doesn't fall into any of the aforementioned aspects, falls in the other aspect. Again the comment can be either positive, negative, or neutral. If there was any contradiction for any comment, we followed the majority voting. The following comment can be considered as an example:

"এই খুনের পিছনে ধর্ম কোন কাজ করেনি, কাজ করেছে রাজনীতি"

The outcome of annotating this comment by 3 annotators is shown in Table 1.

We can see from Table 1 that the comment was marked as negative by all the annotators. For the aspect annotation, it was marked as containing politics aspect by 2 of the annotators and one annotator marked it as containing religion aspect. From majority voting, we finally agreed to annotate this comment as having **Politics** aspect and **Negative** polarity. We didn't face any tie situation. Table 2 shows a sample of BAN-ABSA.

In the final dataset, we have 9009 labeled comments. The dataset contains four aspects and 3 polarities. We tried to balance the dataset as much as possible but in

Table 1 Annotation example of a comment

Comment: এই খুনের পিছনে ধর্ম কোন কাজ করেনি, কাজ করেছে রাজনীতি

Annotator	Aspect	Polarity
A1	Politics	Negative
A2	Religion	Negative
A3	Politics	Negative

Table 2 Dataset sample

Comment	Aspect	Polarity
মেসি চলে যাবে সিটি তে	Sports	Neutral
বার্সেলোনা ওয়াল্ডকাপ জিতবে এইবার	Sports	Positive
ধন্যবাদ, সাধুবাদ ও শুভ কামনা জানাই প্রধান নির্বাচন কমিশনার সাহেব কে	Politics	Positive
এখন মদিনা সনদ আর পাঁচ ওয়াক্ত তাহাজ্জত নামাজ দিয়ে রাষ্ট্র পরিচালিত হবে	Religion	Neutral
বড় আশাছিলো পাক ভারত একটা যুদ্ধ দেখব, তা আর দেখা হলোনা	Politics	Negative
আজ প্রমাণ হলো বাংলাদেশে কোন সংবিধান নেই।	Politics	Negative
আল আমিনের বোলিং তো অসাধারণ	Sports	Positive
মুসলিমের আলোর পথে আনার জন্য তিনি হলেন এক মাত্র ব্যক্তি বর্তমানে।	Religion	Positive
গ্রামীণফোনের ৫০ লাখ ফোরজি গ্রাহক	Others	Neutral
জোর করে রাজ্য জয় করলেন কিন্তু মনটা জয় করতে পারলেন না।	Others	Negative

Table 3 Statistics of the dataset

Aspect	Positive	Negative	Neutral	Subtotal
Politics	506	1684	473	2663
Sports	646	407	527	1580
Religion	958	594	201	1753
Others	509	2036	468	3013
Total				9009

the case of news, people usually throw fewer comments for a piece of good news. Conversely, plenty of comments can be found on a post about any bad incident. The statistics of the dataset are given in Table 3.

3.3 Dataset Analysis

In order to understand the annotators' reliability, we calculated Intra-class Correlation Coefficient (ICC) which was 0.77. We also applied Zipf's law [29] in our dataset. Zipf's law is an empirical law proposed by George Kingsley Zipf. It states that the collection frequency cf_i of the ith most common term in the dataset should be proportional to $1/i$:

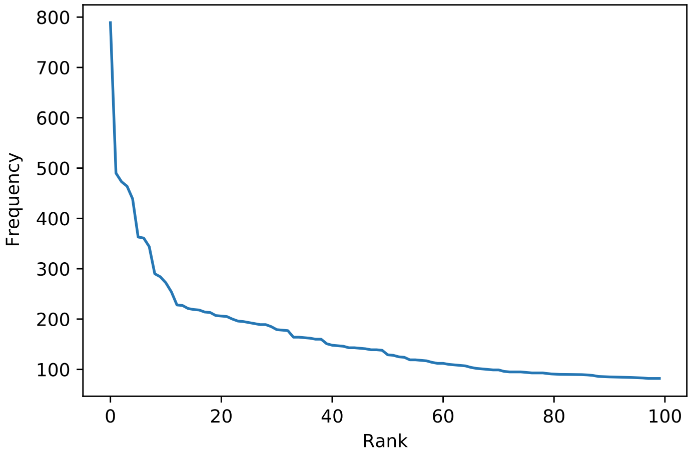

Fig. 1 Word frequency distribution

$$cf_i \propto \frac{1}{i}$$

The application of Zipf's law is shown graphically in Fig. 1. It indicates that the words are not random in our dataset.

4 Experiment

We created a benchmark dataset, BAN-ABSA, to be used in Bengali ABSA. It can be used for both the sub-tasks: aspect extraction and sentiment classification. In the pre-processing step of our experiment, we removed the comments that contained English words. We removed all the punctuation marks. Finally, we tokenized the data. We evaluated some state-of-the-art deep neural network models and some traditional supervised machine learning models to show the baseline for this task. Among several models, we achieved the highest result using Bi-LSTM in both sub-tasks.

4.1 Bi-directional Long Short Term Memory

Recurrent Neural Networks (RNN) are being used in text classification tasks with significant results. To overcome the vanishing gradient problem associated with RNN, a new variant was introduced, LSTM [6]. As for LSTM, information can only be propagated in forward states. As a result, the state is dependent only on past information at any time. But we may need to use forward information as well. Bi-LSTM has

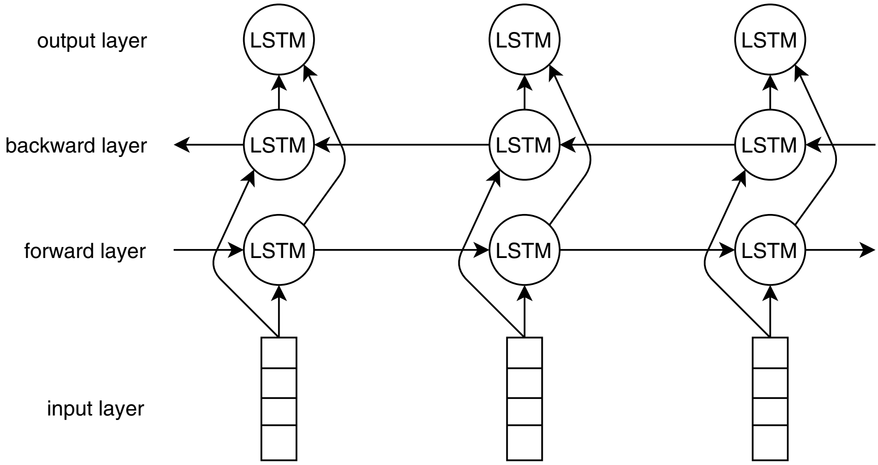

Fig. 2 Bi-LSTM architecture

been introduced to overcome that shortcoming. BiLSTM's architecture incorporates two hidden, opposite direction LSTM layers to completely capture the input context. This is achieved through the splitting of a regular RNN's state neurons into two parts. One is used to capture features of the past using forward states and the other one captures features of the opposite direction using backward states [30]. BiLSTM cells are actually unidirectional LSTM cells, without the reverse states. A simple representation of this model is shown in Fig. 2.

4.2 Result and Discussion

The sub-tasks of ABSA are both multi-label classification problem. Deep neural networks work better than conventional supervised machine learning models in such tasks. The results of our evaluation on the dataset for aspect extraction and sentiment classification is shown in Tables 4 and 5.

Table 4 Performance of aspect extraction on the proposed dataset

Model	Accuracy	F-1 Score	Precision	Recall
SVM	68.53	69.51	73.87	65.64
CNN	**79.09**	74.69	74.61	74.78
LSTM	77.17	77.24	78.11	76.38
Bi-LSTM	78.75	**79.38**	**80.46**	**78.33**

Table 5 Performance of sentiment classification on the proposed dataset

Model	Accuracy	F-1 score	Precision	Recall
SVM	65.02	41.15	45.13	37.82
CNN	**71.48**	60.15	61.89	58.51
LSTM	69.51	61.20	**63.37**	59.17
Bi-LSTM	71.08	**62.30**	62.49	**62.11**

It is clear that Bi-LSTM works better than other deep neural networks in Bengali ABSA. It shows significant F-1 and recall score in both aspect extraction and sentiment classification. Though LSTM works very closely with Bi-LSTM, it still fails to catch some aspects. BiLSTM has better context catching capability because of its bi-directional approach. CNN shows better accuracy in both of the sub-tasks, by achieving accuracy score of 79.09% and 71.48% respectively. Bi-LSTM outperforms CNN by achieving a higher f-1 score of 79.38% and 62.30%, respectively, in the sub-tasks.

To the best of our knowledge, only aspect extraction has been done in Bengali ABSA in [21, 31]. Performance comparison of aspect extraction between Bi-LSTM and other models that were used previously is shown in Table 6. The models were evaluated on our dataset.

SVM, KNN, RF, and CNN models were evaluated for aspect extraction on the dataset proposed in [21] and our dataset. Table 7 shows the comparison of the performances between our dataset and the dataset stated in [21].

It is visible from Table 7 that quality of dataset matters when it comes to performance. The same models perform better in our dataset as our dataset is bigger in size compared to the Cricket and Restaurant dataset and contains more variation.

Some hyper-parameters that we have used to evaluate the models for aspect extraction are given in Table 8.

Table 6 Aspect extraction performance comparison

Model	Accuracy	F-1 Score	Precision	Recall
SVM [21, 31]	68.53	69.51	73.87	65.64
KNN [21, 31]	50.69	47.05	58.30	45.64
RF [21, 31]	65.25	65.10	68.31	63.55
CNN [22]	79.09	74.69	74.61	74.78
Bi-LSTM	**78.75**	**79.38**	**80.46**	**78.33**

Table 7 Performance comparison of dataset proposed in [21] and our dataset

Dataset	Model	F-1 Score	Precision	Recall
Cricket [21]	SVM	0.34	0.71	0.22
	KNN	0.25	0.45	0.21
	RF	0.37	0.60	0.27
	CNN	0.51	0.54	0.48
Restaurant [21]	SVM	0.38	**0.77**	0.30
	KNN	0.42	0.54	0.34
	RF	0.33	0.64	0.26
	CNN	0.64	0.67	0.61
Proposed dataset	SVM	**0.69**	0.74	**0.66**
	KNN	**0.47**	**0.58**	**0.46**
	RF	**0.65**	**0.68**	**0.64**
	CNN	**0.75**	**0.75**	**0.75**

Table 8 Hyper-parameters used in aspect extraction task

Model	Hyper-parameters
SVM	Kernel = rbf Regularization parameter = 1.0
KNN	Neighbours = 7 Weights: uniform Distance metric: minkowski
RF	Number of trees in a forest = 100 All nodes were expanded until all leaves contain less than 2 samples Weighted impurity decrease: $\frac{N_t}{N} * (\text{impurity} - \frac{N_{t_R}}{N_t} * \text{right impurity} - \frac{N_{t_L}}{N_t} * \text{left impurity})$
CNN	Conv layer kernel size: 100, activation function: relu Dense layer units: 6, activation function: softmax Optimizer: adam
Bi-LSTM	LSTM units: 64, activation function = $tanh$ Dense units: 4, activation function = softmax Optimizer: adam

5 Conclusion

Resource barriers hindered many researchers from doing works on Bengali ABSA. We designed a benchmark dataset, BAN-ABSA, which can be used in further ABSA tasks and create more research scope. After several trial and error, we found out that Bi-LSTM works better than many other neural network models. We propose the Bi-LSTM architecture to be used in Aspect-Based Sentiment Analysis. The results may not be too high compared to English ABSA. Modifications can be made in Bi-LSTM and add attention to improve the overall performance. Researches in this field have just been started in the Bengali language. Numerous improvements can be made in this field.

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Chapter 32
Determining the Inconsistency of Green Chili Price in Bangladesh Using Machine Learning Approach

Md. Mehedi Hasan, Md. Rejaul Alam, Minhajul Abedin Shafin, and Mosaddek Ali Mithu

1 Introduction

Chili is greatly beneficial for our health as it provides us with vitamins and phosphorus. It is one of the most cultivated crops in our country, covering an area of about 66,235 ha. The annual production quantity is about 52,215 metric tons [1]. Both summer and winter seasons are suitable for its cultivation. The average yield of green chili is 5–6 metric tons/ha. In rural areas, chili is commonly taken with pantavat (boiled rice in water). Green or dry chilies are the first choice for spices while preparing different types of curries. Considering its demand and market status the price varies now and then. The price of green chili on the 1st January of 2019 was 48 taka/kg but in July the price went up to 145 taka/kg. The sudden rising cost put unprivileged people of our country in much uncertainty. The indefinite nature and ambiguity of the data make the prediction complicated. Some other factors such as weather conditions, storage limitation, productivity, transportation, supply-demand ratio making the prediction even more challenging.

A part of artificial intelligence (AI), Machine learning (ML) allows the system to automatically learn and use that knowledge to improve performance without being explicitly programmed. Einav et al. [2] describes how machines can be efficiently used for prediction purposes. ML has a great impact on financial estimation. Even

Md. M. Hasan · Md. R. Alam · M. Abedin Shafin (✉) · M. A. Mithu
Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh
e-mail: minhajul15-8700@diu.edu.bd

Md. M. Hasan
e-mail: mehedi15-8421@diu.edu.bd

Md. R. Alam
e-mail: rejaul15-9077@diu.edu.bd

M. A. Mithu
e-mail: mosaddek15-8669@diu.edu.bd

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with unstructured data ML can help determine the Future price of a product. To utilize its efficiency and ability we intend to predict green chili price using Machine Learning approach on the chili price dataset. Different kinds of ML libraries like Scikit-Learn, Pandas, Numpy, Matplotlib, Tensorflow, Keras, Pytorch are available tools for ML applications. Different types of feature selection algorithms are applied to make a convenient dataset. There are several predictions giving techniques such as reinforcement learning algorithm, supervised learning algorithm, unsupervised learning algorithm in ML. The supervised algorithm is suitable for our dataset to predict chili prices because the variance of time series data depends on time/seasons and places. We used a classification technique to categorize our data into three classes (high, mid, low). We aim to predict whether the chili price will be low or mid or high at a certain time.

The paper includes the following sections; Literature review is given in Sect. 2. The proposed methodology is specified in Sect. 3. A discussion is held on Experimental result in Sect. 4 and we settle with giving a conclusion and future work in Sect. 5.

2 Literature Review

Machine learning is vastly used for solving issues that is subject to forecasting. For taking steps against the price fluctuation of the products, a lot of work has been done using ML. ML has made this approach much convenient.

Rafieisakhaei [3] worked to discover the relationship between oil price and demand–supply changes. The authors attempted to find a numerical solution by using a model that included the casual loop and mathematical formula. Oil price, oil demand, and oil supply were the main variables. Several types of loops were proposed where the key factors were global oil demand, oil price, and oil supply. They attempted this approach by using regression analysis on the price data. On the other hand, we have acquired our solution by using a Machine Learning approach.

Park et al. [4] has predicted housing prices based on Virginia housing data. The work pattern that the authors followed were assembling datasets, cleaning datasets, and implementing algorithms. They collected data from three different sources. Then they refined the data manually. The dataset consisted of categories naming physical features, public school rating, mortgage contract rate, and others. DaysOnMarket, high or low, were considered as dependent variables. For building the prediction model they used AdaBoost, Ripper, Naïve Bayes, and C4.5. The best result was acquired using Ripper algorithms.

Huang et al. [5] describes how the financial market is a potential sector for ML implementation. Prediction of financial changes is not an easy task due to its inconsistent behavior. The author built the dataset by the collected data from Yahoo and Specific exchange-rate service. They also inquired the movement of NIKKEL225 index. The dataset was split into two parts. First part was used for algorithm implementation and the second part for evaluation and comparison. To determine the accuracy of SVM they examined the performance with Linear Discriminant Analysis, Quadratic Discriminant Analysis, and Elman Backpropagation Neural Network. The best performance came from the combining model.

Similar type of work was done by Sinta et al. [6] to predict the price of rice. The authors used the ensemble K-Nearest Neighbors method on the rice price data of 14 years of Indonesia. They brought up the comparison between K-Nearest Neighbors and ensemble K-Nearest Neighbors. It was shown that KNN is less efficient than ensemble KNN because of time series data. A variation of the training dataset was used to test the accuracy of the model. When the predicted data was compared to the actual data, the results were found very close to the actual price. In contrast to this work, we used a dataset consisting of chili prices for two years. We also used multiple algorithms to get the best prediction.

Tziridis et al. [7] predicted the airfare prices in Greece in this paper. Their methodology comprises data collection, model selection, feature selection, and evaluation. Their source of data was Greek Airline Company. After processing the data, they selected eight features for every flight. They implemented GRNN, MLP, ELM, Regression tree, Random forest Regression tree, Bagging Regression tree, Regression SVM (polynomial, Linear), And Linear Regression. From the algorithms implemented, the Bagging Regression tree provided the best performance giving 87.42% accuracy.

Recently ML has become a broadly used topic in the field of Business Intelligence. For Predicting the price of the product efficiently ML is used. Leung et al. [8] demonstrated that the stock market is a growing sector for the application of ML. They collected historical data from yahoo and used Structural SVM (SSVM) with map inference using minimum graph cuts on the dataset. They labeled the data into two classes positive and negative. For companies, the prices were on the rise they labeled it as positive, and for those, the price was falling negative. After cross-validating they took the test which has accuracy higher than 78%.

By forecasting the future price of products market instability can be removed. Wellman et al. [9] predicted price by historical averaging, competitive analysis, and ML. They showed the significance of ML in the TAG market game. For surveying, the author used TAC02 final rounds. They obtained the EVPP to 47.9%, by Neglecting initial prices.

Similar type of work was done on agricultural fields to predict rice production using the SVM approach by Gandhi et al. [10]. As the population of India constantly increases, they attempt to cope up with rice production demand. In their model, they used the SVM approach, SMO algorithm, and confusion matrix. They filtered the undesired data before algorithms implementation. Followed by this, they applied SVM technique for making labeled training data. They used SMO algorithm for result measurement. After obtaining the results of the proposed model it was tested using confusion matrix.

After going through all the work mentioned above, we have determined that they have some common features and differences in contrast to our research. In context of Bangladesh, no work has been done regarding chili price fluctuation using ML. We prepared our dataset by using the Panda framework. We generated the result by using

KNN, Random Forest, Decision Tree, and Neural network algorithm. As mentioned before no ML approach has been used, so we have predicted the price fluctuation of chili using ML.

3 Methodology

The methodology includes a total of six steps which concludes our research that is shown in Fig. 1. The steps are the following:

3.1 Data Collection

Data collection is very challenging for every research. We have collected our data from the ministry of agriculture website of Bangladesh. Our dataset consists of two parts, the first part used for the training and testing purpose other one used for the prediction purpose. For training and testing part, we used 730 daily price data of the year 2018 and 2019. For prediction part 120 daily price data was used of year, 2020.

3.2 Data Analysis

After data collection, it is very important to analyze this data to identify which algorithm is perfect for this dataset. Table 1 contains a description of our attributes and the reason why we choose these attributes. For data analysis, we primarily focused on location, season, and daily price. After analysis, we understood that our data

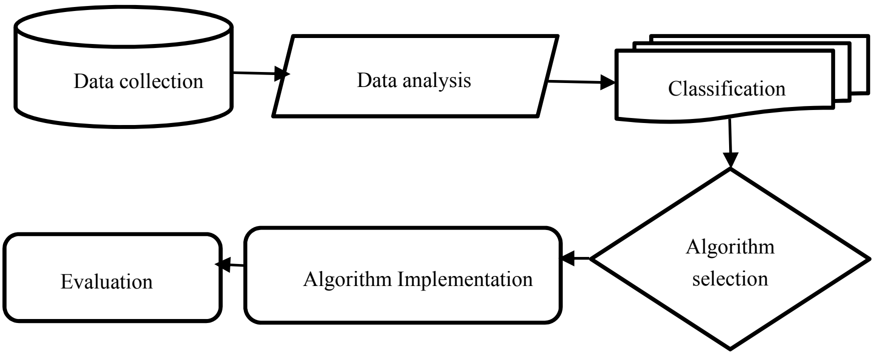

Fig. 1 Methodology diagram

Table 1 Description of attributes

Sl. No.	Attributes	Description
1	Year	In this data set, we noticed that the price of chili is high from July to August every year. So we select year as our attribute
2	Day	As our database contains time series data, we choose the day as our primary attribute
3	Month	Price depends on months. So we take month is one of our attributes
4	Season	Different season contains different weather. There are direct relation between weather and price. We analyzed that for chili price if weather is bad then price is high, and if weather is good, then the price is comparatively low or normal. So we took season as an attribute
5	Location	Green chili is produced more in North Bangle so price is lower than South Bangle where growth rate is comparatively low
6	Price	Our goal attribute is price. We tried to predict the price of chili
7	Category	Category means classification based on price range. We take Category attribute to classify the dataset

is unstructured and time-series data. We were concerned that time-series data is changeable over time. We can also get a clear view from Fig. 2.

In this graph, the X-axis represents the month and Y-axis represents the price. It is the comparison graph for the year 2018 and 2019. We can see that price fluctuation rate is very high during the month of July and August for both years.

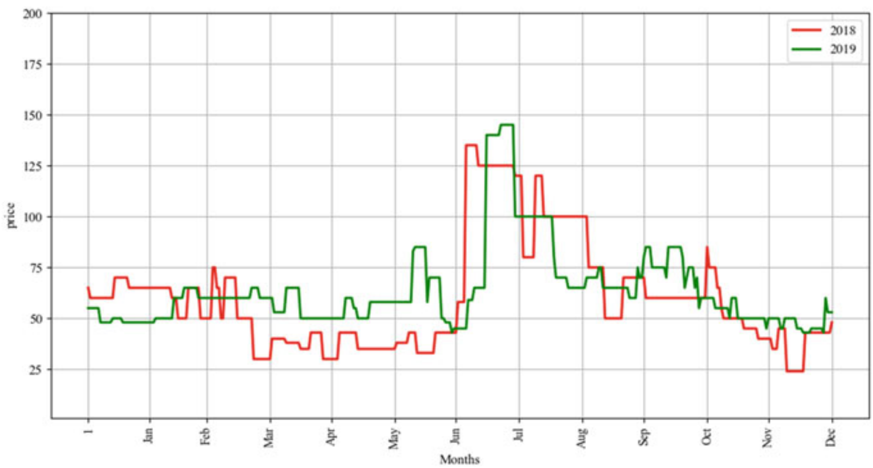

Fig. 2 Monthly price of chili

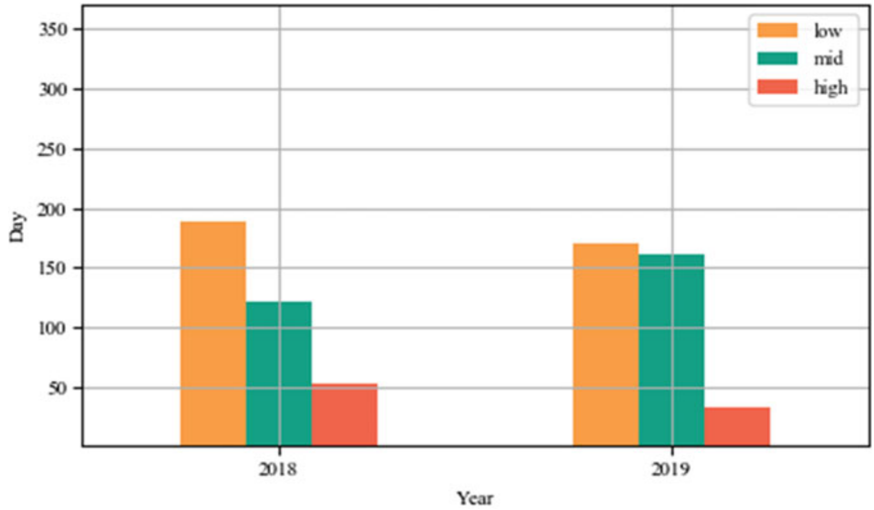

Fig. 3 Classification of prices

3.3 Classification

We divide our dataset into three classes based on chili price. We represent these classes by low, mid, and high. The price range of these three classes 0–59, 60–99, 100, and above correspondingly. Figure 3 represents the classification of our whole dataset. Where *X*-axis contains years and *Y*-axis contains days of each class. The price range during a certain amount of days are displayed in Fig. 3.

3.4 Algorithm Selection

In our work, we focused on classification algorithms because we divided our whole dataset into high, mid, and low classes. We used four different machine learning algorithms, such as KNN, Neural Network, Decision tree, and Random Forest to generate primary accuracy. By this technique, we found the best algorithm with the highest accuracy among them.

3.5 Algorithm Implementation

After applying algorithms, we found Random Forest generated the highest accuracy with an accuracy rate of 95.34% by using 30% data usage rate. Our other algorithms also performed very well. The Random Forest algorithm produced the best performance. We decided to use this algorithm to predict the chili price.

3.6 Evaluation

For our research, firstly we collected our data from a feasible source. Then we evaluated our data by applying four ML algorithms to see which algorithm produces the best performance. Figure 4 represents the highest performance of each algorithm.

Table 2 illustrate the different parameter usage in our algorithms.

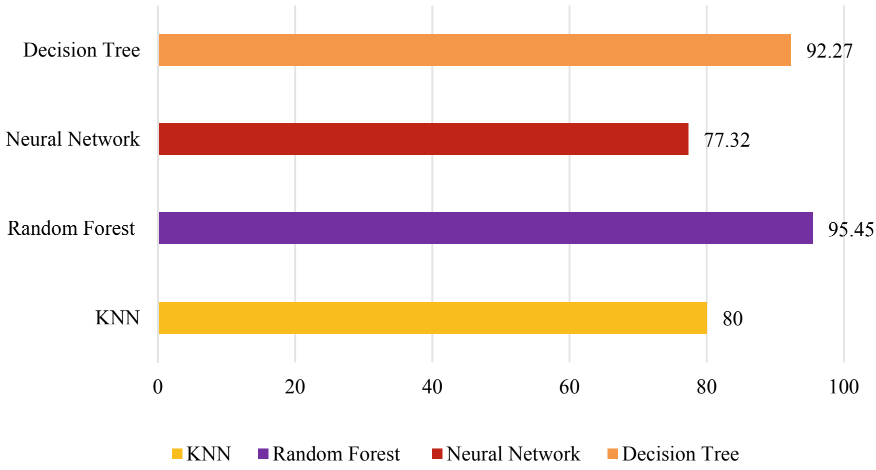

Fig. 4 Highest accuracy algorithms

Table 2 Parameter usage

Algorithms	Details
KNN	$K = 3$, $p = 2$, random_state = 0
Decision tree	Random_state = 0
NN	Max_iter = 1000, alpha = 1
Random forest	n_estimators = 100

4 Experimental Result

In the preprocessed dataset we have applied four different ML algorithms to find the accuracy of our work. As we inserted the accuracy that was generated by these algorithms into Table 3, so we can easily understand, compare them, and make the decision for future research based on their attainment. Five types of training sets are created to calculate the accuracy with the test data usage rate 30, 40, 50, 60, and 70%. We found a pleasant outcome after the analysis of these four algorithms. In our research, the Random forest and decision tree performed closely with high-level accuracy.

On the contrary, KNN and NN were found almost the same performance with a medium level accuracy. Gislason et al. [11] for classification of multisource, Random Forest maintains high dimensional data and supervises a large number of trees. From all the four algorithms, Random forest produced 95.45% accuracy with the 70% training data and 30% test data which is shown in Table 3 by red rectangular border-box. In every column, there is a yellow mark that represents the highest accuracy of those algorithms by different types of test data usage rate. Jin et al. [12] for using model classification and prediction for both induction research and data mining, the decision tree is one of the extraordinary techniques. Second, the highest accuracy generated by the Decision tree algorithm with 92.27% and used 30% test data usage rate. Yang et al. [13] for text categorization application, KNN is considered widely and it has the specialty on statistical and pattern reorganization. In our work, KNN achieved 80% accuracy by 30% test data usage rate. KangaraniFarahani et al. [14] demonstrated that NN is primarily employed for building complex prediction functions dynamically and NN is more harmonious with the de-noising data. The neural network algorithm got 77.32% by 50% test data usage rate.

Figure 5 represents the real and predicted class of first four months of year 2020. Most remarkable thing is, in these four months there were no high price class, and our model also predicted no high price.

Figure 6 illustrated that real price and predicted price of first four months of the year 2020. Red line represents real price and green line represents predicted price, where green line is closed to red line. That means our model predicts much accurately including price prediction.

Table 3 Accuracy table

Test Data usage rate	**Algorithms**			
	KNN	***Neural Network***	***Decision tree***	***Random Forest***
30%	80.00%	74.09%	92.27%	95.45%
40%	77.47%	69.28%	91.47%	94.54%
50%	75.96%	77.32%	91.80%	93.72%
60%	72.73%	73.86%	90.23%	92.73%
70%	69.79%	58.87%	86.74%	92.01%

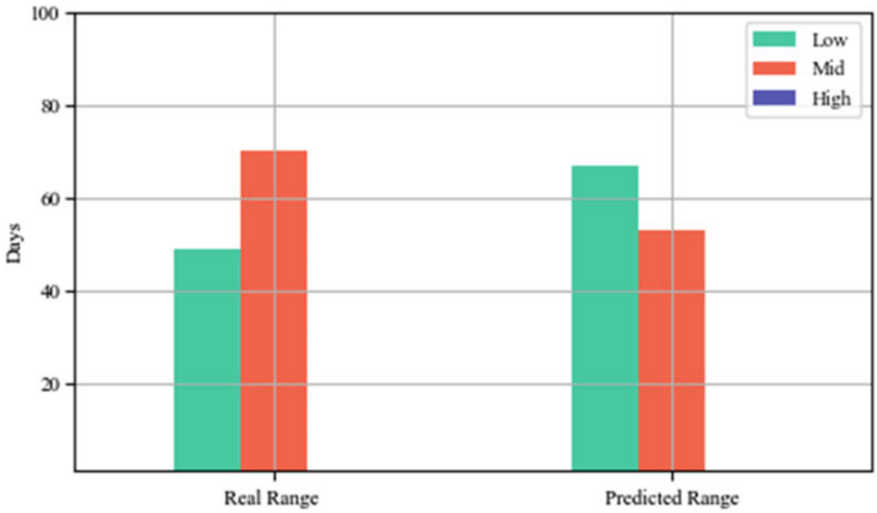

Fig. 5 Comparison between real and predicted class

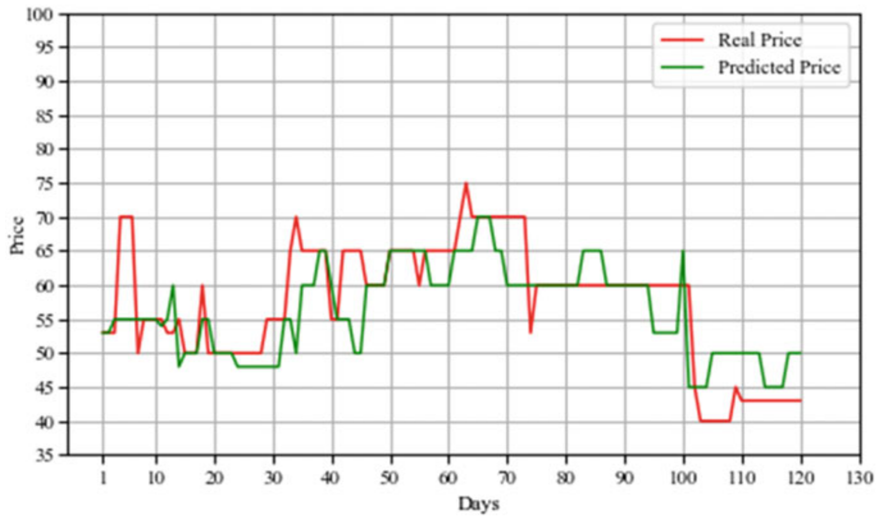

Fig. 6 Comparison between real and forecasting price

5 Conclusion and Future Work

Following the proposed model, we used four of the best preferable ML algorithms to acquire the best outcome. There were some differences in the accuracy level generated by the algorithms. The best result was provided by the Random Forest Algorithm. This algorithm is more suitable than the others for predicting the future price of green chili. By predicting the future price, it is possible to minimize the market imbalance. This prediction can aid the government in taking steps for maintaining the market equilibrium. These results can also provide the information required to attain the balance of chili production.

We attempted to achieve the best possible outcome but there are still some limitations to our work. One of the crucial limitations of our work is the limited number of data. Our collected data was limited to the capital Dhaka only. Also, we have worked with only two years of data. In the future, we hope to increase our data collection by collecting data from all over the country. We will also try to collect data covering a larger time range.

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Chapter 33
Sentiment Analysis of COVID-19 Tweets: How Does BERT Perform?

Kishwara Sadia and Sarnali Basak

1 Introduction

Right when we started feeling at ease with the prosperity and enhancement of the modern world, mankind faced the once-in-a-century pandemic, COVID-19, by spreading the novel Coronavirus worldwide. Here, we are conferring the world where microblogging made the biggest companionship in human life. People are habituated to put anything and everything going on their minds and lives on these platforms such as Twitter, Facebook and Tumblr.

So, the first massive pandemic in mankind after starting to experience social networking has its own impact on modes of communication and exchanging of feelings and information. Along with the analytics of affected people and economy, it is also important to focus on abrupt change in human mood and view.

Sentiment analysis is the domain of analyzing people's views, comments, emotions or opinion on current affairs, products, trends or even social issues in text form. Though it is irrational to think that natural text language can be understood utterly by machines, classification of sentiments is viable with the help of statistical analysis.

In this research, we have analyzed sentiments revealed by people through their microbloggings in Twitter, namely "tweets" having hashtag #coronavirus, #COVID_19, etc., after the locked down was imposed in several countries to prevent the rapid spreading of Coronavirus from March 2020 to May 2020. Later, these sentiments got identified with the overall polarity of texts as positive, negative and neutral.

To analyze these sentiments, we have used a pre-trained model called bidirectional encoder representations from transformers (BERT). BERT is the first representation model based on fine-tuning that showed remarkable performance on large sentence-

K. Sadia (✉) · S. Basak
Department of Computer Science and Engineering,
Jahangirnagar University, Savar, Dhaka, Bangladesh
e-mail: sarnali.cse@juniv.edu

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level and token-level tasks [1]. BERT uses the mechanism of transformer that learns circumstantial relations between words in a text.

The main objective of this paper is to analyze the tweets of users during COVID-19 pandemic and classify them in negative, neutral and positive classes and compare the classification results of various classifiers implemented on the dataset.

The rest of this paper is ordered as follows: Previous and related works have been discussed in Sect. 2 concisely. The entire methodology is exhibited in Sect. 3, and the results are discussed in Sect. 4. Lastly, Sect. 5 concludes the paper.

2 Related Works

Global trends and issues get their own dimensions on social media being the public platform for sharing views. So, the sentiments found in microblogging or in text form got attention for analyzing and classifying.

The uniqueness of Twitter acting as a forum for the global news and issues came to attention when a worldwide celebrity, Michael Jackson died. According to Kim et al., 279,000 tweets were posted within 1 h after the announcement of his death. Though for these findings, they did not use any machine learning approach, it explored the sentiments and emotions of Twitter users very well. They used the lexicon affective norms for English words to score and found out the expressions of sadness [2].

Rosenthal et al. described how in SemEval 2014 participants trained their system in non-sarcastic data but had to test it on different datasets consisting of sarcastic texts from Twitter. This experiment showed a significant result of performance fall from 25 to 70% and established that the system must be adjusted for sarcastic texts [3].

Besides English text from social network sites, several multilingual works have also been done on sentiment analysis. Mahtab et al. classified sentiments expressed on YouTube in Bangla comments. They classified all sentiments into three classes—positive, negative and neutral using machine learning classifiers. They evaluated the performance of those classifiers on metrics of precision, recall and F1-score [4].

Apart from these traditional approaches, we have come across several deep learning techniques for sentiment analysis. Dang et al. used word embedding to transform input data to make ready for deep learning models. They experimented with deep neural network, convolutional neural network and recurrent neural network models to analyze sentiments for microblogging data. They even combined these models and assessed the performance [5].

With the progress in natural language processing (NLP), various models and approaches got introduced over the years. Jacob et al. introduced a new pre-trained bidirectional model called "Bidirectional Encoder Representations from Transformers (BERT)". This pre-trained model can be used for NLP by only fine-tuning with one additional layer. They suggested various hyper-parameters to fine-tune the model according to the dataset [1].

Clark et al. discussed various attention mechanisms of models and applied them to BERT. They suggested how a significant amount of linguistics knowledge can be found in both hidden states and attention maps [6].

Hoang et al. proposed an ABSA model to predict the aspect considering the text by BERT. They classified the whole dataset into two labels -“related” and “unrelated”. Following the same approach, they experimented with sentiment classification analyzing text and context. They proposed a single combined model to classify both aspect and sentiment [7].

Oyebode et al. analyzed COVID-19-related comments from social media and classified them into positive and negative classes after identifying keyphrases. They came up with suggestions to handle various social and political issues following positive themes [8].

Kouzy et al. studied tweets having hashtags and keywords related to COVID-19 for differentiating misinformation and authentic information by statistical analysis [9].

3 Methodology

3.1 Dataset

The dataset for this study has been obtained from Kaggle platform available namely “Month-wise COVID-19 related tweets”. It consists of 12,344 tweets and retweets people posted from March 2020 to May 2020 on Twitter during worldwide COVID-19 with hashtags #coronavirus and #COVID_19 during worldwide COVID-19 [10].

Figure 1 shows the frequent words found in the dataset to understand how the tweets represented the COVID-19 situation.

The dataset provides six attributes, summarized in Table 1.

Numeric value to classify the sentiment has been scored from -1 to 1 where any tweet having a sentiment score more than 0 is considered as positive sentiment and less than 0 is considered as negative. Score 0 indicates the neutral sentiment.

3.2 Preprocessing

For text preprocessing, we had to use a tokenizer that would convert the raw text into tokens (numerical format) for the model. We have used the BERT tokenizer model named “bert-based-cased”.

A batch encoding derived from Python dictionary has been implemented to hold the output of the tokenizer’s encoding method. If special tokens could not be found in the vocabulary, then they were added here.

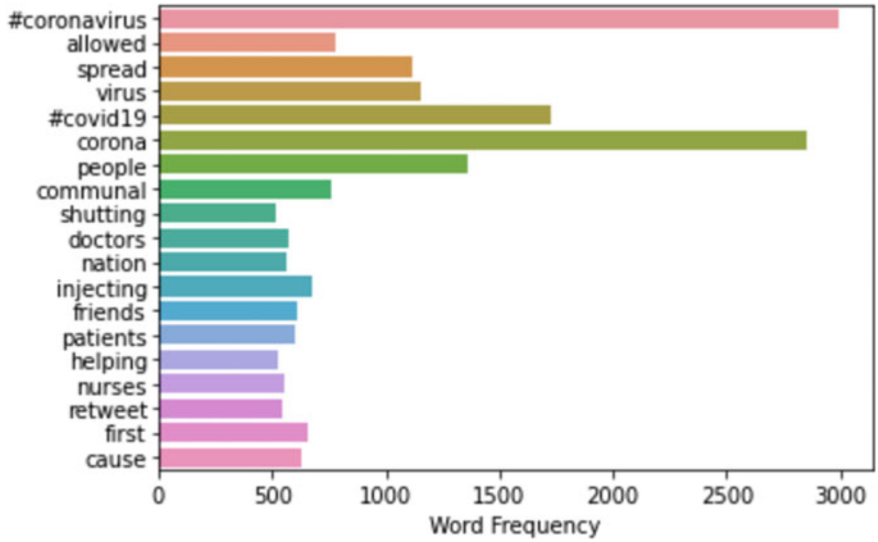

Fig. 1 Frequent words found in the tweets of dataset

Table 1 List of dataset attributes

Attribute	Description
Tweet Id	Id of the posted tweet
Text	Text of the posted tweet by user
Sentiment score (polarity)	Sentiment score from −1 to 1
Month of the tweet	Month when the tweet was posted
Class	Numeric value to classify the sentiment

In case of deep neural network implementation, for tokenizing, universal sentence encoder (USE) was used for encoding sentences. One hot encoding was implemented for categorical variables that could be converted into a form ready to provide a prediction.

We ensured that there remains no missing value or imbalance in the number of tweets based on each class and follows a balanced distribution.

The maximum length of text has been fixed to 280 characters as for each tweet, the maximum character limit is 280.

3.3 Deep Learning Classifiers

Social network has become a hub of public sentiment in text form. Only concentrating on structure and correlations of data is not enough while analyzing text solely based on emotion. Rather it showed better performance in accuracy with several deep learning approaches [11].

The aim of this paper is to determine how a deep learning technique, pre-trained BERT model, performs in sentiment analysis as a classifier. To compare with BERT in performance, we have chosen another deep learning technique—deep neural network (DNN).

In this section, some basic specifications and aspects of these classifiers in sentiment analysis have been discussed.

3.3.1 BERT

Bidirectional encoder representations from transformers (BERT) is a deep learning and pre-trained model developed by Google for natural language processing.

A block diagram of sentiment analysis by BERT is shown by Fig. 2.

BERT reads the whole input text sequence altogether unlike other directional models which do the same task from one direction such as left to right or right to left [6].

To train the language model, during pre-training, BERT uses two strategies—mask language model (MLM) and next sentence prediction (NSP). The first one works by replacing some words of input text sequence with masked tokens. The model then tries to figure out the original word considering the context of the text [13].

An example of BERT MLM got shown in Fig. 3.

Using the next sentence prediction (NSP), BERT attempts to predict if one sentence comes after another in a subsequent approach or randomly.

An example of BERT NSP got shown in Fig. 4.

To improve the performance of BERT for an individual NLP task, it can be fine-tuned with all of its parameters and resulted in faster training time [12].

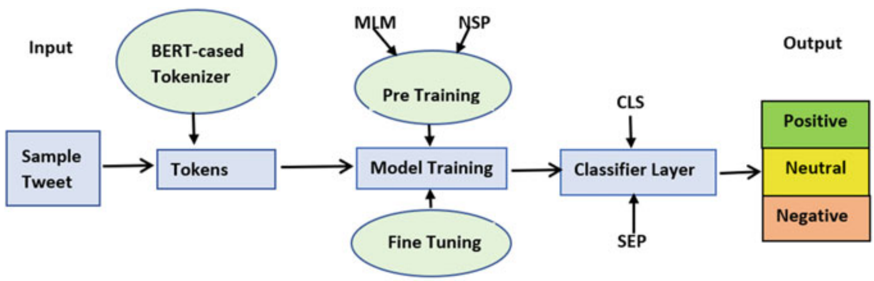

Fig. 2 Block diagram of sentiment analysis by BERT

Fig. 3 Example of BERT MLM

Input: I cannot [MASK1] at home this long. I feel [MASK2]

Labels: [MASK1] = stay; [MASK2] = low

Sentence A= I cannot stay at home this long
Sentence B= I feel low
Label= IsNextSentence

Sentence A= I cannot stay at home this long
Sentence B= She was a brilliant girl
Label= NotNextSentence

Fig. 4 Example of BERT NSP

By simply adding a classification layer to the core BERT model, it can be used in sentiment analysis. To do this, a token called [CLS] is put on the beginning portion of the first sentence. Then a token called [SEP] is placed at the end of other sentences to terminate [13].

3.3.2 Deep Neural Network

Deep neural network (DNN) is a deep learning technique where multiple layers stay between input and output layers. Here, input turns into output through hierarchical manipulation in both linear and nonlinear relationships [14].

3.4 BERT Training and Test Classification

By converting input text into tensor, DNN can provide great performance in sentiment analysis [15]. Embedding technique follows to turn the sentence into embedded vectors for that. TensorFlow Hub provides the encoders. Using Keras, a model is built, and each sentence gets to be passed through it after being encoded as a vector of 512 elements.

3.4.1 Setting Up BERT Pre-trained Model

Being a pre-trained model, BERT treats each text, in our case, that is tweet, as a sequence using BERT for sequence classification class from transformer. Following the tokenizer model, from PyTorch, bert-base-cased model is set up as the model for training the data.

3.4.2 Creating Data Loaders

Batching is required to pass data through the model for training. Data loader helps to automate the batching process. Here, for batching training dataset, we have used random sampler, and for validation dataset, it is sequential sampler. It ensures that each batch of training data comes in random order and provides variation. We experimented with various batch sizes and batch size =16 provided better performance among them for our dataset.

3.4.3 Setting Up Optimizer and Scheduler

To reduce the training and validation loss, deep learning approach comes up with various attributes such as weights and learning rate. Optimizer is used to change that attribute values.

In this paper, we have used AdamW for optimization since it is able to adapt step sizes for each individual weight. As the hyper-parameter for learning rate, we have tried out various learning rate values following BERT official documentation [1]. Among these parameters, we have chosen learning rate 5e-5, performing best on our dataset.

Scheduler is used in organizing processes on time frame. We have created a scheduler with a learning rate which decreases linearly after linear inflation during a warmup period. The number of epochs has been determined 10, so that the model does not get overfitted. The training and testing data have been divided in such a way that we can train 90% of the entire dataset and 10% remains test data.

3.5 Deep Neural Network Training and Test Classification

After preprocessing with USE, a model for sentiment analysis has been built using Keras. The model is formed with two connected hidden layers.

Learning rate, number of epochs and batch remained same as the hyper-parameters for BERT. For regularization, dropout was taken. Activation function ReLU was used as the rectified linear unit.

4 Result

We have trained our dataset following two upgraded approaches—pre-trained BERT model and TensorFlow deep neural network (DNN).

We have tested 1,235 tweets from Twitter where people posted with hashtags #coronavirus and #COVID_19 during lockdown. Among all these tweets, 436 tweets are originally "Positive" tweets, 259 tweets are originally "Negative" tweets, and 540 tweets are originally "Neutral" in class having the highest number of tweets here. Tables 2 and 3 represent the confusion matrices of the test results by means of BERT and DNN, respectively.

This experiment led to the classification report of how the model performed classifying the sentiments as positive, negative and neutral.

Table 2 Confusion matrix of the test result by means of BERT

	Predicted negative	Predicted neutral	Predicted positive
Actual negative	223	24	12
Actual neutral	8	525	7
Actual positive	12	17	407

Table 3 Confusion matrix of the test result by means of DNN

	Predicted negative	Predicted neutral	Predicted positive
Actual negative	207	24	28
Actual neutral	35	468	37
Actual positive	29	43	364

Table 4 Classification report of BERT and DNN

Sentiment	Precision		Recall		F1-score	
	BERT	DNN	BERT	DNN	BERT	DNN
Negative	0.92	0.76	0.86	0.80	0.88	0.78
Neutral	0.93	0.87	0.97	0.87	0.95	0.87
Positive	0.95	0.85	0.93	0.83	0.94	0.84
Weighted average	0.93	0.84	0.92	0.84	0.93	0.84

BERT model showed 92.22% accuracy on our dataset, whereas DNN showed accuracy of 84.12%. Here, we have shown the precision, recall, F1-score and support result of BERT model and DNN model on our dataset in Table 4 .

Here, having a comparatively lower recall score for "Negative" class indicates that BERT and DNN had missed out to predict some of positive labeled tweets in the test set. Similarly, the higher recall value for "Neutral" class shows that it did well to cover all the neutral tweets to label in that class.

Analyzing the F1-score, we have noticed that BERT showed a great job balancing precision and recall score for the dataset in neutral sentiment.

Here, among the 1235 tweets as test data, BERT could correctly predict 407 positive tweets out of 436. It labeled 223 negative tweets correctly out of 259 and 525 neutral tweets out of 540 tweets correctly.

On the other hand, for TensorFlow DNN, we have found that there occurred mislabeling in some cases from precision value. DNN could correctly predict 364 positive tweets out of 436. It correctly labeled 207 negative tweets out of 259 and 468 neutral tweets out of 540 tweets. Similarly, according to recall value, we can say that DNN could not classify negative text well compared to BERT having the same hyper-parameters. It clearly shows BERT model outperformed the DNN approach way ahead.

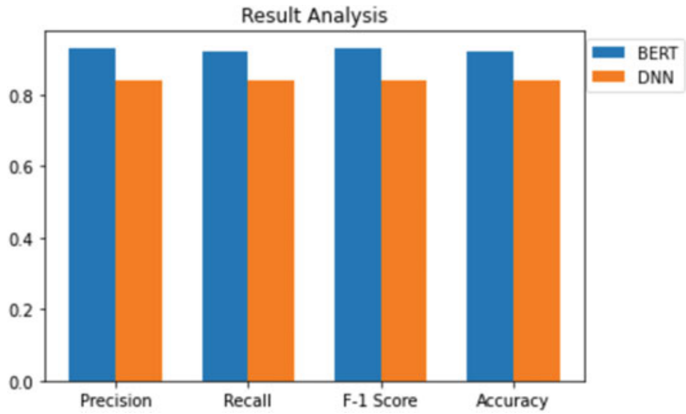

Fig. 5 Result comparison between BERT and DNN

Again in our further research, DNN improved its performance after increasing the training data from 90 to 95%, and by increasing the number of epochs from 10 to 40, it improved its accuracy to 85.43%. So, it is noticeable that with sufficient number of training data and less learning rate, DNN improves its performance.

The graphical representation of the evaluation parameters using different classifiers is shown in Fig. 5.

5 Conclusion and Future Work

Analyzing the precision and recall value in this paper, we recommend that BERT can understand people's emotions more comprehensively. The bidirectional technique of BERT made a great impact on finding the sentiment behind the tweets rather than other unidirectional approaches. Due to ambiguous expression of tweets in the dataset, BERT and DNN could not predict the sentiment in some cases.

Future work can discuss about economic, sociopolitical, educational and political issues from different viewpoints from our findings and can classify better to minimize the knowledge gap.

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Chapter 34
An Intelligent Bangla Conversational Agent: TUNI

Md. Tareq Rahman Joy, Md. Nasib Shahriar Akash, and K. M. Azharul Hasan

1 Introduction

A conversational agent [2, 7] is the way to interact with a machine using natural language like English, Bangla, etc. ELIZA was the first natural language conversation program described by Joseph Weizenbaum in January 1966 [12]. From then, several natural language conversation programs are implemented in the English language which may be in different contexts.

But for Bangla language, there are not any remarkable analysis and research for a man and machine communication system. Again, after one of the popular languages—English, Bangla is the fifth most popular language [13], which is the state language of Bangladesh. So, we have proposed a corpus-based method [14] which establishes a natural language communication system between man and machine in Bangla language.

The machine we have implemented, TUNI, imitates and responses like a psychologist who can communicate with a user through a comfortable terminal-like interface. Though the machine can mimic like a mock psychologist, the editing capabilities of TUNI 'corpus' give the ability to respond in various context.

Bangla language has more complexities than any other language because the pattern of Bangla language is not fixed like other languages as an example—English [1]. Suffixes vary depending on the grammatical person. Even there is a different form of second-person which also causes suffix change. So, the simple approach cannot be enough for Bangla conversational agent.

We have tried with a new approach, pattern matching with Bangla POS tagger [6] and very basic sentiment analysis. With this approach, TUNI can produce natural and syntactically correct replies overcoming those complexities.

Md. T. Rahman Joy (✉) · Md. N. Shahriar Akash · K. M. A. Hasan
Department of Computer Science and Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

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By this method, a corpus is made to match with the input text and respond according to the context. There are many rules available in the corpus that are used to match and extract the information from the input, called decomposition rules. A preprocessing is necessary for making the input text suitable for the decomposition rules.

The preprocessing step normalizes the input text as well as adds POS tags to normalized text. For each decomposition rule, there are some rules which use the parsed information and make ready for the response, called reassembly rules. Using this method, our conversational agent performs well compared to other knowledge-based methods in Bangla. As it is handwritten rules, the responses are relevant to the context and accurate.

Further, the paper is organized into sections like this: Sect. 2 describes the related work in this field, Sect. 3 describes the additional tasks that are needed for our proposed method, Sect. 4 describes the proposed method for our system, Sect. 5 illustrates the performance of our proposed method, Sect. 6 concludes all the section of the paper.

2 Related Works

ELIZA was a domain-based system that uses some semantic rules that simply match the keyword from an extensive collection of rules. And then decompose the input string to remove the semantic conflicts. At last, it navigates through several context-based replies to find a suitable reply to the corresponding input. It was developed using MAD-Slip language that provides a vast collection of data structure that helps in string manipulation. It was based on a SCRIPT, a collection of rules to find the focus of an input given by the user and analyze the semantic conflicts. The editing ability of the SCRIPT gives the system capability of changing the context.

Artificial linguistic Internet computer entity (ALICE) was invented by Richard Wallace in 1995, is a knowledge-based natural language processing system that can communicate with a human by processing the user input using some reasoning heuristically pattern matching rules [11]. It was build based on XML knowledge bases [4]. It matches with some predefined response using a conversational agent that is developed based on artificial intelligence.

Disha [9] is a Bangla health care chatbot. User can provide information about disease symptmops, and Disha can detect the possible disease. It can also store information of user and use them where necessary. It is implemented using several machine learning alogirthms like decision tree [10], SVM [3], Random forest [5], etc.

DOLY [8] is a man and machine communication system that was implemented to give replies to a user query in a human manner for the education system in the Bengali language. It is an AI-based human–computer interaction system where machine learning algorithms are used with the help of Bengali natural language processing (BNLP). It responds to the answer to a question given by a user using the collection

Table 1 Example of POS tag dataset

Word	POS	Description
বড়	adj	Adjective
আমার	p.1	Pronoun — first person
তোমার	p.2	Pronoun — second person
বলতে	v.0	Verb — non finite
বলছে	v.3	Verb — third person
পানি	noun	Noun
না	no	Determinator — No
আজকে	ign	Ignore

of knowledge and finds the desired output. There is a train function adapter that trains the DOLY using the knowledge of replies.

3 Prerequisite Works

For our proposed method, there is a need for some additional concepts. They are described below.

3.1 Bangla Pos Tagging

Though POS tagging is the heart of our proposed method, only a few POS tags are required to detect the pattern of a sentence. Generally, nouns, pronouns, verbs and adjectives are enough for detecting the pattern of a sentence. Additionally, the different form of pronoun and verb is also detected for getting the proper sense of the grammatical person. For adding POS tag to each word of a sentence, a dataset is used. The dataset contains the direct mapping of each word with its POS tag. The POS tag dataset is demonstrated in Table 1.

3.2 Suffix Processing

In Bangla, suffix plays an important role in detecting the person of the word as well as suffix tag is also used to modifying a word properly to find out the stem or root. For adding the suffix tags, there is also a dataset. In the dataset, for each same type suffix, there is a corresponding tag. These tags are helpful to detect the input sentence

Table 2 Example of suffix tag

Suffix Tag	Suffix(es)	Example word
ke	কে, রা, টা, টি, য়, গো, তেই	আকাশটা
er	ের, র, টার	করিমের
e	েই, ই	হাসাই
eto	িত	আলোচিত
null	-	দান

type and to parse the main information. The suffix tag is added with the POS tag. Some suffix tag is described in Table 2. For simplicity, we treated the pronoun as a noun, and it can also have a suffix tag.

3.3 Person Mapping

As we discussed earlier, in Bangla, the suffix of verb changes depending on the grammatical person. As an example, the first person word 'যাচ্ছি' will be changed to a third-person word 'যাচ্ছেন'. Again, the pronoun also changes depending on the speaker, and there is a different subcategory for each person. The word 'যাচ্ছেন' can be in the form of 'যাচ্ছো' or 'যাচ্ছিস'. So, a mechanism is required to handle this type of complex scenario. We introduced a static mapping mechanism that applies to the most type of words. A dataset is made which stores the static mapping of the suffix or person change. For pronouns, direct word mapping like 'আমাকে' to 'তোমাকে' is added there. Some mappings are given in Table 3.

Table 3 Example of person mapping

Suffix/Word		Example	
From	To	From	To
িনি	েননি	শুনিনি	শুনেননি
াম	েন	জানতাম	জানতেন
ো	েন	বলবো	বলবেন
-	েন	চল	চলেন
আমাকে	তোমাকে	-	-

Table 4 Example of PosNeg tag

PosNeg Tag	Description	Example words
p	Positive	ভালো, বেশি, সুখ, বেশি
n	Negative	খারাপ, মৃত্যু, দুঃখ, কম
x	Neutral	আমি, তুমি, ঘুম

3.4 PosNeg Mapping

TUNI can reply more naturally with expressions using some interjections like in Bangla, 'বাহ!', 'সুখবর!' etc. To make a proper expression, the meaning of the input sentence must be evaluated. For our method, we proposed a basic sentiment analyzer that can detect the type of the sentence so that she can make expression accordingly. For detecting if the sentence contains whether positive meaning or negative meaning or neutral meaning is done by checking the presence of some words. The words are stored in the PosNeg dataset. So, for each word in the dataset, there is a corresponding tag that suggests if the word has a positive or a negative meaning. A snip of the dataset is shown in Table 4.

3.5 Sentence Deviation Processing

Breaking down a complex or a compound sentence is necessary for simplifying the information parsing operation. A sentence can be broken down based on some words or punctuations and its position in the sentence. The words or punctuations are stored in the Sentence-Divider dataset. Some contents of the dataset are ।, comma, !, তাহলে,তবুও,যে,যখন,কারন etc.

4 Corpus Structure

A corpus is nothing but a large and structured set of texts. Our corpus contains mainly four pieces of information. They are keyword, the precedence of the keyword, decomposition rules and reassembly rules. A keyword can be a collection of Bangla words or parts-of-speech tags. The words are divided by slash. Each keyword contains a precedence value, which is an integer number that determines the importance of the keyword in a sentence. The precedence values are well observed and statically given by us. Each keyword contains a list of decomposition rules. The decomposition rule is made of some criteria, which can be a POS tag, or a direct Bangla word or both. Multiple POS tags and direct words are separated by a slash. For each decomposition

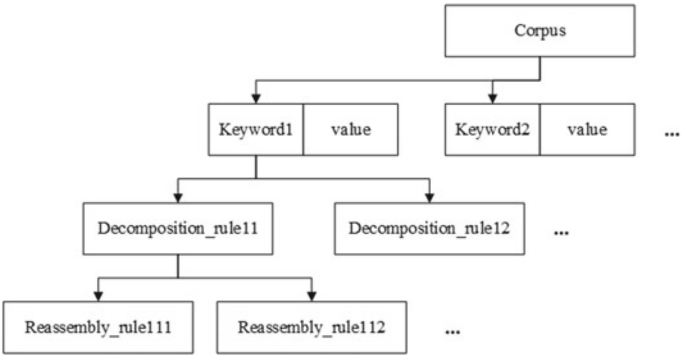


Fig. 1 Corpus structure

rule, there are several reassembly rules. The reassembly rule contains some numerical notation, which specifies the index of the parsed information (Fig. 1).

5 Methodology

After getting the input text, the text is preprocessed, which makes the input text ready for the next step, finding focus. In the finding focus step, the best word which represents the meaning of the sentence is chosen. Then, in the matching step, the preprocessed sentence is checked with all the decomposition rules within the focus. The next step is parsing, where the main information is parsed from the input with the help of the decomposition rule. After that, the reassembly step begins where the parsed information is used to make a relevant reply. The flowchart is shown in Fig. 2.

5.1 Preprocessing

This is the first step of the proposed method. Two basic operations are performed in this step which help to modify the input sentence such that it can be used in the next steps. So, other steps completely depend on this step. For the input sentence, "আমি এখন পড়ালেখা করছি কারন আমি সিজিপিএ বাড়াতে চাই" we can see the step-by-step processing which will help us to illustrate the methodology easily. The two operations are described below:

Removing unnecessary spaces: If the input string contains unnecessary spaces, then all of them are replaced with a single space. If the unnecessary spaces are at the beginning or at the end of the sentence, then they are just removed. Space is added before every punctuation or symbol of the input text.

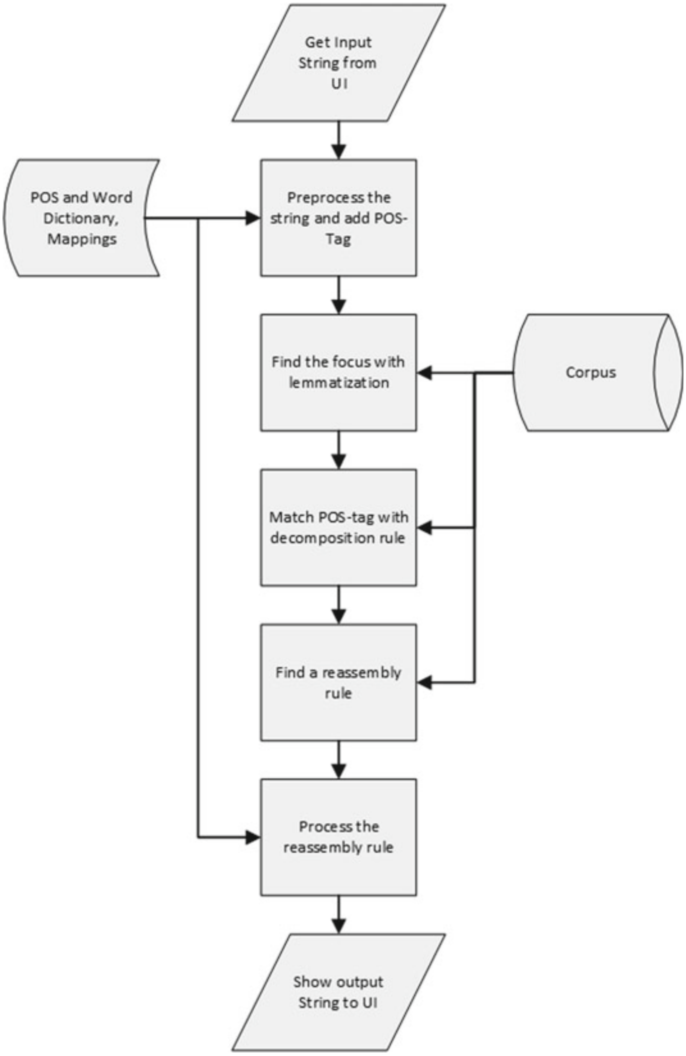


Fig. 2 Flowchart of TUNI

Tokenization: The tokenizer for our context need not to be very advanced. In Bangla, the tokenization process is simple because only space is enough to find out the tokens from the input sentence. The tokens are saved into a list named token-list. For the above example sentence, the token-list is illustrated in Fig. 3.

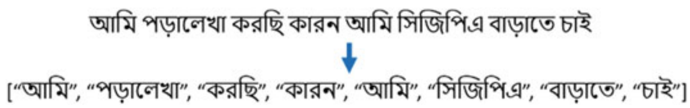


Fig. 3 Tokenization example

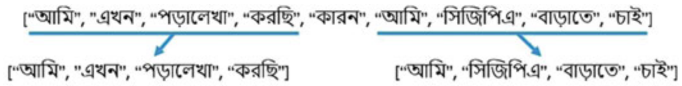


Fig. 4 Sentence breaking into sub-sentence

5.2 Breaking into Sub-sentence

A sentence may be complex or compound, but there can be only one focus in our context. So, breaking down this type of sentence is necessary to find out the focus and the main sub-sentence for meaning parsing. A complex or compound sentence can be broken down based on the presence of some words or punctuation, which are described in the Sentence-Divider dataset. After tokenization, the token-list is divided into sublists depending on the sub-sentence. For our above example, the input sentence is a compound sentence, so it needs to be broken down. The sentence contains the word “কারন”, which is in the Sentence-Divider dataset, and the input sentence will be broken down based on it. After breaking down, the token-list is subdivided into two or more lists. For the previous example, it is illustrated in Fig. 4.

5.3 Adding Tags

For each of the token of the token-list is added with its corresponding POS tag using the POS tagging dataset which is described in Table 1. After adding the POS tags, suffix tag is added with ‘noun’, ‘verb’ and ‘pronoun’ POS tags. Suffix tags are described in Table 2. For our example, the addition of POS and suffix tag is illustrated in Fig. 5.

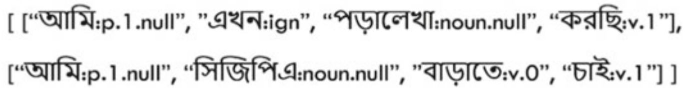


Fig. 5 POS and suffix tag illustration

5.4 Removing Unnecessary Words

Unnecessary words that do not play an important role in the meaning of a sentence can be ignored for the next steps. These type words are annotated with ‘ign’ POS tag in the previous POS tagging step. So, tokens with ‘ign’ POS tag are removed from the token-list. For our example, the token ‘এখন’ is with the ‘ign’ POS tag, and it can be removed from the token-list.

5.5 Finding Focus

Focus is the primary token, which plays a vital role in the meaning of a sentence. In our corpus, there is a precedence value for each keyword. The higher is the value, the more importance in a sentence. This value is set by observing and analyzing a large number of sentences by us. Each token in the generated token-list is given the precedence value from corpus. The exact token may not be found in the corpus. If any token is not found in the exact search, then the stem of the token is searched in the corpus. If the stem is not found, then the POS tag of the token is searched in the corpus. At last, if no POS tag is found corresponding to the word, then precedence value ‘−1’ is set. The search tree and the value assignment are shown in Fig. 6. When values are assigned to all tokens, then the token with the highest precedence value is selected as focus, and the sub-sentence containing the focus is selected as a new token-list. For our previous example, the token ‘চাই’ has the highest precedence value, 4. The selection is illustrated in Fig. 7.

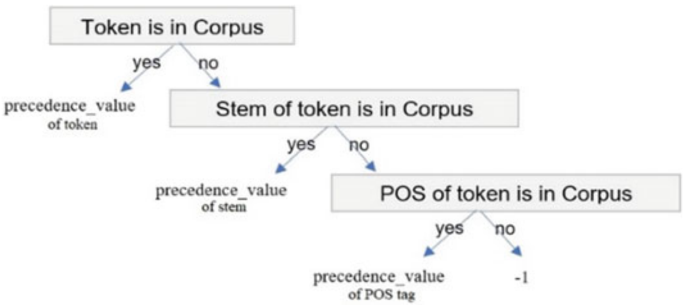


Fig. 6 Keyword search tree

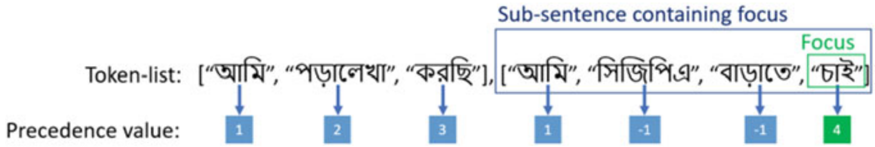


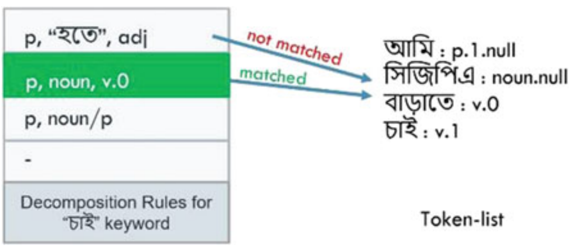
Fig. 7 Focus and sub-sentence selection illustration

5.6 *Matching*

For each keyword in the corpus, there are some decomposition rules, and we can see in Fig. 1. When a focus is selected, all of the decomposition rules are checked with the token-list until one is matched. So, the rule which is matched first is selected for parsing. The decomposition rule is a list of some conditions which must be met with the token-list. Conditions are the combination of POS tags or direct word or PosNeg tag. As an example, if in the decomposition rule, there is a condition ‘v.0’, then to be matched with this rule, the token-list must have a token with ‘v.0’ POS tag. The token-list may have more POS tags than the matched decomposition rule, which are ignored. It is noted that the decomposition rule does not contain any condition regarding the focus. Continuing with the previous example, the matching step is illustrated in Fig. 8. Here, the first rule does not match because the token-list has not any word ‘হতে’. The rest of the rules do match, but the second rule is selected as it is matched first. The last rule‘−’ means the null condition and can be matched with any token-list.

In this step, a simple sentiment of the token-list is evaluated if there is any condition of sentiment in decomposition rule. The sentiment tags or PosNeg tags are shown in Table 4. Every token of the token-list, which is in the PosNeg dataset, is selected for processing. Then, the token with positive tag is set with value ‘+1’, and the negative token is set with ‘−1’. After multiplying them, if the result is ‘+1’, then the input string is considered a positive sentence. Otherwise, it is a negative sentence. When no condition regarding sentiment is in the decomposition rule or no token of the token-list in the PosNeg dataset, then the sentence is not evaluated and set as neutral. For our previous example, no PosNeg tag is in the decomposition rule; hence, no sentiment evaluation is performed.

Fig. 8 Matching illustration



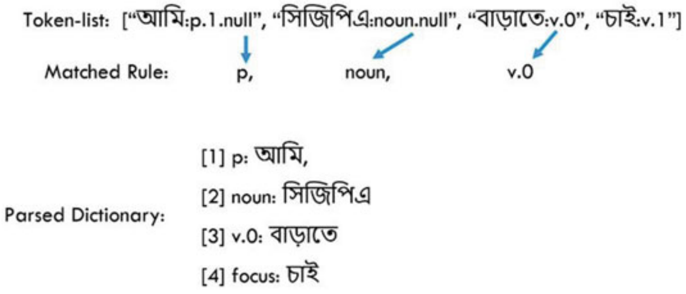


Fig. 9 Parsing illustration

5.7 Parsing

When a decomposition rule is selected, it is selected for parsing the token-list. A dictionary is used to store the parsed tokens corresponding to the conditions of the decomposition rule. So, the key of the dictionary is condition, and the value is token. If there is more than one token for a condition, a list is used as the value. In this case, the order of the tokens in the list follows the order of the token according to the POS tags in the decomposition rule. For the same previous example, the parsing step is illustrated in Fig. 9.

5.8 Reassembly Rule Processing

With the parsed dictionary, a reassembly rule integrates the parsed token to produce meaningful replies within the context. For each decomposition rule, there are several reassembly rules in the corpus shown in Fig. 1. This step is subdivided into three steps:

Rule Selection: Reassembly rule is selected circularly so that TUNI responds more naturally with the same input. A dictionary is maintained which stores the index of reassembly that just has been used. For every response, the index is incremented by the selected decomposition rule.

Placing Token: Reassembly rule contains some numeric notation. They are the index of the token in the parsed dictionary. The notations are replaced by the corresponding tokens. There is a 'foc' keyword which means the matched focus, and it is replaced by focus word. For our example, it is illustrated in Fig. 10, where both '1' and '2' are the numeric notation and replaced by "আমি" and "সিজিপিএ", respectively, from the parsed dictionary.

Evaluating Functions: Evaluation functions can modify parsed information as needed. They are described in Table 5. The 'revPer' and 'rev_all' functions use the person mapping dataset described in Table 3. When all of the functions are evaluated, then the string is ready for output. For our example, function evaluation is illustrated

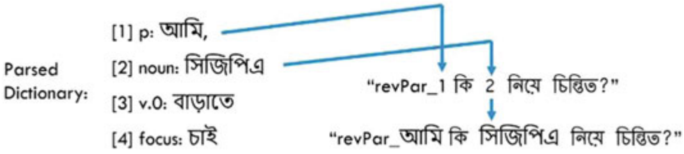
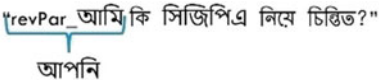


Fig. 10 Placing token illustration

Table 5 Reassembly functions

Function	Description
revPar	Reverses the person form of specified token or word
addSuff[]	Adds or replaces suffix as required
rev_all	Reverses all the person form in a sentence
goto	Selecting replies from another similar keyword

Fig. 11 Function evaluation illustration



in Fig. 11. In this example, only 'revPer' function is available for the reassembly rule which modifies the ' 'আমি'' token into ' 'আপনি' '.

6 Experimental Results

Since there are only a very few conversational agents available in Bengali, we will compare our system with English conversational agent, ELIZA. ELIZA is also a corpus-based system like TUNI. The comparison is shown in Table 6. From the table, we can see for the first case, ELIZA replied with an inaccurate sentence where TUNI replied with an accurate and more natural sentence. For the rest of the cases, we can easily differentiate the responses from ELIZA to TUNI and say that TUNI performs well.

7 Conclusion

In this paper, a new method for Bangla sentence processing and meaning parsing is proposed. With this method, the main complexities in Bangla language like various person forms, various sentence structures, various suffixes, etc., are overcome. Corpus and various datasets have been used for the method. Corpus holds some keywords and rules, and datasets hold tags or mappings. And, the datasets hold

Table 6 Comparison between ELIZA and TUNI

ELIZA	TUNI
YOU: I can't sleep from few days ELIZA: Perhaps you can sleep from few days	আপনিঃ আমি কিছুদিন থেকে ঘুমাতে পারছি না TUNI: কেন আপনি ঘুমাতে পারছেন না?
YOU: an accident has happened with me ELIZA: What does that suggest to you?	আপনিঃ আমার সাথে একটি দুর্ঘটনা ঘটেছে TUNI: কি দুর্ঘটনা ঘটেছে আমি কি জানতে পারি?
YOU: my cousin died in front of me ELIZA: Your cousin died in front of you?	আপনিঃ আমার চাচাতো ভাই আমার সামনে মারা গেছে TUNI: আমি দুঃখিত, আপনার চাচাতো ভাই কি আপনার খুব কাছের ছিল?
YOU: I want to be happy but depression never ends ELIZA: What if you never got to be happy?	আপনিঃ আমি ভাল থাকতে চাই কিন্তু বিষণ্ণতা কাটছে না TUNI: কি হবে যদি ভাল থাকতে না পারেন?
YOU: he wants to buy a car ELIZA: I'm not sure I understand you fully	আপনিঃ সে গাড়ি কিনতে চায় TUNI: কি হবে যদি সে গাড়ি কিনতে না পারে?
YOU: I will kill him ELIZA: That's quite interesting	আপনিঃ আমি তাকে হত্যা করব TUNI: আপনি কাকে হত্যা করবেন?

the dynamic mappings and keywords. An interactive user interface is also implemented for easy communication with TUNI. By using it, a user can type and talk with TUNI. The accuracy of TUNI depends on the POS tag, suffix tag, PosNeg tag and reassembly rules. The more accurate the tags and rules, the more correct and natural replies are given by TUNI.

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Chapter 35
Blockchain-Based Digital Record-Keeping in Land Administration System

Shovon Niverd Pereira, Noshin Tasnim, Rabius Sunny Rizon, and Muhammad Nazrul Islam

1 Introduction

The land administration is one of the most significant branches of any country. The land administration system (LAS) keeps record of real (land) properties, mainly about the location, (historical) ownership, value, and use. The LAS also keeps data of physical, spatial, and topographic characteristics of real property. Thus, the records generated by the LAS are crucial from the social, economic, and political perspective [1]. The LAS records should have long-term availability, transparency, compliance with proper law [1]. Without proper management, the system could be vulnerable to corruption, financial instability, and inaccurate. Again, the LAS exists both in manual form which is paper-based, and in digital form that uses the database management system. Like any other digital systems, cyber-attacks and information leakage are the key threats for digital land management system as well. Thus, in case of digital land management systems, it is utmost importance to maintain the highest level of security against cyber-attack, data tampering, and information leakage.

Data found in LAS regarding physical and legal aspects are often considered to be correct and accurate. The LAS data could get tampered due to data transfer, human error, and in some cases by data abuse [2]. Data transparency, verification, and mutation processes are generally handled by some central authority in LAS [3]. Due to this centralization, the rate of abuse and negligence might increase.

A blockchain is a time-stamped series of immutable record of data that is managed by cluster of computers not owned by any single entity. Blockchain technology is introduced in a paper published in 2008 by person or persons under the alias of Satoshi Nakamoto titled "Bitcoin: A peer-to-peer electronic cash system" [4, 5]. The white paper introduces Bitcoin as trustless electronic cash system which is not con-

S. Niverd Pereira (✉) · N. Tasnim · R. Sunny Rizon · M. Nazrul Islam
Department of Computer Science and Engineering, Military Institute of Science and Technology, Mirpur Cantonment, Dhaka 1216, Bangladesh
e-mail: nazrul@cse.mist.ac.bd

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trolled by a third party. Blockchain technology, a peer-to-peer network system that includes stakeholders as the nodes and represents the data as distributed ledger form. Blockchains are shared, distributed and fault-tolerant database that every participant in the network can share, but no entity can control. Blockchain offers security, interoperability and enhanced archival facility [4, 5]. According to a recent study [6], blockchain can be defined as "*A concept map of Blockchain which represents that a Blockchain consists of blocks containing messages, proof of work, and reference of previous block and stored in shared database, which is able to perform transactions over P2P network maintaining irreversible historical records and transparency*".

Nowadays, Blockchain technology has attracted tremendous interest from a wide range of stakeholders, which include finance, healthcare, utilities, real estate, and government agencies [7–9]. For example, a number of research has been carried out focusing the adoption of blockchain technology in health information in order to obtain enhanced security, interoperability and user trust [10, 11]. While a very few studies show integration of blockchain technology in LAS [1, 12]. However, These studies showed that blockchain stops double spending of transferable entity which is the first and foremost need of LAS. Moreover, the fundamental principles of blockchain and its use in different areas related to record management, introduce several open issues that need to be explored in case of land administration. The key activities of LAS like the process of mutation, access to information, proper archival facility are yet to be examined using the blockchain technologies.

Thus, the objective of this paper is to propose a blockchain-based framework for developing a secured, trusted, and efficient land administration system. To attain this objective, a framework is proposed for LAS considering the relevant properties of blockchain technology and key functionalities of LAS. A prototypical system is also developed based on the proposed framework and then evaluated its performance through simulation.

The organization of the remaining sections are as follows. Section 2 briefly discusses the related work and the summary knowledge derived from the study. The proposed blockchain-based framework for LAS is presented in Sect. 3. Section 4 discusses the simulation procedure and results followed by highlighting the benefits of the proposed framework. ection 5 shows the comparative analysis of our work with existing works and also the summarized research result. Finally, the concluding remarks with future work are discussed in Sect. 6.

2 Related Works

A number of studies have been conducted on IT-based land management system where the database systems and web-based technologies were the key concern to make it automated. For example, Choudhury et al. [13] proposed a web-based land management system for Bangladesh which could scan the paper-based land maps. The scanned images were then converted images into scalable vector graphics format to store in a database. In another work, Khan et al. [14] proposed an automated digital archival system for land registration and keeping records. This system incorporated a central database system and the GPS-based data collection system for digital archive and retrieval for smoothing the land purchase and sale process. Similarly, Talukder et al. [15] introduced digital land management system which included GPS-based land surveying.

A few studies introduced the Blockchain technology in land adminstration systems. For example, in [12] a theoretical representation of a land management system based on Litecoin's public blockchain codebase was discussed that will facilitate to enhance the scalability, privacy, and interoperability. This framework was proposed considering only one type of transaction, i.e., transfer-of-value from one user to another. The proposed model was a permissioned blockchain where only a set of preapproved miners could append land records to the blockchain. They built an idea of private sidechains along with public mainchains to store transaction details. However, this article only reported a conceptual framework and no simulation was carried out. Stevanovi et al. [1] proposed a blockchain-based Land Administration System to increase its transparency, immutability, security, and accountability. They highlighted the main properties of blockchain including transparency, immutability, decentralization and smart contracts; and mapped them into the digital record-keeping system of land management. However, this study could not clarify how the genesis block would be created. Also, the possible majority attack in Proof of Work algorithm was not solved.

Apart from this, Blockchain technology is introduced also in some other fields for record management that includes: patients record management in E-health system, product delivery management, E-voting system, and Banking system. For example, Hanifatunnisa et al. [16] proposed a blockchain-based E-voting System to reduce database manipulation and maintains data integrity. In this article they have introduced, a database recording system on e-voting using permissioned blockchain technology was introduced. The recording of voting results using blockchain algorithm from every place of election ensured the security and transparency. In a study conducted by Ichikawa [10] proposed a framework to develop and evaluate a tamper-resistant m-Health system using blockchain technology which enabled trusted and auditable computation using a decentralized network. Peterson [11] proposed an idea of implementing the blockchain in health information sharing. In this work, a new consensus algorithm was introduced to facilitate data interoperability. The solution was theoretically described with some pseudocode of mining.

However, background study shows the importance of blockchain technology in record management as well as land administration system. In sum, grounding on the literature survey a few concerns have been come out. *Firstly,* a number of studies have been conducted using blockchain for record management system in different fields like health care, E-voting, Banking system. *Secondly,* a limited number of studies have been conducted focusing on digitizing the land management systems; while these studies are primarily focused to minimizing the workloads, save time, and also to enhance security to some extent. *Thirdly,* a very few studies have been conducted focusing to LAS using blockchain. The existing works of land administration system using blockchain have a number of limitations, that includes, for example, most of the proposed model/framework are proposed conceptually and did not conducted any validation study; and all the required properties of land management systems were not considered to propose the framework as well as implementing the Blockchain-based LAS.

3 Proposed Framework

To design a practical system, the proposed system must resemble closely to the already existing LAS. The framework is proposed considering three concerns. Firstly, the key functionalities of the LAS that includes updating land-related information, checking land information, putting in a mutation request, and ownership transfer process. Secondly, the stakeholders of LAS, i.e., the land owner, buyers, authority (representative to manage the LAS) and the government. Thirdly, the properties of the blockchain technology.

3.1 Components of the Framework

The two main components of the proposed framework are Block and Node. Blocks create blockchain and the nodes create the blockchain network. The blocks are the data structures that hold information about any land transaction and changes. In the proposed framework the blocks include,

(a) Previous hash- creates link to previous block;
(b) Seller ID- user ID of the seller of the land of this particular transaction;
(c) Purchaser ID- user id of the purchaser of the same land;
(d) Land ID- the ID of the land in question;
(e) Transaction money- price of the land according to market value and
(f) Hash- unique ID for a particular block.

Again, the nodes are the endpoints of the networks. The stakeholders participating in the network act as the nodes. Nodes can be categorized in two classes: (i) regular nodes, which can only put up mutation requests and cannot participate in the verification process; and (ii) authentication nodes, which are able to authenticate a block, that means it can mine a block which makes them miner nodes. Miners validate new transactions and record them on the global ledger.

3.2 Private and Public Key

Any kind of transaction has to maintain its authenticity. When a user/node initiates a transaction, a block is created with required information. That block is encrypted using the private key of that associated user which can be said as digital signature. Now, if user wants to share information of the block with other user then a public key is generated using the private key which allows intended user to access block's info. The public key expires after sometime to ensure safety.

Smart contract is the program where the behavior of the blockchain is written. Once deployed, the smart contract is immutable and runs unchanged in the network forever. It basically decides the steps required after mining one block. It can directly control the transfer of digital currencies or assets between parties under certain conditions.

3.3 LAS Activities Include in the Framework

The proposed framework is presented in Fig. 1. The key tasks that required for LAS is showed in the data flow diagram in Fig. 2. It depicts the working procedure of the proposed system.

Updating Land Related Information—General information which includes measurement, surrounding, infrastructure, and price related information might need to be changed from time to time. It is very important that alteration is verified thoroughly. To ensure valid and accurate alteration it will be verified by miner nodes using consensus algorithm. Consensus algorithm [17] makes sure that miners are only able to

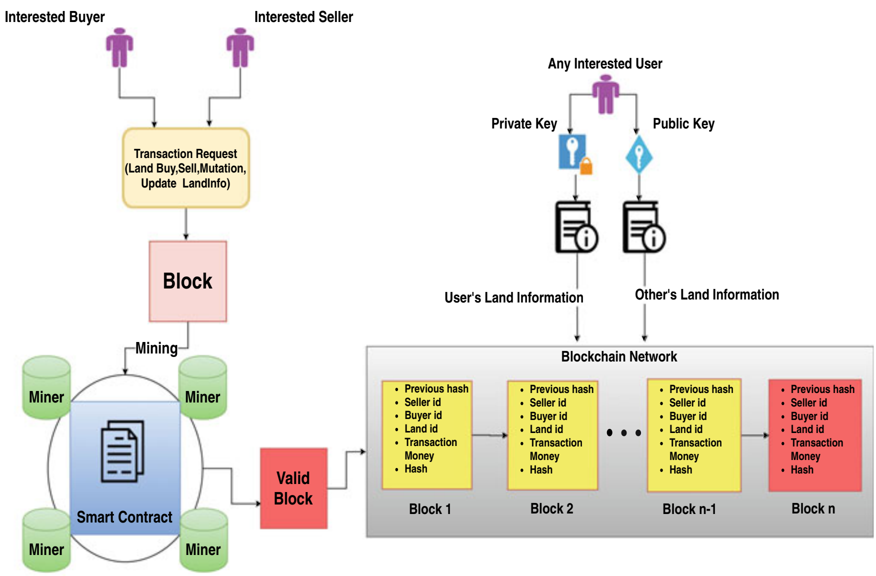

Fig. 1 Proposed framework for blockchain-based LAS

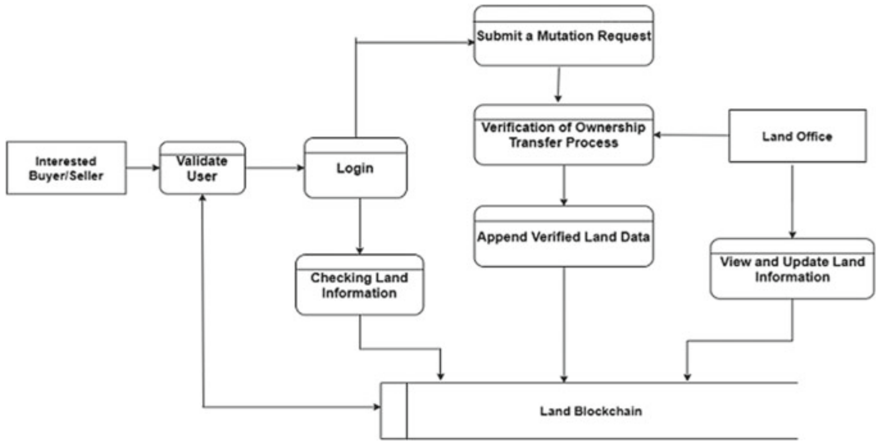

Fig. 2 Data flow diagram of the proposed system

validate a new block of transactions and add it to the blockchain. The changes are verified against the buyer, seller and also the authorities who changed the information has permission or not, the new information is consistent with existing information and local laws accept the change.

Checking Information—Before purchasing a land any buyer would like to check the land related information. In the proposed system, such information will be encrypted by private key. The private key is only available to the owner of the land. Therefore, to give limited access to an interested buyer a public key can be generated using the private key by the owner which will be provided to the buyer. This public key will allow the holder to only have view access to information which will be specified by the land owner.

Submitting the Mutation Request—The mutation request refers to the initiation of ownership transfer process. In the regular system, buyer and seller create a statement mentioning the information required to transfer ownership of a land and present that to the local land administration authority. The authority approves the transfer after extensive verification of all provided information[18]. In the proposed framework, both parties will file a digital request using their respective private key. After that, a block will be created including all the information regarding the land transfer process. Then, the block will be broadcast to the miner nodes for mining.

Ownership Transfer Process—In this process, using consensus algorithm miner nodes verifies the block created containing mutation information and it is checked that whether the has value of the created block is consistent with the existing blockchain. If the hash value is found consistent, the block will be broadcast in the network. Finally, it will be added to the distributed ledger system and the central blockchain.

4 Simulation of the Proposed Framework

The section discusses about the environment settings, coding, and deploying the smart contract.

4.1 Simulation Environment

In order to simulate the proposed framework in local environment the following steps are followed:

1. *Setting up personal blockchain*: A local personal blockchain Ganache was used to simulate the proposed framework. Ganache by default creates ten accounts and work space to simulate the blockchain.
2. *Setting up development environment*: Truffle development environment creates the local environment and the file system for blockchain project. Truffle is the default development environment tool for Ethereum blockhain platform. To setup truffle in Linux operating system the following command is used: bash *npm install -g truffle@5.0.2*
3. *Opening digital wallet*: Digital wallet allows the accounts in the personal blockchain to connect with the browser and finally to the blockchain network. Metamask is a compatible digital wallet for Ethereum.
4. *Creating project and initiating truffle*: The next step is to create the project directory and initiating truffle environment in the directory. Firstly a directory was created under the project name and then in that directory the following command was initiated to set up the file system: bash *truffle init*
5. *Adding project to ganache*: After the project is created it has to be added to Ganache in order to visualize the current state of the blockchain.

4.2 Smart Contract Deployment and Functions

The first step in creating a smart contract is declaring the contract itself. Solidity programming language, which is specialized for blockchain development, treats the smart contract similar to *class* as in object-oriented programming language. After creating the contract, vital operations of the framework such as, updating information, checking information, submitting mutation request, and ownership transfer mentioned in Sect. 3.2 can be included as the member of the contract.

The member methods and block structure of the smart contract are as follows:

i. *Block*: The blocks are basically a type data structure. In solidity it is declared as *struct*. They are created to hold the information of a transaction.

ii. *Posting a free land for sell*: Initially after survey process if any land has no real owner it can be posted for sale by the survey office or any government agency. The sale function is named *createland* and its procedure is presented as pseudo-code in Algorithm 1. The function receives land location and land price and as parameters and after validation of the parameters the record is created and added to the record data structure.

Algorithm 1 Add New Record of Land

function CREATELAND($\_landLocation, \_landPrice$)
  Require $landLocation.length > 0$
  Require $\_landPrice > 0$
  $landCount = landCount + 1$
  $lands[landCount] \leftarrow Add\ New\ Record\ of\ Land$endtate emit $LandCreated$

iii. *Posting a land for sale by its owner*: If the owner of a land wants to sale the land it can be posted. For this a *landforsell* function is called. This function makes the land block available for potential buyers to create the buying block and put up for mining. The pseudo-code of this is presented in algorithm 2. The function takes land id of the of land for sale. The purchase flag is set to false to indicate it is up for sale.

Algorithm 2 Change Land Status For Sell

function LANDFORSELL($landID$)
  Require $Land\ ID\ is\ valid$
  $purchase\ flag \leftarrow false$
  emit $LandforSell$

iv. *Purchase land and mutation*: The purchase land function is the function that actually performs the mutation operation. The mutation process is referred to as purchase land here. Once any buyer puts in a mutation request *purchaseLand* function gets called. The mutation block created by the function is uploaded via metamask for mining in Ethereum. The process of mutation is shown as a pseudocode in Algorithm 3. The method requires *land ID* as parameter. Upon validating the Land ID all the relevant information is fetched from the database and a block is created. That block is emitted to Etheruem for mining and if found valid the mutation is completed and the block is added to the blockchain.

4.3 Ethereum Cost and Gas Fee

Deploying smart contract in Ethereum and triggering any event creates a block. To mine that block and add to the chain upon verification needs some computing power.

Algorithm 3 Purchase Land

function PURCHASELAND(*landID*)
  Require *Land ID is valid*
  CurrentLand ← *Fetch Land from given Land ID*
  Owner ← *Get the Land owner*
  Require *Account Balance land Price*
  Require *Land is not Purchased*
  Require *Seller and Buyer are not same*
  CurrentLand.Owner ← *Change the Owner*
  Transfer the fund
   purchase flag ← *true*
   emitLandPurchased

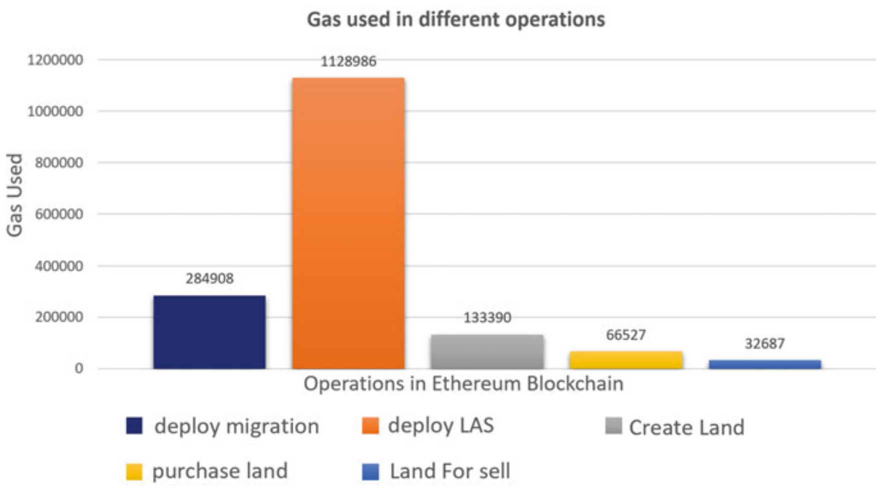

Fig. 3 Gas used for each functions

In Ethereum platform the computing power is measured as *gas fee*. Also, some amount of digital currency is required which is *Ether*.

Computing power for five core operations of our framework has been calculated. They are: deploy migration function, deploy LAS smart contract, create land function, purchase land function, land for sell function.

The histogram in Fig. 3 shows the gas use for each function. The color codes for each function is described in the figure.

Figure 3 showed that deploying the contract consumes the maximum amount of computing power. Compared to that the other functions need considerably lesser amount of computing power. The next histogram in Fig. 4 represents ether costs of the same functions.

As observed in Fig. 3, deploying the contract consumes most power, as a result this is the most costly function. The other functions are less costly and among them adding a new land information costs the most.

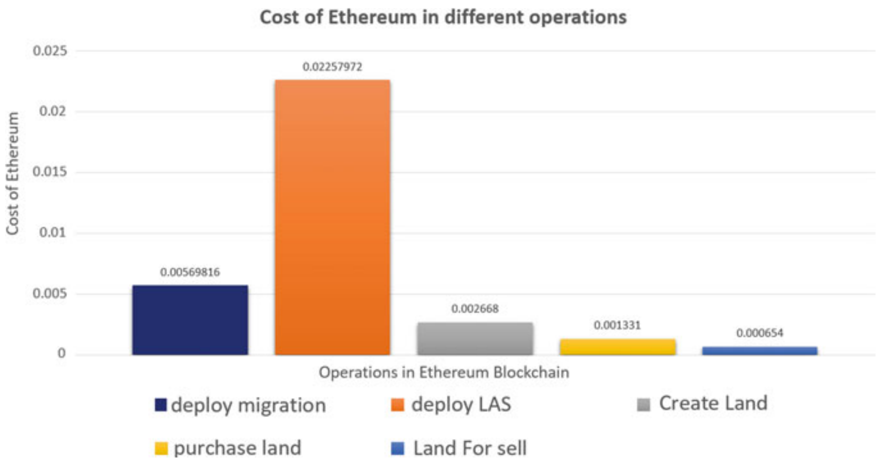

Fig. 4 Ether used for each functions

Fig. 5 Execution for each functions

4.4 *Execution Time*

The graph in Fig. 5 shows the execution time of land for sell, land create, land purchase functions for several test cases in milliseconds. It is evident from the test versus time graph that purchasing land takes the most amount of time and it spans around 900 ms. The second most execution time is taken by creating land which spans around 300–400 ms. Then the lowest execution time is taken by land for sell function.

5 Discussion

In this research, based on the proposed framework a local web application was created using Ganache to host the local blockchain. The computational power and execution time required to execute each function of the proposed LAS is also discussed in prior section. The simulation results showed that the blockchain-based proposed LAS system would bring the following benefits.

First, the proposed framework offers extensive security features to LAS. Consensus algorithm used in the proposed framework ensures that no invalid block is added to the blockchain; and any tampering to the chain will occur a series of change operation which alerts the system immediately. Moreover, since the proposed framework generates private key for each end-user, it also ensures that a user can access only his/her information.

Second, the proposed framework is simulated in an Ethereum blockchain platform which uses proof of work consensus algorithm that ensures consistency among the participating nodes [19]. Thus, any change of information is verified and then updated in every local copy of the blockchain using the proof of work. As a result, the data of the proposed system remains consistent and accurate.

Third, Blockchain also serves as an efficient archival system [20] and enhanced trustworthiness, data integrity and supports multi-factor verification [21], which are the foremost need of land administration system as well. Again, the proposed framework stores change log in the form of blocks in the chain that also ensures the trustworthiness of the system.

Fourth, the computational power needed to mine each block is significantly lower than uploading the smart contract. As the framework allows selected few nodes to be the miner block, thus less computational power is required for mining. The deployment of the contract needs a large amount of computational power, however, this occurs only once in the systems lifetime. For example, as shown in Fig. 3, the *createLand*, *purchase land(mutation)*, *land for sell* showed less computational power.

Finally, some operations like posting a (land) sale notice and creating a new land block showed an average execution time, that is, less than 400 ms. Again, a comparatively higher execution time is showed for the *land purchase(mutation)* (see Fig. 5), it is around one second (1000 ms) that still can be treated as relatively faster. These features can increase the acceptance of the system to end users. Thus, the proposed framework offers great scalability and user reach as well.

6 Conclusion

In this paper, a blockchain-based framework is proposed for land administration system. The functionality includes (a) updating land related information, (b) checking land information, (c) submitting the mutation request, and (d) transferring process of land ownership. The proposed framework is evaluated through simulation. The simulation results showed that a secure, reliable, effective, and efficient land administration system could be developed based on the proposed framework. Though, few existing work focused on the blockchain-based land administration system but these studies did not consider all possible functionalities of LAS as well as did not evaluated through the simulation system, for example, Patil et al. [12] introduced a concept of *landcoin* that includes only the ownership transfer process. Considering all the aspects it can be said that the proposed framework brings uniqueness and contribution to design and develop effective land administration system. However, the performance of the system in real-time transaction is yet to be tested. Again though the memory management of this distributed network is a prime concern, but this research did not explore the memory management. Our future works will focus on these directions.

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Chapter 36
Parkinson's Disease Detection from Voice and Speech Data Using Machine Learning

Anik Pramanik and Amlan Sarker

1 Introduction

Parkinson's disease (PD) is a progressive neurological disorder that exhibits several motor and non-motor dysfunctions [11]. It is one of the most prominent neurological disorder, preceded only by Alzheimer's disease [1]. PD is caused by decay of dopaminergic neurons and Lewy bodies' presence in mid-brain [3]. People with the age of 60 or more are more vulnerable to PD [23]. PD exhibits diverse symptoms including postural instability, tremor, rigidity, dysarthria, hypomimia, dysphagia, shuffling gait, decreased arm swing, micrographia, cognitive impairment, and several other motor and non-motor disability [5, 7]. PD is a growing health care problem. Statistical results show that the number of PD patients is expected to double from 6.2 million cases as of 2015 to 12.9 million cases by 2040 [8]. However, detection of PD is a challenging task as there is no specific diagnostic test for a patient with PD. Patients must be diagnosed according to the clinical symptoms and criteria [20].

Among different procedures for diagnosis of Parkinson's Disease, PD detection based on phonation and speech data has been proven to be very effective as 90% of patients with Parkinson's have shown voice impairments [13]. As symptoms of Parkinson's Disease include dysarthria, dysphagia which causes vocal-muscle disorder, swallowing difficulties, inability in controlling salivary secretion, turbidity, and reduced degree of facial manifestation [5, 7]. These types of dysfunction causes speech disorder, variations in voice intensity, pitch range, articulation rate. So, vocal and acoustic analysis of audio samples is used as a non-invasive procedure for diagnosis of PD by researchers [2, 13, 19]. Furthermore, phonation and speech data analysis is practised in early diagnosis of PD by many researchers [1, 4].

In this literature, a model is proposed to detect Parkinson's Disease using voice and speech signal data in an efficient and robust manner. The features in the data-set

A. Pramanik (✉) · A. Sarker
Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

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include fundamental frequencies, harmonicity variants, time-frequency attributes, wavelet and vocal-based features, and many other speech signal data, all totalling no less than 750 attributes of 252 persons. The higher level of dimensions causes exponential increase in feature space and increases the risk of overfitting the classifier model [24]. Variance Inflation Factor (VIF) was extracted from features to diagnose Multicollinearity in data-set and remove highly correlated attributes [22]. Data was standardized to equalize the magnitude and variability of the attributes [12]. Dimension reduction techniques, such as Principal Component Analysis (PCA), Independent Component Analysis (ICA), were used to reduce the number of features [10, 16]. During classification procedure, k-fold cross-validation was performed to maintain the class distribution proportion as close as original data-set [17]. Different Machine Learning classifiers, such as Support Vector Machine (SVM), Logistic Regression (LR), k-Nearest Neighbour (k-NN), AdaBoost (AdB), Random Forrest (RF) were used in the experiment [6, 9, 14, 15]. Grid search was used for hyper-parameter tuning and optimize the classifiers' performance [21]. This paper contains detailed discussion about how variation of classifiers, their parameters and feature representations affect the performance of the model and thus producing a more accurate, robust, and efficient model. The main contributions of this research can be summarized as follows:

1. A model is proposed to detect Parkinson's Disease using voice and speech signal data.
2. Introduction of intensive data pre-processing to locate underlying and repeating patterns in the features.
3. Usage of different dimension reduction techniques to create a more relevant feature-space.
4. Comparison of five different ML-based model to select the best model for Parkinson's Disease detection.
5. Achievement of highest accuracy in comparison to other research works on same data-set.

Section 2 contains discussion about similar works in this topic. Section 3 gives description of the data-set. Section 4 demonstrates the proposed framework. Details about the used data processing techniques, the ML classifiers and validation techniques are described here. Section 5 discusses the experiment result. The conclusion is drawn in Sect. 6.

2 Literature Review

During recent years, several papers have been published regarding detection of Parkinson's Disease using speech and voice data. An author proposed a model where Neural Networks, Decision Tree, Regression, and DMneural were used for detection of PD and comparative analysis was made [4]. A paper used (LOSO) validation technique on MFCC voice recording samples and SVM classifier to discern between PD patients and healthy people [2]. Another paper describes a two-staged attribute selection and classification model for detection of PD [13]. Ali proposed an early predictive model for PD detection using two-dimensional simultaneous sample and feature selection method [1]. Another author used ensemble learning techniques combining different classifiers for PD detection with accuracy of 86% [19]. There are also some other papers having proposed models with very high accuracy on detection of PD. For instance, a paper combined weighted clustering with Complex Valued Artificial Neural Network (CVANN) in their proposed model which achieved an accuracy of 99.5%. However, despite having high accuracy these experiments show biased results. The reported data-sets used in the experiments had small data points and each subject had multiple voice recording samples [18]. Despite having several papers published on PD detection, improvement is still necessary to provide models with better accuracy, robustness, and efficiency.

3 Dataset

The data-set, on which the proposed model was tested, was publicly available as PD Speech data-set from Department of Neurology in Cerrahpaşa, Faculty of Medicine, Istanbul University. The data-set contained data from 188 PD patients (81 women and 107 men). Their ages ranged between 33–87 and their average age was 65.1. The control group of the data-set contains data from 64 healthy subjects (41 women and 23 men) and their average age being 61.1. As shown in Table 1, the data-set contains baseline features, time-frequency features, Mel Frequency Co-efficient features, and many other speech signal data regarding the subjects. The PD Speech data-set consists of a total 753 speech signal attributes of 252 people.

Table 1 Description of speech signal attribute sets, measurement and number of attributes

Attribute set	Measurements	Number of features
Baseline features	Shimmer, Jitter, FFP, RPDE, DFA, PPE	21
Time frequency features	Bandwidth, intensity, formant	11
Wavelet based features	F_0 related Wavelet attributes	182
Vocal fold features	GQ, GNE, VFER, EMD	22
Mel-frequency cepstral constants (MFCCs)	MFCCs	84
Tunable Q-factor wavelet features	TQWT	432

4 Proposed Framework

In this paper, a model is proposed that efficiently classifies Parkinson's Disease based on PD Speech data-set. The visual illustration of the proposed model is displayed in Fig. 1. Different data-processing techniques, machine learning classifiers, validation approaches were used in the proposed model. The description of those classifiers, methods, their algorithms, parameters are defined below.

4.1 Data Pre-processing

The PD Speech data-set has in total 753 attributes of 252 subjects. That leads to huge feature space for a comparatively smaller number of data-points. So, data pre-processing lies at the heart of high-performing classifier models. The description of all the used methods are given below.

Data Standardization Standardization is a technique that prevents attributes with large values having more precedence over attributes with smaller values. Z-score is a standardization technique that transforms attribute values to standard scores such that the mean and the variance of the attributes become 0 and 1, respectively. Given n dimensional raw data-set $X = [x_{11}, x_{12}, \ldots, x_{mn}]$ with m data-points, Z-score standardization formula can be depicted as:

$$x_{ij} = Z(x_{ij}) = \frac{x_{ij} - \bar{x}_j}{\sigma_j}. \tag{1}$$

In the equation, $\bar{o}_j$ and σ_j denote the sample mean and standard deviation regarding the jth attribute, accordingly.

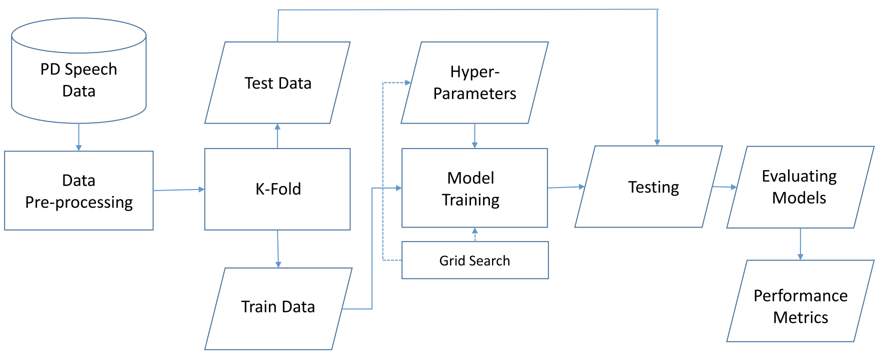

Fig. 1 Block diagram of proposed Parkinson's disease detection model

Multicollinearity Diagnosis Multicollinearity refers to significant correlation and interdependence among attribute values in the data-set. Multicollinearity causes larger confidence intervals, increases error of estimates in the regression model, and leads to inconsistent outcomes of the classifier algorithm. Variance Inflation Factor (VIF) measurement was used to diagnose multicollinearity of each independent variable. VIF indicates how well a feature can be predicted using the rest of the features. VIF of jth attribute can be computed as:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}, \tag{2}$$

where R_j^2 represents the multiple correlation coefficient.

Principal Component Analysis (PCA) PCA is dimension reduction technique such that given an n dimensional standardized data-set Z, PCA outputs a reduced k-dimensional data-set Y.

Steps of the algorithm are as follows:

1. Compute the co-variance matrix of Z as: $\mathrm{COV}(Z) = Z^T Z$.

2. Compute the eigen decomposition of $\mathrm{COV}(Z)$ as PDP^{-1}, where P denotes the eigenvectors matrix and D denotes the diagonal matrix with eigenvalues in the diagonal.

3. Compute the sorted k-dimensional eigenvectors matrix i.e. the projection matrix P^* based on k largest eigenvalues.

4. Use projection matrix P^* to calculate the output i.e. new k-dimensional space as: $Y = P^* Z$.

Independent Component Analysis (ICA) ICA is a method for differentiating a multivariate signal into its fundamental components. Given an n-dimensional data X, ICA outputs a reduced k-dimensional data Y. Steps of the algorithm are as follows:

1. First, whiten the given X by using formula: $X = ED^{-1/2}E^T X$, where, E denotes orthogonal matrix of eigenvectors of covariance of X and D denotes diagonal matrix of eigenvalues.
2. Initially, choose a weight vector randomly as W.
3. Calculate, $W = Xg(w^T X) - g'(w^T X)W$, where g is a non-quadratic derivative function.
4. Then, compute $W = orthogonal(W)$.
5. If converged, terminate the algorithm, else, go to step 3.

4.2 Cross-Fold Validation

The PD Speech data-set was split into 5 folds or groups. One group was used as test data; the rest of the $k - 1$ groups were used as training data. Then, the classifier model was evaluated and scores were recorded. The process was repeated k times with a different fold as the test data each time. The final score of the classier were calculated by combining all scores, as shown equation below:

$$M = \frac{1}{K} \times \sum_{n=1}^{K} C_n \pm \sqrt{\frac{\sum_{n=1}^{K}(C_n - \bar{C})^2}{K-1}}, \tag{3}$$

Here M defines the final performance measurement regarding that classifier and C_n denotes the performance measurement for nth ($1 \le n \le K$) fold.

4.3 Support Vector Machine (SVM)

SVM is a supervised learning technique that classifies data-points using hyperplane. Given, n data-points $(\bar{x}_1, y_1), \ldots, (\bar{x}_n, y_n)$, we define a hyperplane as a set of points satisfying $\bar{x}$ as $\bar{w}\bar{x} - b = 0$. Objective is to find the maximum-margin hyperplane such that:

$$\begin{aligned} \bar{w}\bar{x}_i - b = 0 \ge 1, \quad \text{if} \quad y_i = +1 \\ \bar{w}\bar{x}_i - b = 0 \le 1, \quad \text{if} \quad y_i = -1. \end{aligned} \tag{4}$$

Different kernels (such as rbf, polynomial, sigmoid) are used to separate data points in higher dimensional planes.

4.4 Logistic Regression (LR)

Logistic regression computes the probability of existing data-points within a certain class. Given a data-set with n feature set $(x_1, ...x_n)$, binary target class Y, assume $p = P(Y = 1)$. Then, the linear relationship between log-odds, ℓ and features can be described as:

$$\ell = log_b \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n, \tag{5}$$

where the logarithm base is denoted as b and β_i denotes the parameters of the algorithm. Odds can be calculated by using the exponent of log-odds. For an observation, probability that $Y = 1$ can be calculated as:

$$p = \frac{1}{1 + b^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}}. \tag{6}$$

We can optimize the model by changing regularization parameters and we can also set different solver algorithms (such as lbfgs, sag) to be used in optimization.

4.5 k-Nearest Neighbour (k-NN)

k-NN is a lazy learning and non-parametric algorithm. The algorithm uses a local network, which generates highly adaptive models. Given n-dimensional input data X and target class data Y the algorithm outputs the probability $P \epsilon [0, 1]$ of test data-point, x, where $\sum_i^y P_i = 1$, where y denotes different target classes. Steps of the algorithm are as follows:

1. For the test data-point that is to be classified, compute its distances from each training sample. For euclidean distance, it can be calculated as: $D = \sum_{i=1}^n |X_i - x_i|^2$, where, x denotes the sample that needs to be classified and X denotes the training data point.
2. Store the distances in a set and choose k closest data-points based on the distance set.
3. Estimate the class of test data based on the majority class of k closest Neighbours.

4.6 Random Forrest (RF)

Random forest is an ensemble learning classification method that operates by constructing multiple instances of decision trees and combining the results of those trees to estimate output class. It is based on Bootstrap aggregating (Bagging) technique. Given a n-dimensional training set X with target class Y steps of RF algorithm can be described as follows:

1. Select k features among the n features such that $k = \sqrt{n}$.
2. Select the best node or split-point among k features.
3. Using best-split, split the node into doughter nodes.
4. Repeat steps (1–3) until node size in minimum.
5. Steps (1–4) is decision tree creation. To build the forest, repeat steps (1–4) n times.
6. For test data, calculate the target class prediction for the data of n decision tree. Assign the majority class to the test data.

4.7 AdaBoost (AB)

AdaBoost is a machine learning algorithm in which outputs of a series of sequential learning algorithms are combined to calculate the final output of the classifier. Given n dimensional training set X and target class $Y \epsilon [-1, +1]$, AdaBoost algorithm can be described as follows:

1. Set $D_l(i) = \frac{1}{m}$, where $i = [1, \ldots, m]$.

2. Using distribution D_t where $t = [1, \ldots, T]$, train weak learners. Assume, a boosted classifier as

$$F_T(x) = \sum_{t=1}^{T} f_t(x), \tag{7}$$

where f_t denotes a weak classifier that outputs the class of object x.

3. At each iteration, assign coefficient α_t to the selected weak learner such that training error E_t is minimized. E_t can be described as:

$$E_t = \sum_{i} E[F_{t-1}(x_i) + \alpha_t h(x_i)], \tag{8}$$

where $F_{t-1}(x_i)$ is the boosted classifier that was constructed on the previous training stage. $h(x_i)$ denotes the weak learner output hypothesis for samples in the training set.

5 Experiment Result and Discussion

This section discusses different extensive experiments of the proposed Parkinson's Disease detection model, the performances of ML classifiers and how different data pre-processing methods affected the result of the classifiers.

5.1 Experiment and Result of Data Preprocessing and Transformation

The raw PD Speech data-set contained 753 attributes for 252 subjects. The features of the data-set showed multicollinearity among themselves. To view correlation between features, covariance matrix of vocal ford features is shown in Fig. 3. So, to address the high dimensionality of the data different data processing methods (such as scaling, correlation diagnosis, PCA, ICA) were introduced. Feature reduction techniques, such as PCA, ICA further increased the performance of the classifiers. PCA increased classifiers performance more than ICA for most classifiers, except RF. But, ICA required less number of components than PCA to reach its peak potential. For both algorithms, we experimented over different numbers of components to discover the best fit for different classifiers. Figure 2 visually illustrates the number of features needed to acquire the total variance of the dataset for different classifiers. It can be seen that about (30–60) is enough to get the highest performance for all classifiers. We also combined PCA with correlation diagnosis on data-set which yielded better performances for classifiers such as SVM, LR, AdB.

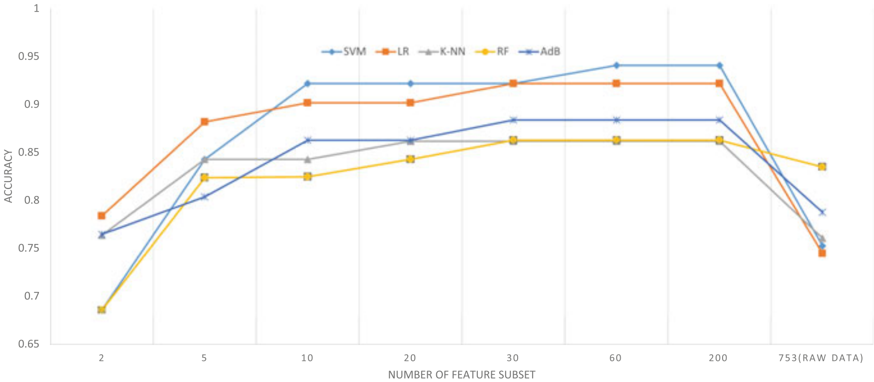

Fig. 2 Accuracy of different classifiers for different number of PCA components and raw data (at the right-most corner)

Fig. 3 Heat-map of Correlation Among Vocal Fold Based Features of PD Speech Data-set

5.2 *Experiment and Result of ML Models*

In this experiment, different ML classifiers with different combinations of hyper-parameters were used to yield the best performance of the model. Table 3 shows the performance of different classifiers on different data representations. Python's Library and API's were used to implement the model and carry out the experiment. For classification on raw data RF showed the best performance. SMV showed the highest overall performance with accuracy 94.1%. For tree-based classifiers (such as: AdB, RF), AdB showed superior highest accuarcy of 90.4%. k-NN showed the

Table 2 The best performing ML model and preprocessing along with tuned hyperparameters with highest possible accuracy

Classifiers	Performances	Best hyperparameters
SVM	$\mathbf{0.941 \pm 0.019}$	gamma=scale kernel=sigmoid C=1
LR	0.924 ± 0.087	penalty=l2 solver=lbfgs max_iter=1000
K-NN	0.863 ± 0.075	n_neighbors=6 weights=uniform algorithm=ball_tree
RF	0.88 ± 0.023	n_estimators=1000 criterion=gini min_samples_split=2
Adb	0.90 ± 0.079	n_estimators=50 algorithm=SAMME.R learning_rate=1

Note The best performing classifier is shown in bold type

lowest peak performance of 86.3%. The hyperparameters of the classifier models are described in Table 2. Our model significantly outperformed previous experiments on the same data-set which achieved the highest accuracy of 86% [19].

6 Conclusion

In this paper, Parkinson's Disease detection was achieved by employing our proposed model on the PD Speech data-set. Extensive data-processing was introduced to improve the quality of data as the data-set contained huge dimensionality with smaller data-points. Inter-attribute dependency was removed, data was standardized, dimensionality reduction method was introduced. Different machine learning classifiers were fine-tuned using different hyper-parameters for optimization. Cross-fold validation were used to diminish the impact of imbalanced data-set. The proposed model significantly outperformed the experiments on the same data-set. For future works, the proposed model can be used as a web application with user-friendly UI. Our proposed model can be used as a widespread tool for prediction of Parkinson's Disease with great efficiency and accuracy.

Table 3 Summary of performance of ML classifiers with data pre-processing techniques, feature reduction methods with number of attributes

Classifiers	Pre-processing	Algorithm	n_features	Performance
SVM	Raw data	N/A	753	0.746 ± 0.146
	Processed data	Corr	466	0.847 ± 0.120
		PCA	45	0.923 ± 0.07
		ICA	7	0.901 ± 0.023
		Corr+PCA	**55**	$\mathbf{0.941 \pm 0.019}$
LR	Raw data	N/A	753	0.745 ± 0.116
	Processed data	Corr	466	0.888 ± 0.084
		PCA	40	0.922 ± 0.07
		ICA	5	0.765 ± 0.091
		Corr+PCA	**55**	$\mathbf{0.924 \pm 0.087}$
K-NN	Raw data	N/A	753	0.761 ± 0.103
	Processed data	Corr	466	0.776 ± 0.123
		PCA	**45**	$\mathbf{0.863 \pm 0.075}$
		ICA	17	0.843 ± 0.103
		Corr+PCA	20	0.854 ± 0.059
RF	Raw data	N/A	753	0.835 ± 0.046
	Processed data	Corr	466	0.831 ± 0.104
		PCA	40	0.852 ± 0.07
		ICA	**17**	$\mathbf{0.881 \pm 0.023}$
		Corr+PCA	30	0.858 ± 0.019
AdB	Raw Data	N/A	753	0.788 ± 0.134
	Processed Data	Corr	466	0.812 ± 0.107
		PCA	40	0.883 ± 0.07
		ICA	20	0.8063 ± 0.023
		Corr+PCA	**45**	$\mathbf{0.904 \pm 0.079}$

Note The best performing approaches were shown in bold type

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Chapter 37
Hate Speech Detection in the Bengali Language: A Dataset and Its Baseline Evaluation

Nauros Romim, Mosahed Ahmed, Hriteshwar Talukder, and Md. Saiful Islam

1 Introduction

Social media has become an essential part of every person's day-to-day life. It enables fast communication, easy access to sharing and receiving ideas and views from worldwide. However, at the same time, this expression of freedom has led to the continuous rise of hate speech and offensive language in social media. Part of this problem has been created by the corporate social media model and the gap between documented community policy and the real-life implication of hate speech [4]. Not only that, but hate speech language is also very diverse [17]. The language used in social media is often very different from traditional print media. It has various linguistic features. Thus, the challenge in automatically detect hate speech is very hard [20].

Even though much work has been done in hate speech prevention in the English language, there is a significant lacking of resources regarding hate speech detection in Bengali social media. Nevertheless, problems like online abuse and especially online abuse toward females are continuously on the rise in Bangladesh [19]. However, for language like Bengali, a low resource language, developing and deploying machine learning models to tackle real-life problems is very difficult, since there is a shortage of dataset and other tools for Bengali text classification [6]. So, the need for research on the nature and prevention of social media hate speech has never been higher.

This paper illustrates our attempt to improve this problem. Our dataset comprises 30,000 Bengali comments from YouTube and Facebook comment sections and has a 10,000 hate speech. We select comments from seven different categories:

N. Romim (✉) · M. Ahmed · H. Talukder · Md. Saiful Islam
Shahjalal University of Science and Technology, Sylhet, Kumargaon 3114, Bangladesh
e-mail: hriteshwar-eee@sust.edu

Md. Saiful Islam
e-mail: saiful-cse@sust.edu

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sports, entertainment, crime, religion, politics, celebrity, TikTok and meme, making it diverse. We ran several deep learning models along with word embedding models such as Word2Vec, FastText, and pretrained BengFastText on our dataset to obtain benchmark results. Lastly, we analyzed our findings and explained challenges to detect hate speech.

2 Literature Review

Much work has been done on detecting hate speech using deep learning models [7, 10]. There have also been efforts to increase the accuracy of predicting hate speech specifically by extracting unique semantic features of hate speech comments [22]. Researchers have also utilized FastText to build models that can be trained on billions of words in less than ten minutes and classify millions of sentences among hundreds of classes [14]. There have also been researches that indicate how the annotator's bias and worldview affect the performance of a dataset [21]. Right now, the state of the art research for hate speech detection reached the point where researches utilize the power of advanced architectures such as transfer learning. For example, in [9], researchers compared deep learning, transfer learning and multitask learning architectures on the Arabic hate speech dataset. There is also research in identifying hate speech from a multilingual dataset using the pretrained state of the art models such as mBERT and xlm-RoBERTa [3].

Unfortunately, very few works have been done on hate speech detection in Bengali social media. The main challenge is the lack of sufficient data. To the best of our knowledge, many of the datasets were around 5000 corpora [5, 8, 12]. There was a publicly available corpus containing around 10,000 corpora, which were annotated into five different classes [2]. Nevertheless, one limitation the authors faced that they could only use sentences labeled as toxicity in their experiments since other labels were low in number. There was also another dataset of 2665 corpus that was translated from an English hate speech dataset [1]. Another research used a rule-based stemmer to obtain linguistic features [8]. Researches are coming out that uses deep learning models such as CNN, LSTM to obtain better results [2, 5, 8]. One of the biggest challenges is that Bengali is a low resource language. Research has been done to create word embedding specifically for a low resource language like Bengali called BengFastText [15]. This embedding was trained on 250 million Bengali word corpus. However, one thing is clear that there is a lack of dataset that is both large and diverse. This paper tries to tackle the problem by presenting a large dataset and having comments from seven different categories, making it diverse. To our knowledge, this is the first dataset in Bengali social media hate speech with such a large and diverse dataset.

3 Dataset

3.1 Data Extraction

Our primary goal while creating the dataset was to create a dataset with different varieties of data. For this reason, comments on seven different categories: *sports, entertainment, crime, religion, politics, celebrity and meme, TikTok and miscellaneous* from YouTube and Facebook were extracted.

We extracted comments from the Facebook public page, Dr. Muhammad Zafar Iqbal. He is a prominent science fiction author, physicist, academic and activist of Bangladesh. These comments belonged to the *celebrity* category. Nevertheless, due to Facebook's restriction on its graph API, we had to focus on YouTube as the primary source of data.

From YouTube, we looked for the most scandalous and controversial topics of Bangladesh between the year 2017–2020. We reasoned that since these were controversial, videos were made more frequently, people participated more in the comment section, and the comments might contain more hate speech. We searched YouTube for videos on keywords related to these controversial events. For example, we searched for renowned singer and actor couple *Mithila-Tahsan divorce*, i.e., the Mithila controversy of 2020. Then, we selected videos with at least 700k views and extracted comments from them. In this way, we extracted comments videos on controversial topics covering five categories: *sports, entertainment, crime, religion and politics.* Finally, we searched for videos that are memes, TikTok and other keywords that might contain hate speech in their comment section. This is our seventh category.

We extracted all the comments using open-source software called FacePager.[1] After extracting, we labeled each comment with a number defining which category it belonged. We also labeled the comments with the additional number to define its *keyword*. For this paper, the *keyword* means the controversial event that falls under a *category.* For example, *mithila* is a keyword that belongs to the *entertainment* category. We labeled every comment with its corresponding category and keyword for future research.

After extracting the comments, we manually checked the whole dataset and removed all the comments made entirely of English or other non-Bengali languages, emoji, punctuation and numerical values. However, we kept a comment if it primarily consists of Bengali words with some emoji, number and non-Bengali words mixed within it. We deleted non-Bengali comments because our research focuses on Bengali hate speech comments, so non-Bengali comments are out of our research focus. However, we kept impure Bengali comments because people do not make pure Bengali comments on social media. So, emoji, number, punctuation and English words are a feature of social media comment. Thus, we believe our dataset can prove to

[1] https://github.com/strohne/Facepager.

be very rich for future research purposes. In the end, we collected a total of 30,000 comments. In a nutshell, our dataset has 30k comments that are mostly Bengali sentences with some emoji, punctuation, number and English alphabet mixed in it. Table 1 shows sample dataset.

3.2 Dataset Annotation

Hate speech is a subjective matter. So, it is quite difficult to define what makes a comment hate speech. Thus, we have come up with some rigid rules. We have based these rules on the community standard of Facebook[2] and YouTube.[3] Bellow, we have listed some criteria with necessary examples:

- Hate speech is a sentence that dehumanizes one or multiple persons or a community. Dehumanizing can be done by comparing the person or community to an insect, object or a criminal. It can also be done by targeting a person based on their race, gender, physical and mental disability.
- A sentence might contain slang or inappropriate language. But unless that slang dehumanizes a person or community, we did not consider it to be hate speech. For example,

  কি বালের মুভি

  Here, the slang word is not used to dehumanized any person. So, it is not hate speech.
- If a comment does not dehumanize a person rather directly supports another idea that clearly dehumanizes a person or a community, this is considered hate speech. For example,

  বোরকা পরে না, ধর্ষিত তো হবেই।

  this sentence supports the dehumanizing act of women, so we labeled it as hate speech.
- If additional context is needed to understand that a comment is a hate speech, we did no consider it to be one. For example, consider this sentence:

  গর্ত যোদ্ধা

  It is a comment taken from Dr. Muhammad Zafar Iqbal's Facebook page. This comment refers to a particular job the haters of Dr. Zafar Iqbal constantly use to attack him in social media. But unless no one mentions an annotator that this comment belongs to this particular Facebook page, that person will have no way of knowing this is actually a hate speech. Thus, these types of comments will be labeled as not hate speech.
- It does not matter if the stand that a hate speech comment takes is right or wrong. Because what is right or wrong is subjective. So if a sentence, without any outside context, dehumanizes a person or community, we considered that to be hate speech.

[2] https://web.facebook.com/communitystandards/.

[3] https://www.youtube.com/howyoutubeworks/policies/community-guidelines/.

Table 1 Sample dataset

Sentence	Hate speech	Category	Keyword
মহিলার বিচার চাই	0	3	1
হ্লারপো তোর কত বড় সাহস?	1	1	6
আমরা পাশে আছি ব্রাদার।	0	2	12
এই গাধার বাচ্চা, কি বলিস তুই	1	3	3
হে আল্লাহ, তার সুস্থতা দা কর	0	6	1

We worked with 50 annotators to annotate the entire dataset. We instructed all annotators to follow our guidelines mentioned above. The annotators are all undergraduate students of Shahjalal University of Science and Technology. Thus, the annotators have an excellent understanding of popular social media trends and seen how hate speech propagates in social media. All comments were annotated three times, and we took the majority decision as the final annotation.

After annotation, we wanted to check the validity of the annotator's annotation. For this reason, we randomly sampled 300 comments from every category. Then, we manually checked each comment's majority decision. Since we, as the authors of this paper, were the ones that set the guideline for defining hate speech and did not participate in the annotation procedure, our checking was a neutral evaluation of the annotator's performance. After our evaluation, we found that our dataset annotation is 91.05% correct.

3.3 Dataset Statistics

Our dataset has a total of 30,000 comments, where there are a total of 10,000 hate speech comments, as evident from Table 2. So, it is clear that our dataset is heavily biased toward not hate speech. If we look closely at each category in Fig. 1, it becomes even more apparent. In all categories, not hate speech comments dominate. But particularly in *celebrity* and *politics*, the number of hate speech is very low, even compared to hate speech in other categories. During data collection, we have observed that there were many hate speech in the *celebrity* section, i.e., in Dr. Muhammad Zafar Iqbal's Facebook page, but they were outside context. As we have discussed before

Table 2 Hate speech comments per category

Hate speech	Not hate speech	Total
10,000	20,000	30,000

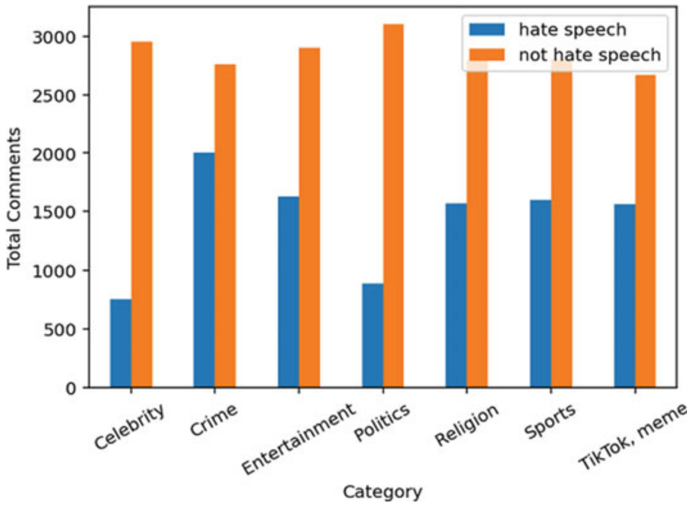

Fig. 1 Distribution diagram of data in each category

in Sect. 3.2, we have only considered texts without context while labeling it as hate speech. So many comments were labeled as not hate speech. For the category *politics*, we have observed that people tend to not attack any person or group directly. Rather they tend to add their own take on the current political environment. So, the number of direct attacks is less in the *politics* category.

When we look at the mean text length in Table 3, we can find a couple of interesting observations. First, we can see that meme comments are very short in length. This makes sense as when a person is posting a comment in a meme video, and he is likely to express his state of mind, requiring a shorter amount sentences. But the opposite is true for the *celebrity* category. This has the longest average text length. This is large because when people comment on Dr. Zafar Iqbal's Facebook page, they add a lot of their own opinion and analysis, no matter the comment is hate speech or not. This shows how unique the comment section of an individual celebrity page can be. Lastly, we see that average hate speech tends to be shorter than not hate speech.

In Table 4, we have compared all state of the art datasets. The table we have included the total number of the dataset and the number of classes the datasets were annotated. As you can see, there are some datasets that have multiple classes. In this paper, we focused on the total number of the dataset and extracted comments from different categories so that we can ensure linguistic variation.

Table 3 Mean text length of the dataset

Category	Mean text length
Sports	75
Entertainment	65.9
Crime	87.4
Religion	71.4
Politics	72.5
Celebrity	134.5
Meme, TikTok and others	56.2
Hate speech	69.59
Not hate speech	84.39

Table 4 Comparison of all state of the art datasets on Bengali hate speech

Paper	Total data	Number of classes
Hateful speech detection in public facebook pages for the Bengali language [12]	5126	06
Toxicity detection on Bengali social media comments using supervised models [2]	10,219	05
A deep learning approach to detect abusive Bengali text [8]	4700	07
Threat and abusive language detection on social media in Bengali language [5]	5644	07
Detecting abusive comments in discussion threads using Naïve Bayes [1]	2665	07
Hate speech detection in the Bengali language: dataset and its baseline evaluation	30,000	02

4 Experiment

4.1 Preprocess

Our 30k dataset had raw Bengali comments with emoji, punctuation and English alphabet mixed in it. We removed all emoji, punctuation, numerical values, non-Bengali alphabet and symbols from all comments for baseline evaluation. After that, we created a train and test set of 24,000 and 6000 comments, respectively. Now, the dataset is ready for evaluation.

4.2 Word Embedding

We have used three word embedding models. They are Word2Vec [16], FastText [13] and BengFastText [18]. To create a Word2Vec model, we used gensim[4] module to train on the 30k dataset. We have used CBoW method for building the Word2Vec model. For FastText, we also used the 30k dataset to create the embedding and used the skip-gram method. The embedding dimension for both models was set to 300. Lastly, BengFastText is the largest pretrained Bengali word embedding based on FastText. But BengFastText was not trained on any YouTube data. So, we wanted to see how it performs on YouTube comments.

4.3 Models

4.3.1 Support Vector Machine (SVM)

We used the support vector machine [11] to determine baseline evaluation. We used the linear kernel and kept all other parameters to its default value.

4.3.2 Long Short-Term Memory (LSTM)

For our experiment, we used 100 LSTM layers, set both dropout and recurrent dropout rate to 0.2 and used 'adam' as the optimizer.

4.3.3 Bidirectional Long Short-Term Memory (Bi-LSTM)

In this case, we used 64 Bi-LSTM layers, with a dropout rate set to 0.2 and optimizer as 'adam.'

4.4 Experimental Setting

We kept 80% of the dataset as a train set and 20% as a test and trained every word embedding with every deep learning algorithm on the train set. For every case, we kept all parameters standard. Epoch and batch size were set to 5 and 64, respectively. Then, we tested all the trained models on the test dataset and measured the accuracy and F-1 score. Below are all types of models, we tested on our dataset.

[4]https://radimrehurek.com/gensim/models/word2vec.html.

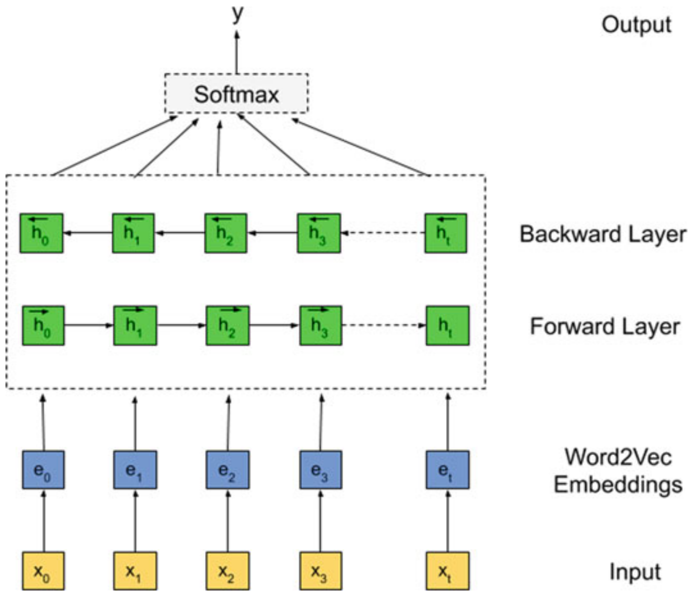

Fig. 2 Deep learning architecture with Word2Vec and Bi-LSTM

- Baseline evaluation: Support Vector Machine (SVM)
- FastText Embedding with LSTM
- FastText Embedding with Bi-LSTM
- BengFastText Embedding with LSTM
- BengFastText Embedding with Bi-LSTM
- Word2Vec Embedding with LSTM
- Word2Vec Embedding with Bi-LSTM (Fig. 2).

4.5 Result

We can observe from Table 5 that all the models achieved good accuracy. SVM achieved the overall best result with accuracy and an F-1 score of 87.5% and 0.911, respectively. But BengFastText with LSTM and Bi-LSTM had relatively the worst accuracy and F-1 score. Their low F-1 score indicates that deep learning models with BengFastText embedding were overfitted the most. BengFastText is not trained on any YouTube data [18]. But our dataset has a huge amount of YouTube comment. This might be a reason for its drop on performance.

Then, we looked at the performance of Word2Vec and FastText embedding. We can see that FastText performed better in terms of accuracy and had a lower F-1 score than Word2Vec. Word2Vec was more overfitted than FastText. FastText has

Table 5 Result of all models

Model name	Accuracy	F-1 score
SVM	**87.5**	**0.911**
Word2Vec + LSTM	83.85	0.89
Word2Vec + Bi-LSTM	81.52	0.86
FastText + LSTM	84.3	0.89
FastText + Bi-LSTM	86.55	0.901
BengFastText + LSTM	81	0.859
BengFastText + Bi-LSTM	80.44	0.857

Bold indicates the best result

one distinct advantage over Word2Vec: It learns from the words of a corpus and its substrings. Thus, FastText can tell ‘love’ and ‘beloved’ are similar words [13]. This might be a reason as to why FastText outperformed Word2Vec.

4.6 Error Analysis

We manually cross-checked all labels of the test set with the prediction of the SVM model. We looked at the false negative and false positive cases and wanted to find which types of sentences the model failed to predict accurately. We found that some of the labels were actually wrong, and the model actually predicted accurately. Nevertheless, there were some unusual cases. For example,

ওরা চাইবো কেন তোর কাছে তুই বিসিবি প্রেসিডেন্ট হয়ে কি বাল ফালাস তোর তো খেয়াল রাখা দরকার ছিল ক্রিকেটারদের ফিসিলিস নিয়ে

This is not a hate speech, but the model predicted it to be hate speech. There are several other similar examples. The reason here is that this sentence contains aggressive words that are normally used in hate speech, but in this case, it was not used to dehumanize another person. However, the model failed to understand that. This type of mistake was common in false negative cases. It demonstrates that words in the Bengali language can be used in a complicated context and it is a tremendous challenge for machine learning models to actually understand the proper context of a word used in a sentence.

5 Conclusion

Hate speech comment in social media is a pervasive problem. So, a lot more research is urgent to combat this issue and ensure a better online environment. One of the biggest obstacles toward detecting hate speech through state of the art deep learning models is the lack of a large and diverse dataset. In this paper, we created a large dataset of 30,000 comments, a 10,000 hate speech. Our dataset has comments from seven different categories making the dataset diverse and abundant. We showed that hate speech comments tend to be shorter in length and word count than not hate speech comments. Finally, we ran several deep learning models with word embedding on our dataset. It showed that when the training dataset is highly imbalanced, the models become overfitted and biased toward not hate speech. Thus, even though overall accuracy is very high, the models cannot predict hate speech well.

However, we believe this is scratching just the surface of solving this widespread problem. One of the biggest obstacles is that there is no proper word embedding for the Bengali language used in social media. There is some word embedding created from newspaper and blog article corpus. Nevertheless, the language used is vastly different from social media language. The main reason is that unlike traditional print media, there is no one to check for grammatical and spelling mistakes. Thus, there are lots of misspelling, grammatical errors, cryptic meme language, emoji, etc. In fact, in our dataset, we found the same word having multiple types of spelling. For example,

জাব, যাব, যাবো, জাবও

Here, they all are the same word. A human brain can understand that these words are the same, but they are different from a deep learning model. Another difference is the use of emoji to convey meaning. Often people will express a specific emotion with the only emoji. Emoji is a recent phenomenon that is absent in blog posts or newspaper articles, or books. Currently, there is no dataset or pretrained model that classifies the sentiment of emoji used in social media.

One of the critical challenges to accurate hate speech detection is to create models that can extract necessary information from an unbalanced dataset to predict the minority class with reasonable accuracy. Our experiment demonstrated that standard deep learning models are not sufficient for this task. Advanced models like mBERT and XLM-RoBERTa can be of great use in this regard as they are trained on a large multilingual dataset and use attention mechanisms. Embedding models based on extensive and diverse social media comment datasets can also be of great help.

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Chapter 38
An Approach Towards Domain Knowledge-Based Classification of Driving Maneuvers with LSTM Network

Supriya Sarker and Md. Mokammel Haque

1 Introduction

With the accelerated industrial revolution, the increasing amount of vehicles forming threats to road safety [1]. Since drivers are responsible for more than 80% of the traffic accident [2, 3] there has a great influence on driver behavior on road traffic security [4]. Advanced driver assistance systems (ADAS) are installed in exclusive vehicles by automotive manufacturers to stabilize safety and competency [5]. Due to the high cost, the deployment of ADAS has been restricted. However, the focus of ADAS is assisting drivers rather than monitoring the driving style. These systems have been developed to rescue and protect people after the accident and successfully lots of lives have been saved, nonetheless cannot able to avoid the accident to take place [6]. Therefore, driver behavior should be considered as one of the most significant aspects for enhancing road traffic safety [7]. It is important for monitoring new drivers and performance evaluation of drivers throughout the training sessions [8]. Besides, driver's behavior can influence the reduction of fuel, i.e., energy consumption and gas emissions.

Alike ADAS, various approaches have proposed that used inertial signals, GPS, video to monitor drivers, and driving behaviors. However, being processed these input through the system exacerbate the system performance and increase computational complexity though it is verified in previous related works that inertial sensors absolutely have the ability to recognize different driving style [9]. Few researchers utilized

S. Sarker (✉) · Md. M. Haque
Department of Computer Science & Engineering, Chittagong University of Engineering & Technology, Chittagong, Bangladesh
e-mail: sarkersupriya7@gmail.com

Md. M. Haque
e-mail: mokammel@cuet.ac.bd

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vehicle information collected through a Controlled Area Network (CAN) using On-Board Diagnostic (OBD) port or Micro Electro Mechanical System (MEMS). Nowadays, MEMS is getting more popular over OBD port because of its compact size, light-weight and lower in energy consumption nature. So, the integration of different sensor units such as accelerometers, gyroscopes, magnetometers becomes easy [10]. Since collecting vehicle information through CAN using OBD port depends on the ability of the private protocol of a particular vehicle and also poses threat to the vehicle and human security [10], hence, MEMS is more appropriate for driving maneuver recognition.

Earlier many researchers follow different fuzzy inference and machine learning techniques to classify driving maneuvers. However, they ignore the analysis of the theoretical base of collected sensors data, hence, prevent the advanced optimization of driving maneuver classification research. While moving, the change of the motion sensor data can be explained with the help of rigid body kinematics [1]. We believe that these data change rules can reinforce to differentiate different maneuvers i.e., events, and therefore, amend the classification algorithm.

The paper introduces the significance of driving maneuver classification of sensor data in Sect. 1. A compressed discussion on the existing approaches is presented in Sect. 2. The theoretical analysis of the physical model with multi-class hypotheses, including deep learning architecture is described in Sect. 3. The experimental environment is being described and results are analyzed in Sect. 4. Finally, Sect. 5 concludes the paper with future directions.

2 Existing Approach

From the literature review, it is obvious that several solutions for driving maneuver recognition or classification have been proposed and developed. We have partitioned the literature review section into two part. The first part contains the approaches that are not related to deep learning time series classification and the second part includes deep learning time series classification approaches.

In [8] the authors proposed a system to classify driving style as aggressive and non-aggressive and type of driving maneuver based on the Dynamic Time Warping (DTW) algorithm and smartphone-based sensor-fusion. In [11] a Fuzzy Inference System (FIS) based mobile tool has been developed that utilized accelerometer, magnetometer, gravity sensor, GPS embedded in drivers' smartphones, and evaluated the overall driving behavior by combining different fuzzy sensing data. However, it was a web-based tool that processes all sensor data in a remote database to score all the driving events based on a few predefined threshold values of each possible event. Ly et al. in [9] proposed a system that used the vehicle's inertial sensors from the CAN bus using SVM and k mean clustering. They imposed few thresholds for lateral, longitudinal acceleration, yaw angular velocity sensors to create an individual driver profile of the driver to ultimately provide proper feedback to reduce the number of dangerous car maneuvers. In [7] drivers were categorized according to four different driving events namely acceleration, braking, turning, and lane changing using smartphone sensory data in a bidirectional vehicle to infrastructure communication system. They established a driver safety index to categorize each driver during the journey. Symbolic Aggregate Approximation (SAX) and Matrix profile method have been used to categorized patterns that appear in time series naturalistic driving data included maneuvers like turning, stopping at intersections, parking, and leaving parking spaces [12]. In [13] the authors concentrated four driving states namely stopped, driving, parking, and parked. They applied Random Forests (RF), Support Vector Machines (SVM), and fuzzy rule-based classifiers on data collected by an accelerometer sensor installed in a smartphone inside the vehicle. In [14] they evaluated the performance and comparative analysis of multiple combinations of machine learning algorithms such as SVM. They used sensors e.g., accelerometer, gyroscope, and magnetometer installed in android smartphones to collect driving data for detecting aggressive driving events. However, only a labeled event dataset has been utilized for classification though there is a lot of unlabeled data that has a potential driving event pattern. In our experiment, we have used the dataset.

A work using the same driving dataset similar to [14] has been introduced in [15]. In this work time series of accelerometer data is employed as inputs of Recurrent Neural Network (RNN) to accomplish driver behavior profiling and investigated the performance of three different types of Recurrent Neural Networks which are Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), and Gated Recurrent Network (GRU). However, they classified drivers' behavior only from accelerometer data, not driving maneuver. By plotting the collected sample dataset, it is clearly shown that only the accelerometer sensor data change rule is not sufficient to classify all the events. Moreover, drivers' behavior is closely related to driving maneuvers. Another researcher in [16] focused on an approach to implement supervised time series classification with Random Forest (RF) and Recurrent Neural Network separately. They introduced RF to classify aggressive events and the result of RF transferred to recognize the type of maneuver. However, a limitation of the implementation is that the labels were selected manually by one person and depend on the labeler's danger perception. We agree with the authors that the assignation of labels should be extended to more criteria to reduce potential bias. We believe that our work will extend a way to find the correct label without manual labeling. In [17] the authors developed a model that consists of three major computational components, distance-based representation of driving context including features of vehicle trajectory and visual features. The model took as an input of videos of the road scene ahead and vehicle sensor data from OBD sensors and the video images of the front view and classified temporal dependencies of five different classes of driving maneuvers namely left turn, right turn, left lane change, right lane change, driving straight using LSTM.

The main contribution made in this research is that (i) We develop a Long Short Term Memory (LSTM) network for classification of driving maneuver from sensor fusion time series data, (ii) Establish hypotheses based on data change rules of time series data during the performance of various driving maneuver discussed in [1], (iii)

Investigate a novel way to reducing manual labeling of unlabeled time series driving dataset.

3 Proposed Methodology

In this work, we model time series sequence data with Long Short Term Memory (LSTM) considering the classification as a multi-class supervised deep learning classification problem. For the multi-class classification purpose, we analyze the physical model of moving vehicles during the performance of different maneuvers and establish hypotheses for a particular class which can be explained by the data change rule of the sensor fusion dataset and fit well with the output classes of maneuvers.

3.1 Analysis of Theoretical Physical Model

Many researchers have studied the dynamic model of vehicles related to the measurement of slip angle, velocity, tire model, etc. In the present research, we only focus on moving subject vehicles disregarding its inner model. During the movement of a vehicle in real traffic conditions, the most common driving maneuvers are moving forward, left and right turning, left, and right lane change. The physical model has been established from the theory of rigid body kinematics [1]. Movement of a vehicle causes changes in its kinematic states such as acceleration, deceleration, angular velocity, etc. which follow specific rules. From the first kinematics formula, the acceleration, $\vec{a}$ in time, t can be defined by Eq. (1). The angular velocity, $\vec{w}$ of a vehicle with radius, $\vec{r}$ can be defined by Eq. (2) and further decomposed in x, y and z dimension. So, we get three acceleration data a_x, a_y, a_z and three angular velocity data w_x, w_y, w_z in three dimensions.

$$\vec{a} = \frac{v_t - v_0}{t} \tag{1}$$

$$\vec{w} = \frac{\vec{v}}{\vec{r}} \tag{2}$$

To establish the driving data change rule, Wu et al. [1] proposed a physical model shown in Fig. 1. As magnetometer sensor data is related to the position of the geographic location of the vehicle and no direct relation with common driving maneuvers has been proved, it is out of our scope of interest. On the basis of the physical model of vehicle motion, the change of sensors' time series data which represents different driving maneuvers is inferred and illustrated in Fig. 2.

In Fig. 1, it is shown that x, y and z-axis data directed towards longitudinal, lateral and angular direction of moving subject vehicle, respectively. When the subject

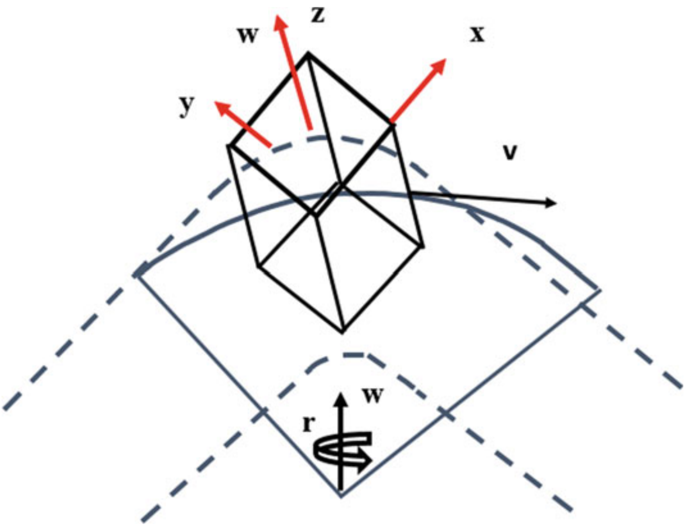

Fig. 1 Physical model of moving vehicle

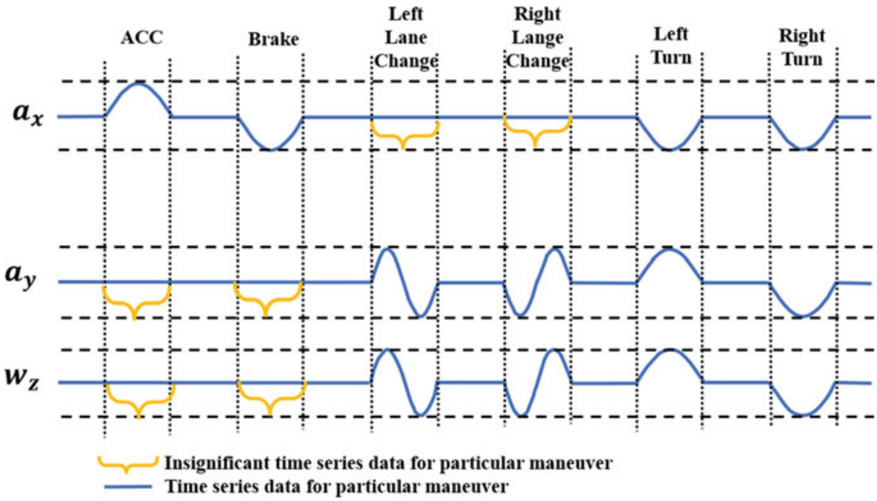

Fig. 2 Illustration of time series data change rule of driving maneuver

vehicle goes straight, there is a significant change in a_x. During the left and right turn, left lane change and right lane change, a lateral displacement with longitudinal displacement of the subject vehicle in left or right direction occurs and as an angular velocity involves with lateral displacement for this type of driving maneuvers which cause changes in a_y and w_z as well. Hence, we focus on a_x, a_y and w_z and data only.

In Eq. (1), if $v_t > v_o$, therefore $a_x > 0$; hence, during the event of acceleration, the time series data initially increases from zero to greater value and finally decreases to a lower value. When $v_t < v_o$, then $a_x < 0$. As a result, while deceleration, commonly known as braking in the transportation system the time series data diminishes from zero to negative values and then increases to zero again.

From Eq. (2) it is clear that $\vec{w} \, \alpha \, \vec{v}$, therefore, change of angular velocity is persistent to acceleration. As at the time of lane change both lateral and angular displacement happen, for left lane change both time series data of a_y and w_z increases from zero and gradually decreases until a negative value, then increases again. The opposite incident occurs for the right lane change.

We plot some sample time series data of labeled maneuvers to validate the data change rules of moving vehicles. The description of the used dataset is provided in Sect. 3.3. When a vehicle moves in longitudinal direction non-aggressively, the a_x data range from -2 to 2 and as angular rotation does not involve here, the change of angular velocity is insignificant. In Fig. 3 Non-aggressive time series data peak ranges from -2 to $+$ 2 shown by red dotted line. By plotting more samples of non-aggressive labeled data, the same range is found. An 'aggressive acceleration' labeled time series data is plotted in Fig. 4 shows that few peaks exceeded $+2$ and reaches $+6$ shown by the red dotted line. The statistical features e.g. mean, variance, and standard deviation reflect substantial change during the maneuver. At the same time, the slopes of the data points are very sharp. This pattern indicates a high gradient and energy at the time of aggressive acceleration. The amount of slope is defined by Eq. (3) and energy is defined by Eq. (4) [1].

$$\text{Slope, } S = \frac{a_{x\,\text{slope}(i)} - a_{x\,\text{slope}(i-1)}}{t_i - t_{i-1}} \tag{3}$$

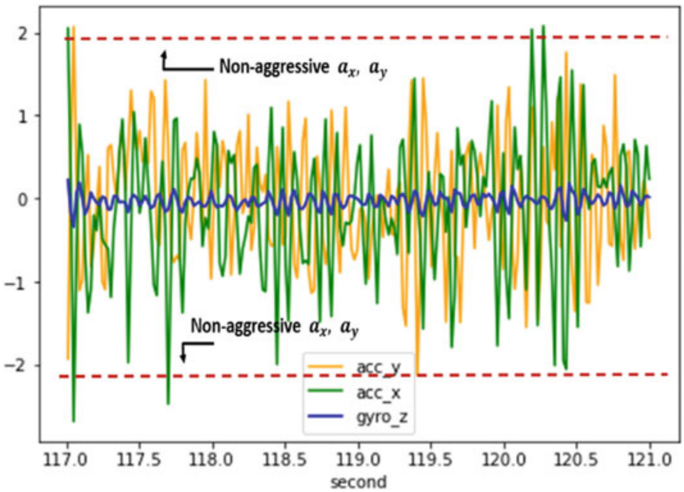

Fig. 3 "Non-aggressive" labeled time series data

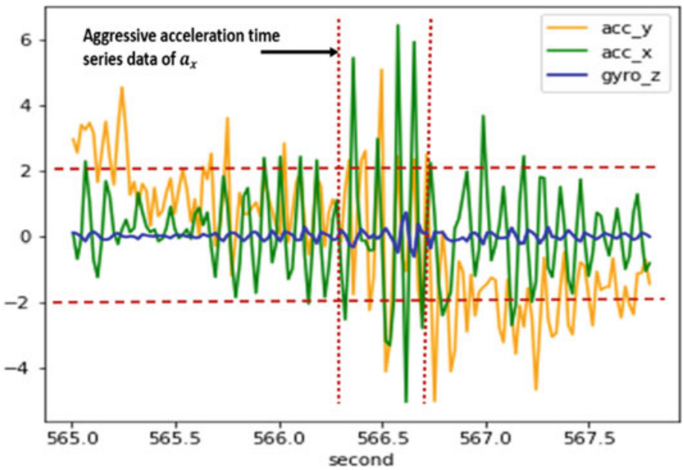

Fig. 4 "Aggressive acceleration" labeled time series data

$$Energy, E = \frac{a_{xi}^2 + a_{x\,i-1}^2 + a_{x\,i-(k+1)}^2}{k} \tag{4}$$

where, $a_{x\,slope(i)}$ = slope of a_x of ith and $(i-1)$th point in a window, $a_{xi} = a_x$ data of ith point in the window, k = size of sliding window.

From Eq. (3) it is clear that slope increases and decreases according to two adjacent data points where Eq. (4) implies that energy increases only in the positive axis. During aggressive braking (in Fig. 5), the value of a_x drops significantly to

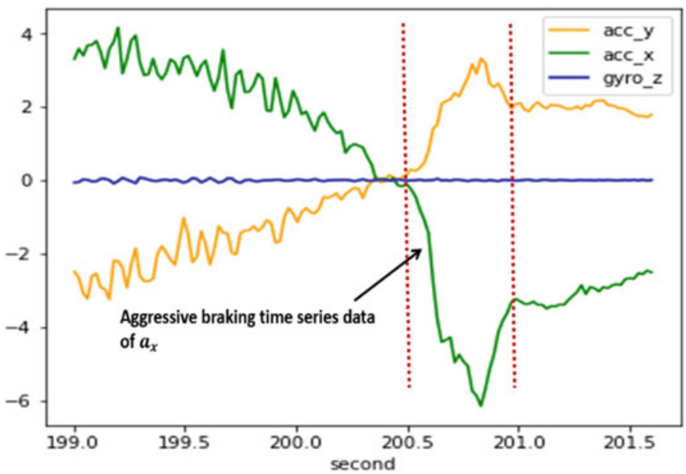

Fig. 5 "Aggressive braking" labeled time series data

a negative value less than −2 and almost −6. As the amount of energy and variance increases both during aggressive acceleration and brake, it cannot make any difference between aggressive acceleration and braking rather indicate an aggressive event. So, we consider a_x, and slope of a_x to differentiate aggressive acceleration and braking to avoid redundancy.

For aggressive Left Turn (LT) and Right Turn (RT), lateral displacement and angular rotation occur at the same time, so data change in a_y and w_z represents these two events. During the aggressive LT, w_z changes from 0 to 1 and for the aggressive RT it ranges from −1 to 0 where during a non-aggressive maneuver, change of w_z ranges from −0.2 to 0.2. In both events the energy of w_z increase from 0.3 during a non-aggressive event. At the moment of LT, the slope remains positive while for RT it becomes negative (see Figs. 8 and 9). There is no significant effect of variance. Hence, we consider a_y, slope and energy of a_y; w_z, slope and energy of w_z to differentiate aggressive LT and RT.

At the moment of Left Lane Change (LLC) and Right Lane Change (RLC), the slope of w_z increases and decreases in greater value than LT and RT, and for both event data point move from one axis to another axis (see Figs. 6 and 7). There is no significant pattern of energy and variance as time series data moves to positive to a negative value and vise-versa for LLC and RLC, respectively; but energy and variance increase for both maneuvers. Therefore, we consider data and slope of a_y and w_z.

While performing one particular maneuver, sometimes the data change pattern can follow patterns of other maneuvers but manual labeling does not show the difference because manual labeling relies on human perceptions. For instance, before or after aggressive braking, the data change follows aggressive acceleration but through human perception, the maneuver is being labeled as aggressive braking only.

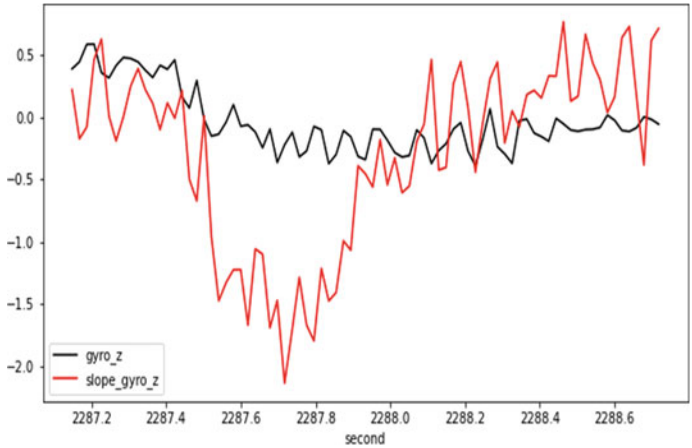

Fig. 6 "Aggressive left lane change" labeled w_z data and slope of w_z

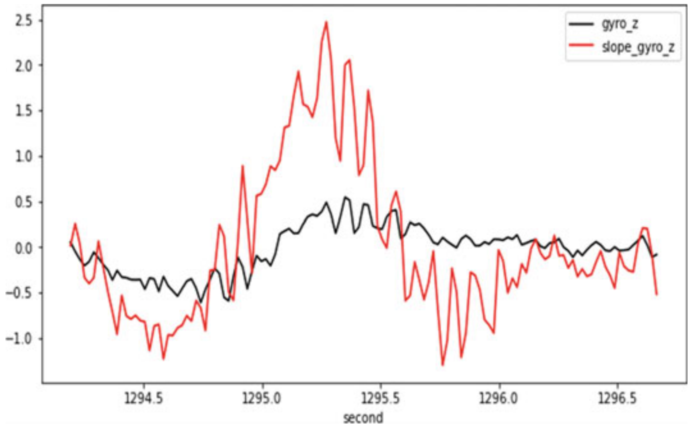

Fig. 7 "Aggressive right lane change" labeled w_z data and slope of w_z

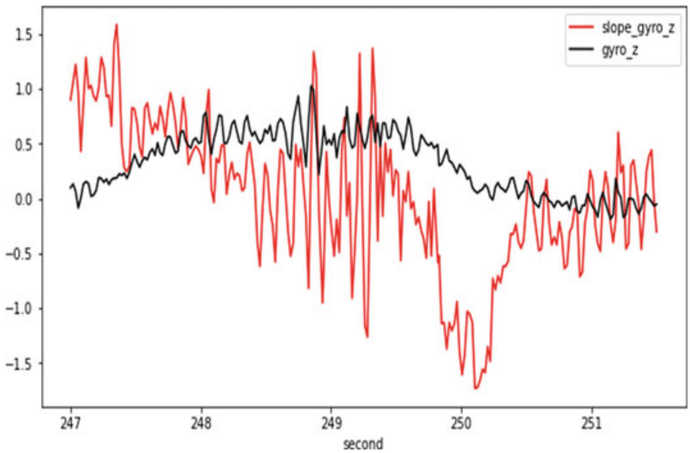

Fig. 8 "Aggressive left turn" labeled w_z data and slope of w_z

Many researchers proposed various classification methods where the time series data change rules have been ignored. We investigate the significance and figure out that the data change rule of a particular maneuver is valuable during feature extraction of sensor fusion time series data. So, we develop some hypotheses from the observations of labeled time series data.

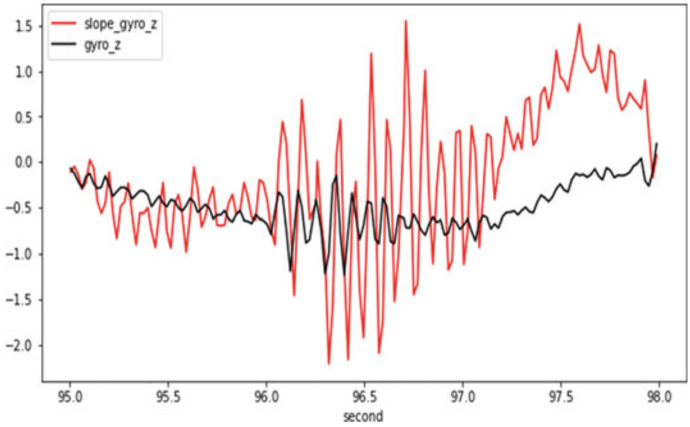

Fig. 9 "Aggressive right turn" labeled w_z data and slope of w_z

3.2 Development of Hypotheses for Driving Maneuver Classes

We develop some hypotheses from the observations of time series data changing patterns for driving maneuver classes on the basis of the physical model.

Class 1: Acceleration

$$\left\{(a_x > 2) \text{ or } (a_{x\,\text{slope}} > 5)\right.; \quad H_1\text{:aggressive\_acceleration}$$

Class 2: Brake

$$\left\{(a_x < -2) \text{ or } (a_{x\,\text{slope}} < -5)\right.; \quad H_2\text{:aggressive\_brake}$$

Class 3: Left Lane Change

$$\left\{\begin{array}{c}(a_y > 2) \text{ or } (a_{y\,\text{slope}} > 5)\\ (w_z < -0.2) \text{ or } (w_{z\,\text{slope}} < -0.6)\end{array}\right.; \quad H_3\text{:aggressive\_left\_lane\_change}$$

Class 4: Right Lane Change

$$\left\{\begin{array}{c}(a_y < -2) \text{ or } (a_{y\,\text{slope}} < -5)\\ (w_z > 0.2) \text{ or } (w_{z\,\text{slope}} > 0.6)\end{array}\right.; \quad H_4\text{:aggressive\_right\_lane\_change}$$

Class 5: Left Turn

$$\left\{\begin{array}{c}(a_y > 2) \text{ or } (a_{y\,\text{slope}} > 5) \text{ or } (a_{y\,\text{energy}} > 0.3)\\ (w_z > 0.2) \text{ or } (w_{z\,\text{slope}} > 0.6) \text{ or } (w_{z\,\text{energy}} > 0.3)\end{array}\right.; \quad H_5\text{:aggressive\_left\_turn}$$

Class 6: Right Turn

$$\left\{\begin{array}{c}\left(a_y < -2\right) \text{ or } \left(a_{y\,\text{slope}} < -5\right) \text{ or } \left(a_{y\,\text{energy}} > 0.3\right) \\ \left(w_z < -0.2\right) \text{ or } \left(w_{z\,\text{slope}} < -0.6\right) \text{ or } \left(w_{z\,\text{energy}} > 0.3\right)\end{array}\right.; H_6\text{:aggressive\_right\_turn}$$

Here, $a_{x\,\text{slope}}$, $a_{y\,\text{slope}}$, $w_{z\,\text{slope}}$ is slope of a_x, a_y and w_z data, respectively. $a_{y\,\text{energy}}$ and $w_{z\,\text{energy}}$ is energy of a_y and w_z data, respectively. Time series data that do not fall into any of the classes will be non-aggressive and belong to Class 7.

3.3 Development of Time Series Classification with Deep Neural Network

Deep learning model such as Recurrent Neural Network, long-short term memory network is being designed for classification of sequential data. Since time series classification requires restoring functional dependencies between the sequences of time series and the finite set of classes after training with a set of known classes [18] we preferred to use LSTM networks for the classification of driving maneuvers.

Time Series Data Collection and Preparation. We perform our experiment using a set of real-world sensor fusion time series data [19]. When two drivers with 15 years of experience performed the different driving maneuvers, a smartphone application has recorded the sensor fusion time series data. During the collection of the sensor fusion time series dataset, the smartphone was sable on the windshield of the subject vehicle. While collecting sensor data, video of the front view of the subject vehicle has been recorded from which later the label of different driving maneuvers had been labeled manually.

The dataset was distributed in multiple files and the labels with start and end time of events were listed in different files for four of each journey with nanosecond timestamp. We prepare the a_x, a_y and w_z data with second. Besides, we add the label for each time step as ground truth label. Among 156,512 time series observations, only 11,077 observations are manually labeled and 145,435 unlabeled observations were unlabeled. We set a rolling window with 20 sample size and extract significant features from a valid window using the hypotheses of Sect. 3.2.

Developing LSTM Model Architecture. The proposed LSTM network architecture consists of one input layer with two LSTM hidden layers and a dense output layer. The architecture of our LSTM is presented in Fig. 10. The 128 output nodes of the first hidden layer are fed into the second hidden layer which generates 128 output nodes. These 128 output nodes are fed into the last dense layer and generate seven output nodes. The last layer must have seven nodes because each node represents one output class.

Wang et al. in [20] evaluated different deep learning models for time series classification and among those this LSTM architecture was one of the outperformed models. A dropout layer with a dropout rate of 0.2 is used after every layer except for

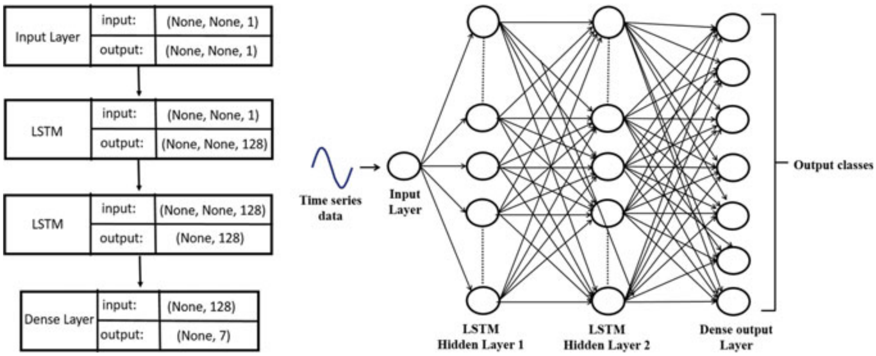

Fig. 10 Architecture of LSTM_128_128_Dense Model for multi-class classification

the input layer. The batch size is (1/10) of the training dataset. The activation function for every layer is the Rectified Linear Unit (ReLU) except for the dense layer. The Softmax activation function is used for the dense layer. As this is a multi-class classification problem, the model uses categorical cross-entropy for loss optimization. The LSTM model is trained during 500 epochs with Adam optimizer.

4 Experimental Results and Analysis

4.1 Experimental Environment Setup

The architecture is implemented using Keras 2.3.0 framework with Tensorflow 2.0 backend in Python 3.8.3. The model is trained on CPU instances with RAM 8 GB and core i-5 processor.

4.2 Results and Discussion

In [20] Wang et al. used Mean-Per-Class Error (MPCE) for multi-class time series classification. We also use MPCE to evaluate our model performance. MPCE is the average of Per Class Error (PCE). PCE of each class is calculated by Eq. (5).

$$\text{PCE} = \frac{1 - \text{accuracy}}{\text{Number of classes}} \tag{5}$$

The calculated MPCE of our model is 0.01386 applying hypotheses and 0.0494 without applying hypotheses. Wang et al. in [20] found MPCE = 0.0407 using the 85 UCR time series datasets.

Table 1 Evaluation score and Per-Class-Error (PCE)

Metric	Score	Class	Per Class accuracy	Per Class Error
Train accuracy	0.9997	Aggressive acceleration	0.985559	0.01444
Test accuracy	0.9720	Aggressive braking	0.992779	0.00722
Train Loss	0.1284	Aggressive LLC	0.995036	0.004963
Test Loss	0.2101	Aggressive RLC	0.953971	0.046028
Precision	0.9748	Aggressive LT	0.995938	0.004061
Recall	0.9602	Aggressive RT	0.995938	0.004061
F1-score	0.9670	Non-aggressive	0.983754	0.016245

Table 2 Comparison between other related papers

Paper	Classifier/model	Accuracy (%)
Proposed work	LSTM	97.20
Paper [16]	RNN	78.59
Paper [15]	LSTM and GRU	> 95
	SimpleRNN	70
Paper [1]	SVM	93.25
	BayesNet	91.1
	Logistic	89.3
	RBF network	85.55
	C4.5	84.75
	Naïve bayes	83.2
	K-NN	82.95

The evaluation scores and Per Class Error is shown in Table 1 and a comparison between other related work the proposed work is shown in Table 2 that indicates LSTM outperforms over other classifiers and deep learning models. The Accuracy and Loss are plotted in Figs. 11 and 12, respectively. The accuracy of the LSTM model without applying hypotheses was 96.2031 and loss 30.42 while test accuracy 97.20 and loss 21.01 with applying hypotheses. Applying hypotheses along with the extracted features set, the LSTM model needs 2 h to be trained with 500 epochs while without hypotheses the model requires 8 h with 2000 epochs to be trained with its best accuracy result for the same feature set. Therefore, our proposed model is well trained by applying data change rules-based hypotheses with ¼ of total training time than without applying hypotheses.

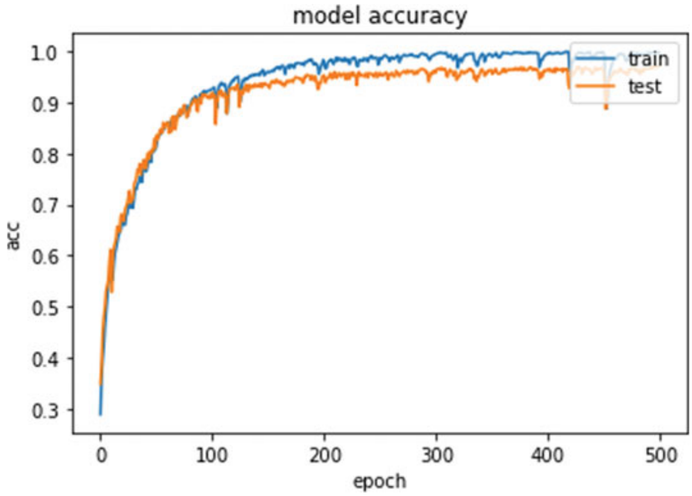

Fig. 11 Plotting train and test accuracy applying hypotheses

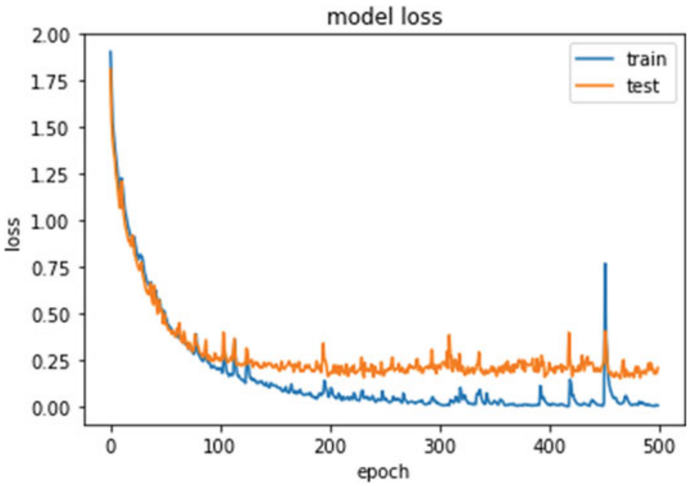

Fig. 12 Plotting train and test loss applying hypotheses

5 Conclusion

The proposed work discusses the significance of sensor fusion data change rules and utilized the idea with deep learning time series multi-class classification of driving maneuvers. We used an LSTM model because it is one of the most appropriate neural network models for the classification of time series data. We propose a few hypotheses which are proven by the experimental results. These significant rules accelerate the feature extraction process from driving data. Moreover, it helps in the automatic labeling of the unlabeled dataset. Our future approach is to develop a tool for generating categorical labels for unlabeled datasets. Besides, we have a plan to optimize the proposed classifier using a grid search.

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Chapter 39
A Novel Intrusion Detection System for Wireless Networks Using Deep Learning

L. Keerthi Priya and Varalakshmi Perumal

1 Introduction

The introduction of Wireless networks has made it convenient for the user to have mobility and improves productivity [8]. The communication among distinct devices utilizes a wide number of wireless networks. These include Wireless Local, Metropolitan, Wide Area Networks and Ad-hoc networks. Attackers can easily break the security of a wireless system and can access highly valued information and can misuse them. There are multiple causes to why the security system of wireless networks is vulnerable to attacks. The first and foremost is the volume of data. Due to commercialization of Internet, the data generated by each user is increasing rapidly. The storage of the data is very much necessary as it is used for providing better services for the user. Data storage has moved to the cloud from databases to satisfy the big data requirement. It is very tedious for the system to handle huge volumes of data. The working procedure of the system has to be revised and tuned to work with big data.

The second reason is that the network packets are not monitored in-depth as a small change in any of the parameters can launch an attack. The detection of such anomalous behavior is necessary so that similar activity can be prevented in the near future [16]. The proposed system utilizes deep learning mechanisms for the detection of malicious traffic. The modeling of highly complex concepts is facilitated using deep learning by utilizing various levels of representation. Machine learning algorithms could be stratified as supervised together with unsupervised learning

L. Keerthi Priya (✉) · V. Perumal
Madras Institute of Technology, Chennai, India

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procedures [4, 6]. Deep learning is of great help on considering IDS, as it helps to build a flexible, expandable and robust model which can be easily deployed for practical applications [7].

2 Existing Works

In [14] the author proposed a pristine deep learning model for recognition of malevolent activity that utilizes an auto-encoder that is non-symmetric. In [15], a neural network was developed by utilizing Gated Recurrent Unit, neural network—Multilayer Perceptron and Softmax function. The author carried out comparison between GRU and LSTM and proved that GRU incorporated neural network is an effective improvement of LSTM. In [11], an ensemble learning procedure for Intrusion detection was proposed that uses existing procedures such as Decision Trees, Naïve Bayes, Artificial Neural Networks to prevent attacks in IoT networks. This technique observed a high-rise in detection rate as well as a low false-positive rate. In [10], diverse machine learning methods were analyzed and a comparison was carried out based on the detection capacity of the models and the disadvantages of the machine learning techniques were discussed. In [12], a model for anomaly detection was proposed for IoT backbone-networks that uses Linear discriminant analysis together with component analysis for dimensional reduction. The classification was performed using a multilayer classification technique which utilizes Naïve Bayes and certainty factor of K-nearest neighbour. In [2], a survey was conducted for the various available internal intrusion detection systems and the systems which uses forensic methods and data mining with the idea that any model that is going to be proposed should be capable of working in real-time. The characteristics that have to be kept in mind while developing any novel IDS for a real-time system have been discussed. In [17], the author proposes a model by combining Deep Belief Neural Network and improved Genetic algorithm. This model was observed to be a compact one and is assumed to diminish the complexity of the structure of neural network.

In [3], an anomaly detection system was proposed which includes the deployment of GRU and LSTM models combined together in open flow controller. This model observed a decent accuracy of 87%. In [18], the author has proposed an unused approach towards detection of malicious activity using linear system theory for routing of re-configurable networks. A high accuracy for detection of attacks had been obtained without increasing packet overhead. In [5], IDS was proposed exclusively for IoT networks, support vector machine was utilized. It has been identified that traditional intrusion detection systems had observed some limitations when applied for IoT. The proposed SVM classifier has been combined with two or three non-complex features and has achieved satisfactory results. Most of the existing intrusion detection models were developed for the environment of various types of computer networks, but these models are not suitable for real-time intrusion detection. They take high training times and yield a high loss value. These models were validated on NSL-KDD dataset. In [1], a two-layer IDS was proposed for wireless sensor networks and mobile adhoc networks where data was collected by dedicated sniffers thereby generating correctly classified instances, that were then sent to super node for classification by linear regression. This model was tested on various extreme circumstances of the network using Random way point (RWP), and Gauss Markov (GM) mobility models. In [9], author provides a description of how deep learning tools are applied to biological data, and describes some challenges in biological data mining.

3 Proposed System

The system presents an Intrusion detection model which consists of a Customized Rotation Forest algorithm used for feature selection and a Gated Recurrent Neural Network for 2-class and 5-class classification of attacks. Figure1 shows detailed architecture diagram of the model. NSL-KDD dataset was utilized for testing cum evaluation of the model propounded.

Two models were built for the classification of attacks. The first model is a 2-class model which classifies the labels as normal or attack. The second model is a 5-class model that classifies the data labels into 4 attack classes namely, Probing, Dos attacks, User to root and Remote to local. The last class is Normal i.e. no attack. The results of both the models were compared and analyzed.

3.1 Feature Selection Using Customized Rotation Forest Algorithm

Feature Selection is a methodology to identify the highly necessary features from the data frames that contribute towards improvement of accuracy attained by the model and reducing the impurity. Customized Rotation forest algorithm was implemented to perform feature selection. The Rotation forest algorithm [13] is an ensemble methodology to build multiple classifier systems. F is the number of features existing in the dataset. The group of features is split into "k" subgroups in a random fashion. Principal Component Analysis will be applied to each of the created subgroups. A new feature set is obtained by combining the principle components obtained for each subgroup.

The Customized rotation forest algorithm uses Linear Discriminant Analysis (LDA) instead of Principal Component analysis (PCA) for creating a new feature set. LDA is supervised learning mechanism because it includes class labels while training whereas PCA excludes class labels and behaves in an unsupervised fashion. Since the proposed IDS is built with a supervised approach, it is better to use LDA than PCA. The main motive of PCA is to maximizing the variance in data.

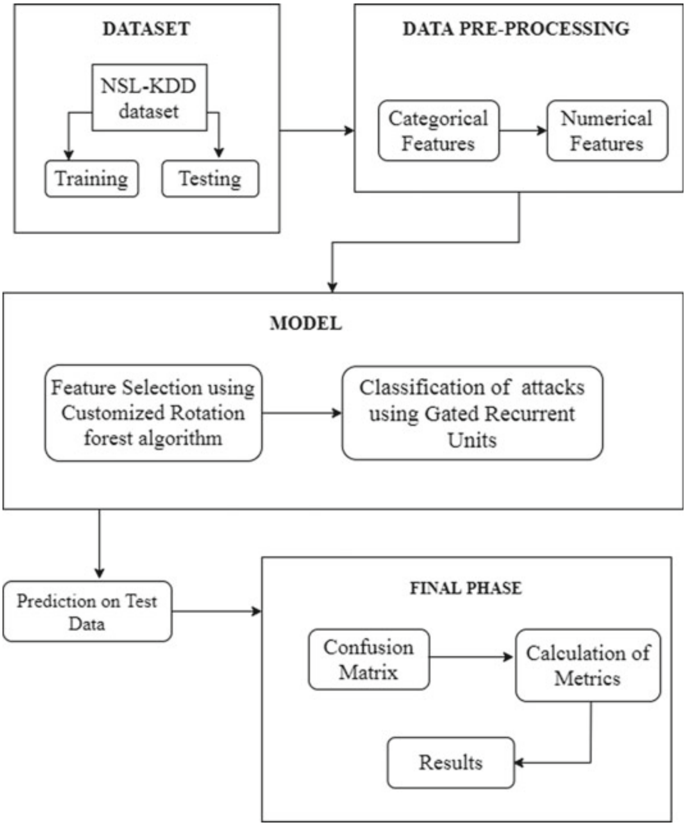

Fig. 1 Architecture of proposed system

Linear Discriminant Analysis, has a series of steps that has to be followed to compute the feature importance of all the features. Firstly, the mean vectors M_1 and M_2 are computed. The score for the separation of the two groups is given by Eq. 1

$$S(B) = \frac{B^T(M_1 - M_2)}{B^T C B} \tag{1}$$

where, B^T is the coefficient vector. Then, the matrix for computing covariance C^1 and C^2 between two groups is calculated. The upcoming step is to compute C which is pooled covariance matrix for two classes. n^1 is the number of records of first class and n^2 is the number of records of second class. It is computed by Eq. 2.

$$C = \frac{n_1 C_1 - n_2 C_2}{n_1 + n_2} \tag{2}$$

The plan of action of Customized Rotation forest algorithm is discussed in the series of steps below.

1. The group of features is split into K disjoint subgroups. Disjoint subgroups help us to achieve high diversity.
2. A non-empty subgroup is selected in a random fashion, from the subgroup of features and 80% of the given data is chosen for training.
3. The LDA algorithm is applied to each of the subgroup of features $F^{(i,j)}$ that consist of M features only. $C^{(-1)}$, the inverse of C—the pooled covariance matrix is computed.
4. The rotation matrix R^i is filled by the components acquired from C^{-1} after applying LDA.
5. The rotation matrix R^i is rearranged to the feature set and a matrix XR^{ai} is created, where X is the dataset.

3.2 Gated Recurrent Unit

Gated Recurrent Unit is a memory cell which is used to store the state of a node. Gated Recurrent Networks were introduced to overcome the difficulties of recurrent neural networks by gating the input signals. The gates are weighted, updated after every iteration. GRU is also considered a variation of LSTM-Long Short Term Memory as both are built similarly and also in some exceptional cases, produce equally amazing results. LSTM has three gates namely, output, input, and forget whereas GRU has update and reset gates. GRU combines the input and forget gates of LSTM. Unlike LSTM, the gated recurrent network does not use a memory cell to control the data flow; it directly makes use of all the hidden states. GRU uses less number of training variables and hence is more optimal than LSTM in terms of execution time and memory consumption.

4 Implementation and Results

KDDCUP'99 and NSL-KDD are widely utilized datasets for attack recognition research-work. The testing set contains 17 modern attack types that aren't included inside training set, the efficacy of the proposed algorithm can be evaluated in recognizing not known or uncommon attacks rightly. The various attacks present in the dataset are categorized and shown in Table 1. After analyzing NSL-KDD dataset, malefic behaviors in network-based intrusions could be differentiated into the ensuing four categories.

- Denial of Service: Legitimate network traffic is shut down due to flooding of illegitimate network packets.
- Probe: Stealing information from a network.

Table 1 Attacks in dataset

Category	Attack type
DoS	Back, land, neptune, pod, smurf, teardrop
Probe	Ipsweep, namp, portsweep, satan
R2L	Guesspassword, imap, multihop, phf, spy, warezclient, warezmaster
U2R	Bufferoverflow, rootkit, perl

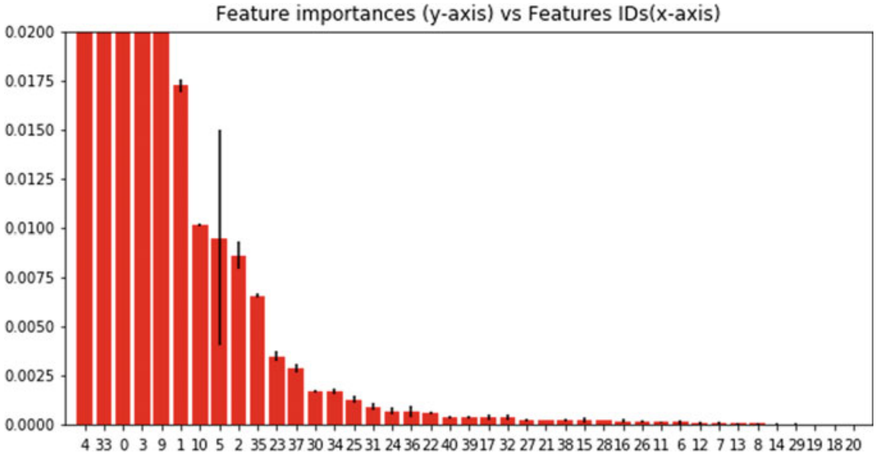

Fig. 2 Feature importance graph for 2-class model

- Remote to Local: Gaining access to a remote machine by hacking the network.
- User to Root: Gaining access to a network as super user (root).

The dataset consists of a training set which consists of 1.25 lakh training records and testing set which consists of 25,195 test records. The categorical data are converted into numerical data and normalized. Each record has 41 features and also a class label. The proposed model uses Customized Rotation forest for selecting important features. Figure 2 is the feature importance graph for 2-class. The x-axis describes the Feature IDs and the y-axis describes the significance of features.

The module was implemented using the Tensor flow platform in Anaconda Environment that uses python programming language. The Tensor flow platform is used for programming various machine learning applications. The scikit-learn library was imported for performing data processing tasks. Total of 25 significant features were chosen from this process. The proposed Intrusion Detection System uses Gated Recurrent Neural Network to classify the attacks into multiple classes. This module is implemented using keras library by importing the GRU module. The model was built using four neurons in input layer with a batch size-32. Adam optimizer, generally is used to improve accuracy by optimizing the parameters.

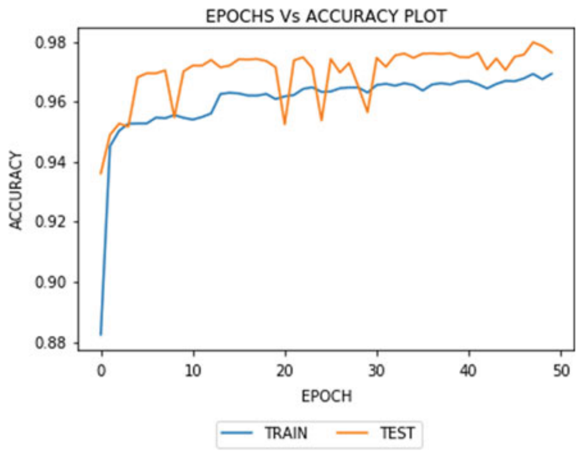

Fig. 3 Accuracy plot for 2-class model

This model classifies the records to 2 classes namely, Normal (1) or Attack (0). This is a model that performs binary classification as the records are split into two classes. Binary cross entropy loss function was used to compute the overall loss of the model. Another model classifies the records to 5 classes from zero to four in order namely, Normal, DoS, Probe, R2L and U2R. Categorical cross entropy loss function was used to compute overall loss of the model.

The 2 models were successfully built and trained on 1.25 lakh training samples and tested on 25,195 samples for 50 epochs. The model consists of 4 units of GRU and a fully connected layer. The 2-class model observed an accuracy of 97.319% and a cross entropy loss as low as 0.1044. The accuracy plot for 50 epochs is displayed in Fig. 3. The results are a living proof that the 2-class model is an efficient model with promising accuracy and loss values.

The other evaluation metrics like Recall, Precision and F1-Score are calculated for the 2-class model, the values for each class are displayed in Table 2. From the values of metrics calculated, it is inferred that the model is highly accurate and reliable as it has high values of recall, precision and F1-Score.

The 5-class model observed an accuracy of 97.478% and cross-entropy loss of 0.1243. The accuracy plot for 50 epochs is displayed in Fig. 4. The loss value is very low and accuracy value is high, which proves that the multi-class model using GRU is an efficient one.

Table 2 Validation metrics of 2-class model

Class	Precision	Recall	F1-score
Attack	0.99	0.95	0.97
Normal	0.96	0.99	0.98

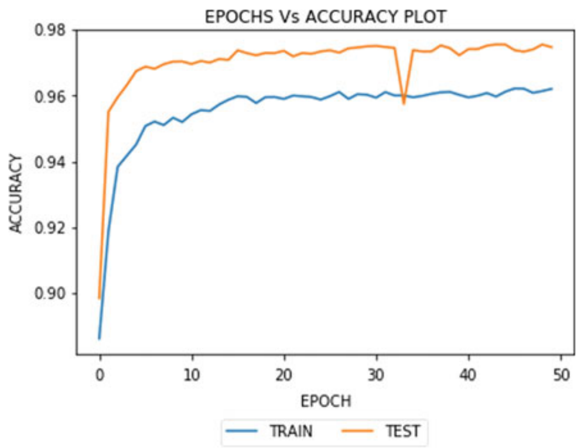

Fig. 4 Accuracy plot for 5-class model

Table 3 Validation metrics of 5-class model

Class	Precision	Recall	F1-score
Normal	0.96	0.99	0.98
DoS	0.99	0.98	0.98
Probe	1.00	0.94	0.97
R2L	0.70	0.75	0.74
U2R	0.56	0.68	0.69

Table 4 Comparison of proposed GRU model with other algorithms

Algorithm	Precision	Recall	F1-score
ANN	55	55	53
RNN	85	85	85
LSTM	95	94	93
GRU	99	95	97

The evaluation metrics were calculated for 5-class model and results are displayed in Table 3. The inference obtained from the values is that the Normal, DoS and Probe were detected correctly with high recall, precision and F1-score values. The R2L and U2R attacks were detected with less values of precision because of the minimum number of records in the dataset that are not sufficient for training a model.

A comparison was carried out between the proposed Gated Recurrent Unit model and the other neural network models like Artificial Neural Network (ANN), Recurrent Neural Network (RNN), Long Short Term Memory (LSTM). The results are tabulated in Table 4. It clearly proves that the proposed system performs better than other neural network algorithms.

5 Conclusion

From the evaluations of the proposed system, it is confirmed that both the models have performed well with promising results. In this paper, issues about recurrent neural network were identified and rectified by proposing a new supervised learning methodology. The proposed model outperforms other previous models by attaining 0.1044 binary cross-entropy loss and 0.1243 categorical cross-entropy loss. These models could be used as a reference to implement a real-time system that detects intrusions to a real-time network and deals with live network traffic and real-time attacks. In future work, the first step of research for improvement would be to capture more data for R2L and U2R attacks and expand the proposed model to detect those attacks with good predictability.

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Chapter 40
A Genetic Algorithm-Based Optimal Train Schedule and Route Selection Model

Md. Zahid Hasan, Shakhawat Hossain, Md. Mehadi Hassan, Martina Chakma, and Mohammad Shorif Uddin

1 Introduction

Railway has been a very popular public transport all over the world. People around the world mostly depend on train journeys for accomplishing their day-to-day activities. So, this public transport is required to be faster and comfortable [1]. Unfortunately, this public demand is merely met by most of the railway transportation systems. Besides, nowadays, train collision and other unexpected occurrence have been a regular issue in train journeys. On the other hand, the demand for transportation is growing more rapidly due to the increase number of population which creates additional traffic overflow. So, to handle this traffic overflow, trains need to be operated properly by considering routes and schedule [2].

Therefore, researches around the world are focusing on optimizing train route selection system as well as optimal scheduling. Garrisi et al. proposed a model for optimizing the schedule of trains on railway networks composed of busy complex stations using genetic algorithm [3]. In another study, genetic algorithm is introduced

Md. Zahid Hasan (✉) · Md. Mehadi Hassan · M. Chakma
Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh
e-mail: zahid.cse@diu.edu.bd

Md. Mehadi Hassan
e-mail: mehadi15-8399@diu.edu.bd

M. Chakma
e-mail: martina15-8354@diu.edu.bd

S. Hossain
Department of Computer Science and Engineering, International Islamic University Chittagong, Chattogram, Bangladesh
e-mail: shakhawat.cse@outlook.com

M. S. Uddin
Department of Computer Science and Engineering, Jahangirnagar University, Dhaka, Bangladesh

M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems,
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for identifying the optimal travel frequency for impacted transportation services including rail, bus, and van, to reschedule each traveling option under the intermodal cooperation model [4].

Fletcher and his co-researchers presented a model combined with a genetic algorithm to optimize system parameters (storage size, charge/discharge power limits, timetable, and train driving style/trajectory) [5]. Shang et al. introduced a model that focuses on the time table optimization for controlling metro rail safely [6]. A transit network is intended by Chai and Liang [7] using NSGA-II algorithm. Again, Perea et al. and Chouhan et al. proposed an enormous design for transportation routing network [8, 9] using genetic algorithm. Wang et al. proposed a train set circulation plan, while Johar et al. developed transit network design using genetic algorithm [10, 11]. A research on ship's route scheduling is done using genetic algorithm by Wijayaningrum et al. [12].

Multi-criteria train routing scheduling and optimization is implemented by Sun et al., and later Dib et al. also worked on the multi-objective train routing network [13, 14]. In the same year, a developed railway scheduling model is postulated by Nirmala and Ramprasad using genetic algorithm [15]. Issue is all of these proposed models bring out solutions for train route selection problems for a specific country and environment. So, route selection problems are hardly dealt in case of Asian countries. However, this paper proposes a model that optimizes train routing approach as well as train schedule by using genetic algorithm which will lead to safe train services.

The brief overview of this paper is composed as follows. Essential steps are demonstrated in Sect. 2 for the better understanding of the genetic algorithm. The numerical experiments are done in Sect. 3. Section 4 is the result analysis part which demonstrates the test cases. And Sect. 5 concludes the paper.

2 Methodology

The genetic algorithm is a robust optimization framework that uses the concept of biological genetic sequence to solve any classical problems. The genetic algorithms perform the optimization operations in several steps. The following section describes the genetic algorithm in four significant steps which is shown in Fig. 1.

Step 1: Initial Population

At the very beginning level, the genetic algorithm needs to be fed with some initial inputs which in the other way is known as initial population (see Fig. 2). These inputs are nothing but a set of sequences of chromosomes which are later updated.

Step 2: Selection of the chromosome or parent according to the fitness function

From different input chromosomes, only the best chromosomes are selected according to their fitness function. Fitness function denotes the strength and non-attacking pair of the given chromosomes. Non-attacking pair of chromosomes can be evaluated using the 8-queen problem or 6-queen problem or so on. In the initial

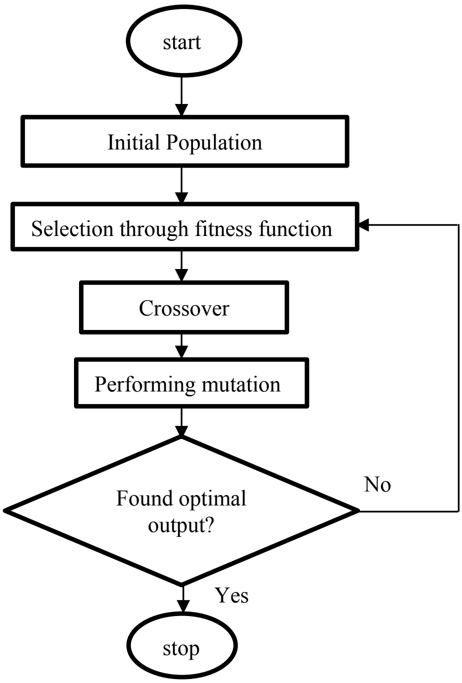


Fig. 1 Flowchart of the proposed optimization model

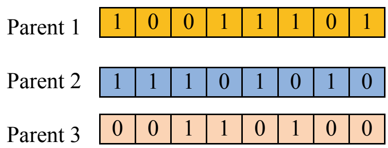


Fig. 2 Initial population

population, three input sets are provided whose fitness functions are calculated as 8%, 6%, 3%. Here, only the parents with 8% and 6% fitness are considered, and the rest one is boycotted due to its insufficient fitness. So, the selected chromosomes are (Fig. 3).

Step 3: Making crossover

Crossover is the most important part to develop the chromosomes. Through crossover, a new offspring or child is developed which may be used to find the optimal output. Crossover is performed between two different parents. The purpose of the crossover is to generate new child/offspring. The process of crossover is shown below (Fig. 4).

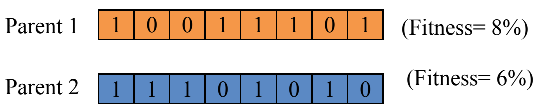


Fig. 3 Fitness function calculation

Fig. 4 Process of crossover

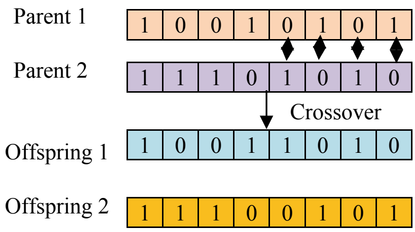
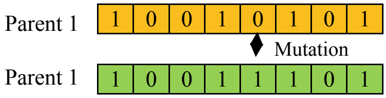


Fig. 5 Process of mutation



Step 4: Performing mutation

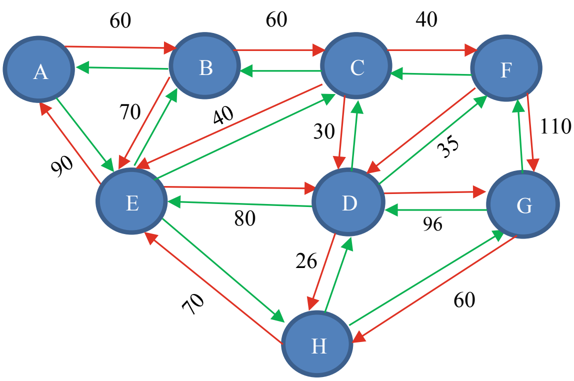
In the genetic algorithm, mutation is performed between two parents by replacing one or more genes of a parent by the genes of other parents. The process of mutation is given in Fig. 5.

3 Numerical Experiments

To understand the practical implementation of the genetic algorithm, a numerical experiment over the Dhaka Metro Rail junction is conducted in this section. The train route in the considered junction is demonstrated using the following directed graph (Fig. 6).

In the considered junctions, the routing points toward different terminals constructing the initial population stage can be stated as,

Fig. 6 Proposed routing model



State 1: BDGDHEEB

State 2: CBGEBDAA

State 3: BDDAEABD

State 4: CBEDCBAC

From the initial population, the non-attacking pair for each of the above state can calculated as follows (Tables 1, 2, 3 and 4).

Based on the number of non-attacking pair, the fitness function for each stated can be calculated as follows (see Table 5):

Because of insufficient non-attacking pair, some states are dropped, i.e., state 4. And the backward and forward route for the rest states can be calculated as follows:

To find the optimal forward routing route, different crossover and mutation is performed as follows over the non-attacking terminals from state 1 and 2 (See Fig. 7).

Table 1 Non-attacking pair calculation for State 1 (BDGDHEEB)

B	D	G	D	H	E	E	B
B							B
	D		D				
					E	E	
		G					
				H			

Train	Non-attacking pair	Non-attacking terminal
q1	DGDHEE	6
q2	BGHEEB	6
q3	BDDHEE	6
q4	BGHEEB	6
q5	BDGDEEB	7
q6	BDGDHB	6
q7	BDGDHB	6
q8	DDHEE	5

Table 2 Non-attacking pair calculation for State 2 (CBGEBDAA)

C	B	G	E	B	D	A	A
						A	A
	B			B			
C							
					D		
			E				
		G					

Train	Non-attacking pair	Non-attacking terminal
q1	GEBDAA	6
q2	GEDAA	5
q3	CBEBAA	6
q4	CBGBDA	6
q5	CGEDAA	6
q6	CBEBAA	6
q7	CBGEBD	6
q8	CBGBD	5

Total = 46/2 = 23 pair

Table 3 Non-attacking pair calculation for State 3 (BDDAEABD)

B D D A E A B

			A		A		
B						B	
	D	D					D
				E			

→

Train	Non-attacking pair	Non-attacking terminal
q1	DAEAD	5
q2	BAEAB	5
q3	AEB	3
q4	BDDEBD	6
q5	BDDAABD	7
q6	BDED	4
q7	DDAED	5
q8	BAEAB	5

Total = 40/2 = 20 pair

Table 4 Non-attacking pair calculation for State 4 (CBEDCBAC)

C B E D C B A

						A	
	B				B		
C				C			C
			D				
		E					

→

Train	Non-attacking pair	Non-attacking terminal
q1	DBA	3
q2	ECAC	4
q3	BC	2
q4	CC	2
q5	B	1
q6	CC	2
q7	CBC	3
q8	BEDBA	5

Total = 22/2 = 11 pair

Table 5 Fitness function calculation

States								Non-attacking pair	Fitness
B	D	G	D	H	E	E	B	24	31%
B	D	G	D	H	E	E	B	23	29%
B	D	G	D	H	E	E	B	20	26%
B	D	G	D	H	E	E	B	11	14%

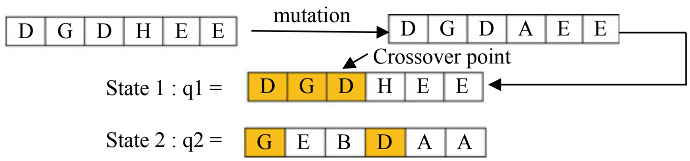


Fig. 7 Crossover and mutation for forward routing

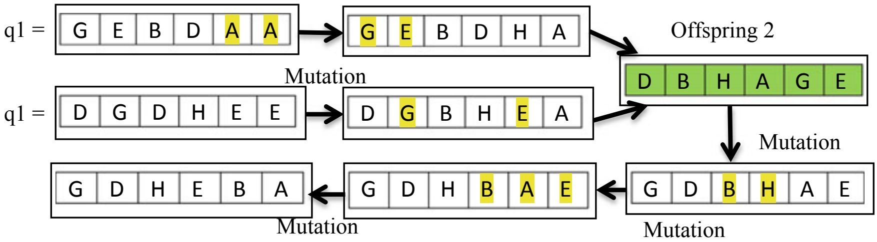


Fig. 8 Crossover and mutation for backward routing

Again working from state 1 and 2: Performing crossover and mutation for backward route calculation with the help of non-attacking pair from chromosomes set 1 and 2 (see Fig. 8).

So, the evaluated forward and backward routing route for train q1,

A	B	E	D	G

Forward route 1:

G	D	H	E	B	A

Backward route 1:

Thus, the forward and backward routing route for the other trains can be evaluated and represented in Table 6.

Deep analysis is performed using genetic algorithm for the railway scheduling, and the final dataset is obtained (see Tables 7 and 8).

4 Result Analysis

The proposed system has been tested for 1124 times with 50 different datasets collected from Bangkok railway station (See Fig. 5). Each dataset contains train movements of five distinct stations for 24 h of 30 days. The datasets were wrangled properly for achieving the maximum accuracy (see Table 9).

Table 6 Forward and backward routing for different route and trains

Train	Route	Forward Route	Backward Route
q2	1	A B C D G	G H E B A
q3	1	A B E C D G	G F C B E A
q4	1	A E H D F G	G F C E B A
q3	1	G F C E B A	G H E C B A
q6	1	A E H G	G H E A
q7	1	A B C F G	G D E B A
q8	1	A E C D G	G H E B A
q1	2	A B C D G	G H E B A
q2	2	A B E D G	G D H E B A
q3	2	A E H D F G	G F C E B A
q4	2	A B E C D G	G F C B E A
q5	2	A E H G	G H E A
q6	2	A B C E D H G	G H E C B A
q7	2	A E C D G	G H E B A
q8	2	A B C F G	G D E B A

The above data was split to train and test the proposed system on a ratio of 60% by 40%. Every time the system suggested the rail route was recorded in a graphical model. The generated graphs were converted into numerical data and then compared to the benchmark ones provided by the Bangkok Railway Authority. The compared results show that the proposed system performs 98.5% accurately in each of the 1124 times. In comparison with another two methodologies such as Tabu search and shifting bottleneck that provide 89% and 91% accuracy for considering the same dataset, respectively (see Table 10), the proposed system is found optimal (Fig. 9).

5 Conclusion

This paper represents an optimized method for train scheduling as well as solving two way railway route selection problem. Railway route selection and management are the important task for providing better service to the passengers. Due to the lack of

Table 7 Planned schedule for metro rail routing

Route index no	Train index no	Forward routing								Backward routing							
		Starting point	Terminals/stations						Designation	Starting point	Terminals/stations						Destination
		A	B	C	D	E	F	H	G	G	H	F	E	D	C	B	A
Route 1	Train q1	6:00	6:14	–	6:50	6:31	–	–	7:13	7:30	8:03	–	8:20	7:53	–	8:37	8:51
	Train q2	6:17	6:29	6:41	6:47	–	–	–	7:06	7:23	7:50	–	8:04	–	–	8:18	8:30
	Train q3	6:34	6:45	7:06	7:12	6:58	–	–	7:30	7:47	–	8:08	8:40	–	8:16	8:27	8:57
	Train q4	6:51	–	–	7:38	7:12	7:46	7:28	8:11	8:28	–	8:53	9:11	–	9:02	9:27	9:41
	Train q5	7:08	7:20	7:40	8:03	7:48	–	8:11	8:37	8:54	9:20	–	9:34	–	9:42	9:54	10:06
	Train q6	7:25	–	–	–	7:42	–	7:56	8:22	8:39	9:05	–	9:19	–	–	–	9:36
	Train q7	7:42	7:57	8:12	–	–	8:22	–	8:50	9:07	–	–	9:51	9:31	–	10:09	10:24
	Trainq8	7:59	–	8:27	8:33	8:18	–	–	8:54	9:11	9:40	–	9:55	–	–	10:10	10:23
Route 2	Train q1	6:00	6:14	6:28	6:35	–	–	–	6:58	7:15	7:47	–	8:04	–	–	8:21	8:35
	Train q2	6:17	6:29	–	6:59	6:43	–	–	7:18	7:35	8:02	–	8:16	7:54	–	8:30	8:42
	Train q3	6:34	–	–	7:12	6:51	7:19	7:04	7:40	7:57	–	8:18	8:34	–	8:26	8:47	8:58
	Train q4	6:51	7:05	7:30	7:37	7:21	–	–	7:59	8:16	–	8:41	9:20	–	8:50	9:04	9:41

(continued)

Table 7 (continued)

Route index no	Train index no	Forward routing								Backward routing							
		Starting point	Terminals/stations						Designation	Starting point	Terminals/stations						Destination
		A	B	C	D	E	F	H	G	G	H	F	E	D	C	B	A
	Train q5	7:08	–	–	–	7:25	–	7:39	8:05	8:22	8:48	–	9:02	–	–	–	9:19
	Train q6	7:25	7:37	7:49	8:12	7:57	–	8:20	8:46	9:03	9:29	–	9:43	–	9:51	10:03	10:15
	Train q7	7:42	–	8:15	8:23	8:05	–	–	8:47	9:04	9:38	–	9:56	–	–	10:14	10:29
	Train q8	7:59	8:12	8:25	–	–	8:34	–	8:58	9:15	–	–	9:53	9:36	–	10:08	10:21

Table 8 Details of train information

Index of train	Max speed/h	Train weight (tons)
Train q1	250	600
Train q2	300	580
Train q3	320	560
Train q4	260	700
Train q5	310	600
Train q6	310	550
Train q7	240	600
Train q9	280	700

Table 9 Sample data table

Date	Time	Terminal 1	Terminal 2	Terminal 3	Terminal 4	Terminal 5	Terminal 6	Terminal 7
01.03.17	4.30	1	1	1	0	0	0	0
01.03.17	4.30	0	0	0	1	1	1	1
01.03.17	4.30	0	0	0	1	1	1	1
01.03.17	4.30	1	1	1	0	0	0	1
01.03.17	4.30	0	0	0	1	1	1	0
01.03.17	4.30	0	1	1	0	0	0	0
01.03.17	4.30	1	0	0	1	1	1	0

Table 10 Accuracy comparison of different algorithms

Tabu search	Shifting bottleneck	Genetic algorithm (proposed)
89%	91%	98.5%

optimal train route selection process, every year a huge number of train clashes occur throughout the world. So, it becomes important to optimize the train route selection process. To optimize the existing train schedule and route selection approaches, it is important to understand the different criteria and aspects of train route schedule-related problems. This works considered Dhaka Metro Rail to analyze the time- and route-related problems and then proposes genetic algorithm-based optimized train route selection approach. The proposed system recommends train route based on current trains' position. The system has been tested with different historical data, and its accuracy is measured 98.5%. Moreover, the accuracy performance of the proposed genetic algorithm for the same dataset is significantly better than other existing methods like Tabu and shifting bottleneck.

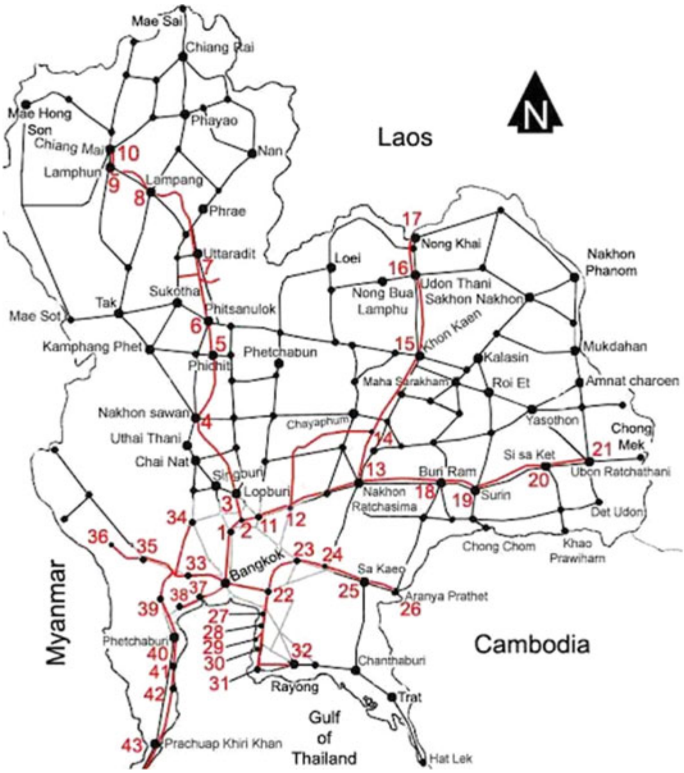


Fig. 9 Bangkok train route

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Chapter 41
Fuzzy Rule-Based KNN for Rainfall Prediction: A Case Study in Bangladesh

Md. Zahid Hasan, Shakhawat Hossain, K. M. Zubair Hasan, Mohammad Shorif Uddin, and Md. Ehteshamul Alam

1 Introduction

Rainfall, the most focused climatic factor has a direct impact on human life. Human's daily activities are mostly controlled by this environmental factor. Agricultural production, construction, and power generation industry consider it as the most important business regulating factor. Forestry, tourism and sports industry always need to consider rainfall probability as a decision-making parameter. Farmers, fishermen and other professionals must concentrate on the probabilistic criteria of this natural phenomenon.

Rainfall is very often responsible for some other natural disasters. Landslides, flooding, mass movements and avalanche are the common outcomes of heavy rainfall [1]. In Bangladesh, rainfall is a common reason behind many other natural disasters [2]. According to the statistics, on 11 June 2007, heavy rainfall caused landslides in Chittagong. That disaster engulfed the people taking shelter around the hilly areas of the city [2]. In that year, 298 people died and a total of 10,211,780 people were badly affected by a devastating flood which was only caused by heavy rainfall [2]. Every

Md. Zahid Hasan (✉) · K. M. Zubair Hasan · Md. Ehteshamul Alam
Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh
e-mail: zahid.cse@diu.edu.bd

Md. Ehteshamul Alam
e-mail: ehteshamul15-9313@diu.edu.bd

S. Hossain
Department of Computer Science and Engineering, International Islamic University Chittagong, Chattogram, Bangladesh
e-mail: shakhawat.cse@outlook.com

M. S. Uddin
Department of Computer Science and Engineering, Jahangirnagar University, Dhaka, Bangladesh

M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems,
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year, the people of Bangladesh have to measure some damages because of sudden rainfall.

Therefore, predicting rainfall has become a crucial issue to prevent measures for many natural disasters and so, Meteorologists all over the world have a great concern about making an appropriate framework for predicting rainfall [3]. In that consequence, a number of methodologies have been proposed at different times to forecast rainfall and almost all of them were conducted manually. But recently, some computer-aided system has been proposed by some researchers [1–4] which are capable of predicting rainfall in most of the cases. A very straightforward comparison among different rainfall prediction algorithms is shown by Kumar and Ramesh and they claimed that KNN provides the maximum prediction accuracy for a certain period of time [4]. The prediction accuracy of KNN decreases for a very big amount of data as this algorithm is unable to capture fuzziness while handling large amount of data. However, the proposed Fuzzy Rule-based KNN approach is capable of handling all types of fuzziness that come from a high-volume dataset and thus, capable of securing the maximum accuracy during rainfall prediction assessment.

The proposed system considers Minimum Temperature, Maximum Temperature, Dew Point Temperature, Humidity, Cloud Amount, Sea Level Pressure, and Wind Speed [5] to construct the knowledge base. Historical datasets are collected from Bangladesh Meteorological Department, Climate Division. A numerical study is provided to prove the accuracy of the proposed system.

2 Related Works

Although rainfall prediction is an old task, its accuracy is still not satisfactory. Researchers have been trying various approaches to predict rainfall accurately. Recently, some machine learning approaches are being used to predict rainfall; most of them are using historical data. Yeon et al. [6] considered temperature, wind direction, speed, gust, humidity, pressure to predict rainfall. The system was developed by using Decision Tree and could count its accuracy up to 99%.

Kannan and Ghosh used Decision Tree and K-Means Clustering [7] to predict rainfall. A 50 years of data set on Temperature, Mean surface level pressure (MSLP), pressure, wind speed, rainfall was precisely prepared and used for this purpose. They implemented both supervised and unsupervised approaches to visualize prediction accuracy of their proposed the framework.

Many researchers believe that rainfall predication can be done accurately if a prediction mode can be built by using Artificial Neural Network while some scientists suggested Modular Artificial Neural Network [8]. Somvanshi et al. (2006) used a 103 years data set on Humidity, Min–Max temperature to achieve the maximum accuracy in rainfall prediction. He combined Auto-Regressive Integrated Moving Average (ARIMA) and Artificial Neural Network (ANN) to make a hybrid rainfall prediction framework [9]. In 2010, Htike and Khalifa proposed Focused Time Delay Neural Network for rainfall prediction [10]. For that, only temperature, solar radiation and evaporation are considered as the decision attributes. But later in 2011, in order to forecast rainfall Geeta and Selvaraj calculated Wind speed, mean temperature, relative humidity, and aerosol values by using Multilayer Back Propagation Neural Network [11]. A year later in 2012 Deshpande showed that Elman neural network performs better than all the remaining methods in case of forecasting rainfall [12]. This system requires the historical data only on rainfall data to predict rainfall accurately [12]. But recently in 2014, Dabhi and Chaudhary claimed that the combination of Wavelet-postfix-GP model and Wavelet ANN provides the maximum prediction accuracy [13].

On the other hand, many researchers opposed all the hybrid modes and suggested to use Regression Models to predict rainfall. Kannan et al. used predicts rainfall from the historical data on Min–Max temperature, wind direction, humidity, and rainfall with the help of Regression model [14]. And, 3 years later, in 2014, Dutta and Tahbilder also agreed with Dabhi on the point that rainfall prediction can be better performed by using regression model [15].

Some researchers depend on some unsupervised method rather than using the traditional supervised method to predict rainfall properly. Vamsi Krishna proposed K-means clustering independently to predict rainfall and showed that, the proposed method is capable of scoring the maximum performance [16]. The system was developed based on cloud status on the sky and the cloud images were handled by Gaussian Mixture Model [16]. Amarakoon used KNN to predict rainfall from Temperature, humidity, precipitation, wind speed [17] where Jan et al. considered wind speed, dew point, seal level, snow depth and rain to predict rainfall by using KNN [18].

In 2017, Fathihah Azahari et al. used sliding window algorithm (SWA) to predict rainfall accurately [19]. This is a new machine learning approach which was first introduced by Kapoor and Bedi to predict weather condition in 2013 [20], capable of providing the maximum prediction accuracy. Some researchers used Support Vector Machine [21] algorithms to predict rainfall where some believe that Support Vector Machine along with Relevance Vector Machine [22] can provide the maximum accuracy in case of rainfall prediction. Nowadays some scientists use deep learning algorithms to predict daily rainfall [23]. Again, some researchers use some advanced machine learning techniques for rainfall forecasting. Kishtawal et al. proposed genetic algorithm to predict rainfall [24]. A few hybrid techniques are also proposed to predict rainfall. Shabib Aftab et al. used a new hybrid technique to predict rainfall which actually a combination of Support Vector Machine (SVM), Naïve Bayes (NB), k-Nearest Neighbor (kNN), Decision Tree (J48), and Multilayer Perceptron (MLP) [25]. Some researcher used both Fuzzy set theory and Rough Set theory at a time to secure the maximum prediction accuracy while predicting rainfall [26].

3 Data

The necessary dataset required for training the proposed system was provided by Bangladesh Meteorological Department, Climate Division. The collected data contains total eight attributes including Minimum Temperature, Maximum Temperature, Dew Point Temperature, Humidity, Cloud Amount, Sea Level Pressure Wind Speed, and Rainfall. A detailed description of the collected data is provided in Table 1 and the nature of the collected data are given in the Table 2.

Bangladesh stays in a monsoon climatic geographic zone and so, the climatic parameters of Bangladesh differs from season to season. In accordance with the variation in the climatic parameters, climatic year is classified into four meteorological seasons (Table 3). Each of these seasons consists of three consequents months.

Bangladesh Meteorological Department (BMD) showed that Rainfall mostly depends on Temperature, Humidity, Cloud Amount, Sea Level Pressure and Wind Speed as well as Wind Direction. To forecast rainfall properly, Temperature needs to be calculated from different aspects say Maximum Temperature, Minimum Temperature and Average Temperature. Again, Dew Point Temperature and Humidity play a vital role for causing rainfall. Besides, Wind Speed and sea level pressure regulate rainfall in most of the cases (Fig. 1).

Although Wind Direction and Temperature are two different climatic issues, researchers argue that Wind Direction is not important to consider as a rain forecasting attribute if Temperature is calculated. In practical, Temperature regulates

Table 1 Data source information

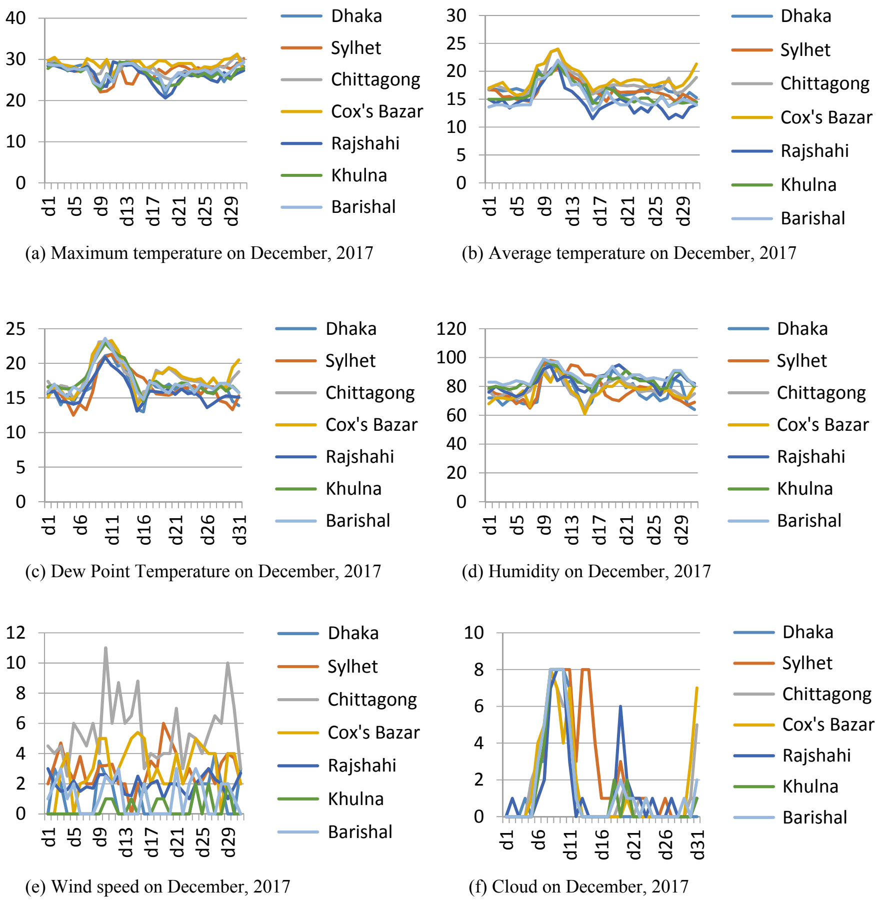
Data source	Bangladesh meteorological department, climate division	
Data collection period	Year	1988–2017
	Month	January–December
	Day	Daily
No. stations		34
Data type		Numeric
Total attribute		9 (Nine)
Dataset format		Microsoft Excel

Table 2 Overview of collected data

Attribute	Type	Measurement unit
Daily minimum temperature	Numeric	°C
Daily maximum temperature	Numeric	°C
Daily average temperature	Numeric	°C
Daily dew point temperature	Numeric	°C
Daily average humidity	Numeric	%
Daily total cloud amount	Numeric	Okta
Daily mean sea level pressure	Numeric	Millibar
Daily prevailing wind speed	Numeric	Knots
Daily total rainfall	Numeric	Millimeter

Table 3 Different outlook variations in metrological seasons in Bangladesh

Meteorological seasons	Months under season	Outlook
Winter	December, January, February	Cool and dry
Pre-monsoon	March, April, May	Humid
Monsoon	June, July, August	Humid and rainy
Post-monsoon	September, October, November	Quiet hot and dry



(a) Maximum temperature on December, 2017

(b) Average temperature on December, 2017

(c) Dew Point Temperature on December, 2017

(d) Humidity on December, 2017

(e) Wind speed on December, 2017

(f) Cloud on December, 2017

Fig. 1 Change of climatic factors at different locations in Bangladesh

the direction of Wind. So, to build the most sophisticated forecasting engine, Wind Direction is not considered in the proposed system.

Again, the Average Temperature provided by the Meteorological Department is not used to train up the system as the Average Temperature is the average value of

Minimum Temperature and Maximum Temperature and it does not have an important impact on rainfall forecasting criteria.

4 Methodology

Fuzzy Rule-based KNN [27] is a hybrid mathematical framework consisting of two basic mathematical theories: Fuzzy Set Theory and K Nearest Neighbor (KNN). Fuzzy Rule-based KNN works with some traditional if–then rules where vagueness of data is captured by Fuzzy Logic and a decision algorithm is formed by KNN based on rules provided by Fuzzy Set.

4.1 Fuzzy Set Theory

Membership Function Representation. Fuzzy set is defined as a membership function where the values assigned to the elements of the universal set fall within a specified range and shows the membership grade of their elements in the set (Table 4).

Most commonly used range of values of membership functions is the unit interval [0, 1].

The membership function of a fuzzy set A is presented as, $A: X \rightarrow [0, 1]$.

There are four basic types of fuzzy set based on data representation. The proposed system uses fuzzy sets for representing linguistic concepts such as low, medium, high or other fuzzy variables shown in Fig. 2 [28].

This type of fuzzy set can be described by a Trapezoidal function. The membership function used in this system can be mathematically defined as:

Table 4 Data Labeling for fuzzy membership function formation

Rainfall (mm)	Fuzzy Linguistic term
<1	No rain
1–10	Light rain
11–22	Moderate rain
23–43	Moderately heavy rain
44–88	Heavy Rain
89–99	Very heavy rain
100–199	Very heavy rain with 100–199 mm
200–299	Very heavy rain with 200–299 mm
>300	Very heavy rain with greater than 300 mm

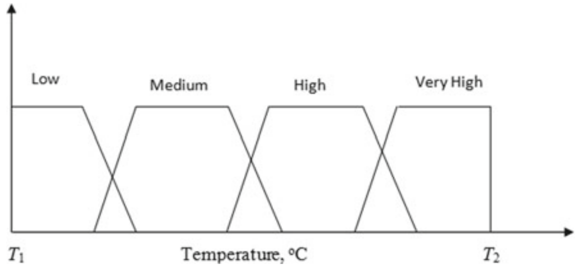


Fig. 2 Temperature in range $[T_1, T_2]$ conceived as Fuzzy variables

$$M(x;\, a,\, b,\, c) = \begin{cases} 0, & (x < a) \text{ or } (x > d) \\ \frac{x-a}{b-\alpha}, & a \le x \le b \\ 1, & b \le x \le c \\ \frac{d-x}{d-c}, & c \le x \le d \end{cases} \tag{1}$$

Here, a is the lower limit and d is the upper limit. B and c are two consecutive support limits where, $a< b < c < d$.

Rule Generation A rule-based system is constructed with some if–then rules and an inference engine. The basic if–then rules are formed from based on a complex analysis of the historical data provided by Bangladesh Meteorological Department. The traditional if- then rules can be presented as:

$$R = \{A,\, V,\, D,\, F\} \tag{2}$$

where, R represents the set of rules. A is the set of attribute and V is its associate values. D defines set of the final decision values where F demonstrates the logical function. In the proposed system, D is represented as a Membership Function. Equation (2) can be elaborately explained as

$$R_i = \left\{A_j,\, V,\, D_k,\, F_b\right\} \tag{3}$$

where, i is the number of rules; $1 \le \text{i} \le \text{n}$ and n is the total number of rules. K is the number of member in a fuzzy membership function. $K = 8$; $b = 2$ ($\wedge$ and $\vee$).

For example,

$$R_1 = \text{if } (A_1\text{is } 0.4) \wedge (A_2\text{is } 12.3) \wedge (A_3\text{is } 3.3) \wedge (A_4\text{is } 0.93) \text{ then } D \text{ is High} \tag{4}$$

Equation (4) can also be represented as,

$$R_1 = \text{if } (A_1,\, 0.4) \wedge (A_2,\, 12.3) \wedge (A_3,\, 3.3) \wedge (A_4,\, 0.93) \text{ then } (D,\, \text{High}) \tag{5}$$

4.2 K Nearest Neighbor

KNN is a supervised learning algorithm mostly used for classification predictive problems [29, 30]. This classification algorithm does not make any theoretical assumption on underlying data. The model structure is formed based on dataset. K nearest neighbor algorithm learns from a preprocessed dataset and classifies new cases using distance functions. Based on dataset distance function can vary. Euclidean distance, Manhattan distance, Minkowski distance can be used for continuous variable where categorical variable requires Hamming distance. However, the proposed system implemented KNN using Euclidean distance [31].

$$\text{Euclidean distance} = \sqrt{\sum_{i=1}^{k}(x_i - y_i)^2} \tag{6}$$

Equation (6) can be more easily expressed when the number of attributes is fixed. For example, if i = 3 then, Eq. (6) becomes,

$$\text{Euclidean distance} = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2} \tag{7}$$

The classification task for any given case is accomplished by the majority votes of its neighbors. If $k = 1$ then the candidate class will be assigned to class of its nearest neighbor.

5 Numerical Study

A numerical example is studied in this section to explain how fuzzy rule-based KNN approach forecast rainfall from a historical dataset. This example also conducts a cross-validation to prove its result accuracy.

The following variables are considered to make the forecasting engine.

1. X_1 = Minimum Temperature;
2. X_2 = Maximum Temperature;
3. X_3= Dew Point Temperature;
4. X_4= Cloud Amount;
5. X_5 = Humidity;
6. X_6 = Sea Level Pressure;
7. X_7 = Wind Speed;
8. X_8 = Rainfall.

To illustrate the forecasting approach, necessary if–then rules are constructed using the observed data collected in November, 2017 from the Ishwardi Station (Tables 5 and 6).

Table 5 Fuzzy Membership Construction

R_1 = if $(X_1, 26.8) \wedge (X_2, 31.7) \wedge (X_3, 26.6) \wedge (X_4, 7) \wedge (X_5, 87) \wedge (X_6, 1003) \wedge (X_7, 1.3)$ then, $(X_8$, Light rain)
R_2 = if $(X_1, 26.8) \wedge (X_2, 32.8) \wedge (X_3, 26.3) \wedge (X_4, 8) \wedge (X_5, 86) \wedge (X_6, 1002.3) \wedge (X_7, 1)$ then, $(X_8$, No rain)
R_3 = if $(X_1, 26.7) \wedge (X_2, 34) \wedge (X_3, 25.6) \wedge (X_4, 6) \wedge (X_5, 86) \wedge (X_6, 1003.7) \wedge (X_7, 2)$ then, $(X_8$, Light rain)
R_4 = if $(X_1, 26.8) \wedge (X_2, 34.8) \wedge (X_3, 26.3) \wedge (X_4, 2) \wedge (X_5, 83) \wedge (X_6, 1004.3) \wedge (X_7, 1.2)$ then, $(X_8$, Light rain)
R_5 = if $(X_1, 27.2) \wedge (X_2, 35.3) \wedge (X_3, 26.1) \wedge (X_4, 6) \wedge (X_5, 80) \wedge (X_6, 1004.5) \wedge (X_7, 1.8)$ then, $(X_8$, No rain)
R_6 = if $(X_1, 26.8) \wedge (X_2, 35.7) \wedge (X_3, 26.4) \wedge (X_4, 5) \wedge (X_5, 84) \wedge (X_6, 1004.9) \wedge (X_7, 1)$ then, $(X_8$, No rain)
R_7 = if $(X_1, 26.8) \wedge (X_2, 35) \wedge (X_3, 26.6) \wedge (X_4, 5) \wedge (X_5, 78) \wedge (X_6, 1005.1) \wedge (X_7, 1.7)$ then, $(X_8$, No rain)
R_8 = if $(X_1, 27.6) \wedge (X_2, 35.5) \wedge (X_3, 26.4) \wedge (X_4, 7) \wedge (X_5, 79) \wedge (X_6, 1005.5) \wedge (X_7, 1.6)$ then, $(X_8$, No rain)
R_9 = if $(X_1, 27.5) \wedge (X_2, 35) \wedge (X_3, 27.3) \wedge (X_4, 7) \wedge (X_5, 84) \wedge (X_6, 1004.4) \wedge (X_7, 2)$ then, $(X_8$, No rain)
R_{10} = if $(X_1, 26.3) \wedge (X_2, 34) \wedge (X_3, 26.4) \wedge (X_4, 8) \wedge (X_5, 87) \wedge (X_6, 1006.3) \wedge (X_7, 1.7)$ then, $(X_8$, Moderate rain)
R_{11} = if $(X_1, 23.5) \wedge (X_2, 30.7) \wedge (X_3, 25.3) \wedge (X_4, 8) \wedge (X_5, 91) \wedge (X_6, 1007.2) \wedge (X_7, 1.3)$ then, $(X_8$, Moderate rain)
R_{12} = if $(X_1, 25) \wedge (X_2, 32.7) \wedge (X_3, 26.8) \wedge (X_4, 5) \wedge (X_5, 87) \wedge (X_6, 1007.2) \wedge (X_7, 1.3)$ then, $(X_8$, Moderately heavy rain)
R_{13} = if $(X_1, 26.5) \wedge (X_2, 34.5) \wedge (X_3, 26.6) \wedge (X_4, 7) \wedge (X_5, 82) \wedge (X_6, 1006.6) \wedge (X_7, 1.5)$ then, $(X_8$, No rain)
R_{14} = if $(X_1, 27) \wedge (X_2, 32) \wedge (X_3, 26.6) \wedge (X_4, 5) \wedge (X_5, 88) \wedge (X_6, 1005.6) \wedge (X_7, 1)$ then, $(X_8$, No rain)
R_{15} = if $(X_1, 26.8) \wedge (X_2, 33.9) \wedge (X_3, 27.3) \wedge (X_4, 3) \wedge (X_5, 85) \wedge (X_6, 1004.2) \wedge (X_7, 1)$ then, $(X_8$, No rain)
R_{16} = if $(X_1, 27.2) \wedge (X_2, 36.2) \wedge (X_3, 27.3) \wedge (X_4, 2) \wedge (X_5, 80) \wedge (X_6, 1002.7) \wedge (X_7, 1)$ then, $(X_8$, No rain)
R_{17} = if $(X_1, 28.2) \wedge (X_2, 34.5) \wedge (X_3, 26.8) \wedge (X_4, 7) \wedge (X_5, 89) \wedge (X_6, 1002.4) \wedge (X_7, 1.5)$ then, $(X_8$, No rain)
R_{18} = if $(X_1, 25) \wedge (X_2, 32.4) \wedge (X_3, 25.8) \wedge (X_4, 6) \wedge (X_5, 87) \wedge (X_6, 1002.7) \wedge (X_7, 1)$ then, $(X_8$, Moderately heavy rain)
R_{19} = if $(X_1, 25.5) \wedge (X_2, 31) \wedge (X_3, 26) \wedge (X_4, 7) \wedge (X_5, 90) \wedge (X_6, 1002.3) \wedge (X_7, 1.7)$ then, $(X_8$, Heavy rain)

Table 6 Distance from different rainfall status

Euclidian Distance	Rainfall Status	Rank	Euclidian Distance	Rainfall Status	Rank
22.71827	Light Rain (1-10 Mm)	24	21.47464	No Rain	15
23.00348	No Rain	25	25.24361	No Rain	29
21.60463	Light Rain (1-10 Mm)	18	21.42405	Moderately Heavy Rain (23-43 Mm)	13
19.69365	Light Rain (1-10 Mm)	3	23.52424	Heavy Rain (44-88 Mm)	27
20.46143	No Rain	7	24.77035	Moderately Heavy Rain (23-43 Mm)	28
21.08886	No Rain	9	23.02477	Moderately Heavy Rain (23-43 Mm)	26
19.66977	No Rain	2	20.39975	Light Rain (1-10 Mm)	6
20.7330	No Rain	8	19.44634	Moderately Heavy Rain (23-43 Mm)	1
22.4232	No Rain	23	22.38683	No Rain	22
21.898	Moderate Rain (11-22 Mm)	21	22.0061	Light Rain (1-10 Mm)	21
21.49372	Moderate Rain (11-22 Mm)	16	21.7	No Rain	19
19.95695	Moderately Heavy Rain (23-43 Mm)	4	21.44854	No Rain	14
20.01225	No Rain	5	21.14025	Light Rain (1-10 Mm)	10
21.80229	No Rain	20	21.15065	Moderate Rain (11-22 Mm)	11
21.27863	No Rain	12	21.55621	Moderate Rain (11-22 Mm)	17

In December 2017, different rainfall regulating parameters are observed as $X_1 = 15$, $X_2 = 28.5$, $X_3 = 15.8$, $X_4 = 0$, $X_5 = 77$, $X_6 = 1013$ and $X_7 = 2.3$ at Ishwardi station. The Euclidian distances on different days are calculated as follows:

Here, the minimum distances are highlighted using green color. Then the final voting is done based on minimum distance. The value of K is considered 8 for this experiment.

From the Table 7, it can be decided that there will be No Rain on 1st December which is correctly judged with the historical data (Figs. 3 and 4).

6 Experimental Result

To visualize the accuracy, the result of the system has been demonstrated against the benchmark result (Table 8).

To validate the system accuracy, the system was tested 200 times with the test data provided by BMD and the accuracy reach up to 97% (Fig. 5).

A number of existing systems are examined in Fig. 6 to validate that the proposed system performs better than all the existing ones. For that, ANN, SVM, Naïve Bayes, Decision Tree, Rough Set Theory, Multilayer Perceptron, Fuzzy set theory, and KNN

Table 7 Voting in KNN

Rain level	No rain	Light rain	Moderate rain	Moderately heavy rain	Heavy rain	Very heavy rain	Extreme rain	Moderate extreme rain	Heavy extreme rain
Votes	4	2	0	2	0	0	0	0	0

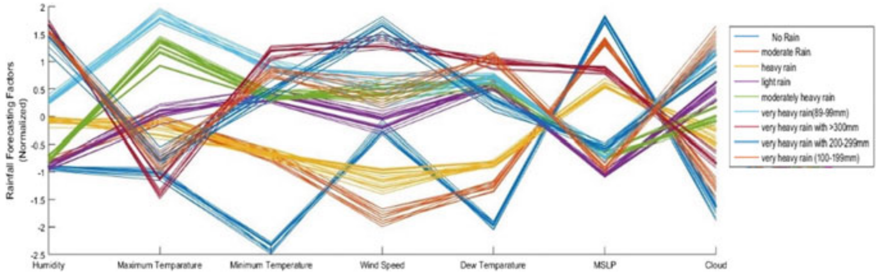


Fig. 3 Graphical representation of the training dataset

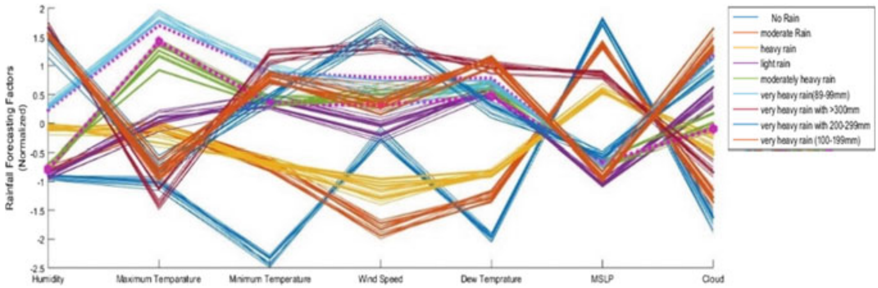


Fig. 4 Graphical representation of the training dataset with 2 test data ("….."denotes test data)

were implemented. BMD data were used to train this system and each system was tested at least 200 times to calculate its accuracy.

The ROC (Receiver Operating Characteristic) curves demonstrate the forecasting accuracy of different rainfall prediction models. The AUC (Area under Curve) explains that the proposed Fuzzy KNN provides the maximum forecasting accuracy which reaches up to 97% (Fig. 5).

7 Conclusion

Rainfall is considered as the most important meteorological phenomena for its direct impact on human life and prediction complexity. A number of systems have been developed to deal with this climatic factor. Many machine learning algorithms were used but none of those systems could forecast rainfall perfectly. However, this paper presents an intelligent system that can forecast rainfall certainly. The proposed system has been developed using Fuzzy KNN. The system was trained with more than 4 lacks data collected from BMD. The Euclidian distance was used to implement KNN approach in this system. This system was tested more than 200 times its forecasting accuracy was found 97%.

Table 8 Comparison between test results and benchmark results

Test run	Predicted results	Benchmark results	Test run	Predicted results	Benchmark results
1	No rain	No rain	16	Moderate rain	Moderate rain
2	No rain	No rain	17	No rain	No rain
3	Light rain	Light rain	18	Moderately heavy rain	Moderately heavy rain
4	Very heavy rain	Very heavy rain	19	No rain	No rain
5	Moderate rain	Moderate rain	20	Very heavy rain with 200–299 mm	Very heavy rain with 200–299 mm
6	Heavy rain	Heavy rain	21	Very heavy rain with 200–299 mm	Very heavy rain with 200–299 mm
7	No rain	No rain	22	Heavy rain	Heavy rain
8	Moderately heavy rain	Moderately heavy rain	23	Heavy rain	Heavy rain
9	Very heavy rain	Very heavy rain	24	Very heavy rain with greater than 300 mm	Very heavy rain with greater than 300 mm
10	Very heavy rain with 100–199 mm	Very heavy rain with 100–199 mm	25	Very heavy rain with 200–299 mm	Very heavy rain with 200–299 mm
11	Very heavy rain with 100–199 mm	Very heavy rain with 100–199 mm	26	Very heavy rain with greater than 300 mm	Very heavy rain with greater than 300 mm
12	Moderate rain	Moderate rain	27	Moderately heavy rain	Moderately heavy rain
13	Moderately heavy rain	Moderately heavy rain	28	Very heavy rain with 200–299 mm	Very heavy rain with 200–299 mm
14	Moderate rain	Moderate rain	29	Moderately heavy rain	Moderately heavy rain
15	Moderately heavy rain	Moderately heavy Rain	30	Moderately heavy rain	Moderately heavy rain

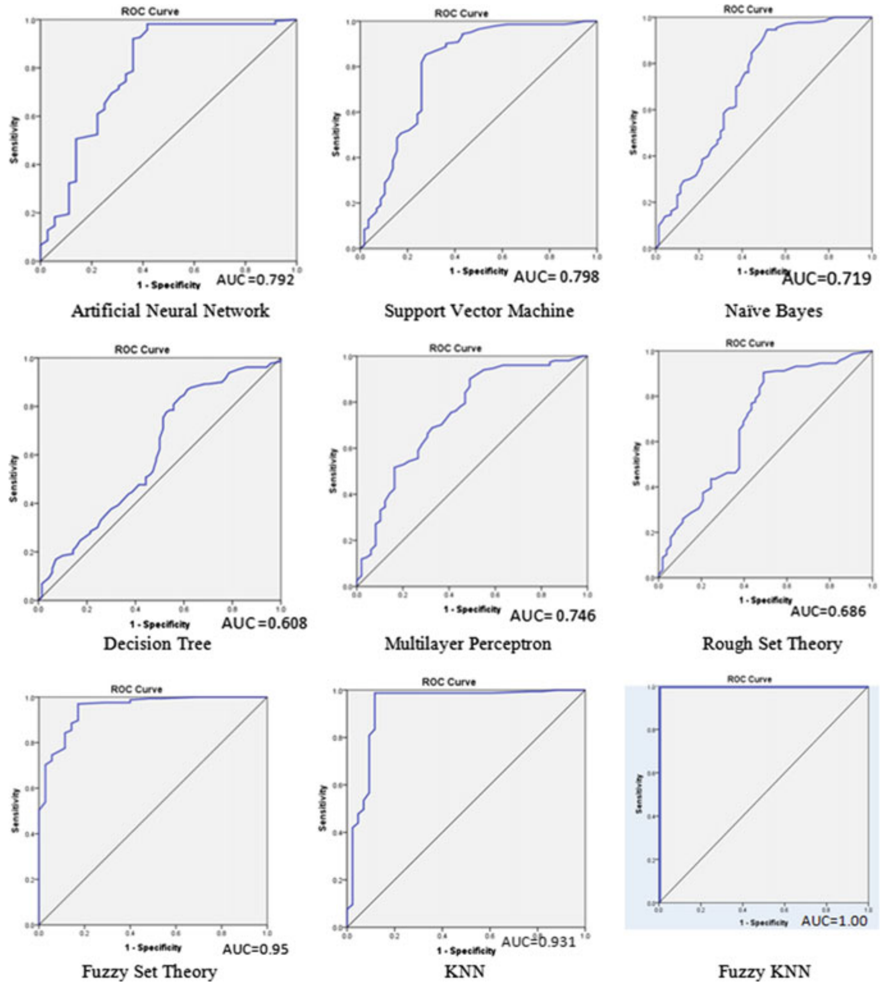


Fig. 5 ROC curves for representing the forecasting accuracy of different methodologies

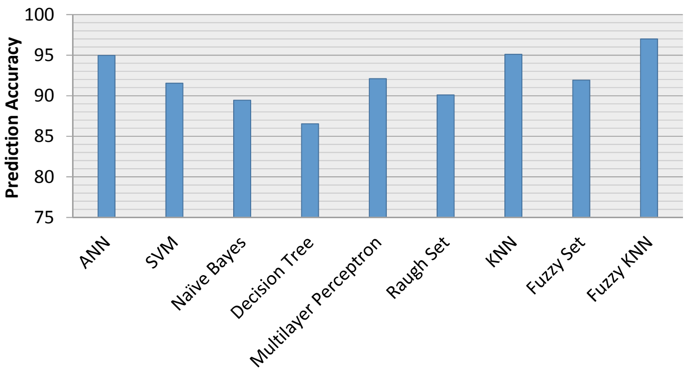


Fig. 6 Comparison among different Rainfall Forecasting Models

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Chapter 42
AdaBoost Classifier-Based Binary Age Group Stratification by CASIA Iris Image

Nakib Aman Turzo and Md. Rabiul Islam

1 Introduction

Thomas Kuhn argued that science progresses by running into problems and then makes a major shift in its paradigms or models. People have seen such a shift as available data becomes richer and possible explanations multiply. Age estimation is the determination of age of person based on various features and can be accomplished through different traits. It is a crucial parameter in archeological and forensic context. Assessing age in archeology offers important information on demographics of population. In addition, in living age estimation is important in migration crisis that takes place in different parts of the world. Artificial intelligence, machine learning and computer vision have been carried out in recent years at very advanced level. Undoubtedly, it is a huge contribution in rapid progress of these studies. Some people hide their personal data like age, gender, etc., and their social nature changes with age. It also depends on behavior of different people. Sometimes, behavior of same age groups is similar and is known as homogeneous behavior. Sometimes, visual information also provides age estimation accurately. Binary classifiers are often used because they possess fairness in the sense of not overly discriminating with respect to a feature deemed sensitive, e.g., age, race, etc. Novel methods were introduced for age estimation in which facial detection and speech signals detection were also included. Age classification is helpful in issuing different permissions at different levels. Nowadays, a captivating topic in iris biometric is being used to resolve age from image of iris. Studies showed that with time, an increase in false non-matching rate due to natural iris ageing. It is more difficult to estimate age on eye snapshot because the tempo at which structure/characteristics of human eye are changed is not very well known. In establishing this biometric research, the community has compromised with earlier findings that iris remains stable throughout life. In this

N. A. Turzo (✉) · Md. Rabiul Islam
Department of Computer Science & Engineering, Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh

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paper, data-driven approach was utilized by using machine learning to elucidate the categorization in age estimation and obtained and implemented CASIA dataset.

2 Literature Review

A new divided and conquer-based method was proposed called fusion of multiple binary age grouping estimation system for estimation of facial age of human. Multiple binary grouping systems were employed. Each face image was classified into one of the two groups. Two models were trained to estimate ages for faces classified into their groups. Two models were trained to estimate ages. Also, the investigation was done on effect of age grouping on performance accuracy [1]. Primary features of the face are determined first in an implementation, and secondary primary features were analyzed. Then, the ratios of babies from adults were being computed. In secondary analysis, a wrinkle geography map is used to guide the detection and measurement of wrinkles [2].

The database for FG-NET aging was released in 2004 to support activities for research which is related to facial aging. Conclusions were related to the type of research carried out in relation to impact of dataset in shaping up the research topic of facial aging, were presented [3]. A new age estimation method was introduced based on the fusion of local features extracted using histogram-based local texture descriptors. Different performances of well-known powerful texture descriptors with improved modification local binary patterns and local phase quantization, which were not analyzed in depth for age estimation, also investigated. Age estimation accuracy of proposed method was better when compared to FG-NET [4].

Age recognition from face image totally relies on reasonable aging description. Aging description needs to be defined with detailed local information. However, it relies highly on the appropriate definition of different aging affiliated textures. Wrinkles are discernible textures in this regard owing to their significant visual appearance in aging human. Local edge prototypic patterns preserve different variations of wrinkle patterns appropriately in representing the aging description [5]. Estimating human age has become an active research. Kernel partial least square methods were introduced. This method has several advantages over other approaches. It can find small number of latent variables and its better than SVM method [6].

Automatic gender and age classification have become relevant to an increasing amount of applications. A simple convolutional net architecture was proposed that could be used even when amount of learning data is limited [7]. A paper proposed an approach to age anticipation from iris images utilizing a combination of a small number of simplest geometric features and more intelligent classifier structure which can get precision to 75% [8].

Anticipation of gender by iris images was also done by utilizing different types of features and by using an intelligent classifier. The gender anticipation accuracy obtained was 90% [9]. A common feature of communicating in online networks happens through short messages. These features make non-standard language texts a little bit complex. To predict age and gender on a corpus of chats, it was determined what features suit best [10].

Recognition of gender is fundamental operation in face analysis. Gender recognition was done on real-life faces using recently built database. Higher precision was obtained by this method [11]. In a research, it was proposed to learn a set of high-level feature representations through deep learning for face verification. It was proposed that DeepID can be effectively learned through multi-class face identification tasks. Proposed features were taken from various face regions to form complementary and over-complete representations. Any state-of-the-art classifiers can be learned from these levels [12].

A detailed outline was made on MRPH a longitudinal face database developed for researchers investigating different facets of adult age. It contributed to several active research areas. This got highlighted in the evaluation of standard face recognition algorithm which illustrated the impact that age progression has on recognition rates [13]. Face recognition has benefitted greatly from many databases. Specific experimental paradigms were provided for which database was suitable [14].

Resting state functional MRI investigations have demonstrated that much of the large-scale functional network architecture supporting motor, sensory and cognitive functions in older pediatric and adult population in present term and prematurely born infants. Support vector regression enabled quantitative estimation of birth gestational age [15]. A recent study showed that the three traits can be estimated simultaneously based on multi-label regression problem. An investigation was done on canonical correlation analysis-based methods including linear CCA, regularized CCA and kernel CCA, and PLS models were compared in solving joint estimation problem and found a consistent ranking of five methods in age, gender estimation [16].

Recent studies human faces in human–computer interaction field revealed significant potentials for designing automatic age estimation systems via face image analysis. Through manifold method of analysis on face images, the dimensionality redundancy of the original image space can be significantly reduced with subspace learning [17]. A paper discussed about different configurations for estimating soft biometric traits. A recent framework was introduced which includes an initial facial detection, the subsequent facial traits description, the data reduction step and final classification step. The algorithmic configurations were featured by different descriptors and different strategies to build the training dataset [18].

A method was proposed for using dynamic methods for age estimation. This showed significant improvement in accuracy in comparing sole use of appearance-based features [19]. There were findings of adult age-related craniofacial morphological changes. The aims were based on two things which were on factors influencing craniofacial aging and general features about aging of head and face [20].

Ahmad Radzi et al. proposed a new system for pattern recognition of human face with the traditional four layers of CNN model. The system worked the facial images with different facial expression poses, occlusions and changing illumination and showed a result with 99.5% accuracy [21]. In another work, for the iris recognition system, they used CNN to extract the deep features. IIT iris dataset and CASIA iris dataset were used, and an accuracy of 99.4% was achieved [22].

3 Methodology

In this research work, from CASIA iris version 4.0, a total 1214 images was selected to predict the age group of people. In the proposed system, we divided the two subsets in a binary group as youth and adult (Fig. 1).

The total system is divided into three parts. The input images were taken for segmentation. The images were segmented to iris circle and pupil circle. The main feature iris–pupil ratio was calculated by using the pupil inner and outer boundary. The segmentation part was implemented by the popular method of Daugman's Integro-Differential Operator. For some images, the iris was not visible due to the imbalanced angle or presence of eyelids. After calculating the ratio, the outcome data was labeling in two groups. For training and testing, the whole dataset was fragmented in different combinations of subsets. The testing part occurred using the popular AI models and compare the results to find the best tuned method.

MATLAB functions used to compute the ratio then create the CSV format for both subsets of data numerical data of visualization. The ratio was labeling to '0'

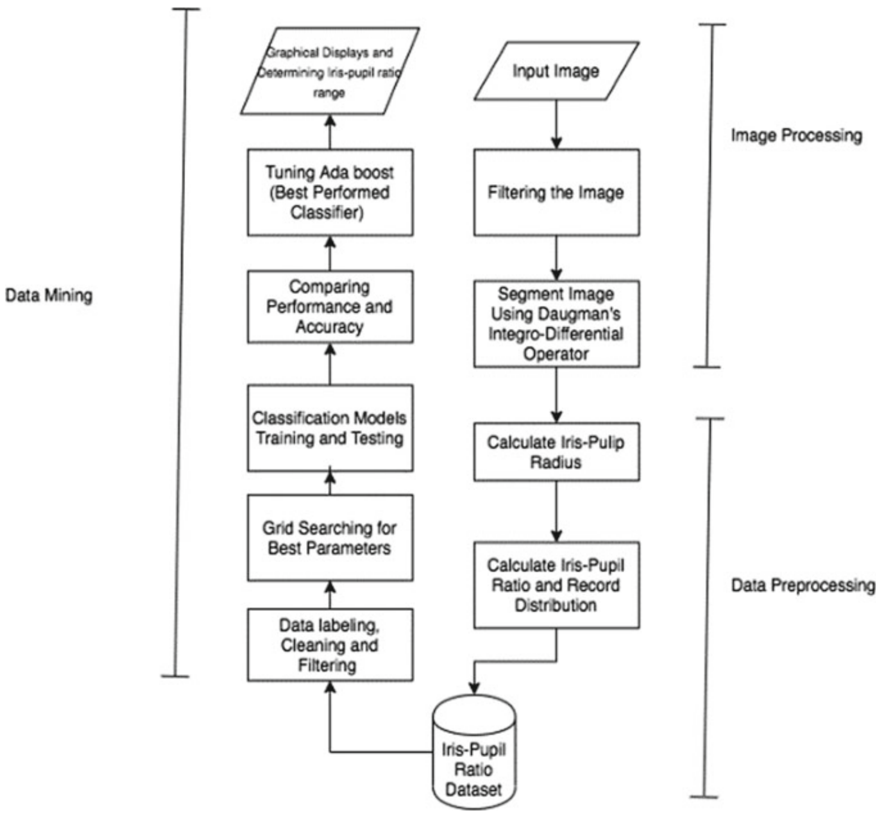

Fig. 1 Diagram of proposed methodology

and ‘1’ of output dataset of iris–pupil ratio uses to train for further classification of age grouping as next part. For classification, data mining is a faster approach to training and testing. For classification algorithms, the inputs were given in a numeric format of data, not in image format. So, it took a shorter time to train the data.

Daugman’s Integro-Differential Operator specially uses to detect the inner and outer boundary of iris as a circular age detector. This operator performs based on illumination difference is maximum between inside and outside of pixels in iris edge circle.

$$\max(r, x_0, y_0)\left|G_\sigma(r) * \frac{\partial}{\partial r} \oint\limits_{r,x_0,y_0} \frac{I(x, y)}{2\pi r}\right| \mathrm{d}s \tag{1}$$

In the equation expressed above, $I(x, y)$ contains an iris image, r is the increasing radius, center coordinates (x_0, y_0), * denotes convolution, G_σ is the Gaussian smoothing function and coordinates (r, x_0, y_0) define the path of contour integration [8]. The line integral part firstly calculates the area of iris without the inner circle (pupil).

Here, in the case of CASIA version 4.0 dataset images, the range of pupil radius is from 16 to 30 mm, and the iris outer boundary is approximately a minimum 20% to maximum 50% radius size than inner boundary [23]. The radius of iris and pupil and ratio is the resided in a dataset. The iris–pupil ratio dataset is then used for classification using various classifiers.

4 Experimental Result and Performance Analysis

Dataset was labeled at first place. For categorization, PyCaret was used. For this investigation, 1214 distinct data of more than 400 subjects were used. In case of this research, images of both left and right eye images were taken into consideration. The whole dataset was divided into two subsets in a binary group as youth and adult, and different representations like ‘0’ for child and ‘1’ for adult were used.

Training and testing dataset were proportioned as 9:1. The experimental results show that among all these data mining techniques, AdaBoost classifier gives the most balanced $F1$ score.

Fourteen varied classifiers were utilized for training and gave following results (Table 1).

From the table, it can be concluded that AdaBoost classifier is giving the highest accuracy, and then, the classifier will be tuned with respect to dataset. An output of approximately 70.83% was obtained from un-tuned version of AdaBoost which was the most accurate and highest value.

To understand the performance of any classification algorithms, the precision and recall value can be calculated from the confusion matrix. From the precision value, we can determine out of all the positive classes we have predicted correctly, how

Table 1 Results accuracy using various classifiers

Model	Accuracy	AUC	Recall	Precision	F1
AdaBoost classifier	**0.7083**	**0.7935**	**0.8210**	0.6337	**0.7116**
Extreme gradient boosting	0.7070	0.7906	0.8122	0.6334	0.7091
Gradient boosting classifier	0.6966	0.7841	0.7708	0.6305	0.6905
CatBoost classifier	0.6966	0.7790	0.7260	0.6402	0.6769
Light gradient boosting machine	0.6939	0.7783	0.6991	**0.6444**	0.6682
Random forest classifier	0.6756	0.7625	0.6602	0.6269	0.6402
K neighbors classifier	0.6731	0.7540	0.6515	0.6305	0.6357
Decision tree classifier	0.6705	0.7507	0.5892	0.6417	0.6096
Extra trees classifier	0.6692	0.7465	0.5951	0.6355	0.6104
Linear discriminant analysis	0.6640	0.7230	0.7197	0.6003	0.6530
Naïve Bayes	0.6550	0.7432	0.8184	0.5760	0.6755
Quadratic discriminant analysis	0.6535	0.7125	0.7291	0.5891	0.6436
Ridge classifier	0.6483	0.0000	0.5950	0.6005	0.5958
Logistic regression	0.6247	0.6882	0.5201	0.5806	0.5464
SVM-linear kernel	0.5635	0.0000	0.5403	0.3673	0.4013

many are actually positive as to how useful the search results are.

$$\text{precision} = \frac{tp}{tp + fp} \tag{2}$$

where t_p and f_p define True Positive and False Positive value.

And the recall is the idea of find of all the positive classes, how much we predicted correctly as to how complete the results are. It should be as high as possible. Here, f_n is called False Negative result.

$$\text{recall} = \frac{tp}{tp + fn} \tag{3}$$

In Fig. 2, features are automatically named according to their index in the input. For instance, the feature '46' radius of iris value is of highest importance for AdaBoost classifier. In case of light gradient boosting classifier, the iris–pupil ratio of '2.5556' is of highest importance.

From Fig. 3, it can be seen that AdaBoost classifier has achieved around 79.22% accuracy with only 80 total misclassification. The tuned results of the classifiers are given in Fig. 4.

From Fig. 5, it can be summarized that AdaBoost classifier resulted in best accuracy due to its high degree of precision. AdaBoost classifier has performed batter that Random Forest Algorithm as it fully considered the weight of each classes.

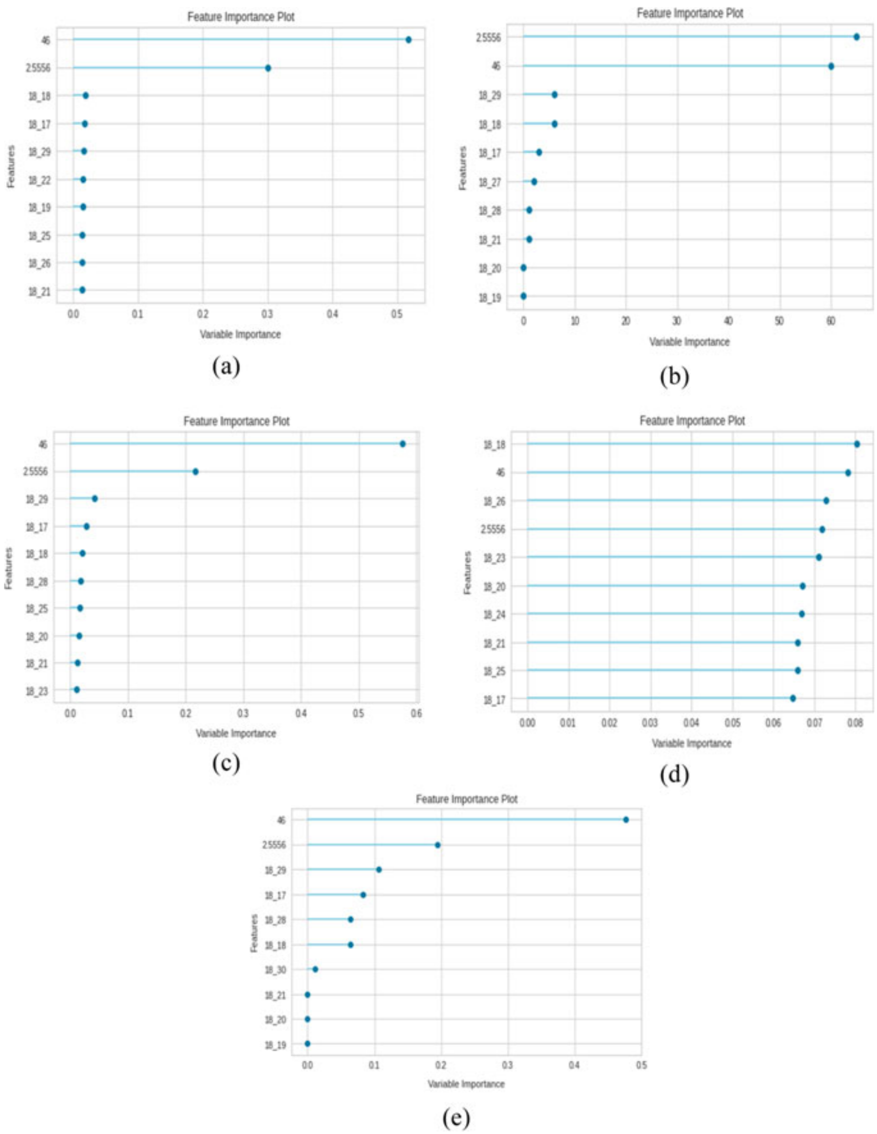

Fig. 2 Feature selection graphs of AI classifiers; **a** AdaBoost, **b** Gradient Boosting, **c** Light Gradient Boosting Machine, **d** Random Forest **e** Extreme Gradient Boosting

The ROC curve of top six classifiers are shown in Fig. 6. The steeper the curve the better the accuracy. Hence, AdaBoost classifier has the steepest curve.

From all of the above analysis, it can be concluded that AdaBoost classifier has been more accurate in classifying age group from iris–pupil ratio. AdaBoost classifier has performed well specially better than the closest competitor different boosting

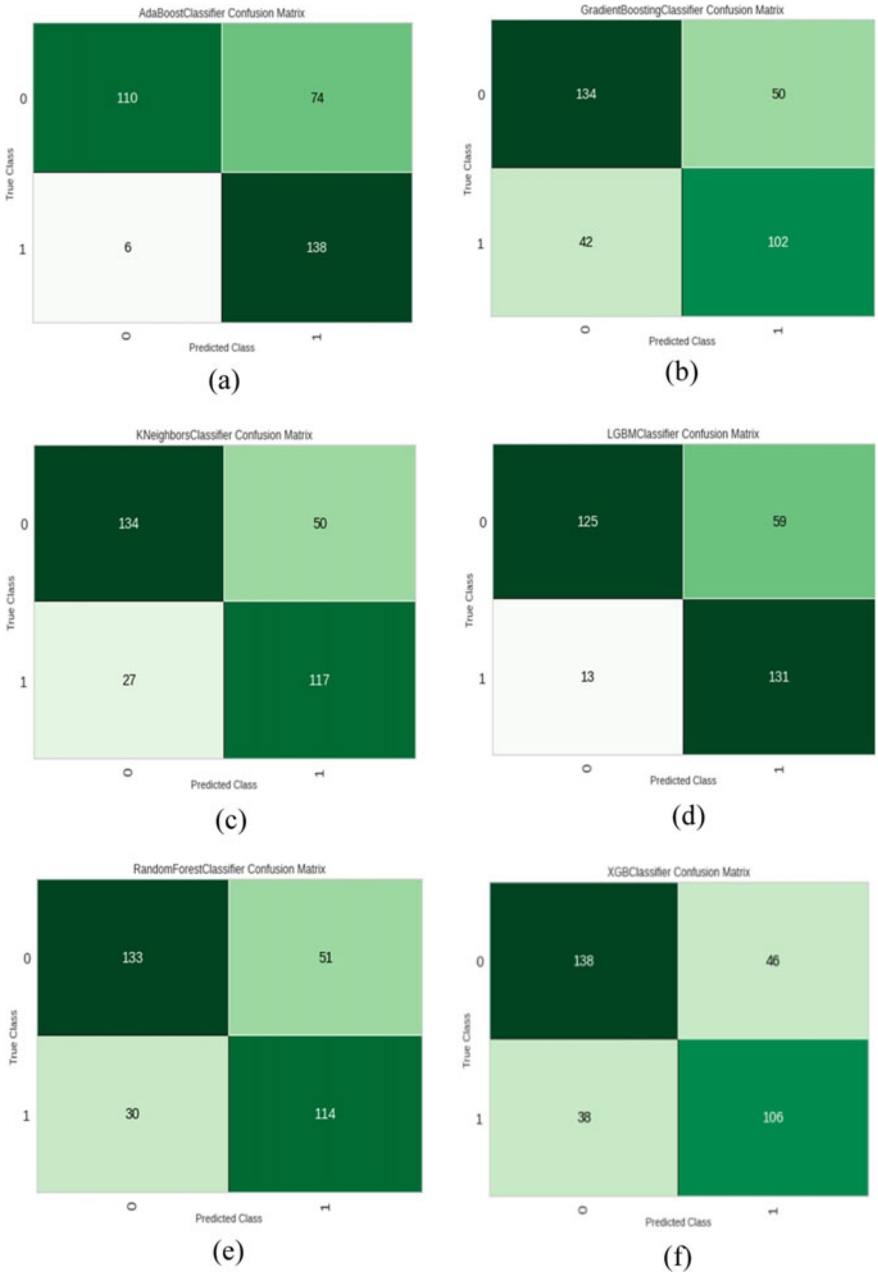

Fig. 3 Confusion Matrix of Classifiers; **a** AdaBoost, **b** Gradient Boosting, **c** K-Nearest Neighbors, **d** Light Gradient Boosting, **e** Random Forest, **f** Extreme Gradient Boosting

	Accuracy	AUC	Recall	Prec.	F1	Kappa
0	0.7532	0.7924	0.9118	0.6596	0.7654	0.5189
1	0.7273	0.7996	0.8824	0.6383	0.7407	0.4683
2	0.8052	0.8420	0.9412	0.7111	0.8101	0.6179
3	0.7922	0.8584	0.9412	0.6957	0.8000	0.5937
4	0.7273	0.8249	0.9412	0.6275	0.7529	0.4745
5	0.7895	0.8259	0.9091	0.6977	0.7895	0.5861
6	0.6974	0.7664	0.9394	0.5962	0.7294	0.4227
7	0.7237	0.7741	0.9394	0.6200	0.7470	0.4694
8	0.7237	0.7833	0.8182	0.6429	0.7200	0.4549
9	0.6579	0.7286	0.9118	0.5741	0.7045	0.3448
Mean	0.7397	0.7996	0.9135	0.6463	0.7560	0.4951
SD	0.0436	0.0370	0.0370	0.0428	0.0332	0.0806

(a)

	Accuracy	AUC	Recall	Prec.	F1	Kappa
0	0.7013	0.7880	0.7353	0.6410	0.6849	0.4035
1	0.6883	0.7497	0.6765	0.6389	0.6571	0.3719
2	0.7532	0.8417	0.6471	0.7586	0.6984	0.4918
3	0.7792	0.8529	0.8235	0.7179	0.7671	0.5591
4	0.6883	0.7808	0.7647	0.6190	0.6842	0.3832
5	0.7500	0.8048	0.7879	0.6842	0.7324	0.5000
6	0.6579	0.7326	0.7576	0.5814	0.6579	0.3274
7	0.7368	0.8097	0.7879	0.6667	0.7222	0.4755
8	0.6184	0.7488	0.5758	0.5588	0.5672	0.2261
9	0.6316	0.7423	0.7647	0.5652	0.6500	0.2791
Mean	0.7005	0.7851	0.7321	0.6432	0.6821	0.4018
SD	0.0513	0.0400	0.0722	0.0623	0.0521	0.1003

(b)

	Accuracy	AUC	Recall	Prec.	F1	Kappa
0	0.7403	0.7904	0.8529	0.6591	0.7436	0.4891
1	0.7273	0.8081	0.7647	0.6667	0.7123	0.4554
2	0.7532	0.8314	0.7353	0.7143	0.7246	0.5012
3	0.7792	0.8557	0.8529	0.7073	0.7733	0.5618
4	0.7273	0.8160	0.8529	0.6444	0.7342	0.4651
5	0.7105	0.7833	0.7273	0.6486	0.6857	0.4190
6	0.7237	0.7326	0.8485	0.6364	0.7273	0.4586
7	0.7105	0.7992	0.7879	0.6341	0.7027	0.4270
8	0.7500	0.7829	0.7576	0.6944	0.7246	0.4965
9	0.6447	0.7206	0.7941	0.5745	0.6667	0.3068
Mean	0.7267	0.7920	0.7974	0.6580	0.7195	0.4580
SD	0.0338	0.0391	0.0484	0.0391	0.0284	0.0636

(c)

	Accuracy	AUC	Recall	Prec.	F1	Kappa
0	0.7013	0.8054	0.7941	0.6279	0.7013	0.4106
1	0.7013	0.7479	0.8235	0.6222	0.7089	0.4142
2	0.7532	0.8321	0.7059	0.7273	0.7164	0.4981
3	0.7792	0.8509	0.9118	0.6889	0.7848	0.5670
4	0.6883	0.7866	0.8235	0.6087	0.7000	0.3905
5	0.7105	0.8118	0.7273	0.6486	0.6857	0.4190
6	0.6842	0.7438	0.8485	0.5957	0.7000	0.3875
7	0.6842	0.8013	0.8182	0.6000	0.6923	0.3834
8	0.7105	0.7710	0.7273	0.6486	0.6857	0.4190
9	0.6184	0.7181	0.7941	0.5510	0.6506	0.2594
Mean	0.7031	0.7869	0.7974	0.6319	0.7026	0.4149
SD	0.0407	0.0396	0.0598	0.0473	0.0322	0.0752

(d)

	Accuracy	AUC	Recall	Prec.	F1	Kappa
0	0.7403	0.7962	0.8235	0.6667	0.7368	0.4860
1	0.7273	0.7640	0.7941	0.6585	0.7200	0.4587
2	0.7662	0.8615	0.7059	0.7500	0.7273	0.5231
3	0.7662	0.8516	0.8529	0.6905	0.7632	0.5374
4	0.7143	0.7931	0.8235	0.6364	0.7179	0.4380
5	0.7500	0.8041	0.8182	0.6750	0.7397	0.5034
6	0.6711	0.7474	0.8485	0.5833	0.6914	0.3641
7	0.6974	0.8062	0.7879	0.6190	0.6933	0.4030
8	0.6842	0.7604	0.7576	0.6098	0.6757	0.3749
9	0.6447	0.7094	0.8529	0.5686	0.6824	0.3142
Mean	0.7162	0.7894	0.8065	0.6458	0.7148	0.4403
SD	0.0392	0.0439	0.0445	0.0512	0.0269	0.0708

(e)

	Accuracy	AUC	Recall	Prec.	F1	Kappa
0	0.6753	0.7798	0.6765	0.6216	0.6479	0.3477
1	0.6494	0.7551	0.6176	0.6000	0.6087	0.2912
2	0.7273	0.8170	0.5882	0.7407	0.6557	0.4348
3	0.7273	0.7982	0.7059	0.6857	0.6957	0.4487
4	0.7273	0.7493	0.7941	0.6585	0.7200	0.4587
5	0.7500	0.8400	0.7273	0.7059	0.7164	0.4930
6	0.5921	0.6987	0.6061	0.5263	0.5634	0.1842
7	0.6842	0.7829	0.6364	0.6364	0.6364	0.3573
8	0.6579	0.7304	0.6061	0.6061	0.6061	0.3037
9	0.6447	0.7157	0.7647	0.5778	0.6582	0.3030
Mean	0.6835	0.7667	0.6723	0.6359	0.6508	0.3622
SD	0.0468	0.0428	0.0690	0.0605	0.0478	0.0911

(f)

Fig. 4 Testing results after tuning different parameters of AI Classifiers; **a** AdaBoost, **b** Gradient Boosting, **c** K-Nearest Neighbors, **d** Light Gradient Boosting Machine, **e** Random Forest, **f** Extreme Gradient Boosting

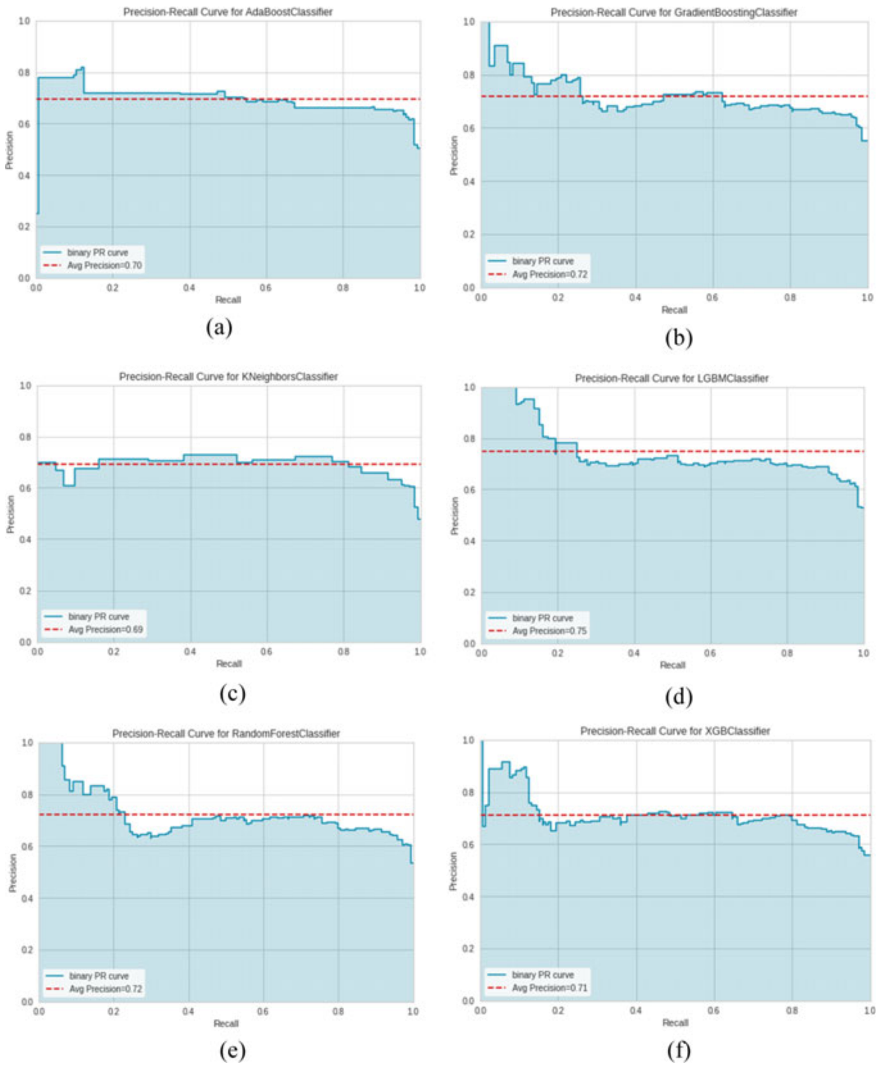

Fig. 5 Precision-recall curves of AI Classifiers; **a** AdaBoost, **b** Gradient Boosting, **c** K-Nearest Neighbors, **d** Light Gradient Boosting Machine, **e** Random Forest, **f** Extreme Gradient Boosting

algorithms due to its different take on creation of weak learner in the iterative process. AdaBoost through slower than most gradient boosting algorithms has given high accuracy as a result.

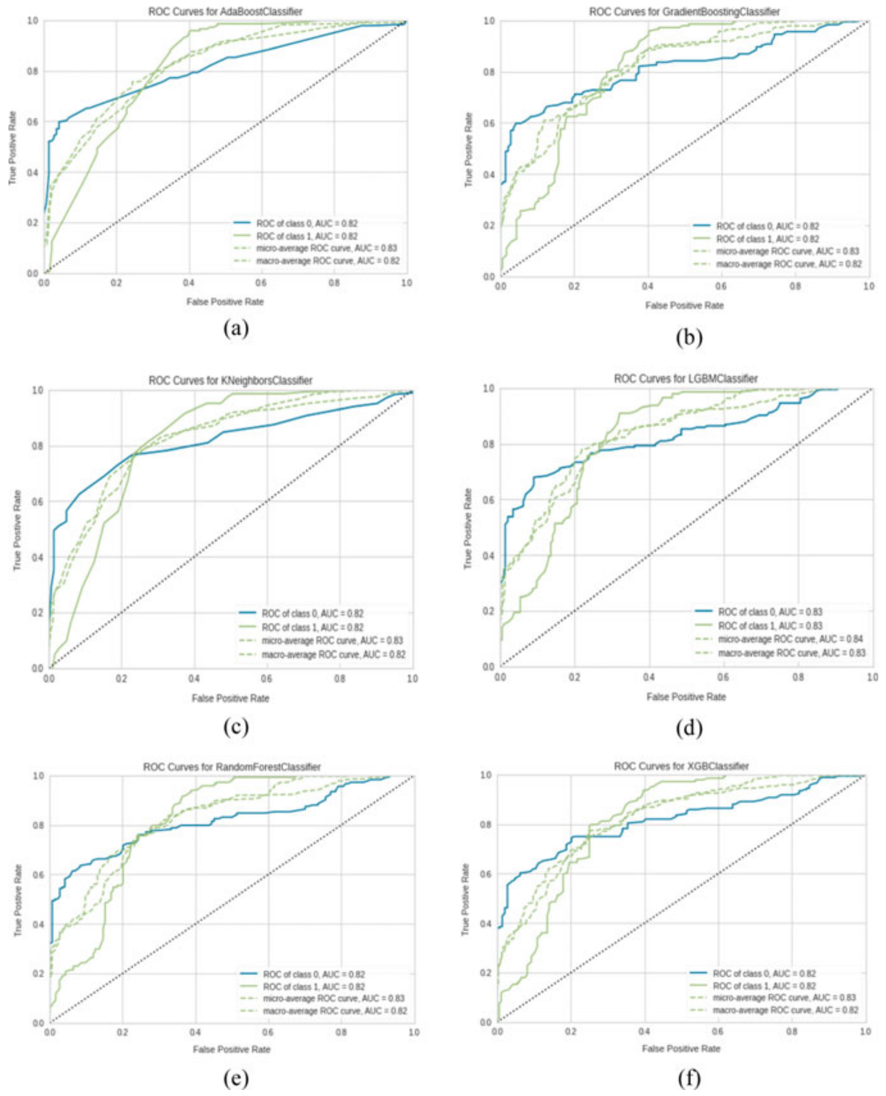

Fig. 6 ROC curves of AI Classifiers;**a** AdaBoost, **b** Gradient Boosting, **c** K-Nearest Neighbors, **d** Light Gradient Boosting Machine, **e** Random Forest, **f** Extreme Gradient Boosting

5 Conclusion

The proposed works find improved accuracy from the previous research using the same feature. The iris–pupil ratio can follow a vital property in template aging application also. From the working dataset, it can be deduced that the iris–pupil ratio range is 1.923–3.353 for younger and 1.929–4.611 for adults. Various categorizers were used for determining the accuracy, but a tuned AdaBoost classifier has proved to be highly precise with about 80.52% accuracy. As from pre-defined studies, we knew that iris–pupil contributes in aging but confusion matrix has showed that it has wrongly classified kids as adults many times due to segmentation errors and non-uniform dataset. As a future dimension, it is intended to do more research on other age groups to determine template aging.

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Chapter 43
Handwritten Indic Digit Recognition Using Deep Hybrid Capsule Network

Mohammad Reduanul Haque, Rubaiya Hafiz, Mohammad Zahidul Islam, Amina Khatun, Morium Akter, and Mohammad Shorif Uddin

1 Introduction

The topographical region which is surrounded by the Indian ocean is known as Indian subcontinent, and it is the home of certain dominant languages. Among them, Hindi (551 million), English (125 Million), Bengali (91 million), Telugu (84 million), Tamil (67 million), etc., are some of the mostly used languages [1]. Majority of people here prefer their mother tongue to read, write and talk with each other. According to a report published by KPMG in 2017 [2], the expected growth of Indian language internet users is about 18% each year, and as a result the total number of people would reach 536 million by 2021. 68% of them prefer digital contents on respective local language than the global language. So the overall Internet ecosystem of contents, applications, social media platforms, etc., needs to be more native language friendly. For these reasons, reading, printed or handwritten digits in any language and converting them to digital media are very crucial and time-consuming task. This is why recognition of handwritten Indic digits plays an active role in their day-to-day life.

A large-scale study has explored on Indic languages, i.e., language translation [3, 4], document categorization [5], handwritten script identification [6], etc. A lot of

M. Reduanul Haque (✉) · R. Hafiz
Department of Computer Science and Engineering,
Daffodil International University, Dhaka, Bangladesh
e-mail: reduan.cse@diu.edu.bd

R. Hafiz
e-mail: rubaiya.cse@diu.edu.bd

M. Zahidul Islam
Department of English, Daffodil International University, Dhaka, Bangladesh
e-mail: zahid.eng@diu.edu.bd

A. Khatun · M. Akter · M. Shorif Uddin
Department of Computer Science and Engineering,
Jahangirnagar University, Savar, Dhaka, Bangladesh

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work has already done with English [7–9], Hindi [10–12], Bengali [13], and Tamil [14] handwritten digit recognition. Leo et al. [15] proposed an artificial neural network and HOG feature-based system to recognize handwritten digit in various south Indian (Malayalam, Devanagari, Hindi, Telugu and Kannada) languages. They got a recognition accuracy of more than 82%. The overall classification rate for the same languages was 83.4%. A multi-language novel structural feature-based handwritten digit recognition system was proposed by Alghazo et al., and it was evaluated on six diverse languages, including Arabic Western, Devanagari, Bangla, Arabic Eastern, Persian and Urdu [16]. Total 65 local structural features were extracted, and among several classifiers random forest achieves the leading outcome with an average identification of 96.73%. Prabhu et al. [17] proposed a seed augment train/transfer (SAT) framework and tested it on real-world handwritten digit dataset of five languages. When a purely synthetic training dataset with 140,000 training samples was employed, they achieved an overall accuracy varying from 60 to 75% for five different languages. They also found that, if the training dataset is augmented with merely 20% of the real-world dataset, the accuracy is shot up by a significant amount.

Alom et al. introduced a deep learning-based handwritten Bangla digit recognition (HBDR) system and evaluated its performance on CMATERdb 3.1.1 (openly accessible database for Bangla numeral image) [18]. They have proposed a method which combines CNN with Gabor features, and it achieved 98.78% classification rate which outperforms the state-of-the-art algorithms for handwritten Bangla digit recognition. Another deep learning-based model was proposed by Ashiquzzaman and Tushar [19]. Their proposed method employed for Arabic numeral recognition was given 97.4% accuracy.

Recently, capsule network (referred to as CapsNet) introduced by Geoffrey Hinton encodes spatial information into features while using dynamic routing [20]. CapsNets has achieved state of the art only on MNIST dataset—a standard dataset of English handwritten digits [21]. From the above literature, it is clear that very few works have been reported for the Indic digit recognition. Diverse shapes, ambiguous handwritten digits and disproportionate cursive handwritings are attributed to some of the main reasons for this slow progress. Furthermore, most of the abovementioned recognition systems fail to converge the desired accuracy when exposed to multinumerals scenario. Hence, an script-independent procedure is needed that yields good classification accuracy. This failure has inspired us to introduce a numeral-invariant handwritten digit identification system for classifying digits written in five popular (i.e., English, Bangla, Devanagari, Tamil and Telugu) scripts. In this paper, we propose a hybrid method and compare the accuracy on top five Indian subcontinent digit datasets as well as explore how this architecture performs on these numerals that are marginally harder in specific ways. Our paper puts forward the following contributions:

1. A hybrid model is developed combining simple artificial neural network with capsule network.
2. Analyze the success and explore whether this method will perform well in more difficult conditions such as noise, color and transformations.

3. Compare the accuracy of the method on top five Indian subcontinent digit datasets and explore how this architecture performs on these numerals.

The forthcoming parts of this paper are fabricated as follows: Step 2 gives the general approach of our suggested scheme. Step 3 contains a brief discussion of the overall experimental results, and, eventually, we have concluded the paper in step 4.

2 Methodology

There are many approaches for handwritten digit recognition which may be broadly classified into two categories: classical approaches (e.g., BoF and support vector machine) and neural-based methods (e.g., simple neural network, deep convolutional neural network, transfer learning and capsule network). A brief description of these techniques is given below.

2.1 Capsule Network

In a capsule network [20], the network learns to render an image inversely; that is by looking at the image it tries to predict the instantiation parameters for that image. Initially, the input image is converted into a block of activations by employing convolution layer and supplied as an input into the primary capsule layer. Dynamic routing between primary capsules is calculated to generate the values of digit capsules. C_{ij} is the coupling coefficients and is used to combine the individual digit capsules which form the final digit capsule as follows.

$$C_{ij} = \frac{\exp(W_{\mathrm{ij}}^{\mathrm{DC}})}{\sum_k \exp(W_{\mathrm{ij}}^{\mathrm{DC}})} \tag{1}$$

A total S_j number of input vectors are processed by jth capsule to produce an activation vector v_j.

$$S_j = \sum_i C_{ij} \hat{U}_{j|i} \tag{2}$$

The resultant squashed combined digit capsule V_j is given by

$$V_j = \frac{\| S_j \|^2}{1+ \| S_j \|^2} \cdot \frac{S_j}{\| S_j \|} \tag{3}$$

s To produce the resultant digit capsule, Eqs. 1–3 are repeatedly performed.

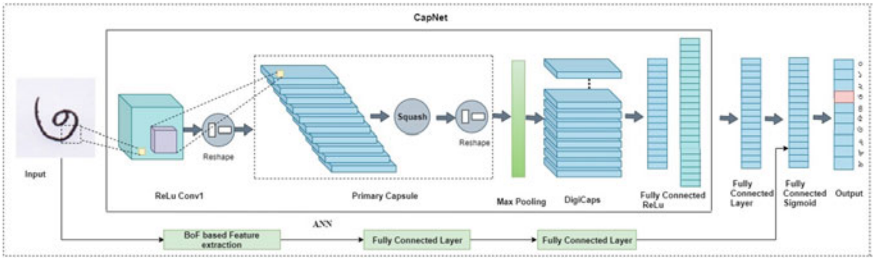

Fig. 1 General structure of our proposed hybrid capsule network

2.2 *Proposed Methodology*

The hybrid model consists of two parts as shown in Fig. 1. The method in the first part is a capsule network, and the second part is a simple artificial neural network with two hidden layers. The inputs to the first and second part are the whole image and the BoF feature vector extracted from images, respectively. The outputs obtained from these two parts are combined using minimum redundancy and maximum relevance (mRMR) [22] feature selection method, which is fed as input to two fully connected layers, and the last layer is a softmax activation function that provides the probability distributions of class predictions. Let $n_1(F_i)$ and $n_2(F_j)$ be the independent slices (slice 1 and slice 2) obtained from BoF ANN and capsule network, respectively. Then the two slices can be combined to feed the third components, $n_{12}(F)$ as given in Eq. 4 of the hybrid model (here, this component is performed as a softmax classifier according to the following combination or orthogonal sum).

$$
\begin{aligned}
n_{12}(F) &= n_1(F_i) \oplus n_2(F_j) \\
&= \sum_{y \in F_j} \sum_{x \in F_i} p(x, y) \log\Big(\frac{p(x, y)}{p_1(x) p_2(y)}\Big)
\end{aligned}
\tag{4}
$$

Here, $p(x, y)$, $p_1(x)$, $p_2(y)$ represents the joint and marginal probability distribution functions.

3 Experimental Results and Discussions

We have experimented our method with five different datasets each of which represents individual language. Some sample digit images of our datasets are shown in Table 1.

Table 1 Digits of different languages

English	0	1	2	3	4	5	6	7	8	9
Bangla	০	১	২	৩	৪	৫	৬	৭	৮	৯
Tamil	0	௧	௨	௩	௪	௫	௬	௭	௮	௯
Hindi	०	१	२	३	४	५	६	७	८	९
Telugu	౦	౧	౨	౩	౪	౫	౬	౭	౮	౯

3.1 Dataset Description

MNIST It is the standard set of normalized and centered images of handwritten digits which contains 60,000 training and 10,000 testing images.

NumtaDB The NumtaDB dataset holds 85,000 images of handwritten Bengali digits. 85% of them are considered as training and remaining 15% are used as test images. This dataset is very difficult to work with because of highly unprocessed and augmented images.

UJTDchar The UJTDchar dataset consists of 100 labeled images of each character in Tamil language.

Devanagari Devanagari dataset comprises images of 47 primary alphabets, 14 vowels, 33 consonants and 10 digits in png format. All the characters are centered within 28×28 pixels.

CMATERdb 3.4.1 We have collected handwritten Telugu numerals from CMATERdb database repository [23, 24].

3.2 Distortions

To find out the robustness of our method on all abovementioned datasets, we have imposed some deformations. As a result, we would be able to figure out the extent to which the model deformation is invariant for recognition.

For all datasets, an alternate, deformed dataset is generated by applying a random affine deformation consisting of

1. Rotation: Rotated image by a uniformly sampled angle within $[-20°, 20°]$
2. Shear: Sheared along x and y axes by uniformly sampling shear parameters within $[-0.2, 0.2]$. (Shear parameters are numbers added to the cross-terms in the 2×3 matrix describing an affine transformation.)
3. Translation: Translated along x- and y-axis by a uniformly sampled displacement parameters within $[-1, +1]$. (Displacement parameters are numbers added to the constant terms in the 2×3 matrix.)
4. Scale: Always scaled image 150%.

Table 2 Overall classification accuracy across five datasets (with and without affine transformations)

Datasets	Normal		Affine	
	CapsNet	Hybrid method	CapsNet	Hybrid method
MNIST	99.2	99.6	74.91	82.5
Tamil	91.8	95.5	72.9	80.05
Telugu	94.2	96.2	76.4	80.2
Hindi	94.8	96.1	75.3	82.9
Bangla	96.2	99.3	74.4	88.9

Table 3 Recognition accuracy under noisy situations (salt and pepper noise with density, d).

Dataset	Capsule network	Hybrid capsule network
MNIST	$d < 0.3$	$d < 0.7$
Telugu	$d < 0.2$	$d < 0.55$
Tamil	$d < 0.35$	$d < 0.73$
Bangla	$d < 0.22$	$d < 0.65$
Hindi	$d < 0.17$	$d < 0.5$

Table 4 Recognition accuracy under noisy situations (Gaussian noise with mean = 0 and variance, v).

Dataset	Capsule network	Hybrid capsule network
MNIST	$v < 0.3$	$v < 0.5$
Telugu	$v < 0.23$	$v < 0.45$
Tamil	$v < 0.25$	$v < 0.38$
Bangla	$v < 0.19$	$v < 0.6$
Hindi	$v < 0.12$	$v < 0.29$

Table 5 Recognition accuracy after random rotation ($-30°$ to $+30°$).

Dataset	CapsNet	Hybrid method
MNIST	77.68	85.53
Tamil	75.8	86.5
Telugu	73.2	80.9
Hindi	77.8	84.1
Bangla	76.2	88.3

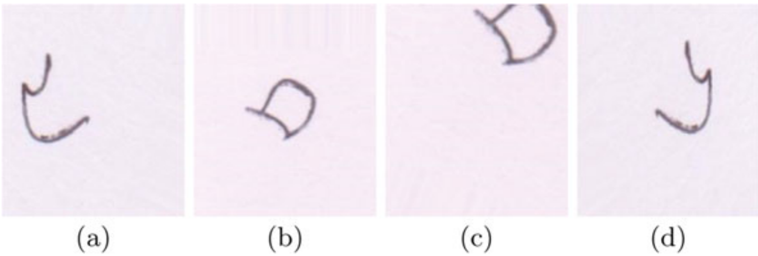

Fig. 2 Examples of images from the NumtaDB dataset. Digit images of **a**, **d** are recognized by all hybrid network correctly; however, failed to recognize **b**, and **c** due to their distortions (rotation and occlusion)

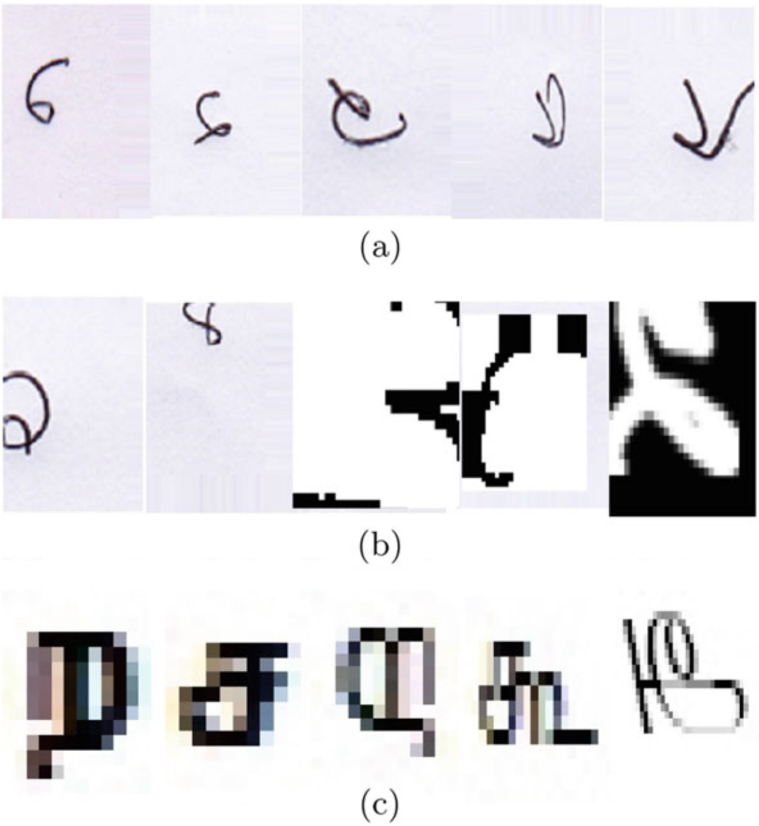

Fig. 3 Some misclassified images from different datasets due to various distortions/deformations

Tables 2, 3, 4 and 5 show the comparative accuracy results of top five Indian subcontinent digit datasets on different conditions, such as with and without affine transformations, with noise as well as random rotation. We can see that when hybrid method is imposed, it gives better accuracy in almost every cases. Few images from the NumtaDB dataset are shown in Fig. 2. Also Fig. 3 shows some misrecognized images from different datasets.

4 Conclusions

Accurate recognition of handwritten digits in real-world scenarios is a challenging task and has drawn considerable attention to the researchers over the last decades. In the work, a hybrid model is developed that combined capsule network and ANN with BoF. Experimental results confirm that the hybrid method gives better result (i.e., most robust) in comparison with other methods in terms of shear, scaling, noisy and rotational situations. Moreover, it gives an overall accuracy of more than 96% that is superior than capsule network.

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Correction to: Statistical Texture Features Based Automatic Detection and Classification of Diabetic Retinopathy

A. S. M. Shafi, Md. Rahat Khan, and Mohammad Motiur Rahman

Correction to:
Chapter 3 in: M. S. Uddin and J. C. Bansal (eds.), *Proceedings of International Joint Conference on Advances in Computational Intelligence*, Algorithms for Intelligent Systems, https://doi.org/10.1007/978-981-16-0586-4_3

The original version of the book was published with incorrect figure 3 in chapter 3. The wrong figure 3 replaced with the revised figure provided by the author. The chapter and book have been updated with the changes.

5	-3	-3
5	0	-3
5	-3	-3

0°

-3	-3	5
-3	0	5
-3	-3	5

45°

-3	-3	-3
5	0	-3
5	5	-3

90°

-3	5	5
-3	0	5
-3	-3	-3

135°

-3	-3	-3
-3	0	-3
5	5	5

180°

5	5	5
-3	0	-3
-3	-3	-3

225°

-3	-3	-3
-3	0	5
-3	5	5

270°

5	5	-3
5	0	-3
-3	-3	-3

315°

Fig. 3 Kirsch's convolution kernels

The updated version of this chapter can be found at
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